
Complex Analysis

— *EYNTKA* Series —

NATHANAEL CHWOJKO-SRAWLEY

October 1, 2026

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0.0.1 Reading the numbering A star marks *bonus material*: a chapter, section or subsection that belongs in the book but that nothing later depends on. Skipping it costs nothing downstream. The star sits on the number, as in §2.7★; in divisions written before this convention it sits at the front of the title instead, as in **Uniform Spaces*. The two mean the same thing.

Summary

On a high-level, Complex analysis is the study of functions $f : D \rightarrow \mathbb{C}$ where $D \subseteq \mathbb{C}$ is a union of open connected set where f shall satisfy the condition of being “complex differentiable”. The initial motivation for considering such functions can be seen as the natural extension of the study of real differentiable functions, however some reader may find it unsatisfying as the reader may wonder why we even need to consider analysis over the complex numbers. A better motivation for complex analysis can be found depending on what is the field of interest of the reader, as complex analysis is the foundation of many branches of mathematics:

1. (algebra) Complex-differentiable functions shall be shown to be the smallest extension of *polynomials* on which we can use our usual tools of analysis. Put in more technical terms, given the uniform-compact topology on the set of polynomials, the completion of this space will give the space of holomorphic functions over the union of connected open sets. In fact, many results in algebraic geometry have an “analytification” whose results lie in complex analysis
2. (geometry) For the reader familiar with manifolds, complex manifolds shall provide foundational objects of study. For example, by examining all closed 1-dimensional complex manifolds (which shall be called *Riemann surfaces* due to Riemann studying them) we shall find that there are only 3 possible geometries (i.e. 3 Riemann metrics) that we can put on a surface: flat, hyperbolic and spherical.
3. (PDE) complex-differentiable functions shall all be *harmonic functions*, and harmonic functions will be the collection of complex-differentiable functions as well as their conjugates. Thus, such functions are the foundation of the study of harmonic functions, and many result in this textbook have generalizations in harmonic analysis.

This book will start by covering the properties of complex numbers, including their arithmetic’s, algebraic, geometric, and analytic properties. It then covers important examples complex function, including fractional linear transformations and complete functions defined on the Riemann sphere (on the 1-point compactification of $\mathbb{C}$). After having defined the elementary properties of $\mathbb{C}$ and complex functions, we move on to introducing the main concept of the book: complex differentiable function. We will explore the consequence of a function being complex-differentiable including harmonic and conformal properties, and give a list of important examples of complex-differentiable functions. After proving all power series are complex-differentiable, we explore the canonical power series: e^x , $\cos(x)$, $\sin(x)$, $\log(x)$ and their related functions. We finish off by generalizing power series to functions that are locally power series, analytic functions, and gain some very important properties that make analytic functions much more powerful than most smooth functions.

Having built-up the notion of complex-differentiability, we shall explore integrating complex functions, more particularly 1-forms $f(z)dz$ where f is complex-differentiable. This will bring us many fruitful results in algebra, geometry, and PDE’s, and will give us results such as Cauchy’s Theorem, Cauchy’s Integral Formula, Maximum Modulus Principle, and Residue Theory.

After covering the fundamentals, we move on to the study of the collection of holomorphic functions. We shall find what are the compact subsets of the space of holomorphic functions, shows that holomorphic functions can factor (as sums or as products), and show some important behaviour that is “unique” to complex analysis, namely Picard’s Theorem.

We shall then move onto studying manifolds endowed with a complex structure. These shall allow for the classification of simply-connected complex 1-manifolds, and pave the way to Riemann geometry.

The following chapter will then analyze the first non simply-connected complex manifold, namely a complex manifold of genus one (more commonly known as an *elliptic curve*).

We shall then take a moment to focus on the fact that complex-differentiable functions satisfy *harmonic operators*, and look what are the consequences of this result (for example, we shall see that many of the results proven work just as well if only working over harmonic functions).

Finally, we shall do a quick overview of what happens when we consider complex analysis in higher dimensions, show how it is very different from real analysis in higher dimensions, and show how many of these ideas lead to the modern developments in algebraic geometry.

Prerequisites

This textbook will assume some basic familiarity with complex numbers and construction of complex number (see for example [18]). All results of a first year analysis class or an advanced calculus class will be assumed; this will include the basics of differential forms (see for example [19]). In chapter 4, familiarity with function spaces and different forms of convergence will be assumed (see for example [24]). In chapter 6, some basic familiarity with differential geometry will be assumed (see for example [23]).

1

Complex Numbers and Functions

In this chapter, we go over how to upgrade the real numbers into the complex numbers. The complex numbers are a wonderful mathematical object to work with: they form the smallest complete algebraically closed characteristic 0 field – about as good with properties in analysis (completeness), algebra (algebraic closure, commutative), and geometry (i can be interpreted as rotation, inner product space giving us angles, norms, and metric), and with a very simple construction from $\mathbb{R}$ (it's extension is only of [real] dimension 2!). These many nice properties provide us a variety of flexibility in the manipulation of complex numbers: we can represent them in polar or Euler form, we can find many useful identities, and we will gain an important tool on dealing with infinities.

What makes complex numbers interesting is the power of that complex functions bring. In complex analysis, a [single-valued] function with domain and codomain $\mathbb{C}$. In this way, the input will be thought of having the properties of $\mathbb{C}$, while the properties of the function shall also be interpreted through $\mathbb{C}$ (since $\mathbb{C}$ shall be it's codomain). It shall become very apparent in the rest of these chapters that complex-differentiable functions will have incredible properties that shall seem very foreign to anyone with intuition from real analysis. In this chapter, we shall start by getting comfortable with interpreting complex linear functions and complex polynomials. We shall then see that it is rather practical to consider continuous functions f which may take ∞ as an input and output ∞ . We shall do this by take the compactification $\mathbb{C}$ and analyzing the result. This shall lead us to a “new” linear function, the fractional linear function.

1.1 Arithmetic's and Conjugate

The rational numbers are the smallest set of numbers (with $\text{char} = 0$ ¹) in which the 4 arithmetic operations $(+, -, \times, \div)$ is well-defined. These make them the ideal starting point in the study of

¹That is, 1 added to itself will never equal to 0

arithmetic's, however, the rationals lack two key properties:

1. polynomials are central to algebra, however not all solutions to polynomials have solutions over $\mathbb{Q}$. We solve this by constructing a larger set of numbers called the *algebraic closure* of $\mathbb{Q}$ and usually denote it $\overline{\mathbb{Q}}$
2. Some sequences of rational numbers do not have a limit even when they seem to approach a limit (i.e. a Cauchy sequence). We rectify this by taking the *completion*, and usually denote it $\mathbb{R}$

These are two important properties, the first in the study of algebra and the second in the study of analysis. The complex numbers are the smallest object that combine these two while preserving the euclidean norm².

The real numbers $\mathbb{R}$ is the completion of $\mathbb{Q}$ with the euclidean metric, making it a very natural field to work with in analysis. However, $\mathbb{R}$ is not algebraically closed, meaning there are polynomials in $\mathbb{R}[x]$ that do not have solutions. The complex numbers rectify this by being the *algebraic closure* of $\mathbb{R}$. Due to this, I will take an algebraic approach to defining the complex numbers. Given the intimate connection between algebraic geometry and complex geometry, this approach can even be considered the more natural starting point for the definition of the complex numbers. To an algebraist, the complex numbers are often defined as the ring $\mathbb{R}[x]$ quotient by the maximal ideal $(x^2 + 1)$

$$\mathbb{C} := \mathbb{R}[x]/(x^2 + 1)$$

with $\bar{x}$ being identified with $\sqrt{-1}$. We may also identify $\bar{x}$ with $-\sqrt{-1}$, this field is sometimes labeled $\overline{\mathbb{C}}$ (the “conjugate” complex field), and we shall temporarily use this notation in this section (the notation shall be used for a more common structure later). The field $\mathbb{C}$ is a degree 2 extension, and so all the elements of $\mathbb{C}$ can be written of the form $a + b\sqrt{-1}$ for the root $\sqrt{-1}$ of $x^2 + 1$. By convention (established by Euler), we will write $i := \sqrt{-1}$, and we will also call the elements of $\mathbb{C}$ *numbers* (or *complex numbers* to be more precise when the context necessitates it). Addition and multiplication is defined as a natural consequence of the extension, that is:

$$(a + bi) + (c + di) = (a + c) + (b + d)i \quad (a + bi)(c + di) = (ac - bd) + (ad + bc)i$$

where 1 and i are the basis elements, and hence all elements of $\mathbb{C}$ can be written as a linear combination of $a + bi$ for $a, b \in \mathbb{R}$. We usually denote the real part of an element $z \in \mathbb{C}$ by $\operatorname{Re}(z)$ and the imaginary part by $\operatorname{Im}(z)$. Since $\mathbb{C}$ is a field, division is defined. As a mnemonic for the computation, if we want to find:

$$\frac{1}{a + bi}$$

then we then multiply this by a fancy 1;

$$\frac{1}{a + bi} \frac{a - bi}{a - bi} = \frac{a - bi}{a^2 + b^2} = \left(\frac{a}{a^2 + b^2} \right) - \left(\frac{b}{a^2 + b^2} \right) i$$

²up to appropriate notion of isomorphism; namely algebraically closed complete field with the euclidean, or “infinity”, norm $|\cdot|$ is isomorphic to $\mathbb{C}$; see the Gelfand-Mazur Theorem for an even more general way in which $\mathbb{C}$ can be seen as universal

It is not difficult, to show that $\mathbb{C}$ is also complete, and so we have that $\mathbb{C}$ is a complete algebraically closed field. In theorem 3.3.3, we will show $\mathbb{C}$ being algebraically closed means any polynomial $p(x) \in \mathbb{C}[x]$ has a root. We are taking this property for granted for now³.

As mentioned earlier, $\mathbb{R}[x]/(x^2 + 1)$ has two possible isomorphisms since $(\bar{x}^2 + 1) = (\bar{x} - 1)(\bar{x} + 1)$. We usually pick $\bar{x} = \sqrt{-1} =: i$ and label the resulting field $\mathbb{C}$, however we could have chosen $-i$. We know from algebra that there is a field isomorphism between $\mathbb{C}$ and $\overline{\mathbb{C}}$, showing the two are indistinguishable as fields. In particular, the isomorphism is:

$$a + bi \mapsto a - bi$$

Since the base field is $\mathbb{R}$, $\mathbb{C}$ and $\overline{\mathbb{C}}$ are even $\mathbb{R}$ -algebra isomorphic. This function is usually denoted $\bar{z}$. Since it's an $\mathbb{R}$ -algebra isomorphism:

$$\overline{zw} = \bar{z}\bar{w} \quad \overline{z + w} = \bar{z} + \bar{w} \quad \overline{rz} = r\bar{z}$$

Importantly, $\bar{z}$ is *not* $\mathbb{C}$ -linear, in particular for any $c \in \mathbb{C}$, $\overline{c \cdot z} = \bar{c} \cdot \bar{z}$. This is perhaps obvious since this is another formulation of the field isomorphism property, however it is in fact very important to remember this in the context of complex differentiation: complex-differentiable functions will be locally $\mathbb{C}$ -linear, and hence conjugation as a function $\mathbb{C} \rightarrow \mathbb{C}$ shall not be complex-differentiable. The function $\bar{z}$ is called the *conjugate* of z . Notice that this is *not* vertical equivalent to the function $z \mapsto -z$, since $-z$ does not reflect across the vertical axis, but *rotates* $\mathbb{C}$ by π degree (think $1 + i \mapsto -1 - i$ and $-1 + i \mapsto 1 - i$). Conjugation has many important algebraic properties, for example, conjugation let's us define many useful functions, such as⁴:

$$z\bar{z} = a^2 + b^2 \in \mathbb{R} \quad z + \bar{z} = 2a \in \mathbb{R}$$

that is, the maps $z \mapsto z\bar{z}$ and $z \mapsto z + \bar{z}$, which both maps $\mathbb{C} \rightarrow \mathbb{R}$. On the other hand,

$$z - \bar{z} = 2bi$$

letting us also isolate the complex part. Since $\bar{\bar{z}} = z$, we get that $\overline{(-)}$ is it's own inverse, which by definition makes it a type of function called an *involution*, which show up a lot in mathematics and usually serve a useful function (ex. if $x^2 = e$ in a group, then the group can be split into a product of groups, or if $A^2 = I$ for a matrix, the matrix is diagonalizable, or if $f^2 = 1$ for some function, then f is bijective, or $\neg\neg A = A$ for some logical statement A). For complex numbers, complex conjugation will give us a useful arithmetic tool that will simplify many identities (some examples of which will be listed bellow). If we have any polynomial $p(x)$ over $\mathbb{C}$, or even any rational function⁵ $p(x)/q(x)$, conjugation distributes over it:

$$\overline{\left(\frac{a_0 + a_1x + a_2x^2 + \dots + a_nx^n}{b_0 + b_1x + b_2x^2 + \dots + b_mx^m} \right)} = \frac{\overline{a_0} + \overline{a_1}x + \overline{a_2}x^2 + \dots + \overline{a_n}x^n}{\overline{b_0} + \overline{b_1}x + \overline{b_2}x^2 + \dots + \overline{b_m}x^m}$$

Three important uses of conjugation are:

1. we can represent the real or imaginary part of a complex number through the following equations:

$$\text{Re}a = \frac{a + \bar{a}}{2} \quad \text{Im}a = \frac{a - \bar{a}}{2i}$$

³This can also be proven using only the Intermediate value theorem if we used some group and field theory; see [22] for an example; or by using the homotopy group; see [20]

⁴Notice that these are the *trace* and *determinant* of the appropriate complex matrix

⁵the numerator and denominator are polynomials

2. Another example that is useful is the ability to represent the inverse of a complex number:

$$z^{-1} = \frac{\bar{z}}{z\bar{z}} \quad (1.1)$$

Naturally, in equation (1.1), the $\bar{z}$ cancel out, but computationally it is easier to understand it in this form since we often treat $\mathbb{C}$ as a vector-space over $\mathbb{R}$ and we want all its elements to be written in the form $a + bi$.

3. If r is a root of the $p(x)$ ($p(r) = 0$), then $\bar{r}$ is a root of $\overline{p(x)}$ (prove this). If $p(x) \in \mathbb{R}[x]$, then r and $\bar{r}$ are roots of $p(x)$.

1.2 Geometry and Analysis

In the following sections, we will explore the geometric properties of complex numbers.

1.2.1 Norms and Inequalities

When extending from the real to the complex numbers, they lose their total ordering (there is no “natural” ordering on them in which any two points have a finite distance and if $a < b$ and $b < a$ then $a = b$). However, $\mathbb{C}$ does a notion of size to every vector given by

$$N(z) = z\bar{z}$$

notice that $N(z) = \overline{N(z)}$ and so its output is real. This is called the *norm form* and is defined for any field extension. It has the two properties of:

$$N(zw) = N(z)N(w) \quad N(z) = 0 \Leftrightarrow z = 0$$

This particular norm has an important property: $N(z) > 0$ if $z \neq 0$, that is it is *positive definite*⁶. Since it is positive definite and has nice linearity properties, we can define an inner product via:

$$\langle v, w \rangle = \operatorname{Re}(z\bar{w})$$

This makes $\mathbb{C}$ is an inner vector space over $\mathbb{R}$ ⁷. We would write the induced norm as:

$$|z| := \langle z, z \rangle = N(z)^{1/2}$$

where the $|\cdot|$ notation is reminiscent of the absolute value notation from $\mathbb{R}$ (notice it in fact it continuously extends $|\cdot|$). One can verify that the axioms of a norm are satisfied by $\|\cdot\|$. Due to historical precedents, the norm is called the *modulus*:

⁶this is not the case for all norms, consider $\mathbb{Q}(\sqrt{2})$ where $N(a + \sqrt{2}) = a^2 - 2b^2$

⁷in particular, a complete normed vector space, making it a Banach space

Definition 1.2.1: Modulus (Norm)

Let $z \in \mathbb{C}$. Then define the function:

$$\|z\| = |z| = \sqrt{a^2 + b^2}$$

which is called the *modulus* of z and can be shown to satisfy the following properties:

1. **Sub-additivity or Triangle Inequality:** $\|x + y\| \leq \|x\| + \|y\|$
2. **Absolute Homogeneity:** $\|sx\| = |s| \|x\|$
3. **Positive definiteness:** If $\|x\| = 0$ then $x = 0$. Note that in tandem with (2) this means that $\|x\| = 0$ if and only if $x = 0$

making it a norm

Since we will be treating $\mathbb{C}$ as “numbers” in it of itself, then we will often abuse notation and write $|z|$ for the norm of z , and call it the *absolute value* or *modulus* of z . This norm naturally defines the metric $|z - w|$, and so the same topology as $\mathbb{R}^2$, and so any continuous function on $\mathbb{R}^2$ is continuous on $\mathbb{C}$. We get some important results about convergent sequences and series from real analysis that carry over to complex analysis that will simplify proofs are:

1. $z_n \rightarrow z$ if and only if $\sum_k^n |z_k| < \infty$ (i.e. (z_n) is convergent if and only if the series of $(|z_n|)$ is absolutely convergent)
2. $\sum_k z_k$ is convergent if $z_k \rightarrow 0$ (but not only if, ex. $z_k = 1/k$)
3. Notice that this also makes the norm *continuous*. Using the limit definition, this means that:

$$\left| \lim_{n \rightarrow \infty} z_n \right| = \lim_{n \rightarrow \infty} |z_n|$$

4. $|z| = |\bar{z}|$ and

$$\max\{|\operatorname{Re}(z)|, |\operatorname{Im}(z)|\} \leq |z| \leq |\operatorname{Re}(z)| + |\operatorname{Im}(z)|$$

Use these inequalities to show that $z_n \rightarrow z$ if and only if the real and imaginary part converge and that Re and Im passes through the limit (i.e. they are continuous)

By consequence of the definition we have:

$$|z|^2 = z\bar{z}$$

which ties the analytic property of the norm to the algebraic property of conjugation. This can make some calculations much simpler. For example, using the definition of an absolute value directly, we get:

$$\begin{aligned} |xy|^2 &= |(ac - bd) + (ad + cd)i|^2 \\ &= (ac - bd)^2 + (ad + cd)^2 \\ &= (ac)^2 - 2acbd + (bd)^2 + (ad)^2 + 2(adbc) + (cd)^2 \\ &= (ac)^2 + (bd)^2 + (ad)^2 + (cd)^2 \\ &= (a^2 + b^2)(c^2 + d^2) \\ &= |x|^2 |y|^2 \end{aligned}$$

This extends to finite products and quotients. If we use the fact that $|z| = z\bar{z}$, we get

$$|ab|^2 = a\bar{b}a\bar{b} = ab\bar{a}\bar{b} = a\bar{a}b\bar{b} = |a|^2|b|^2$$

Using the conjugation definition also allows us to find many more identities in relation to the norm, for example:

$$|z + w|^2 = (z + w)\overline{(z + w)} = (z + w)(\bar{z} + \bar{w}) = z\bar{z} + (z\bar{w} + \bar{z}w) + w\bar{w}$$

which simplifies to

$$|z + w|^2 = |z|^2 + |w|^2 + 2\operatorname{Re}(a\bar{b})$$

and similarly, if we take the difference of the two numbers, we get:

$$|z - w|^2 = |z|^2 + |w|^2 - 2\operatorname{Re}(a\bar{b})$$

Combining these two identities, we get the following interesting result:

$$|z + w|^2 + |z - w|^2 = 2(|z|^2 + |w|^2)$$

which is the *parallelogram law*, showing that this norm induces an *inner product*:

$$\langle z, w \rangle = \frac{\|z + y\|^2 - \|x - y\|^2}{4}$$

Another inequality we get using the absolute the following:

$$-|a| \leq \operatorname{Re} a \leq |a|$$

where equality holds if $b = 0$ and $a \geq 0$. Using this fact, we can combine it with equation $|z + w|^2 = |z|^2 + |w|^2 + 2\operatorname{Re}(a\bar{b})$ to get

$$|z + w|^2 \leq |z|^2 + |b|^2$$

where equality will only hold if $a\bar{b} \geq 0$. If $b \neq 0$, we can re-write it as $|b|^2(a/b) \geq 0$, i.e. $a/b \geq 0$. Repeating this for an arbitrary finite length triangle inequality:

$$|a_1 + a_2 + \cdots + a_n| \leq |a_1| + |a_2| + \cdots + |a_n|$$

we see that $a_i/a_j \geq 0$ for all i, j . In this way, we see how very similar complex numbers are to real numbers (this time, more than vector's in $\mathbb{R}^2$) since this is the property of real numbers!

(there are a couple more inequalities, but I will for now conclude with this one)

$$|a| = |(a - b) + b| \leq |a - b| + |b|$$

moving $|b|$ to the other side, we get

$$|a| - |b| \leq |a - b|$$

Similarly if we do the same for $|b|$: $|b| - |a| \leq |a - b|$. This meaning that either the positive or negative value will both be less than $|a - b|$ and so

$$||a| - |b|| \leq |a - b|$$

Finally, we state the famous Cauchy inequality:

$$\left| \sum a_i b_i \right|^2 \leq \sum |a_i|^2 \sum |b_i|^2$$

We can show that $+$, $\cdot$, and $-$ are all continuous functions, and so our usual limit laws hold. Since conjugation is continuous, we also have:

$$\lim_{n \rightarrow \infty} \overline{z_n} = \overline{\lim_{n \rightarrow \infty} z_n}$$

Exercise 1.2.1

1. Show that $\mathbb{C}$ has no total ordering, meaning that no total ordering $\leq$ exists such that if $x \leq y$, then for all $z \in \mathbb{C}$, $zx \leq zy$.
2. Show that on S^1 , $\frac{1}{z} = \bar{z}$.
3. show that $\mathbb{C}^\times \cong (0, \infty) \times S^1$ as topological groups.
4. Let $\omega \in S^1$. show that $z \mapsto \omega z$ is an isometry of $\mathbb{C}$, that is

$$|\omega z - \omega w| = |z - w|$$

5. Let $\left| \frac{z-1}{z} \right| = 1$. Show that $\operatorname{Re} z = \frac{1}{2}$.

1.2.2 Change of Variables: Polar and Euler Form

Visually, we can picture a complex number as an arrow (or vector) in the complex plane. Note that $\mathbb{C}$ over $\mathbb{R}$ can be visualized as a plane. Usually, the x -axis will be called the real axis, and the y axis will be called the imaginary axis:

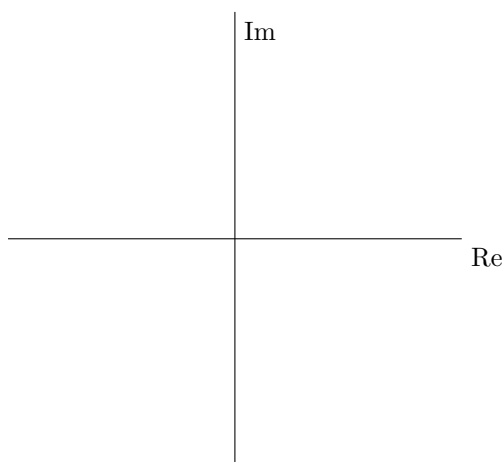


Figure 1.1: Complex Plane

When adding two vectors, it is like moving one vector to the tip of the others, the tip of the combination being the new complex number (equivalently, this comes from the definition of addition in $\mathbb{R}[x]/(x^2 + 1)$). This can also be thought as a line through the following parallelogram:

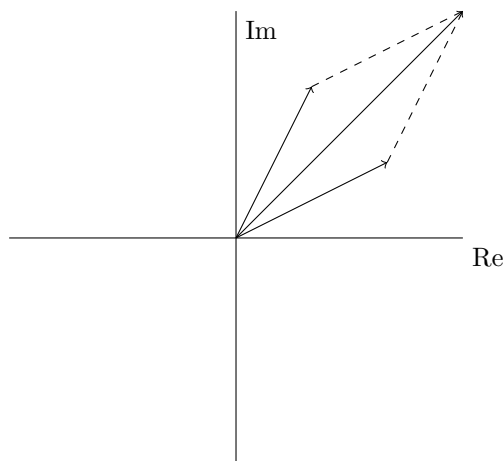


Figure 1.2: Adding Complex Numbers

When it comes to multiplication, we require a little more intuition of what i does in the complex plane. In particular, Notice that $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, $i^5 = i$, so i returns to itself in after multiplying by itself 4 times. Thus, multiplying by i can be thought of as rotating 90 deg or $\pi/2$ times. This intuition can be captured more rigorously: take $\mathbb{C}^\times$ to be set of nonzero complex numbers. Then notice we can represent any element in here by two numbers: $|z|$ and $z/|z|$. The first represents the magnitude, while the second captures the angle. In particular, $|z/|z|| = |z|/|z| = 1$, and so they all lie on the unit circle S^1 . As long as $z \neq 0$, $|z| \neq 0$. We thus can imagine there being a group isomorphism:

$$\mathbb{C}^\times \cong_{\mathbf{Grp}} (0, \infty) \times S^1$$

and indeed there is. First, for any $z/|z|$, using Pythagoras and some high-school trigonometry we can represent this as $\cos(\theta) + i \sin(\theta)$ for appropriate θ . Then we get $x + iy \mapsto (|x + iy|, \cos(\theta) + i \sin(\theta))$. The key is that multiplication preserves angles since:

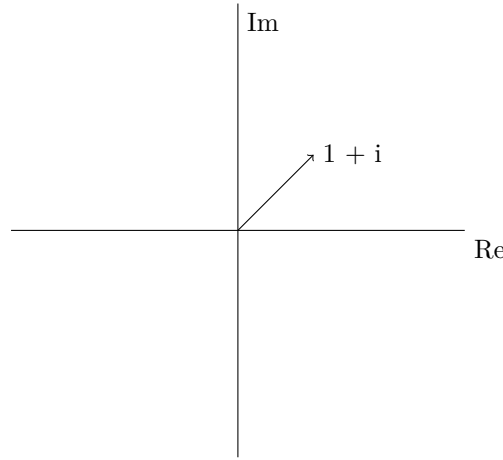
$$\begin{aligned} \cos(\theta_1 + \theta_2) &= \cos(\theta_1) \cos(\theta_2) - \sin(\theta_1) \sin(\theta_2) \\ \sin(\theta_1 + \theta_2) &= \sin(\theta_1) \cos(\theta_2) + \cos(\theta_1) \sin(\theta_2) \end{aligned}$$

Thus, we can represent complex numbers in terms of rotation! For example:

1.

$$\begin{aligned} i &= \cos(\pi/2) + i \sin(\pi/2) \\ 1 &= \cos(0) + i \sin(0) \end{aligned}$$

2. What about representing $1 + i$? Notice that in the complex plane, this would form a triangle:



by Pythagoras, the hypotenuse of the triangle will have length $\sqrt{2}$, which is the norm $1 + i$! Furthermore, since we know the adjacent and the opposite length, we can find the angle by $\tan(b/a)$. Therefore, we get

$$1 + i = \sqrt{2}(\cos(\pi/4) + i \sin(\pi/4))$$

This can be properly formalized as the *change of variables* from Cartesian coordinate functions π_x, π_y to the polar coordinate functions, which for any $z \in \mathbb{C}$:

$$z = |z|(\pm \cos(\theta) \pm i \sin(\theta))$$

Sometimes, the notation is collapsed to $z = |z| \operatorname{cis}(\theta)$ where cis is $\cos + i \sin$. The positive or negative value being added depends in what quadrant the complex number resides in. This also means that numbers in polar representation are defined up to $2k\pi$ rotations, since $2k\pi + \theta = \theta$ in terms of rotation. The value of $|z|$ is called the *magnitude* and the angle θ is called the *argument* (measured by starting on the positive real axis and moving anti-clockwise). The magnitude is usually represented with the variable r , so $a + bi = r(\cos(\theta) + i \sin(\theta))$. We usually write the argument of a complex number as $\arg(z)$. Thus, $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$. By doing some more manipulation, it is not hard to show that

$$z_1/z_2 = (r_1/r_2)(\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2))$$

In other words, $\arg(z_1/z_2) = \arg(z_1) - \arg(z_2)$. This should be ringing a bell: log has these properties. We will return to this in section 2.4.3. In combination with trigonometric identities, these formulas are quite easy to manipulate, for example:

1. $z^{-1} = \frac{1}{(\cos(\theta) + i \sin(\theta))} = r^{-1}(\cos(\theta) - i \sin(\theta)) = r^{-1}(\cos(-\theta) + i \sin(-\theta))$
2. $z^n = r^n(\cos(n\theta) + i \sin(n\theta))$ (known as *Moirve's Formula*)
3. $\sqrt[n]{z} = \sqrt[n]{r}(\cos(\theta/n) + i \sin(\theta/n))$. There are in fact multiple solutions: we need that $n\psi = \theta$. Naturally $\psi = \theta/n$ is a solution, but so is $\psi = \theta/n + 2k\pi/n$ for $0 \leq k \leq n-1$ since

$$n\psi = \theta + 2k\pi = \theta$$

Unfortunately, adding and subtracting does not have a nice representation in polar form, however we can comfortably add and subtract vector in Cartesian form, so this is of no real loss to us. This gives an easy geometric interpretation of adding, subtracting, multiplying, and dividing complex numbers.

Another coordinate representation of complex numbers is *Euler coordinates*. The functions $\cos$ and $\sin$ are defined through an infinite power series that converges everywhere:

$$\cos(x) = 1 - \frac{x^2}{2} + \frac{x^4}{4!} - \dots \quad (1.2)$$

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \quad (1.3)$$

Given appropriate manipulation of these power series, we can make them resemble the power series of e^x :

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + \dots \quad (1.4)$$

which after the appropriate manipulation, we get:

$$|r|e^{\theta i} = |r|(\cos(\theta) + i \sin(\theta))$$

Thus, we have the Euler representation of a complex number as well. We have similar niceties manipulating complex numbers as Euler representations:

1. $zw = |r||s|e^{(\theta+\psi)i}$
2. $z^{-1} = |r|^{-1}e^{-\theta i}$

and so on. Sometimes, they allow for simpler expressions for solutions of polynomials. For example, take the equation $x^n - 1 = 0$ or equivalently $x^n = 1$. By the fundamental theorem of Algebra, $x^n - 1$ has n roots. Since $x^n = 1$, these roots are called the *roots of unity*, and are studied extensively. Given Euler's representation of complex numbers, we know that $1 = e^0$, and so the roots of unity of $x^n - 1$ can be represented as

$$z_k = e^{\frac{2k\pi}{n}} \quad k = 1, 2, \dots, n$$

Notice that $(z_k)^n = 1$ for each k since $2k\pi = 0$. For example, here is the 5th root of unity.

1.2.3 Monodromy

For a proper change of variables from Cartesian to polar, some care must be made so that $\arg(z)$ is in fact continuous and bijective. We have so far defined it as a function $\arg(z) : \mathbb{C}^* \rightarrow \mathbb{R}$ that gives back the angle of z . Note that $\arg(f(0)) = 0$. To see function is not continuous take the parameterization $f : [0, 2\pi] \rightarrow S^1 \subseteq \mathbb{C}^*$ taking $f(\theta) = \cos(\theta) + i \sin(\theta)$. In particular, $\arg(f(\theta)) = \theta$. If $\arg(z)$ were continuous, $\arg(f(\theta))$ would be continuous. Now, notice as $\theta \rightarrow 2\pi$, the output of $f(t)$ approaches 1, which we already know has argument 0, however the angle is still *increasing*. If the function were continuous, the limit would be preserved and would be 2π , however we know that $\arg(f(2\pi)) = \arg(1) = 0$; we must suddenly drop.

More generally as θ ranges over every real number it is not continuous (namely, it jumps when crossing over in the interval $(2\pi - \epsilon, 2\pi + \epsilon)$), and if we make it continuous by limiting the domain it will not be surjective. To make it continuous, we have to make a choice of how we measure angle and

how what interval we pick. The most common is called the *fundamental domain* or the *principal branch* (for reasons that shall be made clear once we study log) with its elements being called the *standard arguments* and is $(-\pi, \pi)$ with:

$$\arg(1) = 0 \quad \arg(i) = \pi/2 \quad \arg(-1) = \pi, \arg(-i) = -\pi/2$$

As mentioned earlier, shrinking the domain will make this function lose surjectivity from $\mathbb{C}^*$. To re-gain surjectivity, we shall eliminate $(-\infty, 0]$ from the domain. There is no way around this due to non-trivial *monodromy*, if we define a different angle function on a different domain, we shall eliminate a different ray from $\mathbb{C}^*$. This phenomena is given a name: we say that the angle function has non-trivial *monodromy*. The angle function defined on $\mathbb{C} \setminus (-\infty, 0]$ for which $\arg(1) = 0$ will be denoted Arg , i.e.

$$\text{Arg} : \mathbb{C}^* \setminus (-\infty, 0] \rightarrow (-\pi, \pi)$$

Though this is a fix, the fact that the function $\theta(z)$ not a continuous function shall give problems when trying to show certain standard functions are continuous. For example, this creates the issue that the function $f : \mathbb{C} \rightarrow \mathbb{C} \quad f(z) = \sqrt{z}$, which is discontinuous. To see this, take:

$$f(z) = \sqrt{|z|} \left(\cos \left(\frac{\theta(z)}{2} \right) + i \sin \left(\frac{\theta(z)}{2} \right) \right)$$

Note we could have defined a second square-root function if we measured the angle by turning clock-wise (which represents the second solution to the equation $f^2 - z = 0$), however the exact same problems shall arise. There is in fact a nice visual argument that can show us this. Let's say we want to make it continuous on the sub-domain S^1 . Let's pick $\sqrt{1} = 1$, and now try to make it continuous on all of S^1 . If we start moving counter-clockwise, we see that the $\sqrt{z}$ must always be half the angle, so we will keep halving the angle but keeping the magnitude. Doing so, we will eventually end up at $\sqrt{-1}$, which by the way we have been choosing our outputs forces us to pick $\sqrt{-1} = i$. If we keep going, we keep halving the angle until we reach back 1, at which point we have $\sqrt{1} = -1$... However we already established that $\sqrt{1} = 1$. This problem cannot be fixed even if we try a different approach. Another insightful way of seeing it that is worth mentioning is if we always take the smallest angle between the positive real axis and our complex number on the unit circle. The argument will remain the same when going counter clockwise towards -1 , but now do this argument but going clock-wise. We again the half the angle every time, but notice that once we reach $\sqrt{-1}$, we are forced to pick $-i$! A visual might be helpful to see this:

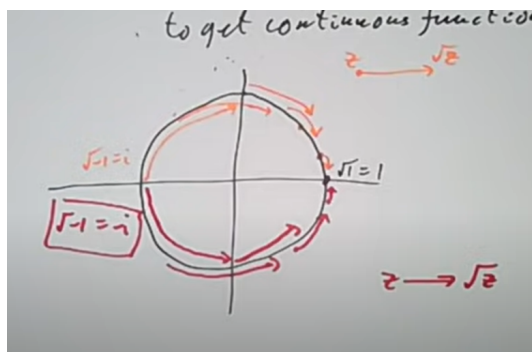


Figure 1.3: Continuity breaks for square root

Hence, $\sqrt{-}$ cannot be continuous on S^1 , and hence more generally on $\mathbb{C}$. We would have to eliminate some points in order to make it continuous, usually the choice of points is $(-\infty, 0)$. The fact that going around in different ways does not give us the same result is, again, called *non trivial monodromy*.

Another approach some authors pick is to keep the entire domain $\mathbb{C}$ and to say that $\sqrt{-}$ is in fact a *multi-valued function*, meaning the output has both possibilities. This is not very desirable. So many of the tools we have worked with (topology, inner product and smoothness to name a notable few) all are based on the idea of function being single valued, we would have to re-think all of these notions to incorporate multi-valuedness and the discontinuity that comes up because of the multi-valuedness!! To make not resort notion of “multi-valuedness”, we will come up with a new domain for which f is in fact single-valued. To make this rigorous is too much at the moment, but the motivating idea is quite interesting. We will do so for $\sqrt{-}$. Take two copies of $\mathbb{C} \setminus (-\infty, 1]$. Label the bottom edge of the first two copies 1 and 3 respectively and the top edges 2 and 4. Now glue 1, 2 and 3, 4 (more formally, take the quotient where these points are in the same equivalent class). Then the resulting shape requires $\mathbb{C}^3$ to embed it without intersection, but we may still visualize it in “in $\mathbb{R}^3 + \mathbb{R}$ for colours”, or $\mathbb{C}^2$, like so:

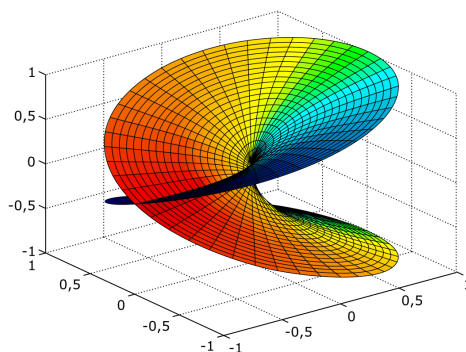


Figure 1.4: Branch Covering for $\sqrt{-}$

Let R represent this surface, known as a *Riemann surface*. We will now lift the function $\sqrt{-}$ to R so that it is a well-defined single-valued function without the need to cut out part of its domain. In particular, notice that R is a covering space for $\mathbb{C}$, usually known as a *branch covering*. Then we define map f from R to $\mathbb{C}$ such that the following diagram commutes:

$$\begin{array}{ccc} R & \xrightarrow{f} & \mathbb{C} \\ \pi \downarrow & \nearrow \sqrt{-} & \\ \mathbb{C} & & \end{array}$$

Then f is now a *single valued* function. This same process will be done for many other function to “recover” a well-defined function. Famously, there is no proper complex log, there are infinitely many outputs. The corresponding branch covering of $\mathbb{C}$ for complex log will be:

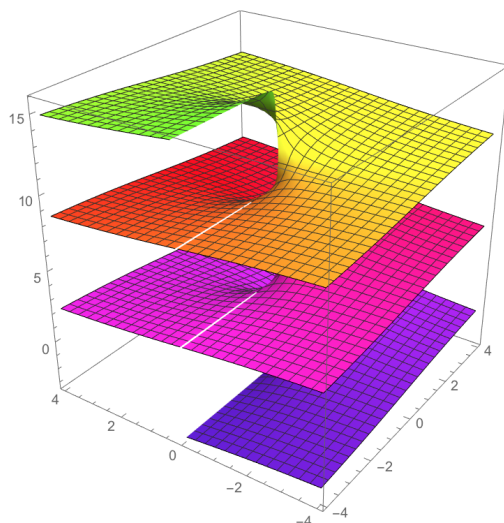


Figure 1.5: Branch Covering for complex log

Exercise 1.2.2

1. If $a, c \in \mathbb{C}$ and $\operatorname{Im} a \bar{c} = 0$, then $c = r \cdot a$ where $r \in \mathbb{R}$
2. Let $z = a + bi \in \mathbb{C}$ and define $\dot{z} = b + ai$. Show that $z \cdot \dot{z}$ is strictly imaginary unless $\dot{z}z = 0$ case it's (naturally) both real and imaginary. Is there a geometric interpretation of this result?

1.2.4 Riemann Sphere: adding a point at infinity

When working with $1/x$, we have a discontinuity at 0, namely since the graph “jumps” at 0. This can be remedied with the 1-point compactification and some differential geometry by taking $f : S^1 \rightarrow S^1$ and taking $z \mapsto \bar{z}$ which extends $1/z$ to all of S^1 , and with the stereographic projection this can be seen as a way of “introducing” $1/x$ in a well-defined manner (we shall make this rigorous in a few paragraphs). However this does not remedy the fact that in coordinates, a connected interval around 0 becomes two disconnected open rays (going to $\pm\infty$), namely when looking restricting back to $\mathbb{R}$, the functions are no longer continuous. Naturally, on S^1 , the map is continuous (even real-differentiable), however we have now lost some of the tools we are accustomed to analyzing $\mathbb{R}$.

We require at least one more dimension to be able to image the image connected⁸. In this way, we shall soon consider the complex function $1/z : \mathbb{C} \rightarrow \mathbb{C}$ to be a continuous function, and even a complex-differentiable function.

In order to consider $1/z$ well-defined function, we need to extend complex numbers. Let us first start with what happens when we do this to $\mathbb{R}$. As mentioned earlier, in $\mathbb{R}$ we can “compactify” $\mathbb{R}$ by adding one point at infinity, $\mathbb{R} \cup \{\infty\}$ (this is called the *one-point compactification*, see my EYNTKA Topology). We will label this as $\mathbb{R}_\infty$. Notice that $\mathbb{R}_\infty$ is homeomorphic to S^1 via the following

⁸in essence, because we require at least two dimensions to embed a circle

homeomorphism: if we take any $z \in S^1$ as a tuple (x, y) , then

$$(x, y) \mapsto \frac{x}{1-y}$$

and $(0, 1) \mapsto \{\infty\}$. This shows that $\mathbb{R}_\infty$ is indeed compact, and furthermore we can try to work to incorporate infinity into our analysis. The first thing we may notice is that we have the problem that $\mathbb{R}_\infty$ loses the well-ordering of $\mathbb{R}$, namely since ∞ is both smaller and larger than any real number, in particular $\infty < 0 < \infty$. However, if we instead take $x \mapsto |x|$, then ∞ is the natural value that appears as either $\lim_{x \rightarrow -\infty} |x|$ or $\lim_{x \rightarrow \infty} |x|$. Thus, it is natural to think of $\mathbb{R}_\infty$ as the codomain of $|\cdot|$, and more generally some norm $\|\cdot\|$ on a real vector-space rather than an element that we can add to the ordering of $\mathbb{R}$, and is “closed” under convergent and divergent sequences which are eventually strictly increasing or decreasing⁹.

Though order is lost, some arithmetic properties are still well-defined (these are shown to be well-defined through the use of arbitrary sequences):

1. $a + \infty = \infty + a = \infty$ for all finite a
2. $b \cdot \infty = \infty \cdot b = \infty$ for $b \neq 0$, including ∞

It is impossible (as shown in 1st year calculus) to define $\infty + \infty$ or $0 \cdot \infty$ since there can be sequence that approach these value’s differently depending on how “fast” each term goes to ∞ or 0 (recall the l’hopital rules). It is however possible to define the convention that $a/0 = \infty$ or $b/\infty = 0$ for $b \neq \infty$. Though there is some arithmetics well-defined in $\mathbb{R}_\infty$, if we transfer over to S^1 via the stereographic homeomorphism, it is much harder to perform arithmetics, and so we would usually take the S^1 interpretation of $\mathbb{R}_\infty$ for it’s geometric benefits.

Let’s now work with $\mathbb{C} \cup \{\infty\}$. Like with the real case, let’s label it $\mathbb{C}_\infty$. Some other common labels are $\widehat{\mathbb{C}}$ and $\overline{\mathbb{C}}$. Just like for $\mathbb{R}$, we are extending $\mathbb{C}$ by a single point instead of a circle of points since we are more so interesting in the “size” of points approaching infinity rather than any sequence converging to some “appropriate” point at infinity (like a $\infty(\cos(\theta) + i \sin(\theta))$). Let’s make precise the relation between $\mathbb{C}_\infty$ and S^2 . Identify $\mathbb{C}_\infty \setminus \{\infty\}$ with the xy -axis in $\mathbb{R}^3$, where the x -axis representing the real component and the y -axis representing the imaginary component. Naturally, no point in $\mathbb{R}^3$ is identified with the point ∞ in $\mathbb{C}_\infty$. Take a unit sphere centered at the origin of $\mathbb{C}_\infty$. Now, starting from the north-pole, project a down through the sphere and into the complex plane. This is a one-to-one correspondence with the xy -axis, omitting the north-pole. If $S^2 = \{(x, y, z) \mid x^2 + y^2 + z^2 = 1, x, y, z \in \mathbb{R}\}$ then

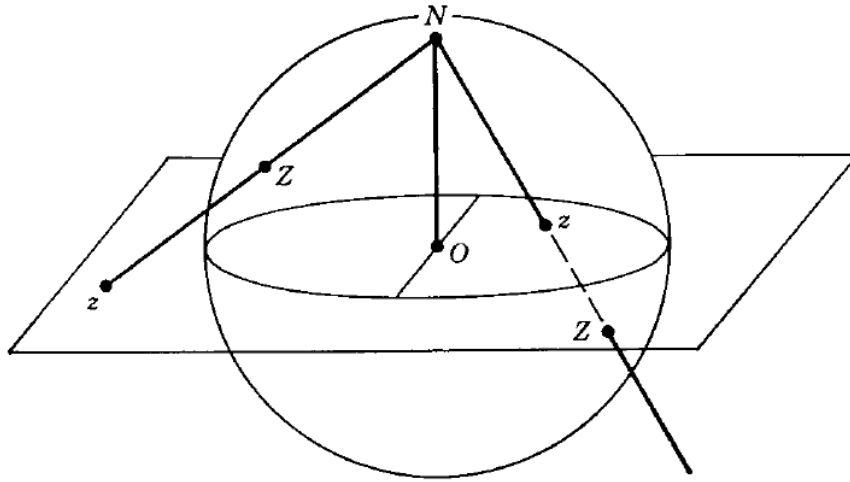
$$z = \frac{x_1 + ix_2}{1 - x_3}$$

By taking the square of the modulus ($|z|^2$), we can re-arrange that equation to get

$$x_1 = \frac{z + \bar{z}}{1 + |z|^2} \quad x_2 = \frac{z - \bar{z}}{i(1 + |z|^2)} \quad x_3 = \frac{|z|^2 - 1}{|z|^2 + 1} \quad (1.5)$$

By letting the pole $(0, 0, 1)$ correspond to the point at infinity, we get that the sphere is homeomorphic to $\mathbb{C}_\infty$ (and hence, we get a nice geometric object representing $\mathbb{C}_\infty$ without needing to “distinguish” a point). This process is called *stereographic projection*, and visually looks like so:

⁹Note that it is not closed for any type of sequence, for example for oscillating sequences



Note that the points where $x_3 < 0$ corresponds to the disk $|z| < 1$, and when $x_3 > 0$ the points correspond to $|z| > 1$ while if $x_3 = 0$ then $|z| = 1$. We may also stereographically project from the south-pole, giving us

$$z' = \frac{x_1 - ix_2}{1 + x_3}$$

Note that z and z' are related by the equation $zz' = 1$ unless z or z' is a pole N or S .

For future purposes, we will quickly say a word on how circles on the Riemann sphere map to $\mathbb{C}_\infty$, namely since this gives a nice visualization of how to think about $\mathbb{C}_\infty$. Any circle *not* passing through the pole maps a circle on the plane. If a point of a circle is at the north pole, then the circle maps to a *line* in $\mathbb{C}_\infty$. To see this, notice that a circle on the Riemann sphere lies in the plane $\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 = \alpha_0$ where we can assume that $\alpha_1^2 + \alpha_2^2 + \alpha_3^2 = 1$ and $0 \leq \alpha_0 < 1$. From equation (1.5), we can re-write this previous equation as:

$$\alpha_1(z + \bar{z}) + \alpha_2 i(z - \bar{z}) = \alpha_3(|z|^2 - 1) = \alpha_0(|z|^2 + 1)$$

or:

$$(\alpha_0 - \alpha_3)(x^2 + y^2) - 2\alpha_1 x - 2\alpha_2 y + \alpha_0 + \alpha_3 = 0$$

When $\alpha_0 \neq \alpha_3$, this is the equation of a circle, and when $\alpha_0 = \alpha_3$, this is the equation of a straight line. Conversely, the equation of a circle and a line can always be written in this form.

Finally, on S^2 the Euclidean distance between two points of $\mathbb{R}^3$ (the length of the chord joining them) gives a finite metric on $\mathbb{C}_\infty$: if (x_1, x_2, x_3) and (x'_1, x'_2, x'_3) lie on S^2 , then

$$(x_1 - x'_1)^2 + (x_2 - x'_2)^2 + (x_3 - x'_3)^2 = 2 - 2(x_1 x'_1 + x_2 x'_2 + x_3 x'_3)$$

Metrics on Riemann Sphere

The rest of this section can be skipped till it is needed. The *chordal distance* between two points on the Riemann sphere (considered as the unit sphere in $\mathbb{R}^3$) means the length of the line segment in $\mathbb{R}^3$ joining the points. The chordal distance induces a metric $d(z, w)$ on $\mathbb{C}$; i.e., if $z, w \in \mathbb{C}$, then $d(z, w)$

is defined as the chordal distance between the points of the Riemann sphere corresponding to z, w by stereographic projection.

Theorem 1.2.2: Chordal metric on Riemann Sphere

For $z \in \mathbb{C}$ let $P(z) \in S^2$ be the point with coordinates (1.5), put $P(\infty) = (0, 0, 1)$, and let $d(z, w) = |P(z) - P(w)|$ be the chordal distance, where $|\cdot|$ is the Euclidean norm of $\mathbb{R}^3$.

1. P is a bijection of $\mathbb{C}_\infty$ onto S^2 , so d is a metric on $\mathbb{C}_\infty$ for which P is an isometry onto S^2 . In particular $(\mathbb{C}_\infty, d)$ is compact.
2. For $z, w \in \mathbb{C}$,

$$d(z, w) = \frac{2|z - w|}{\sqrt{(1 + |z|^2)(1 + |w|^2)}} \quad d(z, \infty) = \frac{2}{\sqrt{1 + |z|^2}}$$

3. With the conventions $1/0 = \infty$ and $1/\infty = 0$, one has $P(1/z) = R(P(z))$ for all $z \in \mathbb{C}_\infty$, where $R(x_1, x_2, x_3) = (x_1, -x_2, -x_3)$. Consequently $d(1/z, 1/w) = d(z, w)$ for all $z, w \in \mathbb{C}_\infty$.

Proof:

For $z \in \mathbb{C}$ write $P(z) = (x_1, x_2, x_3)$ as in (1.5), so that

$$x_1 + ix_2 = \frac{2z}{1 + |z|^2} \quad 1 - x_3 = \frac{2}{1 + |z|^2}$$

Then $x_1^2 + x_2^2 + x_3^2 = (4|z|^2 + (|z|^2 - 1)^2)/(1 + |z|^2)^2 = 1$, so $P(z) \in S^2$.

1. Since $x_3 < 1$, the point $P(z)$ differs from $P(\infty) = (0, 0, 1)$, and the two displayed identities give $z = (x_1 + ix_2)/(1 - x_3)$, so z is determined by $P(z)$. Thus P is injective on $\mathbb{C}_\infty$. It is onto S^2 : given $(y_1, y_2, y_3) \in S^2$ with $y_3 \neq 1$, the number $z = (y_1 + iy_2)/(1 - y_3)$ has $|z|^2 = (1 - y_3^2)/(1 - y_3)^2 = (1 + y_3)/(1 - y_3)$, so $1 + |z|^2 = 2/(1 - y_3)$, and the two displayed identities give $P(z) = (y_1, y_2, y_3)$. Hence $d(z, w) = 0$ only when $P(z) = P(w)$, that is $z = w$, while symmetry and the triangle inequality for d are those of the Euclidean norm. As S^2 is closed and bounded in $\mathbb{R}^3$, it is compact ([8, Closed And Bounded In $\mathbb{R}^n$ Implies Compact]), and so is $(\mathbb{C}_\infty, d)$.

2. Let $z, w \in \mathbb{C}$ and $P(w) = (y_1, y_2, y_3)$. Since $P(z)$ and $P(w)$ are unit vectors, $d(z, w)^2 = 2 - 2P(z) \cdot P(w)$. Here $x_1y_1 + x_2y_2 = \operatorname{Re}((x_1 + ix_2)(y_1 + iy_2)) = 4\operatorname{Re}(z\bar{w})/((1 + |z|^2)(1 + |w|^2))$, so

$$1 - P(z) \cdot P(w) = \frac{(1 + |z|^2)(1 + |w|^2) - 4\operatorname{Re}(z\bar{w}) - (|z|^2 - 1)(|w|^2 - 1)}{(1 + |z|^2)(1 + |w|^2)} = \frac{2|z - w|^2}{(1 + |z|^2)(1 + |w|^2)}$$

using $|z|^2 + |w|^2 - 2\operatorname{Re}(z\bar{w}) = |z - w|^2$. This gives the first formula. For the second, $d(z, \infty)^2 = x_1^2 + x_2^2 + (x_3 - 1)^2 = (1 - x_3^2) + (1 - x_3)^2 = 2(1 - x_3) = 4/(1 + |z|^2)$.

3. For $z \in \mathbb{C} \setminus \{0\}$, the coordinates (1.5) of $P(1/z)$ satisfy

$$\frac{2(1/z)}{1 + |z|^{-2}} = \frac{2\bar{z}}{|z|^2 + 1} = \overline{x_1 + ix_2} \quad \frac{|z|^{-2} - 1}{|z|^{-2} + 1} = \frac{1 - |z|^2}{1 + |z|^2} = -x_3$$

so $P(1/z) = (x_1, -x_2, -x_3) = R(P(z))$. Also $P(1/0) = P(\infty) = (0, 0, 1) = R(0, 0, -1) = R(P(0))$ and $P(1/\infty) = P(0) = R(P(\infty))$. Since R preserves Euclidean distances, $d(1/z, 1/w) = |R(P(z)) - R(P(w))| = |P(z) - P(w)| = d(z, w)$, completing the proof.

This is not to be confused with the geodesic metric which takes the smallest distance between the two points *on* the Riemann sphere:

Definition 1.2.3: Geodesic Metric

$$\rho(z, w) = \arccos(z \cdot w)$$

Finally, in chapter 6, we shall be taking the derivative of many functions on the Riemann sphere, and shall need the following result:

Corollary 1.2.4: Spherical Derivative

Let $\Omega \subseteq \mathbb{C}$ be open, let $f: \Omega \rightarrow \mathbb{C}$ be complex differentiable at a point $z \in \Omega$, and let d be the chordal metric of theorem 1.2.2. Then the limit

$$f^\#(z) := \lim_{w \rightarrow z} \frac{d(f(z), f(w))}{|z - w|}$$

exists, and $f^\#(z) = 2|f'(z)|/(1 + |f(z)|^2)$.

Proof:

For $w \in \Omega$ with $w \neq z$, the formula of theorem 1.2.2(2) for d gives

$$\frac{d(f(z), f(w))}{|z - w|} = \left| \frac{f(w) - f(z)}{w - z} \right| \cdot \frac{2}{\sqrt{(1 + |f(z)|^2)(1 + |f(w)|^2)}}$$

Since f is complex differentiable at z , the difference quotient $(f(w) - f(z))/(w - z)$ tends to $f'(z)$ as $w \rightarrow z$, so the first factor tends to $|f'(z)|$. Writing

$$f(w) - f(z) = \frac{f(w) - f(z)}{w - z} (w - z)$$

the right-hand side tends to $f'(z) \cdot 0 = 0$, so f is continuous at z and $1 + |f(w)|^2$ tends to $1 + |f(z)|^2$. The second factor therefore tends to $2/(1 + |f(z)|^2)$, and the limit of the product is $2|f'(z)|/(1 + |f(z)|^2)$, as claimed.

1.3 Complex Functions

In this section, we will go over how to think about complex functions and emphasize geometric intuitions. For convenience, we will use z, w as complex numbers and x, y as real numbers. a *real function* is a function with codomain $\mathbb{R}$, while a *complex function* has a codomain $\mathbb{C}$. Whether the domain is real, complex, or anything else (ex. a manifold, a vector space, a topological space) is usually deduced from context.

1.3.1 Complex Linear Transformations

Since a function is differentiable if it is locally linear, it is worth taking a moment to understand complex linear functions. Let's start with the simplest $\mathbb{C}$ -linear transformation: a map $f : \mathbb{C} \rightarrow \mathbb{C}$ where

$$f(z) = az \quad a \in \mathbb{C}$$

representing $z = x + yi$ and $a = a + bi$, we get:

$$\begin{aligned} (a + bi)(x + yi) &= (ax - by) + (bx + ay)i \\ \Rightarrow \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} ax - by \\ bx + ay \end{pmatrix} \end{aligned}$$

Notice that this linear translation is a rotation and a scaling. In particular, it *preserved the angles*. A function that is angle-preserving is called *homothetic*. Furthermore, we cannot have any mirror symmetry. For example

$$z \mapsto \bar{z}$$

is homothetic, but it is *not* $\mathbb{C}$ -linear. The way I like to think about it is that if you are in one “mirror-space”, so to speak, then you will always measure the angle going from a fixed choice of either left or right. If you change mirror spaces, then you change which side you measure your angle from. Thus, when we think of angle preserving we will think of angle preserving with respect to a fixed side. From this perspective, $z \mapsto \bar{z}$ maps an angle θ to $2\pi - \theta$, which will not be considered “angle preserving”

Lemma 1.3.1: Real Angle Preserving Then Complex

Let $T : \mathbb{C} \rightarrow \mathbb{C}$ be an $\mathbb{R}$ -linear function that preserves angles. Then there exists an $a \in \mathbb{C}$ such that $T(z) = az$ or $T(z) = a\bar{z}$

Proof:

Since T is homothetic, it must be bijective. Let S be a homothetic function such that $S^{-1}T(1, 0) = (1, 0)$. $S^{-1}T$ is also homothetic, so it must be that $S^{-1}T(0, 1) = (0, c)$ for some $c \neq 0$. Since $S^{-1}T$ is linear, $S^{-1}T(1, 1) = (1, c)$. Since $S^{-1}T$ is angle preserving, it must be that $c = \pm 1$. depending on which this is, we get $T(z) = az$ or $T(z) = a\bar{z}$, completing the proof.

1.3.2 Polynomial and Rational Functions

Polynomial functions (with real coefficients) are easy example of real differentiable functions. So too will complex polynomials. It will turn out that a slight-generalization of these polynomials give a huge example of complex differentiable functions, in particular for allowing the *quotient* of polynomials. Functions that are polynomials over polynomials are called *rational polynomials* or *rational functions*. In this section, we give some common names to notions for rational functions. By the fundamental theorem of algebra, we get

$$P(z) = (z - \alpha_1)(z - \alpha_2) \cdots (z - \alpha_n)$$

for n not necessarily distinct roots (we will prove this theorem in corollary 3.3.5). For now, we will assume differentiation of complex polynomials is identical to that of real polynomials (which will

turn out to be the case, see section 2.1). If $P(\alpha_i) = 0$, we call α_i a zero or root of P . Since the roots need not be distinct, we call the order of a zero the number of roots that are equal. Note that if α has order h for P , then $P^{(h-i)}(\alpha) = 0$ for $1 \leq i \leq h-1$, and $P^{(h)}(\alpha) \neq 0$. Note that if the order of an α is 1, then $P(\alpha) = 0$ but $P'(\alpha) \neq 0$. We call a zero of order 1 a simple zero.

Definition 1.3.2: Rational Function

Let $p(x)$ and $q(x)$ be polynomials over $\mathbb{C}$ where $\gcd(p(x), q(x)) = 1$ (i.e., they do not have any similar roots). Then

$$R(x) : \mathbb{C}_\infty \rightarrow \mathbb{C}_\infty, R(x) = \frac{p(x)}{q(x)}$$

is called a *rational function* or a *Fracitonal Linear Transformation* over $\mathbb{C}_\infty$.

For more information, see https://encyclopediaofmath.org/wiki/Fractional-linear_mapping. We let the codomain be the extended complex numbers in order that the denominator be able to accept 0 values. In particular, we will define

$$R(\alpha) = \frac{p(\alpha)}{q(\alpha)} = \frac{n}{0} = \infty$$

This is another reason to allow only one point of compactification, to allow this extension to be well-defined. This also motivates the name pole: these points get mapped to the north pole of the Riemann sphere. It also allows us to rigorously capture the idea by doing:

$$R(\infty) = \lim_{z \rightarrow \infty} R(z)$$

and so R defined on $\mathbb{C}_\infty$ is indeed well-defined. Since it is possible to plugin $R(\infty)$, it would be convenient to find a way to find zeros or poles at infinity. To find these, we will define

$$R_1(z) = R(1/z)$$

And define $R(\infty) = R_1(0)$. From this, we can define the order of zeros (i.e. roots) or poles at ∞ to be the zeros or poles of the origin of R_1 . Now, given some R :

$$R(z) = \frac{a_0 + a_1z + \cdots + a_nz^n}{b_0 + b_1z + \cdots + b_mz^m}$$

Then

$$R_1(z) = z^{m-z} \frac{a_0z^n + a_1z^{n-1} + \cdots + a_{n-1}z + a_n}{b_0z^m + b_1z^{m-1} + \cdots + b_{m-1}z + b_m}$$

From this we, can see the order of zero's or poles at ∞ . If $m > n$, then the zeroes at infinity has order $m - n$. Conversely, if $m < n$, then the poles at infinity have order $n - m$. If $n = m$, Then we either get

$$R(\infty) = a_n/b_m \neq 0 \quad \text{or} \quad \infty$$

with all the zeros and poles established, we can count the number of zeros or poles a given rational function has, including the zero's or poles at ∞ ; namely by the observations in the above paragraph we have that the number of zeroes/poles is always $\max(n, m)$. Since this is a consistent number, we call this number the *order* of the rational function. The order of a rational function is quite stable. The function $R - a$ has the same number of poles and zeros and hence the same order (meaning any

$R(z) = a$ of order p has p roots). Furthermore, $R - a$ (called a *parallel transition*) keeps the point ∞ stable, while the rational function $1/z$ (called the *inversion*) interchanges 0 and ∞ .

Treating differentiation of rational functions as a formal operation ¹⁰, we also see that the poles are stable under differentiation, which we can see using the quotient rule. As an exercise, check that if a pole of $R(z)$ has order k , then a pole of $R'(z)$ has order $k + 1$.

We may also use differentiation and the inverse function to find the order of a pole: Using $R_1(z)$ to flip the poles and the roots, we can then take the formal derivative until we get a nonzero value.

Finally, we get the following nice consequence of

1.3.3 Möbius Transformation

(see this website: [here](#))

A rational function of particular interest, on which we focus below, is one of order 1. Due to their importance, they are given a name for reference:

Definition 1.3.3: Möbius Transformatino

Let f be a rational function of degree 1

$$S(z) = \frac{\alpha z + \beta}{\gamma z + \delta} \quad \alpha\delta - \beta\gamma \neq 0$$

Then f is called a *Möbius transformation*.

The reason we have $\alpha\delta - \beta\gamma \neq 0$ is so that there is 1 pole and 1 root. Let's say that

$$S(-b/a) = \frac{0}{c\frac{-b}{a} + d} = 0$$

If the denominator was 0 we would get $0/0$, that is the root would not be well defined. That would happen if:

$$c\frac{-b}{a} + d = 0 \quad \Leftrightarrow \quad -cb + ad = 0$$

Hence, we will require that $ad - bc \neq 0$. If $ad - bc = 0$ and $(c, d) \neq (0, 0)$, then the vanishing determinant makes the pairs (a, b) and (c, d) proportional, say $(a, b) = \lambda(c, d)$, and

$$\frac{az + b}{cz + d} = \frac{\lambda(cz + d)}{cz + d} = \lambda$$

wherever $cz + d \neq 0$, a constant function with no pole. Manipulating the equation, we get that the inverse is:

$$S^{-1}(w) = \frac{dw - b}{-cw + a}$$

showing the inverse is also a first order rational function. The key result that we care about these functions is that the pole is sent to a single point. Once we define complex differentiation, we will see that Möbius transformations are the diffeomorphism of $\overline{\mathbb{C}}$, making them worthwhile to study.

¹⁰think of it as a new algebraic operation on the ring of rational functions over $\mathbb{C}$

Already we can see that such a function is certainly a homeomorphism, and hence they belong to $\text{Aut}_{\text{Top}}(\mathbb{C})$.

Given Möbius transformations are such nice function, they should be relatively simple to imagine. Indeed, if $\gamma = 0$, then we have $\frac{\alpha z}{\delta} + \frac{\beta}{\delta} = az + b$ for some constants a and b showing that this Möbius transformation is just scaling and transformation, while if $\gamma \neq 0$ then

$$\begin{aligned} \frac{az + b}{cz + d} &= \frac{az}{cz + d} + \frac{b}{cz + d} \\ &= \frac{a}{c} + \frac{\frac{-ad}{c}}{cz + d} + \frac{b}{cz + d} \\ &= \frac{a}{c} + \frac{bc - ad}{c(cz + d)} \\ &= \frac{a}{c} + \frac{(bc - ad)/c^2}{z + d/c} \end{aligned}$$

showing that in this case a Möbius transformation is the map $z \mapsto 1/(z + d/c)$, followed by multiplication by $(bc - ad)/c^2$ and translation by a/c . This is nice to keep in mind, though when it comes to manipulating the original Möbius transformation it might be hard to see what's going on. Another way we can decompose a Möbius transformation is given if $c \neq 0$:

$$f_1(z) = z + d/c \quad f_2(z) = 1/z \quad f_3(z) = \frac{bc - ad}{c^2}z \quad f_4(z) = z + a/c$$

then

$$f_4 \circ f_3 \circ f_2 \circ f_1(z) = \frac{az + b}{cz + d}$$

The following proposition may help visualize the images of this type of function:

Proposition 1.3.4: Properties Of Möbius Transformations

Let $f(z) = \frac{az+b}{cz+d}$, with $ad - bc \neq 0$, be a Möbius transformation, regarded as a map of $\mathbb{C}_\infty = \mathbb{C} \cup \{\infty\}$ to itself by setting $f(-d/c) = \infty$ and $f(\infty) = a/c$ when $c \neq 0$, and $f(\infty) = \infty$ when $c = 0$. Call a subset of $\mathbb{C}_\infty$ a *generalized circle* if it is a circle in $\mathbb{C}$, or a line in $\mathbb{C}$ together with the point ∞ . Then:

1. f maps every generalized circle onto a generalized circle
2. a Möbius transformation is determined by its values at three distinct points of $\mathbb{C}$, and these values may be any three distinct points of $\mathbb{C}$: for distinct $z_1, z_2, z_3 \in \mathbb{C}$ and distinct $w_1, w_2, w_3 \in \mathbb{C}$ there is exactly one Möbius transformation g with $g(z_k) = w_k$ for $k = 1, 2, 3$
3. if f is not the identity, then f has exactly one or exactly two fixed points in $\mathbb{C}_\infty$

Proof:

Write a point of $\mathbb{C}_\infty$ as a ratio $[\zeta_1 : \zeta_2]$ of a nonzero pair $(\zeta_1, \zeta_2) \in \mathbb{C}^2$, two pairs giving the same point when they are proportional, with $z = [z : 1]$ for $z \in \mathbb{C}$ and $\infty = [1 : 0]$. Then f sends

$[\zeta_1 : \zeta_2]$ to $[a\zeta_1 + b\zeta_2 : c\zeta_1 + d\zeta_2]$: for $z \in \mathbb{C}$ with $cz + d \neq 0$ the pair $(az + b, cz + d)$ represents $(az + b)/(cz + d)$. For $z = -d/c$ the second entry vanishes while $az + b = (bc - ad)/c \neq 0$, which represents ∞ . The pair $(1, 0)$ goes to (a, c) , which represents a/c if $c \neq 0$ and ∞ if $c = 0$. Consequently the composite of two Möbius transformations acts through the product of their coefficient matrices, whose determinant is the product of the two nonzero determinants, so it is again a Möbius transformation, and the matrix inverse gives the inverse map, so f is a bijection of $\mathbb{C}_\infty$.

1. Every generalized circle is the set of $z \in \mathbb{C}$ satisfying an equation

$$A|z|^2 + Bz + \bar{B}\bar{z} + C = 0 \quad A, C \in \mathbb{R}, B \in \mathbb{C}, |B|^2 > AC$$

together with ∞ exactly when $A = 0$, and every such equation arises in this way. For $A \neq 0$ the equation reads $|z + \bar{B}/A|^2 = (|B|^2 - AC)/A^2$, a circle of positive radius. For $A = 0$ it reads $2\operatorname{Re}(Bz) + C = 0$ with $B \neq 0$, a line. Conversely, the circle $|z - z_0| = r$ is $|z|^2 - \bar{z}_0 z - z_0 \bar{z} + |z_0|^2 - r^2 = 0$, and the line through z_0 with nonzero direction v is $\operatorname{Im}(\bar{v}(z - z_0)) = 0$, that is $2\operatorname{Re}(i\bar{v}z) - 2\operatorname{Re}(i\bar{v}z_0) = 0$.

Consider the three maps $T(z) = z + p$, $M(z) = kz$ with $k \neq 0$, and $J(z) = 1/z$ (with $J(0) = \infty$ and $J(\infty) = 0$), each a Möbius transformation. Substituting $z = w - p$, $z = w/k$ and $z = 1/w$ respectively, and multiplying through by $|k|^2$ or by $|w|^2$, shows that z satisfies the equation with coefficients (A, B, C) if and only if its image w satisfies the equation with coefficients

$$(A, B - A\bar{p}, A|p|^2 - 2\operatorname{Re}(Bp) + C) \quad (A, B\bar{k}, C|k|^2) \quad (C, \bar{B}, A)$$

respectively, for $z, w \in \mathbb{C}$ and, in the case of J , $z, w \neq 0$. The three new triples satisfy $|B'|^2 - A'C' = |B|^2 - AC$, $|k|^2(|B|^2 - AC)$ and $|B|^2 - AC$, all positive. The points ∞ and 0 match as well: T and M fix ∞ and keep the coefficient A , while for J the point 0 lies on the first set exactly when $C = 0$, which is exactly when the second set contains ∞ , and symmetrically for ∞ and 0 . Since T , M and J are bijections of $\mathbb{C}_\infty$, each maps every generalized circle onto a generalized circle.

If $c = 0$ then $f(z) = (a/d)z + b/d$ is M followed by T . If $c \neq 0$, consider the four maps f_1, f_2, f_3, f_4 displayed before this proposition, a translation, J , a map of the form M and a translation. For $z \in \mathbb{C} \setminus \{-d/c\}$,

$$f_4(f_3(f_2(f_1(z)))) = f_4\left(\frac{bc - ad}{c^2} \cdot \frac{1}{z + d/c}\right) = \frac{a}{c} + \frac{(bc - ad)/c^2}{z + d/c}$$

which is $(az + b)/(cz + d)$ by the last line of the computation displayed there. The two remaining points also match: $-d/c \mapsto 0 \mapsto \infty \mapsto \infty \mapsto \infty$ and $\infty \mapsto \infty \mapsto 0 \mapsto 0 \mapsto a/c$. So $f = f_4 \circ f_3 \circ f_2 \circ f_1$ on $\mathbb{C}_\infty$. In both cases f is a composite of maps that carry generalized circles onto generalized circles, as we sought to show.

2. Given distinct $z_1, z_2, z_3 \in \mathbb{C}$, put

$$T_z(\zeta) = \frac{(z_1 - z_3)\zeta - z_2(z_1 - z_3)}{(z_1 - z_2)\zeta - z_3(z_1 - z_2)}$$

whose determinant is $(z_1 - z_3)(z_1 - z_2)(z_2 - z_3) \neq 0$. It satisfies $T_z(z_1) = 1$, $T_z(z_2) = 0$ and $T_z(z_3) = \infty$, the last because z_3 is the zero of the denominator. Define T_w by the same formula with w_1, w_2, w_3 in place of z_1, z_2, z_3 . Then $g = T_w^{-1} \circ T_z$ is a Möbius transformation by the first paragraph of this proof, and $g(z_k) = w_k$ for each k because T_w is a bijection of $\mathbb{C}_\infty$ sending w_1, w_2, w_3 to $1, 0, \infty$.

For uniqueness, let g and h be Möbius transformations with $g(z_k) = h(z_k) = w_k$ for each k . Then $\psi = T_z \circ g^{-1} \circ h \circ T_z^{-1}$ is a Möbius transformation fixing $1, 0$ and ∞ . Write $\psi(z) = (pz + q)/(rz + s)$. As $\psi(\infty) = \infty$ we have $r = 0$, and then $s \neq 0$. As $\psi(0) = 0$ we get $q/s = 0$, so $q = 0$. As $\psi(1) = 1$ we get $p/s = 1$. Hence ψ is the identity, so $g^{-1} \circ h$ is the identity and $h = g$.

3. Suppose first that $c = 0$, so that $d \neq 0$, $f(z) = (a/d)z + b/d$ and $f(\infty) = \infty$. A point $z \in \mathbb{C}$ is fixed exactly when $(a/d - 1)z = -b/d$. If $a \neq d$ this has exactly one solution, giving two fixed points in all. If $a = d$ and $b \neq 0$ it has none, and ∞ is the only fixed point. If $a = d$ and $b = 0$, then f is the identity.

Suppose now that $c \neq 0$. Then $f(\infty) = a/c \neq \infty$ and $f(-d/c) = \infty \neq -d/c$, so neither point is fixed. For the remaining $z \in \mathbb{C}$, the equation $f(z) = z$ is equivalent to $az + b = z(cz + d)$, that is

$$cz^2 + (d - a)z - b = 0$$

As $c \neq 0$, the quadratic formula gives one or two solutions in $\mathbb{C}$, according to whether $(d - a)^2 + 4bc$ vanishes. None of them equals $-d/c$, since substituting $z = -d/c$ into the left-hand side gives $(ad - bc)/c \neq 0$. So f has exactly one or exactly two fixed points, completing the proof.

We next analyze the global properties of Möbius transformations. These will be important for two theorems (note that biholomorphic means complex diffeomorphic), the first proved later in this chapter and the second stated here and used without proof:

1. *Riemann mapping theorem*: If U is a non-empty simply connected open subset of the complex number plane $\mathbb{C}$ which is not all of $\mathbb{C}$, then there exists a biholomorphic mapping f from U onto the open unit disk
2. *Uniformization theorem*: Let R be a Riemann surface (a complex 1-manifold) and $U \subseteq R$ a simply connected connected open subset of R . Then U is biholomorphic to one of the following:
 - (a) Riemann sphere (which we will eventually show is $\mathbb{CP}^1$)
 - (b) $\mathbb{C}$
 - (c) $\mathbb{D} = \{z \in \mathbb{C} \mid |z| < 1\}$

In other words, Möbius transformations enter the classification of all simply-connected one-dimensional complex manifolds, of which there are surprisingly only three up to biholomorphism. The following results give the global mapping properties that serve as lemmas for the Riemann mapping theorem. The uniformization theorem is not proved in this book.

Proposition 1.3.5: Möbius Transformation On Upper Half Plane

Let $f(z) = \frac{az+b}{cz+d}$, with $ad - bc \neq 0$, be a Möbius transformation, regarded as a map of $\mathbb{C}_\infty$ to itself as in proposition 1.3.4. Then:

1. if f maps $\mathbb{R} \cup \{\infty\}$ into $\mathbb{R} \cup \{\infty\}$, then f can be written in this form with $a, b, c, d \in \mathbb{R}$
2. if f maps $\mathbb{H}$ onto $\mathbb{D}$, then f must be of the form

$$f(z) = \eta \frac{z - \alpha}{z - \bar{\alpha}}$$

with $|\eta| = 1$ and $\text{Im}\alpha > 0$.

Proof:

1. At most one real number, namely $-d/c$ when $c \neq 0$ and $-d/c \in \mathbb{R}$, is sent to ∞ , so there are distinct $x_1, x_2, x_3 \in \mathbb{R}$ whose images $y_k = f(x_k)$ are finite, hence real by hypothesis, and distinct because f is injective. With the notation of the proof of proposition 1.3.4, the maps T_x and T_y built from x_1, x_2, x_3 and from y_1, y_2, y_3 have real coefficients. The inverse of a Möbius transformation with coefficient matrix $\begin{pmatrix} p & q \\ r & s \end{pmatrix}$ is the one with matrix $\begin{pmatrix} s & -q \\ -r & p \end{pmatrix}$, and a composite acts through the product of the matrices, so $g = T_y^{-1} \circ T_x$ has real coefficients. Since $g(x_k) = y_k = f(x_k)$ for $k = 1, 2, 3$, the uniqueness part of proposition 1.3.4(2) gives $f = g$, so f is written with real coefficients.
2. Since $0 \in \mathbb{D} = f(\mathbb{H})$, there is $\alpha \in \mathbb{H}$ with $f(\alpha) = 0$, so $\text{Im}\alpha > 0$ and $a\alpha + b = 0$. Then $a \neq 0$, since $a = 0$ would force $b = 0$ and $ad - bc = 0$, and

$$f(z) = \frac{a(z - \alpha)}{cz + d}$$

Let $x \in \mathbb{R}$ with $cx + d \neq 0$. For $t > 0$ the point $x + it$ lies in $\mathbb{H}$, so $|f(x + it)| < 1$. The point $x - it$ lies outside $\mathbb{H}$, and f is a bijection of $\mathbb{C}_\infty$ carrying $\mathbb{H}$ onto $\mathbb{D}$, so $f(x - it)$ lies outside $\mathbb{D}$. Letting $t \rightarrow 0$ and using the continuity of f at x , where its denominator does not vanish, gives $|f(x)| \leq 1$ and $|f(x)| \geq 1$, so $|f(x)| = 1$.

If $c = 0$, then f is a nonconstant polynomial of degree 1 and $|f(x)| \rightarrow \infty$ as $x \rightarrow \infty$ along $\mathbb{R}$, contradicting $|f(x)| = 1$. So $c \neq 0$, and with $\eta = a/c$ and $\beta = -d/c$ we have $f(z) = \eta(z - \alpha)/(z - \beta)$, where $\beta \neq \alpha$ because f is not constant. Letting $x \rightarrow \infty$ along $\mathbb{R}$ in $|f(x)| = 1$ gives $|\eta| = 1$, since $|x - \alpha|/|x - \beta| \rightarrow 1$. Hence $|x - \alpha| = |x - \beta|$ for every real $x \neq \beta$. Squaring and expanding,

$$-2x\text{Re}\alpha + |\alpha|^2 = -2x\text{Re}\beta + |\beta|^2$$

for infinitely many real x , so $\text{Re}\alpha = \text{Re}\beta$ and $|\alpha| = |\beta|$, hence $\text{Im}\beta = \pm\text{Im}\alpha$. The sign $+$ would give $\beta = \alpha$, so $\text{Im}\beta = -\text{Im}\alpha$ and $\beta = \bar{\alpha}$. Therefore $f(z) = \eta(z - \alpha)/(z - \bar{\alpha})$ with $|\eta| = 1$ and $\text{Im}\alpha > 0$, as claimed.

To better understand such maps, we will see how they map certain shapes to either $\mathbb{H}$ or $\mathbb{D}$. These are commonly occurring shapes in complex analysis and so knowing where they map is useful, in particular they will be useful when constructing Riemann surfaces via gluing. A circular wedge is a shape of one of these two forms:

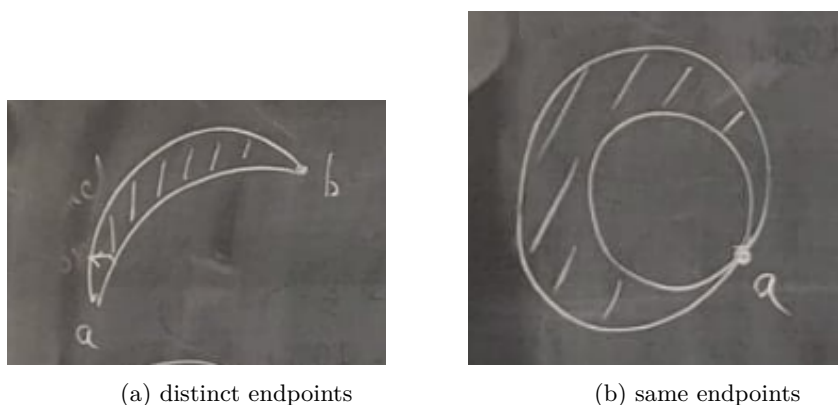
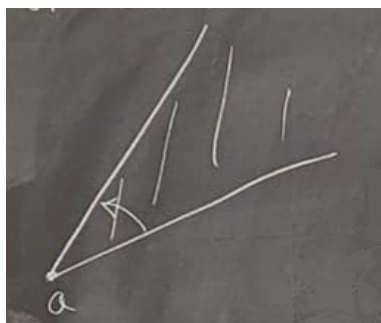


Figure 1.6: circular wedges

For figure (a), map a to 0 and b to infinity likeso:

$$\frac{z-a}{z-b}$$

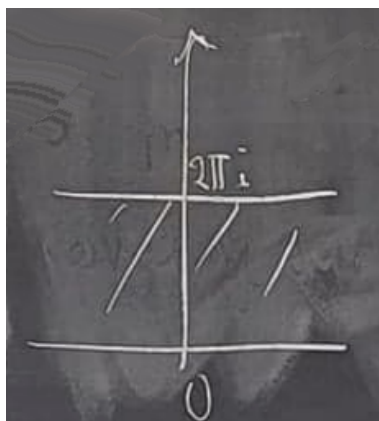
This will stretch out (a) so that it looks like so

Figure 1.7: apply $\frac{z-a}{z-b}$

Now just rotate so that one of the wedge lie on the x -axis, and scalar by an appropriate z^α so that it maps to $\mathbb{H}$. For figure (b), map it by $\frac{1}{z-a}$ so that it looks like so:

Figure 1.8: apply $\frac{1}{z-a}$

then rotate it and dilate it so that we get:



and finally exponentiate it to get $\mathbb{H}$. The last shape we will care to map is $\mathbb{C} - I$ for some open interval I . Without loss of generality say $I = (-1, 1)$. First apply the map $\frac{z+1}{z-1}$ to map $-1 \mapsto 0$, $1 \mapsto \infty$ and $0 \mapsto 0$, then apply the square root function $z^{1/2}$ for your choice of square root function, which will create a $\mathbb{H}$ but with the y -axis, then map that to the disk $\mathbb{D}$ so that we may map it to $\mathbb{H}$. this will make us end up with the equation $w = z - \sqrt{z^2 - 1}$.

1.3.4 Partial Fractions

Every rational function can be represented as a *partial fraction* which splits the rational function as a sum of two rational function with the degree of the numerator being smaller than the degree of the denominator. In order for this to be the case, we need that $R(z)$ has a pole at infinity. Such a decomposition is well-defined since polynomial division is well-defined¹¹.

¹¹If you have taken algebra, recall that $\mathbb{C}[x]$ is a Euclidean Domain

Definition 1.3.6: Partial Fraction Decomposition

Let $R(z)$ be a rational function. Then the resulting partial fraction decomposition will be of the form

$$R(z) = G(z) + H(z)$$

where $G(z)$ is a *polynomial* without a constant term, and $H(z)$ is finite at ∞ . The degree of G is the order of the pole at ∞ . The polynomial G is called the *singular part of R at ∞* .

Example 1.3.7: Partial Fraction Decomposition

here

We can manipulate this equation further to get some nice results that allow for easier integration of more complex functions. Let $b_1, \dots, b_n$ be the poles of $R(z)$. Define a new function $R'_j(\zeta) = R(b_j + 1/\zeta)$. R' has a pole at ∞ . By definition:

$$R'_j(\zeta) = R\left(b_j + \frac{1}{\zeta}\right) = G_j(\zeta) + H_j(\zeta)$$

Or, with some simple change of variables:

$$R(z) = G_j\left(\frac{1}{z - b_j}\right) + H_j\left(\frac{1}{z - b_j}\right)$$

Notice how now $R(z)$ is written in a different form: the polynomial G_j has now $\frac{1}{z - b_j}$ as its indeterminate. The polynomial G_j is called the *singular part of $R(z)$ at b_j* . By construction, $H_j\left(\frac{1}{z - b_j}\right)$ is finite for $z = b_j$. Now, consider the expression

$$R(z) - G_j\left(\frac{1}{z - b_j}\right) + H_j\left(\frac{1}{z - b_j}\right) = 0$$

(words in Authors explaining here on page 32 why we get)

$$R(z) = G(z) + \sum_{i=1}^q G_j\left(\frac{1}{z - b_j}\right)$$

which is the expression used in calculus to simplify integrals. We shall find the generalization of this result in theorem 4.2.7.

Holomorphic Functions

Now that we have studied the arithmetics, geometry, and analytical/topological properties of $\mathbb{C}$, we will move on to studying *differentiable complex functions*. Since $\mathbb{C}$ is homeomorphic to $\mathbb{R}^2$, we see that continuity of $\mathbb{C}$ works identically to that of $\mathbb{R}^2$. We will find that differentiability will produce some very different results, which will in fact greatly restrict which functions are differentiable and give *much* nicer results than the theory of differentiability of real numbers. We will in fact rather quickly answer lot of the “usual” questions that we had to build up a lot of work for in real analysis rather quickly in complex analysis, and we will see that complex-differentiable function are more like the “infinite analog” or “analytic analogue” of polynomials¹

After defining and expanding and exploring complex-differentiable functions and finding some necessary and sufficient conditions for a functions to be complex-differentiable, we will start building up a repertoire of complex-differentiable function. We will show that complex-linear functions are complex-differentiable (as we hope they should be). Then since the sum, product, and quotient of complex-differentiable functions is complex-differentiable, we will get that all polynomials and rational functions are complex-differentiable. The next step would be to explore infinite polynomials, that is power-series. Naturally, power-series need not converge everywhere and so a power-series will only be defined up to domain of convergence which will be well defined by a radius of convergence (with the exception of the border, for which we have to be more careful). Using power-series, we can show that many of our common functions from real analysis are complex-differentiable, most importantly e^x , $\sin$, $\cos$, $\log$. Using the technics we have built up, we can show many properties of these functions, like periodicity or the algebra of the exponential ($e^{x+y} = e^x e^y$).

¹Note that this is not true for real-differentiable functions

2.1 Build-up

let f be a real or complex function. We may ask whether some point a in the codomain satisfies:

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - Df(a)(h)}{\|h\|} = 0$$

The answer will partly depend on the domain and codomain. Let's say $f : \mathbb{C} \rightarrow \mathbb{R}$ is a differentiable function (i.e. the limit above exists for all $z \in \mathbb{C}$). Since we are working in "one-dimension" over $\mathbb{C}$, let's use the usual definition of the derivative:

$$f'(z) = \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h}$$

Remember that the value $f'(z)$ must be independent of how h approaches 0. Let's first approach $h \rightarrow 0$ as only real numbers (i.e., immediately make the imaginary component 0 and keep approaching that way). Then $f'(z)$ is the quotient of two real numbers, and hence will be real. On the other hand, if we $h \rightarrow 0$ from the complex component (i.e. immediately make the real component 0 and keep approaching that way), then $f'(z)$ is the quotient of a real number by an imaginary number. As we know

$$\frac{1}{i} = -i$$

and so the resulting number must be imaginary. The only number to be both real and imaginary is 0, and hence if f is indeed differentiable, then its derivative must be 0 (or f is not differentiable). As a consequence of this, by what we know of integration, the resulting function must be *constant*. This makes sense if we recall $\mathbb{C}$ -linear functions preserve angles, and projecting to $\mathbb{R}$ would certainly not preserve angles locally.

If we switch the order of the domain and co-domain and consider $f : \mathbb{R} \rightarrow \mathbb{C}$, then as the codomain has the same topology as $\mathbb{R}^2$, this case can be thought of as $f : \mathbb{R} \rightarrow \mathbb{R}^2$. Consider that any such function can be represented as $f(t) = x(t) + iy(t)$ for appropriate functions $x : \mathbb{R} \rightarrow \mathbb{R}$ and $y : \mathbb{R} \rightarrow \mathbb{R}$:

$$f(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$$

The element i will act simply as a scalar (or the vector component $(0, 1)$) since it is in the numerator, and so

$$f'(t) = x'(t) + iy'(t)$$

and so f is differentiable if x and y are differentiable. Thus, the case of $f : \mathbb{C} \rightarrow \mathbb{C}$ is truly different than the real case, so the rest of this section comes to figuring the structure of such a function

2.2 Complex Differentiation

To find out the conditions for which a function $f : \mathbb{C} \rightarrow \mathbb{C}$ must be complex-differentiable (which $\mathbb{R}^2$ -differentiable function can be interpreted as $\mathbb{C}$ -differentiable functions), we will define a function to be differentiable, and then find then necessary condition's that this will impose onto f . We will give a special name to functions that are complex-differentiable:

Definition 2.2.1: Holomorphic (Analytic) Function

let $f : \Omega \subseteq \mathbb{C} \rightarrow \mathbb{C}$ be a function with Ω open. Then if for every point $z \in \mathbb{C}$, $f'(z)$ is defined, that is

$$f'(z) = \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h}$$

then f is said to be *holomorphic* or *analytic*

You might recall from calculus that an analytic function is a function such that the function can be represented as a power-series in a local neighborhood at every point. Soon, we will show how this is true for all Holomorphic functions, and in fact is both a necessary and sufficient condition for a holomorphic function, and so the words Analytic and Holomorphic are used interchangeably by many mathematicians.²

The key example of holomorphic functions are:

Example 2.2.2: Holomorphic Functions

1. the function $f(z) = c$ for some $c \in \mathbb{C}$ is a holomorphic function
2. the function $f(z) = z$ is a holomorphic function
3. the function $f(z) = az$ for some $a \in \mathbb{C}$ is a holomorphic function (in other words, $\mathbb{C}$ -linear functions ought to be complex-differentiable).

Holomorphic functions work well under our usual binary operations, and have many properties we would expect of them:

²Note how this is *not* true for real-differentiable functions, consider $f(x) = e^{-1/x}$ when $x > 0$ and 0 otherwise (the derivative grows “too fast”, see cite:REAL-ANALYSIS)

Proposition 2.2.3: Properties of Holomorphic Functions

Let $f : \mathbb{C} \rightarrow \mathbb{C}$, $g : \mathbb{C} \rightarrow \mathbb{C}$ be holomorphic functions. Then

1. $f + g$, fg , $g \circ f$, and zf (for some $z \in \mathbb{C}$) are holomorphic functions (hence, with the result from example 2.2.2, all polynomials are holomorphic) with the same derivative formula as the real equivalent. Similarly, f/g is an holomorphic function, provided that $g(z) \neq 0$ for all $z \in \mathbb{C}$ (i.e. g does not vanish). Hence, all well-defined rational polynomials are holomorphic functions. Sometimes, we abuse notation and say f/g is holomorphic even if $g(z) = 0$ for some points by excluding those points from the domain.^a
2. f is continuous.
3. f is real-differentiable (interpreting f as $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$)
4. There exists functions u, v such that f can be written as $f(z) = u(z) + iv(z)$. Both u and v are continuous. In particular, if $z = x + iy$, then we have $f(z) = u(x, y) + iv(x, y)$. This is called the *standard representation of f* .

^aThis shall be less of an abuse once we define the derivative between $\mathbb{C}_\infty$

Proof:

1. This is the same breaking down the limit trick as seen in elementary calculus

2. Let

$$f(z+h) - f(z) = \frac{h}{h}(f(z+h) - f(z)) = h \cdot \left(\frac{f(z+h) - f(z)}{h} \right)$$

and, evaluating the limit, we get

$$0 \cdot f'(z) = 0$$

and so

$$\lim_{h \rightarrow 0} f(z+h) - f(z) = \Leftrightarrow \lim_{h \rightarrow 0} f(z+h) = f(z)$$

and since this is true for all z , f is continuous

3. Since f is holomorphic, then f' , exists, that is for any $z \in \mathbb{C}$, the limit

$$f'(z) = \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h}$$

exists. Take the absolute value of the right hand side:

$$\left| \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h} \right|$$

since $|\cdot|$ is continuous, it passes through limits:

$$\lim_{h \rightarrow 0} \left| \frac{f(z+h) - f(z)}{h} \right| = \lim_{h \rightarrow 0} \frac{|f(z+h) - f(z)|}{|h|}$$

which, since the topology of $\mathbb{C}$ is identical to that of $\mathbb{R}^2$, this is equivalent to saying that f , if interpreted as $\mathbb{R}^2 \rightarrow \mathbb{R}^2$, is real-differentiable!

4. Recall that every element of $\mathbb{C}$ can be written of the form $a + bi$. Thus, the output of any function $f(z)$ will be of the form $a + bi$. The real and imaginary part are dependent on both a and b , and $f(a + bi) = u(a, b) + iv(a, b)$ is a well-defined function where we have $u, v : \mathbb{R}^2 \rightarrow \mathbb{R}$. Furthermore, since f is holomorphic, it is continuous, and so restricting to only going through the real or imaginary part must also be continuous since projections are continuous, and hence u and v are continuous. Conversely, if u and v are continuous, then so is adding them (multiplying by i is equivalent to specifying a basis element), and so f is continuous.

Notice that since the projection map $\pi : \mathbb{R}^2 \rightarrow \mathbb{R}$ is a differentiable function, both u and v are real-differentiable when interpreted as $u, v : \mathbb{R}^2 \rightarrow \mathbb{R}$. Note that they *cannot* be interpreted as complex-differentiable, since all functions of the form $u, v : \mathbb{C} \rightarrow \mathbb{R}$ are constant, and u, v are certainly not always constant.

The collection of holomorphic function $f : \mathbb{C} \rightarrow \mathbb{C}$ is sometimes denoted $C^\omega(\mathbb{C})$. By the above proposition, $C^\omega(\mathbb{C})$ is a $\mathbb{C}$ -algebra, and is almost a division $\mathbb{C}$ -algebra³. We have yet to establish if we can put any properties on u, v to make f holomorphic. We will give sufficient conditions section 2.2.2. The following proposition we will outline a necessary condition that needs to be put on u, v :

Proposition 2.2.4: Cauchy-Riemann Condition

Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be an holomorphic function. Let $f(z) = f(x + iy) = u(x, y) + iv(x, y)$ be the standard representation of f , so that $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are well-defined. Then

$$\frac{\partial f}{\partial x} = -i \frac{\partial f}{\partial y}$$

Or equivalently, if we represent f as $f(z) = u(z) + iv(z)$, then

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

These equations are called the *Cauchy-Riemann Equations*.

Since we can take the limit any way we want, using the Cauchy-Riemann equation, we have that:

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial f}{\partial x} = -i \frac{\partial f}{\partial y}$$

making it relatively easy to find the complex-derivative of f by simply using real-differentiation. If $a = \frac{\partial u}{\partial x}$ and $b = \frac{\partial u}{\partial y}$, then in matrix form these equations are:

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$

which should strongly remind the reader of complex-linear function from section 1.3.1.

³In section 2.5.2, we shall show how we may loosen the definition a bit to make it into a field

Proof:

Recall that the limit at z must be the same regardless of how h approaches 0. If we approach strictly from the real part, then we will get the partial derivative

$$\frac{\partial f}{\partial x} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

On the other hand, if we approach from a purely imaginary number, we get an intriguing result: let $h = ik$ for some $k \in \mathbb{R}$. Then

$$f'(z) = \lim_{ik \rightarrow 0} \frac{f(z + ik) - f(z)}{ik} = \lim_{ik \rightarrow 0} -i \cdot \frac{f(z + ik) - f(z)}{k} = -i \frac{\partial f}{\partial y} = -i \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}$$

Since f' must be equal no matter which direction we approach from, it must be that:

$$f'(x) = \frac{\partial f}{\partial x} = -i \frac{\partial f}{\partial y}$$

or equivalently, since the real and imaginary part of these equations must agree, we get that:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

as we sought to show

Thus, for a function to be holomorphic this partial differential equation *must* be satisfied! Note that the converse is not true: If we have two functions u, v satisfying the Cauchy-Riemann condition, it does not imply that $f = u + iv$ is holomorphic (as we will show in example 2.2.8). On the other hand, with some added regularity to f (or to it's partials or u, v), the resulting function is complex-differentiable. We will explore more on what regularity condition's we can pose. For now, the simplest we can give is that f is real-differentiable and satisfies C.R. then it is Holomorphic

Proposition 2.2.5: C.R + Real-Differentiable Implies Complex Differentiable

Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be a real-differentiable^a functions satisfying the Cauchy-Riemann equations. Then f is holomorphic.

^aSometimes called Fréchet differentiable due to it's generalization to normed vector spaces

Proof:

Since f is real-differentiable, there exists a matrix A such that

$$F((x, y) + h) = F(x, y) + Ah + o(|h|) \quad (2.1)$$

as $h \rightarrow 0$ with

$$A = \begin{pmatrix} \frac{\partial u}{\partial x}(x, y) & \frac{\partial u}{\partial y}(x, y) \\ \frac{\partial v}{\partial x}(x, y) & \frac{\partial v}{\partial y}(x, y) \end{pmatrix}$$

By the Cauchy-Riemann equations, the matrix A represents the multiplication by the complex number

$$\lambda = \frac{\partial u}{\partial x}(x, y) + i \frac{\partial v}{\partial x}(x, y)$$

and so we can re-write equation (2.1) in complex terms with $z = x + iy$ as

$$f(z + h) = f(z) + \lambda h + o(|h|)$$

as $h \rightarrow 0$. But this implies f is holomorphic, completing the proof.

Holomorphy and the Cauchy-Riemann equations are local conditions, so the two propositions above combine into a single criterion valid on any open set, not just on all of $\mathbb{C}$.

Corollary 2.2.6: Cauchy-Riemann Characterization on an Open Set

Let $U \subseteq \mathbb{C}$ be open and let $f = u + iv$ be a function on U with $u, v \in C^1(U)$. Then f is holomorphic on U if and only if the Cauchy-Riemann equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

hold at every point of U , equivalently $\frac{\partial f}{\partial \bar{z}} = 0$ throughout U .

Proof:

Both holomorphy and the Cauchy-Riemann equations may be checked point by point on U . If f is holomorphic on U then the equations hold at every point by proposition 2.2.4. Conversely, a function with $u, v \in C^1(U)$ is real-differentiable at each point of U , so if the equations hold throughout U then f is holomorphic on U by proposition 2.2.5.

As we know, a function is differentiable if it is locally linear. Thus, we would expect that a holomorphic function is locally $\mathbb{C}$ -linear. As we know from multi-variable calculus, a function f real-differentiable at a produces a tangent map df_a which is $\mathbb{R}$ -linear between tangent spaces. We will show that if $f = u + iv$ is holomorphic, then df_a is $\mathbb{C}$ -linear.

Proposition 2.2.7: Holomorphic, Then Locally Complex Linear

Let f be holomorphic, $a \in \mathbb{C}$, and define df_a to be the tangent map of a real-differentiable function as defined above and $f = u + iv$ be the standard representation. Then u, v satisfy the Cauchy-Riemann condition if and only if df_a is $\mathbb{C}$ -linear, and:

$$\frac{\partial f}{\partial z}(a)(\zeta) = df_a(\zeta) \quad \forall \zeta \in \mathbb{C}$$

Proof:

Let $d_p f : \mathbb{C} \rightarrow \mathbb{C}$ be the tangent map at $p \in \mathbb{C}$. Then as we showed in proposition 2.2.5, the linear approximation satisfies the Cauchy-Riemann equations, which forces our matrix to be of the form representing complex multiplication. But then $d_p f$ is $\mathbb{C}$ -linear.

Example 2.2.8: Holomorphic and non-Holomorphic Functions

1. Let

$$f(x + yi) = \sqrt{|x||y|}$$

Then f is Cauchy-Riemann, but is not Holomorphic (approach from $h = (h_1, h_2) = (t, t)$ and $h = (h_1, h_2) = (t, -t)$). Notice that f is not real-differentiable, showing satisfying the Cauchy-Riemann condition is insufficient for f to be complex-differentiable (which should make sense, recall we needed C^1 for existence of partials implying existence of total derivative)

2. You should check that $f(z) = \bar{z}$ (i.e. the conjugation function) does not satisfy the Cauchy-Riemann. However, $f(z)$ is real-differentiable. Thus, though it looks very complex-differentiable (it's real-differentiable), it cannot be!
3. The function $f(z) = x^2 - y^2 - i2xy = z^2$ will satisfy the Cauchy-Riemann equations and is real-differentiable.
4. More generally, any polynomial

$$f(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$$

satisfies the Cauchy-Riemann equations and are holomorphic, with derivative

$$f'(z) = n a_n z^{n-1} + (n-1) a_{n-1} z^{n-2} + \cdots + a_1$$

5. $R = f/g$ be a ratioanl function. Let $\mathbb{C} - P$ be the domain of R , where P is the set of poles. Then R is holomorphic. R is almost holomorphic on all of $\mathbb{C}$, only failing on a finite set of points. A function which is holomorphic on all but a discrete set of points is called a *meromorphic* functions (from the greek *mero* meaning “part”). We shall show in section 2.5.2 that poles and zeros of a holomorphic functions cannot accumulate^a, making this concept more than a 1-off.
6. Let $f : U \rightarrow \mathbb{C}$ be a holomorphic function. Taking the difference quotient defining f' along real increments shows that $f' = \frac{\partial f}{\partial x}$. Thus, the notion of real and complex differentiation are compatible. If we take $f|_{U \cap \mathbb{R}} : U \cap \mathbb{R} \rightarrow \mathbb{C}$ to be the restriction of f to the real line, then the real-derivative of $f|_{U \cap \mathbb{R}}$ exists and is equal to the restriction of $f' : U \rightarrow \mathbb{C}$ to the real line.
7. All of these functions are not Holomorphic since non satisfy the Cauchy Riemann equations:

$$|z|, \quad \text{Re}z, \quad \text{Im}z, \quad \text{Arg}(z), \quad \bar{z}$$

This can also be seen since any real holomorphic function must be constant (by the observations we have made earlier), but none of these are constant.

We shall in fact be able to classify all holomorphic functions rather precisely. The classification shall be strongly dependent on which open subset $U \subseteq \mathbb{C}$ we are choosing to define our functions, see chapter 4.

^athere always exists an ϵ -ball that contains infinitely many points. It shall imply the entire function is 0

Before continuing onto further explorations of holomorphic functions, a comment must be made: there is often a game in mathematics about how weak of a condition can you impose on your building blocks to make our original condition hold. In this case, we have that $f = u + iv$, and we may ask how weak of conditions can be impose on u and v (or f) to make f Holomorphic. We saw that if f is real-differentiable and satisfies the Cauchy condition, then f is complex-differentiable. In fact, we can assume even more weakly that f is continuous and u, v each have first partial derivatives. This is the *Looman-Menchoff Theorem*, which we shall not prove. We will take a moment to prove a stronger claim than proposition 2.2.5 to practice working with partial derivatives:

Theorem 2.2.9: C^1 + C.R. Then Holomorphic

Let $u, v : \mathbb{R}^2 \rightarrow \mathbb{R}$ be conjugate harmonic functions (or more weakly, u, v are both C^1 and satisfy the Cauchy-Riemann equations). Then:

$$f = u + iv$$

is a holomorphic function.

2.2.10 Note We will later be able to say if and only if to the above theorem by showing that f is Holomorphic, then $f \in C^\infty(\mathbb{C})$.

Proof:

Let $u, v \in C^1(\mathbb{R}^2)$ where u, v satisfy the Cauchy Riemann equations. As we know, the derivative is a linear approximation and so gives rise to the following equation learnt in elementary calculus:

$$u(x + h, y + k) - u(x, y) = \frac{\partial u}{\partial x}h + \frac{\partial u}{\partial y}k + \epsilon_1$$

$$v(x + h, y + k) - v(x, y) = \frac{\partial v}{\partial x}h + \frac{\partial v}{\partial y}k + \epsilon_2$$

where ϵ_1 and ϵ_2 tend to zero faster than h and ik , and so in particular more rapidly than $h + ik$, i.e.

$$\frac{\epsilon_1}{h + ik} \rightarrow 0 \quad \frac{\epsilon_2}{h + ik} \rightarrow 0$$

Using the Cauchy-Riemann equations, we can re-write $f = u + iv$ as:

$$f(z + (h + ik)) - f(z) = \left(\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \right) (h + ik) + \epsilon_1 + i\epsilon_2$$

dividing by $h + ik$ and taking the limit as $(h + ik) \rightarrow 0$, we get:

$$\lim_{(h+ik) \rightarrow 0} \frac{f(z + (h + ik)) - f(z)}{h + ik} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

showing the limit exists. Since u and v also satisfy the Cauchy-Riemann condition, f must be holomorphic, as we sought to show

2.2.1 Relation to Conjugate

Consider a holomorphic function $f : \mathbb{C} \rightarrow \mathbb{C}$. We can interpret such a function as a real-differentiable function $f : \mathbb{R}^2 \rightarrow \mathbb{C}$ that satisfies the Cauchy-Riemann equations: $f(x, y)$. We will analyze such a map and see that it will give us some useful relations between $f(z)$, $f(\bar{z})$ and $\bar{f}(z)$. Since

$$z = x + iy \quad \bar{z} = x - iy$$

we can isolate x and y and get:

$$x = \frac{z + \bar{z}}{2} \quad y = \frac{z - \bar{z}}{2i}$$

Then applying the chain rule imagining we can do real-differentiation, we get the following equations that are important enough to be given a name:

Definition 2.2.11: Wirtinger Derivatives

Let f be a $\mathbb{R}^2$ -differentiable function (*not* necessarily complex-differentiable). Then define the following operations:

$$\frac{\partial}{\partial z} := \frac{1}{2} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) \quad \frac{\partial}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right)$$

We may think of these as operator's that we may apply to any real differentiable function with complex codomain:

$$\frac{\partial}{\partial z}(f) = \frac{\partial f}{\partial z} = \frac{1}{2} \left(\frac{\partial f}{\partial x} - i \frac{\partial f}{\partial y} \right)$$

Let f be a holomorphic function. Taking the limit of the difference quotient defining $f'(z)$ along the real and the imaginary directions gives $\frac{\partial f}{\partial x} = f'$ and $\frac{\partial f}{\partial y} = if'$, which are the Cauchy-Riemann conditions, so:

$$\frac{\partial f}{\partial z} = \frac{1}{2} (f' - i \cdot if') = f' \quad \text{and} \quad \frac{\partial f}{\partial \bar{z}} = \frac{1}{2} (f' + i \cdot if') = 0$$

This shows that we can think of a holomorphic function f as being a function of a *complex variable*, rather than a function of *two real variables* with a complex output (in other word, you can think that the imaginary part of a holomorphic function is essentially determined, up to a constant, by the real part, and vice-versa). We can also think of holomorphic functions as being independent of their conjugate, meaning if we interpret a complex function as being $f(z, \bar{z})$ then we would get $f(0, \bar{z}) = c$ for some constant $c \in \mathbb{C}$ (so we cannot have any conjugate terms in a complex function). Taking the above definition, we have that:

$$\begin{aligned} \frac{\partial}{\partial z} z &= 1 & \frac{\partial}{\partial z} \bar{z} &= 0 \\ \frac{\partial}{\partial \bar{z}} z &= 0 & \frac{\partial}{\partial \bar{z}} \bar{z} &= 1 \end{aligned}$$

Using this, we see that if

$$f(z) = \sum_{n,m=0}^k a_{m,n} z^m \bar{z}^n$$

Then f is holomorphic if and only if $a_{m,n} = 0$ for all $n \neq 0$, hence no complex polynomial which is holomorphic has $\bar{z}$ as a variable. For a real-differentiable function f , define $\bar{f}$ by $\bar{f}(z) = \overline{f(z)}$. Its partial derivatives are $\frac{\partial \bar{f}}{\partial x} = \overline{\frac{\partial f}{\partial x}}$ and $\frac{\partial \bar{f}}{\partial y} = \overline{\frac{\partial f}{\partial y}}$, and conjugating the two operators of definition 2.2.11 gives

$$\begin{aligned}\frac{\partial \bar{f}}{\partial z} &= \frac{1}{2} \left(\frac{\partial \bar{f}}{\partial x} - i \frac{\partial \bar{f}}{\partial y} \right) = \overline{\frac{1}{2} \left(\frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} \right)} = \overline{\frac{\partial f}{\partial \bar{z}}} \\ \frac{\partial \bar{f}}{\partial \bar{z}} &= \frac{1}{2} \left(\frac{\partial \bar{f}}{\partial x} + i \frac{\partial \bar{f}}{\partial y} \right) = \overline{\frac{1}{2} \left(\frac{\partial f}{\partial x} - i \frac{\partial f}{\partial y} \right)} = \overline{\frac{\partial f}{\partial z}}\end{aligned}$$

so taking the conjugate of a function “flips” the behavior of the Wirtinger derivative. In particular, if f is holomorphic, then

$$\frac{\partial \bar{f}}{\partial z} = 0 \quad \text{and} \quad \frac{\partial \bar{f}}{\partial \bar{z}} = \bar{f}'$$

In terms of differential forms, if we take $df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$, then $dz = dx + i dy$ and $d\bar{z} = dx - i dy$. If we re-write, we get

$$dx = \frac{1}{2}(dz + d\bar{z}) \quad dy = \frac{1}{2i}(dz - d\bar{z})$$

giving us df in terms of dz and $d\bar{z}$ to be

$$df = \frac{1}{2} \left(\frac{\partial f}{\partial x} - i \frac{\partial f}{\partial y} \right) dz + \frac{1}{2} \left(\frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} \right) d\bar{z}$$

we can then shorten this by putting in the Wirtinger derivative to get:

$$df = \frac{\partial f}{\partial z} dz + \frac{\partial f}{\partial \bar{z}} d\bar{z}$$

and thus we get

$$df = \frac{\partial f}{\partial z} dz$$

which recovers our original notion of derivative of differential form.

2.2.2 Harmonic Properties and Functions

Let $f = u + iv$ be a holomorphic function. Due to the Cauchy-Riemann equations, u and v have some interesting analytical properties. For the next part, we shall be adding some assumption that are not technically needed. Since we will eventually show that f will be C^∞ , the following discussion is in fact broader than the restrictions we shall put it in, but for now since we have yet to prove f being differentiable once implies infinitely differentiable, we will explicitly assume that the partials of f are at least twice differentiable, meaning the partials of u and v are at least twice differentiable.

Definition 2.2.12: Harmonic Function

Let $f : \mathbb{C} \rightarrow \mathbb{R}$ be an at least twice-differentiable function. Then if f satisfies:

$$\Delta f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$$

Then f is said to be *harmonic*. If f is a multi-valued function, then f is harmonic if each of its components is.

The intuition that I have for harmonic functions is given any point $f(z)$, the “average” of all of its neighbouring points is zero. Think about this in terms of the 2nd derivative for a function from $\mathbb{R} \rightarrow \mathbb{R}$. If the 2nd derivative is zero, then you have a line (as you can check). At any point on the line, the “average” of its neighbouring point is zero. If f'' was not zero, then if you look at point's around it (say for the function x^2 at 0 where $f''(0) > 0$) then the average of the points around it are not zero (in this example, they are positive)!

For a function u of (x, y) write u_x, u_y for its first partial derivatives and $u_{xy} = (u_x)_y$, $u_{yx} = (u_y)_x$.

Lemma 2.2.13: Symmetry of Mixed Partial Derivatives

Let $U \subseteq \mathbb{R}^2$ be open and let $u : U \rightarrow \mathbb{C}$ have continuous partial derivatives of order at most 2. Then $u_{xy} = u_{yx}$ on U .

Proof:

Applying the claim to the real and imaginary parts, we may assume u real valued. Fix $(x, y) \in U$ and $r > 0$ with $[x - r, x + r] \times [y - r, y + r] \subseteq U$, and for $0 < |h| < r$ put

$$\Delta(h) = u(x + h, y + h) - u(x + h, y) - u(x, y + h) + u(x, y)$$

The function $\varphi(s) = u(s, y + h) - u(s, y)$ is differentiable with $\varphi'(s) = u_x(s, y + h) - u_x(s, y)$, so the mean value theorem ([3, The Mean Value Theorem]) gives s_1 between x and $x + h$ with $\Delta(h) = \varphi(x + h) - \varphi(x) = h\varphi'(s_1)$. Applied to $t \mapsto u_x(s_1, t)$, whose derivative is $u_{xy}(s_1, t)$, it gives t_1 between y and $y + h$ with $\varphi'(s_1) = hu_{xy}(s_1, t_1)$. The function $\psi(t) = u(x + h, t) - u(x, t)$ is differentiable with $\psi'(t) = u_y(x + h, t) - u_y(x, t)$, so the mean value theorem gives t_2 between y and $y + h$ with $\Delta(h) = \psi(y + h) - \psi(y) = h\psi'(t_2)$. Applied to $s \mapsto u_y(s, t_2)$, whose derivative is $u_{yx}(s, t_2)$, it gives s_2 between x and $x + h$ with $\psi'(t_2) = hu_{yx}(s_2, t_2)$. Hence $u_{xy}(s_1, t_1) = u_{yx}(s_2, t_2)$, where both points lie within distance $2|h|$ of (x, y) . Letting $h \rightarrow 0$ and using the continuity of u_{xy} and u_{yx} gives $u_{xy}(x, y) = u_{yx}(x, y)$, as claimed.

Lemma 2.2.14: Holomorphic Then Harmonic

Let f be a holomorphic function and represent it as $f(x + iy) = u(x, y) + iv(x, y)$ with u and v twice continuously differentiable. Then u and v must be harmonic:

$$\Delta u = 0 \quad \Delta v = 0$$

Proof:

The second partial derivatives of u and v are continuous, so lemma 2.2.13 applies to u and to v . With the Cauchy-Riemann equations $u_x = v_y$ and $u_y = -v_x$,

$$\Delta u = (u_x)_x + (u_y)_y = (v_y)_x - (v_x)_y = v_{yx} - v_{xy} = 0$$

and $\Delta v = (v_x)_x + (v_y)_y = -(u_y)_x + (u_x)_y = -u_{yx} + u_{xy} = 0$, as claimed.

Thus, if f is holomorphic and u, v are obtained via the standard representation, we have $\Delta u = \Delta v = 0$. This gives us a necessary condition⁴ for a function f to be holomorphic. For future reference we box this result: Functions u, v satisfying the above criterion are actually quite important in PDE's, and so we will label them for future reference:

Definition 2.2.15: Laplace Operator And Harmonic Functions

Let f be a function that is at least twice differentiable. Then the *Laplace operator* on f is defined as:

$$\Delta f = \nabla^2 f = \nabla \cdot \nabla f = \sum_{i=1}^n \frac{\partial^2}{\partial x_i^2} f$$

If $\Delta f = 0$, then f is said to be *harmonic*.

The converse is also true: if u, v satisfy the Cauchy-Riemann equations and are each Harmonic, then we can in fact construct a holomorphic function f via $f := u + iv$. Note that being harmonic does not imply the Cauchy-Riemann equations: $z \mapsto \bar{z}$ is harmonic, but not holomorphic. However, since being harmonic implies being at least C^1 , by proposition 2.2.5 the function is holomorphic.

Definition 2.2.16: Harmonic Conjugate Function

If u is a harmonic function and v is the harmonic function that makes $f = u + iv$ an holomorphic function, then v is called the *harmonic conjugate* of u . In particular, u, v are harmonic functions satisfying the Cauchy-Riemann equations

Example 2.2.17: Finding Conjugate harmonic Function

Let $u(x, y) = x^2 - y^2$. This is a harmonic function, and so we can proceed in finding it's harmonic compliment. Notice that

$$\frac{\partial u}{\partial x} = 2x \quad \frac{\partial u}{\partial y} = -2y$$

Thus

$$\frac{\partial v}{\partial x} = 2y \quad \frac{\partial v}{\partial y} = 2x$$

We can integrate one or the other equation to solve for v . To make an arbitrary choice, let's pick the first one and integrate it with respect to x to get

$$v = 2xy + \varphi(y)$$

⁴but not sufficient, it does not imply the Cauchy-Riemann conditions

where $\varphi(y)$ is some functions based on y . Deriving with respect to y gives us

$$\frac{\partial v}{\partial y} = 2x + \varphi'(y) = 2x$$

telling us that $\varphi'(y) = 0$ so $\varphi(y) = 0$. Thus we get

$$f(z) = f(x + iy) = x^2 + y^2 + i(2xy) = z^2$$

You can now verify that $f(z) = x^2 + y^2 + i2xy$ is indeed holomorphic. Check that this function is actually $f(z) = z^2$. What's the derivative? Remember that you can write it in terms of partial derivatives of u and v . Can you write it in terms of z ?

For reference, here is a list of harmonic conjugates

u	v	$u + iv$
x	y	z
x	$y + 1$	$z + i$
y	$-x$	$-iz$
$x^2 - y^2$	$2xy$	z^2
$e^x \cos(y)$	$e^x \sin(y)$	e^z
$\frac{x}{x^2 + y^2}$	$\frac{-y}{x^2 + y^2}$	$\frac{1}{z}$

Note that for the last example we must exclude the origin from the domain. Being harmonic gives us some important regularity on functions maxima's and minima's:

Theorem 2.2.18: Maximum Principle Of Harmonic Functions

Let U be an open subset of $\mathbb{C}$ and let $u : U \rightarrow \mathbb{R}$ be a harmonic function. Let K be a compact subest of U , and let ∂K be the boundary of K . Then:

$$\sup_{z \in K} u(z) = \sup_{z \in \partial K} u(z)$$

and

$$\inf_{z \in K} u(z) = \inf_{z \in \partial K} u(z)$$

Proof:

We will prove it for sup, since inf follows similarly (note that $-u$ is also harmonic). It is always the case that $\sup_{z \in K} u(z) \geq \sup_{z \in \partial K} u(z)$, so for the sake of contradiction let's say

$$\sup_{z \in K} u(z) > \sup_{z \in \partial K} u(z)$$

As u is continuous and K is compact, u achieves it's maximum at some point, say $z_0 \in K$. By our assumption, z_0 is in the interior. Since z_0 is a local maximum of u , and u is twice differentiable, we must have:

$$\frac{\partial^2 u}{\partial x^2}(z_0) \leq 0 \quad \frac{\partial^2 u}{\partial y^2}(z_0) \leq 0$$

This almost contradicts the harmonicity of u , but it is still possible that both of these partial derivatives vanish. To fix this, we wiggle around an epsilon amount of room to add some convexity. Letting $\epsilon > 0$ be a small number we will choose later, let $u_\epsilon : U \rightarrow \mathbb{R}$ be the modified function

$$u_\epsilon(x + iy) := u(x + iy) + \epsilon(x^2 + y^2)$$

Since K is compact, the function $x^2 + y^2$ is bounded on K . If ϵ is small enough, then by our assumption on u we must also have on u_ϵ the relation:

$$\sup_{z \in K} u_\epsilon(z) > \sup_{z \in \partial K} u_\epsilon(z)$$

By the same argument u_ϵ achieves its maximum at some interior point z_ϵ of K , thus:

$$\frac{\partial^2 u}{\partial x^2}(z_\epsilon) \leq 0 \quad \frac{\partial^2 u_\epsilon}{\partial y^2}(z_\epsilon) \leq 0$$

Since u is harmonic, we must have:

$$\frac{\partial^2 u_\epsilon}{\partial x^2} + \frac{\partial^2 u_\epsilon}{\partial y^2} = \frac{\partial^2 u}{\partial x^2} + 2\epsilon + \frac{\partial^2 u}{\partial y^2} + 2\epsilon = 4\epsilon > 0$$

on U . But that's a contradiction, completing the proof.

Corollary 2.2.19: Maximum Principle For Holomorphic Functions

Let $f : U \rightarrow \mathbb{C}$ be a continuously twice differentiable holomorphic function on an open set U and K a compact subset of U . Then

$$\sup_{z \in K} |f(z)| = \sup_{z \in \partial K} |f(z)|$$

This result is also known as the *maximum modulus principle*

Proof:

For $\theta \in \mathbb{R}$, the function $u_\theta = \operatorname{Re}(e^{i\theta} f)$ is harmonic on U by lemma 2.2.14, applied to the holomorphic function $e^{i\theta} f$, whose real and imaginary parts are twice continuously differentiable. By theorem 2.2.18, $\sup_K u_\theta = \sup_{\partial K} u_\theta \leq \sup_{\partial K} |f|$. Every $w \in \mathbb{C}$ satisfies $|w| = \sup_{\theta \in \mathbb{R}} \operatorname{Re}(we^{i\theta})$, the supremum being attained at $e^{i\theta} = \bar{w}/|w|$ when $w \neq 0$, so

$$\sup_{z \in K} |f(z)| = \sup_{\theta \in \mathbb{R}} \sup_{z \in K} u_\theta(z) \leq \sup_{z \in \partial K} |f(z)|$$

The reverse inequality holds because $\partial K \subseteq K$, K being closed, completing the proof.

There is naturally a *minimum modulus principle* which has the same result but with inf.

Proposition 2.2.20: Consequences Of Harmonic Functions

Let $U \subseteq \mathbb{C}$ be an open subset.

1. If $f : U \rightarrow \mathbb{C}$ is twice continuously differentiable

$$\Delta f = 4 \frac{\partial}{\partial z} \frac{\partial f}{\partial \bar{z}} = 4 \frac{\partial}{\partial \bar{z}} \frac{\partial f}{\partial z}$$

2. If f is a complex polynomial:

$$f(z) = \sum_{\substack{n,m \geq 0 \\ n+m \leq d}} c_{n,m} z^n \bar{z}^m$$

then f is harmonic on $\mathbb{C}$ if and only if $c_{n,m}$ vanishes whenever n and m are both positive (f only contains terms $c_{n,0} z^n$ or $c_{0,m} \bar{z}^m$)

3. if $u : U \rightarrow \mathbb{R}$ is a real polynomial

$$u(x + iy) = \sum_{\substack{n,m \geq 0 \\ n+m \leq d}} a_{n,m} x^n y^m$$

then u is harmonic if and only if it is the real part of a complex polynomial

$$f(z) = \sum_{n=0}^d c_n z^n$$

Proof:

For (1), expand the Wirtinger derivatives (definition 2.2.11):

$$4 \frac{\partial}{\partial z} \frac{\partial f}{\partial \bar{z}} = \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) \left(\frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + i \left(\frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial y \partial x} \right)$$

Since f is twice continuously differentiable, the symmetry of second mixed partial derivatives (lemma 2.2.13) makes the bracketed term vanish, leaving Δf as in definition 2.2.15. Composing the two operators in the other order gives $\Delta f + i \left(\frac{\partial^2 f}{\partial y \partial x} - \frac{\partial^2 f}{\partial x \partial y} \right)$, which equals Δf by the same symmetry.

For (2), the function f is a polynomial in x and y , hence smooth, so (1) applies. The operators $\frac{\partial}{\partial z}$ and $\frac{\partial}{\partial \bar{z}}$ are complex linear combinations of $\frac{\partial}{\partial x}$ and $\frac{\partial}{\partial y}$, so they are linear and satisfy the Leibniz rule. Combining this with the identities $\frac{\partial}{\partial z} z = 1$, $\frac{\partial}{\partial z} \bar{z} = 0$, $\frac{\partial}{\partial \bar{z}} z = 0$, $\frac{\partial}{\partial \bar{z}} \bar{z} = 1$ computed after definition 2.2.11, induction on the exponents gives

$$\frac{\partial}{\partial \bar{z}} (z^n \bar{z}^m) = m z^n \bar{z}^{m-1} \quad \frac{\partial}{\partial z} (z^n \bar{z}^m) = n z^{n-1} \bar{z}^m$$

Hence by (1)

$$\Delta f = 4 \frac{\partial}{\partial z} \frac{\partial f}{\partial \bar{z}} = 4 \sum_{\substack{n,m \geq 1 \\ n+m \leq d}} nm c_{n,m} z^{n-1} \bar{z}^{m-1}$$

If $c_{n,m} = 0$ whenever n and m are both positive, this sum is empty and f is harmonic. For the converse we record a coefficient lemma: a polynomial $P = \sum_{a,b} b_{a,b} z^a \bar{z}^b$ which vanishes identically on a nonempty open set $V \subseteq \mathbb{C}$ has all coefficients zero. Fix $z_0 \in V$ and rewrite P as a polynomial $\sum_{a,b} b'_{a,b} w^a \bar{w}^b$ in $w := z - z_0$ by expanding each monomial binomially. Iterating the two displayed derivative rules, the derivative $\frac{\partial^{j+k} P}{\partial w^j \partial \bar{w}^k}$ evaluated at $w = 0$ equals $j! k! b'_{j,k}$: every monomial $w^a \bar{w}^b$ with $a < j$ or $b < k$ is annihilated, and every one with $(a,b) \neq (j,k)$ surviving differentiation retains a positive power of w or $\bar{w}$ and vanishes at 0. Since P vanishes on a neighbourhood of z_0 , so do all its partial derivatives, whence every $b'_{j,k} = 0$ and $P = 0$ identically on $\mathbb{C}$. Expanding around 0 instead, the same evaluation gives $b_{a,b} = 0$ for all a, b . Now if f is harmonic on $\mathbb{C}$, the lemma applied to $P = \frac{1}{4} \Delta f$ gives $nm c_{n,m} = 0$, hence $c_{n,m} = 0$, whenever $n, m \geq 1$.

For (3), substituting $x = \frac{z+\bar{z}}{2}$ and $y = \frac{z-\bar{z}}{2i}$ and expanding the products writes u as a polynomial

$$u = \sum_{\substack{n,m \geq 0 \\ n+m \leq d}} c_{n,m} z^n \bar{z}^m$$

with complex coefficients $c_{n,m}$. Since u is real valued, $u - \bar{u}$ vanishes on U , and conjugating the expansion term by term gives $\bar{u} = \sum_{n,m} \overline{c_{m,n}} z^n \bar{z}^m$. The coefficient lemma of part (2) applied to $u - \bar{u}$ gives $c_{n,m} = \overline{c_{m,n}}$ for all n, m . In particular $c_{0,0}$ is real.

Suppose u is harmonic on U . The computation of part (2) shows $\frac{1}{4} \Delta u = \sum_{n,m \geq 1} nm c_{n,m} z^{n-1} \bar{z}^{m-1}$ vanishes on the nonempty open set U , so the coefficient lemma gives $c_{n,m} = 0$ whenever $n, m \geq 1$. Using $c_{0,m} = \overline{c_{m,0}}$,

$$u = c_{0,0} + \sum_{n=1}^d c_{n,0} z^n + \sum_{m=1}^d c_{0,m} \bar{z}^m = c_{0,0} + \sum_{n=1}^d (c_{n,0} z^n + \overline{c_{n,0}} \bar{z}^n)$$

so $u = \operatorname{Re}(f)$ for the complex polynomial $f(z) = c_{0,0} + \sum_{n=1}^d 2c_{n,0} z^n$ of degree at most d . Conversely, suppose $u = \operatorname{Re}(f)$ on U with $f(z) = \sum_{n=0}^d c_n z^n$, so that $2u = f + \bar{f}$ where $\bar{f} = \sum_{n=0}^d \overline{c_n} \bar{z}^n$. The displayed derivative rules give $\frac{\partial f}{\partial \bar{z}} = 0$ and $\frac{\partial \bar{f}}{\partial z} = 0$, so by (1), used once in each order,

$$2\Delta u = \Delta f + \Delta \bar{f} = 4 \frac{\partial}{\partial z} \frac{\partial f}{\partial \bar{z}} + 4 \frac{\partial}{\partial \bar{z}} \frac{\partial \bar{f}}{\partial z} = 0$$

and u is harmonic.

We leave with a generalization of the notion of harmonic conjugate. We saw that every harmonic polynomial has at least one harmonic conjugate (up to constant). We may ask whether this holds for more general harmonic functions. If $U = \mathbb{C}$ is the entire plane, this is indeed the case:

Proposition 2.2.21: Entire Harmonic Conjugate On $\mathbb{C}$

Let $u: \mathbb{C} \rightarrow \mathbb{R}$ be a harmonic function with continuous partial derivatives of order at most 2. Then there exists a harmonic conjugate $v: \mathbb{C} \rightarrow \mathbb{R}$ of u . Furthermore, this harmonic conjugate is unique up to constants: if v, v' are two harmonic conjugates of u , then $v' - v$ is a constant function.

Proof:

Write u_x, u_y, u_{xx}, u_{yy} for the partial derivatives of u as a function of $(x, y) \in \mathbb{R}^2$, all continuous by hypothesis. Integrals are oriented, $\int_0^x = -\int_x^0$ for $x < 0$, and every integrand below is continuous, hence integrable ([3, Continuous Functions Are Integrable]). For a continuous function h on $\mathbb{R}$, the function $x \mapsto \int_0^x h$ is differentiable with derivative h : on an interval $[a, b]$ with $a < 0 < b$, additivity ([3, Order Bounds and Additivity over a Subdivision Point]) gives $\int_0^x h = \int_a^x h - \int_a^0 h$, and [3, The Fundamental Theorem of Calculus, First Form] differentiates $x \mapsto \int_a^x h$ on $[a, b]$. Moreover, if h has a continuous derivative h' , then $\int_0^x h' = h(x) - h(0)$ for every real x , by [3, The Fundamental Theorem of Calculus, Second Form] on the interval with endpoints 0 and x .

Step 1: a candidate. Define $v: \mathbb{C} \rightarrow \mathbb{R}$ by

$$v(x, y) = \int_0^y u_x(0, t) dt - G(x, y) \quad G(x, y) = \int_0^x u_y(s, y) ds$$

The first term does not depend on x , and $s \mapsto u_y(s, y)$ is continuous, so the first remark above gives $v_x(x, y) = -u_y(x, y)$. The same remark, applied to the continuous function $t \mapsto u_x(0, t)$, shows that the first term has y -derivative $u_x(0, y)$.

Step 2: differentiating G in y . Fix (x, y) and $\epsilon > 0$, and let I be the closed interval with endpoints 0 and x . The rectangle $I \times [y - 1, y + 1]$ is closed and bounded, hence compact ([8, Closed And Bounded In $\mathbb{R}^n$ Imply Compact]), so the continuous function u_{yy} is uniformly continuous on it ([8, Uniform Continuity Theorem]). Choose $\delta \in (0, 1]$ with $|u_{yy}(s, t) - u_{yy}(s, y)| < \epsilon$ whenever $s \in I$ and $|t - y| < \delta$, and let $0 < |h| < \delta$. For each $s \in I$, the mean value theorem ([3, The Mean Value Theorem]) applied to $t \mapsto u_y(s, t)$ between y and $y + h$ gives a point t_s with $|t_s - y| < |h|$ and

$$\frac{u_y(s, y + h) - u_y(s, y)}{h} = u_{yy}(s, t_s)$$

Hence the continuous function $s \mapsto (u_y(s, y + h) - u_y(s, y))/h - u_{yy}(s, y)$ takes values in $(-\epsilon, \epsilon)$ on I . By linearity of the integral and the order bounds of [3, Order Bounds and Additivity over a Subdivision Point],

$$\left| \frac{G(x, y + h) - G(x, y)}{h} - \int_0^x u_{yy}(s, y) ds \right| \leq \epsilon |x|$$

As ϵ was arbitrary, the partial derivative $G_y(x, y)$ exists and equals $\int_0^x u_{yy}(s, y) ds$.

Step 3: v is a harmonic conjugate. Since u is harmonic, $u_{yy} = -u_{xx}$. The function $s \mapsto u_x(s, y)$ is differentiable with continuous derivative $u_{xx}(s, y)$, so the second remark above gives

$$G_y(x, y) = - \int_0^x u_{xx}(s, y) ds = u_x(0, y) - u_x(x, y)$$

With Step 1, $v_y(x, y) = u_x(0, y) - G_y(x, y) = u_x(x, y)$. Thus $v_x = -u_y$ and $v_y = u_x$ on $\mathbb{C}$. These are continuous, so $u, v \in C^1(\mathbb{C})$ satisfy the Cauchy-Riemann equations and $u + iv$ is holomorphic on $\mathbb{C}$ by corollary 2.2.6. Moreover $v_x = -u_y$ and $v_y = u_x$ have continuous first partial derivatives, so v has continuous second partial derivatives and

$$\Delta v = \frac{\partial}{\partial x} (-u_y) + \frac{\partial}{\partial y} (u_x) = -u_{yx} + u_{xy} = 0$$

by the symmetry of the mixed second partial derivatives of u (lemma 2.2.13). So v is harmonic, and by definition 2.2.16 it is a harmonic conjugate of u .

Step 4: uniqueness. Let v, v' be harmonic conjugates of u and put $w = v' - v$. The Cauchy-Riemann equations for $u + iv$ and $u + iv'$ give $v_x = -u_y = v'_x$ and $v_y = u_x = v'_y$, so $w_x = w_y = 0$ on $\mathbb{C}$. For $(x, y) \in \mathbb{R}^2$, the mean value theorem ([3, The Mean Value Theorem]) applied to $s \mapsto w(s, 0)$ gives $w(x, 0) = w(0, 0)$, and applied to $t \mapsto w(x, t)$ it gives $w(x, y) = w(x, 0)$. Hence $v' - v = w$ is constant, completing the proof.

This proof generalizes for some other domains like rectangles, but is left for when we have built up the notion of contour integrals. In some cases (in particular, when U is not simply connected), harmonic functions will not have harmonic conjugates!

We finish off with the following side-effect that is more relevant for interpreting the growth of a function. The two equalities of the partial derivatives of u and v can be re-written in many forms:

$$|f'(z)|^2 = \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2 = \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} \quad (2.2)$$

which is the Jacobian. This should make sense, as f when interpreted as a real-differentiable function would require the jacobian to be nonzero in order for $f'(z) \neq 0$.

2.2.3 Properties of Holomorphic Functions

In this section, we upgrade some classical results of real-differentiability to complex differentiability taking advantage of the fact that f must satisfy the C.R. Equations. We then show a very important geometrical property of holomorphic functions that we mentioned in section 1.3.1, namely that they must be *conformal*.

Proposition 2.2.22: Derivative Zero Then Constant

Let f be holomorphic on a connected open set $U \subseteq \mathbb{C}$ and $f' = 0$. Then f is constant

Proof:

Since f is holomorphic and $f' = 0$:

$$\frac{\partial f}{\partial x} = -i \frac{\partial f}{\partial y} = 0$$

Thus, $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = 0$ on U . Let $B \subseteq U$ be an open disc with centre $c = c_1 + ic_2$, and let $z = x + iy \in B$. The horizontal segment from c to $x + ic_2$ and the vertical segment from $x + ic_2$ to z lie in B , as each of their points is at distance at most $|z - c|$ from c . Along the first, the real and imaginary parts of f are differentiable functions of the real part with derivative 0, so the mean value theorem ([3, The Mean Value Theorem]) gives $f(x + ic_2) = f(c)$. Along the second, the real and imaginary parts of f are differentiable functions of the imaginary part with derivative 0, and the mean value theorem gives $f(z) = f(x + ic_2)$. So f is constant on every open disc in U . Hence for each $w \in \mathbb{C}$ the set $f^{-1}(w)$ is open, and it is closed in U by continuity, so as U is connected one of these sets is all of U , that is, f is constant, as claimed.

Proposition 2.2.23: Equivalent to Derivative Zero

Let f be holomorphic on a connected open set $U \subseteq \mathbb{C}$. Then:

1. If $|f|$ is constant, then f is constant
2. If $\operatorname{Re} f$ is constant, then f is constant

Note how this contrasts to a real-differentiable function $f : U \subseteq \mathbb{R}^2 \rightarrow \mathbb{C}$, if $|f|$ is constant, this does not imply f is constant (think $U = \mathbb{R}^2 \setminus (0, 0)$ and $(x, y) \mapsto (x, y)/\sqrt{x^2 + y^2} = (x, y)/\|(x, y)\|$), and similarly for $\operatorname{Re} f$ (which we can think of as taking the function $\pi \circ f$ with any nonconstant function f)

Proof:

The Wirtinger derivative $\frac{\partial}{\partial z}$ of definition 2.2.11 is a complex linear combination of $\frac{\partial}{\partial x}$ and $\frac{\partial}{\partial y}$, so it is linear and satisfies the product rule, and the identities computed after that definition give $\frac{\partial f}{\partial z} = f'$ and $\frac{\partial \bar{f}}{\partial z} = 0$ for holomorphic f .

1. Suppose $|f| = c$ on U . If $c = 0$, then $f = 0$. Otherwise $\bar{f}$ has no zeros, and differentiating the constant $|f|^2 = f\bar{f}$ gives

$$0 = \frac{\partial(f\bar{f})}{\partial z} = \frac{\partial f}{\partial z} \bar{f} + f \frac{\partial \bar{f}}{\partial z} = f' \bar{f}$$

so $f' = 0$ on U and f is constant by proposition 2.2.22.

2. Suppose $\operatorname{Re} f$ is constant on U . Since $\operatorname{Re} f = \frac{1}{2}(f + \bar{f})$,

$$0 = \frac{\partial(\operatorname{Re} f)}{\partial z} = \frac{1}{2} \left(\frac{\partial f}{\partial z} + \frac{\partial \bar{f}}{\partial z} \right) = \frac{1}{2} f'$$

so $f' = 0$ on U and f is constant by proposition 2.2.22, completing the proof.

As an exercise show that if $\log |f|$ or $\arg(f)$ is constant, then f is constant.

One important result that is “invisible” in $\mathbb{R}$ but becomes apparent in $\mathbb{C}$ is that (real and complex) differentiable functions preserve angles! Being complex differentiable will also mean the function preserves orientation:

Definition 2.2.24: Conformal Maps

Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be a function. Then f is said to be *conformal* for every z_0 in the domain, f preserves the angles between two curves passing through z_0 and preserves orientation. If f does not preserve orientation, it is called *homothetic* or *angle-preserving*.

A conformal map may be interpreted as preserving the shape of any sufficiently small figure, while possibly rotating and scaling (but not reflecting) it. If f is holomorphic, we will say it's conformable if $Df(a)$ interpreted as a real 2 by 2 matrix is an angle-preserving map and orientation preserving. If it is simply angle-preserving, then *a priori* we do not know if f will flip-flop between orientations. If f is additionally at least real C^1 , then f has to strictly stay within one orientation:

Proposition 2.2.25: C^1 and Angle Preserving Functions

Let $\Omega \subseteq \mathbb{C}$ be a connected open set, $f : \Omega \rightarrow \mathbb{C}$ a real C^1 function with a nonzero Jacobian determinant at every point. Then if f preserves angles at every point of Ω (that is, if it is homothetic), then either:

$$\frac{\partial f}{\partial z} = 0 \quad \text{or} \quad \frac{\partial f}{\partial \bar{z}} = 0$$

where both cannot be zero (or else the jacobian's determinant is zero). In the second case f is holomorphic, in the first case f is anti-holomorphic (locally $\bar{\mathbb{C}}$ -linear)

Proof:

Since f is $\mathbb{R}$ -differentiable, Df exists. Since it is angle preserving, by lemma 1.3.1 Df is either of the form:

$$Df(x)(p) = ap \quad \text{or} \quad Df(x)(p) = a\bar{p}$$

that is, it is either a rotation, or a rotation and a reflection. Consider the sets:

$$\left\{ z \in \Omega \mid \frac{\partial f}{\partial z}(z) = 0 \right\} \quad \left\{ z \in \Omega \mid \frac{\partial f}{\partial \bar{z}}(z) = 0 \right\}$$

These two sets are disjoint, since if both $\frac{\partial f}{\partial z} = \frac{\partial f}{\partial \bar{z}} = 0$, then f a zero Jacobian, contradiction the assumption of the question. Since $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are continuous, the sets

$$\left\{ \frac{\partial f}{\partial z} = 0 \right\} \quad \left\{ \frac{\partial f}{\partial \bar{z}} = 0 \right\}$$

are closed sets. By what we have shown they are disjoint and their union is Ω . By connectivity of Ω , one of these sets must be empty.

This complete the proof. If the second one is empty, then Df is $\mathbb{C}$ -linear and hence f is holomorphic. If the first one is empty, then Df is anti $\mathbb{C}$ -linear (that is $Df_p(az) = \bar{a}Df_p(z)$) and f is called *anti-holomorphic*.

We can re-state the theorem a bit by saying that the Jacobian determinant is nonzero and f is a C^1 angle-preserving function if and only if f is either (strictly) holomorphic or anti-holomorphic. In the above theorem, we have assumed that being locally $\mathbb{C}$ -linear is sufficient to be conformal, and indeed it is. However, we there is a nice proof that goes about this more directly that I think is worth showing to show that it is in fact a very simple result:

Proposition 2.2.26: Holomorphic Then Conformal

Let f be a holomorphic function. Then f is a conformal map.

Proof:

let $\gamma : (-\epsilon, \epsilon) \rightarrow U \subseteq \mathbb{C}$ be some differentiable curve with $\gamma(0) = z_0$ and $f : U \rightarrow \mathbb{C}$ a holomorphic function. We can think of $\gamma'(0)$ as being the velocity vector of a particle passing through z_0 .

Take $f \circ \gamma : (-\epsilon, \epsilon) \rightarrow \mathbb{C}$. By the chain rule:

$$(f \circ \gamma)'(0) = f'(z_0)\gamma'(0)$$

Representing $f'(z_0)$ in polar coordinates we get $f'(z_0) = re^{i\theta}$. Thus, we get:

$$(f \circ \gamma)'(0) = re^{i\theta}\gamma'(0)$$

that is, f transforms the velocity vector by multiplying the speed by a factor of r and rotating it counterclockwise by a fixed angle θ . If we now consider two trajectories γ_1, γ_2 passing through z_0 at $t = 0$, then the map f will preserve the angle between the velocity vector $\gamma_1'(0)$ and $\gamma_2'(0)$ as well as their orientation, completing the proof.

This gives us another way of eliminating non-holomorphic functions. For example $f(x + iy) = x + i(x + y)$ preserves orientation, but not angle, while $f(z) = \bar{z}$ preserves angle, but not orientation. Note too how we in fact have the same angle-preserving phenomena happening for real-differentiable functions $f : \mathbb{R} \rightarrow \mathbb{R}$ for functions of the form $f : I \rightarrow \mathbb{R}$, but this is much less interesting since the only two possible angles between velocity vectors are 0 and π ; this shows the 2-dimensionality of the complex plane makes conformality a much more “rigid” property for complex differentiable functions.

The final result we shall state shall build upon our earlier observation’s that the derivative of f resemble the Jacobi identity, making it tempting to say that holomorphic implies continuously holomorphic. This is indeed the case, by corollary 3.2.7 in section 3.2, whose proof does not use the next theorem. The next theorem assumes that f' is continuous, so from that corollary it applies to every holomorphic f with $f'(z_0) \neq 0$. It is the complex version of the inverse function theorem. This theorem applies to holomorphic functions since a holomorphic function is already real-differentiable, but we would want the inverse to also be complex differentiable:

Theorem 2.2.27: Holomorphic Inverse Function Theorem

Let f be holomorphic in a neighbourhood of z_0 , with f' continuous there and $f'(z_0) \neq 0$, and put $w_0 = f(z_0)$. Then there are open sets $U \ni z_0$ and $V \ni w_0$ such that $f|_U$ is a homeomorphism of U onto V . Its inverse $g : V \rightarrow U$ is holomorphic, and

$$g'(w) = \frac{1}{f'(g(w))} \quad w \in V$$

Proof:

Identify $\mathbb{C}$ with $\mathbb{R}^2$. The partial derivatives of f are $\frac{\partial f}{\partial x} = f'$ and $\frac{\partial f}{\partial y} = if'$, by the Cauchy-Riemann conditions computed after definition 2.2.11, and they are continuous, so f is continuously differentiable as a map of an open subset of $\mathbb{R}^2$ to $\mathbb{R}^2$. Its real differential at z is multiplication by $f'(z) = a + ib$, whose matrix in the standard basis is

$$[f'(z)] = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

of determinant $a^2 + b^2 = |f'(z)|^2$, which is nonzero at z_0 . By [23, Inverse Function Theorem] there are open sets $U \ni z_0$ and $V \ni w_0$ such that $f : U \rightarrow V$ has a differentiable inverse $g : V \rightarrow U$, so

$f|_U$ is a homeomorphism onto V , and for $w = f(z) \in V$

$$Dg(w) = [f'(z)]^{-1} = \frac{1}{a^2 + b^2} \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$

This is the matrix of multiplication by $(a - ib)/(a^2 + b^2) = 1/f'(z)$, so the partial derivatives of g are $\frac{\partial g}{\partial x} = 1/f'(g(w))$ and $\frac{\partial g}{\partial y} = i/f'(g(w))$. They are continuous, since f' and g are, and they satisfy the Cauchy-Riemann equations, so g is holomorphic by corollary 2.2.6, with $g'(w) = \frac{\partial g}{\partial x}(w) = 1/f'(g(w))$, completing the proof.

For the Implicit function theorem, we have the following version, in which the zero set of f is locally the graph of a holomorphic function.

Theorem 2.2.28: Holomorphic Implicit Function Theorem

Let $n \geq 1$, let $W \subseteq \mathbb{C}^n \times \mathbb{C}$ be open, and let $f: W \rightarrow \mathbb{C}$ be continuously differentiable and holomorphic in each of its $n + 1$ variables separately. Write points of W as (x, y) with $x \in \mathbb{C}^n$ and $y \in \mathbb{C}$, and let $(x_0, y_0) \in W$ satisfy $f(x_0, y_0) = 0$ and $\frac{\partial f}{\partial y}(x_0, y_0) \neq 0$. Then there are an open polydisc $U \ni x_0$ in $\mathbb{C}^n$ and an open disc $V \ni y_0$ in $\mathbb{C}$ with $U \times V \subseteq W$, and a continuously differentiable function $g: U \rightarrow V$, holomorphic in each variable separately, with $g(x_0) = y_0$, such that for $(x, y) \in U \times V$, $f(x, y) = 0$ if and only if $y = g(x)$. For $n = 1$ the roles of the two variables may be exchanged: if $\frac{\partial f}{\partial x}(x_0, y_0) \neq 0$, the same conclusion holds with $x = h(y)$ for a holomorphic h .

Proof:

Identify $\mathbb{C}^n \times \mathbb{C}$ with $\mathbb{R}^{2n} \times \mathbb{R}^2$ and regard $f = (\text{Re}f, \text{Im}f)$ as a continuously differentiable map into $\mathbb{R}^2$. For fixed x , the function $y \mapsto f(x, y)$ is complex differentiable, so its real differential at y is $k \mapsto \frac{\partial f}{\partial y}(x, y)k$, multiplication by $\alpha + i\beta = \frac{\partial f}{\partial y}(x, y)$. In the coordinates $(\text{Re}k, \text{Im}k)$ this has matrix with rows $(\alpha, -\beta)$ and (β, α) , of determinant $\alpha^2 + \beta^2 = |\frac{\partial f}{\partial y}(x, y)|^2$, which is nonzero at (x_0, y_0) . By [23, Implicit Function Theorem] there are open sets $A \ni x_0$ in $\mathbb{C}^n$ and $B \ni y_0$ in $\mathbb{C}$ such that for each $x \in A$ there is a unique $g(x) \in B$ with $f(x, g(x)) = 0$, and g is differentiable. Uniqueness gives $g(x_0) = y_0$.

The product neighbourhood. As g is continuous and $\frac{\partial f}{\partial y}$ is continuous, being built from the continuous partial derivatives of f , we may choose an open disc $V \subseteq B$ around y_0 and an open polydisc $U \subseteq A$ around x_0 with $U \times V \subseteq W$, $g(U) \subseteq V$, and $\frac{\partial f}{\partial y} \neq 0$ on $U \times V$. Let $(x, y) \in U \times V$. If $y = g(x)$, then $f(x, y) = 0$. If $f(x, y) = 0$, then $y \in V \subseteq B$, and the uniqueness of $g(x)$ in B gives $y = g(x)$.

Holomorphy of g . Fix $k \in \{1, \dots, n\}$ and all coordinates of a point of U except the k -th, let $x(t)$ be the point of U with k -th coordinate t , and put $\varphi(t) = g(x(t))$, a differentiable function on a disc. Differentiating $f(x(t), \varphi(t)) = 0$ by the chain rule, using that the partial maps of f in its k -th and last variables are complex differentiable, gives

$$\frac{\partial f}{\partial x_k}(x(t), \varphi(t))s + \frac{\partial f}{\partial y}(x(t), \varphi(t))D\varphi(t)(s) = 0 \quad s \in \mathbb{C}$$

So $D\varphi(t)$ is multiplication by $c(t) = -\frac{\partial f}{\partial x_k}(x(t), \varphi(t)) / \frac{\partial f}{\partial y}(x(t), \varphi(t))$, a continuous function of t . Its matrix has rows $(\text{Rec}, -\text{Im}c)$ and $(\text{Im}c, \text{Rec})$, so φ has continuous partial derivatives satisfying the Cauchy-Riemann equations and is holomorphic by corollary 2.2.6. The same formula shows that the partial derivatives of g are continuous, so g is continuously differentiable. For $n = 1$ and $\frac{\partial f}{\partial x}(x_0, y_0) \neq 0$, apply what was proved to $(y, x) \mapsto f(x, y)$, completing the proof.

2.2.4 Polynomial and Rational Functions

As we saw in example 2.2.2, constant and linear functions are holomorphic. By proposition 2.2.3, the sum and product of holomorphic functions is holomorphic. Therefore, all polynomials are holomorphic

$$P(z) = a_0 + a_1z + \cdots + a_nz^n$$

with derivative

$$P'(z) = a_1 + 2a_2z + \cdots + na_nz^{n-1}$$

To not repeat the same restrictions in the following theorems, if we write out a polynomial as we have done for $P(z)$, we will assume $a_n \neq 0$, and 0 is not a polynomial ⁵.

Found this video (commented to compile) :

A defining feature of polynomials is that they are defined by their roots (up to a constant) along with one more point. Having theorems on how the roots behave under differentiation will give us some good insights on polynomial. The following two theorems give geometric insight on the roots:

Theorem 2.2.29: Roots Of Derivative Within Convex Hull (Gauss-Lucas Theorem)

Let P be a polynomial. Then the roots of P' are within the convex hull of the roots of P

Proof:

The proof is broken down into steps:

1. Suppose that $P(z)$ has degree n and zeros $b_1, \dots, b_n$ (each zero listed as many times as multiplicity). Show that

$$\frac{P'(z)}{P(z)} = \sum_{k=1}^n \frac{1}{z - b_k}$$

Proof. This is simply a matter of computation. First, by the fundamental theorem of algebra:

$$P(z) = c(z - b_1) \cdots (z - b_n)$$

then:

$$P'(z) = c \sum_{i=1}^n \prod_{k \neq i} (z - b_k)$$

Then dividing, we get:

$$\frac{P'(z)}{P(z)} = \sum_{k=1}^n \frac{1}{z - b_k}$$

⁵for formal reasons its degree is taken to be $-\infty$, so that $\deg(pq) = \deg p + \deg q$ holds for all polynomials p, q

as we sought to show □

2. Show that if $P'(z) = 0$, then

$$\left(\sum_{k=1}^n \frac{1}{|z - b_k|^2} \right) \bar{z} = \sum_{k=1}^n \frac{\bar{b}_k}{|z - b_k|^2}$$

Proof. If $P'(z) = 0$, then:

$$0 = \sum_{k=1}^n \frac{1}{z - b_k}$$

Now, for each term in the summand, multiply it by $\frac{\bar{z} - \bar{b}_k}{\bar{z} - \bar{b}_k}$ and the summand in two:

$$0 = \sum_{k=1}^n \frac{\bar{z}}{|z - b_k|^2} - \sum_{k=1}^n \frac{\bar{b}_k}{|z - b_k|^2}$$

Moving the values around, we get:

$$\left(\sum_{k=1}^n \frac{1}{|z - b_k|^2} \right) \bar{z} = \sum_{k=1}^n \frac{\bar{b}_k}{|z - b_k|^2}$$

as we sought to show □

3. Deduce that if $P'(z) = 0$, then z lies within the convex hull of the points b_k .

Proof. Let $P'(z_0) = 0$. Then by part (b):

$$\left(\sum_{k=1}^n \frac{1}{|z_0 - b_k|^2} \right) \bar{z}_0 = \sum_{k=1}^n \frac{\bar{b}_k}{|z_0 - b_k|^2}$$

Conjugating both sides and isolating z_0 , we get:

$$z_0 = \frac{\sum_{k=1}^n \frac{b_k}{|z_0 - b_k|^2}}{\left(\sum_{k=1}^n \frac{1}{|z_0 - b_k|^2} \right)} = \sum_{k=1}^n \frac{\frac{b_k}{|z_0 - b_k|^2}}{\left(\sum_{k=1}^n \frac{1}{|z_0 - b_k|^2} \right)} = \left(\sum_{k=1}^n \frac{\frac{1}{|z_0 - b_k|^2}}{\left(\sum_{j=1}^n \frac{1}{|z_0 - b_j|^2} \right)} b_k \right)$$

where the value in the summand is non-negative. Notice here that the coefficients of the b_k 's add up to 1. Since the b_k 's form a convex hull, this is a convex combination, and so z_0 must be in the convex hull, as we sought to show. □

A rational function of particular interest is the Möbius transformation

$$f(z) = \frac{az + b}{cz + d} \quad ad - bc \neq 0$$

Note that since

$$f^{-1}(w) = \frac{dw - b}{-cw + a}$$

then f is in fact *biholomorphic*, meaning it is in fact a symmetry of $\hat{\mathbb{C}}$. In fact, $\text{Aut}(\hat{\mathbb{C}})$ is the collection of all Möbius transformation. If you did not take differential geometry, then the following discussion may be skipped and the statement may be taken for granted for now. Recall that $S^2 \cong \mathbb{CP}^1$ as smooth manifolds. Thinking of S^2 as the Riemann sphere $\hat{\mathbb{C}}$, we see that finding the symmetries of S^2 is finding the symmetries of $\mathbb{CP}^1$, i.e. the automorphism group of complex diffeomorphisms. Recall that $\text{GL}(\mathbb{C}) = \text{Aut}(\mathbb{C})$ and let $\text{GL}(\mathbb{C}) \curvearrowright \mathbb{C}$ in a natural way. We may descend this action to $\text{PGL}(\mathbb{C}) \curvearrowright \mathbb{CP}^1$ by taking $\text{PGL}_2(\mathbb{C}) = \text{GL}_2(\mathbb{C})/Z(\mathbb{C})$ where $Z(\mathbb{C})$ is the center of $\text{GL}_2(\mathbb{C})$, i.e. the collection of all diagonal matrices. To see what this descension looks like, take

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix} \in \text{GL}(\mathbb{C})$$

Then:

$$[z, 1] \begin{pmatrix} a & c \\ b & d \end{pmatrix} = [az + b, cz + d] = \left[\frac{az + b}{cz + d}, 1 \right]$$

which is exactly a Möbius transformation! Thus, we see that the collection of Möbius transformations, which we may label as $\text{PGL}(\mathbb{C})$, are all the automorphisms (i.e. the biholomorphic functions) of $\hat{\mathbb{C}}$.

2.3 Power Series

Since polynomials are holomorphic functions, it is natural to ask whether the completion of polynomials, power series⁶, are also holomorphic functions. The answer was revealed earlier to be yes when we said that another common name for holomorphic functions are analytic functions (since an analytic function is locally given by a convergent power-series).

Definition 2.3.1: Formal Power Series

Let z_0 be any complex number. Then a *formal power series* with complex coefficients around the point z_0 is a formal series of the form

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n$$

for some complex numbers $a_0, a_1, \dots$, with z an indeterminate

These are called “formal” since we require to establish an actual domain and codomain in order for them to be functions, at which point we usually call them power series. Another way of thinking about power series is that they are the completion of the ring of polynomials $\mathbb{C}[z - z_0]$, and hence they are the “formal” result of this completion. We can naturally try and define an evaluation function by replacing z with some value in $\mathbb{C}$ and see if it converges. The following theorem gives us a systemic way of finding which values can be plugged in:

⁶in particular $\varprojlim (k[x]/(x)^n) = k[[x]]$

Theorem 2.3.2: Abel's Theorem

Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be a power series. Then there exists a number R , $0 \leq R \leq \infty$, called the *radius of convergence*, with the following properties:

1. The series converges absolutely for every z with $|z| < R$. If $0 \leq \rho < R$, the convergence is uniform for $|z| \leq \rho$.
2. If $|z| > R$, the terms of the series are unbounded, and the series thus diverges
3. In $|z| < R$, the sum of the series is a holomorphic function. The derivative is obtained by term-wise differentiation, and the derived series has the same radius of convergence.

Note that when $|z| = R$, the result depends on the power-series⁷. This circle is called the *circle of convergence*.

Proof:

Without loss of generality, let $z_0 = 0$ (this is simply shifting the power series over to 0, and does not effect convergence). Recall the important formula by *Hadamards* that relates R to given the coefficients of a power-series:

$$1/R = \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}$$

If $|z| < R$, we can find a ρ such that $|z| < \rho < R$. Then $1/\rho > 1/R$, and by the definition of the limit superior, there exists an n_0 such that $|a_n|^{1/n} < 1/\rho$, meaning

$$|a_n| < \frac{1}{\rho^n} \quad \forall n \geq n_0$$

Thus:

$$|a_n z^n| < \left(\frac{|z|}{\rho}\right)^n \quad \forall n \geq n_0$$

and for any fixed z , the right hand side is a geometric series which for $\rho > 1$ converges by Hadamards formula. To show uniform convergence for $|z| \leq \rho < R$, we take advantage of Weiestrass M -test. Pick a ρ' with $\rho < \rho' < R$. Then we can get for large enough n_0 :

$$|a_n z^n| \leq \left(\frac{\rho}{\rho'}\right)^n \quad n \geq n_0$$

Since the majorant (the power series given by the right hand side) is convergent and has constant terms, by the Weiestrass's M test the power series is uniformly convergent. If $|z| > R$, then we can do the same manipulation we have done before but instead choose $R < \rho < |z|$ to get

$$|a_n z^n| > \left(\frac{|z|}{\rho}\right)^n \quad \forall n \geq n_0$$

Thus, fixing an n_0 , we get that

$$|a_{n_0} z^{n_0}| + |a_{n_0} z^{n_0}| < |a_{n_0} z^{n_0}| + |a_{n_0+1} z^{n_0+1}| + \dots$$

showing the right hand side is unbounded, and all $|z| > R$ diverge.

⁷This problem is related to the same tricky problem of asking when a Fourier series converges

Next, we will show that the derived series $\sum_{n=0}^{\infty} na_n z^{n-1}$ has the same radius of convergence. First, we will show that $\sqrt[n]{n} \rightarrow 1$

Proof. Set $\sqrt[n]{n} = 1 + \delta_n$. Then $\delta_n > 0$, and by the binomial theorem

$$n = (1 + \delta_n)^n > 1 + \frac{1}{2}n(n-1)\delta_n^2$$

Manipulating, we get $\delta_n^2 < 2/n$, and so $\delta_n \rightarrow 0$. □

We now continue the proof. For $|z| < R$, decompose the power-series two:

$$f(z) = \sum_{k=0}^{\infty} a_k z^k = s_n(z) + R_n(z)$$

where

$$s_n(z) = a_1 + a_1 z + \cdots + a_{n-1} z^{n-1}$$

$$R_n(z) = \sum_{k=n}^{\infty} a_k z^k$$

write the “desired” derivative of f as

$$f_1(z) = \sum_{k=1}^{\infty} k a_k z^{k-1} = \lim_{n \rightarrow \infty} s'_n(z)$$

we want to show that $f'(z) = f_1(z)$. To that end, write:

$$= \frac{f(z) - f(z_0)}{z - z_0} - f_1(z_0) \tag{2.3}$$

$$= \left(\frac{s_n(z) - s_n(z_0)}{z - z_0} - s'_n(z_0) \right) + (s'_n(z_0) - f_1(z_0)) + \left(\frac{R_n(z) - R_n(z_0)}{z - z_0} \right) \tag{2.4}$$

where we naturally assume that $z \neq z_0$ and both $|z|, |z_0|$ are $< \rho < R$. Recalling that $a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \cdots + b^{n-1})$, the last term in the above can be rewritten as

$$\sum_{k=n}^{\infty} a_k (z^{k-1} + z^{k-2}z_0 + \cdots + z z_0^{k-1} + z_0^{k-1})$$

since $|z|, |z_0| < \rho$ we have:

$$\left| \frac{R_n(z) - R_n(z_0)}{z - z_0} \right| \leq \sum_{k=n}^{\infty} k |a_k| \rho^{k-1}$$

the right hand side of the above expression is the remainder term of a convergent series, and so we can find a large enough n_0 such that

$$\left| \frac{R_n(z) - R_n(z_0)}{z - z_0} \right| < \frac{\epsilon}{3}$$

The same reasoning can be applied For the second term in equation (2.4) for some $n \geq n_1$. For the 1st term, choose some $n \geq n_0, n_1$. Then by definition of the derivative there exists a $\delta > 0$ such that for $0 < |z - z_0| < \delta$ implies

$$\left| \frac{s_n(z) - s_n(z_0)}{z - z_0} - s'_n(z_0) \right| < \frac{\epsilon}{3}$$

Combining all these equations, we get that:

$$\left| \frac{f(z) - f(z_0)}{z - z_0} - f'_1(z_0) \right| < \epsilon$$

when $0 < |z - z_0| < \delta$. But then that is exactly the condition we need to show that $f'(z_0)$ exists and is equal to $f'_1(z_0)$, completing the proof.

The last reasoning can be repeated indefinitely, meaning a power series with positive radius of convergence has derivatives of all order with the same radius of convergence. Now that we have a radius of convergence, we can define the *power series* of a formal power series to be the function $F : B_R(z_0) \rightarrow \mathbb{C}$ such that

$$F(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

We will most the time not make a distinction between formal power series and power series and call them both power series unless the situation calls for nuance. At this point, it is a good idea to recall some tests to check whether a power series converges:

1. **root test**: if $\limsup_{n \rightarrow \infty} (a_n)^{1/n} < 1$, then the series converges, if > 1 it diverges, and if $= 1$ it is inconclusive.
2. **Ratio test** : if $\lim_{N \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$ then the series converges. if > 1 it diverges, and $= 1$ it is inconclusive.
3. **Comparison test**: If $\sum_{n=0}^{\infty} a_n z^n$ converges and $b_n < a_n$, then $\sum_{n=0}^{\infty} b_n z^n$ converges
4. **Integral test**: If $f[1, \infty) \rightarrow \mathbb{R}_+$ is a non-negative monotonically decreasing function such that $f(n) = a_n$, then if

$$\int_1^{\infty} f(x) dx < \infty$$

then the series also converges, and if it diverges so does the series

5. If $\sum_i^{\infty} |a_n| |z^n|$ converges for some z , so does the power series. If $a_n \not\rightarrow 0$, then the $\sum_n^{\infty} a_n z^n$ for any z does not converge.

We mentioned that what happens at the border is not determined. We in fact get some subtle behavior that is worth going into. Take the power sries $\sum_{n=0}^{\infty} z^n$. Then it has radius of convergence

of 1. if $z \in B_1(0)$, then since the series converges uniformly we have:

$$\begin{aligned} z \sum_{n=0}^{\infty} z^n &= \sum_{n=0}^{\infty} z^{n+1} \\ &= \sum_{n=1}^{\infty} z^n \\ &= \sum_{n=0}^{\infty} z^n - 1 \end{aligned}$$

Thus, with some algebraic manipulation we get the closed form of:

$$\sum_{n=0}^{\infty} z^n = \frac{1}{1-z}$$

It is evident to see that as long as $z \in B_1(0)$ the function $z \mapsto \frac{1}{1-z}$ does not extend continuously to the boundary point $z = 1$ of the disk. However, it *does* extend continuously, even smoothly, to the rest of the boundary. In fact, even more remarkably (though perhaps not so remarkable if we take a moment to think about it), it can be *holomorphically extended* to all of $\mathbb{C} \setminus \{1\}$. However, $\sum_{n=0}^{\infty} z^n$ diverges at every point in the boundary (when $|z| = 1$, the coefficients of z^n of the series do not converge to zero), and evidently it diverges outside the unit ball. Thus, we see that the function that matches the power series can be well-defined much beyond the radius of convergence of the power series. We will prove in the next proposition (proposition 2.3.3) that the power series matching $1/(1-z)$ is unique and hence we cannot find a power series that somehow fixes this.

We can play around with the fact that $\sum_{n=0}^{\infty} z^n$ and $\frac{1}{1-z}$ must be uniquely identified by *formally* identifying the two on all of $\mathbb{C} \setminus \{1\}$. For example, if we plug in $z = 2$, then this identification would lead us to say that:

$$1 + 2 + 2^2 + 2^3 + \dots = -1$$

which under our current understanding of convergence would certainly be absurd. However, there is a way of interpreting this beyond the classical notion of convergence, leading to concepts like the generalised summation methods like the zeta function regularisation. These are discussed further in [2].

(see also Tao's blog [here](#) and [here](#) (example 11))

Proposition 2.3.3: Taylor Expansion Is Unique

Let F, G be two power series centered at z_0 that agree on some neighbourhood U of z_0 . Then the coefficients of each term are equal. In particular, the Taylor series expansion is unique.

Note that if the point around which the power series is centered is different, we can no longer compare coefficients easily. For example, we can see that both $\sum_{n=0}^{\infty} z^n$ and $\sum_{n=0}^{\infty} \frac{1}{2^{n+1}}(z+1)^n$ both converge to $\frac{1}{1-z}$ on $B_1(0)$ but have different coefficients.

Proof:

Write $F(z) = \sum_{n=0}^{\infty} a_n(z-z_0)^n$ and $G(z) = \sum_{n=0}^{\infty} b_n(z-z_0)^n$, and pick $r > 0$ with $B_r(z_0) \subseteq U$. Since F and G agree on U as functions, both series converge at every point of $B_r(z_0)$, so by theorem 2.3.2 both radii of convergence are at least r . By theorem 2.3.2, the sum of a power

series is differentiable on its disk of convergence and its derivative is obtained by term-wise differentiation, the derived series being again a power series centered at z_0 with the same radius of convergence. Applying this k times gives

$$F^{(k)}(z) = \sum_{n=k}^{\infty} n(n-1)\cdots(n-k+1)a_n(z-z_0)^{n-k}$$

on $B_r(z_0)$, and the same formula for $G^{(k)}$ with b_n in place of a_n . At $z = z_0$ every term with $n > k$ vanishes, leaving

$$F^{(k)}(z_0) = k! a_k \quad G^{(k)}(z_0) = k! b_k$$

Since F and G agree on the neighbourhood U of z_0 , and the k -th derivative at z_0 is computed from the values of the function on U alone, their derivatives of every order agree at z_0 . Therefore

$$a_k = \frac{F^{(k)}(z_0)}{k!} = \frac{G^{(k)}(z_0)}{k!} = b_k$$

for every $k \geq 0$. In particular, any power series centered at z_0 that represents a function f on a neighbourhood of z_0 has $f^{(k)}(z_0)/k!$ as its k -th coefficient, so the Taylor expansion of f around z_0 is unique, completing the proof.

(in [this](#) article, exercise 18 gives a way to go back and forth between two power series centered around different points)

As we have seen, the power series (as a function) can be well behaved as one approaches the boundary of the disk of convergence, while being divergent at the boundary. However the converse of this, where the power series converges at the boundary but does not behave well (as a function) as one approaches the boundary, does not occur:

Theorem 2.3.4: Abel's Limit Theorem

If $\sum_0^\infty a_n$ converges, then $f(z) = \sum_0^\infty a_n z^n \rightarrow f(1)$ as z approaches 1 in such a way that

$$\frac{|1-z|}{1-|z|}$$

remains bounded (sometimes known as the *Stolz angle*) or *non-tangent angle approach*.

Geometrically, this condition can be interpreted to mean that as we approach 1, we do so in such a way that the angle our approaching curve is taking is not tangent to the point 1, see the following link [here](#).

Proof:

We may assume $\sum_0^\infty a_n = 0$ by shifting over the sequence and have $\sum_0^\infty a_n$ be a_0 . Writing the partial sum $s_n = a_0 + a_1 + \dots + a_n$, we get

$$\begin{aligned} s_n(z) &= a_0 + a_1 z + \dots + a_n z^n \\ &= s_0 + (s_1 - s_0)z + \dots + (s_n - s_{n-1})z^n \\ &= s_0(1-z) + s_1(z-z^2) + \dots + s_{n-1}(z^{n-1} - z^n) + s_n z^n \\ &= (1-z)(s_0 + s_1 z + \dots + s_{n-1} z^{n-1}) + s_n z^n \end{aligned}$$

Since $s_n z^n \rightarrow 0$, we get the representation

$$f(z) = (1 - z) \sum_0^{\infty} s_n z^n$$

Next we are assuming that $|1 - z| \leq K(1 - |z|)$ for some choice $K \in \mathbb{R}$. Since $s_n \rightarrow 0$, choose m large enough so that $|s_n| < \epsilon$ for $n \geq m$. The remainder of the series $\sum s_n z^n$ from $n = m$ onwards, is dominated by the geometric series

$$\epsilon \sum_m^{\infty} |z|^n = \epsilon \frac{|z|^m}{1 - |z|} < \frac{\epsilon}{1 - |z|}$$

It follows that

$$|f(z)| \leq |1 - z| \left| \sum_0^{m-1} s_k z^k \right| + K\epsilon$$

The term on the right hand side can be made arbitrarily small by choosing z sufficiently close to 1, and so we can conclude that $f(z) \rightarrow 0$ as $z \rightarrow 1$ subject to our given Stolz angle restriction.

We continue now to explore the properties of power series. Important operations we are used to on our functions is adding, scaling, multiplying, inverting, or composing them. We shall see how these operations interact with power series. One important comment must be made: the problem of convergence of the composition of power series is one that is still being investigated. In particular, sufficient conditions for convergence is an area of research, I found this paper ([here](#)). Hence, to avoid such problems when they arise, we may treat our power series as formal power series, the derivative as a formal operation, and define $f(0) := a_0$ (since we are not treating f as a function).

Proposition 2.3.5: Composing Power Series

Let f, g be formal power series. Then $g(f(z))$ is well-defined if $a_0 = 0$.

Proof:

When composing power series, we get:

$$a_1(b_0 + b_1 z + b_2 z^2 + \dots) + a_2(b_0 + b_1 z + b_2 z^2 + \dots)^2 + \dots$$

where each coefficient is an infinite sum, and so we must ask for their convergence conditions. One thing we can do to guarantee this is by setting $a_0 = 0$, which will guarantee that all the coefficients are finite sums and hence well-defined. Note that as we are working with formal power series, there is no issue in re-arranging terms.

Theorem 2.3.6: Inverse Function Theorem On Formal Power Series

Let $f(z)$ be a formal power series. Then there is a formal power series $g(z)$ such that $b_0 = 0$ and $f \circ g = \text{id}$ (where id is the power series $\text{id}(z) = z$) if and only if $a_0 = 0$ and $f'(0) \neq 0$. In this case, g is unique and $g \circ f$ is also the identity. If f has positive radius of convergence, so does g .

Proof:

We want a g such that $f(g(z)) = z$. We want:

$$a_0 + a_1(b_1z + b_2z^2 + \cdots) + a_2(b_1z + b_2z^2 + \cdots)^2 = z$$

expanding and using the method of undetermined coefficients expanding the first two terms we get:

$$a_0 = 0 \quad a_1b_1 = 1$$

This shows us that $a_0 = 0$ and $a_1 = f'(0) \neq 0$ are necessary conditions. To show they are sufficient conditions, we can see that we may deduce all other coefficients given these conditions. For example, we may deduce the coefficient of z^n as $a_0 + a_1g(z) + \cdots + a_ng(z)^n$, so

$$a_1b_n = P_n(a_2, \dots, a_n, b_1, \dots, b_{n-1})$$

Thus, we would start by doing $b_1 = 1/a_1$, and $b_2, b_3, \dots$ are defined recursively (this resulting polynomials is called *Bells polynomial*). With our construction, we have that g satisfies $b_0 = 0$ and $b_1 \neq 0$. We can do the same thing to find the other-side inverse, that is $g(f_1(w)) = w$. These inverses are equal:

$$f_1 = \text{id} \circ f_1 = (f \circ g) \circ f_1 = f \circ (g \circ f_1) = f \circ \text{id} = f$$

where we assumed the associativity of composition, something that can be checked. Finally, for the radius of convergence, we shall delay such considerations for now, noting that when we show that holomorphic functions are analytic we can use the inverse function theorem to estimate the values.

Proposition 2.3.7: Convergence On Operations Of Powerseries

Let f, g be convergent power series with radius of convergence $R(f)$ and $R(g)$. Then $g \circ f$ converges too. In particular, if we take some $r > 0$ so that $\sum_{n=0}^{\infty} |a_n|r^n < R(g)$ (i.e. is less than the absolute power series), then

1. $R(g \circ f) \geq r$
2. $|f(z)| < R(f)$ if $|z| < r$
3. $(g \circ f)(z) = g(f(z))$

Proof:

This is simply finding good bounds using the real power series. If we have

$$(g \circ f)(z) = \sum_p b_p \left(\sum_n a_n z^n \right)^p = \sum_k c_k z^k$$

then taking the absolute value and using the natural generalization of the triangle inequality:

$$\left| \sum_p b_p \left(\sum_n a_n z^n \right)^p \right| \leq \sum_p |b_p| \left(\sum_n |a_n| |z|^n \right)^p = \sum_k \gamma_k |z|^k$$

where we see by the use of the triangle inequality that $|c_k| \leq \gamma_k$. Thus, using the limit comparison test, we get that $(g \circ f)$ converges absolutely if $|z| < r$, telling us that the radius of convergence of $(g \circ f)$ is at least r and that $|f(z)| < R(g)$. The final fact now becomes the fact that we may re-arrange the terms of an absolutely convergent series.

Proposition 2.3.8: Operations On Power Series

Let f, g be power series with radius of convergence R_1 and R_2 :

$$\sum_{n=1}^{\infty} a_n z^n \quad \sum_{n=1}^{\infty} b_n z^n$$

Then:

1. $f + g$ is a power series given by term-wise addition with radius of convergence of at least $\min(R_1, R_2)$ (though it may be much larger)
2. fg is a power series with

$$\left(\sum_k^{\infty} a_n (z - c)^k \right) \left(\sum_k^{\infty} b_k (z - c)^k \right) = \sum_k^{\infty} \left(\sum_i^{\infty} a_i b_{k-i} \right) (z - c)^k$$

3. If $a_0 \neq 0$, then there is a unique power series such that $f(z)g(z) = 1$. If f has a positive radius of convergence, so does g

Proof:

1. here
2. here
3. Without loss of generality we may assume $a_0 = 1$ by dividing by a_0 . Then we may write $f(z) = 1 - h(z)$ where $h(z)$ is now a power series satisfying $h(0) = 0$. Then it is a classical result that

$$\frac{1}{1 - w} = 1 + \sum_{n=1}^{\infty} w^n$$

substituting $w = h(z)$, we get $g(z) = (1 - h(z))^{-1}$. For the radius of convergence, we may use the same trick as used in the inverse function theorem for power series.

2.4 Exponential, Trigonometric, and Logarithmic Function

We now explore 3 important examples of power series which will come up again and again. They will all be based on the single power series which will be known as the *exponential function*.

There are many reasons on why the exponential function is a very nice function, some of them being:

1. It is smooth and defined on all of $\mathbb{C}$
2. It's derivative is itself (even when defined over $\mathbb{C}$)
3. it has no roots; though it's an “infinite polynomial”, there is no value of $z \in \mathbb{C}$ such that $e^z = 0$.
4. The real part of e^z is injective, while e^z shall surject onto $\mathbb{C}^\times$ (as we shall soon show)
5. The exponential shall encapsulate the “rotational” behavior of complex numbers, and shall give us new coordinates in which we may work in (disregarding 0)

While all these facts may already be familiar, I want to draw attention to the 3rd fact, for though it may seem innocuous, it will be the first “infinite polynomial” with no root, a very non-polynomial behavior as polynomials (over $\mathbb{C}$) can be defined by their roots. We shall also see much later that there will be “infinite rational polynomials” polynomials with *transcendental* roots (roots of elements that are not algebraic over a field), the reader already knows these as sin and cos.

For reader that shall be interested in modular forms much down the line, sin and cos are interesting in their own right as being the simplest example of functions that are invariant under a group operation, in this case $\mathbb{Z}$. Thinking of $\mathbb{Z}$ as $\text{GL}_1(\mathbb{Z})$, we will later see that modular forms are invariant under $\text{SL}_2(\mathbb{Z})$ (not $\text{GL}_2(\mathbb{Z})$ for reasons that would be clear once the modular theory is studied) along with some extra natural conditions.

Exponential Function

The exponential function is the function that satisfies the solution to the ODE

$$f'(z) = f(z) \quad f(0) = 1$$

We can solve this by setting

$$\begin{aligned} f(z) &= a_0 + a_1 z + \cdots + a_n z^n + \cdots \\ f'(z) &= a_1 + 2a_2 z + \cdots + na_n z^{n-1} + \cdots \end{aligned}$$

which requires that $a_{n-1} = na_n$ with $a_0 = 1$. Using induction, we get that

$$a_n = \frac{1}{n!}$$

This solution is usually denoted as either e^z or $\exp(z)$

Definition 2.4.1: Exponential Function

Let f be the function such that $f = f'$. Then the solution to this ODE is called the *exponential function*:

$$e^z = 1 + z + \frac{z^2}{2!} + \cdots + \frac{z^n}{n!}$$

it is usually denoted as either e^z or $\exp(z)$.

Notice that $\sqrt[n]{n!} \rightarrow \infty$, and hence e^z converges on the entire complex plane. the exponential function satisfies some nice properties. On immediate inspection, the exponential is its own derivative. We also get a homomorphism $\exp : \mathbb{C} \rightarrow \mathbb{C}^\times$ via

$$e^{a+b} = e^a \cdot e^b$$

To see this, notice that

$$D(e^z e^{c-z}) = e^z \cdot e^{c-z} + e^z \cdot (-e^{c-z}) = 0$$

and so $e^z \cdot e^{c-z}$ is a constant. Note that we have not yet proved the exponential law for complex functions, so we could not just take $e^z e^{c-z} = e^c$. In fact, from this we *can* conclude that $e^z \cdot e^{c-z} = e^c$. Letting $z = a$ and $c = a + b$ gives us our desired result. Another way of doing this is by taking the partial sums of e^{a+b} and doing some algebraic manipulation to get the result, and concluding it works as $n \rightarrow \infty$. We can thus see that we have a homomorphism $r \mapsto e^{ir}$ mapping $\mathbb{R} \rightarrow S^1$. This homomorphism, when restricted to $\mathbb{R}$, has kernel $2\pi\mathbb{Z}$ so $\mathbb{R}/2\pi\mathbb{Z} \cong S^1$. This is in fact a topological homomorphism, and we can consider S^1 having the quotient topology. As an exercise, this topology can be verified to be the same as the subspace topology of $S^1 \subseteq \mathbb{C}$. By our observation that $\mathbb{C}^\times \cong S^1 \times (0, \infty)$, and that the $e^z = e^{x+iy} = e^x e^{iy}$, we in fact get that

$$\exp : \mathbb{C} \rightarrow \mathbb{C}^\times$$

is a surjective homomorphism from $\mathbb{C}$ with addition to $\mathbb{C}^\times$ with multiplication which has the same kernel, and hence the exponential map links addition to multiplication homomorphically! As an aside, This is in fact another way of characterizing the exponential function. As the kernel is $2\pi\mathbb{Z}$, we have that:

$$e^z = e^{2\pi ik} \quad k \in \mathbb{Z}$$

Since $e^z \cdot e^{c-z} = e^c$, $e^z \cdot e^{-z} = e^{-z} \cdot e^z = 1$, implying e^z must have *no roots*, even the n th partial summand always has n roots (hence, an infinite polynomial need not have roots⁸). Another way of seeing this is that the real component of e^z can never be zero, e^z is *never zero*. Choosing $z = x \in \mathbb{R}$ to be real, notice that $e^x > 0$ for $x > 0$. Since e^x and e^{-x} are reciprals, this means that $0 < e^x < 1$ for $x < 0$. Since the series has real coefficients, $e^{\bar{z}}$ is the complex conjugate of e^z . Thus,

$$|e^{iy}|^2 = |e^{iy} \cdot e^{-iy}| \quad |e^{x+iy}| = e^x$$

2.4.1 Trigonometric Functions

Since the addition, composition, multiplication, and divisions of holomorphic functions is holomorphic, we have the following two holomorphic functions:

Definition 2.4.2: Trigonometric Functions

$$\cos(z) = \frac{e^{iz} + e^{-iz}}{2} \quad \sin(z) = \frac{e^{iz} - e^{-iz}}{2i} \quad (2.5)$$

By doing some substitution and computation, we get:

$$\cos(z) = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \dots$$

⁸and it also may have infinitely many as we shall see when defining $\sin(z)$ and $\cos(z)$

$$\sin(z) = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots$$

Using equation (2.5), There is a natural way of linking these formulas to e^z . First, since $e^{x+iy} = e^x e^{iy}$, we only need to expand for e^{iy} :

$$e^{iy} = 1 + iy - i^2 \frac{y^2}{2!} + i^3 \frac{y^3}{3!} - \dots$$

Since e^{iy} is absolutely convergent, we can re-arrange the terms, getting us:

$$e^{iy} = (1 - \frac{y^2}{2!} + \dots) + i(y - \frac{y^3}{3!} + \dots)$$

where the right term on the right hand uses the fact that we can factor constants from converging series. We thus get:

$$e^{iz} = \cos(z) + i \sin(z)$$

Be mindful that for a complex number z , we usually get

$$z = |z|(\cos(\arg(z)) + i \sin(\arg(z)))$$

while for the expression e^{iz} we put z directly into $\cos$ and $\sin$. Using this formula, there is an easy way of remembering the definitions of $\sin$ and $\cos$: take $e^{iz} = \cos(z) + i \sin(z)$, and take $e^{-iz} = \cos(-z) + i \sin(-z) = \cos(z) - i \sin(z)$. Add these two together and divide by 2 to get the definition of $\cos$. Similarly for $\sin$. By doing some algebraic manipulations, we can get:

$$\cos^2(z) + \sin^2(z) = 1$$

Similarly, we get that

$$D(\cos(z)) = -\sin(z) \quad D(\sin(z)) = \cos(z)$$

and we also get the identities

$$\begin{aligned} \cos(a+b) &= \cos(a)\cos(b) - \sin(a)\sin(b) \\ \sin(a+b) &= \cos(a)\sin(b) + \sin(a)\cos(b) \end{aligned}$$

Any other trigonometric function and their properties can now be quickly derived using these (for example, $\tan$, $\cot$, and so forth). An important and counter-intuitive fact to point out is that $\cos$ and $\sin$ are defined on all of $\mathbb{C}$ (since the radius of convergence is ∞) and that the solution to the equation $\cos(z) = w$ has a solution for any $w \in \mathbb{C}$, not just $w \in [-1, 1]$!

Besides these two functions standing out for their importance in trigonometry, they also stand out as being the simplest “infinite polynomials” with an infinite number of roots. Focusing on $\sin(z)$ (since $\cos(z)$ is a translation), we shall show in section 4.2.3 that :

$$\sin(\pi z) = \pi z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)$$

where the 1 in the summation is to insure the elements in the product converges to 1. The reader may say that this is not the simplest possible infinite polynomial (with roots at the integers), would not it be z/n ? The fact that we are considering z^2/n^2 to be the term is because:

$$\prod_{n=1}^{\infty} \left(1 \pm \frac{z}{n}\right)$$

does not converges (as we shall explore later), and requires some correcting terms⁹. In fact, we may consider the πz in multiplying the right hand side and the π term in the sin to be correction terms to insure we may decompose sin into an infinite product.

2.4.2 Periodicity

A function is periodic if there exists a (nonzero) c such that $f(z + c) = f(z)$ for all $z \in \mathbb{C}$. We shall show that e^z has a period, that in particular there exists a nonzero c such that $e^{z+c} = e^z e^c = e^z$. From the properties of e^z , we immediately get if the period exists then $c = iw$ for some real number w . Since

$$D \sin y = \cos(y) \leq y \quad \sin(0) = 0$$

we get by the mean value theorem that $\sin(y) < y$ for $y > 0$. Similarly

$$D \cos(y) = -\sin(y) > -y \quad \cos(0) = 1$$

which gives

$$y > 1 - \frac{y^2}{2}$$

which in turn gives

$$\sin(y) > y - y^3/6$$

and so

$$\cos(y) < 1 - y^2/2 + y^4/24$$

Using this, we get that

$$\cos(\sqrt{3}) < 0$$

Thus, by the Mean value theorem there exists a y_0 between 0 and $\sqrt{3}$ such that $\cos(y_0) = 0$. Since

$$\cos^2(y_0) + \sin^2(y_0) = 1$$

we have that $\sin y_0 = \pm 1$ and that $e^{iy_0} = \pm i$. Hence $e^{4iy_0} = 1$, giving us a period of $4y_0$.

This period is in fact the smallest possible period. Take $0 < y < y_0$. Since

$$y > y(1 - y^2/6) > y/2 > 0$$

we have that $\cos(y)$ is strictly decreasing. Since $\sin(y)$ is positive and $\cos^2(y) + \sin^2(y) = 1$, it follow that $\sin(y)$ is strictly increasing, and hence $\sin(y) < \sin(y_0) = 1$. The double inequality $0 < \sin(y) < 1$ guarantees that e^{iy} is neither ± 1 nor $\pm i$. Thus, $e^{4iy} \neq 1$, and so $4y_0$ is indeed the smallest positive period. Let's say $\omega_0 = 4y_0$.

Let's now say ω is an arbitrary period. We will show that there exists an n st

$$n\omega_0 \leq \omega < (n+1)\omega_0$$

If ω is not equal to $n\omega_0$, then $\omega - n\omega_0$ would be a positive period $< \omega_0$. But we just showed this is impossible, hence every period must be an integral multiple of ω_0 .

We will denote the smallest positive period of e^{iz} by 2π . Important consequence's of the proof we have just done is the following identities:

$$e^{\pi i/2} = i \quad e^{\pi i} = -1 \quad e^{2\pi i} = 1$$

⁹This function, along with the correction terms, shall be the gamma function

2.4.3 Logarithm

The (compositional) inverse of the exponential function is called the *logarithm*, denoted $\log(z)$. In particular we would want that $e^{\log(z)} = z$. Since e^z is never 0 (it's image is $\mathbb{C}^\times$), the number 0 has no logarithm. If we write $z = \log(\omega)$ and choose $\omega \neq 0$, the equation $e^{x+iy} = \omega$ is equivalent (by Euler's formula) to

$$e^x = |\omega| \quad e^{iy} = \frac{\omega}{|\omega|}$$

the first equation has a unique solution $x = \log |\omega|$, that is the real logarithm of the positive number $|\omega|$. The second equation is a complex number with absolute value 1, and thus it has only one solution in the interval $0 \leq y < 2\pi$. It is also satisfied by all y that differ from this solution by an integer multiple of 2π . Thus, every complex number has infinitely many logarithms which differ from each other by multiples of $2\pi i$. The imaginary part of $\log(\omega)$ is called the *argument* and is sometimes denoted $\arg(\omega)$. It is usually seen as the angle (measuring from the right side) between the x axis and $\omega/|\omega|$. By definition, the argument has infinitely many answers, for example:

$$\arg(i) = \{\pi/2, 5\pi/2, 9\pi/2, \dots\}$$

This reflects the periodic nature of e^z , for

$$e^i = e^{i+2\pi} = e^{i+4\pi} = \dots$$

Overall, the logarithm can be decomposed into the family of equations:

Definition 2.4.3: Logarithm

Given $\log |z|$ and $\arg(z)$, we get the following countable family of functions

$$\log(\omega) = \log |\omega| + i \arg(\omega)$$

Naturally, we wish to work with functions. For that, we must choose over which interval we consider $\log$ too properly define the inverse:

Definition 2.4.4: Branch

Let $f(z)$ be a continuous function defined on a connected set Ω . then f is said to be a *branch* of $\log$ if

$$e^{f(z)} = z$$

that is $\exp \circ f = \text{id}$

Notice that once a branch is chosen $\log$ must be a *right inverse*; it cannot be a left inverse for it must *choose* a particular angle to which it maps to (or equivalently, as e^x is surjective but not injective on $\mathbb{C}$, it has a right-inverse but not a left-inverse). For future purposes, we will box the following result:

Lemma 2.4.5: Difference Between Branches

Let $f(z)$ be a branch of $\log(z)$ in a connected open set Ω . Then any other branch will have the form $g(z) = f(z) + 2k\pi i$ for some $k \in \mathbb{Z}$. Conversely, for all $k \in \mathbb{Z}$, $f(z) + 2k\pi i$ is a branch

Proof:

Let f, g be two branches and let

$$h(z) = \frac{f(z) - g(z)}{2\pi i}$$

Since h is continuous on a connected set, we know its image is connected as well. The image of h is in $\mathbb{Z}$ and thus is only one point in the image, implying h must be constant, completing the proof.

When we choose $f(z)$ for which $\arg(1) = 1$, we shall call $f(z)$ the *principal branch*, and denote the argument function as Arg to emphasize we are now using the continuous function with a fixed choice of angle.

2.4.4 Complex Exponentiation

Using $\log$ and $\exp$, we may define z^α .

Definition 2.4.6: General Exponent

Let $a, b \in \mathbb{C}$. Then:

$$a^b := \exp(b \log a)$$

When $z \in \mathbb{R}$, then by convention we will take the real logarithm unless stated otherwise. If a is restricted to positive numbers, then $\log(a)$ shall be real and a^b has a single value. Otherwise, we would consider $\log(a)$ to be the complex logarithm with a^b having infinitely many values, each different from one another by $e^{2\pi i n b}$. There is only a single value if and only if b is an integer n and a^b can be interpreted as a power of a or a^{-1} . If b is a rational number with reduced form p/q , then a^b has exactly q values that can be represented as $\sqrt[q]{a^p}$. We also have the equality of sets (remember these sets are infinite)

$$\log(z_1 z_2) = \log z_1 + \log z_2$$

$$\arg(z_1 z_2) = \arg z_1 + \arg z_2$$

Using this, we can also find inverse sin and cos by solving the equation

$$\cos(z) = \frac{1}{2}(e^{iz} + e^{-iz}) = \omega$$

Manipulating, we get

$$e^{iz} - 2\omega + e^{-iz} = 0 \quad \Leftrightarrow \quad (e^{iz})^2 - 2\omega e^{iz} + 1 = 0$$

This a quadratic equation with e^{iz} as the root $e^{iz} = \omega \pm \sqrt{\omega^2 - 1}$, which implies $iz = \log(\omega \pm \sqrt{\omega^2 - 1})$. Thus:

$$z = \arccos(\omega) = -i \log(\omega \pm \sqrt{\omega^2 - 1})$$

Since $\omega + \sqrt{\omega^2 - 1}$ and $\omega - \sqrt{\omega^2 - 1}$ are reciprocals, we can also write:

$$z = \arccos(\omega) = \pm i \log(\omega + \sqrt{\omega^2 - 1})$$

The inverse of $\sin(z)$ is now easily seen as:

$$\arcsin(\omega) = \frac{\pi}{2} - \arccos(\omega)$$

Proposition 2.4.7: Derivative Of Logarithm

Let $f(z)$ be a branch of $\log(z)$ on a domain Ω . Then $f(z)$ is holomorphic and

$$f'(z) = \frac{1}{z}$$

Proof:

We simply compute:

$$\begin{aligned} & \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{e^{f(z+h)} - e^{f(z)}} \\ &= \lim_{w \rightarrow f(z)} \frac{w - f(z)}{e^w - e^{f(z)}} \\ &\stackrel{!}{=} \frac{1}{e^{f(z)}} \\ &= \frac{1}{z} \end{aligned}$$

where the $\stackrel{!}{=}$ equality comes from the fact that

$$\lim_{w \rightarrow f(z)} \frac{e^w - e^{f(z)}}{w - f(z)}$$

exists since e^z is differentiable, and so the reciprocal exists too with limit $1/e^z$.

We should also find the power series of the log:

Proposition 2.4.8: Log Power Series

For $|z| < 1$, the power series

$$f(z) := \sum_n (-1)^{n+1} \frac{z^n}{n}$$

converges and is equal to the principal branch of $\log(1+z)$

Proof:

It can directly be checked that $f(z)$ and $g(w) = \sum_n \frac{w^n}{n!}$ (which is the series expansion of $e^w - 1$) are inverses of each other, and hence $g(f(z)) = z$. Thus, $e^{f(z)} = z + 1$, which by definition makes $f(z)$ the branch of $\log(z+1)$. Evaluating f at 0 shows that this is a principal branch.

Finally, we take a moment to think about a Riemann surface that will make log single-valued. Consider the covering

$$X = \{(z, w) \in \mathbb{C}^2 \mid z = e^w\}$$

Then $\log$ would just be a mapping to the second coordinate. By the periodicity of $\log$, we see that for any small neighbourhood around $z \in \mathbb{C}$, we have countably many open sets laying above it in X . Furthermore, if we let $\log(1) = 0$, then as we go around counter-clockwise on S^1 with the assumption that $\log$ is continuous, we will get that we would get back to $\log(1) = 2\pi$, and everytime we go around we get another 2π factor: $\log(1) = 2k\pi$. This motivates a visual like so:

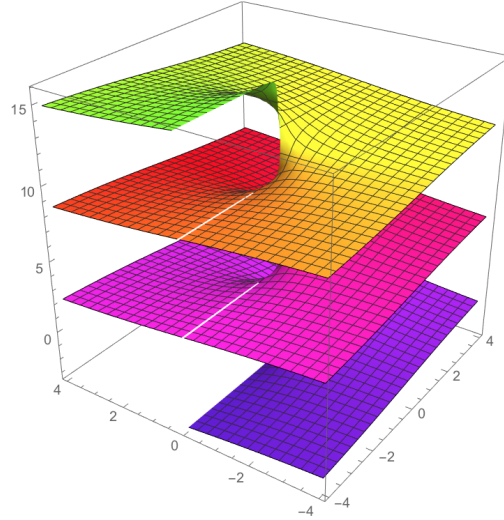


Figure 2.1: Branch Covering for complex $\log$

which we have already briefly mentioned at the end of section 1.2.2.

2.5 Analytic Functions

Since differentiation is a local property, it is natural to talk about functions which are locally a power series. The following is exactly this:

Definition 2.5.1: Analytic Function

Let f be a function on an open set Ω . Then f is said to be *analytic* if every $f(z)$ can be written as a convergent power series centered at some $z_0 \in \Omega$. In particular, for every $z_0 \in \Omega$ there is a convergent power series centered at z_0 such that

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

where z satisfies $|z - z_0| < r$ and r is less than the radius of convergence of the above power series.

Analytic functions have many of the same nice properties as power series. For example, they are

infinitely differentiable, and their integral is easy to find:

$$g(z) = \sum_{n=0}^{\infty} \frac{a_n}{n+1} (z - z_0)^{n+1}$$

which has the same radius of convergence by the root test. The easiest example of analytic functions are power series

Proposition 2.5.2: Power Series Are Analytic

Let $f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$ be a convergent power series with radius of convergence R . Then $f(z)$ is analytic in $|z - z_0| < R$

Proof:

Without loss of generality we may assume $z_0 = 0$ since we are simply translating the entire function to be around 0. Pick any $|r_0| < R$. We shall convert the power series $\sum_n a_n z^n$ into one centered at z_0 . Computing:

$$\begin{aligned} f(z) &= \sum_n a_n (z_0 - (z - z_0))^n \\ &= \sum_{n=0}^{\infty} a_n \sum_{k=0}^n \binom{n}{k} z_0^{n-k} (z - z_0)^k \end{aligned}$$

Since the z in the above is within the radius of convergence at the point 0, we get that the above power series is absolute continuous and hence we may re-arrange the terms:

$$f(z) = \sum_{k=0}^{\infty} \left(\sum_{n=k}^{\infty} a_n \binom{n}{k} z_0^{n-k} \right) (z - z_0)^k$$

If this re-arrangement looks foreign, think of the layer-cake representation for double integrals. This power series respects $|z - z_0| < r$ for any $r < R - |z_0|$, and hence is a valid power series representation, completing the proof.

2.5.1 Principle of Analytic Continuation

Given a domain $\Omega \subseteq \mathbb{C}$ on which a polynomial is defined, we may enlarge Ω to a larger open set, and the polynomial will uniquely be enlarged set, namely as coefficients remain unchanged. Naturally, this generalizes to power series too. What is great is that this uniqueness of extension is true for analytic functions! The key idea is that if all derivatives at a point are zero for a polynomial and a power series, then it is zero for a connected neighbourhood:

Theorem 2.5.3: Principle of Analytic Continuation

Let $f(z)$ be an analytic function in a connected open set Ω and $z_0 \in \Omega$. Then the following are equivalent:

1. $f^{(n)}(z_0) = 0$ for all $n \in \mathbb{N}$
2. f is identically 0 in a neighbourhood of z_0
3. f is 0 in Ω

Proof:

(3) $\Rightarrow$ (1) is immediate, and (1) $\Rightarrow$ (2) comes from Taylor's Theorem. We only require to show (2) $\Rightarrow$ (3).

Let $\omega' = \{z \in \Omega \mid f \equiv 0 \text{ in a neighbourhood of } z \text{ in } \Omega\}$. The set Ω' is nonempty since z_0 is in Ω' . It is also certainly open by definition. If we show Ω' is closed, then it is clopen and hence must be equal to Ω . Let $z \in \overline{\Omega'}$. Since the derivative is continuous, its values can be extended uniquely to its border and hence $f^{(n)}(z) = 0$ for all $n \in \mathbb{N}$. Then, by (1) $\Rightarrow$ (2), f is identically 0 in a neighbourhood of z , and so in fact $z \in \Omega'$. But then $\Omega' = \overline{\Omega'}$, showing it's closed. But then by connectedness $\Omega' = \Omega$.

Note that there are smooth real functions with derivatives zero that are only 0 at a single point, e^{-1/x^2} is an example, and hence this result cannot extend to general smooth functions.

Corollary 2.5.4: Analytic Continuation Of Functions

Let f, g be analytic in a connected open domain Ω and $f = g$ in a neighbourhood of some point. Then $f = g$ in Ω .

Proof:

Simply take the function $f - g$ and 0.

Corollary 2.5.5: Analytic On $\mathbb{C}$, Then Power Series

Let f be an entire analytic function (defined on all of $\mathbb{C}$). Then f is a power series.

Proof:

As f is analytic on $\mathbb{C}$, it is analytic at 0, so take $f(z) = \sum a_n z^n$ on some open neighborhood, U . Represent f on U as g . Then since $f - g$ is 0 in the neighborhood of $\mathbb{C}$, g extends to all of $\mathbb{C}$, completing the proof.

Hence, the extension of an analytic function must be *unique*, we only need two analytic functions to agree on an open neighbourhood, or even better a set of points with an accumulation point in the connected domain (as proposition 2.5.7 below shows, together with theorem 2.5.3), and analytic

function whose domain is all of $\mathbb{C}$ is a power series¹⁰! This is in stark contrast with general smooth functions which may have multiple different possible extensions on their entire domain. For example take

$$f(x) = \begin{cases} 0 & x \leq 0 \\ e^{-1/x^2} & x \geq 0 \end{cases}$$

then f on $(-2, -1)$ agrees with the zero function, but the two will certainly not extend to the same function! Note to that it is possible that the sequence of roots converge on the boundary; for example

$$f(z) = \sin\left(\frac{\pi}{z}\right)$$

has $1/n$ as a sequence of zeros, but is holomorphic on $B_1(1)$ ¹¹.

One interesting algebraic fact we get from this is the following

Corollary 2.5.6: Ring Of Analytic Functions

The ring of analytic functions on an open domain Ω , $\mathcal{A}(\Omega)$ forms an integral domain

Proof:

Let $f, g \in \mathcal{A}(\Omega)$ and consider $fg = 0$. If $f \neq 0$, then f must be nonzero in some open domain. Hence g must be zero in some open domain. But then g is identically 0 in Ω , and hence $g = 0$. But then $\mathcal{A}(\Omega)$ is an integral domain, completing the proof.

This further reflects how analytic functions are like the completion of polynomial rings, and later we shall see more similarities between these rings.

Finally, let us look at the roots of analytic functions. Their local structure mirrors that of polynomials: near a zero, an analytic function factors as a power of $(z - z_0)$ times a nonvanishing analytic function.

Proposition 2.5.7: Order of a Zero and Local Factorization

Let f be analytic in a neighbourhood of z_0 with $f(z_0) = 0$ but f not identically zero. Then there is a unique integer $k \geq 1$, the *order* (or *multiplicity*) of the zero, and an analytic function g in a neighbourhood of z_0 with $g(z_0) \neq 0$, such that

$$f(z) = (z - z_0)^k g(z)$$

Consequently z_0 is an isolated zero: there is an $\epsilon > 0$ with $f(z) \neq 0$ for $0 < |z - z_0| < \epsilon$. In particular the zeros of a nonzero analytic function are isolated, so any compact subset of the domain contains only finitely many of them.

Proof:

Write $f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$ on a neighbourhood of z_0 . Since $f(z_0) = 0$ and f is not identically zero, let k be the smallest integer with $a_k \neq 0$, equivalently the smallest k with $f^{(k)}(z_0) \neq 0$

¹⁰This shall further motivate studying functions with *poles* or *essential singularities* on $\mathbb{C}$, as these will be the only analytic functions that are not power series

¹¹this does not contradict proposition 2.5.7 below, as the accumulation point 0 of these zeros is not in the domain

(theorem 2.5.3). Factoring out the leading power,

$$f(z) = (z - z_0)^k \sum_{n=0}^{\infty} a_{n+k} (z - z_0)^n =: (z - z_0)^k g(z)$$

where g is the sum of a power series with the same radius of convergence, hence analytic, with $g(z_0) = a_k \neq 0$. By continuity $g(z) \neq 0$ on a small enough neighbourhood of z_0 , and $(z - z_0)^k \neq 0$ for $z \neq z_0$, so $f(z) \neq 0$ for $0 < |z - z_0| < \epsilon$ for an appropriate $\epsilon > 0$. The integer k is unique because the power series coefficients are. A nonzero analytic function therefore has isolated zeros, and a compact subset meeting infinitely many of them would carry an accumulation point of zeros, forcing $f \equiv 0$ by theorem 2.5.3, a contradiction.

Doing a coordinate change on the above makes f into quite a simple function: if $\zeta := (z - z_0)g(z)^{1/k}$, then

$$f(z(\zeta)) = \zeta^k$$

This also means that any compact subset of the domain can only contain finitely many zeros.

2.5.2 Meromorphic Functions

Let's now generalize to a quotient of analytic function: $\frac{f(z)}{g(z)}$ where g is not identically 0. If $g(z_0) \neq 0$, then $\frac{f(z)}{g(z)}$ is well-defined and analytic in a neighbourhood of z_0 which we see by taking fg^{-1} in proposition 2.3.8. If $g(z_0) = 0$, then either g is identically zero in a neighbourhood around z_0 , in that case the coefficients in the Taylor expansion around z_0 of g are all zero, or it is not identically zero, in which case there is a k such that $g^{(n)}(z_0) = 0$ for $n \leq k$.

Then we may use an limit argument on polynomials to show that z_0 is a root of f and g , in particular we can re-write $f(z) = (z - z_0)^k f_1(z)$ and $g(z) = (z - z_0)^l g_1(z)$ where $f_1(z_0), g_1(z_0) \neq 0$. Notice how this is not necessarily possible if f is not analytic, take for example a smooth bump function with compact support and try factoring out $(x - x_0)^n$ outside the support. Since we can do this for analytic functions, we have:

$$\frac{f(z)}{g(z)} = (z - z_0)^{k-l} \frac{f_1(z)}{g_1(z)}$$

If $k \geq l$, then $\frac{f}{g}$ extends to be analytic at z_0 . If $k < l$, then z_0 is a pole of $\frac{f}{g}$ of order $l - k$, and we get

$$\lim_{z \rightarrow z_0} \left| \frac{f(z)}{g(z)} \right| = \infty$$

Notice that the above limit is for any path¹², and hence it is natural to extend $\frac{f}{g}$ as a function on the Riemann sphere. Due to this non-analytic behavior on only a discrete set (or similarly, analytic behavior on S^2) we label them:

¹²This shall be more nuanced when working in $\mathbb{C}^n$, see chapter 8

Definition 2.5.8: Meromorphic Functions

A *meromorphic function* in an open set Ω is a function that is well-defined and analytic in the complement of a discrete set and expressible in a neighbourhood of any point in Ω as a quotient of analytic function $\frac{f}{g}$ where g is not identically 0.

Meromorphic functions from a *field*: they are the field of fractions of the analytic functions which form an integral domain. The key here was the fact that analytic functions have roots, we shall see in chapter 3, section 3.4.1 that the poles can be thought of as determining most of the “important” information.

***Analytic Continuation**

The next concept we will tackle is known as *analytic continuation*. Say we have a real differentiable function $f : (0, 1) \rightarrow \mathbb{R}$. If I ask you to tell me what is $f(-1)$, then you will say this is obviously undefined. We may extend f smoothly to have it defined at -1 , but there are certainly many ways of doing this. In the complex case, we will get that if f is determined to be complex-differentiable in some connected region, then it automatically determines uniquely on any larger connected open set, that is there is always a unique holomorphic extension to a larger connected open set. The most infamous analytic continuation is that of the *Riemann-zeta function*

$$\zeta(s) = \sum_{k=1}^{\infty} \frac{1}{k^s}$$

As of writing this, there is a million dollars to them that could show that all the zero's of $\zeta(s)$ are either real or have real part $1/2$:

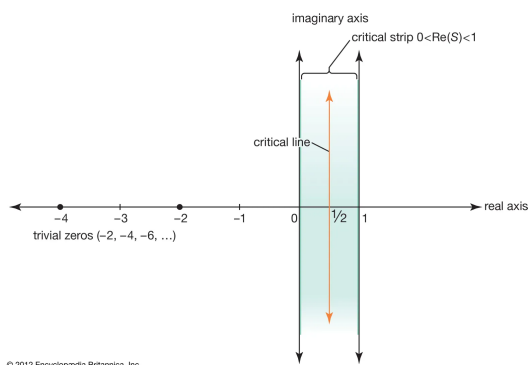


Figure 2.2: Visualizing where the critical points lie

More can be found in [21].

Exercise 2.5.1

1. Show that the ring of analytic functions is a Bézout domain, namely all finitely generated ideals are principle.

2. Show that the ring of analytic function is a GCD domain (may require infinite products for this)
3. Let f, g be analytic functions on U that agree a non-isolated set of points S . Then $f = g$ on U
4. Let f be analytic and $f(z_0) = a$ for some point z_0 . Show that for a sufficiently small neighborhood around z_0 , there is no points w such that $f(w) = a$.

Complex Forms: Integration, Growth, Singularities

As we mentioned earlier, we can already define the Lebesgue integral on $\mathbb{C}$ (by the Borel σ -algebra given by the euclidean topology on $\mathbb{R}^2$) giving us notion of integration of the 2D regions in $\mathbb{C}$. Thus, we may instead study the case of a path γ when the complex 1-form has a holomorphic coefficient, that is:

$$\omega = f(z)dz = f(z)(dx + idy) \quad f \in C^\omega(\Omega)$$

In studying these, we shall find that such forms have the property of being *closed* or *conservative* (the complex counter-part of the Fundamental Theorem of Calculus, FTC), which equivalently shall show us that the integral of any path is path independent (on connected open sets).

Once we show that $f(z)dz$ is conservative given f is holomorphic, we shall show the important consequence that if f is once complex-differentiable, it is infinitely complex differentiable, and even better it is analytic. Furthermore, using proposition 2.3.2 it can be deduced that analytic functions are holomorphic, giving us the following important equivalence:

$$\text{Holomorphic} \iff \text{Analytic}$$

This is completely false for real-differentiable function: consider any “pathological” (Lebesgue) integrable function f . Then $F(x) = \int_0^x f(t)dt$ is (real) differentiable by the FTC. However, it can be far away from being twice differentiable. The fact that the Cauchy-Riemann equations force this relation is thus at first hand certainly quite a surprising result, but it will come down to the fact that this PDE equation forces enough regularity on f so that it is *harmonic*¹.

After understanding the case of $f(z)dz$ where f is holomorphic, we shall take a moment to examine some of the “dynamics” of holomorphic functions by defining some important inequalities the

¹As a consequence, all harmonic function are indeed analytic

coefficients that the power-series expansion must satisfy. These results are technically a result that can be done on harmonic functions, but it is very intuitive to see the consequence of these results on holomorphic functions. For example, we shall be able to prove the infamous *Liouville's Theorem* which states that all bounded entire holomorphic functions must be constant (a result which is false for smooth functions, consider e^{-x^2}).

We shall then return to investigating integrals look at the case where f is meromorphic: functions that are allowed to be undefined at certain points with “nice enough” singularities (the case of having not “nice enough” singularities will be covered in section 4.4). It will turn out that in this case, $f(z)dz$ is not as trivial, in particular $f(z)dz$ does not always have a primitive and f is not always conservative. However, f will still have many nice properties and will be relatively simple to integrate using residue theory. Residue theory will give us many integration tools allowing us to:

1. evaluating $\int_0^\infty \frac{\sin(x)}{x} dx$ and get $\frac{\pi}{2}$
2. Showing that

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$$

3.1 Integration Review

(Tao's blog can also be helpful: [here](#))

As mentioned, we need to specify a path on which we are integrating. A path (or curve) will be a continuous function $\gamma : [a, b] \rightarrow U \subseteq \mathbb{C}$. It is differentiable if γ is differentiable. The notation $-\gamma$ will represent reversing the path, that is

$$-\gamma(x) = \gamma(a + b - x)$$

Now consider $f : U \rightarrow \mathbb{C}$. Just like we have done for Riemann integrals, we can take any path γ in a plane and split it up via linear connections, then take their sum:

$$\sum_{i=1}^n f(z_i)(z_{i+1} - z_i)$$

where $z_i \in \gamma([a, b])$. Intuitively, we can then take the limit of this value² to define $\int_\gamma f(x)dz$. This limit would exist in cases where, say, f is continuous. If you remember your theorem's from calculus, when the integral exists we can also think of this as taking the supremum over all possible path approximations. Overall, this notion gives a good intuition on what is the integral, but not a good way of computing it. We can find a good computational tool if the path γ is C^1 . In this case, we can perform a change of variables. By letting $dz = \gamma'(x)$, then $dz = \gamma'(x)dx$, which is saying that

$$z_{i+1} - z_i \approx \gamma'(x)(x_{i+1} - x_i)$$

and this approximation becomes better as the partition gets finer. Then the integral becomes:

$$\int_a^b f(\gamma(x))\gamma'(x)dx$$

²which can be shown to be independent of partition

But now, we have this integral back down along the real axis, and so like we mentioned earlier we can split it along the real and imaginary parts. This integral is also well-defined since γ' is continuous. Sometimes, we will want γ to have some sharp points, which in derivative language means it has critical points. This is fine, since it will be only a zero-measure set amount of points, and so we would split γ into a piece-wise differentiable path and define the integral to be the sum over the pieces.

Another way of looking at this is by coming back to vector-valculus. Given $dz = dx + idy$ and $f = u + iv$, we get

$$\begin{aligned}\int_{\gamma} f(z)dz &= \int_{\gamma} (u + iv)(dx + idy) \\ &= \int_{\gamma} (udx - vdy) + \int_{\gamma} (vdx + udy)\end{aligned}$$

This will be important when we use Green's theorem in a later proof (TBD).

The integral is independent of the re-paramaterization up to change of direction: if $\gamma' : [a, b] \rightarrow U$ defines the same path, the integral is the same, but if $\gamma' = -\gamma$, then that would change the sign of the integral. Here are some important properties we will want to reference later

Proposition 3.1.1: Properties Of Line Integrals

Let $U \subseteq \mathbb{C}$ be open, let $f, g: U \rightarrow \mathbb{C}$ be continuous, and let $\gamma: [a, b] \rightarrow U$ be a piecewise C^1 path.

1. If $\gamma_1: [a, b] \rightarrow U$ and $\gamma_2: [b, c] \rightarrow U$ are piecewise C^1 paths with $\gamma_1(b) = \gamma_2(b)$, and $\gamma_1 \cup \gamma_2: [a, c] \rightarrow U$ is the path equal to γ_1 on $[a, b]$ and to γ_2 on $[b, c]$, then $\int_{\gamma_1} f(z)dz + \int_{\gamma_2} f(z)dz = \int_{\gamma_1 \cup \gamma_2} f(z)dz$.
2. $\int_{\gamma} (f + g)dz = \int_{\gamma} fdz + \int_{\gamma} gdz$.
3. If $h: [\alpha, \beta] \rightarrow \mathbb{C}$ is continuous, then $\left| \int_{\alpha}^{\beta} h(t)dt \right| \leq \int_{\alpha}^{\beta} |h(t)|dt$.
4. $\left| \int_{\gamma} f(z)dz \right| \leq M|\gamma|$, where M is an upper bound for $|f|$ on $\gamma([a, b])$ and $|\gamma| = \int_a^b |\gamma'(t)|dt$ is the length of γ .

Proof:

Choose a subdivision $a = t_0 < t_1 < \dots < t_n = b$ such that γ is C^1 on each $[t_{i-1}, t_i]$, with one-sided derivatives at the endpoints. By definition,

$$\int_{\gamma} f(z)dz = \sum_{i=1}^n \int_{t_{i-1}}^{t_i} f(\gamma(t))\gamma'(t) dt$$

where the integral of a continuous complex-valued function is the integral of its real part plus i times the integral of its imaginary part. This value does not depend on the subdivision. Inserting one further point s into $[t_{i-1}, t_i]$ replaces one term by two terms with the same sum, by the additivity over a subdivision point in [3, Order Bounds and Additivity over a Subdivision Point] applied to the real and imaginary parts, and any two subdivisions have a common refinement. The length $|\gamma|$ is computed with the same subdivision.

1. Take subdivisions of $[a, b]$ and of $[b, c]$ adapted to γ_1 and to γ_2 . Together they form a subdivision of $[a, c]$ adapted to $\gamma_1 \cup \gamma_2$, which is continuous because $\gamma_1(b) = \gamma_2(b)$, and on each of its subintervals $\gamma_1 \cup \gamma_2$ coincides with γ_1 or with γ_2 . Splitting the sum defining $\int_{\gamma_1 \cup \gamma_2} f(z) dz$ into the terms coming from $[a, b]$ and those coming from $[b, c]$ gives $\int_{\gamma_1} f(z) dz + \int_{\gamma_2} f(z) dz$.
2. On each $[t_{i-1}, t_i]$ we have $(f + g)(\gamma(t))\gamma'(t) = f(\gamma(t))\gamma'(t) + g(\gamma(t))\gamma'(t)$, so [3, Linearity of the Integral], applied to the real and imaginary parts, splits each term of the sum, and summing over i gives the claim.
3. Let $I = \int_{\alpha}^{\beta} h(t) dt$ and choose $\theta \in \mathbb{R}$ with $e^{-i\theta} I = |I|$. Linearity of the integral gives $e^{-i\theta} I = \int_{\alpha}^{\beta} e^{-i\theta} h(t) dt$, and taking real parts,

$$|I| = \int_{\alpha}^{\beta} \operatorname{Re}(e^{-i\theta} h(t)) dt \leq \int_{\alpha}^{\beta} |h(t)| dt$$

by $\operatorname{Re} w \leq |w|$ and the order bound of [3, Order Bounds and Additivity over a Subdivision Point].

4. By (3), applied to $h(t) = f(\gamma(t))\gamma'(t)$ on $[t_{i-1}, t_i]$, and since $|f(\gamma(t))| \leq M$, each term satisfies

$$\left| \int_{t_{i-1}}^{t_i} f(\gamma(t))\gamma'(t) dt \right| \leq M \int_{t_{i-1}}^{t_i} |\gamma'(t)| dt$$

Summing over i and applying the triangle inequality gives $\left| \int_{\gamma} f(z) dz \right| \leq M|\gamma|$, as claimed.

Lemma 3.1.2: Paths In Connected Set

Any two points in a connected open set $\Omega \subseteq \mathbb{R}^2$ can be joined by a piece-wise C^1 curve in Ω

Proof:

Let $a \in \Omega$ and let E be the set of $b \in \Omega$ joined to a by a piecewise C^1 curve in Ω . The constant curve shows $a \in E$. Let $b \in \Omega$ and let $B \subseteq \Omega$ be an open disc around b , which exists as Ω is open. Every point $c \in B$ is joined to b by the segment from b to c , which lies in the convex set B . If $b \in E$, following a curve from a to b by that segment joins a to c , so $B \subseteq E$. If $b \notin E$, then no $c \in B$ lies in E , as a curve from a to c followed by the segment from c to b would join a to b , so $B \subseteq \Omega \setminus E$. Hence E and $\Omega \setminus E$ are open. As Ω is connected and $E \neq \emptyset$, we get $E = \Omega$, as claimed.

Many of our results will necessitate the property of being simply connected, which says our space does not have a hole (or more technically that all closed paths are nullhomotopic). We shall give an example of what can go wrong to see why we care:

Example 3.1.3: Complex Integration not Simply Connected

1. We will integrate z^n for different $n \in \mathbb{Z}$'s and different paths where $\gamma(0) = \gamma(1) = 1$. Naturally, if the path is constant the $\int_{\gamma} z^n = 0$. Let's instead take the path $\gamma(\theta) = e^{2\pi i \theta}$ so

that it forms a circle around 0. Then, by also changing $z^n = e^{inx}$ and doing a change of variables we get:

$$\begin{aligned}\int_{\gamma} z^n dz &= \int_0^{2\pi} e^{inx} i e^{ix} dx \\ &= i \int_0^{2\pi} e^{i(n+1)x} dx \\ &= i \left(\frac{e^{i(n+1)x}}{n+1} \Big|_0^{2\pi} \right)\end{aligned}$$

we now get to a scenario where the value we get depends on n . If $n \neq -1$ we get that the integral is equal to 0. If $n = -1$, the integral becomes:

$$i \int_0^{2\pi} 1 dx = 2\pi i - 0 = 2\pi i$$

which is not zero! Looking back at our function, we see that our integral is $\int_{\gamma} 1/z dz$. This is essentially exactly a definition of the real $\log(x)$. In the complex case, notice that $1/z$ is not defined at $z = 0$, meaning the domain of this function is $\mathbb{C} \setminus \{0\}$, in particular this is *not a simply connected set*. This is a good example of what goes wrong if your set is not simply connected.

However, $1/z^n$ for $n > 1$ has a zero integral even though is defined on a non-simply connected set. Furthermore, we shall see that we cannot holomorphically extend $1/z^n$ (for $n > 0$) to $\mathbb{C}$, and so we may have conservative functions even on non-simply connected sets.

The geometric intuition behind why $1/z$ has “non-trivial monodromy” while $1/z^n$ for $n > 1$ does not is because $1/z$ is the “critical function” where there is just enough “unwinding” to not cancel out. If we take z^n , then it is easy to visualize: every time we make an n th turn (so $1/n$ turns), we turn around 1, so after 1 turn we turned around n times.

For $n = 1$, 1 turn gets us 1 counter-clockwise turn. What happens at -1 is exceptional precisely because we are taking a circle (hence parameterized by $e^{i\theta}$) and integrating against $e^{-i\theta}$, and hence what would usually just be 1 turn will actually cancel out and create an integration state that is not canceled out. For higher negative powers, this cancellation comes back.

This also means that when we are between -1 and 1 , we may experience some non-trivial monodromy, this is like the “critical strip” for monodromy. For example, Let us also do $\sqrt{z} = z^{1/2} = e^{1/2 \ln z}$. Then:

$$\begin{aligned}\int_{\gamma} z^{1/2} dz &= \int_0^{2\pi} e^{3/2 ix} dx \\ &= i \left(\frac{e^{\frac{3i}{2}x}}{3/2} \Big|_0^{2\pi} \right) \\ &= \frac{-4}{3} i\end{aligned}$$

Differential Forms Review

Differential forms are the natural tool to use to hold geometric information when integrating, which in our case would be paths. In this section, a map $\gamma : [a, b] \rightarrow \Omega$ where $\Omega \subseteq \mathbb{R}^2$ will be C^1 or piecewise C^1 , and $\gamma(t) = (x(t), y(t))$. A 1-form will be written as

$$\omega = Pdx + Qdy$$

with P, Q being (real or complex) continuous functions on Ω . Then define:

$$\int_{\gamma} \omega = \int_a^b \gamma^* \omega = \int_a^b F(t) dt$$

where $F(t) = P(\gamma(t))x'(t) + Q(\gamma(t))y'(t)$, which we get from the pullback:

$$\gamma^*(Pdx + Qdy) = \gamma^*(P)\gamma^*(dx) + \gamma^*(Q)\gamma^*(dy)$$

where $\gamma^*(P) = P \circ \gamma$ and $\gamma^*(dx) = d(\gamma^*x) = d(x \circ \gamma) = d(x(t)) = x'(t)$. As usual, reparametrization of the curves does not affect the value of the curve up to a sign: if $u : [c, d] \rightarrow [a, b]$ is a strictly positive diffeomorphism so that we have a new path $t \mapsto \gamma(u(t))$, then if this new path is γ' , then:

$$\int_{\gamma'} \omega = \int_c^d f(\gamma(u(t)))u'(t)dt = \int_a^b f(u)du = \int_{\gamma} \omega$$

If it is strictly negative then $\int_{\gamma'} \omega = -\int_{\gamma} \omega$. a diffeomorphism can only be strictly positive or negative since if its derivative is 0 at any point it is not a diffeomorphism. If we have a piecewise curve, then we define:

$$\int_{\gamma} \omega = \sum_{i=1}^n \int_{\gamma_i} \omega$$

If $\gamma(a) = \gamma(b)$ (if the endpoints are the same), we shall say that γ is a closed curve. In this case, we may start by integrating at any point (show this if you are not comfortable with it). One of the nicest 1-forms are those of the form $\omega = dF$. These are particularly nice to compute. By the FTC:

$$\int_{\gamma} dF = \int_a^b F'(\gamma(t))dt = F(\gamma(b)) - F(\gamma(a))$$

which we can see visually as thinking of the graph F defining the surface for which gravity “pulls down” on, and then dF represents the “potential” energy given by walking up or down this surface. Then the overall potential energy is only dependent on your height, giving us the above formula. If we return to the same point, then we get zero net potential energy, hence

$$\oint_{\gamma} dF = 0$$

Since we are trying to define integration of 1-forms $f(z)dz$, it would be nice if they always have an antiderivative just like the real 1-forms $f(x)dx$. We lead up to this by developing the theory for general complex 1-forms:

Definition 3.1.4: Primitive

given a 1-form ω , a *primitive* of ω is a C^1 function $F : \Omega \rightarrow \mathbb{C}$ such that

$$\omega = dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy$$

Outside of complex, such forms are usually called *exact*.

Note that this gives another quick proof that if Ω is connected and dF is zero then F is constant (which in the gravity interpretation can be thought as never gaining any potential energy). Using this, we see that if $\omega = dG$ for some other function, then $d(F - G) = 0$, showing that F and G differ by a constant and hence primitives are unique up to a constant. Not all complex 1-forms have a primitive (as we shall see after the next proposition). The next proposition gives the exact property needed for complex 1-forms to have a primitive:

Proposition 3.1.5: Criterion for Primitive Existing

Let $\omega = P dx + Q dy$ be a 1-form on an open set $\Omega \subseteq \mathbb{C}$.

1. If F is a primitive of ω , then $\int_{\gamma} \omega = F(\gamma(b)) - F(\gamma(a))$ for every piecewise C^1 path $\gamma : [a, b] \rightarrow \Omega$.
2. ω has a primitive if and only if $\oint_{\gamma} \omega = 0$ for every piecewise C^1 closed curve γ in Ω .

Proof:

1. Let $a = t_0 < \dots < t_n = b$ be such that $\gamma = (x, y)$ is C^1 on each $[t_{i-1}, t_i]$. There, as F is C^1 (definition 3.1.4), the chain rule gives

$$\frac{d}{dt} F(\gamma(t)) = \frac{\partial F}{\partial x}(\gamma(t))x'(t) + \frac{\partial F}{\partial y}(\gamma(t))y'(t) = P(\gamma(t))x'(t) + Q(\gamma(t))y'(t)$$

which is the continuous integrand of $\int_{t_{i-1}}^{t_i} \gamma^* \omega$. By the fundamental theorem of calculus ([3, The Fundamental Theorem of Calculus, Second Form], applied to real and imaginary parts), the integral over the i -th piece is $F(\gamma(t_i)) - F(\gamma(t_{i-1}))$, and summing over i gives $F(\gamma(b)) - F(\gamma(a))$.

2. If $\omega = dF$ and γ is closed, then (1) gives $\oint_{\gamma} \omega = 0$. Conversely, suppose the integral over every closed curve vanishes. Integrals over a concatenation of paths add, by definition, and traversing a path backwards changes the sign of the integral, by the substitution $s = a + b - t$. Let W be a connected component of Ω , which is open, and fix $p \in W$. The points of W joined to p by a polygonal path in W form an open set, discs being convex, and so do the remaining points of W . As W is connected, every point of W is joined to p by a polygonal path. For $z \in W$ define $F(z) = \int_{\gamma} \omega$, where γ is such a path from p to z . F is well-defined: if δ is another path from p to z , then γ followed by δ reversed is a closed curve, so the integrals over γ and δ are equal. Let $z = (x, y) \in W$, and let $h \neq 0$ be real and so small that the segments from z to $z + h$ and to $z + ih$ lie in W . Appending

these segments to a path from p to z gives

$$F(x+h, y) - F(x, y) = \int_x^{x+h} P(t, y) dt \quad F(x, y+h) - F(x, y) = \int_y^{y+h} Q(x, t) dt$$

Dividing by h and letting $h \rightarrow 0$, the fundamental theorem of calculus ([3, The Fundamental Theorem of Calculus, First Form]) and the continuity of P and Q give $\frac{\partial F}{\partial x} = P$ and $\frac{\partial F}{\partial y} = Q$.

These are continuous, so F is a C^1 function on Ω with $dF = \omega$, completing the proof.

It would be nice if we can check a subset of curves. For example, if we can only check all rectangles paths are zero, we would have a much easier time computing. In general this is not possible: we will require our domain to be *simply connected*. On a simply connected surface, like the open disk, this is a sufficient condition

Proposition 3.1.6: Existence Of Primitive On Disk

Let D be an open disc. If $\int_{\gamma} \omega = 0$ whenever γ is the boundary of a rectangle contained in D with sides parallel to the axes, then ω has a primitive in D

The key is that every rectangle must be defined, and hence our space cannot have any holes.

Proof:

Write $\omega = Pdx + Qdy$ with P, Q continuous, and let (x_0, y_0) be the center of the disk. Let $(x, y) \in D$. The points (x, y_0) and (x_0, y) are no farther from the centre than (x, y) , so the four corners of the rectangle with opposite corners (x_0, y_0) and (x, y) lie in D , and so does the rectangle, D being convex. Let γ_1 be the path from (x_0, y_0) horizontally to (x, y_0) and then vertically to (x, y) , and γ_2 the path from (x_0, y_0) vertically to (x_0, y) and then horizontally to (x, y) . If $x \neq x_0$ and $y \neq y_0$, then γ_1 followed by γ_2 reversed is the boundary of the rectangle, traversed in one of its two directions, so by assumption $\int_{\gamma_1} \omega = \int_{\gamma_2} \omega$, and otherwise the two paths coincide. We may now define

$$F(x, y) = \int_{\gamma_1} \omega = \int_{x_0}^x P(t, y_0) dt + \int_{y_0}^y Q(x, t) dt = \int_{y_0}^y Q(x_0, t) dt + \int_{x_0}^x P(t, y) dt$$

the last expression being $\int_{\gamma_2} \omega$. In the second expression only the last integral depends on y , and in the third only the last one depends on x . By the fundamental theorem of calculus ([3, The Fundamental Theorem of Calculus, First Form], applied to real and imaginary parts), the continuity of P and Q gives

$$\frac{\partial F}{\partial x}(x, y) = P(x, y) \quad \frac{\partial F}{\partial y}(x, y) = Q(x, y)$$

These partial derivatives are continuous, so F is a C^1 function with $dF = \omega$ (definition 3.1.4), completing the proof.

Thus, instead of checking on all curves, it suffices to check for all γ that are boundary of rectangles, or in particular sufficiently small rectangles, if Ω is an open disk.

As we mentioned, this works since there are no holes on open disks. On spaces that have holes, the above argument breaks down, and we can find a form ω that does not have a primitive (dz/z on

$\mathbb{C} \setminus \{0\}$ is an example of this by proposition 3.1.5). For forms of the form $f(z)dz = f(z)(dx + idy)$, we will fortunately still have that it has *local primitives* for any simply connected subset (this a version of Poincaré’s lemma). We thus define the following:

Definition 3.1.7: Closed Form

Let $\omega = Pdx + Qdy$ be a 1-form on an open set Ω . Then we say that ω is *closed* if for any point $z \in \Omega$, there is an open neighbourhood in which ω has a primitive.

Another more common definition in the world of geometry is that a form is a closed form if $d\omega = 0$. We can always assume we can pick a small enough neighbourhood around $z \in D$ to make it a disk, hence a form ω is closed if $\int_{\gamma} \omega = 0$ whenever γ is the boundary of a rectangle contained in a disk D . In general, a closed differential form on a disk has a primitive, but a closed differential form on Ω need not have a primitive as we saw in example 3.1.3. In the example, we had the form

$$\frac{dz}{z} = \frac{dx + idy}{x + iy} = \frac{xdx + ydy}{x^2 + y^2} + i \frac{xdy - ydx}{x^2 + y^2}$$

Let’s consider a new form which is the imaginary part of this form:

$$w = \frac{xdy - ydx}{x^2 + y^2}$$

This is a closed form in the plane with the origin excluded, and has no primitive since if γ is the unit circle then

$$\int_{\gamma} \frac{xdy - ydx}{x^2 + y^2} = 2\pi$$

If we inspected this form more closely, we would see that it’s equal to $d \arctan\left(\frac{y}{x}\right)$, which is a *many-valued function*, showing us how ω fails to have a primitive.

If ω is a closed form, it only has local primitives. We may wonder if instead of finding a global primitive, we find a primitive given a chosen path γ . The following makes precise what this may mean:

Definition 3.1.8: Primitive Along A Path

Let $\gamma : [a, b] \rightarrow \Omega$ be a path contained in an open set Ω and let ω be a closed form in Ω . A continuous function $f : [a, b] \rightarrow \mathbb{C}$ is called a *primitive of ω along γ* if for any $\tau \in [a, b]$, there exists a primitive F of ω in a neighbourhood of the point $\gamma(\tau) \in \Omega$ such that

$$F(\gamma(t)) = f(t)$$

for t near enough τ .

To understand the above intuitively, we shall give an example. Recall that $\int_{\gamma} dz/z$ has is nonzero if γ is a circle, hence it is not a primitive form (however it is closed since it is locally a primitive). Letting $\omega = dz/z$, if we take $f(t) = \int_0^t \gamma^* \omega$ where $\gamma(t) = e^{2\pi i t}$, then $f(0) = 0$ and $f(2\pi) = 2\pi i$. If ω was a primitive, we would have $f(0) = f(2\pi) = 0$, i.e. we would be on a single surface, or the graph of a function, thus f would represent the “height” at the point $f(t)$. However, this is not the case, as

$\gamma(0) = \gamma(2\pi)$. This is probably some foreshadowing, but we may see f as hinting at the existence of a *Riemann surface* on which dz/z does have a primitive in some correct generalization.

Note that if ω had a global primitive, then a loop γ would have $f(\gamma(0)) = f(\gamma(1))$, while this is not necessarily the case for local primitive functions.

Theorem 3.1.9: Primitive Along A Path

Let ω be a closed form on an open set $\Omega \subseteq \mathbb{C}$ and let $\gamma : [a, b] \rightarrow \Omega$ be a path, required only to be continuous. Then a primitive of ω along γ exists, and any two such primitives differ by an additive constant. If γ is moreover piecewise C^1 and f is a primitive of ω along γ , then

$$\int_{\gamma} \omega = f(b) - f(a)$$

Proof:

Uniqueness. Let f_1 and f_2 be primitives of ω along γ and fix $\tau \in [a, b]$. Choose primitives F_1, F_2 of ω on neighbourhoods of $\gamma(\tau)$ with $F_j(\gamma(t)) = f_j(t)$ for t near τ , and shrink both neighbourhoods to a common open disc V around $\gamma(\tau)$. On V we have

$$d(F_1 - F_2) = \omega - \omega = 0$$

and V is connected, so $F_1 - F_2$ is a constant on V . Hence $f_1 - f_2$ takes a single value on a neighbourhood of τ in $[a, b]$. Since τ was arbitrary, $f_1 - f_2$ is locally constant, so for each $u \in \mathbb{C}$ the set $(f_1 - f_2)^{-1}(u)$ is both open and closed in $[a, b]$. The segment $[a, b]$ is connected, so one of these sets is all of $[a, b]$, which says that $f_1 - f_2$ is constant.

Existence. Fix $\tau \in [a, b]$. As ω is closed, definition 3.1.7 gives an open disc $D_{\tau} \subseteq \Omega$ centred at $\gamma(\tau)$ on which ω has a primitive, and continuity of γ produces a relatively open interval $I_{\tau} \ni \tau$ in $[a, b]$ with $\gamma(I_{\tau}) \subseteq D_{\tau}$. The sets I_{τ} cover the compact metric space $[a, b]$, so the cover has a Lebesgue number $\lambda > 0$ by [8, Lebesgue Number Lemma]. Take any subdivision

$$a = t_0 < t_1 < \cdots < t_{n+1} = b$$

of mesh smaller than λ . Each $[t_i, t_{i+1}]$ then lies inside a single I_{τ} , so $\gamma([t_i, t_{i+1}])$ lies inside an open disc $U_i \subseteq \Omega$ on which ω has a primitive F_i .

Now correct the F_i in order of increasing i . Given that $F_0, \dots, F_i$ have been fixed, the point $\gamma(t_{i+1})$ lies in U_i and in U_{i+1} , so $U_i \cap U_{i+1}$ is a nonempty intersection of two open discs and is therefore convex, hence connected. On it $d(F_{i+1} - F_i) = 0$, so $F_{i+1} - F_i$ is a constant c_i there. Replacing F_{i+1} by $F_{i+1} - c_i$ leaves a primitive of ω on U_{i+1} and makes it agree with F_i on $U_i \cap U_{i+1}$. With the n corrections made, put

$$f(t) = F_i(\gamma(t)) \quad t \in [t_i, t_{i+1}]$$

At a shared endpoint t_{i+1} the two available formulas agree, because F_i and F_{i+1} agree at $\gamma(t_{i+1})$. So f is well defined, and it is continuous on each $[t_i, t_{i+1}]$ as a composite of continuous maps. A function on $[a, b]$ that is continuous on each of finitely many closed subintervals covering $[a, b]$ and takes matching values where they overlap is continuous, so f is continuous.

Condition (P). If τ is not one of the interior division points $t_1, \dots, t_n$, then τ belongs to exactly one $[t_i, t_{i+1}]$, and $F = F_i$ on the neighbourhood U_i of $\gamma(\tau)$ satisfies $F(\gamma(t)) = f(t)$ for every t of

that subinterval. If instead $\tau = t_{i+1}$ with $1 \leq i+1 \leq n$, take $F = F_i$ on the open set $U_i \cap U_{i+1}$, which is a neighbourhood of $\gamma(\tau)$, and restrict to those t near τ with $\gamma(t) \in U_i \cap U_{i+1}$. For such t to the left of τ we get $f(t) = F_i(\gamma(t)) = F(\gamma(t))$, and for such t to the right the correction gives $f(t) = F_{i+1}(\gamma(t)) = F_i(\gamma(t)) = F(\gamma(t))$. No differentiability of γ was used anywhere above.

The integral formula. Let γ be piecewise C^1 and let f be a primitive of ω along γ . Refine the subdivision built above by adjoining the finitely many parameters at which γ fails to be C^1 , keeping for each new subinterval the disc and primitive of the old subinterval containing it. On $[t_i, t_{i+1}]$ the function $t \mapsto F_i(\gamma(t))$ is a primitive of ω along $\gamma|_{[t_i, t_{i+1}]}$, witnessed at every parameter by F_i on U_i , so the uniqueness already proved makes it differ from $f|_{[t_i, t_{i+1}]}$ by a constant. Writing $\omega = Pdx + Qdy$ and $\gamma(t) = (x(t), y(t))$, the chain rule gives

$$\frac{d}{dt}F_i(\gamma(t)) = \frac{\partial F_i}{\partial x}(\gamma(t))x'(t) + \frac{\partial F_i}{\partial y}(\gamma(t))y'(t) = P(\gamma(t))x'(t) + Q(\gamma(t))y'(t)$$

which is the integrand of $\int_{t_i}^{t_{i+1}} \gamma^* \omega$. By the fundamental theorem of calculus and the cancellation of that constant in a difference,

$$\int_{\gamma|_{[t_i, t_{i+1}]}} \omega = F_i(\gamma(t_{i+1})) - F_i(\gamma(t_i)) = f(t_{i+1}) - f(t_i)$$

Summing over i telescopes the right-hand side to $f(b) - f(a)$, which is the stated formula.

3.1.10 Primitive of $\log(z)$ We may use this theorem to show things like any closed path which does not pass through the origin for the integral

$$\frac{1}{2\pi i} \int_{\gamma} \frac{dz}{z}$$

is an integer, since the primitive along the path will be a branch of $\log(z)$, and so we will get $f(b) - f(a)$, which will be the difference between two branches of $\log(z)$ at $\gamma(a) = \gamma(b)$, which we showed in lemma 2.4.5 is $2k\pi i$, and hence we have an integer.

The form dz/z is rather important as we saw it does not have a primitive, but in a real sense it is the “only” complex form that does not have a primitive. It is reasonable to think that any other form that does not have a primitive stems from this form.

Relation to Homotopies

Since holes are the main problem in defining a primitive, we shall take some time showing that up to being “stuck” or “wound” around a hole, the value of the Ingram is path independent. The natural concept to introduce then is that of homotopy. Let $I = [0, 1]$. In this section, we show that integrals of paths in closed forms are homotopy invariant.

Definition 3.1.11: Homotopy

Let $\gamma_0 : I \rightarrow D$ and $\gamma_1 : I \rightarrow D$ be two path with the same initial and final points. Then these paths are *homotopic* (in D) with fixed endpoints if there exists a continuous map $(t, u) \rightarrow \delta(t, u)$ of $I \times I$ into D st

$$\begin{cases} \delta(t, 0) = \gamma_0(t) & \delta(t, 1) = \gamma_1(t) \\ \delta(0, u) = \gamma_0(0) = \gamma_1(0) & \delta(1, u) = \gamma_0(1) = \gamma_1(1) \end{cases}$$

Two closed paths γ_0, γ_1 are *homotopic as closed paths* if there is a continuous map $(t, u) \rightarrow \delta(t, u)$ of $I \times I$ into D st

$$\begin{cases} \delta(t, 0) = \gamma_0(t) & \delta(t, 1) = \gamma_1(t) \\ \delta(0, u) = \delta(1, u) & \text{for all } u \end{cases}$$

we say a closed path is null homotopic if it is homotopic to a constant function.

We shall show that on a closed form, two homotopic paths (with fixed points or that are closed) have the same value. We first upgrade theorem 3.1.9 to homotopies

Definition 3.1.12: Primitive Along Homotopies

Let $(t, u) \rightarrow \delta(t, u)$ be a continuous mapping of a rectangle

$$a \leq t \leq b \quad a' \leq u \leq b'$$

into the open set D and let ω be a closed form in D . A *primitive of ω following the mapping δ* is a continuous function $f(t, u)$ in the rectangle satisfying the property that for any point (τ, ν) of the rectangle, there exists a primitive F of ω in a neighbourhood of $\delta(\tau, \nu)$ such that

$$F(\delta(t, u)) = f(t, u)$$

at any point (t, u) sufficiently near (τ, ν)

Lemma 3.1.13: Primitive Along Homotopies

There always exists an f as given in definition 3.1.12, and it is unique up to addition of a constant.

Proof:

Let $R = [a, b] \times [a', b']$, and let δ and ω be as in definition 3.1.12. Recall from the discussion after definition 3.1.4 that a function F with $dF = 0$ on a connected open set is constant there.

Uniqueness. Let f_1 and f_2 be primitives of ω following δ and fix $(\tau, \nu) \in R$. Choose primitives F_1, F_2 of ω on neighbourhoods of $\delta(\tau, \nu)$ with $F_k(\delta(t, u)) = f_k(t, u)$ for (t, u) near (τ, ν) , and shrink both neighbourhoods to a common open disc V around $\delta(\tau, \nu)$. On V we have $d(F_1 - F_2) = \omega - \omega = 0$, so $F_1 - F_2$ is constant on V . By continuity of δ , every (t, u) near enough (τ, ν) has $\delta(t, u) \in V$, so $f_1 - f_2$ is constant on a neighbourhood of (τ, ν) in R . Hence for each $c \in \mathbb{C}$

the set $(f_1 - f_2)^{-1}(c)$ is open, and it is closed by continuity. The rectangle R is convex, hence connected, so one of these sets is all of R , that is, $f_1 - f_2$ is constant.

Existence: a grid. Since ω is closed (definition 3.1.7), each $(\tau, \nu) \in R$ has an open disc $U \subseteq D$ around $\delta(\tau, \nu)$ on which ω has a primitive, and $\delta^{-1}(U)$ is a relatively open neighbourhood of (τ, ν) in R . These neighbourhoods cover the compact metric space R , so the cover has a Lebesgue number $\lambda > 0$ by [8, Lebesgue Number Lemma]. Choose subdivisions $a = t_0 < t_1 < \dots < t_m = b$ and $a' = u_0 < u_1 < \dots < u_n = b'$ so fine that each rectangle $R_{ij} = [t_{i-1}, t_i] \times [u_{j-1}, u_j]$ has diameter less than λ . Then $\delta(R_{ij})$ lies in an open disc $U_{ij} \subseteq D$ on which ω has a primitive F_{ij} .

Existence: the strips. Fix j and correct the F_{ij} in order of increasing i . The set $U_{ij} \cap U_{i+1,j}$ contains $\delta(\{t_i\} \times [u_{j-1}, u_j])$, so it is a nonempty intersection of two open discs, hence convex and connected, and $F_{i+1,j} - F_{ij}$ is constant on it. Subtracting that constant from $F_{i+1,j}$ makes $F_{i+1,j} = F_{ij}$ on $U_{ij} \cap U_{i+1,j}$. Then

$$g_j(t, u) = F_{ij}(\delta(t, u)) \quad (t, u) \in R_{ij}$$

is well defined on the strip $S_j = [a, b] \times [u_{j-1}, u_j]$, because on a shared edge $t = t_i$ the point $\delta(t_i, u)$ lies in $U_{ij} \cap U_{i+1,j}$, where the two formulas agree. It is continuous on S_j , being continuous on each of finitely many closed rectangles covering S_j and agreeing on their overlaps.

Restricted to the edge $u = u_j$, the function $t \mapsto g_j(t, u_j)$ is a primitive of ω along the path $t \mapsto \delta(t, u_j)$ in the sense of definition 3.1.8. Near a parameter τ with $t_{i-1} < \tau < t_i$ the primitive F_{ij} on U_{ij} witnesses this, and near a division point $\tau = t_i$ the primitive F_{ij} on $U_{ij} \cap U_{i+1,j}$ does, since $F_{i+1,j} = F_{ij}$ there. The same holds for $t \mapsto g_{j+1}(t, u_j)$ on the bottom edge of S_{j+1} , with the primitives $F_{i,j+1}$.

Existence: gluing the strips. Correct the strips in order of increasing j . By the uniqueness part of theorem 3.1.9, the two primitives $g_j(\cdot, u_j)$ and $g_{j+1}(\cdot, u_j)$ along the same path differ by a constant. Subtracting it from g_{j+1} and from every $F_{i,j+1}$ keeps the agreements $F_{i+1,j+1} = F_{i,j+1}$ obtained above and makes $g_{j+1}(t, u_j) = g_j(t, u_j)$ for every t . After these corrections, put $f = g_j$ on S_j . This is well defined on the shared edges, and continuous on R , being continuous on each of finitely many closed strips covering R and agreeing on their overlaps. Moreover $F_{i,j+1} = F_{ij}$ on $U_{ij} \cap U_{i,j+1}$. This set contains $\delta(t_i, u_j)$, as (t_i, u_j) lies in both R_{ij} and $R_{i,j+1}$, so it is a nonempty intersection of two open discs, hence connected, and $F_{i,j+1} - F_{ij}$ is constant on it. Evaluating at $\delta(t_i, u_j)$, the constant is $g_{j+1}(t_i, u_j) - g_j(t_i, u_j) = 0$.

Existence: the defining property. Let $(\tau, \nu) \in R$, let Q be the set of index pairs (i, j) with $(\tau, \nu) \in R_{ij}$, at most four of them, and let W be the intersection of the discs U_{ij} for $(i, j) \in Q$, an open set containing $\delta(\tau, \nu)$. Two pairs of Q differing in one index give primitives that agree on W by the two agreements established above. Two pairs of Q differing in both indices occur only when (τ, ν) is a common corner, and then a third pair of Q differs from each in one index. So all F_{ij} with $(i, j) \in Q$ coincide on W , and we call this common function F . The union of the rectangles not containing (τ, ν) is closed and misses (τ, ν) , so by continuity of δ there is a neighbourhood N of (τ, ν) in R meeting only the rectangles R_{ij} with $(i, j) \in Q$ and satisfying $\delta(N) \subseteq W$. For $(t, u) \in N$ choose $(i, j) \in Q$ with $(t, u) \in R_{ij}$. Then $f(t, u) = F_{ij}(\delta(t, u)) = F(\delta(t, u))$. Hence f is a primitive of ω following δ , completing the proof.

Theorem 3.1.14: Closed Forms And Homotopic Paths

Let $D \subseteq \mathbb{C}$ be open and let $\gamma_0, \gamma_1: I \rightarrow D$ be two piecewise C^1 paths which are homotopic with fixed end points, or two piecewise C^1 closed paths which are homotopic as closed paths (definition 3.1.11). Then if ω is any closed form in D ,

$$\int_{\gamma_0} \omega = \int_{\gamma_1} \omega$$

Proof:

Let $\delta: I \times I \rightarrow D$ be the homotopy and let f be a primitive of ω following δ , which exists by lemma 3.1.13. For each fixed $u \in I$, restricting the defining property of f to the segment $I \times \{u\}$ shows that $t \mapsto f(t, u)$ is a primitive of ω along the path $t \mapsto \delta(t, u)$. Restricting it to $\{0\} \times I$ and to $\{1\} \times I$ shows in the same way that $u \mapsto f(0, u)$ and $u \mapsto f(1, u)$ are primitives of ω along the paths $u \mapsto \delta(0, u)$ and $u \mapsto \delta(1, u)$. Since $\gamma_0 = \delta(\cdot, 0)$ and $\gamma_1 = \delta(\cdot, 1)$ are piecewise C^1 , theorem 3.1.9 gives

$$\int_{\gamma_0} \omega = f(1, 0) - f(0, 0) \quad \int_{\gamma_1} \omega = f(1, 1) - f(0, 1)$$

If the endpoints are fixed, the path $u \mapsto \delta(0, u)$ is constant, equal to $p = \gamma_0(0)$. If F is a primitive of ω near p , the constant function with value $F(p)$ is a primitive of ω along this constant path, so by the uniqueness part of theorem 3.1.9 the function $u \mapsto f(0, u)$ is constant and $f(0, 0) = f(0, 1)$. The path $u \mapsto \delta(1, u)$ is constant too, equal to $\gamma_0(1)$, so the same reasoning makes $u \mapsto f(1, u)$ constant and $f(1, 0) = f(1, 1)$. Hence the two integrals agree.

If instead γ_0 and γ_1 are closed and δ is a homotopy of closed paths, then $\delta(0, u) = \delta(1, u)$ for every u , so $u \mapsto f(0, u)$ and $u \mapsto f(1, u)$ are primitives of ω along the same path. By the uniqueness part of theorem 3.1.9 there is a constant c with $f(1, u) - f(0, u) = c$ for every $u \in I$. Therefore

$$\int_{\gamma_0} \omega = f(1, 0) - f(0, 0) = c = f(1, 1) - f(0, 1) = \int_{\gamma_1} \omega$$

as we sought to show.

Definition 3.1.15: Simply Connected

Let D be a set. We say that D is *simply connected* if D is connected and any closed path in D is nullhomotopic.

Simply connected sets are the exact condition needed to guarantee a closed form is a primitive:

Theorem 3.1.16: Closed Form Has Primitive In Simply Connected Set

Let ω be a closed differential form on a simply connected open set D (i.e. ω has local primitives). Then ω has a global primitive in D

Proof:

By theorem 3.1.14, we have $\int_{\gamma} \omega = 0$ for any closed paths contained in D , which by proposition 3.1.5 means that ω has a primitive in D .

This means that the closed form dz/z on any simply connected open set of $\mathbb{C}$ not containing zero has a primitive. This primitive would be a branch of $\log(z)$.

Winding Number of Closed Path

We keep bringing up the 1-form dz/z as our example of a closed non-primitive form. We may slightly expand this to a family of such 1-forms by translating:

$$\frac{dz}{z-a}$$

The reason for this particular closed (but not primitive) form is two fold:

1. they show up in the limit definition:

$$\lim_{z \rightarrow a} \frac{f(z) - f(a)}{z - a}$$

We shall in proceeding sections integrate the above value to great effect, and so understanding the behavior of the denominator will be very important.

2. We shall show that all holomorphic functions are locally equal to series of the form $\sum_{n \in \mathbb{Z}} a_n z^n$. These fail to be primitive exactly when $n = -1$, hence due to the rigidity of holomorphic functions, they only fail represent primitive forms whenever $a_{-1} \neq 0$.

The value the integral

$$\int_{\gamma} \frac{dz}{z-a}$$

Is dependnet on the path γ and where how often it “winds”

Definition 3.1.17: Winding Number Of Closed Path

Let γ be a closed path in $\mathbb{C}$ and let $a \in \mathbb{C}$ such that a is not in the image of γ . Then the *winding number* of γ with respect to a (sometimes also called the *index* of γ with respect to a) is defined to be the value of the integral

$$I(\gamma, a) := \frac{1}{2\pi i} \int_{\gamma} \frac{dz}{z-a}$$

By box 3.1.10, applied to the closed path $\gamma - a$, $I(\gamma, a)$ is an integer. To find the winding number, we must find some continuous complex-valued function $f(t)$ defined for $0 \leq t \leq 1$ st

$$e^{f(t)} = \gamma(t) - a$$

which then gives us

$$I(\gamma, a) = \frac{f(1) - f(0)}{2\pi i} = n$$

where n depends on the number of times γ winds. Some notable properties of the winding number are

1. If a is fixed, $I(\gamma, a)$ remains constant when the closed path γ is continuously deformed without passing through a .
2. If we fixed the closed path γ and vary a in the complement of the image of γ , then $I(\gamma, a)$ is a locally constant function with respect to a . Hence $I(\gamma, a)$ is a function of a which is constant in each connected component in the complement of the image of γ .
3. If $\gamma(I)$ is contained in a simply connected open set Ω where the image doesn't contain a , then $I(\gamma, a) = 0$.
4. If γ is a circle described in the positive sense (so that $I(\gamma, 0) = 1$), then $I(\gamma, a) = 0$ for a outside the circle, and $I(\gamma, a) = 1$ for a inside the circle.

The winding number has a few interesting mapping consequences

Proposition 3.1.18: Image Of Circle

Let $f : \mathbb{D}_r \rightarrow \mathbb{R}^2$ be a continuous function from $\mathbb{D}_r = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq r^2\}$ and let γ be the restriction of f to $x^2 + y^2 = r^2$. If a point a of the plane does not belong to the image of γ and if $I(\gamma, a) \neq 0$, then f takes the value of a at at least one point in the open disk $x^2 + y^2 < r^2$.

Proof:

For the sake of contradiction, let's say this was not the case. Then there is a continuous homotopy from γ to the concentric circle of radius 0. Thus, $\int_{\gamma} \frac{dz}{z-a} = 0$, contradicting our hypothesis, as we sought to show.

(Product of paths, p. 64 Cartan)

(idk if I should add information about oriented boundaries here)

3.2 Holomorphic Forms

Recall that in 1-dimensional real integration (whether it is signed definite, or unsigned definite like Lebesgue, or antiderivative giving a function), there is always a simple choice for the area of integration, with only two choices of orientation if we required such information. For example, given the usual Riemann integration, if we take

$$\int_a^b f(x) dx$$

there is only one direction we can go from a to b ³. In 2 dimensions, there are many possible paths we can take between two points. We have seen in calculus that the natural setting in which to think about integrating such paths are integration over differential forms:

$$\int_{\gamma} \omega$$

where $\omega = f_1 dx + f_2 dy$ for some (at least continuous, usually C^1) function f and a (at least piece-wise continuous, usually piece-wise C^1) path γ . We saw that under the right condition, $\int_{\gamma} \omega$ is path independent (on a connected set), and other times there exists an F such that $\omega = dF$. If f has the first property, it is said to be *conservative*, and if f has the second property it is said to have a primitive. If f is holomorphic, then we are *guaranteed* that $f dz$ ($dz = dx + idy$) is conservative, and if the domain is simply connected then it even has a primitive! In a sense, this can be seen as a feature that would be greatly lacking if it was not the case: By the FTC this is true of f is real differentiable, and so we would certainly expect a complex equivalent of the FTC. On $\mathbb{R}$, a non simply-connected set is simply not a connected set, which makes the introduction of requiring a simply connected set a novelty in the case of $\mathbb{C}$ which is not “visible” in the case of $\mathbb{R}$.

We now bring our discussion to 1-forms of the form $\omega = f(z)dz$ where f is a *holomorphic function*. The first fundamental result is that on simply connected open sets, ω has a primitive, hence generalizing the FTC for complex integration of curves. Since this is the generalization of FTC, this result can also be seen as a special case of the generalized Stokes theorem.

Theorem 3.2.1: Cauchy’s (Integral) Theorem

Let $U \subseteq \mathbb{C}$ be an open set, and let $f : U \rightarrow \mathbb{C}$ be a holomorphic function. Let $\gamma : [a, b] \rightarrow U$ be a smooth closed curve. If γ is homotopic to a constant curve, then

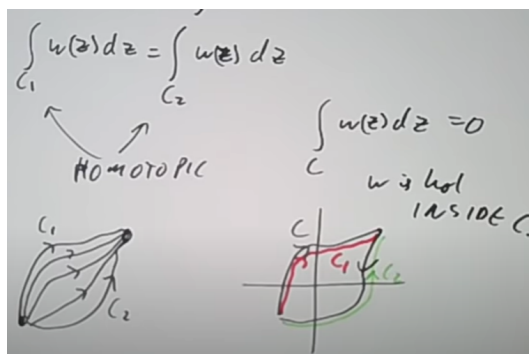
$$\int_{\gamma} f(z)dz = 0$$

Equivalently, if γ_1 and γ_2 are homotopic, then:

$$\int_{\gamma_1} f(z)dz = \int_{\gamma_2} f(z)dz$$

Another way of stating this theorem is that if $f(z)$ is holomorphic in an open set D of the complex plane, then $f(z)dz$ is a closed form in D (where $dz = dx + idy$). To see why we may either take a closed form to be zero or two paths with the same fixed point, you may stare at this image:

³Granted, we may change the speed at which we get there, but the change of variables theorem gives the value does not change



There are many proofs of this, depending on how strong of assumptions we have. Here is a quick intuitive proof if we also have access to Green's Theorem. Recall that Green's theorem states that for a closed path γ :

$$\int_{\gamma} f(x, y)dx + g(x, y)dy = \iint_{\text{int}(\gamma)} \left(-\frac{\partial f}{\partial y} + \frac{\partial g}{\partial x} \right) dxdy$$

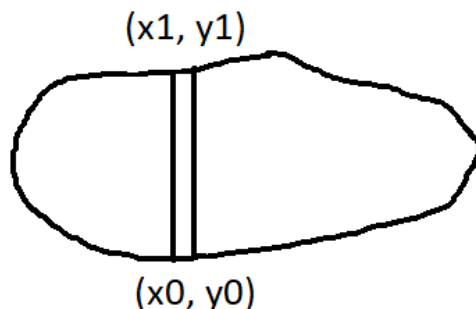
that is, we may integrate over the boundary of γ , or we integrate over the interior region. We will recall the sketch of the proof of Green's theorem on a convex region for simplicity. We may simplify further by just showing

$$\int_{\gamma} f(x, y)dx = \iint_{\text{int}(\gamma)} -\frac{\partial f}{\partial y} dxdy$$

since we would do the same proof for the g component and sum the two. Take some convex region:



Take some vertical strip, and select two points on the bottom and top border:



Now, by the fundamental theorem of calculus, we get that:

$$\int_{y=y_0}^{y_1} \frac{\partial f}{\partial y} dy = f(y_1) - f(y_0)$$

Now, take the above equation, and integrate both sides with respect to x :

$$\int \int_{\text{int}(\gamma)} \frac{\partial f}{\partial y} dy = \int_x \int_{y=y_0}^{y_1} \frac{\partial f}{\partial y} dy = - \int_{\gamma} f(x, y) dx$$

where the integral is negative because we integrated anti-clockwise, if we integrated clockwise we would get a positive. But this is the result. Then for most cases of Cauchy's theorem, we would have a finite union of convex regions, which suffices for our purposes.

Proof:

Let γ be a closed path and f a holomorphic function. With Green's Theorem, we simply compute:

$$\begin{aligned} \int_{\gamma} f(z) dz &= \int_{\gamma} (u + iv)(dx + idy) \\ &= \int_{\gamma} u dx - v dy + i \int_{\gamma} u dy + v dx \\ &= \int \int_{\text{int}(\gamma)} \left(\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} \right) dx dy + \int \int_{\text{int}(\gamma)} \left(-\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) dx dy \\ &\stackrel{!}{=} 0 \end{aligned}$$

where the $\stackrel{!}{=}$ inequality comes by using the Cauchy Riemann equations!

Using Wirtinger derivatives, we may compress the entire theorem into the following one-liner:

$$\int_{\gamma} f(z) dz = \int \int_{\text{int}(\gamma)} \frac{\partial f}{\partial \bar{z}} d\bar{z} dz = 0$$

where the first equality is green's theorem and the rest is the compressed version of what we just did.

see this video for where the proof comes from: (Richard Broached in [here](#))

The theorem can be proven by just assuming f is holomorphic without yet knowing the partials are continuous.

Lemma 3.2.2: Goursat's Lemma

Let f be holomorphic on an open set $D \subseteq \mathbb{C}$, and let $R \subseteq D$ be a closed rectangle with sides parallel to the axes. Then

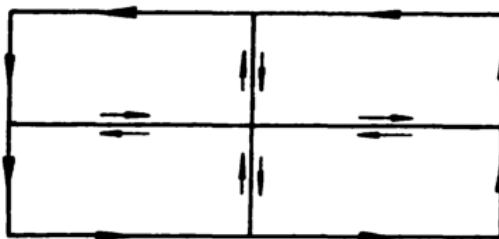
$$\int_{\partial R} f(z) dz = 0$$

where ∂R is the boundary of R , traversed once.

Proof:

For a closed rectangle $Q \subseteq D$ with sides parallel to the axes, let $p(Q) = \int_{\partial Q} f(z) dz$, with ∂Q traversed counterclockwise. Traversing a path backwards changes the sign of the integral, by the substitution $s = a + b - t$, so it suffices to show that $p(R) = 0$. Let d and L be the diameter and the perimeter of R .

Step 1: subdivision. Bisecting the sides of Q cuts it into four closed rectangles $Q_1, \dots, Q_4$, each with half the diameter and half the perimeter of Q :



Each half of the two inner segments of the subdivision is a side of exactly two of the Q_j , whose counterclockwise boundaries traverse it in opposite directions, so the two integrals over it cancel. The remaining sides of the Q_j make up ∂Q , so by proposition 3.1.1(1)

$$p(Q) = p(Q_1) + p(Q_2) + p(Q_3) + p(Q_4)$$

and $|p(Q_j)| \geq \frac{1}{4}|p(Q)|$ for at least one j .

Step 2: a nested sequence. Put $R^{(0)} = R$, and for $k \geq 1$ let $R^{(k)}$ be one of the four rectangles of Step 1 for $Q = R^{(k-1)}$ with $|p(R^{(k)})| \geq \frac{1}{4}|p(R^{(k-1)})|$. Then

$$|p(R^{(k)})| \geq 4^{-k}|p(R)|$$

and $R^{(k)}$ has diameter $2^{-k}d$ and perimeter $2^{-k}L$. The sets $R^{(k)}$ are nonempty, closed and nested, with diameters tending to 0, in the complete metric space $\mathbb{C} = \mathbb{R}^2$ ([5, Completeness of Finite Products and of $\mathbb{R}^n$]). So they have a common point z_0 ([5, Cantor's Intersection Theorem]), which lies in $R \subseteq D$.

Step 3: the estimate. Put $\epsilon(z) = f(z) - f(z_0) - f'(z_0)(z - z_0)$, a continuous function on D .

As f is complex differentiable at z_0 , for every $\eta > 0$ there is $\delta > 0$ with $|\epsilon(z)| \leq \eta|z - z_0|$ whenever $|z - z_0| < \delta$. The constant 1 and the function $z - z_0$ are the complex derivatives of the polynomials z and $(z - z_0)^2/2$, so the forms dz and $(z - z_0)dz$ have primitives on $\mathbb{C}$, and their integrals over the closed path $\partial R^{(k)}$ vanish (proposition 3.1.5). Hence

$$p(R^{(k)}) = \int_{\partial R^{(k)}} \epsilon(z) dz$$

Fix $\eta > 0$, with δ as above, and take k with $2^{-k}d < \delta$. Every point of $R^{(k)}$ lies within $2^{-k}d$ of $z_0 \in R^{(k)}$, so $|\epsilon(z)| \leq \eta 2^{-k}d$ on $\partial R^{(k)}$, and the length estimate (proposition 3.1.1(4)) gives

$$4^{-k}|p(R)| \leq |p(R^{(k)})| \leq \eta 2^{-k}d \cdot 2^{-k}L = 4^{-k}\eta dL$$

So $|p(R)| \leq \eta dL$ for every $\eta > 0$, and $p(R) = 0$, completing the proof.

Proof of theorem 3.2.1 from lemma 3.2.2:

Let $\Delta \subseteq U$ be an open disc. By lemma 3.2.2, $\int_{\partial R} f(z)dz = 0$ for every closed rectangle $R \subseteq \Delta$ with sides parallel to the axes, so $f(z)dz$ has a primitive on Δ (proposition 3.1.6). Every point of U is the centre of such a disc, so $f(z)dz$ is closed in U (definition 3.1.7). If the closed curve γ is homotopic, as a closed path, to a constant path c , then theorem 3.1.14 gives $\int_{\gamma} f(z)dz = \int_c f(z)dz = 0$, the derivative of c being 0. If γ_1 and γ_2 are homotopic, the same theorem gives $\int_{\gamma_1} f(z)dz = \int_{\gamma_2} f(z)dz$, completing the proof.

Corollary 3.2.3: Local Primitive For Holomorphic Functions

Let f be a holomorphic function in an open neighbourhood $\Omega \subseteq \mathbb{C}$. Then f has a holomorphic primitive on every open disc contained in Ω . In particular f locally has a primitive which is holomorphic, and $f(z)dz$ is a closed form.

Proof:

Let $D \subseteq \Omega$ be an open disc. By lemma 3.2.2, $\int_{\partial R} f(z)dz = 0$ for every closed rectangle $R \subseteq D$ with sides parallel to the axes. By proposition 3.1.6, the form $f(z)dz = f dx + if dy$ has a primitive F on D , so that $\frac{\partial F}{\partial x} = f$ and $\frac{\partial F}{\partial y} = if$. These partial derivatives are continuous and satisfy $\frac{\partial F}{\partial x} = -i\frac{\partial F}{\partial y}$, so F is holomorphic on D (corollary 2.2.6), with complex derivative $\frac{\partial F}{\partial x} = f$. Every point of Ω is the centre of such a disc, which gives the remaining claims.

For technical purposes and that of our next theorem, we shall temporarily require a slightly more general result to Cauchy's Theorem

Corollary 3.2.4: Cauchy Theorem On All But line of Points

Let $f(z)$ be a continuous function in an open set D which is holomorphic at every point of D except perhaps at the points of a line Δ parallel to the real axis. Then the form $f(z)dz$ is closed. In particular, if f is holomorphic at any point of D except perhaps at some isolated points, then the form $f(z)dz$ is closed

Proof:

This is simply an epsilon argument bounding away the bad points with a negligible extra area.

We want to show that $\int_{\gamma} f(z)dz$ is zero for the boundary γ of any rectangle contained in D . This is obvious if the rectangle does not intersect the line Δ , so suppose it has a side contained in Δ . Let $u, u+a, u+ib, u+a+ib$ be the four corners of the rectangle, and $u, u+a$ being on the line Δ ; a and b are real and without loss of generality we may assume $b > 0$. Let $R(\epsilon)$ be the rectangle with corners

$$u + i\epsilon, u + a + i\epsilon, u + ib, u + a + ib$$

with $\epsilon > 0$. The integral $\int f(z)dz$ is zero on the boundary of $R(\epsilon)$, and as $\epsilon \rightarrow 0$, this integral tends to the integral round the boundary γ of the rectangle R . Certainly the limit passes through since we are dealing with compact sets, and hence $\int_{\gamma} f(z)dz = 0$.

Finally, let's say Δ intersects the rectangle without containing the entire boundary (i.e. it intersects it at two points). Then the line Δ splits R into two rectangles R', R'' , and we may repeat the previous argument to show that $\int f(z)dz$ is zero around the boundaries of both R' and R'' , and so the sum of these rectangles is equal to the integral $\int_{\gamma} f(z)dz$, completing the proof.

This next theorem combines many results of holomorphic functions that at first seems to state a fact that should be generally true, but is in fact quite particular to our current build-up:

Theorem 3.2.5: Cauchy's Integral Formula

Let f be a holomorphic function in an open set Ω . Let $a \in \Omega$ and let γ be a closed path of Ω which does not pass through a and which is homotopic to a point in Ω . Then

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z-a} dz = I(\gamma, a) f(a)$$

where $I(\gamma, a)$ denotes the winding number of the closed path γ with respect to a . In particular, if γ winds around a once, then:

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z-a} dz = f(a)$$

Proof:

Let $g : \Omega \rightarrow \mathbb{C}$ be the function in Ω defined by

$$g(z) = \begin{cases} \frac{f(z)-f(a)}{z-a} & z \neq a \\ f'(a) & z = a \end{cases}$$

which is continuous by the definition of the derivative. Then g is holomorphic at any point of Ω except the point a . By corollary 3.2.4 we thus have for any path γ that is nullhomotopic:

$$\int_{\gamma} g(z)dz = \int_{\gamma} \frac{f(z)-f(a)}{z-a} dz = \int_{\gamma} \frac{f(z)}{z-a} dz - \int_{\gamma} \frac{f(a)}{z-a} dz = 0$$

which implies:

$$\int_{\gamma} \frac{f(z)}{z-a} = \int_{\gamma} \frac{f(a)}{z-a}$$

For the right hand side, by definition:

$$\int_{\gamma} \frac{f(a)}{z-a} dz = f(a) \int_{\gamma} \frac{1}{z-a} dz = 2\pi i I(\gamma, a) f(a)$$

Thus, substituting back in we get and moving terms around we get:

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z-a} dz = I(\gamma, a) f(a)$$

If γ winds around a once, then:

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z-a} dz = f(a)$$

completing the proof.

Some important observations are in order:

1. There is no analogy for this in $\mathbb{R}$. Any path in $\mathbb{R}$ that is closed will cannot go around the singularity $x-t$. Even if we tried to fix this somehow, we would get $0 = \int_0^0 f(t)/(x-t) dt = f(x)$, which is simply not true, there are many smooth non-constant functions.
2. We cannot generalize this for $\mathbb{R}^n$ differentiation since we cannot divide ($\mathbb{R}^n$ is not a field⁴). In fact, the only complete division algebras are $\mathbb{R}, \mathbb{C}, \mathbb{H}$, and $\mathbb{O}$, and only $\mathbb{R}$ and $\mathbb{C}$ are fields, hence Cauchy's Integral theorem is a particularity of the structure of $\mathbb{C}$ and of holomorphic functions (as we used the holomorphicity of f).
3. Notice how γ can be *any* path as long as it's nullhomotopic, while $f(a)$ is fixed. This shows the rigidity the global nature of holomorphic functions!
4. If $g(z) = \frac{f(z)}{z-z_0}$ with f holomorphic on Ω , and γ is as in the theorem with $I(\gamma, z_0) = 1$, then we can immediately evaluate:

$$\int_{\gamma} g(z) dz = 2\pi i f(z_0)$$

We shall return to this with the residue theorem (theorem 3.5.4)

The key result we may conclude with this theorem is that in an open disc, any holomorphic function can be expanded as a power series

Theorem 3.2.6: Taylor Expansion Of Holomorphic Function

Let $f(z)$ be a holomorphic function in the open disc $|z| < \rho$. Then f can be expressed as a power series in this disk. In particular, there exists a power series $p(x) = \sum_i^{\infty} a_n x^n$ whose radius of convergence is $r \geq \rho$ and whose sum $p(z)$ is equal to $f(z)$ for $|z| < \rho$.

⁴In particular, there is not field extension from $\mathbb{R}$ that has degree greater than 2

Proof:

Let $r < \rho$. We shall find a power series which uniformly convergent to $f(z)$ for $|z| \leq r$. By the uniqueness of the power series expansion, this power series is independent of our choice of r .

Choose an r_0 such that $r < r_0 < \rho$. Then by theorem 3.2.5, by taking γ to be the radius of the circle r_0 centered at 0 going counter-clockwise (i.e. in the positive direction), then

$$f(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(t)}{t-z} dt$$

for $|z| \leq r$. The function $1/(t-z)$ can be expanded to a series for $|z| < |t|$, in particular

$$\frac{1}{t-z} = \frac{1/t}{1-z/t} = \frac{1}{t} \left(1 + \frac{z}{t} + \frac{z^2}{t^2} + \cdots + \frac{z^n}{t^n} + \cdots \right)$$

Thus, we have

$$f(z) = \frac{1}{2\pi i} \int_{\gamma} \sum_{n=0}^{\infty} z^n \frac{f(t)}{t^{n+1}} dt$$

The series converges uniformly for $|z| \leq r$ and $|t| = r_0$. Thus, we may integrate term by term and we obtain a uniformly convergent series for $|z| \leq r$:

$$\begin{aligned} f(z) &= \int_{\gamma} \frac{1}{2\pi i} \sum_{n=0}^{\infty} z^n \frac{f(t)}{t^{n+1}} dt \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \int_{\gamma} \frac{f(t)}{t^{n+1}} dt \right) z^n \\ &= \sum_{n=0}^{\infty} a_n z^n \end{aligned}$$

where the coefficient are given by the integral formula

$$a_n = \frac{1}{2\pi i} \int_{|t|=r_0} \frac{f(t)}{t^{n+1}} dt$$

as we sought to show.

Corollary 3.2.7: Holomorphic, Then Analytic

Let f be a holomorphic function on a domain Ω . Then f is analytic

Proof:

For any point $a \in \Omega$ of the domain of the holomorphic function f , by theorem 3.2.6 we may write f as a power series, and hence it is locally a power series, and hence it is analytic.

Corollary 3.2.8: Entire, Then Power Series

Let f be holomorphic on $\mathbb{C}$. Then f is a power series.

Proof:

In theorem 3.2.6, we may take $r \in \mathbb{R}_+$ since $\rho = \infty$. Note that we could have established this as well by using analytic continuity.

As a consequence, every holomorphic function is smooth, and every derivative is itself holomorphic. Comparing the two ways of writing the Taylor coefficients about z_0 , through derivatives and through the Cauchy integral, gives an integral formula for every derivative.

Corollary 3.2.9: Cauchy Integral Formula for Derivatives

Let f be holomorphic in an open set containing the closed disc $|t - z_0| \leq r_0$. Then f is infinitely complex-differentiable, and for every $n \geq 0$,

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{|t-z_0|=r_0} \frac{f(t)}{(t-z_0)^{n+1}} dt$$

the circle being traversed once in the positive direction.

Proof:

By theorem 3.2.6, applied at the centre z_0 , the function f is analytic, hence infinitely differentiable, and its Taylor coefficients about z_0 are $a_n = f^{(n)}(z_0)/n!$. That same theorem also gives these coefficients as the integrals

$$a_n = \frac{1}{2\pi i} \int_{|t-z_0|=r_0} \frac{f(t)}{(t-z_0)^{n+1}} dt$$

Equating the two expressions for a_n and multiplying by $n!$ gives the formula.

Using the fact that holomorphic functions must be analytic, we can prove the converse of Cauchy's theorem, showing us that all closed forms on $\Omega \subseteq \mathbb{C}$ are holomorphic:

Theorem 3.2.10: Morera's Theorem

Let $f(z)$ be a continuous function in an open (connected) set $\Omega \subseteq \mathbb{C}$. If the differential form $f(z)dz$ is closed, then the function $f(z)$ is holomorphic in D

Proof:

$f(z)dz$ has a holomorphic primitive g locally, and by definition $g'(z) = f(z)$. Since $f = g'$ is the derivative of a holomorphic function, it is itself holomorphic. Since Ω is connected, we see that $g' = f$ on Ω . Hence, f is holomorphic.

Thus, when Ω is also simply connected, we have characterized the 1-forms ω on $\mathbb{R}^2$ that are closed, namely they are of the form $\omega = f(z)dz$ for holomorphic $f(z)$. From this perspective, the Cauchy-

Riemann equations are a geometric property: they guarantee that if ω is exact so that there exists an f such that $df = \omega$, then f is holomorphic.

3.2.11 On Conservative Vector Fields A conservative vector field on $\mathbb{R}^2$ corresponds to a differential form $Pdx + Qdy$. It is conservative if

$$\frac{dP}{dx} = \frac{dQ}{dy}$$

which is only one of the Cauchy-Riemann conditions, as the reader may already know this is not enough to guarantee there exists an f such that $df = Pdx + Qdy$ (i.e. to guarantee exactness). However, on a *simply connected domain* by Poincaré's lemma^a guarantees these two are the same thing. Due to this, Morera's theorem without the "simply" connected condition is not a converse ($1/zdz$ being the canonical counter-example).

^asee [23]

Corollary 3.2.12: Holomorphic On All But Some Points, Then Holomorphic

Let f be continuous on D and holomorphic at all points of D except perhaps at points of some set Δ . Then f is holomorphic on all points of D .

Proof:

Without loss of generality we may suppose Δ is parallel to the real axis by rotating if necessary. Then we have by corollary 3.2.4 that $f(z)dz$ is closed. Then by Morera's theorem f is holomorphic at all points of D , completing the proof.

From this perspective, corollary 3.2.4 only seemed like a generalization, however it was important to establish for technical reasons. As a consequence, if f fails to be holomorphic at an isolated point, f cannot be continuous at that point. Let's say x is the point at which f is not holomorphic and not continuous. We shall see in proposition 3.4.6 that if f is bounded around x , then x holomorphically extends to x , and hence it has to be that f is *unbounded* at x , i.e. it has to be undefined. We shall explore the properties of such points in section 3.4.

Using corollary 3.2.12, we may holomorphically extend a function in the way that we are about to describe, this type of extension is known as *Schwarz' principle of symmetry* or *Schwarz Reflection Principal*.

Proposition 3.2.13: Schwarz Principle Of Symmetry

Let D be a non-empty, connected, open set D which is symmetric with respect to the real axis. Let D^+ be the intersection of D with the half-plane $y \geq 0$ and D^- be the intersection of D with the half-plane $y \leq 0$. Let's say $f(z)$ is continuous on D^+ , and holomorphic on the points of D^+ where $y > 0$. Then we may holomorphically extend f to all of D .

Proof:

Consider $g(z)$ defined on D^- like so

$$g(z) = \overline{f(\bar{z})}$$

By our discussion in section 2.2.1, g is holomorphic on all D^- where $y < 0$. Then the function $h(z)$ which is $f(z)$ on D^+ and $g(z)$ on D^- is continuous on D and holomorphic on all of D except perhaps the points on the real axis. Thus, by corollary 3.2.12, it is holomorphic on all of D .

Exercise 3.2.1

1. Let f be a holomorphic function defined on a simply connected open subset Ω and $f(z) \neq 0$ for all $z \in \Omega$. Show that there exists a $g(z)$ such that $f(z) = e^{g(z)}$.

Given the same condition as above, show that for all n , there is a holomorphic function g such that $f = g^n$.

3.3 Analytic Properties: Growth of Holomorphic Forms

Let f is holomorphic in an open set Ω . Then as we just saw, $f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$ converges in some ball of radius r around z_0 . Then any power series is determined by its coefficient, which can partly be studied by looking at their growth which is the goal of this section.

We know that the coefficients of this power series can be recovered by taking derivatives of f , namely

$$a_n = \frac{1}{n!} f^{(n)}(z_0)$$

We shall show two very important results:

1. that the growth of these coefficient is bounded in an important way. If you know some real analysis, then since f has been shown to be analytic, it is “obvious” since a smooth function which has bounded derivative growth rate is analytic (see, for examples, Pughs real analysis or EYNTKA real Analysis I)
2. that $f(z)$ is always the average of a the integral of a circle of points (it satisfies the man value property, the integral equivalent of Harmonic)

These are analytical results, and shall even have connection to harmonic analysis (which we shall touch on). Hence, this section can be considered analyzing holomorphic functions from an analysis lens. We start by how the coefficients of the Taylor series are bounded:

Theorem 3.3.1: Cauchy Inequalities

Let f be a holomorphic function in Ω , let $z_0 \in \Omega$, and let $R > 0$ be such that $f(z) = \sum a_n(z - z_0)^n$ for $|z - z_0| < R$. Then for each coefficient we have the bound, for each $0 < r < R$:

$$\left| \frac{f^{(n)}(z_0)}{n!} \right| = |a_n| \leq \frac{M(r)}{r^n} \quad n \in \mathbb{N}$$

where $M(r) = \sup_{\theta} |f(z_0 + re^{i\theta})|$. These inequalities are called the *Cauchy inequalities*.

Proof:

Pick $z_0 \in \Omega$ and write the Taylor series expansion at z_0

$$f(z) = \sum_n a_n(z - z_0)^n$$

valid for $|z - z_0| < R$. Then for $0 < r < R$, the input may be written as $z = z_0 + re^{i\theta}$, giving us

$$f(z_0 + re^{i\theta}) = \sum_n a_n r^n e^{in\theta} \quad (3.1)$$

Fixing r and allowing θ to vary, $g_r(\theta) = f(z_0 + re^{i\theta})$ is a periodic function with respect to θ and the right hand side converges uniformly when θ varies and r is fixed (since the circle is compact). In fact equation (3.1) is the fourier expansion of f . Since the series converges uniformly, we may integrate term by term the following usual integral from fourier analysis:

$$\begin{aligned} \frac{1}{2\pi} \int_0^{2\pi} e^{-ip\theta} f(z_0 + re^{i\theta}) d\theta &= \frac{1}{2\pi} \int_0^{2\pi} \sum_{n \geq 0} a_n r^n e^{i(n-p)\theta} d\theta \\ &= \sum_{n \geq 0} \frac{1}{2\pi} \int_0^{2\pi} a_n r^n e^{i(n-p)\theta} d\theta \quad \text{abs. convergent} \\ &= \sum_{n \geq 0} \frac{1}{2\pi} a_n r^n \int_0^{2\pi} e^{i(n-p)\theta} d\theta \\ &\stackrel{!}{=} a_p r^p \end{aligned}$$

for the $\stackrel{!}{=}$ equality on the right hand side, we see that integrating against $n \neq p$ gives 0, while integrating against $p = n$ gives 2π , which cancels with $1/2\pi$, and hence we get a single constant giving us:

$$a_p r^p = \frac{1}{2\pi} \int_0^{2\pi} e^{-ip\theta} f(z_0 + re^{i\theta}) d\theta \quad (3.2)$$

which holds for all $0 < r < R$. If $R > 1$, taking $r = 1$ gives a simple formula for the coefficients:

$$a_p = \frac{1}{2\pi} \int_0^{2\pi} e^{-ip\theta} f(z_0 + e^{i\theta}) d\theta \quad \left[\text{if } z_0 = 0: = \frac{1}{2\pi} \int_0^{2\pi} e^{-ip\theta} f(e^{i\theta}) d\theta \right]$$

The same identity also follows from the Cauchy integral formula for derivatives. For $r < R$ the closed disc $|t - z_0| \leq r$ lies in the open disc $|t - z_0| < R$, on which f is the sum of its power series,

hence holomorphic with $a_p = f^{(p)}(z_0)/p!$ by p term-wise differentiations (theorem 2.3.2). Then corollary 3.2.9 with the circle $|t - z_0| = r$, parametrized by $t = z_0 + re^{i\theta}$ with $dt = ire^{i\theta}d\theta$, gives

$$\begin{aligned} a_p &= \frac{1}{2\pi i} \int_{|t-z_0|=r} \frac{f(t)}{(t-z_0)^{p+1}} dt = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(z_0 + re^{i\theta})}{r^{p+1}e^{i(p+1)\theta}} ire^{i\theta} d\theta \\ &= \frac{1}{2\pi r^p} \int_0^{2\pi} e^{-ip\theta} f(z_0 + re^{i\theta}) d\theta \end{aligned}$$

which is (3.2). With (3.2), we can upper bound the value of a_n : Let $M(r) = \sup_{\theta} |f(z_0 + re^{i\theta})|$ where r is fixed. Then the right hand side of equation (3.2) becomes

$$|a_p r^p| = \left| \frac{1}{2\pi} \int_0^{2\pi} e^{-ip\theta} f(z_0 + re^{i\theta}) d\theta \right| \leq \frac{1}{2\pi} \int_0^{2\pi} |e^{-ip\theta}| M(r) d\theta = \frac{2\pi}{2\pi} M(r) = M(r)$$

Hence

$$|a_p| \leq \frac{M(r)}{r^p} \quad p \in \mathbb{N}, \quad 0 < r < R$$

and since $a_p = f^{(p)}(z_0)/p!$ (theorem 2.3.2), this is the claimed inequality.

Due to their importance, we shall put a box around the definition:

Definition 3.3.2: Cauchy Inequalities

Let f be a holomorphic function with $f(z) = \sum a_n(z - z_0)^n$ for $|z - z_0| < R$, and let $M(r) = \sup_{\theta} |f(z_0 + re^{i\theta})|$. Then for each coefficient we have the family of bounds:

$$|a_n| \leq \frac{M(r)}{r^n} \quad n \in \mathbb{N}, \quad 0 < r < R$$

or equivalently:

$$\left| \frac{f^{(n)}(z_0)}{n!} \right| \leq \frac{M(r)}{r^n} \quad n \in \mathbb{N}, \quad 0 < r < R$$

which vary with for all $r < R$. These inequalities are called the *Cauchy inequalities*.

The Cauchy inequalities can be used to great effect as they force the bounds on the growth of f , in particular each derivative $f^{(n)}$ is bounded in growth by values of f , and the limitation becomes “weaker” as n gets bigger. Note that if is just f smooth, it is not guaranteed that these bounds are finite, in fact if the bounds are finite as they are in Cauchy’s inequalities then a real smooth function is analytic, showing that the real limitor behind analyticity is rate of growth (see Pugh’s Real Analysis). As practice, it should be shown that if f is holomorphic in a disk $|z| < R$, and that if $|z| \leq r < R$, then:

$$|f^{(n)}(z)| \leq \frac{M(r)n!}{(R-r)^n}$$

This limitations on the growth of coefficients puts lots of important restraint on holomorphic functions, perhaps the most famous such example is the following theorem:

Theorem 3.3.3: Liouville's theorem

Let f be a holomorphic function on $\mathbb{C}$. Then if f is bounded, f is constant

Notice how this is certainly not true for $\mathbb{R}$: $\sin, \cos, e^{-1/x^2}$ are all smooth on $\mathbb{R}$ and bounded but certainly not constant. Notice further how this shows the importance of the domain being $\mathbb{C}$: e^{1/x^2} can be complexified to e^{1/z^2} which is unbounded on $\mathbb{C}$, but *is* bounded on a strip $[-i, i] \times \mathbb{R}$. We shall also find functions that are bounded on the upper half plane $\mathbb{H}$ ($\frac{1}{z+i}$ is an example), showing that it is not simply that the function is unbounded.

Proof:

Since f is holomorphic in $\mathbb{C}$, it is a power series, hence express f in terms a of a power series around 0. By the discussion above, each coefficient a_n are bounded by Cauchy's inequalities.

$$|a_n| \leq \frac{M(r)}{r^n}$$

Since f is bounded, $M(r) \leq M$ for some positive M independent of r , and so we have

$$|a_n| \leq \frac{M}{r^n}$$

for any $r \in \mathbb{R}$. But then that means means that the above inequality holds true for all r , and so it must be that $a_n = 0$. since this is true for all $n \geq 1$, we see that f is in fact constant

One famous consequence of this is that any complex polynomial must have a root

Theorem 3.3.4: D'Alembert's Theorem

Let $p(x)$ be a non-constant complex polynomial defined on $\mathbb{C}$. Then $p(x)$ has at least one root.

Proof:

For the sake of contradiction, assume $p(x)$ is non-constant and $p(z) \neq 0$ for any $z \in \mathbb{C}$. Then $1/p(z)$ is holomorphic on the whole plane. It is bounded, since

$$p(z) = a_0 + a_1 z + \cdots + a_n z^n = z^n \left(\frac{a_0}{z^n} + \frac{a_1}{z^{n-1}} + \cdots + \frac{a_{n-1}}{z} + a_n \right)$$

for $a_n \neq 0$ tends to infinity as $|z|$ tends to infinity, so there is compact disc outside of which $|1/p(z)|$ is bounded. On the other hand $|1/p(z)|$ is bounded inside the compact disc since it is a continuous function. Hence $1/p(z)$ is bounded in the whole plane and hence is constant by Liouville's theorem. But that implies $p(z)$ is constant, contradicting our hypothesis, and completing the proof.

Geometrically, this means that all complex complex polynomials are *surjective functions*, which contrasts with the real case (ex. $f(x) = x^2$).

Corollary 3.3.5: Complex Polynomials are Surjective

Let $p(z)$ be a polynomial of degree $n \geq 1$ on $\mathbb{C}$ with leading coefficient c . Then p has exactly n roots counted with multiplicity, the multiplicity of a root being its order as a zero (proposition 2.5.7), and

$$p(z) = c \prod_{k=1}^n (z - a_k)$$

where $a_1, \dots, a_n$ are the roots, each listed as many times as its multiplicity. As a consequence, for every $w \in \mathbb{C}$ the equation $p(z) = w$, that is $p(z) - w = 0$, has exactly n solutions counted with multiplicity, so p is surjective. These n solutions are distinct points unless $w = p(\zeta)$ for one of the at most $n - 1$ roots ζ of p' . In this sense p is an n -to-1 function.

Proof:

Step 1: the factorization. We argue by induction on n . By theorem 3.3.4, p has a root a . Writing $p(z) = \sum_{k=0}^n c_k z^k$ with $c_n = c$, and using $z^k - a^k = (z - a)(z^{k-1} + z^{k-2}a + \dots + a^{k-1})$ for $k \geq 1$,

$$p(z) - p(a) = \sum_{k=1}^n c_k (z^k - a^k) = (z - a)q(z)$$

where q is a polynomial of degree $n - 1$ with leading coefficient c . If $n = 1$, then $q = c$ and $p(z) = c(z - a)$. If $n \geq 2$, the induction hypothesis factors q , and hence p , in the stated form. Grouping equal factors gives $p(z) = c \prod_{j=1}^r (z - b_j)^{m_j}$ with $b_1, \dots, b_r$ distinct and $m_1 + \dots + m_r = n$. The b_j are exactly the roots of p , because a product of complex numbers vanishes only when one of its factors does. Near b_j we have $p(z) = (z - b_j)^{m_j} g_j(z)$ with $g_j(z) = c \prod_{i \neq j} (z - b_i)^{m_i}$, a polynomial with $g_j(b_j) \neq 0$, so the uniqueness in proposition 2.5.7 shows that the order of the zero b_j is m_j . Hence the multiplicities of the roots add up to n .

Step 2: the equation $p(z) = w$. Let $w \in \mathbb{C}$. As $n \geq 1$, the polynomial $p - w$ has degree n and leading coefficient c , so by Step 1 it has exactly n roots counted with multiplicity, and in particular at least one. Let ζ be such a root, of multiplicity m , and write $p(z) - w = (z - \zeta)^m g(z)$ with g a polynomial and $g(\zeta) \neq 0$, as in Step 1. The product rule gives

$$p'(z) = m(z - \zeta)^{m-1} g(z) + (z - \zeta)^m g'(z)$$

so $p'(\zeta) = g(\zeta) \neq 0$ when $m = 1$, while $p'(\zeta) = 0$ when $m \geq 2$. Thus the n roots of $p - w$ are distinct unless some solution of $p(z) = w$ is a root of p' , that is, unless $w = p(\zeta)$ for a root ζ of p' . If $n = 1$, then $p' = c$ has no roots. If $n \geq 2$, then p' has degree $n - 1 \geq 1$ and so, by Step 1, at most $n - 1$ distinct roots, which give at most $n - 1$ exceptional values of w , completing the proof.

3.3.6 On curves and compact Riemann surfaces This result has far reaching geometric consequences when we look at compact Riemann surfaces; namely any holomorphic non-constant map $f : X \rightarrow Y$ between compact Riemann surfaces must be surjective. By a result of Serre ([14]) There is an equivalence of categories between compact Riemann surfaces and smooth algebraic curves over the complex numbers, hence any non-constant smooth map between smooth algebraic curves is surjective, and with a bit of work (see [1] or [22]) this naturally generalizes to non-smooth curves over $\mathbb{C}$ (more generally over $\bar{K}$ for some field K , $\text{char}(K) \neq 0$).

Corollary 3.3.7: Derivative and Injectivity

Let f be a holomorphic function on an open set $\Omega \subseteq \mathbb{C}$ and let $z_0 \in \Omega$. Then $f'(z_0) = 0$ if and only if f is injective on no neighbourhood of z_0 .

This contrasts starkly with the real case, for example x^3 is injective but has zero-derivative at 0.

Proof:

If $f'(z_0) \neq 0$, then by the holomorphic inverse function theorem (theorem 2.2.27) f is injective on some neighbourhood of z_0 . Conversely, suppose $f'(z_0) = 0$, let W be a neighbourhood of z_0 , and let $D \subseteq W \cap \Omega$ be an open disc around z_0 . If f is constant on D , it is not injective on $D \subseteq W$. Otherwise the analytic function $f - f(z_0)$ (corollary 3.2.7) is not identically zero on the connected set D , so by theorem 2.5.3 it vanishes identically on no neighbourhood of z_0 , and proposition 2.5.7 gives $f(z) - f(z_0) = (z - z_0)^k g(z)$ near z_0 with g holomorphic and $g(z_0) \neq 0$. Here $k \geq 2$, since $k = 1$ would give $f'(z_0) = g(z_0) \neq 0$.

Choose an open disc $D' \subseteq D$ around z_0 on which g has no zeros. The holomorphic function g'/g has a holomorphic primitive ℓ on the disc D' (corollary 3.2.3), that is, $\ell' = g'/g$. Then $(ge^{-\ell})' = 0$, so $ge^{-\ell}$ is a nonzero constant (proposition 2.2.22), and after adding a constant to ℓ we have $g = e^\ell$ on D' . The function $\varphi(z) = (z - z_0)e^{\ell(z)/k}$ satisfies $\varphi^k = f - f(z_0)$ on D' , $\varphi(z_0) = 0$ and $\varphi'(z_0) = e^{\ell(z_0)/k} \neq 0$. By theorem 2.2.27, φ maps an open neighbourhood $U \subseteq D'$ of z_0 homeomorphically onto an open set $V \ni 0$. Choose $\epsilon > 0$ with $\{\zeta \mid |\zeta| < \epsilon\} \subseteq V$ and w with $0 < |w| < \epsilon^k$. By corollary 3.3.5, the equation $\zeta^k = w$ has k distinct solutions $\zeta_1, \dots, \zeta_k$, since w is not the value 0 that ζ^k takes at the only root of its derivative. Each has $|\zeta_j| = |w|^{1/k} < \epsilon$, so $\zeta_j \in V$, and the points $z_j = (\varphi|_U)^{-1}(\zeta_j)$ are $k \geq 2$ distinct points of $U \subseteq W$ with $f(z_j) = f(z_0) + w$. So f is not injective on W , as we sought to show.

Using the Cauchy inequalities, we see that there are relatively few holomorphic functions in each growth class ($O(f(n))$). The following gives two simple examples of this:

Proposition 3.3.8: Bounding Growth Of Holomorphic Function

Let $f(z)$ is an entire function (i.e. holomorphic in $\mathbb{C}$). Then:

1. If $|f(z)| < 1 + |z|^{1/2}$, f is constant
2. If $|f(z)| < 1 + |z|^n$, f is a polynomial of degree $\deg(f) \leq n$

Proof:

1. By the Taylor expansion of Holomorphic functions, let

$$f(z) = \sum a_n z^n$$

where

$$a_n = \frac{1}{2\pi i} \int_{|z|=r} \frac{f(t)}{t^{n+1}} dt$$

Since f is entire, f agrees with this power series on all of $\mathbb{C}$ (we may use analytic continuity or directly use the Taylor expansion theorem). Then, bounding the coefficients, we get:

$$\begin{aligned} |a_n| &= \left| \frac{1}{2\pi i} \int_{|z|=r} \frac{f(t)}{t^{n+1}} dt \right| \\ &\leq \frac{1}{2\pi} \int_{|z|=r} \left| \frac{f(t)}{t^{n+1}} \right| dt \\ &\leq \frac{1}{2\pi} \int_{|z|=r} \frac{1+t^{1/2}}{t^{n+1}} dt \\ &= \frac{2\pi r(1+r^{1/2})}{2\pi r^{n+1}} \\ &= \frac{1+r^{1/2}}{r^n} \end{aligned}$$

since this holds as $r \rightarrow \infty$, we see that $|a_n| = 0$, hence f is constant. Since $f(0) = a_0$, we get that $|f(0)| \leq 1$, so f is a constant function in with image in $[-1, 1]$, completing the proof.

2. By the Taylor expansion of Holomorphic functions, let

$$f(z) = \sum a_n z^n$$

where

$$a_n = \frac{1}{2\pi i} \int_{|z|=r} \frac{f(t)}{t^{n+1}} dt$$

Then

$$\begin{aligned} |a_n| &= \left| \frac{1}{2\pi i} \int_{|z|=r} \frac{f(t)}{t^{n+1}} dt \right| \\ &\leq \frac{1}{2\pi} \int_{|z|=r} \left| \frac{f(t)}{t^{n+1}} \right| dt \\ &\leq \frac{1}{2\pi} \int_{|z|=r} \frac{1+t^k}{t^{n+1}} dt \\ &= \frac{2\pi r(1+r^k)}{2\pi r^{n+1}} \\ &= \frac{1+r^k}{r^n} \end{aligned}$$

Hence, if $n > k$, we see that the limit as $r \rightarrow \infty$ goes to 0, and so we see that the Taylor expansion of $f(z)$ must be

$$f(z) = \sum_{k=0}^n a_n z^n$$

and so it's a polynomial. Since f is entire, by analytic continuity it must be that f is equal to this polynomial everywhere, hence f is a polynomial completing the proof.

Recall that holomorphic functions are harmonic, and that harmonic functions satisfy the maximum modulus principle (theorem 2.2.18) which allows us to upgrade the bounds given by the Cauchy inequality to a bound on the growth of f . If $n = 0$, then the Cauchy inequality along with the maximum modulus principle shows that:

$$|a_0| = |f(z_0)| \leq \frac{\sup_{z \in \partial \overline{D}_{z_0}} |f(z)|}{r^0} = \sup_{z \in \partial \overline{D}_{z_0}} |f(z)| = \sup_{z \in \overline{D}_{z_0}} |f(z)|$$

or written differently:

$$|f(z)| \leq \sup_{\theta} |f(z_0 + re^{i\theta})| \quad |z| \leq r < R$$

The maximum modulus principle will extend this result to the entire disk:

Corollary 3.3.9: Cauchy Inequalities And Maximum Modulus Principle

Let f be holomorphic at z_0 and $M(r) = \sup_{\theta} |f(z_0 + re^{i\theta})|$. Then

$$|f(z)| \leq M(r) \quad \forall |z| \leq r$$

Proof:

By the maximum modulus principle, the maximum of $|f(z)|$ in some compact neighbourhood is achieved on the boundary. let $f(z_0)$ be the point such that $|f(z_0)|$ is the maximum, then $|f(z_0)| = M(r)$, hence

$$|f(z)| \leq M(r) \quad |z| \leq r$$

More generally,

$$|f^{(n)}(z_0)| \leq \frac{\sup_{\theta} (f(re^{i\theta}))}{r^n} \quad n \in \mathbb{N}, \forall r < R$$

completing the proof.

3.3.1 Mean Value Property and Schwarz' Lemma

We now examine the behavior of $f(z)$ instead of the growth of its derivative, which corresponds to the case where $n = 0$. We get the following family of equalities:

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta}) d\theta \quad r < R$$

which varies for all r less than the radius of convergence: $r < R$. Since $z^0 = 1$, this can equivalently be written as:

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta}) d\theta$$

This means that $f(z_0)$ is the average (the mean value) on the any circle centered at z_0 of radius r (reflecting the harmonic property of holomorphic function discussed in section 2.2.2). Thus, if S_{z_0} is a closed disk centered at z_0 that is contained in an open set D_{z_0} centered at z_0 and f is holomorphic in the interior of S_{z_0} , the value of f at the centre of S_{z_0} is its mean value of the boundary of S_{z_0} . This is a rather powerful property, and so we distinguish it in hopes of finding it in other sorts of functions⁵

Definition 3.3.10: Mean Value Property

Let f be a complex-valued continuous function. Then f is said to respect the *mean value property* if for all $r \in \mathbb{R}$

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} f(z + re^{i\theta}) d\theta$$

As mentioned in section 2.2.2, all harmonic functions have this property, and in fact all functions satisfying the mean value property are harmonic, but not all harmonic functions are holomorphic (easy counter-example is $z \mapsto \operatorname{Re} z$). We shall explore more this generalization in chapter 7. For our purposes with holomorphic functions, the mean value property shall allow us to upgrade Cauchy's inequalities to hold true for a set of points, not just the single point at which the power series is centered. For this, we require the following key property of functions satisfying the mean value property:

Theorem 3.3.11: Maximum Modulus Principle via Averaging

Let f be a continuous complex-valued function in an open set D of the plane $\mathbb{C}$. If f has the mean value property and $|f|$ has a relative maximum at a point $a \in D$ (i.e. $|f(z)| \leq |f(a)|$ for any z sufficiently close to a), then f is constant in some neighbourhood of a

Proof:

If $f(a) = 0$, then the result is immediate, so suppose that $f(a) \neq 0$. By multiplying f by a complex constant if necessary, we can reduce the theorem to the case where $f(a)$ is real and > 0 . Define:

$$M(r) = \sup_{\theta} |f(a + re^{i\theta})|$$

by maximality hypothesis, we have that for sufficiently small $r \geq 0$, we have $M(r) \leq f(a)$. By the mean value property,

$$f(a) = \frac{1}{2\pi} \int_0^{2\pi} f(a + re^{i\theta}) d\theta \leq M(r) \tag{3.3}$$

thus $f(a) \leq M(r)$, and so $f(a) = M(r)$. Thus, the following function is always ≥ 0 for sufficiently small $|z - a| = r$

$$g(z) = \operatorname{Re}(f(a) - f(z))$$

⁵see chapter 7

and $g(z) = 0$ if and only if $f(z) = f(a)$. By equation (3.3), the mean value of $g(z)$ on the circle $|z - a| = r$ is zero. Since g is continuous and ≥ 0 , this means that g is identically zero on this circle, and so $f(z) = f(a)$ when $|z - a| = r$ when r is sufficiently small. Since this is true for all r smaller, we have that f is constant on a sufficiently small neighbourhood, completing the proof.

If D is a connected set, then we can conclude that there is no functions satisfying the mean value property that is locally constant without being constant:

Corollary 3.3.12: Harmonic and max Attained in Interior, then Constant

Let D be a bounded, connected, open subset of $\mathbb{C}$ and f be a complex-valued continuous function defined on $\overline{D}$ having the mean value property in D . Then if M is the upper bound of f and $|f(a)| = M$ at some point $a \in D$, then f is constant

Proof:

Recall by theorem 2.2.18 that $\sup_{z \in \overline{D}} |f(z)| = \sup_{z \in \partial \overline{D}} |f(z)| = M$. Let's say that there is a $a \in D$ such that $|f(a)| = M$. Then by theorem 3.3.11 f is constant in some neighbourhood of a . This set is certainly open, and by doing a limit argument we see that it is also certainly closed, and it contains a meaning it's non-empty, and so it must be all of $\overline{D}$. Hence $f(z) = f(a)$ for all $z \in \overline{D}$

As we see, the Maximum modulus principle puts more strong constraints on the growth of holomorphic functions. The following theorem is one application of this, and shows how it can even force linearity of a function in the right situation (this can be thought of as an extension of proposition 3.3.8)

Theorem 3.3.13: Schwarz' Lemma

Let f be a holomorphic function in the disk $\mathbb{D} = \{z \in \mathbb{C} \mid |z| < 1\}$.

1. If $f(0) = 0$ and $|f(z)| \leq M$ for all $z \in \mathbb{D}$, for a constant M , then $|f(z)| \leq M|z|$ for all $z \in \mathbb{D}$
2. If moreover $|f(z_0)| = M|z_0|$ for some $z_0 \neq 0$ in $\mathbb{D}$, then in fact

$$f(z) = \lambda z$$

for some λ such that $|\lambda| = M$

3. (Schwarz-Pick) If $f(\mathbb{D}) \subseteq \mathbb{D}$, then for all $z, w \in \mathbb{D}$

$$\left| \frac{f(z) - f(w)}{1 - \overline{f(w)}f(z)} \right| \leq \left| \frac{z - w}{1 - \overline{w}z} \right| \quad |f'(w)| \leq \frac{1 - |f(w)|^2}{1 - |w|^2}$$

Proof:

1. By Taylor's expansion theorem (theorem 3.2.6), $f(z) = \sum_{n=0}^{\infty} a_n z^n$ for $|z| < 1$, and the coefficient a_0 is zero since $f(0) = 0$. Thus $g(z) = \sum_{n=1}^{\infty} a_n z^{n-1}$ is holomorphic in $\mathbb{D}$, a power series with the same radius of convergence, and $g(z) = f(z)/z$ for $z \neq 0$. Since

$|f(z)| \leq M$ by assumption, we have

$$|g(z)| \leq \frac{M}{|z|} = \frac{M}{r} \quad |z| = r$$

for $0 < r < 1$. The function g is infinitely differentiable, being analytic (corollary 3.2.7), so the maximum modulus principle (corollary 2.2.19), applied on the closed disc $|z| \leq r$, shows that this bound holds for all $|z| \leq r$. Fixing $z \in \mathbb{D}$, we get $|f(z)| \leq M|z|/r$ for any r with $|z| \leq r < 1$. In the limit as $r \rightarrow 1$, we get

$$|f(z)| \leq M|z|$$

which establishes (1).

2. By (1), $|g| \leq M$ on $\mathbb{D} \setminus \{0\}$, hence on $\mathbb{D}$ by continuity, and $|g(z_0)| = M$. Let $|z_0| < r < 1$. The holomorphic function g has the mean value property (definition 3.3.10) on the disc $|z| < r$, by the $p = 0$ case of (3.2) on circles around its points that lie in $\mathbb{D}$, and the modulus of g on the closed disc $|z| \leq r$ attains its upper bound M at the interior point z_0 , so g is constant on that disc by corollary 3.3.12. This holds for every such r , so $g = \lambda$ on $\mathbb{D}$, with $|\lambda| = |g(z_0)| = M$. But then $f(z) = \lambda z$, completing the proof of (2).

3. For $a \in \mathbb{D}$ let $\varphi_a(z) = \frac{z-a}{1-\bar{a}z}$, holomorphic on $\mathbb{D}$ as $|\bar{a}z| < 1$. Expanding both sides gives

$$|1 - \bar{a}z|^2 - |z - a|^2 = (1 - |a|^2)(1 - |z|^2)$$

so φ_a maps $\mathbb{D}$ into $\mathbb{D}$. Direct computation gives $\varphi_{-a}(\varphi_a(z)) = z$, $\varphi_a(a) = 0$ and $\varphi'_a(z) = (1 - |a|^2)/(1 - \bar{a}z)^2$. Fix $w \in \mathbb{D}$ and put $h = \varphi_{f(w)} \circ f \circ \varphi_{-w}$, a holomorphic map of $\mathbb{D}$ into $\mathbb{D}$ with $h(0) = \varphi_{f(w)}(f(w)) = 0$. By (1) with $M = 1$, $|h(\zeta)| \leq |\zeta|$ for $\zeta \in \mathbb{D}$. Taking $\zeta = \varphi_w(z)$, for which $\varphi_{-w}(\zeta) = z$, gives $|\varphi_{f(w)}(f(z))| \leq |\varphi_w(z)|$, the first inequality. The bound $|h(\zeta)/\zeta| \leq 1$ gives $|h'(0)| \leq 1$, and by the chain rule

$$h'(0) = \varphi'_{f(w)}(f(w)) f'(w) \varphi'_{-w}(0) = \frac{f'(w)(1 - |w|^2)}{1 - |f(w)|^2}$$

which gives the second inequality, completing the proof.

This theorem is in fact more powerful than it first may seem. Theorem 6.1.10 in section 6.1 will show that all simply connected proper subsets of $\mathbb{C}$ are biholomorphic to the unit disk. Hence, this theorem gives us a blueprint for bounding conditions on a larger family of functions!

Exercise 3.3.1

1. Show that *not* every power series has a root
2. Let φ be a continuous function, γ an oriented boundary of a compact set K , and $D = \mathbb{C} \setminus \{\Gamma\}$ and define on D :

$$f(z) = \int_{\Gamma} \frac{\varphi(t)}{t - z} dt$$

Show that for $a \in D$, f can be expanded as a Taylor series with radius $\rho = \inf_{\gamma \in \Gamma} |\gamma - a|$, making f analytic. In this way, the above formula is a transformation from the set of continuous function to a set of analytic functions!

3.3.2 Jensen's Formula and the Borel-Carathéodory Theorem

Two further growth results relate the size of a holomorphic function on a circle to its behaviour inside. The first, Jensen's formula, ties the mean of $\log |f|$ on a circle to the zeros of f enclosed. The second, the Borel-Carathéodory theorem, bounds $|f|$ on an interior circle by the size of $\operatorname{Re} f$ on a larger one, a substitute for the maximum modulus principle when only the real part is under control.

Theorem 3.3.14: Jensen's Formula

Let f be holomorphic on a neighbourhood of the closed disc $|z| \leq R$ with $f(0) \neq 0$, and let $a_1, \dots, a_N$ be the zeros of f in $|z| < R$, repeated according to multiplicity, with none on $|z| = R$. Then

$$\log |f(0)| = - \sum_{k=1}^N \log \frac{R}{|a_k|} + \frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta$$

Proof:

First suppose f has no zeros in $|z| \leq R$. The open set where f is defined and nonzero contains the compact disc $|z| \leq R$, hence a disc $|z| < R'$ with $R' > R$, the distance from the compact disc to the closed complement being positive. Then f'/f is holomorphic on that disc and has a holomorphic primitive h there (corollary 3.2.3). Since $(fe^{-h})' = 0$ the function fe^{-h} is a nonzero constant, so after adjusting h by an additive constant $f = e^h$ is a holomorphic logarithm. Then $\log |f| = \operatorname{Re} h$ is the real part of a holomorphic function, and the mean value property of holomorphic functions (definition 3.3.10, the $n = 0$ Cauchy coefficient) gives $h(0) = \frac{1}{2\pi} \int_0^{2\pi} h(Re^{i\theta}) d\theta$. Taking real parts yields Jensen's formula with empty sum.

In general, put

$$B(z) = f(z) \prod_{k=1}^N \frac{R^2 - \overline{a_k} z}{R(z - a_k)}$$

Each factor $\frac{R^2 - \overline{a_k} z}{R(z - a_k)}$ has a pole at a_k cancelling the corresponding zero of f , and its own zero $R^2/\overline{a_k}$ lies outside the disc since $|R^2/\overline{a_k}| = R^2/|a_k| > R$. Hence B is holomorphic and nonvanishing on a neighbourhood of the closed disc. On $|z| = R$ one has $z\overline{z} = R^2$, so

$$|R^2 - \overline{a_k} z| = |z\overline{z} - \overline{a_k} z| = |z| |\overline{z} - \overline{a_k}| = R |z - a_k|$$

and therefore $|B(z)| = |f(z)|$ for $|z| = R$. Applying the zero-free case to B ,

$$\log |B(0)| = \frac{1}{2\pi} \int_0^{2\pi} \log |B(Re^{i\theta})| d\theta = \frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta$$

Finally $B(0) = f(0) \prod_k \frac{R^2}{R(-a_k)} = f(0) \prod_k \left(-\frac{R}{a_k}\right)$, so $\log |B(0)| = \log |f(0)| + \sum_k \log \frac{R}{|a_k|}$. Substituting and rearranging completes the proof.

Theorem 3.3.15: Borel-Carathéodory Theorem

Let f be holomorphic on a neighbourhood of the closed disc $|z| \leq R$, and for $0 < r < R$ set $A(R) = \sup_{|z|=R} \operatorname{Re} f(z)$. Then

$$\sup_{|z|=r} |f(z)| \leq \frac{2r}{R-r} A(R) + \frac{R+r}{R-r} |f(0)|$$

Proof:

Consider first the case $f(0) = 0$. Since e^f is holomorphic and $|e^{f(z)}| = e^{\operatorname{Re} f(z)}$, the maximum modulus principle (corollary 3.3.12) gives $|e^{f(z)}| \leq \sup_{|w|=R} |e^{f(w)}| = e^{A(R)}$, hence $\operatorname{Re} f(z) \leq A(R)$ for all $|z| \leq R$, and in particular $A := A(R) \geq \operatorname{Re} f(0) = 0$. If $A = 0$ then $\operatorname{Re} f \equiv 0$ on the disc, which forces f constant, hence $f \equiv 0$ and the bound is trivial, so assume $A > 0$. The function

$$\varphi(z) = \frac{f(z)}{2A - f(z)}$$

is holomorphic on $|z| \leq R$, because $\operatorname{Re}(2A - f) = 2A - \operatorname{Re} f \geq A > 0$ keeps the denominator away from 0, and $\varphi(0) = 0$. Writing $f = \sigma + i\tau$ with $\sigma \leq A$,

$$|f|^2 - |2A - f|^2 = (\sigma^2 + \tau^2) - ((2A - \sigma)^2 + \tau^2) = 4A(\sigma - A) \leq 0$$

so $|\varphi(z)| \leq 1$ on the disc. Applying Schwarz' lemma (theorem 3.3.13) to $z \mapsto \varphi(Rz)$ on the unit disc gives $|\varphi(z)| \leq |z|/R$. Solving $f = 2A\varphi/(1 + \varphi)$ and using $|1 + \varphi| \geq 1 - |\varphi|$,

$$|f(z)| \leq \frac{2A|\varphi(z)|}{1 - |\varphi(z)|} \leq \frac{2A(r/R)}{1 - r/R} = \frac{2Ar}{R-r} \quad (|z| = r)$$

which is the claim when $f(0) = 0$. In general apply this to $g = f - f(0)$: since $\operatorname{Re} g = \operatorname{Re} f - \operatorname{Re} f(0)$ one has $\sup_{|z|=R} \operatorname{Re} g \leq A(R) + |f(0)|$, and then

$$\sup_{|z|=r} |f| \leq |f(0)| + \sup_{|z|=r} |g| \leq |f(0)| + \frac{2r}{R-r} (A(R) + |f(0)|) = \frac{2r}{R-r} A(R) + \frac{R+r}{R-r} |f(0)|$$

completing the proof.

Combined with Cauchy's inequalities (theorem 3.3.1), the Borel-Carathéodory bound on $|f|$ over the circle $|z| = \frac{1}{2}(R+r)$ controls every derivative $f^{(n)}$ on $|z| \leq r$ in terms of $A(R)$ and $|f(0)|$. This is the form in which the theorem bounds a logarithmic derivative f'/f from a bound on $\operatorname{Re} \log f$.

3.4 Singularities and Meromorphic Forms

So far, we have been studying holomorphic functions without much concern for singularities (for example, $f(z) = 1/z$ is holomorphic except at $z = 0$ where it has a singularity). Furthermore, our integration techniques work all up to homotopy, and Cauchy's Theorem applies to simply connected domains. This leaves out many important functions, as we have shown that rational functions are (up to the roots of the denominators) holomorphic functions. As we have seen with $1/z$, the behavior

of holomorphic functions around singularities changes. When it comes to studying the singularities, Taylor series start getting limited. We thus extend the notion of Taylor series in effect in insure a good tool for studying singularities, namely we shall add negative terms to our power series to deal with poles:

Definition 3.4.1: Laurent Series

A *Laurent series* is a formal infinite series of the form:

$$\sum_{n \in \mathbb{Z}} a_n x^n$$

where n can vary accros all integers.

Any Laurent series is assocaited with two formal power series

$$\sum_{n=0}^{\infty} a_n x^n \quad \sum_{n=-1}^{-\infty} a_n x^{-n}$$

If ρ_1 was the radius of convergence of the first and $1/\rho_2$ was the radius of convergence of the second one, then the original laurent series would have an *annulus of convergence* $\rho_2 < z < \rho_1$.

Proposition 3.4.2: Laurent Series Is Holomorphic

Let f be a Laurent series convergent in $\rho_2 < z < \rho_1$. Then f is holomorphic and the laurent series converges uniformly in the anulus $\rho_2 \leq z \leq \rho_1$ and absolutely in $\rho_2 < z < \rho_1$.

Proof:

Let's say that we have convergent Laurent series:

$$f_1(z) = \sum_{n=0}^{\infty} a_n z^n \quad \text{for} \quad |z| < \rho_1 \quad (3.4)$$

$$f_2(z) = \sum_{n=-1}^{-\infty} a_n z^n \quad \text{for} \quad |z| > \rho_2 \quad (3.5)$$

We start by showing that f_2 is holomorphic with respect to z . Putting $z = \frac{1}{u}$, we get

$$g(u) = \sum_{n=0}^{\infty} a_{-n} u^n$$

which is holomorphic for $|u| < 1/\rho_2$ and it's derivative is given by

$$g'(u) = \sum_{n=0}^{\infty} n a_{-n} u^{n-1}$$

Then differentiating and using the chain rule, we get

$$f'_2(z) = \frac{-1}{z^2} g'(1/z) = \sum_{n=-1}^{-\infty} n a_n z^{n-1}$$

Thus, the series f_2 is differentiable term by term for $|z| > \rho_2$, and hence holomorphic. From now on in the proof, assume $\rho_2 < \rho_1$. Then the sum of the series

$$f(z) = \sum_{n \in \mathbb{Z}} a_n z^n$$

is holomorphic in the annulus $\rho_2 < |z| < \rho_1$ and its derivative $f'(z)$ is the sum of the series $\sum_n a_n z^{n-1}$ given by differentiating term by term. Note that Laurent series converge uniformly in the annulus $r_2 \leq |z| \leq r_1$ where

$$\rho_2 < r_2 < r_1 < \rho_1$$

Finally, for uniqueness, in the uniformly convergent part of the annulus, if we integrate

$$f(re^{i\theta}) = \sum_{n \in \mathbb{Z}} a_n r^n e^{in\theta}$$

with respect to θ , we get just like before the integral formula

$$a_n r^n = \frac{1}{2\pi} \int_0^{2\pi} e^{-in\theta} f(re^{i\theta}) d\theta$$

and thus given a function f , the coefficients a_n of the Laurent expansion (when it exists) is determined uniquely by the above relation.

Definition 3.4.3: Laurent Expansion

Let f be a function. Then f is said to have a *Laurent expansion* in the annulus $\rho_2 < |z| < \rho_1$ if there is a Laurent series $\sum_{n \in \mathbb{Z}} a_n z^n$ which converges in this annulus and that is equal to $f(z)$ at each point in the annulus.

Naturally, by our previous comment, if f is locally a Laurent series, f is holomorphic in the annulus and uniformly converges in $r_2 \leq |z| \leq r_1$ where $\rho_2 < r_2 < r_1 < \rho_1$.

Theorem 3.4.4: Laurent Expansion Of Holomorphic Functions

Let $0 \leq \rho_2 < \rho_1 \leq \infty$ and let f be holomorphic in the annulus $A = \{z \mid \rho_2 < |z| < \rho_1\}$. Then f is representable as a Laurent expansion in A :

$$f(z) = \sum_{n \in \mathbb{Z}} a_n z^n \quad a_n = \frac{1}{2\pi i} \int_{|t|=r} \frac{f(t)}{t^{n+1}} dt$$

for $z \in A$, where $\rho_2 < r < \rho_1$, the circle $|t| = r$ is described in the positive sense, and a_n does not depend on r . The series converges uniformly on every closed annulus $r_2 \leq |z| \leq r_1$ with $\rho_2 < r_2 < r_1 < \rho_1$.

Proof:

For $n \in \mathbb{Z}$, the function $t \mapsto f(t)t^{-n-1}$ is holomorphic on A , so the form $f(t)t^{-n-1}dt$ is closed in A (corollary 3.2.3). Two circles $|t| = r$ and $|t| = r'$ in A , described in the positive sense, are homotopic as closed paths in A through the circles of radii $(1-s)r + sr'$, $s \in [0, 1]$, so theorem 3.1.14 shows that a_n does not depend on r . The same argument applies to every function holomorphic on A .

Choose r_1, r_2 with $\rho_2 < r_2 < r_1 < \rho_1$, and then r'_1, r'_2 such that

$$\rho_2 < r'_2 < r_2 < r_1 < r'_1 < \rho_1$$

Let γ_1 and γ_2 be the circles with center 0 and radii r'_1 and r'_2 , in the positive direction. Fix z with $r_2 \leq |z| \leq r_1$, and let $h(t) = (f(t) - f(z))/(t - z)$ for $t \in A$, $t \neq z$, and $h(z) = f'(z)$. Near z , $f(t) = \sum_{n \geq 0} b_n(t - z)^n$ by theorem 3.2.6, so $h(t) = \sum_{n \geq 1} b_n(t - z)^{n-1}$ there, and h is holomorphic on A . By the previous paragraph, $\int_{\gamma_1} h(t)dt = \int_{\gamma_2} h(t)dt$.

For the first circle, we have $|t| = r'_1$ and $|z| \leq r_1 < r'_1$, so we can write the series expansion

$$\frac{1}{t - z} = \sum_{n=0}^{\infty} \frac{z^n}{t^{n+1}}$$

which converges uniformly when t describes the circle of center 0 and radius r'_1 , by comparison with the geometric series in r_1/r'_1 . For the second circle, we have

$$|t| = r'_2 \quad |z| \geq r_2 > r'_2$$

and so we have

$$\frac{1}{t - z} = -\frac{1}{z} \frac{1}{1 - t/z} = -\sum_{n=-1}^{-\infty} \frac{z^n}{t^{n+1}}$$

which converges uniformly when t describes the circle of radius r'_2 , by comparison with the geometric series in r'_2/r_2 . These convergences are also uniform in z on $r_2 \leq |z| \leq r_1$. On a circle $|t| = \rho$ described in the positive sense, $\int t^k dt = 0$ for $k \neq -1$, as t^k has the primitive $t^{k+1}/(k+1)$ on $\mathbb{C} \setminus \{0\}$, while $\int t^{-1} dt = \int_0^{2\pi} i d\theta = 2\pi i$. Integrating the two series term by term, which uniform convergence allows by the length estimate of proposition 3.1.1, gives

$$\frac{1}{2\pi i} \int_{\gamma_1} \frac{dt}{t - z} = 1 \quad \frac{1}{2\pi i} \int_{\gamma_2} \frac{dt}{t - z} = 0$$

As $\int_{\gamma_k} h(t)dt = \int_{\gamma_k} \frac{f(t)}{t - z} dt - f(z) \int_{\gamma_k} \frac{dt}{t - z}$, the equality of the integrals of h becomes

$$f(z) = \frac{1}{2\pi i} \int_{\gamma_1} \frac{f(t)}{t - z} dt - \frac{1}{2\pi i} \int_{\gamma_2} \frac{f(t)}{t - z} dt$$

Replacing $\frac{1}{t - z}$ in the two integrals by the two series and integrating term by term again, we get

$$\frac{1}{2\pi i} \int_{\gamma_1} \frac{f(t)}{t - z} dt = \sum_{n=0}^{\infty} a_n z^n \quad - \frac{1}{2\pi i} \int_{\gamma_2} \frac{f(t)}{t - z} dt = \sum_{n=-1}^{-\infty} a_n z^n$$

with $a_n = \frac{1}{2\pi i} \int_{\gamma_1} \frac{f(t)}{t^{n+1}} dt$ for $n \geq 0$ and $a_n = \frac{1}{2\pi i} \int_{\gamma_2} \frac{f(t)}{t^{n+1}} dt$ for $n < 0$, which are the coefficients of the statement by the first paragraph. Combining these two, we get

$$f(z) = \sum_{n \in \mathbb{Z}} a_n z^n$$

for $r_2 \leq |z| \leq r_1$, the two series converging uniformly there, as the bounds on their terms do not depend on z . Every point of A lies in such a closed annulus, completing the proof.

Proposition 3.4.5: Holomorphic Function Decomposition In Annulus

Let f be a holomorphic function in the annulus $\rho_2 < |z| < \rho_1$. Then there exists holomorphic functions $f_1(z)$ in the disk $|z| < \rho_1$ and a holomorphic function $f_2(z)$ for $|z| > \rho_2$ such that

$$f(z) = f_1(z) + f_2(z)$$

Furthermore, this decomposition is unique if f_2 tends to 0 as $|z|$ tends to infinity.

Proof:

Let $f(z) = \sum_{n \in \mathbb{Z}} a_n z^n$ be the Laurent expansion of f and let

$$f_1(z) = \sum_{n=0}^{\infty} a_n z^n \quad f_2(z) = \sum_{n=-1}^{-\infty} a_n z^n$$

Certainly we have found the desired decomposition and $|f_2(z)|$ tends to zero as $|z|$ tends to infinity. Now, suppose that

$$f(z) = g_1(z) + g_2(z)$$

is another such decomposition where g_2 decays to 0 as $|z| \rightarrow \infty$. We will show that $f_1 = g_1$ and $f_2 = g_2$. Let h be the holomorphic function which is equal to $f_1 - g_1$ for $|z| < \rho_1$ and equal to $g_2 - f_2$ for $|z| > \rho_2$. This function is holomorphic on $\mathbb{C}$ and tends to 0 as $|z| \rightarrow \infty$. By the maximum modulus principle or Liouville's theorem, the function h must be identically 0, but then we get $f_1 = g_1$ and $f_2 = g_2$, showing uniqueness and completing the proof.

An immediate use of Laurent expansion is showing that e^{-1/z^2} is *not* analytic. Indeed, the Laurent series converges everywhere except $z = 0$. There is this nice visual from wikipedia for this:

https://en.wikipedia.org/wiki/Laurent_series#/media/File:Expinvsq_lau_GIF.gif

Note that by using the above decomposition, the Cauchy inequalities expand to the Laurent expansion, namely

$$|a_n| \leq \frac{M(r)}{r^n} \quad n \in \mathbb{N}$$

3.4.1 Study of Singularities

So why did we just make sure the Laurent series is well-defined? The power of Laurent series is that we can expand Laurent series around singularities. For example, we cannot center a power

series around 0 for $1/z$, but we can “center” a Laurent series around 0. Let’s now use our new local expansion of f in terms of Laurent series to study singularities. Let f be a holomorphic on the punctured disk $\mathbb{D}$ punctured at 0. When can f be holomorphically extended so that f is holomorphic on all of $\mathbb{D}$? In $\mathbb{R}^2$, it is certainly the case that there exists smooth function that cannot be extended, consider $\|x\|$ on the punctured disk. Such an extension is certainly unique (by analytic continuation, or simply because any continuous function is uniquely extended to its boundary), and it certainly has to be bounded in a neighbourhood of the puncture. For complex function’s being bounded is actually the exact condition needed for the extension to be holomorphic:

Proposition 3.4.6: Riemann Removable Singularity Theorem

Let f be a holomorphic function on $\mathbb{D} \setminus \{0\}$. Then f extends holomorphically to 0 if and only if f is bounded in some neighbourhood of 0

Proof:

If f extends holomorphically to 0, the extension is continuous at 0, so f is bounded near 0. Conversely, suppose that $|f(z)| \leq M$ for $0 < |z| < \rho$, where $0 < \rho \leq 1$. Take the Laurent expansion of the holomorphic function f on the punctured disk $0 < |z| < 1$ (theorem 3.4.4):

$$f(z) = \sum_{n \in \mathbb{Z}} a_n z^n \quad a_n = \frac{1}{2\pi i} \int_{|t|=r} \frac{f(t)}{t^{n+1}} dt$$

where r may be any number in $(0, 1)$. For $0 < r < \rho$, the length estimate of proposition 3.1.1 applied to the circle of length $2\pi r$ gives

$$|a_n| \leq \frac{1}{2\pi} \cdot \frac{M}{r^{n+1}} \cdot 2\pi r = \frac{M}{r^n}$$

For $n < 0$, the right-hand side tends to 0 as $r \rightarrow 0$, so $a_n = 0$. Thus, the Laurent expansion of f reduces to the power series $\sum_{n \geq 0} a_n z^n$, which converges at every point of $0 < |z| < 1$, so its radius of convergence is at least 1, and its sum is holomorphic on $\mathbb{D}$ (theorem 2.3.2). This sum agrees with f off 0, so it is the required extension, completing the proof.

Corollary 3.4.7: Holomorphic Cannot be Dominated

Let f, g be entire holomorphic functions. If $|f| \leq |g|$ on $\mathbb{C}$, then it must be that $f = cg$ for some constant c (in other words, there is no holomorphic functions that “dominate”).

Proof:

Consider $\frac{f}{g}$. Then $\left| \frac{f}{g} \right| = \frac{|f|}{|g|} \leq 1$. If h is entire, we may apply Liouville’s Theorem. For all points $g^{-1}(0)$, the inequality just established shows that its bounded, hence a removable discontinuity and hence we may holomorphically extend h to it. But then h is entire, and hence by Louisville theorem constant: $f/g = \lambda$ showing $f = \lambda g$, completing the proof.

If we cannot extend, we define the following:

Definition 3.4.8: Isolated Singularity

Let $f(z)$ be a holomorphic function in the punctured disk $0 < |z| < \rho$. The origin 0 is said to be an *isolated singularity* of f if the function f cannot be extended to a holomorphic function on the entire disk $|z| < \rho$.

Note that the singularity is isolated since analytic functions only have isolated singularities. Certainly, if not all negative coefficient of the Laurent expansion of f are zero, then 0 is an isolated singularity. This splits into two cases

1. There are only finitely many $n < 0$ such that $a_n \neq 0$. In this case, there is a positive n such that $z^n f(z) = g(z)$ is a holomorphic function in some neighbourhood around the origin. Thus, $f(z) = g(z)/z^n$ is *meromorphic* (a quotient of holomorphic functions) in some neighbourhood of the origin. Hence, in a neighborhood of the singularity, f is holomorphic on the Riemann sphere.
2. There are infinitely many negative integers such that $a_n \neq 0$. In this case, $f(z)$ is not a meromorphic function in a neighborhood around the singularity.

These lead to the following definition:

Definition 3.4.9: Pole And Essential Singularities

If f cannot be extended to a holomorphic function at the origin but is meromorphic, then 0 is a *pole* of f . If f is not meromorphic on $\mathbb{D}$, then 0 is an *essential singularity* of f .

Naturally, since poles are involved, this leads us to talk about the Riemann sphere, and define holomorphic functions on the Riemann sphere. The way we define a function f to be holomorphic on S^2 is how we do it for [complex] manifolds: let $f : S^2 \rightarrow \mathbb{C}$, and take the coordinate charts eliminating the north and south pole respective by:

$$z = \frac{x + iy}{1 - u} \quad z' = \frac{x - iy}{1 + u}$$

Then if $D \subseteq S^2$ is an open subset, f is holomorphic on D if and only if for any $p \in D$ that is not the north pole, f is holomorphic in coordinates z , and for any $p \in D$ that is not the south pole, f is holomorphic in coordinates z' (i.e., f represented in our two charts is holomorphic). We will say that $f : \mathbb{C} \rightarrow \mathbb{C}$ is “holomorphic at infinity” if by doing a change of variables $z' = 1/z$ and a neighborhood $|z| > r$, f is holomorphic at 0 in $|z'| < 1/r$. Similarly f is meromorphic at infinity and has an isolated essential singularity at infinity if the function $f(1/z)$ is meromorphic or has an isolated essential singularity at 0.

3.4.2 A Word on Essential Singularities

A lot of behaviour of holomorphic functions have parallels with rational polynomials. Essential singularities are one of the new properties that holomorphic functions can have: notice that singularity can have infinite order, but by theorem 2.5.3 a root cannot have infinite order. Essential singularities can be thought of as the “bad” case, or the case introduced because we are allowed to take “infinite rational polynomials”. Their behavior will be noticeably different.

One way to detect if we have an essential singularity is to check that small enough neighborhood around the origin map to small enough neighbors around infinity, where a small neighborhood of infinity means compliment large neighbors around 0. This result can be thought of as a “lack of convergence” when switching over to $z' = 1/z$ coordinates:

Theorem 3.4.10: Casorati-Weierstrass Theorem

If 0 is an isolated essential singularity of a holomorphic function f on the punctured disk $0 < |z| < \rho$, then for any $\epsilon > 0$, the image of the punctured disk $0 < |z| < \epsilon$ under f is everywhere dense in the plane $\mathbb{C}$

Note that the case where z_0 is an essential singularity is naturally reduced to the case when $z_0 = 0$ by replacing z by $z - z_0$. To check if it is at infinity we take $f(1/z)$.

Proof:

Suppose for the sake of contradiction that there exists a disc centered at a of radius $r > 0$ which is outside the image of the punctured disk $0 < |z| < \epsilon$ under f . Then

$$|f(z) - a| \geq r \quad 0 < |z| < \epsilon$$

Take the function $g(z) = \frac{1}{f(z) - a}$. Then g is holomorphic and bounded in the punctured disc $0 < |z| < \epsilon$. By proposition 3.4.6 this function can be extended to a holomorphic function on $|z| < \epsilon$. We denote the extension by the same letter, g . Re-arranging g , we get

$$f(z) = a + \frac{1}{g(z)}$$

showing that f is meromorphic on $|z| < \epsilon$, being the sum of meromorphic functions^a, which contradicts the hypothesis that 0 is an essential singularity of $f(z)$.

^aNote that $1/g(z)$ is meromorphic, as either $g(z_0) \neq 0$ giving a removable singularity or $g(z) = 0$ which is a zero of finite order, and so $1/g(z)$ is a pole of finite order

This should show why $\lim_{z \rightarrow 0} f(z)$ cannot be well-defined, even on the Riemann sphere: no matter how small we make the neighbourhood a neighbors around 0, we will never force a convergent sequence in the codomain as the image of the open neighborhood is always almost all of $\mathbb{C}$ ⁶. In fact, a stronger result known as Picard's theorem holds: the image under f of every punctured disc around an isolated essential singularity is the whole plane $\mathbb{C}$ or the plane $\mathbb{C}$ with one missing point. Its proof needs more tools, and it appears as theorem 4.4.6.

One might like to have an example of a holomorphic function with an essential singularity:

Example 3.4.11: Essential Singularity

1. Take $e^{1/z} = \sum_{n=0}^{\infty} \frac{1}{n!} \frac{1}{z^n}$ which is holomorphic in the punctured plane $z \neq 0$ and has an isolated essential singularity at the origin since the coefficient of $1/z^n$ is nonzero for all $n \geq 0$. This function never take the value 0. Any other value *is* taken for any z in $0 < |z| < \epsilon$.
2. On the Riemann sphere, we can as usual do a change of coordinates $z' = 1/z$ to bring ∞ to

⁶The following visual shows how this is not the case for poles: [demos graph](#)

0 and work with it. Then f has a pole singularity at infinity if $f(z) = \sum_n^\infty a_n z^n$ in $|R| > z$ and only finitely many coefficients are nonzero.

3. There can be more than one essential singularity, for example $f(z) = e^{1/z} + e^{\frac{1}{1-z}}$ has an essential singularity at 0 and 1.

Finally, essential singularities can also be thought of as not behaving well when looking at converging series

Proposition 3.4.12: Singularities and Convergence

Let f be holomorphic in $0 < |z - a| < \epsilon$. Then:

1. a is a removable singularity, that is, f extends to a holomorphic function on $|z - a| < \epsilon$, if and only if $\lim_{z \rightarrow a} f(z)$ exists in $\mathbb{C}$
2. a is a pole if and only if $\lim_{z \rightarrow a} |f(z)| = \infty$, that is, $\lim_{z \rightarrow a} 1/f(z) = 0$
3. a is an essential singularity if and only if neither $\lim_{z \rightarrow a} f(z)$ exists in $\mathbb{C}$ nor $\lim_{z \rightarrow a} |f(z)| = \infty$

Proof:

Poles and essential singularities are as in definition 3.4.9, with the disc $|z - a| < \epsilon$ in place of $\mathbb{D}$.

1. If f extends to a holomorphic function F on $|z - a| < \epsilon$, then F is continuous at a , so $\lim_{z \rightarrow a} f(z) = F(a)$. Conversely, suppose $\lim_{z \rightarrow a} f(z) = L \in \mathbb{C}$. Then $|f| \leq |L| + 1$ on some punctured disc $0 < |z - a| < \delta$ with $\delta \leq \epsilon$, so $w \mapsto f(a + \delta w)$ is holomorphic and bounded on $\mathbb{D} \setminus \{0\}$. By proposition 3.4.6 it extends holomorphically to $\mathbb{D}$, so f extends holomorphically to $|z - a| < \delta$, and together with f itself this is a holomorphic extension to $|z - a| < \epsilon$.
2. Suppose $|f(z)| \rightarrow \infty$ as $z \rightarrow a$, and choose $\delta \in (0, \epsilon]$ with $|f(z)| \geq 1$ for $0 < |z - a| < \delta$. Then $h = 1/f$ is holomorphic on this punctured disc and $h(z) \rightarrow 0$, so by (1) it extends to a holomorphic function H on $|z - a| < \delta$ with $H(a) = 0$. The function H is not identically zero, since h has no zeros. Thus $f = 1/H$ near a , a quotient of analytic functions (corollary 3.2.7) with denominator not identically zero, and f is holomorphic at the other points, so f is meromorphic on $|z - a| < \epsilon$. It does not extend holomorphically across a , since an extension would be bounded near a . Hence a is a pole.

Conversely, suppose a is a pole. Then near a we can write $f = g/h$ off the zeros of h , with g and h analytic on a disc around a and h not identically zero, and g is not identically zero either, as otherwise f would extend by 0. By proposition 2.5.7, applied to g and to h when they vanish at a , we can write $g = (z - a)^k g_1$ and $h = (z - a)^l h_1$ with integers $k, l \geq 0$ and g_1, h_1 analytic near a and nonzero at a . By continuity g_1 and h_1 have no zeros on some disc $|z - a| < \rho$, and there

$$f(z) = (z - a)^{k-l} \frac{g_1(z)}{h_1(z)} \quad 0 < |z - a| < \rho$$

If $k \geq l$, the right-hand side is holomorphic on $|z - a| < \rho$ and extends f across a , which is

impossible for a pole. So $l > k$, and $1/f(z) = (z - a)^{l-k}h_1(z)/g_1(z) \rightarrow 0$ as $z \rightarrow a$, that is, $|f(z)| \rightarrow \infty$.

3. By definition 3.4.8 and definition 3.4.9, if a is not a removable singularity, then it is a pole when f is meromorphic near a and an essential singularity otherwise. A removable singularity is not essential either, since a holomorphic extension F makes $f = F/1$ meromorphic near a . So a is an essential singularity exactly when it is neither removable nor a pole, which by (1) and (2) means that neither $\lim_{z \rightarrow a} f(z)$ exists in $\mathbb{C}$ nor $|f(z)| \rightarrow \infty$, as we sought to show.

3.5 Integrating Meromorphic Forms: Residue Theorem

Cauchy's theorem required that two paths be homotopic. Sometimes, we can "fix" the fact that two paths are not homotopic by extending the domain to be simply connected. By proposition 3.4.6, analytic continuity, and the fact that singularities of analytic functions are isolated, we have a generally good idea of when this is possible, that is we may always extend f to be holomorphic up to removable singularities, leaving us with poles and essential singularities. We would now like to find the integral of a function even if two paths are not homotopic, that is they are impeded by a singularity. This leads us to *Residue Theory*, the subject of most of the remainder of this chapter. Residue theory can be thought of extending the result of the integral of $1/(z - z_0)$ as far as possible:

Theorem 3.5.1: Cauchy's Theorem On An Annulus

Let $f(z)$ be holomorphic in an annulus $\rho_2 < |z| < \rho_1$ centered at the origin. Then if γ is a closed path contained in the annulus, then

$$\frac{1}{2\pi i} \int_{\gamma} f(z) dz = I(\gamma, 0) a_{-1}$$

where $I(\gamma, 0)$ is the winding number of the path γ with respect to origin 0 and a_{-1} is the coefficient of $1/z$ in the Laurent expansion of f

Proof:

f has a Laurent expansion (theorem 3.4.4), and so we may write:

$$f(z) = \frac{a_{-1}}{z} + g(z) \quad g(z) = \sum_{n \neq -1} a_n z^n$$

The image of γ is compact, so it lies in a closed annulus $r_2 \leq |z| \leq r_1$ with $\rho_2 < r_2 < r_1 < \rho_1$, on which the Laurent series converges uniformly. Each z^n with $n \neq -1$ has the primitive $z^{n+1}/(n+1)$ on $\mathbb{C} \setminus \{0\}$, so its integral over the closed path γ is 0 (proposition 3.1.5). Thus, by uniform convergence and the length estimate of proposition 3.1.1:

$$\begin{aligned} \int_{\gamma} f(z) dz &= a_{-1} \int_{\gamma} \frac{1}{z} dz + \sum_{n \neq -1} \int_{\gamma} a_n z^n dz \\ &= a_{-1} \int_{\gamma} \frac{1}{z} dz \end{aligned}$$

Thus, since $\frac{1}{2\pi i} \int_{\gamma} \frac{dz}{z}$ is the winding number $I(\gamma, 0)$ (definition 3.1.17), we get:

$$\int_{\gamma} f(z) dz = 2\pi i a_{-1} I(\gamma, 0)$$

In particular, if γ winds around once in the positive sense, we get

$$\int_{\gamma} f(z) dz = 2\pi i a_{-1}$$

completing the proof.

Note that if f is holomorphic in $|z| < \rho_1$, then $a_{-1} = 0$ (since f has a Taylor expansion), and hence this theorem properly generalizes Cauchy's Theorem.

Definition 3.5.2: Residue

Let f be a function defined on $0 < |z| < r$ with an isolated singularity at 0 (which may be either a pole or an essential singularity). When γ is a closed path in some neighbourhood of 0 which does not pass through 0, then the coefficient a_{-1} of the Laurent expansion is called the *residue* of the function f at the singular point 0, and is written as

$$\text{Res}(f, 0) := a_{-1}$$

More concretely, we can write:

$$\text{Res}(f, \infty) = \lim_{R \rightarrow \infty} \frac{-1}{2\pi i} \int_{|z|=R} f(z) dz \stackrel{s=1/z}{=} \lim_{R \rightarrow \infty} \frac{-1}{2\pi i} \int_{|s|=1/R} \frac{f(1/s)}{s^2} ds = \text{Res}\left(\frac{-f(1/z)}{z^2}, 0\right)$$

So if $f(z)$ is holomorphic for $|z| > r$ and $z = \frac{1}{z'}$, then

$$f(z) dz = -\frac{1}{z'^2} f\left(\frac{1}{z'}\right) dz'$$

Definition 3.5.3: Residue At The Point At Infinity

Let f be holomorphic in $|z| > r$. Then the residue of the point at infinity is the residue of the function $-\frac{1}{z'^2} f\left(\frac{1}{z'}\right)$ at the point $z' = 0$. If $\sum_{n \in \mathbb{Z}} a_n z^n$ is the Laurent expansion of $f(z)$ in a neighbourhood of the point at infinity, the residue of f at infinity is $-a_{-1}$, and is written as

$$\text{Res}(f, \infty) = -a_{-1}$$

see [here](#)

The following theorem generalizes theorem 3.5.1 for the general case of integrating a holomorphic functions with multiples poles or essential singularities. In it, and wherever this book speaks of the *oriented boundary* Γ of a compact set A , the set A is a region with corners in the plane or in the Riemann sphere, in the sense of [23, Region with Corners], and Γ is its boundary, each face oriented by the outward-normal-first rule of that book. For a disc, a rectangle or an annulus, this orients the outer boundary counterclockwise and an inner boundary circle clockwise.

Theorem 3.5.4: Residue Theorem

Let $\Omega \subseteq S^2$ be an open subset of the Riemann sphere and let f be a holomorphic function in Ω except perhaps at isolated points which are singularities of f . Let Γ be the oriented boundary of a compact subset $A \subseteq \Omega$ and suppose that Γ does not pass through any singularities of f or the point at infinity. Then only a finite number of singularities z_k are contained in A , and

$$\int_{\Gamma} f(z) dz = 2\pi i \left(\sum_k \text{Res}(f, z_k) \right)$$

where $\text{Res}(f, z_k)$ denotes the residue of the function f at the point z_k . The summation extends over all singularities $z_k \in A$ including the point at infinity if it qualifies

Proof:

A point of Ω at which f is holomorphic has a neighbourhood free of singularities, and a singularity has a neighbourhood containing no other singularity. So every point of the compact set A has a neighbourhood containing at most one singularity, finitely many of these neighbourhoods cover A , and A contains only finitely many singularities. None lies on Γ , so they lie in the interior of A . We shall split the proof into two cases: When the point at infinity is in A , and when it is not.

The integral theorem. If f is holomorphic on an open set $M \subseteq \mathbb{C}$ containing a compact region with corners A' , then $\int_{\partial A'} f(z) dz = 0$. Indeed f is infinitely differentiable, being analytic (corollary 3.2.7), so $f dz$ is a smooth 1-form on M , and $d(f dz) = \frac{\partial f}{\partial \bar{z}} d\bar{z} \wedge dz = 0$ since f satisfies the Cauchy-Riemann equations. Applied to the real and imaginary parts, [23, Stokes' Theorem on a Region with Corners] gives $\int_{\partial A'} f(z) dz = \int_{A'} d(f dz) = 0$.

If the point at infinity is not in A , then A is a compact set of $\mathbb{C}$. Each singular point z_k is the center of a closed disc S_k in the interior of A , and we can choose small enough radii so that these discs are disjoint (since there are finitely many of them). Let γ_k be the boundary of the disc S_k described in the positive sense and let $\delta_k = S_k \setminus \gamma_k$. Let

$$A' = A - \bigcup_k \delta_k$$

This is again a compact region with corners, which agrees with A near Γ and near each γ_k is the outside of a disc. Its oriented boundary is Γ together with the circles γ_k traversed clockwise, the outward normal of A' along γ_k pointing toward z_k . The function f is holomorphic on an open set containing A' , so the integral theorem gives

$$\int_{\Gamma} f(z) dz = \sum_k \int_{\gamma_k} f(z) dz$$

and for each integral, theorem 3.5.1 applied on a punctured disc around z_k , where the positively oriented circle γ_k has winding number 1 around z_k , gives

$$\int_{\gamma_k} f(z) dz = 2\pi i \text{Res}(f, z_k)$$

Thus:

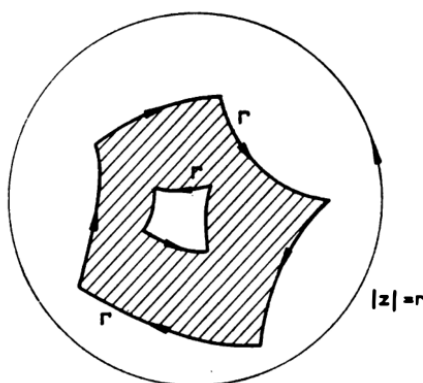
$$\int_{\Gamma} f(z)dz = \sum_k 2\pi i \text{Res}(f, z_k)$$

completing the proof in the case of A not containing the point at infinity.

Now let A contain the point at infinity, which then lies in the interior of A as it is not on Γ . Let $r \in \mathbb{R}_{>0}$ be such that the neighbourhood $\{z \mid |z| \geq r\} \cup \{\infty\}$ of the point at infinity lies in the interior of A and $f(z)$ is holomorphic in it (the point at infinity being excluded). Let

$$A' = A - \{z \in \mathbb{C} \mid |z| > r\}$$

This will leave us with a region like so where the shaded region is the compliment of A



The set A' is a compact region with corners in $\mathbb{C}$, and its oriented boundary is Γ together with the circle $|z| = r$ described in the positive sense, the outward normal of A' there pointing away from 0. Applying what we have proved in the first case to A' , we get:

$$\int_{\Gamma} f(z)dz + \int_{|z|=r} f(z)dz = 2\pi i \sum_k \text{Res}(f, z_k)$$

where the sum on the right hand side is over all singularities z_k in A *except* the point at infinity. Now, f has a Laurent expansion $\sum_n a_n z^n$ on $|z| > r'$ for some $r' < r$ (theorem 3.4.4), and theorem 3.5.1 together with the definition of the residue at infinity (definition 3.5.3) gives

$$\int_{|z|=r} f(z)dz = 2\pi i a_{-1} = -2\pi i \text{Res}(f, \infty)$$

Thus, substituting and moving around we get

$$\int_{\Gamma} f(z)dz = 2\pi i \left(\text{Res}(f, \infty) + \sum_k \text{Res}(f, z_k) \right)$$

which is exactly the relation we were seeking, completing the proof.

We shall show how to use the residue theorem to compute many integrals without the explicit need

to find anti-derivatives. Before we move on to giving many useful tools, we immediately point out on consequence of the Residue theorem when $A = S^2$ is the entire Riemann sphere.

Corollary 3.5.5: Residue Theorem On Riemann Sphere

let $A = S^2$ be the Riemann sphere.

$$\sum_k \text{Res}(f, z_k) = 0$$

Proof:

Let $\Gamma = \emptyset$ in theorem 3.5.4, so $\int_{\Gamma} f(z)dz = 0$.

For example, rational functions are holomorphic at all but a finite number of isolated singularities, and so the sum of residues of a rational function (including the residue at infinity) is zero.

3.5.6 Residue theorem on Riemann Surface These results generalize well to 1-dimensional complex manifolds, and the result on the Riemann sphere generalizes to compact Riemann surfaces.

3.5.1 Calculating Residues

Let f have a simple pole z_0 (pole of multiplicity 1) that is not at infinity. Then we may write

$$f(z) = \frac{1}{z - z_0} g(z)$$

where g is holomorphic in a neighborhood of z_0 and $g(z_0) \neq 0$. Let

$$g(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

be its Taylor expansion at z_0 . Then the Laurent expansion of f at z_0 is (by uniqueness):

$$f(z) = \frac{a_0}{z - z_0} + \sum_{n=1}^{\infty} a_n (z - z_0)^{n-1}$$

Since $a_0 = g(z_0)$, we see that $b_{-1} = g(z_0)$ at $f(z)$, $z \neq z_0$. Thus:

$$\text{Res}(f, z_0) = \lim_{\substack{z \rightarrow z_0 \\ z \neq z_0}} (z - z_0) f(z) = \lim_{\substack{z \rightarrow z_0 \\ z \neq z_0}} g(z) = g(z_0)$$

Hence, we simply have to find g and plug in z_0 . Let's now consider $f = P/Q$ where P and Q are holomorphic in a neighbourhood of z_0 and z_0 is a simple zero of Q with $P(z_0) \neq 0$. Then:

$$f(z) = \frac{1}{z - z_0} \frac{P(z)}{Q_0(z)}$$

Now $\frac{P(z)}{Q_0(z)}$ is holomorphic in a neighbourhood of z_0 , so it has power series:

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n$$

Hence:

$$f(z) = \frac{a_0}{z - z_0} + \sum_{n=1}^{\infty} a_n (z - z_0)^{n-1}$$

Hence

$$\text{Res}(f, z_0) = a_0 = \frac{P(z_0)}{Q_0'(z_0)} = \frac{P(z_0)}{Q'(z_0)}$$

Example 3.5.7: Application I

Let $f(z) = \frac{e^{iz}}{z^2 + 1}$, which has two simple poles at $z = \pm i$. By our above observations, we see that we must find $P/Q' = \frac{1}{2z} e^{iz}$ so that:

$$\text{Res}(f, i) = -\frac{i}{2e} \quad \text{Res}(f, -i) = \frac{ie}{2}$$

Let's now examine the case of multiple poles so that

$$f(z) = \frac{1}{(z - z_0)^k} g(z)$$

where $g(z)$ is holomorphic in an open neighbourhood of z_0 where $g(z_0) \neq 0$. Repeating the above computations for finding the Laurent series of f , we get:

$$f(z) = \frac{a_0}{(z - z_0)^k} + \cdots + \frac{a_{k-1}}{z - z_0} + a_k + \sum_{n=k+1}^{\infty} a_n (z - z_0)^{n-k}$$

Hence, $\text{Res}(f, z_0) = a_{k-1}$ where a_{k-1} is the coefficient of the Taylor expansion of g . Hence, to find the residue, we simply have to partly do the Taylor expansion of g at z_0 .

Example 3.5.8: Application II

Let $f(z) = \frac{e^{iz}}{z(z^2 + 1)^2}$. We will find $\text{Res}(f, i)$. Then:

$$f(z) = \frac{1}{(z - i)^2} \frac{e^{iz}}{z(z + i)^2} = \frac{1}{(z - i)^2} g(z)$$

For computational simplicity, let $t = z - i$ so that $z = i + t$ and we expand around 0 instead of i . We shall find the coefficient of t in the Taylor expansion of:

$$g(t) = \frac{e^{i(i+t)}}{(i+t)(2i+t)^2}$$

To so do so, we write down a partial Taylor expansion of the 3 terms in $h(t)$. In our case, it suffices

to find the degree 1 term:

$$\begin{aligned} e^{i(i+t)} &= e^{-1}(1+it+\cdots) \\ (i+t)^{-1} &= -i(1-it)^{-1} = -i(1+it+\cdots) \\ (2i+t)^{-2} &= -\frac{1}{4}\left(1-\frac{i}{2}t\right)^{-2} = -\frac{1}{4}(1+it+\cdots) \end{aligned}$$

Hence, multiplying all the power series we get:

$$g(t) = \frac{i}{4e}(1+3it+\cdots)$$

Thus, taking the 2nd order element we get:

$$\text{Res}(f, i) = -\frac{3}{4e}$$

3.5.2 Argument Principle and Consequences

The above examples demonstrate how important roots and poles of meromorphic functions are in finding their integrals. We can reverse this, and use integrals to find the roots and poles of a meromorphic, since the integral is nonzero if there is a root in the denominator. If we are clever with our function, the value of the integral will be the multiplicity of the roots. The whole argument turns on one local computation, isolated first as a lemma: the residue of a logarithmic derivative reads off the order of a zero or pole.

Lemma 3.5.9: Residue of the Logarithmic Derivative

Let f be meromorphic in a neighbourhood of z_0 and not identically zero, and write $f(z) = (z - z_0)^k g(z)$ with g holomorphic and nonzero at z_0 , where $k > 0$ if z_0 is a zero of order k , $k < 0$ if z_0 is a pole of order $-k$, and $k = 0$ otherwise. Then near z_0

$$\frac{f'}{f} = \frac{k}{z - z_0} + \frac{g'}{g}$$

with g'/g holomorphic at z_0 , so f'/f has at worst a simple pole at z_0 , with

$$\text{Res}\left(\frac{f'}{f}, z_0\right) = k$$

In particular the residue of f'/f at z_0 is the order of the zero of f there, or minus the order of the pole. For the log-derivative $-f'/f$ the residue is $-k$.

Proof:

Taking the logarithmic derivative of $f(z) = (z - z_0)^k g(z)$,

$$\frac{f'}{f} = \frac{k}{z - z_0} + \frac{g'}{g}$$

Since g is holomorphic and nonzero at z_0 , the term g'/g is holomorphic there, so the only contribution to the residue is the simple pole $k/(z - z_0)$, giving $\text{Res}(f'/f, z_0) = k$, as claimed.

Theorem 3.5.10: Argument Principle

Let $f(z)$ be a meromorphic function which is not constant in an open set D and let Γ be the oriented boundary of a compact set K contained in D . Suppose that the function f has no poles on Γ and does not take the value α on Γ . Then:

$$\frac{1}{2\pi i} \int_{\Gamma} \frac{f'(z)}{f(z) - \alpha} dz = Z - P \quad (3.6)$$

where Z denotes the sum of the orders of multiplicity of the roots contained in K of the equation

$$f(z) - \alpha = 0 \quad \text{i.e.} \quad f(z) = \alpha$$

and P denotes the sum of the orders of multiplicity of the poles of f at α contained in K .

Proof:

At each point z_0 inside K where $f - \alpha$ has a zero or a pole, write $(f - \alpha)(z) = (z - z_0)^k g(z)$ with g holomorphic and nonzero at z_0 , so that k is the order of the zero (if $k > 0$) or minus the order of the pole (if $k < 0$). Since $(f - \alpha)' = f'$, lemma 3.5.9 gives

$$\text{Res}\left(\frac{f'}{f - \alpha}, z_0\right) = k$$

Summing over the finitely many such z_0 in K , the total residue is $Z - P$. By the Residue theorem, for Γ as described in the theorem,

$$\frac{1}{2\pi i} \int_{\Gamma} \frac{f'(z)}{f(z) - \alpha} dz = Z - P$$

completing the proof.

If f is holomorphic, then the above equation (3.6) becomes:

$$\frac{1}{2\pi i} \int_{\Gamma} \frac{f'(z)}{f(z) - \alpha} dz = Z$$

Hence, this integral both finds zeros of holomorphic functions and their multiplicity. One important result of this theorem is the stability of the “degree” or size of fibers of meromorphic functions:

Proposition 3.5.11: Stability of Roots

Let z_0 be a root of order k of the equation $f(z) = a$, f being a non-constant holomorphic function in some neighborhood of z_0 . For any sufficiently small enough U of z_0 , and for any b sufficiently near to a and not equal to a ($b \neq a$), the equation $f(z) = b$ has exactly k simple solutions in U .

Proof:

Let Γ be a circle centered at z_0 with sufficiently small radius so that z_0 is the only solution to the equation $f(z) = a$ contained in the closed disc bounded by γ . Furthermore, suppose that the radius of γ is sufficiently small so that $f'(z) \neq 0$ at any point of the disc except the center z_0 . Consider the integral:

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z) - b} dz$$

Then the above remains constant when b varies in a connected component of the compliment of the image of γ under f . Thus, for any b sufficiently near to a , we have

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z) - b} dz = \frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z) - a} dz = k$$

Thus, the equation $f(z) = b$ has exactly k roots in the interior of γ , if each root is counted with its order of multiplicity. But for b sufficiently near to a but $\neq a$, the root of the equation $f(z) = b$ are all simple because the derivative of $f'(z) \neq 0$ at any point of z sufficiently near to z_0 and $\neq z_0$, completing the proof.

Corollary 3.5.12: Augment Argument Principle

Let $f(z)$ be a meromorphic function which is not constant in an open set D and let Γ be the oriented boundary of a compact set K contained in D . Suppose that the function f has no poles on Γ and does not take the value α on Γ . Then if $p(z)$ is a polynomial:

$$\frac{1}{2\pi i} \int_{\Gamma} \frac{f'(z)p(z)}{f(z) - \alpha} dz = \sum_{\substack{w \\ \text{zero or pole}}} n_w p(w) \quad (3.7)$$

where the sum runs over the zeros and poles w of $f - \alpha$ contained in K , and n_w is the order of the zero at w , or minus the order of the pole at w .

Proof:

Write $F(z) := \frac{f'(z)p(z)}{f(z) - \alpha}$. The function $f - \alpha$ is meromorphic in D , and it is not identically zero on any component of D meeting K : such a component would contain boundary points of K , which lie on Γ , where $f \neq \alpha$ by hypothesis. Its zeros in K are therefore isolated by proposition 2.5.7, and its poles, which are the poles of f , are isolated by the definition of a meromorphic function. At every other point of D the function F is holomorphic, being a product and quotient of holomorphic functions with nonvanishing denominator, p being entire. So F is holomorphic in D outside isolated singularities.

At each zero or pole w of $f - \alpha$ in K , write $(f - \alpha)(z) = (z - w)^{n_w} g(z)$ with g holomorphic and nonzero at w , so that n_w is the order of the zero (if $n_w > 0$) or minus the order of the pole (if $n_w < 0$). Since $(f - \alpha)' = f'$, lemma 3.5.9 applied to $f - \alpha$ gives

$$\frac{f'(z)}{f(z) - \alpha} = \frac{n_w}{z - w} + \frac{g'(z)}{g(z)}$$

with g'/g holomorphic at w . Multiplying by the polynomial p and splitting off the value $p(w)$,

$$F(z) = \frac{n_w p(w)}{z - w} + n_w \frac{p(z) - p(w)}{z - w} + p(z) \frac{g'(z)}{g(z)}$$

The difference quotient $\frac{p(z) - p(w)}{z - w}$ is again a polynomial, since $z^k - w^k = (z - w)(z^{k-1} + z^{k-2}w + \dots + w^{k-1})$ for each monomial of p . The last two terms are therefore holomorphic at w , and the residue of F at w is read off the simple pole:

$$\text{Res}(F, w) = n_w p(w)$$

Finally, Γ is the oriented boundary of the compact set $K \subseteq D$ and passes through no singularity of F : on Γ the function f has no poles and does not take the value α by hypothesis. The Residue theorem (theorem 3.5.4) applies, its statement providing also that only finitely many singularities of F lie in K , and yields

$$\frac{1}{2\pi i} \int_{\Gamma} \frac{f'(z)p(z)}{f(z) - \alpha} dz = \sum_{\substack{w \\ \text{zero or pole}}} \text{Res}(F, w) = \sum_{\substack{w \\ \text{zero or pole}}} n_w p(w)$$

the sums running over the zeros and poles of $f - \alpha$ in K , completing the proof.

Corollary 3.5.13: Argument Principle On Polynomials

Let $P(z)$ be a polynomial of degree $n \geq 1$, and let $a_1, \dots, a_n$ be its roots, each listed as many times as its multiplicity (corollary 3.3.5). Then for every $z \in \mathbb{C}$ with $P(z) \neq 0$,

$$\frac{P'(z)}{P(z)} = \sum_{k=1}^n \frac{1}{z - a_k}$$

Note that this can be proven directly with no machinery at all, however it is a good litmus test in understanding of the machinery we have built-up.

Proof:

Let $b_1, \dots, b_r$ be the distinct roots of P , with multiplicities $m_1, \dots, m_r$, so that the right-hand side equals $\sum_{j=1}^r m_j/(z - b_j)$. Consider the rational function

$$F(z) = \frac{P'(z)}{P(z)} - \sum_{j=1}^r \frac{m_j}{z - b_j}$$

which is holomorphic on $\mathbb{C} \setminus \{b_1, \dots, b_r\}$.

The singularities are removable. Fix j . By proposition 2.5.7, $P(z) = (z - b_j)^{m_j} g(z)$ near b_j with g holomorphic and $g(b_j) \neq 0$. Lemma 3.5.9, the residue lemma on which the argument principle (theorem 3.5.10) rests, gives

$$\frac{P'(z)}{P(z)} = \frac{m_j}{z - b_j} + \frac{g'(z)}{g(z)}$$

near b_j , with g'/g holomorphic at b_j . The simple pole of P'/P at b_j is therefore cancelled by the

term $m_j/(z - b_j)$: near b_j the function F equals $g'/g - \sum_{i \neq j} m_i/(z - b_i)$, which is holomorphic at b_j , so F extends holomorphically across b_j . Hence F extends to an entire function, still written F .

The extension vanishes. Write $P(z) = \sum_{k=0}^n c_k z^k$ with $c_n \neq 0$. Dividing numerator and denominator of $P'(z)/P(z)$ by z^n , the numerator $\sum_{k=1}^n k c_k z^{k-1-n}$ tends to 0 and the denominator $\sum_{k=0}^n c_k z^{k-n}$ tends to $c_n \neq 0$ as $|z| \rightarrow \infty$, so $P'(z)/P(z) \rightarrow 0$. Each $m_j/(z - b_j)$ tends to 0 as well, so $F(z) \rightarrow 0$ as $|z| \rightarrow \infty$. In particular F is bounded outside some closed disc, and it is bounded on that disc by continuity, so F is a bounded entire function and hence constant by theorem 3.3.3. The constant is $\lim_{|z| \rightarrow \infty} F(z) = 0$. Therefore $P'(z)/P(z) = \sum_{j=1}^r m_j/(z - b_j) = \sum_{k=1}^n 1/(z - a_k)$ wherever $P(z) \neq 0$, as claimed.

The following theorem gives another way of finding roots of a function in the case of one polynomial bounding another. By the harmonicity of holomorphic functions, it will suffice to check on the boundary of the region in which we are checking for zeros:

Theorem 3.5.14: Rouché's Theorem

Let f, g be holomorphic function on K with closed contour ∂K (that has no self-intersection). Then if on ∂K

$$|g(z)| < |f(z)|$$

Then f and $f + g$ have the same number of roots in K , including multiplicity.

Proof:

By doing some substitution, it is equivalent to prove that if $|f(z) - g(z)| < |f(z)|$ on ∂K , then $f(z)$ and $g(z)$ have the same number of roots in K . Since $|f(z)|$ must be strictly greater than $|f(z) - g(z)|$ on ∂K , it follows that $|f(z)| > 0$ on the boundary. Thus dividing both sides of the previous equation by $f(z)$ we get that on ∂K :

$$\left| 1 - \frac{g(z)}{f(z)} \right| < 1$$

Thus, the function $F(z) := \frac{g(z)}{f(z)}$ takes values in $|z - 1| < 1$ to the unit disk centered at 1. Consider now the argument principle:

$$\frac{1}{2\pi i} \int_{\partial K} \frac{F'}{F} dz = Z - P$$

Since $F = g/f$, we see that the number of zeros is the number of zeros of g , and the number of poles is the number of zeros of f (namely, since f and g are holomorphic in Ω). Thus, if we show this integral is zero, we have completed the proof. To show this, first notice that by definition ∂K is a disjoint union of closed curves, and hence the above integral splits into the sum of closed curve γ of ∂K . Next, we may do the substitution $w = F(z)$ to get

$$\frac{1}{2\pi i} \int_{\gamma} \frac{F'}{F} dz = \frac{1}{2\pi i} \int_{F \circ \gamma} \frac{dw}{w}$$

which is the integral of the index, hence to find the value of the above integral it suffices to find how many times $F \circ \gamma$ winds around 0. Since $F \circ \gamma$ lies in $|z - 1| < 1$, the path *never* winds

around the origin, and hence:

$$\frac{1}{2\pi i} \int_{F \circ \gamma} \frac{dw}{w} = 0$$

hence, the number of zeros of f and g are the same in K , as we sought to show.

The next result we shall prove using the Argument principle is that all holomorphic functions are in fact open maps.

Theorem 3.5.15: Complex Open Mapping Theorem

Let $f : U \rightarrow V$ be a non-constant holomorphic function in a connected open set U . Then f is an open map.

Not to be confused with the functional analysis open mapping theorem for Linear operators on Banach spaces. In particular, f need not be surjective.

Proof:

Let $z_0 \in U$, $y_0 = f(z_0)$. We shall find an open neighborhood for y_0 in $f(U)$. First, as $f(z_0) - y_0 = 0$, we have that under the right arrangement:

$$\int_{\gamma} \frac{f'(z)}{f(z) - y_0} dz = \rho \geq 1$$

Consider now:

$$g(w) = \int_{\gamma} \frac{f'(z)}{f(z) - w} dz$$

Then g is certainly holomorphic, and by proposition 3.5.11 in a small enough open neighborhood of y_0 it is constant, and hence $g(w) = \rho$, but that implies the equation $f(z_1) = w$ is satisfied in a sufficiently small open neighborhood of y_0 showing it is open, as we sought to show.

It is very informative to see that there is an alternative more elementary proof requiring only the inverse function theorem:

Proof:

It suffices to show that the image of every sufficiently small open ball is open. Let $f : U \rightarrow V$ be a non-constant holomorphic function and pick $z \in U$. Then if $f'(z) \neq 0$, then by the holomorphic inverse function (theorem 2.2.27), there exists a local inverse which shows that the image of sufficiently small open balls are open, so say $f'(z) = 0$. Since shifting is a biholomorphism, we may without loss of generality assume that $z = 0$. What we shall do is write $f(z)$ in a different form, namely:

$$f(z) = (zh(z))^n$$

where n will be the order of the root and $h(0) \neq 0$. Then the function on the inside will have an inverse in a small enough neighbourhood by the Inverse Function Theorem, and from there we shall be able to derive our desired result.

Let's say the order of the zero is n . Then we may expand the Taylor series of f and re-write it as:

$$f(z) = cz^n(1 + f_1(z))$$

where $c \neq 0$ and f_1 is holomorphic with $f_1(0) = 0$. Next, define:

$$f_2(z) = c^{1/n}(1 + f_1(z))^{1/n}$$

where we choose some branch. Then f_2 is holomorphic at the origin and, most importantly, $f_2(0) \neq 0$. With this function defined, noticed that we may write f as:

$$f(z) = (zf_2(z))^n$$

Since $(zf_2(z))' = f_2(z) + zf_2'(z)$, $zf_2(z)$ has a nonzero derivative at 0! Thus, by the Inverse Function Theorem (theorem 2.2.27), there exists a local inverse around 0, say g . Then taking $g \circ w^{1/n}$ by choosing a branch of $w^{1/n}$, we get on a sufficiently small neighbourhood:

$$g(w^{1/n}) = z$$

But this means that f is biholomorphic in some small neighborhood of 0, and hence the image of a sufficiently small open ball around 0 is open, completing the proof.

Notice how this breaks in the real case since there does not exist an inverse $z^{1/n}$.

Corollary 3.5.16: Proper Holomorphic Map Is Surjective

Let $U, V \subseteq \mathbb{C}$ denote domains (connected open sets) in $\mathbb{C}$ and let $f : U \rightarrow V$ be a holomorphic mapping. Suppose that f is *proper* ($f^{-1}(K)$ is compact for compact K). Then $f(U) = V$

Proof:

If we show that f is closed, then f maps clopen sets to clopen sets, and so $f(A) = B$. Let's say $f(x_n) \rightarrow y$. Then

$$S = \{f(x_n) \mid n \in \mathbb{N}\} \cup \{y\}$$

is compact, hence $f^{-1}(S)$ is compact, in particular

$$\{x_n \mid n \in \mathbb{N}\} \subseteq f^{-1}(S)$$

By compactness, there exists a convergent subsequence say $x_{n_k} \rightarrow x$. By continuity $f(x_{n_k}) \rightarrow f(x) = y$. Hence, the image of a closed set is closed, and by connectedness we have the proof.

Since we are working with metric spaces, it is equivalent to say that $f : X \rightarrow Y$ is proper if whenever a sequence (x_n) escapes to infinity⁷, then $(f(x_n))$ escapes to infinity. From this, we can conclude polynomials where $U = V = \mathbb{C}$ are proper maps⁸. Other examples include the trigonometric functions.

Using the open mapping theorem, we may generalize theorem 3.4.10 to open neighborhood of infinity:

⁷For every $S \subseteq X$, only finitely many elements of (x_n) are in S

⁸The choice of domain/codomain matters, for example consider $U = B_1(0)$ and $V = B_2(0)$

Proposition 3.5.17: Neighborhood At Infinity

Let $U \subseteq \mathbb{C}$ be open and connected, and let $f : U \rightarrow \mathbb{C}$ be holomorphic with a pole at $a \in \mathbb{C}$ of arbitrary order. Then for any $\epsilon > 0$ where $P = B_\epsilon(a) \setminus \{a\} \subseteq U$, there exists $r > 0$ such that

$$\{z \in \mathbb{C} \mid |z| > r\} \subseteq f(P)$$

Proof:

Let $g(z) = \frac{1}{f(z)}$. Then $\lim_{z \rightarrow a} g(z) = 0$, hence g has a removable singularity at a and hence is defined. By the open mapping theorem, any open neighborhood at infinity U of a maps to an open set $g(U) = \{1/f(z) \mid z \in U\}$. Hence, there exists a $\delta > 0$ such that

$$\{z \in \mathbb{C} \mid |z| < \delta\} \subseteq g(U)$$

Letting $r = 1/\delta$ completes the proof.

Finally, Runge's approximation theorem allows us to approximate holomorphic functions (on not necessarily simply connected domains) by rational functions.

Theorem 3.5.18: Runge's Approximation Theorem

Let $f : \Omega \rightarrow \mathbb{C}$ be a holomorphic function on an open set Ω and $K \subseteq \Omega$ a compact subset. Then if A is a set containing at least one complex number from each bounded connected component of $\mathbb{C} \setminus K$, then there exists a sequence $(r_n)_{n \in \mathbb{N}}$ of rational function which converges uniformly to f on K such that all the poles of each function r_n are in A

Proof:

If K is empty there is nothing to prove, so assume $K \neq \emptyset$. Let $\mathcal{B}$ be the set of functions $K \rightarrow \mathbb{C}$ that are uniform limits on K of rational functions whose poles all lie in $A \setminus K$. Such rational functions are continuous, hence bounded, on the compact set K , and so are their uniform limits. So $\mathcal{B}$ contains the constants and is closed under sums and products: if $g_n \rightarrow g$ and $h_n \rightarrow h$ uniformly on K , then $|g_n| \leq \sup_K |g| + 1$ on K for n large, and

$$|g_n h_n - gh| \leq |g_n| |h_n - h| + |h| |g_n - g| \leq \left(\sup_K |g| + 1 \right) |h_n - h| + \sup_K |h| |g_n - g|$$

tends to 0 uniformly on K . It is also closed under uniform limits on K : if $g_k \in \mathcal{B}$ and $g_k \rightarrow g$ uniformly, choosing for each k a rational r_k with poles in $A \setminus K$ and $\sup_K |g_k - r_k| < 1/k$ gives $r_k \rightarrow g$ uniformly. It suffices to show that $f|_K \in \mathcal{B}$.

Step 1: a Cauchy representation. Let $\eta > 0$ with 2η less than the distance from K to $\mathbb{C} \setminus \Omega$ (any $\eta > 0$ if $\Omega = \mathbb{C}$). Cover $\mathbb{C}$ by the closed squares of side η with vertices in $\eta\mathbb{Z} + i\eta\mathbb{Z}$, and let $Q_1, \dots, Q_M$ be those meeting K , finitely many as K is bounded. Each point of Q_j lies within $\sqrt{2}\eta < 2\eta$ of a point of K , so $Q_j \subseteq \Omega$. Let ∂Q_j be the boundary of Q_j traversed in the positive sense. Fix z in the interior of some Q_{j_0} , so that $z \notin Q_j$ for $j \neq j_0$. The power series of f at z (theorem 3.2.6) gives $f(\zeta) = f(z) + (\zeta - z)h(\zeta)$ near z with h holomorphic, so the function $\zeta \mapsto f(\zeta)/(\zeta - z)$, holomorphic on $\Omega \setminus \{z\}$, has residue $f(z)$ at z (definition 3.5.2), the singularity

being removable when $f(z) = 0$. The residue theorem (theorem 3.5.4) applied to the compact sets $Q_j \subseteq \Omega$ gives

$$\frac{1}{2\pi i} \int_{\partial Q_j} \frac{f(\zeta)}{\zeta - z} d\zeta = \begin{cases} f(z) & j = j_0 \\ 0 & j \neq j_0 \end{cases}$$

Summing over j , every edge shared by two of the squares appears twice with opposite orientations, and reversing the orientation of a path changes the sign of an integral, so these contributions cancel. Let Σ be the finite set of remaining oriented edges. Then

$$f(z) = \frac{1}{2\pi i} \sum_{\sigma \in \Sigma} \int_{\sigma} \frac{f(\zeta)}{\zeta - z} d\zeta \quad (3.8)$$

No edge $\sigma \in \Sigma$ meets K : a point of K on σ lies in both closed squares adjacent to σ , so both meet K , both are among the Q_j , and σ would be a shared edge. Hence the union $|\Sigma|$ of these edges is a compact subset of Ω at positive distance from K . The right-hand side of (3.8) is continuous in z on $\mathbb{C} \setminus |\Sigma|$: on $|\Sigma| \times \overline{B}$, for a closed disc $\overline{B}$ missing $|\Sigma|$, the integrand is continuous on a compact set, hence uniformly continuous ([8, Uniform Continuity Theorem]), and proposition 3.1.1(4) turns a uniformly small change of the integrand into a small change of the integral. The left-hand side f is continuous too. A point of K lies in some Q_j and is a limit of interior points of Q_j , so (3.8) holds for every $z \in K$.

Step 2: Riemann sums. Let $\sigma \in \Sigma$ run from p to q , parametrized by $\sigma(t) = p + t(q - p)$ for $t \in [0, 1]$, and put $\varphi(t, z) = f(\sigma(t))(q - p)/(\sigma(t) - z)$. This function is continuous on the compact set $[0, 1] \times K$, hence uniformly continuous ([8, Uniform Continuity Theorem]). For $N \geq 1$ the Riemann sum

$$S_N(z) = \frac{1}{N} \sum_{l=1}^N \varphi\left(\frac{l}{N}, z\right) = \sum_{l=1}^N \frac{f(\sigma(l/N))(q - p)}{N(\sigma(l/N) - z)}$$

is a linear combination of functions $1/(z - b)$ with $b \in |\Sigma| \subseteq \mathbb{C} \setminus K$, and

$$\begin{aligned} \left| \int_{\sigma} \frac{f(\zeta)}{\zeta - z} d\zeta - S_N(z) \right| &= \left| \sum_{l=1}^N \int_{(l-1)/N}^{l/N} \left(\varphi(t, z) - \varphi\left(\frac{l}{N}, z\right) \right) dt \right| \\ &\leq \sup_{|t-s| \leq 1/N, z \in K} |\varphi(t, z) - \varphi(s, z)| \end{aligned}$$

since, by proposition 3.1.1(3) and the order bound of [3, Order Bounds and Additivity over a Subdivision Point], each of the N integrals has modulus at most $1/N$ times that supremum. By uniform continuity the supremum tends to 0 as $N \rightarrow \infty$. By (3.8), f is therefore a uniform limit on K of linear combinations of functions $z \mapsto 1/(z - b)$ with $b \in \mathbb{C} \setminus K$. Since $\mathcal{B}$ contains the constants and is closed under sums, products and uniform limits, it remains to show that $z \mapsto 1/(z - b)$ lies in $\mathcal{B}$ for every $b \in \mathbb{C} \setminus K$.

Step 3: moving a pole. Let $b \in \mathbb{C} \setminus K$ with $1/(z - b) \in \mathcal{B}$, let $d_b > 0$ be the distance from b to K , and let $|b' - b| < d_b$. For $z \in K$ put $w = (b' - b)/(z - b)$, so $|w| \leq \kappa := |b' - b|/d_b < 1$. The identity $1/(1 - w) - \sum_{n=0}^N w^n = w^{N+1}/(1 - w)$ gives

$$\left| \frac{1}{z - b'} - \sum_{n=0}^N \frac{(b' - b)^n}{(z - b)^{n+1}} \right| = \left| \frac{1}{z - b} \cdot \frac{w^{N+1}}{1 - w} \right| \leq \frac{\kappa^{N+1}}{d_b(1 - \kappa)}$$

for every $z \in K$, and the bound tends to 0 as $N \rightarrow \infty$. The partial sums are polynomials in $1/(z - b)$, hence in $\mathcal{B}$, so $1/(z - b') \in \mathcal{B}$.

Step 4: every pole. Let V be a connected component of $\mathbb{C} \setminus K$, open because $\mathbb{C} \setminus K$ is open and discs are connected, and let S be the set of $b \in V$ with $1/(z - b) \in \mathcal{B}$. By Step 3, S contains with each b the disc of radius d_b around b , which lies in V , so S is open. If $b \in V$ is a limit of points $b_k \in S$, then for k large $|b_k - b| < d_b/2$ and $d_{b_k} \geq d_b - |b_k - b| > |b_k - b|$, so $b \in S$ by Step 3 applied at b_k . Thus S is open and closed in the connected set V . It is nonempty: if V is bounded, it contains a point $a \in A$, which lies outside K as $V \subseteq \mathbb{C} \setminus K$, and $1/(z - a)$ is itself rational with its pole in $A \setminus K$. If V is unbounded, let $\rho = \max_K |z|$. The set $\{w \mid |w| > \rho\}$ is connected and misses K , so it lies in one component, which is then the only unbounded one, namely V . Pick $b_0 \in V$ with $|b_0| > 2\rho$. For $z \in K$, the same identity with $w = z/b_0$, $|w| \leq 1/2$, gives

$$\left| \frac{1}{z - b_0} + \sum_{n=0}^N \frac{z^n}{b_0^{n+1}} \right| \leq \frac{1}{|b_0|} \cdot \frac{(1/2)^{N+1}}{1/2}$$

where the bound tends to 0 as $N \rightarrow \infty$, and the partial sums are polynomials, which lie in $\mathcal{B}$, so $b_0 \in S$. Hence $S = V$. Every $b \in \mathbb{C} \setminus K$ lies in some component, so $1/(z - b) \in \mathcal{B}$ for all of them, and therefore $f|_K \in \mathcal{B}$. Choosing r_n rational with poles in $A \setminus K \subseteq A$ and $\sup_K |f - r_n| < 1/n$ gives the required sequence, completing the proof.

Corollary 3.5.19: Runge's Approximation By Polynomials

In the setting of theorem 3.5.18, if $\mathbb{C} \setminus K$ is connected, for instance if K is a closed disc, then A may be taken to be empty and each r_n is a polynomial.

Proof:

The set $\mathbb{C} \setminus K$ is unbounded, as K is bounded, so when it is connected it has no bounded connected component. The empty set therefore meets every bounded component, and theorem 3.5.18 with $A = \emptyset$ gives rational functions $r_n \rightarrow f$ uniformly on K with no poles in $\mathbb{C}$. Writing such an r_n as a quotient p/q of polynomials in lowest terms, the denominator q has no root, so q is constant by theorem 3.3.4, and r_n is a polynomial, as claimed.

3.5.3 Method of Residues: Solving Real Integrals

We now introduce some integration tricks. Note that there is no general way of solving integrals, we just gained some more tools.

First type (I)

Consider an integral of the form

$$I = \int_0^{2\pi} R(\sin(t), \cos(t)) dt$$

where $R(x, y)$ is a rational function without a pole on the circle $x^2 + y^2 = 1$. Let $z = e^{it}$ so that we can write:

$$\sin(t) = \frac{e^{it} - e^{-it}}{2i} = \frac{1}{2i} \left(z - \frac{1}{z} \right) \quad \cos(t) = \frac{e^{it} + e^{-it}}{2} = \frac{1}{2} \left(z + \frac{1}{z} \right)$$

Thus:

$$\int_0^{2\pi} R(\sin(\theta), \cos \theta) d\theta = -i \int_{|z|=1} R \left(\frac{1}{2} \left(z + \frac{1}{z} \right), \frac{1}{2i} \left(z - \frac{1}{z} \right) \right) \frac{dz}{z} = 2\pi \sum_{|z|<1} \text{Res}(g(z))$$

where after our substitution we had:

$$\frac{1}{iz} R \left(\frac{1}{2i} \left(z - \frac{1}{z} \right), \frac{1}{2} \left(z + \frac{1}{z} \right) \right)$$

and so we get:

$$I = 2\pi \sum \text{Res} \left\{ \frac{1}{z} R \left(\frac{1}{2i} \left(z - \frac{1}{z} \right), \frac{1}{2} \left(z + \frac{1}{z} \right) \right) \right\}$$

Example 3.5.20: Residue Method I

Consider

$$\int_0^{2\pi} \frac{1}{a + \sin(t)} dt$$

where $a \in \mathbb{R}_{>1}$. Decomposing, we get

$$\begin{aligned} \int_0^{2\pi} \frac{1}{a + \sin t} dt &= \int_{\gamma} \frac{1}{a + \frac{1}{2i}(z - z^{-1})} \frac{dz}{z} \\ &= \int_{\gamma} \frac{2i}{z^2 + 2iz - 1} dz \end{aligned}$$

Hence, our residue in the unite circle is:

$$I = 2\pi \sum \text{Res} \frac{2i}{z^2 + 2iaz - 1}$$

The only pole containd in the unit disc is

$$z_0 = -ia + i\sqrt{a^2 - 1}$$

by factoring the denominator and evaluating it at the pole, we get that it's residue is $\frac{i}{z_0 + ia} = \frac{1}{\sqrt{a^2 - 1}}$. Thus:

$$I \int_0^{2\pi} \frac{1}{a + \sin(t)} dt = \frac{2\pi}{\sqrt{a^2 - 1}}$$

Second Type (II)

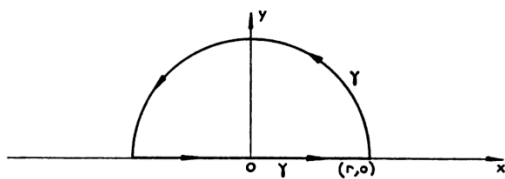
Our next type of integral will be of the form:

$$I = \int_{-\infty}^{\infty} R(x) dx \tag{3.9}$$

where R is a rational function without a real pole. Naturally, we need to assume the integral is convergent, which is equivalent (i.e. it is a necessary and sufficient condition) that the principal part of $R(x)$ at infinity is of the form $\frac{1}{x^n}$ for integer $n \in \mathbb{N}_{\geq 2}$, or stated another way if R has a zero of order at least two at $\pm\infty$. Equivalently, we must have:

$$\lim_{|x| \rightarrow \infty} xR(x) = 0$$

To calculate integrals of the form in equation (3.9), we will integrate the function $R(z)$ in the complex plane with the boundary of a half-disc centered at 0 with radius r and situated in the half-plane $y \geq 0$:



For a sufficiently large r , $R(z)$ is holomorphic on the boundary, and so by the Residue theorem $\int_{\gamma} R(z)dz$ is equal to the sum of the residues of the poles of R inside γ . We thus have the following breakdown of the integral:

$$\int_{-r}^r R(x)dx + \int_{\delta(r)} R(z)dz = 2\pi i \sum \text{Res}(R(z)) \quad (3.10)$$

where $\delta(r)$ is the rest of the unit circle without the $[-r, r]$ interval, i.e. the upper half circle centered at 0 with radius r described in the positive sense. Our goal now is to show that as $r \rightarrow \infty$, the second integral in equation (3.10) tends to 0. This will leave us with

$$\int_{-\infty}^{\infty} R(x)dx = 2\pi i \sum \text{Res}(R(z))$$

or, if we were going in the lower half plane, we would get

$$\int_{-\infty}^{\infty} R(x)dx = -2\pi i \sum \text{Res}(R(z))$$

where we sum over all the poles in the lower half plane.

Lemma 3.5.21: Residue Method Lemma I

Let $f(z)$ be a continuous function defined in the sector $\theta_1 \leq \theta \leq \theta_2$ and r, θ denote the modulus and the argument of z . Then if

$$\lim_{|z| \rightarrow \infty} zf(z) = 0 \quad (\theta_1 \leq \arg(z) \leq \theta_2)$$

Then $\int f(z)dz$ extended over the arc of the circle $|z| = r$ contained in the sector tends to 0 as $r \rightarrow \infty$.

Proof:

Let $M(r)$ be the maximum of $|f(z)|$ on $|z| = r$. Then

$$\left| \int_{\delta(r)} f(z) dz \right| \leq M(r) \int_{\delta(r)} dz = r(\theta_2 - \theta_1) \xrightarrow{r \rightarrow \infty} 0$$

since we can do this on the sector from 0 to π , we see that the sector has no effect on the integral.

Notice that we have the same result when $z \rightarrow 0$ if $zf(z) \rightarrow 0$:

Lemma 3.5.22: Residue Method Lemma II

Let $f(z)$ be a continuous function defined in the sector $\theta_1 \leq \theta \leq \theta_2$ and r, θ denote the modulus and the argument of z . Then if

$$\lim_{|z| \rightarrow 0} zf(z) = 0 \quad (\theta_1 \leq \arg(z) \leq \theta_2)$$

Then $\int f(z) dz$ extended over the arc of the circle $|z| = r$ contained in the sector tends to 0 as $r \rightarrow 0$.

Example 3.5.23: Residue Method II

Let

$$I = \int_0^\infty \frac{1}{1+x^6} dx$$

First, the domain of integration is not from $-\infty$ to ∞ , but this is fine since the integral is even, and so we have:

$$I = \frac{1}{2} \int_{-\infty}^\infty \frac{1}{1+x^6} dx$$

The complexification $1/(1+z^6)$ has 6 poles, all on the unit circle, three in the upper half plane and three in the lower. Sticking to the upper-half plane:

$$e^{i\frac{\pi}{6}} \quad e^{i\frac{3\pi}{6}} \quad e^{i\frac{5\pi}{6}}$$

Calculating the residues at each pole we get:

$$\text{Res}(f, z_1) = \lim_{z \rightarrow z_1} (z - z_1) f(z)$$

$\frac{1}{6z^5} = \frac{-z}{6}$ (since $z^5 = -1$). Thus:

$$\begin{aligned} I &= \frac{1}{2} \int_{-\infty}^\infty \frac{1}{1+x^6} dx \\ &= -\frac{\pi i}{6} \left(e^{i\frac{\pi}{6}} + e^{i\frac{3\pi}{6}} + e^{i\frac{5\pi}{6}} \right) \\ &= \frac{\pi}{6} (2 \sin(\pi/6) + 1) \\ &= \frac{\pi}{3} \end{aligned}$$

Third Type (III)

We next study integrals of the form

$$I = \int_{-\infty}^{\infty} f(x)e^{ix} dx$$

where f is holomorphic in a neighbourhood of each point of the closed half plane $y \geq 0$, except perhaps at a finite number of points. By taking the real and imaginary part, this is the same as solving:

$$I = \int_{-\infty}^{\infty} f(x) \cos(x) dx \quad I = \int_{-\infty}^{\infty} f(x) \sin(x) dx$$

Note that it is important to convert this problem into an exponential one since $\sin$ and $\cos$ are unbounded in the complex plane. We shall split up this integral into the cases where the singularities or on the real axis and when they are not on the real axis. Let's start with the case where the singularities are *not* on the real axis. Then the integral

$$\int_{-r}^r f(x)e^{ix} dx$$

is well-defined, and as $r \rightarrow \infty$, if f does not have a pole of order 2 at infinity ($zf(z) \rightarrow 0$), we naturally get

$$\int_{-\infty}^{\infty} f(x)e^{ix} dx = 2\pi i \sum_{\text{Im}(z)>0} \text{Res}(R(z)e^{iz})$$

if the integral is convergent. This time, it will not be able to take the equivalent path in the lower half plane since e^{iz} grows quickly in the lower half plane (recall $|e^{iz}| = e^{-y}$). Due to the presence of e^{iz} , we may weaken the speed of convergence of $R(z)$:

Lemma 3.5.24: Residues Method Lemma II

Let $f(z)$ be a function defined in a sector of the half-plane $y \geq 0$. If

$$\lim_{|z| \rightarrow \infty} f(z) = 0$$

Then $\int_{|z|=r} f(z)e^{iz} dz$ contained in the sector tends to 0 as $r \rightarrow \infty$.

Proof:

Let $z = re^{i\theta}$ and $M(r) = \max_{\theta \in S} |f(re^{i\theta})|$ where S is the sector in the upper half plane. Then:

$$\left| \int f(z)e^{iz} dz \right| \leq M(r) \int_0^\pi e^{-r \sin(\theta)} r d\theta$$

I claim that the integral on the right hand side is bounded by π . Indeed, first notice that

$$\int_0^\pi \int_0^\pi e^{-r \sin(\theta)} r d\theta = 2 \int_0^{\pi/2} e^{-r \sin(\theta)} r d\theta$$

Next, for $0 \leq \theta \leq \pi/2$, we have $\frac{2}{\pi} \leq \frac{\sin(\theta)}{\theta} \leq 1$. Thus:

$$\int_0^{\pi/2} e^{-r \sin(\theta)} r d\theta \leq \int_0^{\pi/2} e^{-\frac{2}{\pi} r \theta} r d\theta \leq \int_0^{\infty} e^{-\frac{2}{\pi} r \theta} r d\theta = \frac{\pi}{2}$$

Sinec $f(z) \xrightarrow{|z| \rightarrow \infty} 0$, $M(r) \rightarrow 0$, completing the proof.

Proposition 3.5.25: Evaluating Type III Integrals

If $\lim_{|z| \rightarrow \infty} f(z) = 0$ for $\text{Im}(z) \geq 0$, then

$$\int_{-\infty}^{\infty} f(x) e^{ix} dx = 2\pi i \sum \text{Res}(f(z), e^{iz}) \quad (3.11)$$

where the summation is over the singularities of $f(z)$ contained in the upper half-plane $y > 0$.

Proof:

This is now an immediate consequence of the above lemma and our previous observations.

Example 3.5.26: Residue Method III

1. Consider

$$\int_0^{\infty} \frac{\cos(x)}{x^2 + 1} dx = \frac{1}{2} \text{Re} \left(\int_{-\infty}^{\infty} \frac{e^{ix}}{x^2 + 1} dx \right)$$

Then it is equal to

$$\pi i \sum \text{Res} \left(\frac{e^{iz}}{z^2 + 1} \right)$$

where the summation goes over the poles in the upper half-plane. Since there is only the pole $z = i$, it is simple, and hence it's residue is $\frac{1}{2ei}$. Thus:

$$\int_0^{\infty} \frac{\cos(x)}{x^2 + 1} dx = \frac{\pi}{2e}$$

2. Using our current results, we may integrate integrals of the form $\int_{-\infty}^{\infty} f(x) e^{-ix} dx$. Note that we would have to integrate in the *lower* half-lane instead of the upper, for $|e^{-iz}|$ is bounded in the lower half-plane. More generally, an integral of the form

$$\int_{-\infty}^{\infty} f(x) e^{ax} dx$$

where $a \in \mathbb{C}$ can be evaluated in the half-plane where $|e^{az}| \leq 1$. To see this, consider $\int_0^{\infty} \frac{\cos(mx)}{x^2 + 1}$. Then it is equal to

$$\frac{1}{2} \text{Re} \int_{-\infty}^{\infty} \frac{e^{imx}}{2x+1} dx$$

Then substituting $z = mx$, we get:

$$\begin{aligned}\int_{-\infty}^{\infty} \frac{e^{imx}}{x^2 + 1} dx &= \int_{-\infty}^{\infty} \frac{e^{iz}}{(z/m)^2 + 1} \frac{dz}{m} \\ &= \int_{-\infty}^{\infty} \frac{me^{iz}}{z^2 + m^2} dz\end{aligned}$$

Hence, we have a pole at $z = im$, and so

$$\text{Res}\left(\frac{me^{iz}}{z^2 + m^2}, im\right) = \frac{me^{-m}}{2im}$$

Hence, simplifying and multiplying by $\frac{1}{2}2\pi i$, we get

$$\int_0^{\infty} \frac{\cos(mx)}{x^2 + 1} = \frac{\pi}{2e^m}$$

3. Now, what about:

$$\int_{-\infty}^{\infty} f(x) \sin^n(x) dx \quad \int_{-\infty}^{\infty} f(x) \cos^n(x) dx$$

For such an expression, we can convert $\sin^n(x)$ and $\cos^n(x)$ into a rational function of $\sin(mx)$ and $\cos(mx)$ by repeated use of double or half-angle identities.

Let's now say that $f(z)$ has singularities on the real axis. For simplicity, let's say it is at the origin (the proof for other points is a translation of this special case). What we shall do is modify our path of integration to bypass the singularity:



Let's say that the smaller semi-circle has path $\gamma(\epsilon)$. Then:

Lemma 3.5.27: Residues Method Lemma III

Let $z = 0$ be a simple pole of $g(z)$. Then

$$\lim_{\epsilon \rightarrow 0} \int_{\gamma(\epsilon)} g(z) dz = \pi i \text{Res}(g, 0)$$

Proof:

First, we have $g(z) = \frac{a}{z} + h(z)$ where h is holomorphic at the origin and $a = h(0)$. The integral $\int_{\gamma(\epsilon)} h(z)$ tends to 0 as $\epsilon \rightarrow 0$, and the integral $\int_{\gamma(\epsilon)} \frac{a}{z} dz$ tends to πia , but that complete the proof.

Thus in general we have

$$f(z)e^{iz} = \frac{a}{z} + R_0(z)$$

where $a = \text{Res}(R(z)e^{iz}, 0)$. Overall, we have:

$$\int_{-\infty}^{\infty} R(x)e^{ix} = 2\pi i \sum_{\text{Im}(z)>0} \text{Res}(R(z)e^{iz}) + \pi i \sum_{\text{Im}(z)=0} \text{Res}(R(z)e^{iz})$$

Example 3.5.28: Residue Method IV

Let

$$I = \int_0^{\infty} \frac{\sin(x)}{x} dx = \frac{1}{2} \text{Im} \int_{-\infty}^{\infty} \frac{e^{iz}}{z} dz$$

Then, since 0 is a simple pole, we see that the residue at 0 of the integrand on the right hand side is 1. Hence:

$$\int_{-\infty}^{\infty} \frac{e^{iz}}{z} = 1\pi i = \pi i$$

taking the imaginary part of the above and dividind by two we get:

$$\int_0^{\infty} \frac{\sin x}{x} dx = \frac{1}{2}\pi = \frac{\pi}{2}$$

Fourth Type (IV)

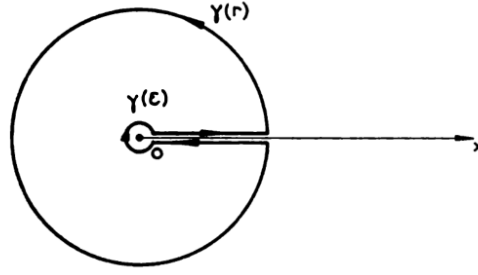
When integrating real valued function, we often come across fractional functions such that $x^{1/2}$. In the complex case, such functions are multi-valued, and hence we must take extra care when working with them. Thus, the forth type of integral we will considier is:

$$I = \int_0^{\infty} \frac{R(x)}{x^{\alpha}} dx$$

where $\alpha \in (0, 1)$, and $R(x)$ is a rational fuction with no poles in the positive real axis $x \geq 0$. It is clear the integral converges in the lower limit. For it to converge in the upperlimit, the principal part of $R(x)$ at infinity must be of the form $\frac{1}{x^n}$ where $n \geq 1$. In other words, it is necessary and sufficient that:

$$\lim_{x \rightarrow \infty} R(x) = 0$$

To find this integral, we naturally first complexify to $f(z) = \frac{R(z)}{z^{\alpha}}$ defined on the plane with positive real axis $x \geq 0$ excluded: let Ω be this set. Since $\alpha \in (0, 1)$, we have to specify the branch of z^{α} in Ω : we shall take the branch with argument between in $(0, 2\pi)$. With this, we shall integrate on the following region:



Let's label this path $\delta(r, \epsilon)$. This path goes from $\epsilon > 0$ to $r > 0$, then circles around 0 with radius r in the positive sense, then goes down the real axis from r to ϵ , and finally circles around 0 with radius ϵ . Now, considering the integral

$$\int_{\delta(r, \epsilon)} \frac{R(z)}{z^\alpha} dz$$

we see it is equal to the sum of residues of the poles of $\frac{R(z)}{z^\alpha}$ contained in Ω , if r has been chosen to be sufficiently large and ϵ sufficiently small. Let's now show that the contour on the two circles go to zero. We first decompose the integral: since we are circling around a point that has a non-trivial index, notice that after letting $f(z) = R(z)/z^\alpha$, we can decompose this integral into:

$$\int_{\delta(r, \epsilon)} f(z) dz = \int_{\gamma(r)} f(z) dz + \int_{\gamma(\epsilon)} f(z) dz + (1 - e^{-2\pi i \alpha}) \int_{\epsilon}^r f(x) dx$$

where $\gamma(r)$ and $\gamma(\epsilon)$ are the two circles centered at 0. Now, since the argument of z is bounded, $zf(z)$ tends to 0 when $z \rightarrow 0$, and when $|z| \rightarrow \infty$. Thus, the integral over $\gamma(r)$ and $\gamma(\epsilon)$ tends to 0 as $r \rightarrow \infty$ and $\epsilon \rightarrow 0$ (this is lemma 3.5.21 and 3.5.22). Thus, at the limit, we have

$$(1 - e^{2i\pi\alpha})I = 2\pi i \sum \text{Res} \left(\frac{R(z)}{z^\alpha} \right) \quad (3.12)$$

which we can now use to calculate the value of I

Example 3.5.29: Residue Method V

Let

$$I = \int_0^\infty \frac{1}{x^\alpha(1+x)} dx$$

for $\alpha \in (0, 1)$. Then $R(z) = \frac{1}{1+z}$ which only has one pole at $z = -1$. Using our normal method of residue computations:

$$\text{Res}\left(\frac{1}{z^\alpha(1+z)}, -1\right) = e^{-\pi i \alpha}$$

Thus, by equation 3.12 we get:

$$(1 - e^{-2\pi i \alpha}) \int_0^\infty \frac{1}{x^\alpha(x+1)} dx = 2\pi i \sum_{\mathbb{C} \setminus [0, \infty)} \text{Res}\left(\frac{1}{z^\alpha(1+z)}\right) = 2\pi i \frac{1}{e^{\pi i \alpha}}$$

Simplifying, we get:

$$\int_0^\infty \frac{1}{x^\alpha(1+x)} dx = \frac{2\pi i}{e^{\pi i \alpha}(1 - e^{-2\pi i \alpha})} = \frac{2\pi i}{e^{\pi i \alpha} - e^{-\pi i \alpha}} = \frac{\pi}{\sin(\pi \alpha)}$$

Fifth Type (V)

The last type of integral we will consider is of the form:

$$\int_0^\infty R(x) \log(x) dx$$

where R is a rational function with no pole on the positive real axis $x \geq 0$, and $\lim_{x \rightarrow \infty} xR(x) = 0$ so that the integral converges. Let Ω be the same as for the fourth type and let $\delta(r, \epsilon)$ be the same path of integration. The branch of $\log(z)$ we choose takes arguments of z between in $(0, 2\pi)$.

For reasons we shall see soon, we shall be integrating $R(z)(\log(z))^2$, this will allow us to put $R(z)$ and $R(z) \log(z)$ in relation. Just like before, the integrals along $\gamma(r)$ and $\gamma(\epsilon)$ tend to 0 as $r \rightarrow \infty$ and $\epsilon \rightarrow 0$ by lemma 3.5.21 and 3.5.22. When the argument of z is 2π , we have

$$\log(z) = \log(|z|) + 2\pi i$$

Letting $x = |z|$, we have:

$$\int_0^\infty R(x)(\log(x))^2 dx - \int_0^\infty R(x)(\log(x) + 2\pi i)^2 dx = 2\pi i \sum \text{Res} (R(z)(\log(z))^2)$$

Thus:

$$-2 \int_0^\infty R(x) \log(x) dx - 2\pi i \int_0^\infty R(x) dx = \sum \text{Res} (R(z)(\log(z))^2)$$

For a general rational function, this is about as good as we could get in the general case. However, if the coefficients of $R(z)$ are *real*, or it takes real values for real x , we can separate the real and imaginary parts of the above equation to obtain the two relations:

$$\int_0^\infty R(x) \log(x) dx = -\frac{1}{2} \text{Re} \left(\sum \text{Res} (R(z)(\log(z))^2) \right) \quad (3.13)$$

$$\int_0^\infty R(x) x = -\frac{1}{2} \text{Im} \left(\sum \text{Res} (R(z)(\log(z))^2) \right) \quad (3.14)$$

where the summation extends over all the poles of the rational function $R(z)$ in Ω

Example 3.5.30: Residue Method VI

We shall evaluate

$$I = \int_0^\infty \frac{\log(x)}{(1+x)^3} dx$$

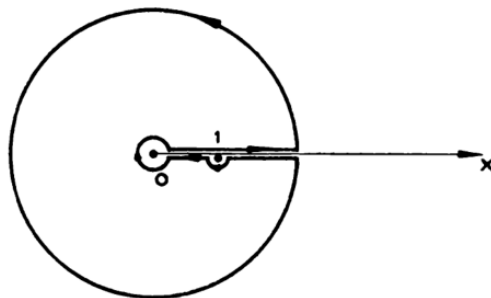
The residue of $(\log(z))^2/(1+z)^2$ at $z = -1$ is equal to the coefficient of t^2 in the partial Taylor expansion of $(i\pi + \log(1-t))^2$. Computing, we get $1 - i\pi$, and so

$$\int_0^\infty \frac{\log(x)}{(1+x)^3} dx = -\frac{1}{2}$$

Note that if we integrate $R(z) \log(z)$ on $\delta(r, \epsilon)$, then we would get

$$\int_0^\infty R(x) dx = - \sum \text{Res}(R(z) \log(z))$$

We may still get similar results if there is a simple pole on the real axis, say $x = 1$ for simplicity. In this case, we can still make sense of $\int_0^\infty R(x) \log(x) dx$ since $\log(z)$ has a simple zero at 1. It will be necessary to modify the contour of integration to go around 1, so that it looks like so:



It should be proven that on this path:

$$\int_0^\infty R(x) \log(x) dx = \pi^2 \text{Re}(\text{Res}(R, 1)) - \frac{1}{2} \text{Re} \left(\sum \text{Res}(R(z) (\log(z))^2) \right)$$

where the summation goes over all poles of $R(z) (\log(z))^2$ except $z = 1$.

Exercise 3.5.1

1. Show that $\int_0^\infty \frac{\log(x)}{x^2-1} = \frac{\pi^2}{4}$.

Let $f(z) = z^n + a_{n-1}z^{n-1} + \dots + a_1z + a_0$. Show that all the roots of f lie in a circle of radius $\sqrt{1 + |a_{n-1}|^2 + \dots + |a_1| + |a_0|}$ (hint: Rouché's Theorem on z^n and the rest)

Analysis of Holomorphic Functions

With the exposition of integration of holomorphic functions, we prove a couple more important results on the representation and the convergence of holomorphic functions. We shall strongly be relying on some basic notions of functional analysis in this section, see [24] for more information.

4.1 Topology of Holomorphic Functions

Recall the following definitions:

Definition 4.1.1: Space Of Holomorphic Functions

Let D be an open subset of $\mathbb{C}$. Then $\mathcal{H}(D)$ be the set of all holomorphic functions from D to $\mathbb{C}$.

Naturally, this forms a complex vector-space even a complex algebra, for adding and multiplying by a constant $\mathbb{C}$ gives another holomorphic function, and multiplying two holomorphic functions is again a holomorphic function. We are interested in the construction of new holomorphic functions given a sequence of holomorphic functions. As we have implicitly been using every-time we created a power-series convergence, we have a particular notion of convergence that will result in the limit being holomorphic:

Definition 4.1.2: Convergence On Compact Sets

Let $f_n \in C(D)$ be a sequence continuous functions. Then we say that $f_n \rightarrow f$ on compact sets if and only if for each compact subset $K \subseteq D$, $f_n \rightarrow f$ uniformly on K .

This is indeed a sufficient notion for convergence: By Morera's Theorem, if holomorphic functions

$f_n \rightarrow f$ uniformly, then f is holomorphic (see Corollary 4.1.4). The following immediately follows:

Lemma 4.1.3: Uniform Convergence On Compact Sets

let $(f_n) \subseteq \mathcal{H}(D)$ be a sequence of holomorphic functions such that $f_n \rightarrow f$ uniformly on compact sets. Then f is holomorphic

Proof:

By Morera's theorem it suffices to show that $\int_\gamma f = 0$. But γ is the boundary of a compact set K , and by uniform convergence on compact sets we have

$$\int_\gamma f = \int_\gamma f|_K = \int_\gamma \lim_n f_n|_K = \lim_n \int_\gamma f_n|_K = 0$$

completing the proof.

Naturally, if we strengthen the condition to uniform convergence, we get the same result. This result is mentioned to compare to the case of real differentiation:

Lemma 4.1.4: Weiestrass Convergence Theorem

Let $f_n \rightarrow f$ be a sequence of holomorphic function converging uniformly to f on an open disk $\mathbb{D}$. Then f is holomorphic

Note how there is no mention of how the derivatives of f_n converge, which is usually required for the equivalent result for real differentiable functions to converge to real differentiable functions.

Proof:

Since $\mathbb{D}$ is connected, for any closed path in $\mathbb{D}$ we have

$$\int_\gamma f_n(z) dz = 0$$

Then by uniform convergence:

$$\int_\gamma f(z) dz = \int_\gamma \lim_{n \rightarrow \infty} f_n(z) dz = \lim_{n \rightarrow \infty} \int_\gamma f_n(z) dz = 0$$

Thus, by Morera's Theorem f is holomorphic, completing the proof.

Uniform convergence on compact sets is weaker than uniform convergence. The reason for this weakening is necessary because uniform convergence is in fact too strong of a notion for convergence of holomorphic functions.

Example 4.1.5: Uniform Convergence Need Not Exist

Let $f_n = \sum_k^n z^k$. Then we know this converges pointwise to $\sum_k^\infty \frac{1}{1-z}$ on $|z| < 1$. However, this

does *not* converge uniformly on $|z| < 1$. The problem is that the error term:

$$|f_n - f| = |z^n + z^{n+1} + \dots| = \left| \frac{z^n}{1 - z} \right|$$

does not uniformly shrink for $z < 1$, in fact for arbitrary $z = 1 - \epsilon$ we may make the error term as large as we want. This problem has already showed up before and was dealt with: recall in Abel's Theorem (theorem 2.3.2) that we were careful to pick $r < |z|$ and then insured convergence on $\rho \leq r$. This is precisely because of this limitation.

For our case, notice that if we fix any particular r , we may in fact bound the error term in such a way that does not depend on z . For example, if we take $|z| \leq 1/2$, then

$$\left| \frac{z^n}{1 - z} \right| \leq \frac{(1/2)^n}{1/2} \xrightarrow{n \rightarrow \infty} 0$$

and more generally for any $|z| \leq 1 - \epsilon$ for $\epsilon > 0$, we see that the error term tends to zero.

The fact that holomorphic functions are preserved on uniform convergence for compact sets is intuitive since being holomorphic is a local condition¹. In this way, we may ask if being locally uniform convergence is sufficient, that for any point we may choose a neighbourhood for which the functions converge uniformly. Since $\mathbb{C}$ is a locally compact space, this is in fact equivalent: namely any point is contained in a compact neighbourhood and hence it is covered by finitely many open neighbourhoods, giving us the equivalence.

Beyond sums and limits, holomorphic functions can also be assembled by integration. Many of the central functions of analysis, such as the Gamma function, the Laplace and Fourier transforms, and the integral representations that drive analytic number theory (see [2]), arise as integrals

$$f(s) = \int_X g(x, s) d\mu(x)$$

of a family of holomorphic functions $g(x, \cdot)$ over a measure space. The Morera argument of lemma 4.1.3 adapts to this setting: where uniform convergence let us exchange the limit with the path integral, an integrable dominating function lets us exchange the two integrals. The measure theory required, namely measurability, the Fubini-Tonelli theorem, and the dominated convergence theorem, is developed in [25].

¹Though it has a lot of fascinating global consequences, nevertheless the check for a function being holomorphic is still a point-wise check

Theorem 4.1.6: Holomorphy Of Parameter Integrals

Let (X, μ) be a σ -finite measure space, let $\Omega \subseteq \mathbb{C}$ be open, and let $g : X \times \Omega \rightarrow \mathbb{C}$ be a function satisfying:

1. for each $s \in \Omega$, the function $x \mapsto g(x, s)$ is measurable;
2. for each $x \in X$, the function $s \mapsto g(x, s)$ is holomorphic on Ω ;
3. for each $s_0 \in \Omega$, there is a neighbourhood $U \subseteq \Omega$ of s_0 and a function $h \in L^1(\mu)$ such that $|g(x, s)| \leq h(x)$ for all $s \in U$ and all $x \in X$.

Then

$$f(s) = \int_X g(x, s) d\mu(x)$$

is a well-defined holomorphic function on Ω , the function $x \mapsto \frac{\partial g}{\partial s}(x, s)$ is in $L^1(\mu)$ for each $s \in \Omega$, and we may differentiate under the integral sign:

$$f'(s) = \int_X \frac{\partial g}{\partial s}(x, s) d\mu(x)$$

Proof:

First, f is well-defined: given $s \in \Omega$, applying hypothesis (3) at $s_0 = s$ gives an $h \in L^1(\mu)$ with $|g(x, s)| \leq h(x)$ for all x , and since $x \mapsto g(x, s)$ is measurable by hypothesis (1), $g(\cdot, s) \in L^1(\mu)$ and so $f(s)$ is a finite complex number.

Next, we show f is continuous. Fix $s_0 \in \Omega$, let U and h be as in hypothesis (3), and choose $\rho > 0$ small enough that $\overline{D} = \{s \in \mathbb{C} \mid |s - s_0| \leq \rho\} \subseteq U$. Let (s_n) be any sequence with $s_n \rightarrow s_0$; discarding finitely many terms (which changes neither limit below), we may assume every s_n lies in $\overline{D}$. For each $x \in X$ we have $g(x, s_n) \rightarrow g(x, s_0)$, since holomorphic functions are continuous, and $|g(\cdot, s_n)| \leq h$ for every n . Thus by the dominated convergence theorem ([25, Dominated Convergence Theorem (DCT)]):

$$f(s_n) = \int_X g(x, s_n) d\mu(x) \rightarrow \int_X g(x, s_0) d\mu(x) = f(s_0)$$

Since this holds for every sequence $s_n \rightarrow s_0$ and $\mathbb{C}$ is a metric space, f is continuous at s_0 .

We now show that f is holomorphic on the open disk $D = \{s \in \mathbb{C} \mid |s - s_0| < \rho\}$; since holomorphy is a local condition and s_0 was arbitrary, this proves f is holomorphic on Ω . The disk D is an open connected set and f is continuous on it (s_0 above was arbitrary, so continuity holds at every point of Ω), so by Morera's theorem (theorem 3.2.10) it suffices to show the form $f(s)ds$ is closed on D , and by proposition 3.1.6 together with definition 3.1.7 it in turn suffices to show

$$\int_{\partial R} f(s) ds = 0$$

for every rectangle $R \subseteq D$ with sides parallel to the axes. Fix such an R and parametrize its boundary by a piecewise-linear closed path $\gamma : [0, 1] \rightarrow D$. Then γ' exists and satisfies $|\gamma'(t)| \leq L$ for some constant L , except at the four parameter values $t_1, \dots, t_4$ corresponding to the corners,

where we set $\gamma'(t_i) = 0$. Unwinding the definition of the path integral and of f :

$$\int_{\partial R} f(s) ds = \int_0^1 f(\gamma(t)) \gamma'(t) dt = \int_0^1 \left[\int_X g(x, \gamma(t)) \gamma'(t) d\mu(x) \right] dt$$

We want to exchange the two integrals using the Fubini-Tonelli theorem, for which we must check that $G(x, t) := g(x, \gamma(t)) \gamma'(t)$ is measurable with respect to the product σ -algebra on $X \times [0, 1]$, where $[0, 1]$ carries the Lebesgue measure λ (note both μ and λ are σ -finite). To that end, define

$$G_n(x, t) = \sum_{k=0}^{n-2} g(x, \gamma(k/n)) \chi_{[k/n, (k+1)/n)}(t) \gamma'(t) + g(x, \gamma((n-1)/n)) \chi_{[(n-1)/n, 1]}(t) \gamma'(t)$$

Each summand is a product of a measurable function of x (hypothesis (1)) with a measurable function of t , so each G_n is measurable with respect to the product σ -algebra. Moreover, $G_n \rightarrow G$ pointwise: at the corner values $t = t_i$ both sides vanish, and for any other (x, t) the left endpoint $a_n(t)$ of the step interval containing t satisfies $t - a_n(t) \leq 1/n$, so $\gamma(a_n(t)) \rightarrow \gamma(t)$ by continuity of γ , hence $g(x, \gamma(a_n(t))) \rightarrow g(x, \gamma(t))$ by continuity of $g(x, \cdot)$. Thus G is a pointwise limit of product-measurable functions, hence product-measurable. Since $\gamma(t) \in \partial R \subseteq D \subseteq U$ for all t , we also have the domination $|G(x, t)| \leq Lh(x)$, so by the Tonelli half of [25, Fubini-Tonelli Theorem]:

$$\int |G| d(\mu \times \lambda) = \int_0^1 \left[\int_X |G(x, t)| d\mu(x) \right] dt \leq L \int_X h d\mu < \infty$$

Hence $G \in L^1(\mu \times \lambda)$, and the Fubini half of the same theorem lets us exchange the iterated integrals:

$$\begin{aligned} \int_{\partial R} f(s) ds &= \int_0^1 \left[\int_X G(x, t) d\mu(x) \right] dt = \int_X \left[\int_0^1 g(x, \gamma(t)) \gamma'(t) dt \right] d\mu(x) \\ &= \int_X \left[\int_{\partial R} g(x, s) ds \right] d\mu(x) \end{aligned}$$

For each fixed x , the function $g(x, \cdot)$ is holomorphic on Ω and ∂R is homotopic to a point inside the convex set D , so Cauchy's theorem (theorem 3.2.1) gives $\int_{\partial R} g(x, s) ds = 0$. Therefore $\int_{\partial R} f(s) ds = \int_X 0 d\mu = 0$ for every rectangle $R \subseteq D$, and f is holomorphic.

It remains to differentiate under the integral sign. Fix $s_0 \in \Omega$, keep U and h as above, and this time choose $r > 0$ small enough that $\{s \in \mathbb{C} \mid |s - s_0| \leq 2r\} \subseteq U$. We first check that the derivative is measurable in x : for fixed $s \in \Omega$, we may write

$$\frac{\partial g}{\partial s}(x, s) = \lim_{n \rightarrow \infty} n(g(x, s + 1/n) - g(x, s))$$

since the limit defining the complex derivative may be taken along the particular sequence $1/n \rightarrow 0$, and $s + 1/n \in \Omega$ for n large as Ω is open. Thus $x \mapsto \frac{\partial g}{\partial s}(x, s)$ is a pointwise limit of measurable functions, hence measurable. We next bound the derivative: let $|s - s_0| < r$. The open set U contains the compact disc $\{w \in \mathbb{C} \mid |w - s_0| \leq 2r\}$, hence a disc $|w - s_0| < 2r + \epsilon$ with $\epsilon > 0$, the distance from a compact set to a disjoint closed set being positive. This disc contains $\{w \in \mathbb{C} \mid |w - s| < r + \epsilon\}$, on which $g(x, \cdot)$ is therefore the sum of its Taylor series

about s (theorem 3.2.6). So the Cauchy inequalities (theorem 3.3.1), with $R = r + \epsilon > r$, applied to the first coefficient give

$$\left| \frac{\partial g}{\partial s}(x, s) \right| \leq \frac{\sup_{\theta} |g(x, s + re^{i\theta})|}{r} \leq \frac{h(x)}{r}$$

where the last inequality holds because $|s + re^{i\theta} - s_0| \leq |s - s_0| + r < 2r$ places $s + re^{i\theta}$ inside U . In particular, taking $s = s_0$, the function $x \mapsto \frac{\partial g}{\partial s}(x, s_0)$ is measurable and dominated by $h/r \in L^1(\mu)$, hence is itself in $L^1(\mu)$.

Now take any s with $0 < |s - s_0| < r$. The function $g(x, \cdot)$ is a primitive of the form $\frac{\partial g}{\partial s}(x, w)dw$, so integrating along the straight segment $[s_0, s]$ (which lies in the disk $|w - s_0| < r$) gives, by proposition 3.1.5(1):

$$g(x, s) - g(x, s_0) = \int_{[s_0, s]} \frac{\partial g}{\partial s}(x, w)dw$$

Bounding this path integral by the length $|s - s_0|$ of the segment times the supremum of the integrand, and using the bound on the derivative just established:

$$\left| \frac{g(x, s) - g(x, s_0)}{s - s_0} \right| \leq \sup_{w \in [s_0, s]} \left| \frac{\partial g}{\partial s}(x, w) \right| \leq \frac{h(x)}{r}$$

Let (s_n) be any sequence with $0 < |s_n - s_0| < r$ and $s_n \rightarrow s_0$, and set

$$\varphi_n(x) = \frac{g(x, s_n) - g(x, s_0)}{s_n - s_0}$$

Each φ_n is measurable by hypothesis (1); $\varphi_n(x) \rightarrow \frac{\partial g}{\partial s}(x, s_0)$ for every x by the definition of the complex derivative; and $|\varphi_n| \leq h/r \in L^1(\mu)$ by the inequality above. Hence by the dominated convergence theorem once more:

$$\frac{f(s_n) - f(s_0)}{s_n - s_0} = \int_X \varphi_n(x) d\mu(x) \rightarrow \int_X \frac{\partial g}{\partial s}(x, s_0) d\mu(x)$$

We have already shown that f is holomorphic, so the difference quotients on the left converge to $f'(s_0)$, and since limits are unique:

$$f'(s_0) = \int_X \frac{\partial g}{\partial s}(x, s_0) d\mu(x)$$

completing the proof.

Example 4.1.7: The Gamma Function Is Holomorphic

Let $\Omega = \{s \in \mathbb{C} \mid \operatorname{Re}(s) > 0\}$ be the right half-plane, let $X = (0, \infty)$ with the Lebesgue measure, and let

$$g(x, s) = x^{s-1}e^{-x} := e^{(s-1)\log(x)}e^{-x}$$

where $\log$ is the ordinary real logarithm of $x > 0$. We check the three hypotheses of theorem 4.1.6. For fixed s , the function $x \mapsto x^{s-1}e^{-x}$ is continuous on $(0, \infty)$, hence measurable. For fixed x , the function $s \mapsto e^{(s-1)\log(x)}e^{-x}$ is the exponential composed with an affine map, hence entire. For the domination, fix s_0 with $\sigma_0 = \operatorname{Re}(s_0) > 0$ and take the neighbourhood $U =$

$\{s \in \mathbb{C} \mid \sigma_0/2 < \operatorname{Re}(s) < \sigma_0 + 1\}$. For $s \in U$:

$$|x^{s-1}e^{-x}| = x^{\operatorname{Re}(s)-1}e^{-x} \leq h(x) := \left(x^{\sigma_0/2-1} + x^{\sigma_0}\right)e^{-x}$$

since $x^{\operatorname{Re}(s)-1} \leq x^{\sigma_0/2-1}$ for $x \leq 1$ and $x^{\operatorname{Re}(s)-1} \leq x^{\sigma_0}$ for $x \geq 1$. The dominating function h is in $L^1(0, \infty)$: on $(0, 1]$ we have $h(x) \leq x^{\sigma_0/2-1} + 1$, and $\int_0^1 x^{\sigma_0/2-1} dx = 2/\sigma_0$ converges since $\sigma_0 > 0$; on $[1, \infty)$ we have $h(x) \leq 2x^{\sigma_0}e^{-x} \leq Ce^{-x/2}$ for some constant C (as $x^{\sigma_0}e^{-x/2}$ is continuous and tends to 0, hence bounded), which is integrable. Theorem 4.1.6 now tells us that the *Gamma function*

$$\Gamma(s) = \int_0^\infty x^{s-1}e^{-x} dx$$

is holomorphic on the right half-plane, and that its derivative is $\Gamma'(s) = \int_0^\infty x^{s-1} \log(x) e^{-x} dx$.

Since we have a convergence, we may naturally ask if it comes from a topology to better understand what type of space we are working with (is it a metric space? a Fréchet space? A Banach space? what are its compact sets? and so forth)². For those who have seen some functional analysis, the following will be seen as the natural answer:

Definition 4.1.8: Topology On $\mathcal{H}(D)$

Take $\mathcal{H}(D)$, and for $g \in \mathcal{H}(D)$, a compact set $K \subseteq D$ and $\epsilon > 0$ define

$$V_g(K, \epsilon) = \left\{ f \in \mathcal{H}(D) \mid \sup_{z \in K} |f(z) - g(z)| < \epsilon \right\}$$

and $V(K, \epsilon) := V_0(K, \epsilon)$. A subset of $\mathcal{H}(D)$ is open in the *uniform-compact* topology if it contains, around each of its points g , some set $V_g(K, \epsilon)$. The same formula, with the space $C(D)$ of continuous functions $D \rightarrow \mathbb{C}$ in place of $\mathcal{H}(D)$, defines the uniform-compact topology on $C(D)$.

These sets can equivalently be described using the collection of seminorms $\rho_K(f) = \sup_{z \in K} |f(z)|^a$, as $V_g(K, \epsilon)$ is the set of f with $\rho_K(f - g) < \epsilon$.

^aEach ρ_K is a seminorm. It need not be a norm, even on $\mathcal{H}(D)$: for $K = \{z\}$ it vanishes at every f with $f(z) = 0$.

Naturally, this topology can be defined on continuous functions or even L^p functions. Uniform convergence on compact sets translate to convergence in $\mathcal{H}(D)$ in this topology by insuring that $f_n \rightarrow f$ if and only if for all $\epsilon > 0$, there exists an n large enough so that

$$f_n \in V_f(K, \epsilon) = \left\{ g \in \mathcal{H}(D) \mid \sup_K |g - f| < \epsilon \right\}$$

or equivalently there exists an n large enough so that $f - f_n \in V(K, \epsilon)$. We shall call the collection $V(K, \epsilon)$ the *fundamental neighbourhoods*. If you have done some functional analysis, you

²Notes that some forms of convergence, like convergence in measure, do not come from topological notions, and hence this is not immediately intuitive. On the other hand, uniform convergence comes from a topology, so uniform on compact sets seem like an intuitive generalization

may recognize this is a Fréchet space³, and hence the this collection of (semi)norms induced an invariant metric. we can infact explicitly figure out this metric. Note that the metric that is defined must induce the same topology, name uniform convergence on compact subspaces, and that the construction of this metric is just the special case of the same construction for general Fréchet spaces:

Proposition 4.1.9: Invariant Metric

Let $D \subseteq \mathbb{C}$ be open. Then D has a *compact exhaustion*, that is, compact sets $D_1 \subseteq D_2 \subseteq \dots$ with each D_i contained in the interior of D_{i+1} , $\bigcup_i D_i = D$, and every compact subset of D contained in some D_i . Given a compact exhaustion $\{D_i\}$ of D ,

$$d(f, g) = \sum_{i=1}^{\infty} 2^{-i} \min \left\{ 1, \max_{D_i} |f - g| \right\}$$

is a translation-invariant metric on the space $C(D)$ of continuous functions $D \rightarrow \mathbb{C}$, hence also on $\mathcal{H}(D)$, and it induces the uniform-compact topology: a set is open for d if and only if it contains, around each of its points g , some set $V_g(K, \epsilon)$ with $K \subseteq D$ compact and $\epsilon > 0$. In particular $f_n \rightarrow f$ for d if and only if $f_n \rightarrow f$ uniformly on every compact subset of D .

Proof:

A maximum over an empty D_i is read as 0.

Existence of an exhaustion. If $D = \mathbb{C}$ take $D_i = \{z \mid |z| \leq i\}$. Otherwise let

$$D_i = \{z \in \mathbb{C} \mid |z| \leq i, \text{dist}(z, \mathbb{C} \setminus D) \geq 1/i\}$$

The distance to the nonempty set $\mathbb{C} \setminus D$ is continuous, being 1-Lipschitz by [5, Distance to a Set Is 1-Lipschitz], so D_i is closed, and it is bounded, hence compact ([8, Closed And Bounded In $\mathbb{R}^n$ Implies Compact]). A point at positive distance from $\mathbb{C} \setminus D$ lies in D , so $D_i \subseteq D$. The open set $\{w \mid |w| < i + 1, \text{dist}(w, \mathbb{C} \setminus D) > 1/(i + 1)\}$ contains D_i and lies in D_{i+1} , so D_i lies in the interior of D_{i+1} . Each $z \in D$ has $\text{dist}(z, \mathbb{C} \setminus D) > 0$, as $\mathbb{C} \setminus D$ is closed, so $z \in D_i$ for i large, and $\bigcup_i D_i = D$. In both cases, a compact $K \subseteq D$ is covered by the increasing open sets $\text{int}(D_{i+1})$, so finitely many of them cover it, and then the largest of those does, giving $K \subseteq D_N$ for some N .

The metric. Each maximum exists, $|f - g|$ being continuous on the compact D_i , and $d(f, g) \leq \sum_i 2^{-i} = 1$. Each term is symmetric in f and g , so $d(f, g) = d(g, f)$, and $d(f, f) = 0$. If $d(f, g) = 0$ then $|f - g|$ vanishes on every D_i , hence on D . For $a, b \geq 0$ we have $\min\{1, a + b\} \leq \min\{1, a\} + \min\{1, b\}$, the right-hand side being at least 1 when $a \geq 1$ or $b \geq 1$, and equal to $a + b$ otherwise. Combined with $\max_{D_i} |f - h| \leq \max_{D_i} |f - g| + \max_{D_i} |g - h|$ and summing, this gives the triangle inequality. Finally $(f + h) - (g + h) = f - g$, so d is translation-invariant.

The topology. Let $g \in C(D)$. Given $r > 0$, choose N with $2^{-N} < r/2$. If $\max_{D_N} |f - g| < r/2$,

³Note that any open subset of $\mathbb{C}$ has a countable compact exhaustion

then $\max_{D_i} |f - g| < r/2$ for $i \leq N$ as $D_i \subseteq D_N$, so

$$d(f, g) < \frac{r}{2} \sum_{i=1}^N 2^{-i} + \sum_{i>N} 2^{-i} < \frac{r}{2} + \frac{r}{2} = r$$

Thus the d -ball of radius r around g contains $V_g(D_N, r/2)$. Conversely, given $V_g(K, \epsilon)$, choose N with $K \subseteq D_N$ and put $r = 2^{-N} \min\{1, \epsilon\}$. If $d(f, g) < r$, then $2^{-N} \min\{1, \max_{D_N} |f - g|\} \leq d(f, g) < 2^{-N} \min\{1, \epsilon\}$, which forces $\max_{D_N} |f - g| < \min\{1, \epsilon\} \leq \epsilon$, so $f \in V_g(K, \epsilon)$. Thus every $V_g(K, \epsilon)$ contains a d -ball around g . A set therefore contains a d -ball around each of its points if and only if it contains a set $V_g(K, \epsilon)$ around each of its points g , which is the claim about the topology. Applying the two inclusions to the balls around f shows that $d(f_n, f) \rightarrow 0$ if and only if, for every compact K and $\epsilon > 0$, eventually $\sup_K |f_n - f| < \epsilon$, as we sought to show.

We give this metric a name:

Definition 4.1.10: Metric of Compact Convergence

The metric

$$d(f, g) = \sum_{i=1}^{\infty} 2^{-i} \min \left\{ 1, \max_{D_i} |f - g| \right\}$$

of proposition 4.1.9 is called the *metric of compact convergence* associated with the compact exhaustion $\{D_i\}$. It is not to be confused with the chordal metric of theorem 1.2.2.

The main take-away from us is that $\mathcal{H}(D)$ is a *metric space*, meaning we have the results we usually have on metric spaces, which we shall do when finding the compact subsets of $\mathcal{H}(D)$.

Convergence of Derivatives, Injective, and non-zero Functions

Since holomorphic functions converging uniformly on compact sets is a sufficient for the limit to be holomorphic, it is perhaps intuitive that the derivative converges to the derivative of f , that is if $f_n \rightarrow f$ on compact sets, then $f'_n \rightarrow f'$, on compact sets:

Proposition 4.1.11: Convergence of Derivative Of Holomorphic Functions

Let the sequence of holomorphic functions $f_n \rightarrow f$ uniformly on compact sets. Then $f'_n \rightarrow f'$ uniformly on compact sets.

This is not true for smooth functions, for example take x^2 converging to $|x|$ on $[-1, 1]$; the derivative x does not uniformly converge to $\text{sgn}(x)$. This means that this is particular to harmonic functions, in particular we must take advantage of the analytic expansion:

Proof:

Let $K \subseteq D$ be a compact set, without loss of generality a compact disk. Let $K \subsetneq C$ be a slightly larger compact disk and take $\gamma = \partial C$. Consider:

$$f(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(t)}{t - z} dt$$

then differentiating, we get (by differentiating under the integral sign):

$$f'(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(t)}{(t-z)^2} dt$$

Then by uniform convergence:

$$f'(z) = \lim_n \frac{1}{2\pi i} \int_{\gamma} \frac{f_n(t)}{(t-z)^2} dt = \lim_n f'_n(z)$$

from this, uniformness on compact sets follows with some bounding arguments.

There is another proof that directly uses Cauchy inequalities that is insightful enough to put down:

Proof:

(alternative)

We will show that if f is holomorphic on $|z| < r_0 + \epsilon$ for small ϵ and $|f(z)| \leq M$ for all $|z| \leq r_0$, then:

$$|f'(z)| \leq M \frac{r_0}{(r_0 - r)^2} \quad |z| \leq r < r_0$$

First, in $|z| \leq r_0$ we get

$$f(z) = \sum_k a_k z^k$$

by Cauchy's inequalities we get $|a_k| \leq \frac{M}{(r_0)^k}$. On the other hand, differentiating the above we get

$$f'(z) = \sum_k k a_k z^{k-1}$$

for $|z| \leq r < r_0$. Substituting, we get

$$|f'(z)| \leq \frac{M}{r_0} \sum_k \frac{k r^{k-1}}{(r_0)^{k-1}}$$

What's left to show is that $\sum_k n \left(\frac{r}{r_0}\right)^{k-1}$ converges. Since kt^{k-1} is the derivative of t^n , then $\sum_k kt^{k-1}$ is the derivative of $\sum t^k = \frac{1}{1-t}$. Thus

$$\sum_k k \left(\frac{r}{r_0}\right)^{k-1} = \frac{1}{(1 - r/r_0)^2}$$

Substituting, we are left with:

$$|f'(z)| \leq \frac{M}{r_0} \frac{1}{(1 - r/r_0)^2} = M \frac{r_0}{(r_0 - r)^2}$$

But now, we have the same bounds on the derivative, from which we can deduce the Taylor expansion has the same coefficients, completing the proof.

Note that by using the Maximum modulus principle, we can reduce the proof of uniform convergence on compact sets to uniform convergence on the boundary of the compact set, since $|f_m(z) - f_n(z)|$ would attain its maximum in a compact set K on the boundary of K .

Due to the restrictive nature of holomorphic functions, they can have a unique analytic continuation, and by taking the log of an analytic function we may count the number of roots of a holomorphic function. These two facts lead to the following theorem:

Proposition 4.1.12: Hurwitz Theorem

Let $(f_n) \subseteq \mathcal{H}(D)$ where D is a connected open set. Then if $f_n(z) \neq 0$ for all $z \in D$ and $f_n \rightarrow f$ uniformly on compact sets, then either $f(z) \neq 0$ for all $z \in D$, or f is identically zero.

This is not true for the real case: consider $f_n(x) = x^2 + 1/n$. The constants $f_n = 1/n$ show that the alternative $f \equiv 0$ does occur.

Proof:

The limit f is holomorphic by lemma 4.1.3. Suppose f is not identically zero and $f(z_0) = 0$ for some $z_0 \in D$. By theorem 2.5.3 on the connected set D , the analytic function f (corollary 3.2.7) does not vanish identically near z_0 , so by proposition 2.5.7 there is $r > 0$ such that the closed disc $B = \{z \mid |z - z_0| \leq r\}$ lies in D and z_0 is the only zero of f in B . Let γ be the circle $|z - z_0| = r$, the oriented boundary of B , and let $m = \min_\gamma |f| > 0$. By proposition 4.1.11, $f_n \rightarrow f$ and $f'_n \rightarrow f'$ uniformly on γ , so $|f_n| \geq m/2$ on γ for n large, and then

$$\left| \frac{f'_n}{f_n} - \frac{f'}{f} \right| \leq \frac{|f'_n - f'|}{|f_n|} + \frac{|f'| |f - f_n|}{|f_n| |f|} \leq \frac{2}{m} |f'_n - f'| + \frac{2 \max_\gamma |f'|}{m^2} |f - f_n|$$

on γ , which tends to 0 uniformly. By the argument principle (theorem 3.5.10, with $\alpha = 0$ and the non-constant f), $\frac{1}{2\pi i} \int_\gamma \frac{f'_n}{f_n} dz$ is the number of zeros of f_n in B counted with multiplicity, which is at least 1, while $\int_\gamma \frac{f'_n}{f_n} dz = 0$ by the residue theorem (theorem 3.5.4), since f'_n/f_n is holomorphic on D , f_n having no zeros. By the length estimate of proposition 3.1.1, the second integrals converge to the first as $n \rightarrow \infty$, which is impossible. Hence f has no zeros unless it vanishes identically, as claimed.

The key intuition is that if we extend f_n analytically, we can get at most countably many zeros which are all isolated. Then as $f_n \rightarrow f$ uniformly on compact sets, these zero's move around (they cannot suddenly “appear” like in $f_n(x) = x^2 + 1/n$ since there is a unique analytic continuation for any holomorphic function). Since D is open, if $f(z) = 0$, then for ϵ small enough there would be an n such that f_n would have a zero in the open disk (the zeros would have to follow a path). Thus, this theorem essentially shows that the zero's of a function already “exist”, even if not in D .

We get yet another property of functions that is preserved under the limit that is usually not preserved for non-harmonic functions, again taking advantage of the argument principle:

Proposition 4.1.13: Injective, The Limit Injective

Let $f_n \subseteq \mathcal{H}(D)$ be a sequence of injective holomorphic functions. Then if $f_n \rightarrow f$ uniformly on compact sets, f is injective or constant.

Note how this is false for the real case, imagine

$$f(x) = \begin{cases} 0 & x < 0 \\ e^x - 1 & x \geq 0 \end{cases}$$

and a locally uniform sequence of injective smooth functions $f_n \rightarrow f$ converging to f

Proof:

For the sake of contradiction let's say $f(z_1) = f(z_2) = a$. Then for γ containig a and $f(z) \neq a$ on γ :

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z) - a} \geq 2$$

but:

$$2 \leq \frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z) - a} = \frac{1}{2\pi i} \int_{\gamma} \lim_n \frac{f'_n(z)}{f_n(z) - a} = \lim_n \frac{1}{2\pi i} \int_{\gamma} \frac{f'_n(z)}{f_n(z) - a} \leq 1$$

a contradiction.

We summarize these results in more theoretical real-analysis language:

Corollary 4.1.14: Properties Of Holomorphic And Continuous Function

Let D be an open connected set and $C(D)$, $\mathcal{H}(D)$ the collection of [complex] continuous and holomorphic function on D . Then giving $C(D)$ the uniform-compact topology

1. $C(D)$ is a complete space
2. $\mathcal{H}(D) \subseteq C(D)$ is a closed subspace
3. The mapping $\mathcal{H}(D) \rightarrow \mathcal{H}(D)$ mapping $f \mapsto f'$ is continuous

Proof:

All these statments are just reformulations of what we have just shown. (1) and (2) are immediate. To see (3), if $f_n \rightarrow f$, then $f'_n \rightarrow f'$, hence if $F(f) = f'$,

$$\lim_n F(f_n) = \lim_n f'_n = f' = F(f) = F(\lim_n f_n)$$

giving continuity.

4.1.1 Compact Subsets of $\mathcal{H}(D)$

One of the important properties of the topology of $\mathcal{H}(D)$ is that it has the Heine-Borel property: compact subsets are closed and bounded. Equivalently, since $\mathcal{H}(D)$ is a metric space, we know that a set is compact if and only if it is sequentially compact; that every sequence has a convergent subsequence. In this section, we shall show that closed and *locally bounded* implies compact in $\mathcal{H}(D)$. This makes finding compact subsets rather easy, and if we can find a compact subset, then we know every sequence has a convergent subsequence, letting us very easily obtain the existence of

a holomorphic function (in compact sets). We start with explicitly laying out what it means for a subset of $\mathcal{H}(D)$ to be bounded in its topology:

Definition 4.1.15: Bounded and locally bounded in $\mathcal{H}(D)$

Let $A \subseteq \mathcal{H}(D)$ be a subset. Then A is said to be *bounded* if for any neighbourhoods $V(K, \epsilon)$ of the origin, there is a finite number $\lambda > 0$ such that $A \subseteq \lambda V(K, \epsilon)$.

A set $A \subseteq \mathcal{H}(D)$ is said to be *locally bounded* if for every point $z \in D$, there exists a neighbourhood U of z such that $A|_U = \{f|_U \mid f \in A\}$ is bounded.

Note that a set is locally bounded iff it is locally bounded on compact sets.

To understand this bounding condition explicitly, the relation $A \subseteq \lambda V(K, \epsilon)$ can be thought of as

$$\sup_{\substack{f \in A \\ z \in K}} |f(z)| \leq \lambda \epsilon$$

Thus, a set A of holomorphic functions on D bounded if and only if for every $K \subseteq D$, there exists a $M_A(K) \in \mathbb{R}_{>0}$ such that

$$\sup_{\substack{f \in A \\ z \in K}} |f(z)| \leq M_A(K)$$

In other words, every function in A is *uniformly bounded on compact subsets*. Now, if A is bounded, certainly $\overline{A}$ is bounded (recall that convergence in this topology is uniform convergence on compact subsets). Thus, we immediately get:

Lemma 4.1.16: Bounded Operator On Holomorphic Functions

The map $f \mapsto f'$ from $\mathcal{H}(D)$ to itself takes bounded sets to bounded sets.

Proof:

Follows since $f'_n \rightarrow f'$ and f' is bounded by M where $|f(z)| < M$ for $|z| < r_0$ times some proportion of r (see the alternative proof of proposition 4.1.11, page 168), namely if f is holomorphic on $|z| < r_0 + \epsilon$ for small ϵ and $|f(z)| \leq M$ for all $|z| \leq r_0$, then:

$$|f'(z)| \leq M \frac{r_0}{(r_0 - r)^2} \quad |z| \leq r < r_0$$

With this established, we prove the easy direction:

Proposition 4.1.17: Compact In Holomorphic, Then Closed And locally Bounded

Let $A \subseteq \mathcal{H}(D)$ be compact. Then A is closed and locally bounded

Proof:

As we demonstrated earlier, $\mathcal{H}(D)$ is metrizable, hence any compact subset of $\mathcal{H}(D)$ is closed by the usual topological argument. Hence, we must show that A is bounded (the metric result only guarantees *totally bounded*). Let K be a compact subset of D , and take the (continuous) norm

$\mathcal{H}(D) \rightarrow \mathbb{R}$

$$f \mapsto \|f\|_K = \sup_{z \in K} |f(z)|$$

Since it's continuous, it takes compact sets to compact sets, so $\|A\|_K \subseteq \mathbb{R}$ is bounded, namely each $f \in A$ are uniformly bounded on K . Since this is true for any $K \subseteq D$, the set A is indeed locally bounded in $\mathcal{H}(D)$, as we sought to show.

Note that what has been proved also works for $C(D)$ (The space of continuous functions), for we may define the exact same topology. The converse though shall use properties specific to holomorphic functions. Due to historical reasons, precompact sets are given another name:

Definition 4.1.18: Normal Family

Let $A \subseteq \mathcal{H}(D)$. Then if A is precompact (it's closure is compact), A is called a *normal family* of functions.

More generally, we may define normal families for any type of space, usually function spaces such as $C(D)$. To proceed, we require some classical result from functional analysis.

Arzela-Ascoli

First, recall a result from real analysis that in a metric space, a set K is compact if and only if every sequence has a convergent subsequence (see [5]). This shall be the basis of how we prove (and use) compactness. Next, recall that compact sets of continuous functions are characterized by the following:

Definition 4.1.19: Uniform Equicontinuous Family

Let $\mathcal{F} \subseteq C(D)$ be a set of continuous functions. Then $\mathcal{F}$ is said to be *uniformly equicontinuous* on $E \subseteq D$ if

$$\forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } \forall f \in \mathcal{F}, \forall z, w \in E, |z - w| < \delta \implies d(f(z), f(w)) < \epsilon$$

Note that if $\mathcal{F} = \{f\}$, then $\mathcal{F}$ is uniformly equicontinuous if and only if f is uniformly continuous; more generally $\mathcal{F}$ is an uniformly equicontinuous family if and only if each function is uniformly continuous with the same δ . Sometimes, the term uniform is dropped and we just say the family is equicontinuous, however this can cause confusion we we just want a family of functions be be simply continuous with the same delta. Since we are in a metric space, compactness is akin to sequential compactness:

Proposition 4.1.20: Normality Through Convergent Subsequences

Let $\Omega \subseteq \mathbb{C}$ be open, give $C(\Omega)$ the uniform-compact topology, and let $\mathcal{F} \subseteq C(\Omega)$. Then $\mathcal{F}$ is normal in Ω , that is, its closure in $C(\Omega)$ is compact, if and only if every sequence $(f_n) \subseteq \mathcal{F}$ has a subsequence which converges uniformly on every compact subset $K \subseteq \Omega$.

Proof:

Let d be the metric of proposition 4.1.9 for a compact exhaustion of Ω . It induces the uniform-compact topology, and convergence for d is uniform convergence on every compact subset. Write $\overline{\mathcal{F}}$ for the closure of $\mathcal{F}$, and assume $\mathcal{F} \neq \emptyset$, the empty case being immediate.

Normal implies the sequential condition. Suppose $\overline{\mathcal{F}}$ is compact and let $(f_n) \subseteq \mathcal{F}$. Call $g \in \overline{\mathcal{F}}$ a cluster point of (f_n) if every d -ball around g contains f_n for infinitely many n . If no point of $\overline{\mathcal{F}}$ were a cluster point, each $g \in \overline{\mathcal{F}}$ would lie in a ball containing f_n for only finitely many n , and finitely many of these balls would cover $\overline{\mathcal{F}}$, which contains every f_n . Then $\mathbb{N}$ would be a finite union of finite sets, which is absurd. So there is a cluster point g , and choosing $n_1 < n_2 < \dots$ with $d(f_{n_k}, g) < 1/k$ gives a subsequence converging to g for d , that is, uniformly on every compact subset.

The sequential condition implies normal. The limit g of a subsequence converging uniformly on every compact subset is continuous, since each point of Ω has a closed disc around it in Ω on which g is a uniform limit of continuous functions ([5, Uniform Convergence Preserve's Continuity]). So $g \in C(\Omega)$ and the subsequence converges to g for d .

First, every sequence (h_n) in $\overline{\mathcal{F}}$ has a subsequence converging for d to a point of $\overline{\mathcal{F}}$. By [5, Characterizations of the Closure] there are $f_n \in \mathcal{F}$ with $d(f_n, h_n) < 1/n$. By hypothesis some subsequence f_{n_k} converges to a $g \in C(\Omega)$, and then $d(h_{n_k}, g) \leq 1/n_k + d(f_{n_k}, g) \rightarrow 0$. As $\overline{\mathcal{F}}$ is closed, $g \in \overline{\mathcal{F}}$ by [5, Closed Sets Contain Their Limits].

Hence the metric space $\overline{\mathcal{F}}$ is complete: a Cauchy sequence in it has a convergent subsequence, so it converges by [5, A Cauchy Sequence with a Convergent Subsequence Converges]. It is also totally bounded. Otherwise there is an $\epsilon > 0$ such that no finitely many d -balls of radius ϵ centred in $\overline{\mathcal{F}}$ cover $\overline{\mathcal{F}}$, and choosing $h_1 \in \overline{\mathcal{F}}$ and then each $h_{n+1} \in \overline{\mathcal{F}}$ outside the balls of radius ϵ around $h_1, \dots, h_n$ produces a sequence with $d(h_j, h_k) \geq \epsilon$ for $j \neq k$. No subsequence of it is Cauchy, so none converges, contradicting the previous paragraph. By [5, Compactness Of Metric Space (Generalized Heine-Borel)] the metric space $\overline{\mathcal{F}}$ is compact. As d induces the topology of $C(\Omega)$, this says that $\mathcal{F}$ is normal, completing the proof.

What is important for us is to characterize normal sets. The following two results from real analysis give us important conditions towards this goal. The first allows us to characterize local boundedness as a point-wise condition.

Lemma 4.1.21: Characterizing Local Boundedness

Let Ω be a domain and let $\mathcal{F} \subseteq C(\Omega)$ be uniformly equicontinuous on every compact subset of Ω . Then the following are equivalent:

1. There exists some $z_0 \in \Omega$ where $\{f(z_0) \mid f \in \mathcal{F}\}$ is bounded
2. For every $z \in \Omega$, $\{f(z) \mid f \in \mathcal{F}\}$ is bounded
3. $\mathcal{F}$ is locally bounded: every $z \in \Omega$ has a neighbourhood on which the functions of $\mathcal{F}$ are uniformly bounded

Proof:

1. (1) *implies* (2). Assume (1), let $M = \sup_{f \in \mathcal{F}} |f(z_0)|$, and let $z \in \Omega$. By lemma 3.1.2 there is a path $\gamma: [0, 1] \rightarrow \Omega$ from z_0 to z , and its image E is compact by [8, Continuity and Compactness]. Equicontinuity on E gives $\delta > 0$ with $|f(p) - f(q)| < 1$ for all $f \in \mathcal{F}$ and $p, q \in E$ with $|p - q| < \delta$. The path γ is uniformly continuous on $[0, 1]$ ([8, Uniform Continuity Theorem]), so there is an integer $N \geq 1$ with $|\gamma(s) - \gamma(t)| < \delta$ whenever $|s - t| \leq 1/N$. For every $f \in \mathcal{F}$,

$$|f(z)| \leq |f(z_0)| + \sum_{k=1}^N |f(\gamma(\frac{k}{N})) - f(\gamma(\frac{k-1}{N}))| \leq M + N$$

so (2) holds.

2. (2) *implies* (3). Assume (2) and let $z \in \Omega$. Choose $r > 0$ with the closed disc $E = \{w \mid |w - z| \leq r\}$ contained in Ω , take $\delta > 0$ from equicontinuity on E for the tolerance 1, and let U be the open disc of radius $\min\{r, \delta\}$ around z . For $w \in U$ and $f \in \mathcal{F}$ we get $|f(w)| \leq |f(z)| + 1 \leq \sup_{f \in \mathcal{F}} |f(z)| + 1$, which is finite by (2). So (3) holds.
3. (3) *implies* (1). Assume (3). A domain is nonempty, so take any $z_0 \in \Omega$. The functions of $\mathcal{F}$ are uniformly bounded on a neighbourhood of z_0 , in particular at z_0 , which is (1), completing the proof.

Theorem 4.1.22: Arzela Ascoli Theorem

Let $\Omega \subseteq \mathbb{C}$ be open and $\mathcal{F} \subseteq C(\Omega)$. Then $\mathcal{F}$ is normal iff:

1. $\mathcal{F}$ is uniformly equicontinuous on every compact set $E \subseteq \Omega$
2. (pointwise relatively compact) For any $z \in \Omega$, the set $A_z = \{f(z)\}$ lies in a compact subset of the codomain (importantly, it is bounded).

Since the codomain is $\mathbb{C}$ (meaning compact sets are closed and bounded), condition (2) is equivalent to $f(z)$ being bounded.

Proof:

- ($\Leftarrow$) Assume (1) and (2). For a compact $E \subseteq \Omega$, take $\delta > 0$ from equicontinuity on E for the tolerance 1, and cover E by finitely many open discs of radius δ centred at points $z_1, \dots, z_p \in E$. Every $w \in E$ lies within δ of some z_j , so $|f(w)| \leq \max_j \sup_{f \in \mathcal{F}} |f(z_j)| + 1 =: C_E$ for all $f \in \mathcal{F}$, a finite bound by (2).

Let $(f_n) \subseteq \mathcal{F}$ and let $D_1 \subseteq D_2 \subseteq \dots$ be a compact exhaustion of Ω (proposition 4.1.9). On the compact metric space D_i , the real parts $\operatorname{Re} f_n$ are continuous, bounded by C_{D_i} and equicontinuous, since $|\operatorname{Re} f(s) - \operatorname{Re} f(t)| \leq |f(s) - f(t)|$. By [5, Arzela-Ascoli Theorem], any subsequence of (f_n) has a further subsequence along which $\operatorname{Re} f_n$ converges uniformly on D_i . Applying the same theorem to the imaginary parts along that subsequence gives a further subsequence along which f_n itself converges uniformly on D_i . Choose recursively index

sequences $(n_{1,k})_k$, along which f_n converges uniformly on D_1 , and $(n_{i+1,k})_k$, a subsequence of $(n_{i,k})_k$ along which f_n converges uniformly on D_{i+1} . Put $m_k = n_{k,k}$. As $(n_{k+1,j})_j$ is a subsequence of $(n_{k,j})_j$, we have $m_{k+1} \geq n_{k,k+1} > n_{k,k} = m_k$, so (f_{m_k}) is a subsequence of (f_n) . For each i , $(m_k)_{k \geq i}$ is a subsequence of $(n_{i,k})_k$, so f_{m_k} converges uniformly on D_i . Every compact $K \subseteq \Omega$ lies in some D_i , so (f_{m_k}) converges uniformly on every compact subset, and $\mathcal{F}$ is normal by proposition 4.1.20.

($\Rightarrow$) Assume $\mathcal{F}$ is normal, so that $\overline{\mathcal{F}}$ is compact. For $z \in \Omega$, the evaluation $h \mapsto h(z)$ is continuous on $C(\Omega)$, since $|h(z) - g(z)| < \epsilon$ for $h \in V_g(\{z\}, \epsilon)$. By [8, Continuity and Compactness] it carries $\overline{\mathcal{F}}$ onto a compact subset of $\mathbb{C}$, which contains A_z . This is (2).

For (1), let $E \subseteq \Omega$ be compact and $\epsilon > 0$. The sets $V_g(E, \epsilon/3)$ with $g \in \overline{\mathcal{F}}$ are open in $C(\Omega)$, since $V_g(E, \epsilon/3)$ contains $V_h(E, \epsilon/3 - \sup_E |h - g|)$ around each of its points h , and they cover $\overline{\mathcal{F}}$. So finitely many of them, centred at $g_1, \dots, g_m$, cover $\overline{\mathcal{F}}$. Each g_j is uniformly continuous on the compact set E ([8, Uniform Continuity Theorem]), so there is a $\delta > 0$ with $|g_j(z) - g_j(w)| < \epsilon/3$ for every j and all $z, w \in E$ with $|z - w| < \delta$. For $f \in \mathcal{F}$ choose j with $\sup_E |f - g_j| < \epsilon/3$. Then for such z, w ,

$$|f(z) - f(w)| \leq |f(z) - g_j(z)| + |g_j(z) - g_j(w)| + |g_j(w) - f(w)| < \epsilon$$

so $\mathcal{F}$ is uniformly equicontinuous on E , as we sought to show.

4.1.2 Montel's Theorem

We shall now focus on compact subsets of $\mathcal{H}(D)$. Note that the 1st condition of Arzela-Ascoli (for complex functions) implies that if A is normal, then the functions of A are uniformly bounded on every compact set.

This implication is in fact sufficient for holomorphic functions. The following theorem characterizes normal families in terms of sets and in terms of sequences:

Theorem 4.1.23: Montel's Theorem

Let $D \subseteq \mathbb{C}$ be open and let $A \subseteq \mathcal{H}(D)$ be a family of functions.

1. A is normal if and only if the functions in A are uniformly bounded on every compact set $K \subseteq D$, that is they are locally uniformly bounded. When D is connected, this holds if and only if $A' = \{f' \mid f \in A\}$ is locally bounded, every point of D having a neighbourhood on which the functions of A' are uniformly bounded, and there exists a $z_0 \in D$ such that $\{f(z_0) \mid f \in A\}$ is bounded.
2. If D is connected, A is normal and $z_0 \in D$, then a sequence $(f_k) \subseteq A$ is convergent (in the topology of $\mathcal{H}(D)$) if and only if for each $n \geq 0$, the sequence of n th derivatives $f_k^{(n)}(z_0)$ has a limit.

Proof:

The closure of A in $C(D)$ consists of limits of sequences in A , by [5, Characterizations of the Closure] applied to the metric of proposition 4.1.9, and such limits are holomorphic by lemma 4.1.3. So the closures of A in $C(D)$ and in $\mathcal{H}(D)$ coincide, and A is normal in the sense of definition 4.1.18 exactly when it is normal as a subset of $C(D)$.

A derivative estimate. Let f be holomorphic on D , let $w \in D$ and $r > 0$ with the closed disc $|t - w| \leq r$ contained in D , and let M bound $|f|$ on that closed disc. Then corollary 3.2.9 with $n = 1$ and the length estimate of proposition 3.1.1 give

$$|f'(w)| = \left| \frac{1}{2\pi i} \int_{|t-w|=r} \frac{f(t)}{(t-w)^2} dt \right| \leq \frac{1}{2\pi} \cdot \frac{M}{r^2} \cdot 2\pi r = \frac{M}{r}$$

Moreover, if $|f'| \leq M'$ on a disc containing the segment $[w, z]$, then f is a primitive of the closed form $f'(\zeta)d\zeta$, so theorem 3.1.9 gives $f(z) - f(w) = \int_{[w,z]} f'(\zeta)d\zeta$, and proposition 3.1.1 gives $|f(z) - f(w)| \leq M'|z - w|$.

1. Suppose A is normal. For a compact $K \subseteq D$, the map $h \mapsto \sup_K |h|$ is continuous on $\mathcal{H}(D)$, as it changes by less than ϵ on $V_g(K, \epsilon)$, so it is bounded on the compact closure of A ([8, Continuity and Compactness]). Hence the functions of A are uniformly bounded on K .

Conversely, suppose the functions of A are uniformly bounded on every compact set. Let $K \subseteq D$ be compact, and choose $r > 0$ with $3r$ less than the distance from K to $\mathbb{C} \setminus D$ (any $r > 0$ if $D = \mathbb{C}$). The set L of points at distance at most $2r$ from K is closed and bounded, hence compact, and lies in D , so $M = \sup_{f \in A, t \in L} |f(t)|$ is finite. For w at distance at most r from K , the closed disc $|t - w| \leq r$ lies in L , so the derivative estimate gives $|f'(w)| \leq M/r$ for all $f \in A$. If $z, w \in K$ and $|z - w| < r$, the segment $[w, z]$ lies in the disc $|t - w| < r$, all of whose points are within r of K , so $|f(z) - f(w)| \leq (M/r)|z - w|$. Given $\epsilon > 0$, the choice $\delta = \min\{r, \epsilon r/(M + 1)\}$ therefore shows that A is uniformly equicontinuous on K . The sets $\{f(z) \mid f \in A\}$ are bounded, $\{z\}$ being compact, so A is normal by theorem 4.1.22 and the first paragraph of this proof.

Now let D be connected. If A is uniformly bounded on compact sets, then for $w \in D$ take a closed disc K around w inside D . The argument of the previous paragraph bounds $|f'|$ uniformly on K , which contains a neighbourhood of w , so A' is locally bounded, and $\{f(z_0) \mid f \in A\}$ is bounded for any z_0 . Conversely, suppose A' is locally bounded and $\{f(z_0) \mid f \in A\}$ is bounded. Let $K \subseteq D$ be compact. Each $z \in K$ has a radius $r_z > 0$ and a bound M_z with the disc $|t - z| < 2r_z$ inside D and $|f'| \leq M_z$ there for all $f \in A$. Finitely many of the discs $|t - z_j| < r_{z_j}$ cover K , and we let $r = \min_j r_{z_j}$ and $M' = \max_j M_{z_j}$. If $z, w \in K$ with $|z - w| < r$ and $|z - z_j| < r_{z_j}$, then the segment $[w, z]$ lies in the disc $|t - z_j| < 2r_{z_j}$, so $|f(z) - f(w)| \leq M'|z - w|$. Thus A is uniformly equicontinuous on every compact subset of the domain D , and lemma 4.1.21 shows that every point has a neighbourhood on which the functions of A are uniformly bounded. Finitely many such neighbourhoods cover a given compact set, so A is uniformly bounded on every compact set.

2. If $f_k \rightarrow f$ in $\mathcal{H}(D)$, then proposition 4.1.11, applied n times, gives $f_k^{(n)} \rightarrow f^{(n)}$ uniformly on compact sets, and in particular $f_k^{(n)}(z_0) \rightarrow f^{(n)}(z_0)$ for every n .

Conversely, suppose $c_n = \lim_k f_k^{(n)}(z_0)$ exists for every $n \geq 0$. As A is normal, proposition 4.1.20 and lemma 4.1.3 show that every subsequence of (f_k) has a further subsequence converging uniformly on compact sets to a holomorphic function. Let g be such a limit for some subsequence. By the implication proved above, applied to that subsequence, $g^{(n)}(z_0) = c_n$ for every n . Let d be the metric of proposition 4.1.9. If $d(f_k, g) \not\rightarrow 0$, there are $\epsilon > 0$ and a subsequence (f_{l_j}) with $d(f_{l_j}, g) \geq \epsilon$ for all j . A further subsequence of it converges to a holomorphic h , and again $h^{(n)}(z_0) = c_n = g^{(n)}(z_0)$ for every n . The function $g - h$ is analytic (corollary 3.2.7) on the connected set D with all derivatives zero at z_0 , so $g = h$ by theorem 2.5.3. This contradicts $d(f_{l_j}, g) \geq \epsilon$ along a subsequence converging to $h = g$. Hence $f_k \rightarrow g$ in $\mathcal{H}(D)$, as we sought to show.

A famous result of this is akin to the *uniform boundedness principle*: if a sequence $(f_k) \subseteq \mathcal{H}(D)$ is uniformly bounded on every compact subset of D and converges pointwise on D , then it converges uniformly on compact subsets. Indeed, by theorem 4.1.23 and proposition 4.1.20 every subsequence of (f_k) has a further subsequence converging in $\mathcal{H}(D)$, and every such limit equals the pointwise limit, so all these limits coincide. If (f_k) did not converge to that common limit f , some subsequence would stay at distance at least $\epsilon > 0$ from f in the metric of proposition 4.1.9, and a further subsequence of it would converge to f , a contradiction.

Corollary 4.1.24: Normal, then Derivative Normal

Let $A \subseteq \mathcal{H}(D)$ be a normal family. Then:

$$A' = \{f' \mid f \in A\}$$

is a normal family

Proof:

The closure $\overline{A}$ of A is compact. The map $f \mapsto f'$ is continuous on $\mathcal{H}(D)$: this space is metric (proposition 4.1.9), convergence in it is uniform convergence on compact sets, and if $f_n \rightarrow f$ in this sense then $f'_n \rightarrow f'$ in the same sense (proposition 4.1.11), which is continuity by [5, Three Characterizations of Continuity]. So it carries $\overline{A}$ onto a compact set ([8, Continuity and Compactness]) containing A' . The space $\mathcal{H}(D)$ is metric (proposition 4.1.9), hence Hausdorff, so this compact set is closed ([8, compact subspaces of Hausdorff Spaces are closed]) and contains the closure of A' . A closed subset of a compact set is compact ([8, closed subsets of compact spaces are compact]), so A' is normal, as claimed.

Next, let D be non-empty and connected, and let $(f_k) \subseteq \mathcal{H}(D)$ be a locally bounded sequence such that, for some $z_0 \in D$, each sequence $(f_k^{(n)}(z_0))_k$ converges. Then the sequence converges uniformly on every compact subset of D , by theorem 4.1.23(2). The reason is that if f, g are limits of two subsequences of (f_k) , then $f^{(n)}(z_0) = g^{(n)}(z_0)$ for all $n \geq 0$, so $f = g$ by analytic continuation. The same argument applies when (f_k) converges pointwise on a set with an accumulation point in D , as two such limits agree on that set, hence everywhere by proposition 2.5.7 and theorem 2.5.3.

Finally, we show that local boundedness is preserved under the derivative, which is something that must be assumed for the real case but follows in the complex case:

Corollary 4.1.25: Local Boundedness preserved under derivative

A locally bounded family of holomorphic functions has locally bounded derivative

Proof:

This follows from Cauchy's representation of the derivative:

$$f'(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(t)}{(t-z)^2} dt$$

Thus, if γ is the boundary of a closed disk in Ω of radius r , then $|f'(z)| \leq 4M/r$, in the concentric disc of radius $r/2$, and M is the bound of $|f|$ on C , showing it is locally bounded.

4.2 Factorization of Holomorphic Functions

In this section, explore the convergence of series of holomorphic and meromorphic functions. A driving factor in the convergence of series is generalizing the notion of characterizing polynomials by their roots. Given a polynomial, we may take it to be a product of monomials, which using partial fraction decomposition we can turn into a finite sum. We may try the same technique with meromorphic functions, identifying the roots and poles, and then defining a series, the “partial fraction decomposition”, that let's us analyze the function in terms of our better known analysis techniques. Many functions, such as our usual trigonometric functions, logarithm, and exponential function shall fall under this category. However, whole new types of functions will be found that shall capture really important geometric, number theoretic, PDE, or algebraic information in ways that are still being studied today. For example, the study of modular forms (generalization of Weierstrass p -functions that will be an example of an infinite partial fraction) are closely linked to number theory, and the Riemann zeta-function is closely linked to the distribution of primes. From the perspective of complex analysis, these are just any other functions like the trigonometric functions (granted not usually entire or holomorphic). Hence, knowing how to identify properties and work with such functions is a useful skill to possess.

In many ways, this part of complex analysis is the continuation of usual Calculus; the study of Taylor series generalizes to the study of analytic, hence holomorphic, functions. Having essentially exhausted the theory of entire holomorphic functions (the study of power series), we study not necessarily entire holomorphic functions (with more lenient growth factors) and meromorphic functions (which may have “ramification” around singularities, that is new behavior such that those of $1/z$). These are naturally interpreted as infinite rational polynomials. Though it may seem like an easy step up, it turns out to still have a lot of mysteries! For example, the Riemann-zeta function will be seen to be a meromorphic function defined via meromorphic convergence (as we will demonstrate soon), however we have yet (as of writing these notes) to fully understand the nature of their poles! Hence, this can be thought of as a natural continuation of calculus!

One more insight that is worth mentioning is the link to number theory and modular forms. The closed form of power series hold a (countably) infinite amount of numerical information within a

single compact expression. For example, if we take the sequence $(1, 1, \dots, 1, \dots)$, then this can be associated to

$$q + q^2 + \dots + q^n + \dots$$

Factoring q and noticing the rest is the usual $1/(1-q)$ when $|q| < 1$, we see that this sequence can be represented as $q/(1-q)$. If we take the sequence $(1, 2, 3, \dots, n, \dots)$, and take:

$$q + 2q^2 + \dots + nq^n + \dots$$

and again factor the q , notice that the result series is the derivative of $1/(1-q)$, and hence $q/(1-q)^2$ holds information about the sequence $(1, 2, 3, \dots, n, \dots)$. Let us link this to modular forms with the following insight that shall not be proved. Consider the equation $y^2 + y = x^3 - x^2$. We can take solutions to this mod p for primes p . Counting the number of solutions, for each prime, we get the sequence:

$$(4, 4, 4, 9, 10, 9, \dots)$$

It shall not be proven here, but this sequence is in some sense “fudged” by the prime, and it is more accurate to “normalize” to p -# of solutions; this number shall also grow but hover around 0. This gives us a new sequence:

$$(-2, -1, 1, -2, 1, 4, \dots)$$

Now, consider the power series given by the infinite product:

$$q(1-q)^2(1-q^{11})^2(1-q^2)^2(1-q^{22})^2(1-q^3)^2(1-q^{33})^2 \dots$$

Expanding this, we get:

$$q - 2q^2 - q^3 + 2q^4 + q^5 - 2q^7 - 2q^9 - 2q^{10} + q^{11} - 2q^{12} + 4q^{13} + \dots$$

Notice that the prime powers of each of these coefficients match the normalized solutions! The modularity theorem (the theorem resulting from the Taniyama-Shimura conjecture) states that this happens for any curve that is an elliptic curve, which for all intent and purposes of this motivating section they are curves of the form $y^2 = x^3 + ax + b$ ⁴, and links them to a special type of power series called a *modular form*. This was the key for proving the infamous *Fermat Last Theorem* (that every elliptic curve is *modular*).

4.2.1 Series of Meromorphic Functions

Starting with the definition of meromorphic convergence. We require some a little more technical than “a sequence of meromorphic functions” as we must insure our poles never cluster:

Definition 4.2.1: Meromorphic Convergence

Let (f_n) be a sequence of meromorphic function on $\Omega \subseteq \mathbb{C}$. Then $\sum_n^\infty f_n$ converges uniformly (resp. uniformly and absolutely) on $X \subseteq \Omega$ if for all but finitely many terms in all compact subset $K \subseteq X$ have no poles on X , and eliminating the terms with poles forms a uniformly (resp uniformly and absolutely) convergent series of holomorphic functions in Ω

⁴More precisely, they are curves of ‘genus one’, which the curve in the above example is one

For a sequence $\sum_n f_n$ of meromorphic functions with all but finitely many of them having poles in K , we may split it as:

$$\sum_{n=1}^{n \leq n_0} f_n + \sum_{n > n_0}^{\infty} f_n$$

where for all $n > n_0$, f_n is holomorphic. The left sum in the above equation is meromorphic, being a finite sum of meromorphic function, and the right sum is a holomorphic function since it is a uniformly converging series of holomorphic function on U . Note that the meromorphic function does not depend on the choice of n_0 by the independence of associativity.

Theorem 4.2.2: Convergence Of Meromorphic Function

Let (f_n) be a sequence of meromorphic functions. Then if $\sum_n f_n$ converges uniformly on compact subset of Ω , then $f = \sum_n f_n$ is meromorphic on Ω and $\sum f'_n$ converges uniformly to f' on Ω

Proof:

$f = \sum_n f_n$ is already meromorphic since for a given division:

$$\sum_{n \leq n_0} f_n + \sum_{n > n_0} f_n$$

we have the right sum converges uniformly to some holomorphic f_0 , and the left sum is a meromorphic function, say g . Then $f = g + f_0$ is meromorphic.

For the derivative, if we consider

$$f = \sum_{n \leq n_0} f'_n + \left(\sum_{n > n_0} f_n \right)'$$

where since the left sum is a finite sum we may pass the derivative through, and for the right sum by proposition 4.1.11 we can bring the derivative in as well, but that give

$$f' = \sum_n f'_n$$

as we sought to show.

Example 4.2.3: Series Of Meromorphic Functions

Consider

$$\sum_{n \in \mathbb{Z}} \frac{1}{(z - n)^2}$$

We shall show that this series uniformly converges on all compact subsets of $\mathbb{C}$. First, notice that:

$$f(z + 1) = f(z)$$

Hence, it suffices to show these converge on strips:

$$S_{x_0, x_1} = \{z \in \mathbb{C} \mid x_0 \leq \operatorname{Re} z \leq x_1\}$$

where $x_0, x_1 \in \mathbb{R}$. Then for any such strip, there are only a finite number of integers n where $x_0 \leq n \leq x_1$; hence the series

$$\sum_{n < x_0} \left| \frac{1}{(z - n)^2} \right|$$

is bounded by $\sum_{n < x_0} \frac{1}{(x_0 - n)^2}$, and hence the partial series is uniformly convergent on this strip. Similarly, the partial series

$$\sum_{x_1 < n} \frac{1}{(z - n)^2}$$

is also bounded and hence uniformly converges too. Thus, removing a suitable finite number of terms from the series, we are left with a series of holomorphic functions which uniformly converge in the strip, and hence it converges holomorphically.

What is this function? Let $f(z)$ represent this series. Then we know it is a meromorphic function defined on all of $\mathbb{C}$. The function f has period 1, that is

$$f(z + 1) = f(z)$$

since

$$\sum_n \frac{1}{(z + 1) - n)^2} = \sum_m \frac{1}{(z - m)^2}$$

where $m = n + 1$. The poles of f are the integers $z = n$, and they are all double poles. The residue of these poles are zero since, in some neighbourhood of $z = n$

$$f(z) = \frac{1}{(z - n)^2} + g(z)$$

where g is holomorphic. From this, we shall deduce the following

Proposition 4.2.4: Important Meromorphic Series

Let $f(z) = \sum_{n \in \mathbb{Z}} \frac{1}{(z - n)^2}$. Then

$$f(z) = \left(\frac{\pi}{\sin(\pi z)} \right)^2$$

Proof:

Let $z = x + iy$. First, note that $f(z)$ uniformly tends to zero as $|y| \rightarrow \infty$, that is for all $\epsilon > 0$, there exists an a such that $|y| \geq a$ implies $|f(z)| < \epsilon$. To see this, let z remain in the strip $x_0 \leq x \leq x_1$ and $|y| \geq 0$ for $a > 0$. As we have seen, this series is uniformly convergent to a holomorphic function as $|y| \rightarrow \infty$, where as the term tends to 0 uniformly with respect to the x strip. Thus, the sum of the series uniformly tends to 0 as $|y| \rightarrow \infty$ with respect to the x strip. But $f(z)$ has period 1, so by applying the above argument to a strip of width at least 1, we get that $f(z)$ tends to 0 as $|y| \rightarrow \infty$ absolutely with respect to x .

Now, consider $g(z) = \left(\frac{\pi}{\sin(\pi z)} \right)^2$. It has the following properties:

1. It is meromorphic in $\mathbb{C}$ and has period 1 (immediate)

2. it has double poles at all integers $z = n \in \mathbb{Z}$ (expand the Taylor series) with principal parts $\frac{1}{(z-n)^2}$
3. $g(z)$ tends to 0 absolutely with respect to x as $|y| \rightarrow \infty$

To see the third property, recall that $|\sin(\pi z)|^2 = \sin^2(\pi x) + \sinh^2(\pi y)$, hence $|\sin(\pi z)|$ uniformly tends to infinity with respect to x as $|y|$ tends to infinity.

Finally, we shall show that $f(z) - g(z)$ is bounded with constant zero to show they are equal. The function $f - g$ is holomorphic in $\mathbb{C}$ since f, g have the same poles and same principal parts. To show it's bounded, consider a strip $x_0 \leq x \leq x_1$. It is bounded for $|y| \leq a$ (since it's on a compact set), and hence $f - g$ is bounded on the whole plane by periodicity. Thus by Liouville's theorem it is constant. Since $f - g$ tends to 0 as $|y| \rightarrow \infty$, it must be that the constant is 0, showing equality.

This proof shows a general strategy to finding when a series is equal to a function:

1. Check for the principal part (all terms with negative exponents in the power series expansion)
2. The convergence in the limits
3. The periodicity

A famous consequence of this identity is the evaluation of the even zeta values, read off by comparing the Laurent expansions of its two sides at the origin.

Proposition 4.2.5: Even Zeta Values $\zeta(2)$ and $\zeta(4)$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \qquad \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$$

Proof:

Expand both sides of the identity $\left(\frac{\pi}{\sin(\pi z)}\right)^2 = \sum_{n \in \mathbb{Z}} \frac{1}{(z-n)^2}$ (proposition 4.2.4) in powers of z near $z = 0$ and compare coefficients. On the right, splitting off the $n = 0$ term and expanding $\frac{1}{(z-n)^2} = \frac{1}{n^2} \sum_{m \geq 0} (m+1)(z/n)^m$ for $n \neq 0$,

$$\sum_{n \in \mathbb{Z}} \frac{1}{(z-n)^2} = \frac{1}{z^2} + \sum_{n \neq 0} \frac{1}{n^2} + 3z^2 \sum_{n \neq 0} \frac{1}{n^4} + \cdots = \frac{1}{z^2} + 2\zeta(2) + 6\zeta(4)z^2 + \cdots$$

the odd powers cancelling in the pairs $n \leftrightarrow -n$. On the left, from $\sin(\pi z) = \pi z \left(1 - \frac{(\pi z)^2}{6} + \frac{(\pi z)^4}{120} - \cdots\right)$ one gets $\frac{\pi}{\sin(\pi z)} = \frac{1}{z} + \frac{\pi^2}{6}z + \frac{7\pi^4}{360}z^3 + \cdots$, and squaring,

$$\left(\frac{\pi}{\sin(\pi z)}\right)^2 = \frac{1}{z^2} + \frac{\pi^2}{3} + \frac{\pi^4}{15}z^2 + \cdots$$

Matching the constant terms gives $2\zeta(2) = \pi^2/3$, that is $\zeta(2) = \pi^2/6$. Matching the coefficients of z^2 gives $6\zeta(4) = \pi^4/15$, that is $\zeta(4) = \pi^4/90$.

The Laurent-series comparison behind proposition 4.2.5 extends to give $\zeta(2k)$ for every k from the same identity. The next expansion instead produces a new series from an old one:

Proposition 4.2.6: Tangent Function As Meromorphic Series

Let

$$f(z) = \frac{1}{z} + \sum_{0 \neq n \in \mathbb{Z}} \left(\frac{1}{z-n} + \frac{1}{n} \right)$$

Then

$$f(z) = \frac{\pi}{\tan(\pi z)}$$

Proof:

First, this series uniformly converges on compact sets. It's poles are at $z = n$ for $n \in \mathbb{Z}$ which are all simple poles with residue 1. Note that:

$$f'(z) = \frac{-1}{z^2} - \sum_{n \neq 0} \frac{1}{(z-n)^2} = - \left(\frac{\pi}{\sin(\pi z)} \right)^2 = \frac{d}{dz} \left(\frac{\pi}{\tan(\pi z)} \right)$$

Hence, $f(z) - \frac{\pi}{\tan(\pi z)}$ is constant. Since $f(-z) = -f(z)$, we see that f is odd, and hence the constant must be zero.

Note that the series can be re-arranged by pairing together the term for n and $-n$:

$$\left(\frac{1}{z-n} + \frac{1}{n} \right) + \left(\frac{1}{z+n} - \frac{1}{n} \right) = \frac{2z}{z^2 - n^2}$$

Hence, we equivalently have:

$$\frac{1}{z} + \sum_{n \geq 1} \frac{2z}{z^2 - n^2} = \frac{\pi}{\tan(\pi z)}$$

As a final one, it can be shown that

$$\sum_{n \in \mathbb{Z}} \frac{(-1)^n}{(z-n)^2} = \frac{\pi^2}{(\sin(\pi z))(\tan(\pi z))}$$

Using this one, we can use some differentiation to show that

$$\frac{1}{z} \sum_{n \geq 1} (-1)^n \frac{2z}{z^2 - n^2} = \frac{\pi}{\sin(\pi z)}$$

If you have done some representation theory, these functions can be thought of as a sort of “averaging” similar to what is given in Maschke’s Theorem (recall the formula $\pi = \frac{1}{n} \sum_{g \in G} g \pi_i g^{-1}$). Though this connection is not of much use to us, this idea of somehow averaging shall come back later when look to find the class of all meromorphic functions define on $\mathbb{C}$.

4.2.2 Meromorphic Series Decomposition

On S^2 , a meromorphic function is rational, so there are in particular only finitely many poles. In this section, we classify all meromorphic functions on $\mathbb{C}$. On $\mathbb{C}$, there can be infinitely many poles. Given some prescribed list of poles (and of principal parts), can we always find a meromorphic function with given poles? In particular, given a set of poles $\{b_k\} \subseteq \mathbb{C}$, $\lim_{k \rightarrow \infty} b_k = \infty$ and $\{p_k(z)\}$ is a polynomial without constant term, can we find a meromorphic function with poles b_k and principal parts $p_k\left(\frac{1}{z-b_k}\right)$? The answer is yes! This is not immediately obvious given the strong restrictions placed on meromorphic functions. The following theorem demonstrates how to construct such a function.

Theorem 4.2.7: Mittag-Leffler Theorem

Any meromorphic function on $\mathbb{C}$ can be written of the form:

$$f(z) = \sum_{k=1}^{\infty} \left(P_k \left(\frac{1}{z-b_k} \right) - p_k(z) \right) + g(z)$$

where $p_k(z)$ are the chosen so that the sum converges uniformly on compact sets and $g(z)$ is an entire function

Proof:

Assume $b_k \neq 0$. Then $p_k((z-b_k)^{-1})$ is holomorphic in $|z| < |b_k|$, so we can expand the Taylor series at 0. Let $p_k(z)$ be the sum of the first n_k terms where n_k is chosen so that

$$\left| p_k \left(\frac{1}{z-b_k} \right) - p_k(z) \right| \leq \frac{1}{2^k} \quad |z| \leq \frac{|b_k|}{2}$$

Then we will show that $\sum (p_k((z-b_k)^{-1}) - p_k(z))$ converges uniformly and absolutely on $|z| \leq r$ for any r . For this, choose m such that $|b_k| > 2r$ if $k \geq m$. Then

$$\sum_{k=m}^{\infty} (p_k((z-b_k)^{-1}) - p_k(z))$$

converges uniformly and absolutely on $|z| \leq m$ by comparison with $\sum \frac{1}{2^k}$.

Finally, any meromorphic function with these poles and principal parts differ from one another by a holomorphic function, which gives the final form and completing the proof.

4.2.8 Mittag-Leffler Problem This theorem is the solution of the Mittag-Leffler Problem on the Riemann surface $\mathbb{CP}^1$. The Mittag-Leffler problem is the following. Let η_i be a Laurent principal part. Then under what condition does there exist an $f \in \mathcal{C}(C)$ (for a Riemann surface C) such that in a neighbourhood of p_i , f has η_i as a Laurent principal part? More on this in chapter 6, as well as its connection to the Riemann-Roch Theorem in [1].

4.2.3 Infinite Products

Every polynomial can be represented as a product of linear terms. In the following we generalize this idea, leading us to the notion of defining a function to be an infinite product of holomorphic functions. We shall decompose the trigonometric functions as product of roots, and shall also create new functions using this method. We shall show that all entire functions can be factorized into an (infinite) product of linear terms with correction factors up to an entire function.

We start with the element-wise definition of infinite product:

Definition 4.2.9: Infinite Product

Let (a_i) be a sequence. Then we say that $\prod_i^\infty a_i$ exists if $p_k = \prod_i^k a_i$ converge, that is

$$\prod_i^\infty a_i = \lim_k \prod_i^k a_i = \lim_k p_k = p$$

More generally, we say the product *almost converges* if the product converges after removing finitely many terms (we may consider there the “roots”)

Certainly, we require that $p_k \rightarrow 1$ if all $a_k \neq 0$. If $a_n \neq 0$ for all n we can write $u_n = 1 + a_n$, and for n sufficiently large,

$$\sum_n \log(1 + a_n)$$

is well-defined where we take the principal branch of $\log$. If $s_n = \sum_k^n \log(1 + a_k)$, then $p_n = e^{s_n}$ is well-defined.

Lemma 4.2.10: Converges Of Partial Series, Then Convergence Of Partial Products

Let s_n, p_n be defined as above. Then if s_n converges if and only if p_n converges.

Proof:

If s_n converges, then certainly p_n converges. Conversely, suppose p_n converges. The problem is that $\log(n)$ is a many-valued function, or in particular we must be careful with choosing the appropriate branch (the appropriate imaginary part). Let $p = \lim_n p_n$. Choose a branch by taking $\log p = \log |p| + i \arg(p)$. Then consider

$$\log p_n = \log |p_n| + i \arg p_n \quad \arg p_n \in (\arg p - \pi, \arg p + \pi)$$

Then we get $s_n = \log p_n + 2\pi i k_n$, $k_n \in \mathbb{Z}$. Then:

$$\log(1 + a_{n+1}) = s_{n+1} - s_n = \log(p_{n+1}) - \log(p_n) + 2\pi i(k_{n+1} - k_n)$$

For n large enough, we have $\arg(1 + a_n), \arg(p_n - p), \arg(p_{n+1} - p)$ are all $< 2\pi/3$. Thus, $k_{n+1} = k_n$ since they must all be in $\mathbb{Z}$. Finally, setting $k = k_n$ for large enough n , we get

$$s_n \rightarrow \log p + 2\pi i k$$

showing convergence the other way, completing the proof.

Definition 4.2.11: Product Absolute Convergence

Let $(a_n) \subseteq \mathbb{C}$. Then $\prod a_n$ converge absolutely if

$$\sum_n \log(1 + a_n)$$

converges absolutely.

This is equivalent to saying that $\sum a_n$ converges absolutely, since (by using the Taylor expansion)

$$\lim_{z \rightarrow 0} \frac{\log(1+z)}{z} = 1 \implies \left| \frac{|\log(1+a_n)|}{|a_n|} - 1 \right| < \epsilon$$

hence

$$(1 - \epsilon)|a_n| < |\log(1 + a_n)| < (1 + \epsilon)|a_n|$$

showing equivalence. Hence, $\prod_n (1 + a_n)$ converges absolutely if and only if $\sum_n a_n$ converges absolutely.

With convergence of infinite products established, we now define the convergence of the infinite functions:

Definition 4.2.12: Infinite Product of Functions

Let (f_n) be a sequence of continuous functions defined on $D \subseteq \mathbb{C}$. Then we say that $\prod_n f_n$ converges uniformly (resp. uniformly and absolutely) on compact sets $K \subseteq D$ if:

1. As $n \rightarrow \infty$, f_n uniformly tends to 1 on K . In particular, for n sufficiently large, $|f_n - 1| < \epsilon$ on K . Hence, $\sum_n \log f_n$ converges uniformly.
2. the series is said to converge *absolutely* if $\sum_n \log f_n$ converge absolutely.

For a succinct condition, let $f_n = 1 + u_n$; then condition (1) says that u_n converges uniformly to 0 in K , and when u_n is small, $\log f_n$ and u_n are equivalent to the first order, hence we see that $\sum_n u_n$ converges uniformly on K . Thus we get that $\prod f_n$ converges uniformly if and only if $\sum u_n$ converges uniformly:

$$\prod_n f_n \iff \sum_n u_n$$

Proposition 4.2.13: Uniform Product Convergence Is Holomorphic

Let $(f_n) \subseteq \mathcal{H}(D)$ be a sequence of nonzero holomorphic functions. Then if $\prod_n f_n$ converges uniformly,

$$f = \prod_n f_n$$

is holomorphic, and we can write for any $p \in \mathbb{N}$

$$f = f_1 f_2 \cdots f_p \left(\prod_{n>p} f_n \right) \quad (4.1)$$

The set of zeros of f is the union of the sets of zeros of the functions f_n , the order of multiplicity of a zero of f is equal to the sum of orders of multiplicity given by each f_n

Proof:

f is holomorphic since $\prod_k^n f_k$ uniformly converges to f (the product of holomorphic functions is holomorphic). The decomposition is immediate. Since $f_n \rightarrow 1$ uniformly, for n sufficiently large f_n has no zeros in U , hence the zeros of f is the union of the set of zeros of (f_m) for $m < n$, and similarly for their multiplicity.

Proposition 4.2.14: Logarithmic Derivative Converges

Given $\prod f_n \rightarrow f$ uniformly and absolutely on D , the series $\sum_n f'_n/f_n$ of meromorphic functions converges uniformly on compact subsets of D and its sum is the logarithmic derivative of f'/f .

Proof:

Let U be a pre-compact (relatively compact) open subset of D . Define

$$g_p = \exp \left(\sum_{n>p} \log f_n \right)$$

which is holomorphic for sufficiently large p (since f_n is nonzero for sufficiently large p). Then by equation (4.1) and the above, we can take the derivative of $\log(f)$ (which can be done component-wise) and we have

$$\frac{f'}{f} = \sum_{n \leq p} \frac{f'_n}{f_n} + \frac{g'_p}{g_p} = \sum_{n \leq p} \frac{f'_n}{f_n} + \sum_{n>p} \frac{f'_n}{f_n}$$

Note that $\sum_{n>p} \frac{f'_n}{f_n}$ converges absolutely on compact subsets of D since by assumption the series $\sum_{n>p} \log f_n$ converges absolutely to $\log g_p$ meaning it's derivative converge uniformly absolutely, hence the series of derivatives of these logarithm converges (absolutely on compact subsets) to the derivative g'_p/g_p . Hence, the above equation becomes:

$$\frac{f'}{f} = \sum_n \frac{f'_n}{f_n}$$

as we sought to show.

Notice how the use of the logarithmic derivative made it very easy to take the derivative, as it is linear in the finite sums, while the product would require keeping track of the product rule!

Example 4.2.15: Infinite Product

1. We shall try to write $\sin(\pi z)$ as an infinite product. Let

$$f(z) = z \prod_{n \geq 1} \left(1 - \frac{z^2}{n^2}\right)$$

This converges uniformly on compact subsets of $\mathbb{C}$ since $\sum_n z^2/n^2$ converges uniformly on compact sets. Hence, $f(z)$ is holomorphic on $\mathbb{C}$ and its zeros are simple and have exist at the integers.

Taking the logarithmic derivative, we get

$$\frac{f'(z)}{f(z)} = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2}$$

but the term on the right hand side has already been studied, namely:

$$\frac{\pi}{\tan(\pi z)} = \frac{(\sin(\pi z))'}{\sin(\pi z)}$$

Hence, $f'/f = g'/g$, meaning taking the integral we get:

$$\frac{f(z)}{z} = c \frac{\sin(\pi z)}{z}$$

What remains is to determine c . We know that $f(z)/z \rightarrow 1$ as $z \rightarrow 0$, and since $\sin(\pi z)/z$ has π as the limit, we get $c = 1/\pi$, giving us

$$\frac{\sin(\pi z)}{\pi z} = \prod_n \left(1 - \frac{z^2}{n^2}\right)$$

2. (Γ -function) Consider

$$\begin{aligned} g_n(z) &= z(1+z) \left(1 + \frac{z}{2}\right) \cdots \left(1 + \frac{z}{n}\right) \frac{1}{n^z} \\ &= \frac{z(z+1)(z+2) \cdots (z+n)}{n!} n^{-z} \end{aligned}$$

For $n \geq 2$, we have

$$\frac{g_n(z)}{g_{n-1}(z)} = \left(1 + \frac{z}{n}\right) \left(1 - \frac{1}{n}\right)^z = f_n(z)$$

3. (Weiestrass infinite product)

Infinite Fundamental Theorem of Algebra

Some basic classification questions should be asked, for example can we always find a holomorphic function with certain given zeros? If there are finitely many zeros (given the sequence of zeros do not converge, since zeros of holomorphic functions are isolated), then polynomial would be the answer, however what about countably many zeros? Can we find all such entire functions? First, if $g(z)$ is entire, then $f(z) = e^{g(z)}$ remains entire and has no zeros, and f'/f is the derivative of the entire function $g(z)$. If $a_1, a_2, \dots, a_n \in \mathbb{C}$ (counted for multiplicity), then we may define for sum $m \geq 0$ and entire holomorphic function g :

$$f(z) = z^m e^{g(z)} \prod_k^n \left(1 - \frac{z}{a_k}\right)$$

If $n = \infty$, then we can do a similar trick as when the Weierstrass $\wp$ function: take the Taylor series expansion of the log

$$\log \left(1 - \frac{z}{a_k}\right) = \frac{-z}{a_k} - \frac{1}{2} \left(\frac{z}{a_k}\right)^2 - \frac{1}{3} \left(\frac{z}{a_k}\right)^3 - \dots$$

Finally, we must eliminate enough terms for our series to converge. The culmination of all of these steps is given in the following:

Theorem 4.2.16: Weierstrass Factorization Theorem

Let $(a_n) \subseteq \mathbb{C}$ where $\lim_{n \rightarrow \infty} a_n = \infty$. Then there is an entire function with zeros at each a_n . The most general form of such a function is:

$$z^m e^{g(z)} \prod_k^\infty \left[\left(1 - \frac{z}{a_k}\right) e^{p_k(z)} \right]$$

where

$$p_k(z) = \frac{z}{a_k} + \frac{1}{2} \left(\frac{z}{a_k}\right)^2 + \dots + \frac{1}{m_k} \left(\frac{z}{a_{m_k}}\right)^{m_k}$$

with $m_k \in \mathbb{Z}$ chosen so the series converges uniformly and absolutely on compact sets

Conversely, if f is an entire holomorphic function with zeros a_n where $\lim_{n \rightarrow \infty} a_n = \infty$, then f is of the above form.

Note that since this is an iff, this is a generalization of the Fundamental Theorem of Algebra!

Note that the z^m term is to signify a root at the origin, the $e^{p(x)}$ is the correction factors to insure convergence, and the $e^{g(z)}$ is the “similarity” factor, that is if we find holomorphic functions with the same roots, then they are equal up to a $e^{g(z)}$

Proof:

Take $(a_n) \subseteq \mathbb{C}$ where $\lim_{n \rightarrow \infty} a_n = \infty$ and let

$$p_k(x) = \frac{z}{a_k} + \frac{1}{2} \left(\frac{z}{a_k}\right)^2 + \dots + \frac{1}{m_k} \left(\frac{z}{a_{m_k}}\right)^{m_k}$$

be the coefficient of the exponent in the product. Then it suffices to show that

$$\prod_k \left[\left(1 - \frac{z}{a_k} \right) e^{p_k(z)} \right]$$

converges, which is equivalent to

$$\sum_{k=1}^{\infty} \log \left(1 - \frac{z}{a_k} \right) + p_k(z) = \sum_{k=1}^{\infty} g_k(z)$$

and $\log$ is chosen so that $\log(1 - z/a_k) + p_k(z)$ is the principal branch of $\log$ of the corresponding function in the product. Choose $g_k(z)$ so that the imaginary part is between $-\pi$ and π . Taking the Taylor series of $\log(1 - z/a_k)$, we get

$$\log \left(1 - \frac{z}{a_k} \right) = -\frac{z}{a_k} - \frac{1}{2} \left(\frac{z}{a_k} \right)^2 - \dots$$

so choose

$$p_k(z) = \frac{z}{a_k} + \left(\frac{z}{a_k} \right)^2 + \dots + \frac{1}{m_k} \left(\frac{z}{a_k} \right)^{m_k}$$

where m_k will be chosen soon to be large enough for the series to converge. To find m_k , take $|z| \leq r$ and consider a_k where $|a_k| > r$. Then

$$g_k(z) = \frac{-1}{m_k + 1} \left(\frac{z}{a_k} \right)^{m_k+1} - \frac{1}{m_k + 2} \left(\frac{z}{a_k} \right)^{m_k+2} - \dots$$

So:

$$|g_k(z)| \leq \frac{1}{m_k + 1} \left(\frac{r}{|a_k|} \right)^{m_k+1} \left(1 - \frac{r}{|a_k|} \right)^{-1}$$

Now we choose m_k so that

$$\sum_{k=1}^{\infty} \frac{1}{m_k + 1} \left(\frac{r}{|a_k|} \right)^{m_k+1}$$

converges (for example, choose $m_k = k$). Then $|g_k(z)| \rightarrow 0$, in the part between $(-\pi, \pi)$ for k large enough. Thus $\sum_k g_k(z)$ is uniformly and absolutely convergent in $|z| \leq r$, showing that the product is well-defined. Let's label such a product $P(z)$ for the next part of the proof.

For the converse, Given an entire holomorphic function f with (infinite) roots a_n where $\lim_{n \rightarrow \infty} a_n = \infty$, construct the product $P(z)$ as just done. Then consider:

$$h(z) = \frac{f(z)}{P(z)}$$

As the zero's cancel out, we get that $h(z)$ is an entire holomorphic that is nowhere zero. Thus, by exercise 3.2.1(3.2.1[3.2.1(1)]) we can define $\log(h(z)) = g(z)$ where $g(z)$ is entire, and so $h(z) = e^{g(z)}$. Then:

$$f(z) = h(z)P(z) = e^{g(z)}P(z)$$

giving us our decomposition, completing the proof.

Thus, it was proved that we can find functions with poles and zeros at fixed locations. This allows us to construct some entire functions with many well-controlled properties. The following shows that we may express any meromorphic function in a “global manner” as a quotient of two entire holomorphic functions:

Corollary 4.2.17: Meromorphic Quotient of Entire Function

Every meromorphic function on $\mathbb{C}$ is a quotient of 2 entire functions

Proof:

Let $h(z)$ be a meromorphic function on $\mathbb{C}$. Then we may choose an entire function $g(z)$ whose zeros are the poles of $h(z)$ (with multiplicity). Hence, $g(z)h(z) = f(z)$ is an entire holomorphic. Thus

$$h(z) = \frac{f(z)}{g(z)}$$

completing the proof.

This next theorem shows we can be even more precise: given some [isolated] sequence of points, we may choose corresponding points in $\mathbb{C}$ (with multiplicity), and find a holomorphic function with the appropriate values:

Corollary 4.2.18: Determining Holomorphic Function given Sequence

Given $(a_k), (b_k) \subseteq \mathbb{C}$, $\lim_{k \rightarrow \infty} a_k = \infty$, $m_k \in \mathbb{N}$, then there is an entire function $f(z)$ such that every a_k is a root of order m_k of $f(z) = b_k$

Proof:

Near a_k , $f(z)$ looks like $b_k = (z - a_k)^{m_k} \ell(z)$ where $\ell(z)$ is something nonzero near a_k . Now, there is an entire function $g(z)$ with zeros of order $m_k + 1$ at a_k . Now take $f(z) = g(z)H(z)$ where

$$H(z) = \left(\frac{b_k}{g(z)} + h(z) - \frac{b_k}{g(z)} \right)$$

and where $h(z)$ is a meromorphic function with poles a_k with principal part $\frac{1}{z - a_k}$ plus the principal at a_k of $\frac{b_k}{g(z)}$.

Corollary 4.2.19: Factorization Results

1. Let f denote an entire function (not identically zero). Let n be a natural number. Prove that there exists an entire function g such that $f = g^n$ if and only if every zero of f has order divisible by n
2. Denote $Z(f)$ the set of zeros of a function f . Prove that if f_1, f_2 are entire functions such that $Z(f_1) \cap Z(f_2) = \emptyset$, then there exist entire function g_1, g_2 such that $f_1 g_1 + f_2 g_2 = 1$

Proof:

1. If $f = g^n$ on $\mathbb{C}$, then if $f(z) = 0$, $g(z)^n = 0$, showing that the order of the pole must be divisible by n .

Conversely, suppose that f has zeros with order divisible by n : $a_1, a_2, \dots, a_k$ with order $n_1, n_2, \dots, n_k$. Define g_1 to be an entire holomorphic function with zeros at a_i with order n_i/n . Then:

$$f/(g_1^n) = g_2'$$

is a nonzero entire function. By homework 1 question 4, we have that there exists a g_2 such that $g_2' = (g_2)^n$. Hence:

$$f = (g_1^n g_2^n) = (g_1 g_2)^n = g^n$$

as we sought to show.

2. Let f_1, f_2 be entire functions with disjoint roots. We want to find entire functions g_1, g_2 such that $f_1 g_1 + f_2 g_2 = 1$. Define h to be the entire function with roots from f_1 of the same order and roots of f_2 at 1 of the same order. Define:

$$g_1 := h/f_1$$

g_1 is entire since the multiplicity of the poles of f_1 match the multiplicity of the roots of h at each pole of f_1 . Next, define:

$$g_2 := \frac{1 - f_1 g_1}{f_2} = \frac{1 - f_1(h/f_1)}{f_2} = \frac{1 - h}{f_2}$$

Then g_2 is entire since for each root of f_2 , the numerator has the same root with the same order. Hence, we have:

$$f_1 g_1 + f_2 \left(\frac{1 - f_1 g_1}{f_2} \right) = f_1 g_1 + 1 - f_1 g_1 = 1$$

as we sought to show.

4.2.4 Application: Gamma Function

While $\sin$ (and $\cos$) have zeros at all the integers, we shall introduce a (rather simple) function which has zeros at all the natural numbers. With the technics we have given above, we see that such a function could be

$$G(z) = \prod_{n=1}^{\infty} \left(1 + \frac{z}{n} \right) e^{-z/n}$$

with the correction term $e^{-z/n}$ insuring convergence, and has roots at all the negative integers. Naturally, $G(-z)$ has roots at all the positive integers. Using $G(z)$ and $G(-z)$, we rather simply construct a function with poles at all the integers, namely:

$$zG(z)G(-z) = \frac{\sin(\pi z)}{\pi}$$

By construction, G has some nice properties, for example $G(z-1)$ has the same zeros as $G(z)$ along with a zero at the origin. Thus, by using theorem 4.2.16 that

$$G(z-1) = ze^{\gamma(z)}G(z)$$

For some entire function $\gamma(z)$. To find it, we take the logarithmic of both sides giving us:

$$\sum_{n=1}^{\infty} \left(\frac{1}{z-1+n} - \frac{1}{n} \right) = \frac{1}{z} + \gamma'(z) + \sum_{n=1}^{\infty} \left(\frac{1}{z+n} - \frac{1}{n} \right)$$

Notice that the left hand side can be manipulated like so:

$$\begin{aligned} \sum_{n=1}^{\infty} \left(\frac{1}{z-1+n} - \frac{1}{n} \right) &= \frac{1}{z} - 1 + \sum_{n=1}^{\infty} \left(\frac{1}{z+n} - \frac{1}{n+1} \right) \\ &= \frac{1}{z} - 1 + \sum_{n=1}^{\infty} \left(\frac{1}{z+n} - \frac{1}{n+1} \right) + \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right) \end{aligned}$$

Moving the terms around, we see that $\gamma'(z) = 0$, hence γ is a constant function, giving us:

$$G(z-1) = e^{\gamma}G(z)$$

Defining $H(z) = e^{\gamma}G(z)$, we get:

$$H(z-1) = zH(z)$$

4.2.5 Application: Riemann-Zeta Function

Definition 4.2.20: Riemann Zeta Function

The *Riemann-zeta function* is a complex function given by the series

$$\zeta_{\mathbb{Q}}(s) = \zeta(s) := \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \frac{1}{1-p^{-s}}$$

and is defined on $\operatorname{Re}(s) > 1$.

Using the integral test, we see that this function converges for $\operatorname{Re} s > 1$ (recall the complex part only adds rotation information). Let us prove the second equality of this definition:

Proposition 4.2.21: Euler's Product

For $\operatorname{Re}(s) > 1$, we have:

$$\sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \frac{1}{1-p^{-s}}$$

where the product runs over the primes in increasing order and is absolutely convergent, and $\zeta(s) \neq 0$ for $\operatorname{Re}(s) > 1$.

Proof:

Fix s with $\sigma = \operatorname{Re}(s) > 1$, so that $|n^{-s}| = n^{-\sigma}$. The series $\sum_n n^{-\sigma}$ converges by the integral test ([3, The Integral Test]), as $\int_1^\infty x^{-\sigma} dx = 1/(\sigma - 1)$. For a prime p we have $p^{-\sigma} \leq 2^{-\sigma} < \frac{1}{2}$, so the geometric series gives $\sum_{e \geq 0} p^{-es} = \frac{1}{1-p^{-s}}$. For $N \geq 2$ let S_N be the set of positive integers all of whose prime factors are at most N , and put

$$P_N(s) = \prod_{p \leq N} \frac{1}{1-p^{-s}}$$

Step 1: $P_N(s) = \sum_{n \in S_N} n^{-s}$. For $E \geq 1$, let $S_{N,E}$ be the set of $n \in S_N$ in which every prime appears with exponent at most E . Expanding the finite product of finite sums and using unique factorization, by which each $n \in S_{N,E}$ arises from exactly one choice of exponents,

$$\prod_{p \leq N} \sum_{e=0}^E p^{-es} = \sum_{n \in S_{N,E}} n^{-s}$$

As $E \rightarrow \infty$ the left-hand side tends to $P_N(s)$. On the right, a number of $S_N \setminus S_{N,E}$ has a prime exponent larger than E , so it is at least 2^{E+1} , and the difference from $\sum_{n \in S_N} n^{-s}$ is at most $\sum_{n \geq 2^{E+1}} n^{-\sigma}$, which tends to 0.

Step 2: convergence. Every $n \leq N$ lies in S_N , so

$$|\zeta(s) - P_N(s)| = \left| \sum_{n \notin S_N} n^{-s} \right| \leq \sum_{n > N} n^{-\sigma}$$

which tends to 0 as $N \rightarrow \infty$. Hence the partial products converge to $\zeta(s)$. The factors are $1 + a_p$ with $a_p = p^{-s}/(1-p^{-s})$, and $|a_p| \leq p^{-\sigma}/(1-p^{-\sigma}) \leq 2p^{-\sigma}$, so $\sum_p |a_p|$ converges and the product converges absolutely (definition 4.2.11 and the remark after it).

Step 3: $\zeta(s) \neq 0$. For $0 \leq x < 1$ we have $1+x \leq 1/(1-x)$, as $(1+x)(1-x) \leq 1$. With $|1-p^{-s}| \leq 1+p^{-\sigma}$ and Step 1 at the real point σ , this gives

$$\frac{1}{|P_N(s)|} \leq \prod_{p \leq N} (1+p^{-\sigma}) \leq \prod_{p \leq N} \frac{1}{1-p^{-\sigma}} = P_N(\sigma) = \sum_{n \in S_N} n^{-\sigma} \leq \zeta(\sigma)$$

So $|P_N(s)| \geq 1/\zeta(\sigma)$ for all N , and by Step 2, $|\zeta(s)| \geq 1/\zeta(\sigma) > 0$, completing the proof.

Theorem 4.2.22: Meromorphic Extension Of Zeta Function (Analytic continuation)

$\zeta(s)$ extends meromorphically to $\operatorname{Re}(s) > 0$ with a pole at $s = 1$, namely to:

$$\zeta(s) = \frac{1}{s-1} + \varphi(s)$$

where $\varphi(s)$ is a holomorphic function on $\operatorname{Re}(s) > 0$.

Proof:

let us start by showing we can write:

$$\zeta(s) = \frac{1}{s-1} + \varphi(s)$$

for a holomorphic function $\varphi(s)$ on $\operatorname{Re}(s) > 0$. To see this, take:

$$\begin{aligned} \zeta(s) - \frac{1}{s-1} &= \sum_{n \geq 1} n^{-s} - \int_1^\infty x^{-s} dx \\ &= \sum_{n \geq 1} \left(n^{-s} - \int_n^{n+1} x^{-s} dx \right) \quad \text{abs. convergence} \\ &= \sum_{n \geq 1} \int_n^{n+1} (n^{-s} - x^{-s}) dx \end{aligned}$$

Define $\varphi_n(s) = \int_n^{n+1} (n^{-s} - x^{-s}) dx$. Then these functions are holomorphic on $\operatorname{Re}(s) > 0$. To show $\varphi(s)$ is holomorphic, for any fixed s_0 such that $\operatorname{Re}(s_0) > 0$, and $x \in [n, n+1]$, we get:

$$\begin{aligned} |n^{-s_0} - x^{-s_0}| &= \left| \int_n^x s_0 t^{-s_0-1} dt \right| \\ &\leq \int_n^x \frac{|s_0|}{t^{s_0+1}} dt \\ &= \int_n^x \frac{|s_0|}{t^{1+\operatorname{Re}(s_0)}} dt \\ &\leq \frac{|s_0|}{n^{1+\operatorname{Re}(s_0)}} \end{aligned}$$

Hence,

$$|\varphi_n(s_0)| \leq \int_n^{n+1} | \leq \frac{|s_0|}{n^{1+\operatorname{Re}(s_0)}}$$

From this bound, we have that for $\epsilon = \operatorname{Re}(s_0)/2$, on $U = B_\epsilon(s_0)$ and each $n \geq 1$:

$$\sup_{s \in U} |\varphi_n(s)| \leq \frac{|s| + \epsilon}{n^{1+\epsilon}} =: M_n$$

Then we get:

$$\sum_n M_n = (|s_0| + \epsilon) \zeta(1 + \epsilon)$$

which converges. Thus, the series $\sum_n \varphi_n$ converges absolutely on $\operatorname{Re}(s) > 0$. By the Weiestrass M -test and uniqueness of analytic extension, $\sum_n \varphi_n$ converges to a function

$$\varphi(s) = \zeta(s) - \frac{1}{s-1} \quad \operatorname{Re}(s) > 0$$

as we sought to show.

Corollary 4.2.23: Residue Of Riemann-Zeta Function

$$\operatorname{Res}_{s=1} \zeta(s) = 1$$

By proposition 4.2.21, we see that $\zeta(s)$ has no roots for $\operatorname{Re}(s) > 1$. This result can be extended to $\operatorname{Re}(s) \geq 1$, which will be important for the prime number theorem. We require the following lemma

Lemma 4.2.24: Mertens Lemma

For any $x, y \in \mathbb{R}$ with $x > 1$

$$|\zeta(x)^3 \zeta(x + iy)^4 \zeta(x + 2iy)| \geq 1$$

This is a “trick” that was found by Merten. For a longer proof but one with more intuition, MIT notes gives a proof by Hadamard (p.149).

Proof:

As $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ for $\operatorname{Re}(s) > 1$:

$$\log |\zeta(s)| = - \sum_p \log |1 - p^{-s}| = - \sum_p \operatorname{Re} \log(1 - p^{-s}) = \sum_p \left(\sum_{n \geq 1} \frac{\operatorname{Re}(p^{-ns})}{n} \right)$$

Plugging in $s = x + iy$, we get:

$$\log |\zeta(x + iy)| = \sum_p \left(\sum_{n \geq 1} \frac{\cos(ny \log p)}{np^{nx}} \right)$$

And so:

$$\log |\zeta(x)^3 \zeta(x + iy)^4 \zeta(x + 2iy)| = \sum_p \sum_{n \geq 1} \frac{3 + 4 \cos(ny \log p) + \cos(2ny \log p)}{np^{nx}}$$

noting that $\cos(2\theta) = 2 \cos^2(\theta) - 1$, we get:

$$3 + 4 \cos(\theta) + \cos(2\theta) = 2(1 + \cos(\theta))^2 \geq 0$$

Taking $\theta = ny \log p$, we get:

$$\log |\zeta(x)^3 \zeta(x + iy)^4 \zeta(x + 2iy)| \geq 0 \Rightarrow |\zeta(x)^3 \zeta(x + iy)^4 \zeta(x + 2iy)| \geq 1$$

completing the proof.

Theorem 4.2.25: Riemann-zeta No Zeros On $x = 1$ Line

The Riemann-zeta function $\zeta(s)$ has no zeros on $\operatorname{Re}(s) \geq 1$

Proof:

By proposition 4.2.21, $\zeta(s)$ on $\operatorname{Re}(s) > 1$ has no zeros. Now, suppose that $\zeta(1 + iy) = 0$ for some $y \in \mathbb{R}$. Then it must be that $y \neq 0$ as $\zeta(s)$ has a pole at $s = 1$, and $\zeta(s)$ does not have a pole at $1 + 2iy \neq 1$ by theorem 4.2.22. It follows that we must have:

$$\lim_{x \rightarrow 1^+} |\zeta(x)^3 \zeta(x + iy)^4 \zeta(x + 2iy)| = 0$$

As $\zeta(s)$ has a simple pole at $s = 1$, a zero at $1 + iy$, and no pole at $1 + 2iy$. But this contradicts lemma 4.2.24, completing the proof.

The extension of ζ to the entire complex plane, which explains the common undergraduate “prank” of $1 + 2 + 3 + \cdots = -1/12$ as the value $\zeta(-1)$, is carried out in [2]. For now, let us continue on to using the zeta function for some truly incredible results.

4.3 Meromorphic Sequences and Families

Naturally, we often work with meromorphic functions, which can be interpreted as holomorphic functions on $\overline{\mathbb{C}}$, or equivalently $\mathbb{CP}^1 \cong S^2$. We shall see that such functions exhibit different behaviour, for example on $\mathbb{CP}^1$ we have that e^z is not meromorphic as it has an essential singularity at infinity. In fact:

Proposition 4.3.1: Meromorphic Function On Riemann Sphere

Let $\mathbb{CP}^1$ be the Riemann sphere. Then the meromorphic functions are the rational functions, and the only holomorphic functions are constant.

Proof:

Let f be holomorphic on all of $\mathbb{CP}^1$. Its image is compact, being the continuous image of a compact space, and so is closed and bounded. It is open unless f is constant, by the Open Mapping Theorem applied in the chart $\mathbb{C}$ and in a chart around ∞ . A nonempty subset of $\mathbb{C}$ that is compact and open would be a nonempty clopen subset, proper because it is bounded, of the connected space $\mathbb{C}$, so f is constant.

Now let f be meromorphic on $\mathbb{CP}^1$. Its poles are isolated, and an infinite set of them would accumulate in the compact $\mathbb{CP}^1$ at a point that is neither a pole nor a point of holomorphy, so there are finitely many. Let $p_1, \dots, p_k$ be those lying in $\mathbb{C}$ and let P_j be the principal part of f at p_j , a polynomial in $(z - p_j)^{-1}$ without constant term. Then $R = \sum_j P_j$ is rational and vanishes at ∞ . If ∞ is a pole of f , of order m , let $Q(z) = b_1 z + \cdots + b_m z^m$ collect its principal part there, read in the coordinate $s = 1/z$; otherwise set $Q = 0$. Then $f - R - Q$ is holomorphic on $\mathbb{C}$, each finite pole having had its principal part removed and Q being entire, and it is holomorphic at ∞ , since R vanishes there and $f - Q$ has no pole there by the choice of Q . So $f - R - Q$ extends holomorphically across ∞ , hence is constant by the previous paragraph, and $f = R + Q + c$ is

rational.

In the other direction, a rational function in lowest terms is holomorphic off the zeros of its denominator, with a pole at each of those and possibly at ∞ , so it is meromorphic on $\mathbb{CP}^1$.

When we are working with (simply) connected open subsets $\Omega \subseteq \mathbb{CP}^1$, we have many more meromorphic functions.

The euclidean metric cannot be extended to $\overline{\mathbb{C}}$ since there is no finite distance from any point of $\mathbb{C}$ to ∞ , however we may use a metric on the sphere (either the chordal metric or the spherical metric):

Definition 4.3.2: Chordal Metric

Let $\mathbb{CP}^1 = \overline{\mathbb{C}}$ be the Riemann sphere. Then the *chordal metric* on $\mathbb{CP}^1$ is defined as:

$$d(z, w) = \frac{2|z - w|}{\sqrt{1 + |z|^2} \sqrt{1 + |w|^2}}$$

This metric represents the smallest distance between two points as though the Riemann sphere was embedded in $\mathbb{R}^3$ with radius 1. This following (equivalent) metric follows the shortest path *on* the sphere

Definition 4.3.3: Spherical Metric

Let $\mathbb{CP}^1 = \overline{\mathbb{C}}$ be the Riemann sphere. Then the *spherical metric* on $\mathbb{CP}^1$ is defined as:

$$d(x, y) = \cos^{-1}(x \cdot y)$$

where $x \cdot y$ is the dot product of x and y

These two are equivalent: if two points of the unit sphere make the angle $\theta \in [0, \pi]$ at its centre, their chordal distance is $2 \sin(\theta/2)$ and their spherical distance is θ , and $\frac{2}{\pi}\theta \leq 2 \sin(\theta/2) \leq \theta$, by the concavity of $\sin$ on $[0, \pi/2]$. Hence each can be used to show any result the reader may want about convergence.

Let us now generalize our result on convergent holomorphic functions to Meromorphic functions (i.e. holomorphic functions on $\mathbb{CP}^1$). Note how it is important for convergence of meromorphic function that we use the chordal (or spherical) metric or else we cannot converge at poles. A meromorphic function f on an open set $\Omega \subseteq \mathbb{C}$ is regarded as a map $\Omega \rightarrow \mathbb{C}_\infty$ taking the value ∞ at its poles, and d denotes the chordal metric on $\mathbb{C}_\infty$ of theorem 1.2.2. The map f is continuous for d : at a point which is not a pole this follows from $d(a, b) \leq 2|a - b|$, a consequence of theorem 1.2.2(2), and at a pole p from $|f(z)| \rightarrow \infty$ as $z \rightarrow p$ (proposition 3.4.12(2)), which gives $d(f(z), \infty) = 2/\sqrt{1 + |f(z)|^2} \rightarrow 0$.

Theorem 4.3.4: Hurwitz's theorem For Meromorphic Functions

Let $\Omega \subseteq \mathbb{C}$ be open, let f_n be meromorphic on Ω , and suppose that $f_n \rightarrow f$ uniformly on every compact subset of Ω with respect to the chordal metric d , for some map $f: \Omega \rightarrow \mathbb{C}_\infty$.

1. If $z^* \in \Omega$ and $f(z^*) \neq \infty$, there are a closed disc $B \subseteq \Omega$ centred at z^* , a constant C and an n_0 such that for $n \geq n_0$ the functions f_n have no poles on B , $|f_n| < C$ and $|f| < C$ on B , the function f is holomorphic on the interior of B , and $f_n \rightarrow f$ uniformly on B in the Euclidean metric.
2. If $z^* \in \Omega$ and $f(z^*) = \infty$, there are a closed disc $B \subseteq \Omega$ centred at z^* and an n_0 such that for $n \geq n_0$ the functions f_n have no zeros on B , the functions $1/f_n$ are holomorphic on the interior of B , and $1/f_n \rightarrow 1/f$ uniformly on B in the chordal metric, where $1/\infty = 0$. The function $1/f$ is holomorphic near z^* , with value 0 at z^* . In particular, each point of Ω has a neighbourhood on which f or $1/f$ is holomorphic.
3. If Ω is connected, then the limit f is a meromorphic function on Ω or identically equal to ∞ .

Proof:

The map f is continuous, as each point of Ω has a compact disc neighbourhood on which it is a uniform limit of continuous maps ([5, Uniform Convergence Preserve's Continuity]). Let $a \in \mathbb{C}$, put $\epsilon = d(a, \infty)/2 = 1/\sqrt{1+|a|^2}$ and $C = 2/\epsilon$, using the formulas of theorem 1.2.2(2). If $d(w, a) < \epsilon$, then $d(w, \infty) \geq d(a, \infty) - d(w, a) > \epsilon$, so $w \neq \infty$ and $2/\sqrt{1+|w|^2} > \epsilon$, whence $|w| < C$. And for $w_1, w_2 \in \mathbb{C}$ with $|w_1|, |w_2| \leq C$, the formula for d gives $|w_1 - w_2| \leq \frac{1+C^2}{2} d(w_1, w_2)$.

1. Let $a = f(z^*)$ and ϵ, C as above. Choose a closed disc $B \subseteq \Omega$ centred at z^* with $d(f(z), a) < \epsilon/2$ on B , and n_0 with $d(f_n(z), f(z)) < \epsilon/2$ for $z \in B$ and $n \geq n_0$. Then $d(f_n(z), a) < \epsilon$ and $d(f(z), a) < \epsilon$, so $f_n(z)$ and $f(z)$ are finite and of modulus less than C on B . In particular f_n has no poles on B , and $|f_n - f| \leq \frac{1+C^2}{2} d(f_n, f)$ tends to 0 uniformly on B . The f_n with $n \geq n_0$ are holomorphic on the interior of B , so f is holomorphic there by lemma 4.1.3.
2. Choose a closed disc $B \subseteq \Omega$ centred at z^* with $d(f(z), \infty) < 1$ on B , and n_0 with $d(f_n(z), f(z)) < 1$ for $z \in B$, $n \geq n_0$. Then $d(f_n(z), \infty) < 2 = d(0, \infty)$, so f_n has no zeros on B , and $1/f_n$ is holomorphic on the interior of B , its singularities at the poles of f_n being removable (proposition 3.4.12). Since inversion preserves d (theorem 1.2.2(3)), $1/f_n \rightarrow 1/f$ uniformly on B , and $1/f(z^*) = 0$, so (1) applied to these functions on the interior of B shows that $1/f$ is holomorphic near z^* . With (1), every point of Ω has a neighbourhood on which f or $1/f$ is holomorphic.
3. Let S be the set of points of Ω near which f is the constant ∞ . The set S is open. If $z \in \Omega$ lies in the closure of S , then $f(z) = \infty$ by continuity, so $1/f$ is holomorphic on a disc around z by (2), and it vanishes on the nonempty open part of that disc lying in S . As $1/f$ is analytic (corollary 3.2.7), it vanishes on the whole disc (corollary 2.5.4), and $z \in S$. So S is open and closed in the connected set Ω , and $S = \Omega$ or $S = \emptyset$. If $S = \emptyset$, then near each point either f is holomorphic, or $1/f$ is holomorphic and not identically zero, with isolated zeros (proposition 2.5.7), so that $f = 1/(1/f)$ is a quotient of analytic functions. Hence f is meromorphic on Ω (definition 2.5.8), completing the proof.

Corollary 4.3.5: Holomorphic and Chordal Metric

Let $\Omega \subseteq \mathbb{C}$ be a connected open set and let $(f_n) \subseteq \mathcal{H}(\Omega)$ be a sequence of holomorphic functions such that $f_n \rightarrow f$ uniformly on compact sets with respect to the chordal metric. Then the limit is a holomorphic function or identically equal to ∞ .

Proof:

By theorem 4.3.4(3), f is meromorphic on Ω or the constant ∞ . In the first case, suppose that $f(z^*) = \infty$ for some z^* . Take B and n_0 from theorem 4.3.4(2). For $n \geq n_0$, the functions $1/f_n$ are holomorphic on the interior of B and have no zeros there, since f_n has no poles. By theorem 4.3.4(1), applied at z^* to $1/f_n \rightarrow 1/f$, they converge uniformly on a closed disc $B' \subseteq B$ around z^* to $1/f$, which vanishes at z^* . The poles of the meromorphic function f are isolated, so $1/f$ is not identically zero near z^* . This contradicts Hurwitz's theorem (proposition 4.1.12) on the interior of B' . Hence f takes no value ∞ , and it is holomorphic, as claimed.

4.3.1 Normal Families of Meromorphic Functions

The codomain is now extended to include the point at infinity, so that a sequence tending to infinity counts as convergent. As before, a meromorphic function is regarded as a continuous map into $(\mathbb{C}_\infty, d)$.

Definition 4.3.6: Normal Family On Riemann Sphere

Let $\Omega \subseteq \mathbb{C}$ be open. A family $\mathcal{F}$ of meromorphic functions on Ω is said to be *normal* (in the chordal metric) if every sequence in $\mathcal{F}$ has a subsequence which converges, uniformly with respect to the chordal metric d on every compact subset of Ω , to some map $\Omega \rightarrow \mathbb{C}_\infty$. The limit may be the constant ∞ : this is the case of a subsequence tending uniformly to ∞ on every compact set.

On compact sets containing no pole, the chordal and Euclidean metrics are equivalent, so the two notions of uniform convergence differ only near poles.

4.3.7 Note The derivatives of the functions in a normal family need not form a normal family. The functions $f_n(z) = n(z^2 - n)$ on $\mathbb{C}$ form a normal family, since $f_n \rightarrow \infty$ uniformly on every compact set. Their derivatives $f'_n(z) = 2nz$ do not: $f'_n(z) \rightarrow \infty$ for $z \neq 0$ while $f'_n(0) = 0$, so no subsequence has a continuous limit.

By corollary 4.1.25, if $A \subseteq \mathcal{H}(D)$ is locally bounded, so is A' . On a connected D the converse holds when $\{f(z_0) \mid f \in A\}$ is bounded for one z_0 (theorem 4.1.23), and the constants $f_k \equiv k$ show that the pointwise bound is needed. For families of meromorphic functions, which may have poles, the role of the derivative is taken by a different “type” of derivative, which measures the rate of change in the chordal metric:

Definition 4.3.8: Spherical Derivative

Let $\Omega \subseteq \mathbb{C}$ be open and let f be meromorphic on Ω . Then the *spherical derivative* of f at $z \in \Omega$ is

$$f^\#(z) := \lim_{w \rightarrow z} \frac{d(f(z), f(w))}{|z - w|}$$

where d is the chordal metric.

For readers familiar with Riemannian geometry, $f^\#(z)$ is the operator norm of the differential of $P \circ f$ at z , where the plane carries the Euclidean metric and the sphere S^2 the metric g induced from $\mathbb{R}^3$ (see [7]):

$$f^\#(z) = \sup_{v \in \mathbb{R}^2, |v|=1} \|d(P \circ f)_z(v)\|_g$$

By construction, the spherical derivative is *real* and *non-negative*. The limit exists, and it can be computed from f' :

Proposition 4.3.9: Computing the Spherical Derivative

Let f be meromorphic on an open set $\Omega \subseteq \mathbb{C}$.

1. The limit defining $f^\#(z)$ exists at every $z \in \Omega$. If z is not a pole of f , then $f^\#(z) = 2|f'(z)|/(1+|f(z)|^2)$. If z is a pole, then $1/f$ extends holomorphically to a neighbourhood of z with value 0 at z , and $f^\#(z) = 2|(1/f)'(z)|$.
2. $f^\#$ is continuous on Ω .
3. For $a \in \mathbb{C}$ and $b \in \mathbb{C} \setminus \{0\}$, the function $g(\zeta) = f(a + b\zeta)$, defined where $a + b\zeta \in \Omega$, has $g^\#(\zeta) = |b| f^\#(a + b\zeta)$.
4. If $U \subseteq \Omega$ is open and $1/f$ is meromorphic on U , then $(1/f)^\# = f^\#$ on U .

Proof:

Item (4) comes first, since (1) uses it. By theorem 1.2.2(3), $d(1/a, 1/b) = d(a, b)$ for all $a, b \in \mathbb{C}_\infty$, so the quotients whose limits define $(1/f)^\#(z)$ and $f^\#(z)$ coincide, which is (4).

1. At a point z which is not a pole, f is holomorphic on a disc around z , and corollary 1.2.4 gives the limit and the formula. At a pole z , $|f(w)| \rightarrow \infty$ as $w \rightarrow z$ (proposition 3.4.12(2)), so $h = 1/f$ tends to 0 and extends holomorphically to a disc U around z with $h(z) = 0$ (proposition 3.4.12(1)). By (4), $f^\#(z) = h^\#(z)$, and corollary 1.2.4 applied to h gives $h^\#(z) = 2|h'(z)|/(1+|h(z)|^2) = 2|h'(z)|$.
2. On a disc containing no pole, $f^\# = 2|f'|/(1+|f|^2)$ is continuous, f' being holomorphic. Near a pole z , with h and U as in (1) and U chosen so small that h has no zeros on U other than possibly z , (4) gives $f^\# = h^\# = 2|h'|/(1+|h|^2)$ on U , which is continuous.
3. For $w \neq \zeta$ we have $|(a + bw) - (a + b\zeta)| = |b||w - \zeta|$, so

$$\frac{d(g(\zeta), g(w))}{|\zeta - w|} = |b| \frac{d(f(a + b\zeta), f(a + bw))}{|(a + b\zeta) - (a + bw)|}$$

and letting $w \rightarrow \zeta$ gives $g^\#(\zeta) = |b|f^\#(a + b\zeta)$, completing the proof.

Lemma 4.3.10: Möbius Transformations of the Sphere

Let $T(z) = \frac{az+b}{cz+d}$, with $ad - bc \neq 0$, be a Möbius transformation, regarded as a map of $\mathbb{C}_\infty$ to itself as in proposition 1.3.4.

1. T is a bijection of $\mathbb{C}_\infty$, and its inverse is the Möbius transformation $T^{-1}(w) = \frac{dw-b}{-cw+a}$.
2. There is a constant L with $d(T(z), T(w)) \leq L d(z, w)$ for all $z, w \in \mathbb{C}_\infty$. Consequently, if maps g_n converge uniformly to G on a set with respect to d , then $T \circ g_n \rightarrow T \circ G$ uniformly on that set.
3. If f is meromorphic on an open set U and $T \circ f$ does not take the value ∞ , then $T \circ f$ is holomorphic on U .
4. For distinct $c_1, c_2, c_3 \in \mathbb{C}_\infty$ there is a Möbius transformation T with $T(c_1) = 0$, $T(c_2) = 1$ and $T(c_3) = \infty$.
5. Let f be meromorphic on a punctured disc $V = \{z \mid 0 < |z - z_0| < \delta\}$, and suppose that $T \circ f$ is holomorphic on V with a removable singularity or a pole at z_0 . Then f extends to a meromorphic function on $|z - z_0| < \delta$.

Proof:

1. Write points of $\mathbb{C}_\infty$ in homogeneous coordinates $[\zeta_1 : \zeta_2]$, ratios of nonzero pairs in $\mathbb{C}^2$ up to proportionality, with $z = [z : 1]$ for $z \in \mathbb{C}$ and $\infty = [1 : 0]$. For $z \in \mathbb{C}$ with $cz + d \neq 0$ we have $[az + b : cz + d] = [T(z) : 1]$. If $c \neq 0$, then $a(-d/c) + b = (bc - ad)/c \neq 0$, so $[a(-d/c) + b : 0] = \infty = T(-d/c)$. Finally $[a : c]$ is a/c if $c \neq 0$ and ∞ if $c = 0$, which is $T(\infty)$. So T is the map $[\zeta_1 : \zeta_2] \mapsto [a\zeta_1 + b\zeta_2 : c\zeta_1 + d\zeta_2]$ induced by the invertible matrix A with rows (a, b) and (c, d) . The matrix with rows $(d, -b)$ and $(-c, a)$ is $(ad - bc)A^{-1}$, so it induces the inverse map, and by the same computation it induces $w \mapsto \frac{dw-b}{-cw+a}$, whose determinant $da - bc$ is nonzero.
2. For $c \in \mathbb{C}$ let $\tau_c(w) = w + c$, for $\lambda \in \mathbb{C} \setminus \{0\}$ let $\delta_\lambda(w) = \lambda w$, with $\tau_c(\infty) = \delta_\lambda(\infty) = \infty$, and let $J(w) = 1/w$. By theorem 1.2.2(3), J preserves d . For $z \in \mathbb{C}$, $|z| \leq |z + c| + |c|$ gives $1 + |z|^2 \leq 2(1 + |c|^2)(1 + |z + c|^2)$, and $(1 + |z|^2)/(1 + |\lambda|^2|z|^2) \leq \max(1, |\lambda|^{-2})$. With the formulas of theorem 1.2.2(2), these give

$$d(\tau_c(z), \tau_c(w)) \leq 2(1 + |c|^2) d(z, w) \quad d(\delta_\lambda(z), \delta_\lambda(w)) \leq \max(|\lambda|, |\lambda|^{-1}) d(z, w)$$

for $z, w \in \mathbb{C}$, and also when $w = \infty$, the two ratios then being $\sqrt{(1 + |z|^2)/(1 + |z + c|^2)} \leq 2(1 + |c|^2)$ and $\sqrt{(1 + |z|^2)/(1 + |\lambda|^2|z|^2)} \leq \max(|\lambda|, |\lambda|^{-1})$. If $c = 0$, then $T = \tau_{b/d} \circ \delta_{a/d}$. If $c \neq 0$, then $T = \tau_{a/c} \circ \delta_{(bc-ad)/c^2} \circ J \circ \tau_{d/c}$, by the computation before proposition 1.3.4, and the two sides also agree at $-d/c$ and at ∞ , both sending them to ∞ and a/c . A composition of maps satisfying such bounds satisfies one with the product of the constants, and $d(T(g_n(z)), T(G(z))) \leq L d(g_n(z), G(z))$ gives the last claim.

3. Let $z_0 \in U$. If $f(z_0) \in \mathbb{C}$, then $cf(z_0) + d \neq 0$, since otherwise $T(f(z_0)) = \infty$, so $T \circ f = (af + b)/(cf + d)$ is holomorphic near z_0 . If z_0 is a pole of f , then $c \neq 0$, since

$T(\infty) = \infty$ when $c = 0$, and near z_0 the function $h = 1/f$ is holomorphic with $h(z_0) = 0$ (proposition 3.4.12). There $T \circ f = (a + bh)/(c + dh)$, the two sides agreeing at z_0 because $T(\infty) = a/c$. The denominator is $c \neq 0$ at z_0 , so $T \circ f$ is holomorphic near z_0 .

4. If $c_1, c_2, c_3 \in \mathbb{C}$, take $T(z) = \frac{(c_2 - c_3)(z - c_1)}{(c_2 - c_1)(z - c_3)}$, of determinant $(c_2 - c_3)(c_2 - c_1)(c_1 - c_3)$. If $c_1 = \infty$, $c_2 = \infty$ or $c_3 = \infty$, take respectively $\frac{c_2 - c_3}{z - c_3}$, $\frac{z - c_1}{z - c_3}$ or $\frac{z - c_1}{c_2 - c_1}$, of determinants $c_3 - c_2$, $c_1 - c_3$ and $c_2 - c_1$. All of these are nonzero, and the conventions of proposition 1.3.4 give the values 0, 1, ∞ at c_1, c_2, c_3 .
5. By proposition 3.4.12, $T(f(z))$ has a limit $w_0 \in \mathbb{C}_\infty$ as $z \rightarrow z_0$, so by (1) and (2), applied to T^{-1} , $f(z)$ tends to $v_0 = T^{-1}(w_0)$ in the chordal metric. Suppose first that $v_0 \in \mathbb{C}$, and put $\eta = d(v_0, \infty)/2 > 0$. There is $r \in (0, \delta]$ with $d(f(z), v_0) < \eta$, and hence $d(f(z), \infty) > \eta$, for $0 < |z - z_0| < r$. On that punctured disc f is finite, so holomorphic, and $2/\sqrt{1 + |f(z)|^2} > \eta$ shows that $|f| < 2/\eta$. By proposition 3.4.6, applied to $z \mapsto f(z_0 + rz)$, f extends holomorphically to $|z - z_0| < r$. If instead $v_0 = \infty$, choose $r \in (0, \delta]$ with $d(f(z), \infty) < d(1, \infty)$, that is $|f(z)| > 1$, for $0 < |z - z_0| < r$. On that punctured disc f has no zeros, so $1/f$ is holomorphic there, its singularities at the poles of f being removable (proposition 3.4.12), and $|1/f| < 1$. By proposition 3.4.6, $1/f$ extends to a holomorphic function h on $|z - z_0| < r$, which is not identically zero. Then $f = 1/h$ is a quotient of analytic functions near z_0 , with z_0 an isolated zero of h at most (proposition 2.5.7). In both cases f extends to a meromorphic function on $|z - z_0| < \delta$ (definition 2.5.8), completing the proof.

Lemma 4.3.11: Chordal Limits of Meromorphic Functions

Let $U \subseteq \mathbb{C}$ be open, let g_n be meromorphic on U , and suppose that $g_n \rightarrow G$ uniformly on every compact subset of U with respect to the chordal metric d , for some map $G: U \rightarrow \mathbb{C}_\infty$. On each component of U , the map G is meromorphic or the constant ∞ (theorem 4.3.4(3)).

1. Put $G^\# = 0$ on the components of U on which G is the constant ∞ . Then $g_n^\# \rightarrow G^\#$ uniformly on every compact subset of U .
2. If U is connected, $c \in \mathbb{C}_\infty$, and no g_n takes the value c , then either G does not take the value c or G is the constant c .

Proof:

1. Fix $z^* \in U$. Suppose first $G(z^*) \neq \infty$. Take B, C and n_0 from theorem 4.3.4(1), so that $|g_n| < C$ and $|G| < C$ on B for $n \geq n_0$, and let B' be the concentric closed disc of half the radius. As $g_n \rightarrow G$ uniformly on B and these are holomorphic on its interior, $g'_n \rightarrow G'$ uniformly on B' (proposition 4.1.11). On B' , by proposition 4.3.9(1) and the inequality $|1/(1+s) - 1/(1+t)| \leq |s-t|$ for $s, t \geq 0$,

$$|g_n^\# - G^\#| \leq 2||g'_n| - |G'|| + 2|G'| ||G|^2 - |g_n|^2| \leq 2|g'_n - G'| + 4C \max_{B'} |G'| |G - g_n|$$

which tends to 0 uniformly on B' . If instead $G(z^*) = \infty$, take B and n_0 from theorem 4.3.4(2), so that for $n \geq n_0$ the functions $1/g_n$ are holomorphic on the interior of B and converge uniformly on B to $1/G$, with $1/G(z^*) = 0$. The case just treated, applied to $1/g_n \rightarrow 1/G$, gives $(1/g_n)^\# \rightarrow (1/G)^\#$ uniformly on a closed disc B' around z^* , and

$(1/g_n)^\# = g_n^\#$ and $(1/G)^\# = G^\#$ there by proposition 4.3.9(4), the second also when G is the constant ∞ near z^* , both sides being 0. Thus every point of U has a closed disc neighbourhood on which $g_n^\# \rightarrow G^\#$ uniformly. Finitely many of their interiors cover a given compact subset of U , so the convergence is uniform on it.

2. If $c = \infty$, the g_n are holomorphic, and G is holomorphic or the constant ∞ by corollary 4.3.5. Now let $c \in \mathbb{C}$. The map $T(w) = 1/(w - c)$ is a Möbius transformation, so $1/(g_n - c) = T \circ g_n \rightarrow T \circ G = 1/(G - c)$ uniformly on compact subsets by lemma 4.3.10(2). The functions $1/(g_n - c)$ omit ∞ , as the g_n omit c , so they are holomorphic (lemma 4.3.10(3)), and by the first case $1/(G - c)$ omits ∞ or is the constant ∞ , that is, G omits c or is the constant c , as claimed.

Theorem 4.3.12: Marty's Theorem

Let $\Omega \subseteq \mathbb{C}$ be open and let $\mathcal{F}$ be a family of meromorphic functions on Ω . Then $\mathcal{F}$ is normal (definition 4.3.6) if and only if

$$\mathcal{F}^\# = \{f^\# \mid f \in \mathcal{F}\}$$

is locally bounded, or equivalently, for each compact $K \subseteq \Omega$ there exists a constant M_K such that:

$$\sup_{z \in K} f^\#(z) \leq M_K \quad \forall f \in \mathcal{F}$$

Proof:

The two forms of the condition agree: closed discs are compact, and a compact set is covered by finitely many open sets on which $\mathcal{F}^\#$ is uniformly bounded.

Sufficiency, Step 1: a Lipschitz bound. Suppose the condition holds, let $B \subseteq \Omega$ be a closed disc, $M = M_B$, $f \in \mathcal{F}$, and $z_1 \neq z_2$ in B . Put $\sigma(t) = z_1 + t(z_2 - z_1)$ and $c(t) = P(f(\sigma(t)))$ for $t \in [0, 1]$, with P as in theorem 1.2.2. The map $P \circ f$ is continuously differentiable from Ω to $\mathbb{R}^3$: away from the poles, P is given by the smooth formulas (1.5) and f is holomorphic, and near a pole $P \circ f = R \circ P \circ (1/f)$ with $1/f$ holomorphic and R linear, by theorem 1.2.2(3). So c is continuously differentiable, and as $|c(t+s) - c(t)| = d(f(\sigma(t+s)), f(\sigma(t)))$ and $|\sigma(t+s) - \sigma(t)| = |s||z_2 - z_1|$,

$$|c'(t)| = \lim_{s \rightarrow 0} \frac{|c(t+s) - c(t)|}{|s|} = f^\#(\sigma(t)) |z_2 - z_1| \leq M |z_2 - z_1|$$

by proposition 4.3.9(1). Let e be a unit vector with $c(1) - c(0) = |c(1) - c(0)|e$. The real function $t \mapsto e \cdot c(t)$ has continuous derivative $e \cdot c'(t) \leq |c'(t)|$, so the fundamental theorem of calculus and the order bounds ([3, The Fundamental Theorem of Calculus, Second Form], [3, Order Bounds and Additivity over a Subdivision Point]) give

$$d(f(z_1), f(z_2)) = |c(1) - c(0)| = \int_0^1 e \cdot c'(t) dt \leq M |z_2 - z_1|$$

Step 2: normality. Let $K \subseteq \Omega$ be compact. Cover K by finitely many open discs $D_1, \dots, D_p$ whose closures B_j lie in Ω , let λ be a Lebesgue number of the cover of the compact metric space K by the sets $D_j \cap K$ ([8, Lebesgue Number Lemma]), and let $M = \max_j M_{B_j}$. If $w_1, w_2 \in K$ and

$|w_1 - w_2| < \lambda$, then both lie in some D_j , and Step 1 gives $d(f(w_1), f(w_2)) \leq M|w_1 - w_2|$ for every $f \in \mathcal{F}$. Write $P \circ f = (\varphi_1, \varphi_2, \varphi_3)$ and put $u_f = \varphi_1 + i\varphi_2$ and $v_f = \varphi_3$. These are continuous complex-valued functions on Ω , bounded by 1, and $|u_f(w_1) - u_f(w_2)|$ and $|v_f(w_1) - v_f(w_2)|$ are at most $|P(f(w_1)) - P(f(w_2))| = d(f(w_1), f(w_2))$. So the families $\{u_f \mid f \in \mathcal{F}\}$ and $\{v_f \mid f \in \mathcal{F}\}$ are uniformly equicontinuous on every compact set and pointwise bounded, hence normal in $C(\Omega)$ by theorem 4.1.22. Given $(f_n) \subseteq \mathcal{F}$, proposition 4.1.20 gives a subsequence along which u_{f_n} converges uniformly on compact sets, and a further subsequence along which v_{f_n} does as well. Along it, $P \circ f_n$ converges uniformly on compact sets to a continuous map $\Phi: \Omega \rightarrow \mathbb{R}^3$ with values in the closed set S^2 . With $G = P^{-1} \circ \Phi$, which makes sense by theorem 1.2.2(1), we get $d(f_n(z), G(z)) = |P(f_n(z)) - \Phi(z)| \rightarrow 0$ uniformly on compact sets. Hence $\mathcal{F}$ is normal.

Necessity. Suppose $\mathcal{F}$ is normal but $\sup_{f \in \mathcal{F}} \sup_K f^\# = \infty$ for some compact $K \subseteq \Omega$. Choose $f_n \in \mathcal{F}$ and $z_n \in K$ with $f_n^\#(z_n) \rightarrow \infty$, and pass to a subsequence with $z_n \rightarrow z^* \in K$ and $f_n \rightarrow G$ uniformly on compact sets in the chordal metric. Let $B \subseteq \Omega$ be a closed disc centred at z^* . By lemma 4.3.11(1), applied with $U = \Omega$, $f_n^\# \rightarrow G^\#$ uniformly on B , and $G^\#$ is continuous on B , either by proposition 4.3.9(2) or because it vanishes, so it is bounded on B . Hence the $f_n^\#$ are uniformly bounded on B for n large, while $z_n \in B$ for n large and $f_n^\#(z_n) \rightarrow \infty$. This contradiction completes the proof.

Exercise 4.3.1

1. Show that $(g \circ f)^\#(z) = f^\#(g(z))|g'(z)|$

If $f^\#(z) \neq 0$, then f is locally 1 to 1.

$f^\# = (1/f)^\#$. Use this to show that $f^\#$ is a continuous function on all of Ω (including if it contains a pole).

4.4 Picard's Theorems and Essential Singularities

In this final section, we shall take a look at the behaviour of Essential singularities, a phenomena that can be thought of as fundamentally analytic and separates complex analysis from algebraic geometry. In the process, we shall find some new conditions for normal families.

First, let us link the following two seemingly unrelated results:

Theorem 4.4.1: Great Picard's Theorem Equivalence

The following two statements are equivalent:

1. If f is holomorphic on a punctured disc $0 < |z - z_0| < \delta$ and has an essential singularity at z_0 , then there is at most one $\lambda \in \mathbb{C}$ such that the equation $f(z) = \lambda$ has only finitely many solutions in some punctured disc $0 < |z - z_0| < \epsilon$ with $\epsilon \leq \delta$. Equivalently, f takes every other complex value at infinitely many points of every such punctured disc.
2. If f is meromorphic on $0 < |z - z_0| < \delta$ and f omits three distinct values of $\mathbb{C}_\infty$, then f extends to a meromorphic function on $|z - z_0| < \delta$.

Proof:

- (1 $\Rightarrow$ 2) Assume (1), and let f be meromorphic on $V = \{z \mid 0 < |z - z_0| < \delta\}$ and omit the distinct values $c_1, c_2, c_3 \in \mathbb{C}_\infty$. Let T be a Möbius transformation with $T(c_1) = 0$, $T(c_2) = 1$ and $T(c_3) = \infty$ (lemma 4.3.10(4)). Then $g = T \circ f$ omits ∞ , 0 and 1 , since T is a bijection (lemma 4.3.10(1)), so g is holomorphic on V (lemma 4.3.10(3)). The equations $g(z) = 0$ and $g(z) = 1$ have no solutions in V , so by (1) the point z_0 is not an essential singularity of g . Hence it is a removable singularity or a pole of g , and f extends to a meromorphic function on $|z - z_0| < \delta$ by lemma 4.3.10(5).
- (2 $\Rightarrow$ 1) Assume (2), and let f be holomorphic on $0 < |z - z_0| < \delta$ with an essential singularity at z_0 . Suppose that two distinct values $\lambda_1, \lambda_2 \in \mathbb{C}$ each have only finitely many solutions of $f(z) = \lambda_j$ in some punctured disc around z_0 . These finitely many solutions lie at positive distance from z_0 , so there is $\epsilon > 0$ such that f omits λ_1 and λ_2 on $0 < |z - z_0| < \epsilon$. Being holomorphic, f also omits ∞ there. By (2), f extends to a meromorphic function on $|z - z_0| < \epsilon$, so z_0 is a removable singularity or a pole of f , which contradicts the hypothesis. Hence at most one value has only finitely many solutions in some punctured disc, which is (1), completing the proof.

We shall prove the 2nd statement. The original proof was given in the early 20th century, but the proof we shall give is a rather recent one that will rely on a result first proved in the 1970s which shall give a more straightforward approach:

Lemma 4.4.2: Zalcman's Lemma

Let $\mathcal{F}$ be a family of meromorphic functions on an open set $\Omega \subseteq \mathbb{C}$. Then $\mathcal{F}$ is *not normal* (definition 4.3.6) if and only if there exist $f_n \in \mathcal{F}$, points $a_n \in \Omega$ converging to a point of Ω , and numbers $\rho_n > 0$ with $\rho_n \rightarrow 0$, such that the functions

$$g_n(\zeta) = f_n(a_n + \rho_n \zeta)$$

converge, uniformly on compact subsets of $\mathbb{C}$ with respect to the chordal metric, to a non-constant meromorphic function g on $\mathbb{C}$ with

$$g^\#(\zeta) \leq 1 = g^\#(0) \quad \zeta \in \mathbb{C}$$

Here g_n is defined where $a_n + \rho_n \zeta \in \Omega$, a set which contains any given compact subset of $\mathbb{C}$ once n is large.

Interestingly, we are giving a criterion for *failure* of normality, not by a lack of convergence condition, but *by* a convergence condition. The idea is that $g(z)$ is *not* constant, the point being that if the f_n converged, then because the ρ_n 's are converging to 0, the g_n 's would converge to a constant function. Hence, normality would mean convergence to a constant.

Proof:

The condition implies that $\mathcal{F}$ is not normal. Suppose that f_n , $a_n \rightarrow a \in \Omega$, ρ_n and g are as described, and that $\mathcal{F}$ is normal. Choose $r > 0$ with $B = \{z \mid |z - a| \leq r\} \subseteq \Omega$. By Marty's theorem (theorem 4.3.12) there is an M with $f^\# \leq M$ on B for every $f \in \mathcal{F}$. For n large, $|a_n - a| + \rho_n < r$, so g_n is defined on $|\zeta| < 1$, and proposition 4.3.9(3) gives

$g_n^\#(0) = \rho_n f_n^\#(a_n) \leq \rho_n M$. By lemma 4.3.11(1), applied on the disc $|\zeta| < 1$ to the g_n with n large, $g_n^\#(0) \rightarrow g^\#(0) = 1$, while $\rho_n M \rightarrow 0$. This is a contradiction.

If $\mathcal{F}$ is not normal, the condition holds. By Marty's theorem there are a compact $K \subseteq \Omega$, functions $f_n \in \mathcal{F}$ and points $b_n \in K$ with $f_n^\#(b_n) \rightarrow \infty$. Passing to a subsequence, $b_n \rightarrow b \in K$. Choose $r > 0$ with $B = \{z \mid |z - b| \leq r\} \subseteq \Omega$, and discard the finitely many n with $|b_n - b| \geq r/2$. The function $z \mapsto (r - |z - b|)f_n^\#(z)$ is continuous on B (proposition 4.3.9(2)), so it attains its maximum M_n at some $a_n \in B$, and

$$M_n \geq (r - |b_n - b|)f_n^\#(b_n) \geq \frac{r}{2}f_n^\#(b_n) \xrightarrow{n \rightarrow \infty} \infty$$

Discarding finitely many more n , we have $M_n > 0$. The function vanishes on $|z - b| = r$, so $|a_n - b| < r$ and $f_n^\#(a_n) > 0$. Put

$$\rho_n = \frac{1}{f_n^\#(a_n)} = \frac{r - |a_n - b|}{M_n} \leq \frac{r}{M_n}$$

so that $\rho_n \rightarrow 0$. If $|\zeta| < M_n$, then $|a_n + \rho_n \zeta - b| < |a_n - b| + \rho_n M_n = r$, so $g_n(\zeta) = f_n(a_n + \rho_n \zeta)$ is defined and meromorphic on $|\zeta| < M_n$. For such ζ , the point $z = a_n + \rho_n \zeta$ satisfies $(r - |z - b|)f_n^\#(z) \leq M_n$ and $r - |z - b| \geq r - |a_n - b| - \rho_n |\zeta| > 0$, so proposition 4.3.9(3) gives

$$g_n^\#(\zeta) = \rho_n f_n^\#(a_n + \rho_n \zeta) \leq \frac{\rho_n M_n}{r - |a_n - b| - \rho_n |\zeta|} = \frac{r - |a_n - b|}{r - |a_n - b| - \rho_n |\zeta|} = \frac{1}{1 - |\zeta|/M_n}$$

and $g_n^\#(0) = \rho_n f_n^\#(a_n) = 1$.

Fix $R > 0$. The g_n with $M_n \geq 2R$ are meromorphic on $|\zeta| < R$ with $g_n^\# \leq 2$ there, so by Marty's theorem they form a normal family on $|\zeta| < R$. Applying this with $R = 1, 2, 3, \dots$ and passing to successive subsequences, the diagonal subsequence converges uniformly on compact subsets of each disc $|\zeta| < R$, and hence on every compact subset of $\mathbb{C}$, to a map $g: \mathbb{C} \rightarrow \mathbb{C}_\infty$. By theorem 4.3.4(3), applied on each disc $|\zeta| < R$, the restriction of g to that disc is meromorphic or the constant ∞ , and by lemma 4.3.11(1), $g_n^\# \rightarrow g^\#$ pointwise, with $g^\# = 0$ in the second case. As $g_n^\#(0) = 1$, the second case does not occur, so g is meromorphic on $\mathbb{C}$ with $g^\#(0) = 1$ and $g^\#(\zeta) \leq \lim_n 1/(1 - |\zeta|/M_n) = 1$. In particular g is not constant (proposition 4.3.9(1)). Finally, the a_n lie in the compact disc $B \subseteq \Omega$, so a further subsequence converges to a point of Ω , completing the proof.

Example 4.4.3: Lacking Normality

The family $\mathcal{F} = \{f_n\}$ with $f_n(z) = z^n$ is normal on $\mathbb{D}$, where $f_n \rightarrow 0$, and on $\mathbb{C} \setminus \overline{\mathbb{D}}$, where $f_n \rightarrow \infty$, uniformly on compact sets in both cases. It is not normal on the disc $|z| < 2$. To see this, let $a_n = 1$ and $\rho_n = 1/n$ in lemma 4.4.2. Then

$$f_n(a_n + \rho_n \zeta) = \left(1 + \frac{\zeta}{n}\right)^n \rightarrow e^\zeta = g(\zeta)$$

uniformly on compact subsets of $\mathbb{C}$. Indeed, the coefficient of ζ^k in $e^\zeta - (1 + \zeta/n)^n$ is $\frac{1}{k!}(1 - \prod_{j=0}^{k-1} (1 - j/n))$ for $k \leq n$ and $\frac{1}{k!}$ for $k > n$, and in both cases it lies between 0 and $\frac{k(k-1)}{2n k!}$, so

$|e^\zeta - (1 + \zeta/n)^n| \leq R^2 e^R / (2n)$ for $|\zeta| \leq R$. Uniform convergence in the Euclidean metric implies it in the chordal metric, as $d(a, b) \leq 2|a - b|$. With $\zeta = x + iy$,

$$g^\#(\zeta) = \frac{2|g'(\zeta)|}{1 + |g(\zeta)|^2} = \frac{2e^x}{1 + e^{2x}} \leq 1 = g^\#(0)$$

since $1 + e^{2x} - 2e^x = (1 - e^x)^2 \geq 0$.

Theorem 4.4.4: Montel's Big Theorem

Let $\Omega \subseteq \mathbb{C}$ be open and let $c_1, c_2, c_3 \in \mathbb{C}_\infty$ be distinct. A family of meromorphic functions on Ω , none of which takes any of the values c_1, c_2, c_3 , is normal in the chordal metric (definition 4.3.6).

In particular, a family of holomorphic functions on Ω omitting two fixed values of $\mathbb{C}$ is normal, since its members also omit ∞ . The condition is sufficient but not necessary: the family consisting of the single function $z \mapsto z$ on $\mathbb{C}$ is normal, and it omits only the value ∞ .

Proof:

Step 1: reductions. Let $\mathcal{F}$ be the family, and let T be a Möbius transformation with $T(c_1) = 0$, $T(c_2) = 1$ and $T(c_3) = \infty$ (lemma 4.3.10(4)). For $f \in \mathcal{F}$, the function $T \circ f$ omits ∞ , 0 and 1, as T is a bijection, so it is holomorphic on Ω (lemma 4.3.10(1),(3)). If $\{T \circ f \mid f \in \mathcal{F}\}$ is normal, then so is $\mathcal{F}$: if $T \circ f_n \rightarrow H$ uniformly on compact sets in the chordal metric, then $f_n = T^{-1} \circ T \circ f_n \rightarrow T^{-1} \circ H$ in the same way (lemma 4.3.10(1),(2)). So we may assume that every $f \in \mathcal{F}$ is holomorphic and omits 0 and 1. By Marty's theorem (theorem 4.3.12), $\mathcal{F}$ is normal once $\mathcal{F}^\#$ is bounded on a neighbourhood of each point of Ω . If $D \subseteq \Omega$ is an open disc on which the restrictions of the members of $\mathcal{F}$ form a normal family, then Marty's theorem on D bounds $\mathcal{F}^\#$ on a neighbourhood of the centre of D . These restrictions belong to the family $\mathcal{N}_0$ of all holomorphic functions on D which omit 0 and 1, and a subfamily of a normal family is normal. So it suffices to fix an open disc D and show that $\mathcal{N}_0$ is normal.

Step 2: square roots. For $m \geq 0$ let $\mathcal{N}_m$ be the family of holomorphic functions on D which omit 0 and every 2^m -th root of unity $e^{2\pi i k / 2^m}$, $k = 0, \dots, 2^m - 1$. For $m = 0$ this is the family of Step 1. Let $f \in \mathcal{N}_m$. As f has no zeros, f'/f is holomorphic on the disc D , so it has a holomorphic primitive ℓ on D (corollary 3.2.3), that is, $\ell' = f'/f$. Then $(fe^{-\ell})' = 0$, so $fe^{-\ell}$ is a nonzero constant e^c (proposition 2.2.22), and replacing ℓ by $\ell + c$ we get $f = e^\ell$. Put $h = e^{\ell/2}$, so that $h^2 = f$. The function h has no zeros, and if $h(z) = \zeta$ with $\zeta^{2^{m+1}} = 1$, then $f(z) = \zeta^2$ satisfies $(\zeta^2)^{2^m} = 1$, so $f(z)$ is a 2^m -th root of unity, which f omits. Hence $h \in \mathcal{N}_{m+1}$.

Step 3: if $\mathcal{N}_{m+1}$ is normal, so is $\mathcal{N}_m$. For $z, w \in \mathbb{C}$, the inequality $1 + |z|^4 \geq (1 + |z|^2)^2 / 2$ and theorem 1.2.2(2) give

$$d(z^2, w^2) = \frac{2|z - w||z + w|}{\sqrt{(1 + |z|^4)(1 + |w|^4)}} \leq d(z, w) \frac{2(|z| + |w|)}{\sqrt{(1 + |z|^2)(1 + |w|^2)}} \leq 4d(z, w)$$

and $d(z^2, \infty) = 2/\sqrt{1 + |z|^4} \leq 2\sqrt{2}/(1 + |z|^2) \leq \sqrt{2}d(z, \infty)$. So the squaring map $S(w) = w^2$, $S(\infty) = \infty$, satisfies $d(S(z), S(w)) \leq 4d(z, w)$ on $\mathbb{C}_\infty$. Given $(f_n) \subseteq \mathcal{N}_m$, Step 2 gives $h_n \in \mathcal{N}_{m+1}$ with $h_n^2 = f_n$. A subsequence of (h_n) converges uniformly on compact subsets of D in the chordal metric, to H say, and then the corresponding $f_n = S \circ h_n$ converge in the same way to

$S \circ H$.

Step 4: Zalcman limits. Suppose that $\mathcal{N}_0$ is not normal. By Step 3 and induction on m , no $\mathcal{N}_m$ is normal. Fix m . Zalcman's lemma (lemma 4.4.2) gives a non-constant meromorphic function g_m on $\mathbb{C}$ with $g_m^\# \leq 1 = g_m^\#(0)$, which is the limit, uniformly on compact subsets of $\mathbb{C}$ in the chordal metric, of functions $\zeta \mapsto f_n(a_n + \rho_n \zeta)$ with $f_n \in \mathcal{N}_m$. For each $R > 0$, these functions are holomorphic on $|\zeta| < R$ once n is large, and they omit ∞ , 0 and the 2^m -th roots of unity. As $g_m^\#(0) = 1$, the function g_m is not constant on any disc around 0 (proposition 4.3.9(1)), so lemma 4.3.11(2), applied on the discs $|\zeta| < R$, shows that g_m omits all these values as well. So g_m is entire, has no zeros, and omits every 2^m -th root of unity.

Step 5: a limit of the g_m . Since $g_m^\# \leq 1$ for all m , Marty's theorem shows that $\{g_m \mid m \geq 0\}$ is normal on $\mathbb{C}$. Let g be the limit, uniformly on compact subsets of $\mathbb{C}$ in the chordal metric, of a subsequence (g_{m_j}) . By theorem 4.3.4(3), g is meromorphic on $\mathbb{C}$ or the constant ∞ , and by lemma 4.3.11(1), $g_{m_j}^\# \rightarrow g^\#$ pointwise, with $g^\# = 0$ in the second case. As $g_{m_j}^\#(0) = 1$, the function g is meromorphic with $g^\#(0) = 1$, so it is not constant (proposition 4.3.9(1)). For each $M \geq 0$, the g_{m_j} with $m_j \geq M$ omit ∞ , 0 and every 2^M -th root of unity, since a 2^M -th root of unity is also a 2^{m_j} -th root of unity. By lemma 4.3.11(2), so does g .

Step 6: the contradiction. So g is entire and non-constant, and it omits the set E of all 2^M -th roots of unity, $M \geq 0$. The set E is dense in the unit circle S^1 : given $\theta \in \mathbb{R}$ and M , some integer k has $|\theta - 2\pi k/2^M| \leq \pi/2^M$, and then $|e^{i\theta} - e^{2\pi i k/2^M}| \leq \pi/2^M$. The image $g(\mathbb{C})$ is open (theorem 3.5.15), so $g(\mathbb{C}) \cap S^1$ is an open subset of S^1 missing the dense set E , and hence it is empty. The image $g(\mathbb{C})$ is connected, as $\mathbb{C}$ is, and it is the union of the disjoint open sets $g(\mathbb{C}) \cap \{w \mid |w| < 1\}$ and $g(\mathbb{C}) \cap \{w \mid |w| > 1\}$, so it lies in one of them. If $|g| < 1$ on $\mathbb{C}$, then g is constant by Liouville's theorem (theorem 3.3.3). If $|g| > 1$ on $\mathbb{C}$, then $1/g$ is entire with $|1/g| < 1$, so $1/g$ is constant, and so is g . Both contradict that g is not constant. Hence $\mathcal{N}_0$ is normal, which completes the proof.

Theorem 4.4.5: Great Picard's Theorem

If f is meromorphic in a punctured disk

$$0 < |z - z_0| < \delta$$

and f omits three distinct values of $\mathbb{CP}^1$, then f extends to a meromorphic function on $|z - z_0| < \delta$.

Proof:

Let T be a Möbius transformation sending the three omitted values to 0, 1 and ∞ (lemma 4.3.10(4)). Then $g = T \circ f$ omits 0, 1 and ∞ , so it is holomorphic on the punctured disc (lemma 4.3.10(1),(3)). By lemma 4.3.10(5), it suffices to show that z_0 is a removable singularity or a pole of g . Replacing g by $z \mapsto g(z_0 + \delta z/2)$, we may assume that $z_0 = 0$ and that g is holomorphic on $\Omega = \{z \mid 0 < |z| < 2\}$.

For $n \geq 1$ put $g_n(z) = g(2^{-n}z)$, a holomorphic function on Ω which omits 0, 1 and ∞ . By

Montel's big theorem (theorem 4.4.4), a subsequence (g_{n_k}) converges uniformly on compact subsets of Ω in the chordal metric, to G say. As the g_n omit ∞ and Ω is connected, lemma 4.3.11(2) shows that G omits ∞ or is the constant ∞ .

Case 1: G omits ∞ . By theorem 4.3.4(1), each point of the circle $C = \{z \mid |z| = 1\}$ has a closed disc neighbourhood on which $g_{n_k} \rightarrow G$ uniformly in the Euclidean metric, and finitely many of these discs cover C . Hence $g_{n_k} \rightarrow G$ uniformly on C , and with $M = \max_C |G|$ there is a k_0 with $|g_{n_k}| \leq M + 1$ on C for $k \geq k_0$, that is, $|g(z)| \leq M + 1$ on $|z| = 2^{-n_k}$. The function g is infinitely differentiable, being analytic (corollary 3.2.7). For $k \geq k_0$, the maximum principle (corollary 2.2.19), applied to g on the closed annulus $2^{-n_{k+1}} \leq |z| \leq 2^{-n_k}$, whose boundary consists of two such circles, gives $|g| \leq M + 1$ on the annulus. These annuli cover $0 < |z| \leq 2^{-n_{k_0}}$, so g is bounded near 0, and 0 is a removable singularity of g (proposition 3.4.6).

Case 2: G is the constant ∞ . The function $1/g$ is holomorphic on Ω , as g has no zeros, and it omits 0, 1 and ∞ . Since inversion preserves the chordal metric (theorem 1.2.2(3)), $1/g_{n_k} \rightarrow 0$ uniformly on compact subsets of Ω . Case 1, applied to $1/g$ along the same subsequence, shows that 0 is a removable singularity of $1/g$. So $1/g(z)$ has a limit in $\mathbb{C}$ as $z \rightarrow 0$, and by proposition 3.4.12, 0 is a removable singularity of g if the limit is nonzero and a pole if it is zero, completing the proof.

Theorem 4.4.6: Picard's Theorem

Let f be holomorphic on a punctured disc $0 < |z - z_0| < \delta$ with an essential singularity at z_0 . Then f takes every complex value, with at most one exception, at infinitely many points of every punctured disc $0 < |z - z_0| < \epsilon$ with $\epsilon \leq \delta$. In particular, f maps every such punctured disc onto $\mathbb{C}$ or onto $\mathbb{C}$ with one point removed.

Proof:

By theorem 4.4.5, statement (2) of theorem 4.4.1 holds, so statement (1) holds, and it is the first claim. The image of a punctured disc $0 < |z - z_0| < \epsilon$ therefore contains every complex value except at most one, and it lies in $\mathbb{C}$ because f is holomorphic, which gives the second claim.

Theorem 4.4.7: Picard's Little Theorem

Any non-constant entire function omits at most 1 value.

Proof:

Let f be entire and non-constant, and suppose that f omits two distinct values $a, b \in \mathbb{C}$. The function $g(w) = f(1/w)$ is holomorphic on $0 < |w| < 1$ and omits a, b and ∞ , so by Great Picard's theorem (theorem 4.4.5) it extends to a meromorphic function on $|w| < 1$. Near 0 this extension is a quotient p/q of analytic functions with q not identically zero (definition 2.5.8), and $q(w) = w^k u(w)$ with $k \geq 0$ and $u(0) \neq 0$ (proposition 2.5.7 if $q(0) = 0$). So there are C and $r_0 > 0$ with $|g(w)| \leq C|w|^{-k}$ for $0 < |w| \leq r_0$, that is, $|f(z)| \leq C|z|^k$ for $|z| \geq 1/r_0$. By theorem 3.2.6, applied on discs of every radius, $f(z) = \sum a_n z^n$ for all $z \in \mathbb{C}$, and the Cauchy inequalities (theorem 3.3.1) give $|a_n| \leq Cr^k/r^n$ for every $r \geq 1/r_0$. Letting $r \rightarrow \infty$ shows that $a_n = 0$ for $n > k$, so f is a polynomial, of degree at least 1 because f is not constant. By

corollary 3.3.5, f takes every value in $\mathbb{C}$, contradicting that it omits a , which completes the proof.

I like to think of the existence of the omitted value as a consequence of the essential singularity having no parallel for roots. For example, an “infinite” root by theorem 2.5.3. This need not happen, for example this does not happen for $\cos(z)$ and $\sin(z)$ which are surjective⁵.

Corollary 4.4.8: Consequences of Picard Little Theorem

1. Any entire holomorphic function which omits more than 1 points must be constant.
2. Any meromorphic function on $\mathbb{C}$ which omits more than 2 points of $\mathbb{C}_\infty$ must be constant.

Proof:

1. This is the contrapositive of Picard's little theorem (theorem 4.4.7).
2. Let f omit the distinct values $c_1, c_2, c_3 \in \mathbb{C}_\infty$, and let T be a Möbius transformation with $T(c_1) = 0$, $T(c_2) = 1$ and $T(c_3) = \infty$ (lemma 4.3.10(4)). Then $T \circ f$ omits 0, 1 and ∞ , so it is entire (lemma 4.3.10(1),(3)) and omits two values of $\mathbb{C}$. By part (1) it is constant, and so is $f = T^{-1} \circ T \circ f$, completing the proof.

The reader should keep e^z as the canonical example of a holomorphic function that omits 1 point, and $\frac{e^z}{e^z - 1}$ as the canonical example of a meromorphic function that omits 2 points. Note that $\frac{e^z}{e^z - 1}$ has infinitely many simple poles, at $z = 2\pi in$ for each n , since near 0 it can be written as

$$\frac{1 + z + \frac{z^2}{2} + \cdots}{z + \frac{z^2}{2} + \frac{z^3}{3!} + \cdots} = \frac{1}{z} \cdot \frac{e^z}{1 + \frac{z}{2} + \frac{z^2}{3!} + \cdots}$$

and the function has period $2\pi i$.

Exercise 4.4.1

1. Show that ze^z is surjective on $\mathbb{C}$
2. Let f, g be nonconstant holomorphic function. Show there does not exist $n > 2$ such that $f^n + g^n = 1$
3. Show the same as above if f, g are meromorphic function with no poles in common
4. If $n = 2$, show that there exists functions where $f^2 + g^2 = 0$ and that all such solutions are of the form $f(z) = \sin(h(z))$, $g(z) = \cos(h(z))$ for some entire function $h(z)$.

⁵Here I like to think of their periodicity insuring this does not happen

Elliptic Functions and Weierstrass p-function

As we saw, periodic functions such that $\sin$ and $\cos$ are all easy to understand holomorphic functions. They are also fundamental functions as they form an orthonormal basis for $L^2(\mathbb{T})$, allowing for their extension into Fourier Analysis. When working over $\mathbb{C}$, we can imagine the interest in studying doubly periodic functions and how these may behave, in particular functions that are invariant under a lattice transformation. Originally linked to the study of ellipses (hence the name *elliptic*), these functions eventually greatly evolved and now play an important role in linking two seemingly disparate areas of mathematics

Harmonic Analysis and Number Theory

Due to this important connection, it is worth-while taking a moment to explore these types of functions. Unfortunately, we will not have time to explore modular forms, but this section shall start providing some of the background material for them.

5.1 Elliptic Functions

A natural way to construct the torus is by taking $\mathbb{C}$ and taking its quotient by a complete lattice Λ , 2 $\mathbb{Z}$ -dimension subgroup spanned by $\{\omega_1, \omega_2\}$ that $\mathbb{R}$ -linearly (this is the universal covering that is also a local isometry)¹; denote the torus by $\mathbb{C}/\Lambda$. Then a function is well defined on $\mathbb{C}/\Lambda$ if it respects the quotient, in particular if:

$$f(z + \alpha\omega_1 + \beta\omega_2) = f(z) \quad \forall \alpha, \beta \in \mathbb{Z}$$

¹The important of the $\mathbb{R}$ -linear independence is to avoid situations of choosing $(1, 1)$ and $(\sqrt{2}, \sqrt{2})$ which are $\mathbb{Z}$ -linearly independent, but in $\mathbb{R}$ are on the same line; we want them to be in different dimensions of $\mathbb{C}$

By some basic topology, any $\mathbb{C}$ -valued continuous function on $\mathbb{C}/\Lambda$ must respect this property. We give this property a name:

Definition 5.1.1: Doubly Periodic Functions

Let e_1, e_2 be two complex numbers that are $\mathbb{R}$ -linearly independent, that is $e_1 \neq 0$ and $e_2/e_1 \notin \mathbb{R}$. Then we may form a lattice $\Gamma = \{n_1 e_1 + n_2 e_2 \mid n_1, n_2 \in \mathbb{Z}\}$, giving us a discrete subgroup of $\mathbb{C}$. Then we say that a function $f(z)$ defined on $\mathbb{C}$ is *doubly periodic* and has Γ as period, if:

$$f(z + n_1 e_1 + n_2 e_2) = f(z) \quad \forall z \in \mathbb{C}$$

Using induction, this is equivalent to the condition

$$f(z + e_1) = f(z) \quad f(z + e_2) = f(z)$$

Now, if $z_0 \in \mathbb{C}$ is any complex number, we may consider the closed parallelogram bounded by

$$z_0 \quad z_0 + e_1 \quad z_0 + e_2 \quad z_0 + (e_1 + e_2)$$

which we may label as:

$$P = \{z + 0 + t_1 e_1 + t_2 e_2 \mid t_1, t_2 \in [0, 1]\}$$

this parallelogram is called the *parallelogram of periods* or the *fundamental domain* with first vertex z_0 . Let's say f is a meromorphic functions on $\mathbb{C}$ that has Γ as its group of periods and we choose z_0 in such a way that f has no poles on the boundary γ of the parallelogram of periods with z_0 as its first vertex. Does such a function exist? Naturally, smooth such functions exist, so the interesting questions is are there any holomorphic functions that are doubly periodic? A doubly periodic meromorphic function on $\mathbb{C}$ is called an *elliptic function*.

Proposition 5.1.2: Holomorphic Elliptic Functions are Constant

Let f be a holomorphic function on $\mathbb{C}$ that is doubly periodic with respect to a lattice Γ . Then f is constant. Equivalently, an elliptic function with no poles is constant.

Proof:

The closed parallelogram of periods P is compact, so the continuous function $|f|$ is bounded on P . By double periodicity every point of $\mathbb{C}$ is congruent modulo Γ to a point of P , and $|f|$ takes equal values at congruent points, so $|f|$ is bounded on all of $\mathbb{C}$. A bounded entire function is constant by Liouville's theorem (theorem 3.3.3).

Hence a nonconstant doubly periodic function must be meromorphic rather than holomorphic. How would we find such a function? The easier way is to start with a meromorphic function g , and take its "average", that is

$$f(z) = \sum_{n,m} g(z + n e_1 + m e_2)$$

as long as the series converges absolutely (so that we have no worry of the order of summation), this is doubly periodic. To make our life easier, let's only consider rational functions for now to at least show the existence of elliptic functions. To roughly see what conditions we need on g so that the series is uniformly convergent, draw out a lattice in $\mathbb{C}$, and take a circle of radius R and $R + 1$.

Then the number of points between these two rings that belongs in the lattice is roughly a constant c times R , that is $cR + 2\pi$:

$$(\pi \cdot R^2 + \pi R^1 + 2\pi) - \pi \cdot R^2$$

Thus, if $g(z) = 1/z^\alpha$, we see that that when evaluating at any point, the functions grows by about:

$$\sum_{R \in \mathbb{N}} \frac{cR}{R^\alpha}$$

Then this converges when $\alpha > 2$:

$$g(z) \leq \frac{C}{z^\alpha} \quad \alpha > 2$$

We may take g to be any rational function of degree ≥ -3 . This means that we can easily find an elliptic function with 3 poles (ex. $g(z) = \frac{1}{z-a} \frac{1}{z-b} \frac{1}{z-c}$). The question for 1 and 2 poles shall soon be addressed.

With the existence of elliptic functions established, we give a lemma that shall simplify comparing elliptic functions:

Lemma 5.1.3: Comparing Elliptic Function Using Poles

Let f_1, f_2 be two elliptic functions for the same lattice, neither identically zero. Then if the poles and zeros match, with their multiplicities, then $f_1 = f_2$ up to a constant

Proof:

The quotient f_1/f_2 is meromorphic on $\mathbb{C}$ and doubly periodic. At a common zero or pole z_0 , of order k as a zero or $-k$ as a pole for both, write $f_j(z) = (z - z_0)^k g_j(z)$ with g_j holomorphic and nonzero at z_0 , as in lemma 3.5.9. Then $f_1/f_2 = g_1/g_2$ near z_0 , which is holomorphic there. So f_1/f_2 extends to a doubly periodic entire function, which is constant by proposition 5.1.2, as claimed.

As we are working with a meromorphic function, we should find some information about its roots/poles. We can consider the integral $\int_\gamma f(z) dz$. Integrating around the boundary of a fundamental domain is the source of every structural constraint on an elliptic function, and its first consequence is that the residues over a period parallelogram must cancel.

Proposition 5.1.4: Sum of Residues of an Elliptic Function is Zero

Let f be an elliptic function with period lattice Γ , and let P be a fundamental parallelogram whose boundary γ carries no pole of f . Then the sum of the residues of f at its poles inside P is zero:

$$\sum_{p \in P} \text{Res}_p(f) = 0$$

Proof:

Interchanging e_1 and e_2 if necessary, we may assume $\text{Im}(e_2/e_1) > 0$, so that the path γ from the first vertex z_0 through $z_0 + e_1$, $z_0 + e_1 + e_2$ and $z_0 + e_2$ back to z_0 is the oriented boundary of P . By the Residue theorem (theorem 3.5.4) $\int_\gamma f(z) dz = 2\pi i \sum_{p \in P} \text{Res}_p(f)$, so it suffices to

show $\int_{\gamma} f(z) dz = 0$. Writing γ as the four sides of P , and pairing the two edges related by the period e_2 with the two related by e_1 ,

$$\begin{aligned} \int_{\gamma} f(z) dz &= \int_0^1 [f(z_0 + te_1) - f(z_0 + te_1 + e_2)] e_1 dt \\ &\quad + \int_0^1 [f(z_0 + e_1 + te_2) - f(z_0 + te_2)] e_2 dt \end{aligned}$$

By periodicity $f(z_0 + te_1 + e_2) = f(z_0 + te_1)$ and $f(z_0 + e_1 + te_2) = f(z_0 + te_2)$, so both integrands vanish and $\int_{\gamma} f(z) dz = 0$, as claimed.

The same cancellation, applied to f'/f in place of f , controls the zeros and poles of f directly.

Proposition 5.1.5: Zeros and Poles of Doubly Periodic Functions

Let $f(z)$ be a non-constant meromorphic function in $\mathbb{C}$ with Γ as group of periods.

1. For every countable set $E \subseteq \mathbb{C}$, there are parallelograms of periods whose boundary meets no point of E . In particular, there are parallelograms of periods whose boundary carries no zero and no pole of f .
2. If the boundary of a parallelogram of periods P carries no pole of f , the poles of f in P , listed with multiplicity, correspond bijectively to the classes modulo Γ of the poles of f , with their multiplicities. So their number does not depend on P . The same holds for the zeros of f when the boundary of P carries no zero.
3. For a parallelogram of periods whose boundary carries no zero and no pole of f , the number of zeros of f contained in it is equal to the number of poles contained in it, both counted with multiplicity.

Proof:

1. Let the first vertex z_0 move along a line L parallel to neither e_1 nor e_2 . A point q lies on the side $z_0 + [0, 1]e_1$ exactly when z_0 lies on the segment $q - [0, 1]e_1$, which meets L at most once, and likewise for the other three sides. So each point of E lies on the boundary for at most four positions of z_0 on L , and all but countably many positions give a boundary missing E . The zeros and poles of f are isolated, so every compact set contains finitely many of them, and they form a countable set.
2. Let z_0 be the first vertex of P . Every $z \in \mathbb{C}$ is congruent modulo Γ to exactly one point of $Q = \{z_0 + t_1e_1 + t_2e_2 \mid t_1, t_2 \in [0, 1)\}$, obtained by reducing the real coordinates of $z - z_0$ in the basis e_1, e_2 modulo 1. Points of Q not on the boundary of P lie in its interior, and two distinct points of the interior are never congruent. By periodicity, congruent points are poles of the same order. Since no pole lies on the boundary, the representative in Q of each class of poles lies in the interior of P , and the poles of f in P are pairwise non-congruent interior points. This gives the bijection, and the argument for the zeros is the same with zeros in place of poles.
3. Differentiating $f(z + e_j) = f(z)$ gives $f'(z + e_j) = f'(z)$, so f'/f is an elliptic function, f

being non-constant and hence not identically zero. Its poles in P are the zeros and poles of f , none of them on the boundary, and its residue there is the order of the zero, or minus the order of the pole (lemma 3.5.9). By proposition 5.1.4 applied to f'/f , these residues add up to zero, which is the claim.

Since the number of poles in a parallelogram of periods whose boundary carries no pole does not depend on the parallelogram (proposition 5.1.5(2)), it is called the number of poles of f in a period parallelogram.

Corollary 5.1.6: Number Of Poles Reflects Mapping Property

Let f be a non-constant elliptic function with m poles in a period parallelogram, counted with multiplicity. Then for every $c \in \mathbb{C}$, the equation $f(z) = c$ has exactly m solutions, counted with multiplicity, in any parallelogram of periods whose boundary carries no pole of f and no solution of $f(z) = c$.

Proof:

Let P be such a parallelogram. The function $f - c$ is a non-constant elliptic function with the same poles as f , with the same multiplicities, and its zeros are the solutions of $f(z) = c$. By proposition 5.1.5(3) applied to $f - c$, the number of solutions in P , counted with multiplicity, equals the number of poles of f in P . By proposition 5.1.5(2), as the boundary of P carries no pole, that number is the number m of poles of f in a period parallelogram, as claimed.

Proposition 5.1.7: Relation between function and Group

Let $f(z)$ be a non-constant meromorphic function in the whole plane $\mathbb{C}$ having Γ as its group of periods, and let $a \in \mathbb{C}$. Let P be a parallelogram of periods whose boundary carries no pole of f and no solution of $f(z) = a$, denote by α_i the roots in P of the equation $f(z) = a$ and by β_i the poles of f in P , each listed as often as its multiplicity. Then:

$$\sum_i \alpha_i \equiv \sum_i \beta_i \pmod{\Gamma}$$

In particular, the class modulo Γ of the sum of the roots of $f(z) = a$ is independent of a .

Proof:

Interchanging e_1 and e_2 if necessary, we may assume $\text{Im}(e_2/e_1) > 0$, so that the path γ from the first vertex z_0 of P through $z_0 + e_1$, $z_0 + e_1 + e_2$ and $z_0 + e_2$ back to z_0 is the oriented boundary of P . Put $F = f'/(f - a)$, an elliptic function with no poles on γ . Then corollary 3.5.12, applied in $\mathbb{C}$ to the closed parallelogram and to the polynomial $z \mapsto z$, gives

$$\frac{1}{2\pi i} \int_{\gamma} z F(z) dz = \sum_{\substack{p \\ \text{poles and zeros of } f-a}} n_p \cdot p = \sum_i \alpha_i - \sum_i \beta_i \quad (5.1)$$

where n_p is the order of the zero of $f - a$ at p , or minus the order of the pole, the poles of $f - a$ being those of f . Let $\sigma_1(t) = z_0 + te_1$ and $\sigma_2(t) = z_0 + te_2$ for $t \in [0, 1]$. The sides of γ are, in order,

σ_1 , then $\sigma_2 + e_1$, then $\sigma_1 + e_2$ reversed, then σ_2 reversed. Since $F(z + e_1) = F(z + e_2) = F(z)$,

$$\begin{aligned} \int_{\sigma_1} zF(z)dz - \int_{\sigma_1+e_2} zF(z)dz &= \int_0^1 (\sigma_1(t) - \sigma_1(t) - e_2)F(\sigma_1(t))e_1 dt = -e_2 \int_{\sigma_1} F(z)dz \\ \int_{\sigma_2+e_1} zF(z)dz - \int_{\sigma_2} zF(z)dz &= \int_0^1 (\sigma_2(t) + e_1 - \sigma_2(t))F(\sigma_2(t))e_2 dt = e_1 \int_{\sigma_2} F(z)dz \end{aligned}$$

Adding the two lines gives the integral over γ , so

$$\frac{1}{2\pi i} \int_{\gamma} zF(z)dz = -\frac{e_2}{2\pi i} \int_{\sigma_1} F(z)dz + \frac{e_1}{2\pi i} \int_{\sigma_2} F(z)dz$$

For $k = 1, 2$, the chain rule gives

$$\frac{1}{2\pi i} \int_{\sigma_k} \frac{f'(z)}{f(z) - a} dz = \frac{1}{2\pi i} \int_0^1 \frac{(f \circ \sigma_k)'(t)}{f(\sigma_k(t)) - a} dt = I(f \circ \sigma_k, a)$$

the winding number of the path $f \circ \sigma_k$ around a (definition 3.1.17). This path is closed, as $f(z_0 + e_k) = f(z_0)$, and it avoids a , so $I(f \circ \sigma_k, a)$ is an integer n_k (box 3.1.10). With (5.1) we get

$$\sum_i \alpha_i - \sum_i \beta_i = n_2 e_1 - n_1 e_2 \in \Gamma$$

so the sums of the roots and of the poles are in the same class modulo Γ . By the correspondence of proposition 5.1.5(2), the class of $\sum_i \beta_i$ is the sum of the classes modulo Γ of the poles of f , with multiplicity, so it depends neither on a nor on P , which gives the last claim.

Corollary 5.1.8: Single Simple Pole Elliptic Function Does Not Exist

There does not exist an elliptic function with a single simple pole, that is, with exactly one pole in a fundamental parallelogram counted with multiplicity.

Proof:

Suppose f were one, and take a parallelogram of periods whose boundary carries no zero and no pole of f (proposition 5.1.5(1)). It contains one simple pole b , and by proposition 5.1.5(3) exactly one zero a , which is simple. By proposition 5.1.7 with the value 0, we have

$$a - b = ne_1 + me_2$$

for some integers n, m . By periodicity $f(b) = f(a) = 0$, which is impossible at the pole b , completing the proof.

Overall, elliptic functions must satisfy the following two properties:

$$\sum n_p = 0 \quad \sum pn_p = ne_1 + me_2$$

These will turn out to be sufficient conditions to satisfy to be an elliptic function, meaning they are completely determined by these algebraic limitations. We shall also find an elliptic function with two pole in the following section.

5.2 Weierstrass $\wp$ -function

The following function can be thought of as the building block of elliptic functions: all elliptic functions will be a rational function of $\wp$ and $\wp'$ (theorem 5.2.6). Let $\Gamma = \{ne_1 + me_2 \mid e_1, e_2 \in \mathbb{C}\}$ where e_1, e_2 are linearly independent. We shall show we can associate to such a group a function with many interesting properties.

First, the function

$$\sum_{\omega \in \Omega} \frac{1}{(z - \omega)^2}$$

is not absolutely convergent as we saw before. Instead, we will modify it so that it shrinks fast enough so that it *is* absolutely convergent:

Proposition 5.2.1: Weierstrass $\wp$ -Function

Let $\Omega \subseteq \mathbb{C}$ be a discrete subgroup with two generators. Then the series

$$\wp(z) = \frac{1}{z^2} + \sum_{\omega \in \Omega \setminus \{0\}} \left(\frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right)$$

is uniformly convergent on compact subsets of $\mathbb{C}$

Note that this is no longer the “averaging” of a rational functions, but an affine translation of a holomorphic function. We shall first show it converges (hence well-defined) and then show it is indeed an elliptic function. To prove convergence, we first require the following lemma:

Lemma 5.2.2: Weierstrass Function Lemma

The series

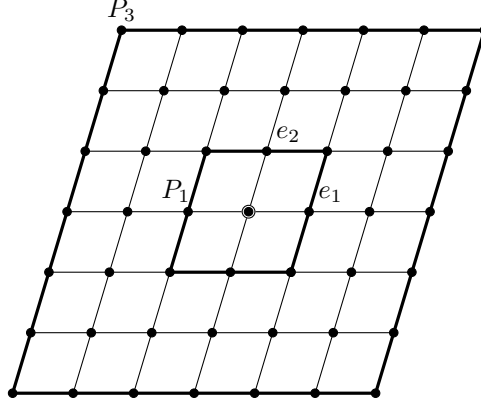
$$\sum_{\substack{\omega \in \Omega \\ \omega \neq 0}} \frac{1}{|\omega|^3}$$

converges

Proof:

For each $n \in \mathbb{N}_{>0}$, consider the parallelogram P_n formed by the points $z = t_1 e_1 + t_2 e_2$ where

the real numbers t_1, t_2 respect $\sup(|t_1|, |t_2|) = n$, take for example:



In each border of P_n , there are $8n$ points of Ω , and the distance between each of these points and 0 is $\geq kn$ for some fixed $k > 0$ (the smallest distance from 0 to the points of P_1). Then the sum of $\frac{1}{|\omega|^3}$ over the points of P_n is bounded by $\frac{8n}{kn^3}$, hence:

$$\sum_{\omega \neq 0} \leq \sum_{n \geq 1} \frac{8}{k^3 n^2}$$

and since $\frac{1}{n^2}$ is convergent, by the comparison test so is our desired series.

Proof:

Of Weierstrass $\wp$ Function Convergence

We will show $\wp$ converges on $|z| \leq r$. Then $|\omega| \geq 2r$ for all but finitely many ω , thus for all but finitely many terms:

$$\left| \frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right| = \left| \frac{2\omega(z - \omega^2)}{\omega^2(\omega - z)^2} \right| = \frac{|z(2 - \frac{z}{\omega})|}{|\omega^3| |1 - \frac{z}{\omega}|^2} \leq \frac{r(5/2)}{|\omega|^3(1/4)} = \frac{10r}{|\omega|^3}$$

since $|z| \leq r$. But then, by our above lemma, we see that the series converges uniformly on the disc, and hence converges uniformly on compact sets, as we sought to show.

Definition 5.2.3: Weierstrass Function

A *Weierstrass function* $\wp$ is a meromorphic function which is the sum of a series as in the above proposition depending on a choice of discrete group Ω with two generators.

Proposition 5.2.4: Weierstrass Function is Elliptic

The Weierstrass function $\wp$ is an elliptic function.

Proof:

First, $\wp$ is an even function of z , since:

$$\wp(-z) = \frac{1}{z^2} + \sum_{\omega \neq 0} \left(\frac{1}{(z+\omega)^2} - \frac{1}{\omega^2} \right) = \frac{1}{z^2} + \sum_{\omega \neq 0} \left(\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} \right)$$

where we use the fact that the sum sums over negative values. Furthermore, by proposition 4.1.11, we have that

$$\wp'(z) = -2 \sum_{\omega \in \Omega} \frac{1}{(z-\omega)^3}$$

Notice that the derivative is Ω -periodic and is odd:

$$\wp'(z+\omega) = \wp'(z) \quad \wp'(-z) = -\wp'(z)$$

Therefore, $\wp'(z+e_i) - \wp'(z) = 0$. Hence, we get:

$$\wp(z+e_i) - \wp(z) = c$$

By linearity, it suffices to show that $\wp(z+e_i) = \wp(z)$ for $i \in \{1, 2\}$. If we plug in $z = e_i/2$ and take advantage of evenness, we get

$$0 = c$$

and hence it is indeed periodic, as we sought to show.

Overall, The Weierstrass $\wp$ -function is a doubly periodic meromorphic function with period Ω , with poles at the points of Ω , each pole having order 2 and principal part $\frac{1}{(z-\omega)^2}$. Note to by the evenness of $\wp$ we have that $\wp(\pm a) = b$.

Weierstrass functions are important in many ways. The following identify is usually considered the most important property of Weierstrass functions due to their connection to curves of genus one in algebraic geometry:

Theorem 5.2.5: Elliptic Curve Of Weierstrass Function

Let $\wp$ be a Weierstrass p -function. Then there exists constants g_2 and g_3 such that

$$(\wp')^2 = 4\wp^3 - g_2\wp - g_3$$

This can be intuitively seen like so: since $\wp$ has a pole of order 2, $\wp'$ has a pole of order 3. Hence, there is some relation between $(\wp')^2$ and $\wp^3$ up to some massaging:

Proof:

Recall that two doubly-periodic functions are equal up to a constant if they have the same poles and principal parts, and if then it may be shown that the constant is zero, the two functions are equal. This essentially means that we can construct many important identities by “cancelling out the poles” so to speak. Thus, start with:

$$\wp(z) = \frac{1}{z^2} + a_2 z^2 + a_4 z^4 + \dots$$

Then

$$\begin{aligned}\wp(z) &= -2z^{-3} + 2a_2z + 4a_4z^3 + \cdots \\ \wp'(z) &= -2z^{-3} + 2a_2z + 4a_4z^3 + \cdots \\ \wp'(z)^2 &= 4z^{-6} + 8a_2z^2 - 16a_4 + \cdots \\ \wp(z)^3 &= z^{-6} + 3a_2z^{-2} + 3a_4\end{aligned}$$

Thus:

$$\wp'(z)^2 - 4\wp(z)^3 = -20a_2z^{-2} - 28a_4 + z^2(\cdots)$$

Thus, we get that the function $\wp'(z)^2 - 4\wp(z)^3 + 20a_2\wp(z) + 28a_4$ is holomorphic in some neighbourhood of the origin and hence has no poles and is must be holomorphic and hence constant as it is bounded. As it is zero at the origin, it must be zero. We thus get:

$$\wp'(z)^2 = 4\wp(z)^3 - 20a_2\wp(z) - 28a_4$$

which can be thought of as plugging in $(\wp, \wp')$ into the algebraic curve:

$$y^2 = 4x^3 - 20a_2x - 28a_4$$

This curve has some nice properties, for one all it's poles are always distinct, that is the equation

$$0 = 4x^3 - 20a_2x - 28a_4$$

has 3 distinct roots. Later, we shall see that a_2 and a_4 are *modular forms* when interpreted as function on the upper half plane (see [6]).

Let $r = \min \{|\omega| \mid 0 \neq \omega \in \Omega\}$. Let $G_n = \sum_{0 \neq \omega \in \Omega} \omega^{-n}$. Then the Laurent expansion around $0 < |\lambda| < r$ is:

$$\wp(z) = \frac{1}{z^2} + \sum_n (2n+1)G_{2n+2}z^{2n}$$

It is easy to find the roots of $\wp'$: if the period is $1, \tau$ (which, up to a biholomorphism all elliptic functions have this period, where $\tau = \omega_2/\omega_1$), then the roots of $\wp'$ are

$$1/2, \tau/2, (1+\tau)/2$$

However, the roots of $\wp$ are in general hard to determine. If $\wp(a) = 0$ and a is a double zero, then

$$a = \frac{\omega_1 + \omega_2}{2}$$

Besides this case, it is rather difficult to find the root, and starts going into the theory of modular forms (the modular form $j(\tau)$ is involved).

Theorem 5.2.6: Elliptic Function Rational Function Of Weierstrass

Let f be an elliptic function. Then f is a rational polynomial of the Weierstrass function and it's derivative:

$$f = R(\wp, \wp')$$

Proof:

1. Any even elliptic function $f(z)$ can be written in the form

$$f(z) = c \prod_{k=1}^n \frac{\wp(z) - \wp(a_k)}{\wp(z) - \wp(b_k)}$$

(where c is a constant and $\wp(z)$ denotes the Weierstrass $\wp$ -function with the same periods), provided that 0 is neither a zero nor a pole of f . Conclude that every even elliptic function f can be written $f = R(\wp)$ where R is a rational function

Proof. If f is a constant, then $f(z) = c$, so assume f is non-constant.

First, we shall find some properties of the roots in the case where $f = \wp$. In particular, if $a \not\equiv -a \pmod{\Omega}$, then a is a simple zero of $\wp$, and if $a \equiv -a \pmod{\Omega}$, then in the fundamental domain:

$$a \in \left\{ \frac{\omega_1}{2}, \frac{\omega_2}{2}, \frac{\omega_1 + \omega_2}{2} \right\} \quad (5.2)$$

and a is a root of order 2. To see this, we need to show that $\wp'(a) = 0$. Since $\wp$ is odd and doubly periodic, we see that:

$$-\wp'\left(\frac{\omega_1}{2}\right) = \wp'\left(-\frac{\omega_1}{2}\right) = \wp'\left(\frac{\omega_1}{2}\right)$$

Hence, $\wp'\left(\frac{\omega_1}{2}\right) = 0$. Similarly for $\frac{\omega_2}{2}$ and $\frac{\omega_1 + \omega_2}{2}$. Since $\wp'$ has order 3 (since $\wp$ has order 2), these must all be simple zeros of $\wp'$. Hence a is a simple within the fundamental domain if it is not equal to one of these values, and is of order 2 if it is equal to one of these values. More generally, if f is an even elliptic function, then for some root a , we may divide f by $(\wp - \wp(a))^n$ for appropriate n that will make the resulting function have the root be of order 2 or 3, at which point we repeat the argument.

Next, two elliptic functions are equal up to a constant if they share the same roots/poles: if f_1, f_2 share their roots and pole, f_1/f_2 is holomorphic, hence bounded in $\mathbb{C}$ (due to periodicity), hence constant.

Thus, it suffices to show that the given expression f has the same number of roots/poles as the expression on the right hand side. Let f be an elliptic function with period group Ω , and let $a_1, a_2, \dots, a_k$ be the roots of f within the fundamental domain (counting multiplicity, and $a_i \neq 0$ for all i) with corresponding poles $b_1, b_2, \dots, b_k$ (recall the number of roots and poles match) where $b_i \neq 0$ for all i . If each a_k/b_k is a simple root/pole of $\wp - \wp(a_k)$ (resp. $\wp - \wp(b_k)$), then we may simply take:

$$g(z) = \frac{f(z)}{\prod_{k=1}^n \frac{\wp(z) - \wp(a_k)}{\wp(z) - \wp(b_k)}}$$

giving us a bounded holomorphic function, hence for some constant $c \in \mathbb{C}$:

$$f(z) = c \prod_{k=1}^n \frac{\wp(z) - \wp(a_k)}{\wp(z) - \wp(b_k)}$$

If one of the a_k/b_k is in the set in equation (5.2), then that means that there is an i, j such that $a_i = a_j$ (resp. $b_i = b_j$). In this case, since $\wp - \wp(a_k)$ has order 2, we do not need to divide by $\wp - \wp(a_k)$ twice (or else we would over count).

Notice that even if we change the number products in the numerator/denominator, there *must* be the same number of products in the denominator/numerator, or else $f(z)$ would have a pole/zero at 0, contradicting the given conditions of f . Thus, we once again end up with:

$$f(z) = c \prod_{k=1}^n \frac{\wp(z) - \wp(a_k)}{\wp(z) - \wp(b_k)}$$

as we sought to show. $\square$

2. Show that every odd elliptic function f can be written $f = \wp' R(\wp)$ where R is a rational function

Proof. If f is odd, then since $\wp'$ is odd, $g = f/\wp'$ is even ($f(-z)/\wp'(-z) = f(z)/\wp'(z)$). Then we may apply the above to get g in terms of a rational function of $\wp$, which gives us

$$f = \wp' R(\wp)$$

as we sought to show. $\square$

3. show that every elliptic function f can be written $f = R(\wp, \wp')$, where R is rational

Proof. Note that

$$f = \frac{f(z) - f(-z)}{2} + \frac{f(z) + f(-z)}{2}$$

where the first term on the right hand side is odd and the 2nd term is even. Then $f = \wp' R_1(\wp) + R_2(\wp) = R(\wp, \wp')$, as we sought to show. $\square$

Theorem 5.2.7: Additive Law For Weierstrass Function

Let Ω be the fundamental parallelogram and $u, v \in \Omega$. Then:

$$\wp(u+v) = -\wp(u) - \wp(v) + \frac{1}{4} \left(\frac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)} \right)^2$$

Proof:

First, we will show that for all $a, b \in \mathbb{C}$, the function

$$\wp' - a\wp(z) - b$$

has 3 zeros in a period parallelogram, and their sum equals a period. Then, we will show that if $u, v \in \mathbb{C}$, $u \pm v \not\equiv 0 \pmod{\Gamma}$, then we can find a, b such that the function above has zeros at

$u, v, -u - v$ and deducate the additive theorem: if $u + v + w = 0$, then

$$\det \begin{pmatrix} \wp(u) & \wp'(u) & 1 \\ \wp(v) & \wp'(v) & 1 \\ \wp(w) & \wp'(w) & 1 \end{pmatrix} = 0$$

Proof. First, note that the $\wp'$ has a pole of order 3 and $\wp$ at each $\omega \in \Omega$ and $a\wp$ has a pole of order 2 at each $\omega \in \Omega$. Since the principal parts differ, the poles do not cancel out, and hence there are three poles in the period parallelogram, meaning there are 3 zeros in the period parallelogram (by using the argument principle as we did in class). Since the sums of the poles and the zeros are $0 \bmod \Omega$, and the sums of the poles are $0 \bmod \Omega$ (since they are all in Ω), we see that the sum of the zeros are $0 \bmod \Omega$, i.e. the sum of the poles are in Ω .

Now, choose $u, v \in \mathbb{C}$. We will want to find a, b such that

$$\wp'(u) - a\wp(u) - b = 0 \quad \wp'(v) - a\wp(v) - b = 0$$

If u, v are zeros of this equation, then since $u + v + w \in \Omega$, it must be that $w \equiv -u - v \bmod \Omega$. If $-u - v \equiv 0 \bmod \Omega$, then $u + v \equiv 0 \bmod \Omega$, so $-u - v$ would also be a *pole*, but that contradicts the fact that it's a *root*.

Hence, let's say $u + v \not\equiv 0 \bmod \Omega$. By linear-algebra manipulation, we get:

$$a = \frac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)} \quad b = \wp'(v)\wp(u) - \wp(v)\wp'(u)$$

where $\wp(u) - \wp(v) \neq 0$ since . Finally, if $u + v + w = 0$, then notice that

$$\begin{pmatrix} \wp(u) & \wp'(u) & 1 \\ \wp(v) & \wp'(v) & 1 \\ \wp(w) & \wp'(w) & 1 \end{pmatrix} \begin{pmatrix} -a \\ 1 \\ b \end{pmatrix} = 0$$

showing the matrix cannot be full rank, hence the determinant is zero. $\square$

We now prove the main result. Recall that

$$\wp'(z)^2 = 4\wp(z)^3 - 28a_2\wp(z) - 20a_4$$

and that $a = \frac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)}$. Since $\wp'(u) = a\wp(u) + b$, plugging it into the equation we get:

$$a^2\wp(u)^2 + 2ab\wp(u) + b^2 = 4\wp(u)^3 - 28a_2\wp(u) - 20a_4$$

$$\Leftrightarrow$$

$$0 = 4\wp(u)^3 - a^2\wp(u)^2 - 28a_2\wp(u) - 20a_4 - 2ab\wp(u) - b^2$$

Then we know by properties of polynomials that the the sum of the roots will be equal to $a^2/4$, in particular

$$\wp(u + v) + \wp(u) + \wp(v) = \frac{1}{4} \left(\frac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)} \right)^2$$

where $\wp(-u - v) = \wp(u + v)$ by evenness of $\wp$. Re-arranging, we get:

$$\wp(u + v) = -\wp(u) - \wp(v) + \frac{1}{4} \left(\frac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)} \right)^2$$

as we sought to show.

Theorem 5.2.8: Elliptic Functions With Same Period

Let f, g be two elliptic functions with the same period. Then the two are linked by an algebraic relation:

$$f = P(g)$$

Proof:

This is really cool! How to link two elliptic functions with the same period:

[https://math.stackexchange.com/questions/822173/
an-algebraic-relation-between-any-two-elliptic-functions-with-the-same-periods](https://math.stackexchange.com/questions/822173/an-algebraic-relation-between-any-two-elliptic-functions-with-the-same-periods)

6

Complex Manifolds and Riemann Surfaces

We shall now analyze complex functions from a more geometric perspective. We shall start by classifying the 3 possible “geometries” on complex space. What it means to have distinct “geometries” is something that would be covered rigorously in a class in Riemann Geometry (see [7]), however the concepts are immediately accessible and insightful.

After seeing the 3 possible geometries, we shall move onto solving the problem of functions such as $\sqrt{z}$ and $\log(z)$ being multi-valued functions. We shall give an introduction to *Riemann surfaces* and see how these allow us make these properly defined holomorphic functions. We shall conclude by showing that elliptic curves give rise to a natural Riemann surface.

Finally, we shall take a moment to classify all 1-dimensional closed complex manifolds. For 1-dimensional real manifolds, there is only $\mathbb{R}$ and S^1 up to diffeomorphism. This is not so simple for 1-dimensional complex manifolds and requires some more information such as the universal covering (see [23]). However, we can classify all closed complex manifolds which shall be determined by an invariant calls the *genus*.

6.1 Riemann Mapping Theorem

When working with holomorphic functions, we usually work with either:

1. a proper (simply) connected subset $\Omega \subsetneq \mathbb{C}$
2. all of $\mathbb{C}$
3. the compactification $\overline{\mathbb{C}} \cong \mathbb{CP}^1 \cong S^2$

Since being differentiable is a local condition, we may focus on the simply connected cases of $\mathbb{C}$, simply connected proper subset of $\mathbb{C}$, and its compactification. What we shall show in this section is that up to biholomorphism, the only simply connected subsets are:

1. D , the unit disk
2. $\mathbb{C}$, the complex plane
3. $\mathbb{CP}^1$, the complex projective line, equivalently the Riemann sphere.

This will allow us to classify all Biholomorphisms of simply connected subsets of $\overline{\mathbb{C}}$, and is the beginning of Riemann geometry, namely this classifies all simply connected Riemann surfaces¹

6.1.1 Classifying Automorphisms

We start by showing that $\mathbb{C}$ and D are distinct up to biholomorphism:

Theorem 6.1.1: Plane And Disk Not Biholomorphic

The plane $\mathbb{C}$ and the open disk $D = \{z \mid |z| < 1\}$ are *not* biholomorphic

Proof:

Let $f : \mathbb{C} \rightarrow D$ be a holomorphic map. Then this is a bounded map, hence since f is entire it must be constant, thus it cannot be bijective (namely it's not injective), hence cannot be biholomorphic.

Thus, it is meaningful to classify automorphisms of $\mathbb{C}$ and D distinctly. Starting with $\mathbb{C}$:

Proposition 6.1.2: Automorphisms Of $\mathbb{C}$

$$G_{\mathbb{C}} := \text{Aut}_{\text{Holo}}(\mathbb{C}) = \{az + b \mid a, b \in \mathbb{C}, a \neq 0\}$$

Proof:

Consider the type of pole f has at infinity, it must either be essential or a pole, hence either:

1. f has an essential singularity at infinity
2. f is a polynomial

The first one is impossible since the function would not be injective: if f were injective, then $\text{im } f$ of $|z| > 1$ does not meet the image of f under $|z| < 1$, this image is open and non-empty. Hence $|z| > 1$ is not *dense* in the whole plane, hence by Weierstrass theorem (theorem 3.4.10), the point at infinity is *not* an essential singularity, thus f must be a polynomial. But then by D'Alembert (theorem 3.3.4) if the degree is greater than 1, then $f(z) = w$ has n distinct roots (except a small number of w), but f must be injective, hence this is impossible. Thus, the only possibility is to linear functions, completing the proof.

¹We shall also show that this classifies all possible “geometries” in $\mathbb{R}^2$, see [23]

Lemma 6.1.3: Stabilizers And Transitive Groups

Let $D \subseteq \mathbb{CP}^1$ and $G \leq \text{Aut}_{\text{Holo}}(D)$. Then if:

1. G is transitive in D
2. there is at least one point of D whose stabiliser is contained in G

Then $G = \text{Aut}_{\text{Holo}}(D)$.

Proof:

Let $S \in \text{Aut}_{\text{Holo}}(D)$ and $z_0 \in D$ be a point whose stabiliser is contained in G . Since G is transitive, there is a $T \in G$ such that

$$T(z_0) = S(z_0)$$

Hence, the transformation $T^{-1} \circ S \in \text{Aut}_{\text{Holo}}(D)$ leave the point z_0 fixed, and hence belongs in G ; $T^{-1} \circ S \in G$. Thus

$$T \circ (T^{-1} \circ S) = S \in G$$

showing equality, as we sought to show.

Note that if we take the automorphisms of $\mathbb{C}$ and consider the compactification of $\mathbb{C}$, then if $a = 1$, it has a fixed point at infinity, while if $a \neq 1$ then

$$z = \frac{b}{1-a}$$

is a fixed point. Note that $G_{\mathbb{C}}$ is a *transitive group* on the plane, that is for any pair (z_1, z_2) , $z_1, z_2 \in \mathbb{C}$, there exists at least one $f \in G_{\mathbb{C}}$ such that $f(z_1) = z_2$.

Theorem 6.1.4: Automorphisms Of Riemann Sphere

$$\text{Aut}_{\text{Holo}}(\mathbb{CP}^1) = \left\{ \frac{az+b}{cz+d} \mid ad-bc \neq 0 \right\}$$

Proof:

First, any fractional linear transformation takes the Riemann sphere to itself (recall proposition 1.3.4), and since $ad - bc \neq 0$:

$$z = \frac{dz-b}{-cw+a}$$

is the inverse. Certainly this map is holomorphic, hence it is biholomorphic. I claim that that all automorphisms of $\mathbb{CP}^1$ are of this form. Indeed, let $G \leq \text{Aut}_{\text{Holo}}(\mathbb{CP}^1)$ be the subgroup consisting of FLT's, and consider the elements of G which are FLT's leaving the point at infinity fixed, i.e. $c = 0$ and $d \neq 0$, without loss of generality we may suppose $d = 1$ (since multiplying all the constant by a constant does not change the transformation). Then these elements can be written in the form $w = az + b$. Then we see that G is certainly transitive on $\mathbb{CP}^1$ and contain the stabiliser of the point at infinity, hence by the above lemma must be the entire automorphism group, as we sought to show.

For those with an interest in algebraic geometry, if V is a k -vector space of dimension $n + 1$ and $P^n(V)$ its projectivization (so $\mathbb{CP}^1 = P^1(\mathbb{C}^2)$), then the automorphisms of $P^n(V)$ as an algebraic variety are $\mathrm{GL}_{n+1}(k)/k^*$. If $n = 1$ and $k = \mathbb{C}$, these are exactly the Möbius transformations.

Thus, our earlier study of FLT's was in fact the study of the automorphisms of $\mathbb{CP}^1$!

Corollary 6.1.5: Automorphisms Of Upper Half Plane

$$\mathrm{Aut}_{\mathbf{Holo}}(\mathbb{H}) = \left\{ \frac{az+b}{cz+d} \mid a, b, c, d \in \mathbb{R}, ad - bc > 0 \right\}$$

and the stabilizer of i in $\mathrm{Aut}_{\mathbf{Holo}}(\mathbb{H})$ consists of the maps

$$z \mapsto \frac{z \cos(\theta/2) + \sin(\theta/2)}{-z \sin(\theta/2) + \cos(\theta/2)} \quad \theta \in \mathbb{R}$$

Proof:

Let G be the set on the right.

Step 1: G is a subgroup of $\mathrm{Aut}_{\mathbf{Holo}}(\mathbb{H})$. Let $T(z) = \frac{az+b}{cz+d}$ with $a, b, c, d \in \mathbb{R}$ and $ad - bc > 0$. For $z \in \mathbb{H}$ we have $cz + d \neq 0$, since $c = d = 0$ is excluded, and

$$\mathrm{Im}T(z) = \frac{\mathrm{Im}((az+b)(c\bar{z}+d))}{|cz+d|^2} = \frac{(ad-bc)\mathrm{Im}z}{|cz+d|^2}$$

so T maps $\mathbb{H}$ holomorphically into $\mathbb{H}$. If $T_1, T_2 \in G$ have coefficient matrices A_1, A_2 , then multiplying the numerator and the denominator of $T_1(T_2(z))$ by the denominator of $T_2(z)$ shows that $T_1 \circ T_2$ is the Möbius transformation with matrix $A_1 A_2$, which has real entries and determinant $\det A_1 \det A_2 > 0$. The matrix with rows $(d, -b)$ and $(-c, a)$ has determinant $ad - bc > 0$, and its product with the matrix of T is $(ad - bc)$ times the identity, so $w \mapsto \frac{dw-b}{-cw+a}$ is an element of G inverse to T . Hence every element of G is an automorphism of $\mathbb{H}$, and G is a subgroup.

Step 2: G is transitive on $\mathbb{H}$. For $\alpha + \beta i \in \mathbb{H}$, the map $z \mapsto \beta z + \alpha$ lies in G and sends i to $\alpha + \beta i$. Composing such a map with the inverse of another, any point of $\mathbb{H}$ is sent to any other.

Step 3: the stabiliser of i is contained in G . Let $C(z) = \frac{z-i}{z+i}$. Since $|z-i| < |z+i|$ exactly when $\mathrm{Im}z > 0$, C maps $\mathbb{H}$ into $\mathbb{D}$, and $C^{-1}(w) = \frac{i(1+w)}{1-w}$ maps $\mathbb{D}$ into $\mathbb{H}$, as $\mathrm{Im}C^{-1}(w) = \frac{1-|w|^2}{|1-w|^2}$. So C is a biholomorphism of $\mathbb{H}$ onto $\mathbb{D}$ with $C(i) = 0$. Let S be an automorphism of $\mathbb{H}$ with $S(i) = i$. Then $\varphi = C \circ S \circ C^{-1}$ is an automorphism of $\mathbb{D}$ with $\varphi(0) = 0$. Schwarz' lemma (theorem 3.3.13) applied to φ and to φ^{-1} gives $|\varphi(w)| \leq |w|$ and $|w| = |\varphi^{-1}(\varphi(w))| \leq |\varphi(w)|$, so $|\varphi(w)| = |w|$ on $\mathbb{D}$, and its second part gives $\varphi(w) = e^{i\theta}w$ for some $\theta \in \mathbb{R}$. Multiplying the numerator and the denominator by $e^{-i\theta/2}$ in the middle expression below,

$$S(z) = C^{-1}(e^{i\theta}C(z)) = \frac{i((z+i) + e^{i\theta}(z-i))}{(z+i) - e^{i\theta}(z-i)} = \frac{z \cos(\theta/2) + \sin(\theta/2)}{-z \sin(\theta/2) + \cos(\theta/2)}$$

which lies in G , its determinant being $\cos^2(\theta/2) + \sin^2(\theta/2) = 1$.

By lemma 6.1.3, $G = \text{Aut}_{\text{Holo}}(\mathbb{H})$. Each of the maps in the statement fixes i , since $i \cos(\theta/2) + \sin(\theta/2) = i(\cos(\theta/2) - i \sin(\theta/2))$, and Step 3 shows that every automorphism fixing i is one of them, as claimed.

This also shows we may extend automorphisms of $\mathbb{H}$ to automorphisms of $\mathbb{CP}^1$, something that, before the classification, is not obvious. With the automorphisms of $\mathbb{H}$ classified, we may easily get the automorphisms of the unit disk D by composing with $\frac{z-i}{z+i}$.

Theorem 6.1.6: Automorphisms Of Unit Disk

$$\text{Aut}_{\text{Holo}}(D) = \left\{ e^{i\theta} \frac{z - z_0}{1 - \bar{z}_0 z} \mid \theta \in \mathbb{R}, |z_0| < 1 \right\}$$

Proof:

For $|z_0| < 1$ put $\varphi_{z_0}(z) = (z - z_0)/(1 - \bar{z}_0 z)$. Its denominator does not vanish on D , since $|\bar{z}_0 z| < 1$ there, so φ_{z_0} is holomorphic on D . For $|z| < 1$,

$$|1 - \bar{z}_0 z|^2 - |z - z_0|^2 = 1 + |z_0|^2 |z|^2 - |z|^2 - |z_0|^2 = (1 - |z|^2)(1 - |z_0|^2) > 0$$

the cross terms $-2\text{Re}(\bar{z}_0 z)$ cancelling, so $|\varphi_{z_0}(z)| < 1$ and φ_{z_0} maps D into D . A direct substitution gives

$$\varphi_{z_0}(\varphi_{-z_0}(w)) = \frac{(w + z_0) - z_0(1 + \bar{z}_0 w)}{(1 + \bar{z}_0 w) - \bar{z}_0(w + z_0)} = \frac{w(1 - |z_0|^2)}{1 - |z_0|^2} = w$$

for $w \in D$, and replacing z_0 by $-z_0$ gives $\varphi_{-z_0}(\varphi_{z_0}(z)) = z$. So φ_{z_0} is a holomorphic bijection of D with holomorphic inverse φ_{-z_0} , that is, $\varphi_{z_0} \in \text{Aut}_{\text{Holo}}(D)$. The rotation $z \mapsto e^{i\theta} z$ is also an automorphism of D , so every map $e^{i\theta} \varphi_{z_0}$ of the stated form lies in $\text{Aut}_{\text{Holo}}(D)$.

Conversely, let $f \in \text{Aut}_{\text{Holo}}(D)$ and put $z_0 = f^{-1}(0) \in D$. Then $g = f \circ \varphi_{-z_0}$ is an automorphism of D with $g(0) = f(z_0) = 0$, and so is g^{-1} . Both map D into D and fix 0, so Schwarz' lemma (theorem 3.3.13(1) with $M = 1$) gives $|g(z)| \leq |z|$ and $|g^{-1}(w)| \leq |w|$ for all $z, w \in D$. Taking $w = g(z)$ in the second inequality gives $|z| \leq |g(z)|$, so $|g(z)| = |z|$ for every $z \in D$. In particular $|g(z_1)| = |z_1|$ at any $z_1 \neq 0$ in D , and theorem 3.3.13(2) gives $g(z) = \lambda z$ with $|\lambda| = 1$, say $\lambda = e^{i\theta}$. Therefore

$$f(z) = g(\varphi_{z_0}(z)) = e^{i\theta} \frac{z - z_0}{1 - \bar{z}_0 z}$$

which is of the stated form, completing the proof.

6.1.2 Riemann Mapping Theorem

We shall now build up to showing that all simply connected proper open subsets $U \subsetneq \mathbb{C}$ are isomorphic to the unit D .

Lemma 6.1.7: Riemann Mapping Theorem Lemma I

Let $D \subsetneq \mathbb{C}$ be simply connected and open. Then there exists an isomorphism of D onto a bounded open set of $\mathbb{C}$.

Proof:

We shall construct an explicit isomorphism. By assumption, there exists a point $a \notin D$. Take $\log(z - a)$ in D . Since D is simply connected, we may choose a branch $g(z)$. Then g is injective in D since if $g(z_1) = g(z_2)$, then:

$$e^{g(z_1)} = e^{g(z_2)} \implies z_1 - a = z_2 - a$$

giving us $z_1 = z_2$. Now, choose $z_0 \in D$. Then g takes the values of a disc E centered at $g(z_0)$ into D . Translating this disk by $2\pi i$, we get a disk which has no points in common with the image of D by g , since e^g is injective. Then:

$$f(z) = \frac{1}{g(z) - g(z_0) - 2\pi i}$$

is holomorphic, injective, and *bounded* in D . Thus, we get an isomorphism of an open such that D onto a bounded set of $\mathbb{C}$, as we sought to show.

Thus, we may assume that D is bounded. By shrinking D enough via the biholomorphism λz and translating, we may, suppose that $0 \in D$ and D is contained in the unit disk $|z| < 1$, which is something we shall assume from here on out in this section

Lemma 6.1.8: Riemann Mapping Theorem Lemma II

If two complex numbers $u, v \in \mathbb{C}$ satisfy $\operatorname{Re}(u) < 0$ and $\operatorname{Re}(v) < 0$, then:

$$\left| \frac{v - u}{v + \bar{u}} \right| < 1$$

Proof:

Write $u = a + ib$ and $v = c + id$ with $a, b, c, d \in \mathbb{R}$, so that $a < 0$ and $c < 0$. Then $\bar{u} = a - ib$, hence

$$v - u = (c - a) + i(d - b) \quad v + \bar{u} = (c + a) + i(d - b)$$

Since $a < 0$ and $c < 0$ we have $c + a < 0$, so $v + \bar{u} \neq 0$ and the quotient is defined. The two imaginary parts agree, so comparing moduli reduces to the real parts:

$$|v + \bar{u}|^2 - |v - u|^2 = (c + a)^2 - (c - a)^2 = 4ac > 0$$

a and c being of the same sign. Hence $|v - u| < |v + \bar{u}|$, which is the claimed inequality.

Lemma 6.1.9: Riemann Mapping Theorem Lemma III

Let $\Omega \subsetneq \mathbb{C}$ be simply connected and bounded, containing 0 and contained in $|z| < 1$. Let $A \subseteq \mathcal{H}(\Omega)$ be a subset of injective functions where

1. $f(0) = 0$
2. $|f(z)| < 1$ for all $z \in \Omega$

Then $\text{im} f$ is exactly the unit disk if and only if $|f'(0)|$ is maximum in the set of values which it takes as f describes A

Proof:

If the image of f is the unit disk, then these two conditions are certainly satisfied, so let's show that these conditions are sufficient.

For the sake of contradiction, let's say that there is a a in the unit disk such that $a \notin \text{im}(f)$. Then I claim there exists a g such that

$$|f'(0)| < |g'(0)|$$

To see this, let

$$F(z) = \log \left(\frac{f(z) - a}{1 - \bar{a}f(z)} \right)$$

This function is holomorphic and injective in Ω . The value of $f(z)$ are in the unit disk, and the values of $\frac{f(z)-a}{1-\bar{a}f(z)}$ are also in the unit disk, hence $F(z)$ has real part < 0 . Naturally, we have chosen some branch of the logarithm for $F(z)$, which we can do since D is simply connected. Now, consider:

$$g(z) = \frac{F(z) - F(0)}{F(z) + \overline{F(0)}}$$

which is holomorphic and injective in $\text{im}(f)$. Then $G(0) = 0$, and furthermore by lemma II $|g(z)| < 1$. Hence, $g \in A$, and:

$$g'(0) = \frac{F'(0)}{F(0) + \overline{F(0)}} \quad F'(0) = \left(\bar{a} - \frac{1}{a} \right) f'(0)$$

Thus:

$$\frac{|g'(0)|}{|f'(0)|} = \frac{1 - a\bar{a}}{2|a| \log \left| \frac{1}{a} \right|}$$

now, to show that $|g'(0)| > |f'(0)|$, it suffice to show that

$$\frac{1 - t^2}{t} - 2 \log(1/t) > 0 \quad 0 < t < 1$$

but this is rather easy to see: the left hand side is a function of t whose derivative is < 0 thus it is strictly decreasing on $0 < t \leq 1$, and is equal to 0 for $t = 1$, and is > 0 for $0 < t < 1$. But then we get the result.

Theorem 6.1.10: Riemann Mapping Theorem

Let $\Omega \subsetneq \mathbb{C}$ be a simply connected proper subset of $\mathbb{C}$. Then Ω is isomorphic to the unit disc.

Proof:

By lemma III, we need only prove that there exists a function $f \in A$ such that the upper bound of $|f'(0)|$ is attained. Let B be the set of all $f \in A$ such that $|f'(0)| \geq 1$. Then the set B is non-empty since $f(z) = z$ is in B . The set B is also bounded in $\mathcal{H}(D)$ since $|f(z)| < 1$ for all $z \in D$ and $f \in B$. If we also show that B is *closed*, then we would get that it is compact. Given B is compact, then the mapping which associates the real number $|f'(0)|$ to each $f \in B$ is a continuous mapping, which then must attain its upper bound, which gives the desired result.

To see that it is compact, we shall use Montel's Theorem (theorem 4.1.23). Let f be a holomorphic function which is a limit of $f_n \in B$. Then

$$f(0) = \lim f_n(0) = 0$$

Next, since the derivative $f'_n \rightarrow f'$, we get that

$$|f'(0)| = \lim_n |f'_n(0)| \geq 1$$

Thus, f is *not* constant on D . Next, f is the limit of injective function, and hence is injective. Since $|f_n(z)| < 1$ for all $z \in D$, we have $|f(z)| \leq 1$. Since $|f(z)| = 1$ is impossible for any $z \in D$ (by the maximum modulus principal, since f is not constant), we have that f satisfies all the condition to be in B , that is

$$f \in B$$

but this then proves the result, completing the proof.

Corollary 6.1.11: Unique Mapping

Let $\Omega \subsetneq \mathbb{C}$ be a simply connected proper subset of $\mathbb{C}$. Then for each $z_0 \in \Omega$, there exists a unique holomorphic function $f(z)$ over Ω where $f(z_0) = 0$ and $f'(z_0) > 0$, such that $f(z)$ is biholomorphic onto the open disk.

Proof:

The uniqueness comes quickly, for if f_1, f_2 were two such functions, then $S = f_1 \circ f_2^{-1}$ would be an injective map from the unit disk to itself. We classified all biholomorphisms, hence we know what such a map would be. Given $S(0) = 0$, $S'(0) \in \mathbb{R}$, and $S'(0) > 0$, the only possibility is that $S(z) = z$, i.e. $S = \text{id}$, hence $f_1 = f_2$.

Corollary 6.1.12: Dimension of Biholomorphisms

Let $\Omega \subsetneq \mathbb{C}$ be a proper simply connected open subset. Then the collection of biholomorphisms $f : \Omega \rightarrow D$ form a 3-dimensional real manifold. If $g'(0) > 0$, then we get a 2-dimensional real manifold; namely Ω itself.

Proof:

By corollary 6.1.11, we can choose $f(a) = 0 \in \Omega$, giving us 2-dimensions worth of points, and a rotation η to make the derivative real, giving us another two dimensions. Hence we have the 3-dimensional manifold $\Omega \times S^1$ that parameterizes all biholomorphisms.

Finally, we can ask the question about the behaviour at the boundary. Recall that the boundary behaviour is in general very unpredictable, for example $1/(1-z)$ has can be extended to all of $\mathbb{C}$ except at 1, while $\sum_1 z^{n^2}$ cannot at all be extended to the boundary; there is a great multitude of behaviour in-between these two scenarios (see theorem 2.3.4 for one of the few results we have). However, when working with biholomorphism we that the boundary behaviour is very predictable:

Theorem 6.1.13: Carathéodory Boundary Theorem

Let $f : \Omega \rightarrow D$ be a biholomorphism from a simply connected open set $\Omega \subsetneq \mathbb{C}$ to the unit disk D . Then f extends homomorphically to $\bar{f} : \bar{\Omega} \rightarrow \bar{D}$ if and only if $\partial\Omega$ is a Jordan curve^a.

^ahomeomorphic to S^1

This proof is beyond the scope of this of this book; the curious reader can checkout [12] or [10]. Note that being just a homeomorphism is not enough: the unit square $[0, 1]^2$ minus the “comb” $\{1/n\} \times [0, 1 - 1/n]$ for all $n \in \mathbb{N}$ has a homeomorphism map that will map the deleted “comb” to the boundary, but the map cannot be extended to the boundary, i.e. to the comb (a sequence on the comb will map to the boundary, while a sequence between the teeth of the comb will not; the idea then is to create a sequence in $[0, 1]^2$ whose convergence is not preserved after extending to $\bar{f}$). In this textbook, we shall look at a more limited version of this theorem in the next section where the boundary is an n -gon.

In chapter 8, we shall see that once we are in dimension $n \geq 2$, the Hartogs extension theorem holds (theorem 8.2.1): if $\Omega \subseteq \mathbb{C}^n$ is open with $n \geq 2$ and $K \subseteq \Omega$ is a compact subset such that $\Omega \setminus K$ is connected, then every holomorphic function on $\Omega \setminus K$ extends uniquely to a holomorphic function on all of Ω .

Notice how this is false in 1-dimension, say $f(z) = 1/z$ on the punctured disk; this result can be thought of as the beginning of the study of multi-dimensional complex functions as its own study.

Exercise 6.1.1

1. We have found how to factor all holomorphic functions on $\mathbb{C}$ and $\mathbb{CP}^1$. Using the Riemann mapping theorem, it suffices to find how to factor functions on D (develop Blaschke products).

6.1.3 Application: Schwarz-Christoffel Mappings

One consequence of the Riemann mapping theorem that seems totally implausible is that Ω is biholomorphic to regular n -gons, even though it seem like the angles may change at the corners. In fact, it is not too difficult to explicitly describe the biholomorphism, which is what we aim to achieve in this section.

Definition 6.1.14: Schwarz–Christoffel Mapping

A Schwarz-Christoffel mapping is a holomorphic function from Ω or $\mathbb{H}$ onto a simple polygon.

Consider an set $\Omega \subseteq \mathbb{C}$ that is a regular n -g with vertices $z_1, z_2, \dots, z_n$ where $z_{n+1} = z_1$, and the angel between each is

$$\arg \frac{z_{k-1} - z_k}{z_{k+1} - z_k} = \alpha_k \pi \quad (0 \leq \alpha_k < 2)$$

For future purposes, we also introduce

$$\beta_k \pi = (1 - \alpha_k) \pi \quad -1 < \beta_k < 1$$

since

$$\beta_1 + \dots + \beta_n = 2$$

and the polynomial is convex if $\beta_i > 0$ for all i .

Theorem 6.1.15: Schwarz-Christoffel Formula

Let $F(w) = z$ be the map from $|w| < 1$ that conformally maps onto the polygons with angles $\alpha_k \pi$ (for $k \in \{1, \dots, n\}$). Then such a map is of the form:

$$F(w) = C \int_0^w \prod_{k=1}^n (t - w_k)^{-\beta_k} dt + C'$$

where $\beta_k = 1 - \alpha_k$, the w_k are points on the unit circle $|w| = 1$, and C, C' are complex constants.

Proof:

Let Ω be the interior of the polygon, and let $f : \Omega \rightarrow D$ be the map given by the Riemann mapping theorem. We shall show that F as defined in theorem 6.1.15 is the inverse. Without loss of generality we shall compose D to conformally map to the upper-half plane.

For each α_k , define:

$$g_{\alpha_k}(\zeta) = f(z_k + \zeta^{\alpha_k})$$

This function is invertible around 0 (as we can extend to the half-plan by the reflection principle). Thus, write:

$$\zeta = \sum_{n=0}^{\infty} b_n (w - w_k)^n$$

Since we expanded around $\zeta = 0$, we have $b_0 = 0$. The derivative is non-zero by the definition of g_{α_k} , hence we have:

$$\zeta = \sum_{n=1}^{\infty} b_n (w - w_k)^n = (w - w_k) \left(\sum_{n=0}^{\infty} b_{n+1} (w - w_k)^n \right)$$

Let the 2nd product on the right hand side be $g_k(w)$. By construction, $g_k(w)$ is non-zero around $w = w_k$. As $z = z_k + \zeta^{\alpha_k}$, we have:

$$z = F(w) = z_k + (w - w_k)^{\alpha_k} g_k(w)^{\alpha_k}$$

To simplify the following derivative, re-label $g_k(w)^{\alpha_k}$ as $g_k(w)$ (so that we do not keep applying the chain rule). Now, taking the derivative of this we get:

$$F'(w) = \alpha_k(w - w_k)^{\alpha_k - 1} g_k(w)^{\alpha_k} + (w - w_k)^{\alpha_k} g'_k(w)$$

By replacing $\beta_k = 1 - \alpha_k$, we re-write as :

$$F'(w)(w - w_k)^{\beta_k} = \alpha_k g_k^{\alpha_k}(w) + \alpha_k(w - w_k)g'_k(w)$$

Then $F'(w)(w - w_k)^{\beta_k}$ is non-zero around w_k , and $F'(w)$ is nonzero way from the vertices (as f is conformal at these points). Thus, define the holomorphic function:

$$H(w) = F'(w) \prod_{k=1}^n (w - w_k)^{\beta_k}$$

which vanishes in a neighbourhood of $\overline{D}$.

Now, $H(w)$ is in fact constant, which shall allow us to conclude

$$F'(w) = c \prod_{k=1}^n (w - w_k)^{-\beta_k}$$

Indeed, we first show that $\text{Arg}H(w)$ is constant on S^1 . To see this, first observe that:

$$\text{Arg} \left(\frac{d}{d\theta} F(e^{i\theta}) \right) = \text{Arg}(F'(e^{i\theta})) + \left(\theta + \frac{\pi}{2} \right)$$

To see that the left hand side is constant, consider $e^{i\theta}$ between w_k and w_{k+1} . Then $F(e^{i\theta})$ describes a straight line, so:

$$F(e^{i\theta}) = \alpha t(\theta) + \beta$$

Then:

$$\text{Arg} \left(\frac{d}{d\theta} F(e^{i\theta}) \right) = \text{Arg}(\alpha' t(\theta)) = \text{Arg}(\alpha)$$

Furthermore, $\text{Arg}(e^{i\theta} - w_k) = \frac{\theta}{2} + c$. But then:

$$\text{Arg}H(e^{i\theta}) = -\theta + \left(\sum_{k=1}^n \beta_k \right) \frac{\theta}{2} + c$$

As $\sum \beta_k = 2$, we just get that it is constant! Therefore, by the mean value property, $H(0)$ is this constant. Then by corollary 3.3.9, H is constant on the entire closed disk, hence:

$$F'(w) = c \prod_{k=1}^n (w - w_k)^{-\beta_k}$$

By integrating, we get our desired result.

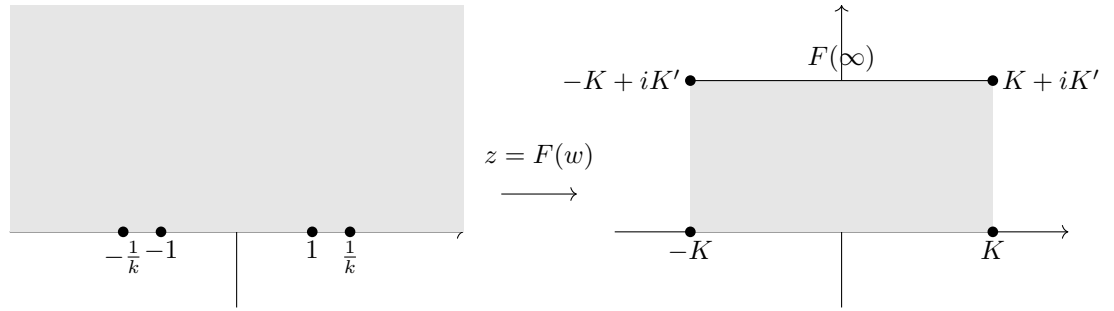
Example 6.1.16: Mapping Onto Rectangle

1. Take $\mathbb{H}^+$, and identify the points $-\frac{1}{k}, -1, 1, \frac{1}{k}$. As $\alpha = 1/2$, $\beta = 1/2 - 1 = -1/2$. Then:

$$F(w) = \int_0^w \frac{1}{\sqrt{t-1}\sqrt{t-\frac{1}{k}}\sqrt{t+\frac{1}{k}}\sqrt{t+1}} dt = \int_0^w \frac{1}{\sqrt{(1-t^2)(1-k^2t^2)}} dt$$

Let us find where the points map to. For the first one, we shall integral from 0 to 1.

$$K = \int_0^1 \frac{1}{\sqrt{(1-t^2)(1-k^2t^2)}} dt \quad K' = \int_1^\infty \frac{1}{\sqrt{(1-t^2)(1-k^2t^2)}} dt$$



Note that this map extends to a map on the entire plane as we can reflect the square to make it a doubly-periodic function.

2. Let us do a triangle with angles $\alpha_1\pi, \alpha_2\pi, \alpha_3\pi$. Then $\beta_i = \alpha_i - 1$ with points $0, 1, \infty$ on $\mathbb{H}^+$. Then we get:

$$z = F(w) = \int_0^w t^{\beta_1} (t-1)^{\alpha_2} dt$$

We can omit the point at infinity^a. We can ask if we can again make the function defined on the entire plane by tiling. The easiest thing to do is tile on integers, we require rational rotations. For that we need:

$$\alpha_k\pi = \frac{2\pi}{2n_k} = \frac{1}{n_k}$$

for integers n_k . As the angles must sum to 2π , we shall need these integers to satisfy:

$$\frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3} = 1$$

The only such integers will be $(3, 3, 3)$, $(2, 4, 4)$, $(2, 3, 6)$.

^aFor reasons I'm not entirely sure unfortunately

6.2 Complex Curves

In this section, we take a look at how to make functions like $\log$ and z^α for fractional α well-defined by finding a good domain. This technique will require the use of manifold theory; the notion of manifolds will be taken as prerequisite for this section, see EYNTKA Differential Geometry, [23], for a refresher. We shall be focusing on 1-dimensional complex manifolds. For reference, we shall put the definition of a complex manifold

Definition 6.2.1: Complex Manifold

Let $n \geq 1$. A map $F = (F_1, \dots, F_m)$ from an open subset of $\mathbb{C}^n$ to $\mathbb{C}^m$ is *holomorphic* if it is continuous and each F_k is holomorphic in each variable separately, the other variables being held fixed. A space X is said to be a *complex manifold* of dimension n if it is paracompact, Hausdorff, and has a *complex atlas*: a covering by open sets U_i with homeomorphisms φ_i of U_i onto open subsets of $\mathbb{C}^n$, called *charts*, such that every transition map $\varphi_j \circ \varphi_i^{-1}: \varphi_i(U_i \cap U_j) \rightarrow \varphi_j(U_i \cap U_j)$ is holomorphic.

A one-dimensional complex manifold is called a *complex curve* or *Riemann Surface* (usually a Riemann surface must also be connected).

A map $f: M \rightarrow N$ between complex manifolds is said to be holomorphic if it is continuous and each representative $\psi \circ f \circ \varphi^{-1}$, for charts φ of M and ψ of N , is holomorphic where it is defined. If a holomorphic inverse exists, then it is said to be *biholomorphic* or *isomorphic* and we write $M \cong N$.

It is sometimes desirable to distinguish a holomorphic function between open sets of $\mathbb{C}$ to itself and between complex manifolds. In this case, functions between complex manifolds will be called *holomorphic maps*.

Example 6.2.2: Complex Manifolds

1. $\mathbb{C}$ is trivially a complex manifold
2. $\mathbb{C}/\mathbb{Z}$ is a complex manifold. Let $\pi: \mathbb{C} \rightarrow \mathbb{C}/\mathbb{Z}$ be the quotient map, with the quotient topology. The map π is open, since $\pi^{-1}(\pi(V)) = \bigcup_{n \in \mathbb{Z}} (V + n)$ for $V \subseteq \mathbb{C}$. For every open set $V \subseteq \mathbb{C}$ on which π is injective, such as a disc of radius $1/2$, the restriction $\pi|_V$ is a homeomorphism onto an open set, and its inverse is a chart. A transition map between two such charts sends z to $z + n$, where the integer n is a continuous, hence locally constant, function of z , so it is holomorphic. The map $x + iy \mapsto (e^{2\pi i x}, y)$ induces a homeomorphism of $\mathbb{C}/\mathbb{Z}$ onto $S^1 \times \mathbb{R}$, so $\mathbb{C}/\mathbb{Z}$ is Hausdorff and paracompact.
3. The Riemann sphere $\mathbb{CP}^1 = \mathbb{C}_\infty$ is a complex manifold, with the two charts $z \mapsto z$ on $\mathbb{C}_\infty \setminus \{\infty\}$ and $z \mapsto 1/z$ on $\mathbb{C}_\infty \setminus \{0\}$, where $1/\infty = 0$. The transition map is $z \mapsto 1/z$ on $\mathbb{C} \setminus \{0\}$, which is holomorphic, and $\mathbb{C}_\infty$ is homeomorphic to S^2 (theorem 1.2.2), hence compact and Hausdorff. Read on S^2 , these charts are the two stereographic projections.
4. Let Ω be a lattice in $\mathbb{C}$, that is, a subgroup $\{n_1\omega_1 + n_2\omega_2 \mid n_1, n_2 \in \mathbb{Z}\}$ with ω_1, ω_2 linearly independent over $\mathbb{R}$, and consider the canonical projection $\pi: \mathbb{C} \rightarrow \mathbb{C}/\Omega$. As in (2), π is open, and the inverses of the restrictions of π to open sets $V \subseteq \mathbb{C}$ on which π is injective

form an atlas, whose transition maps are translations $z \mapsto z + \omega$ with $\omega \in \Omega$ locally constant, hence holomorphic. Two points $\pi(z) \neq \pi(w)$ have the disjoint neighbourhoods $\pi(B(z, \epsilon))$ and $\pi(B(w, \epsilon))$, where $2\epsilon > 0$ is the distance from $z - w$ to the closed discrete set Ω , so $\mathbb{C}/\Omega$ is Hausdorff. It is compact, being the image under π of a closed parallelogram of periods, hence paracompact. So $\mathbb{C}/\Omega$ is a compact complex manifold. This is a compact manifold which has a “hole” since it can be thought to be constructed like T^2 .

Though Γ_1, Γ_2 may be two doubly periodic groups and they produce two tori $\mathbb{C}/\Gamma_1$ and $\mathbb{C}/\Gamma_2$, we shall see that it need not be the case that they are biholomorphic. This is part of the starting point of the study of *modular curves*, see [4].

5. Show that the only holomorphic function on the Riemann sphere are constant functions (hint: proposition 4.3.1 classified all meromorphic functions).

The first thing to see is if the results of complex functions translate to maps between complex manifold. Since holomorphic functions are analytic, their global behavior is essentially defined locally due to analytic continuation. The same is true for holomorphic maps:

Proposition 6.2.3: Analytic Continuity For Holomorphic Maps

Let $f, g : M \rightarrow N$ be holomorphic mappings. Then the set $U \subseteq M$ of points on which f and g coincide in a neighborhood is both open and closed

Proof:

By definition, U is open, hence it must be shown to be closed. Let a lie in the closure of U . Then $f(a) = g(a)$, since f and g are continuous, agree on U , and N is Hausdorff. Choose a chart ψ of N around $f(a)$, and a chart φ of M around a whose domain is mapped by f and by g into the domain of ψ , which is possible by continuity. Let n be the dimension of M . The image of φ contains a polydisc $P = P_1 \times \cdots \times P_n$ centred at $\varphi(a)$, on which the representatives $F = \psi \circ f \circ \varphi^{-1}$ and $G = \psi \circ g \circ \varphi^{-1}$ are holomorphic. The neighbourhood $\varphi^{-1}(P)$ of a meets U , so $F = G$ on a polydisc $Q = Q_1 \times \cdots \times Q_n \subseteq P$. For $0 \leq k \leq n$ let $P^{(k)}$ be the set of $w \in P$ with $w_j \in Q_j$ for every $j > k$, so that $P^{(0)} = Q$. Suppose $F = G$ on $P^{(k-1)}$, let $w \in P^{(k)}$, and put $h(\zeta) = F(w_1, \dots, w_{k-1}, \zeta, w_{k+1}, \dots, w_n) - G(w_1, \dots, w_{k-1}, \zeta, w_{k+1}, \dots, w_n)$ for $\zeta \in P_k$. Each component of h is holomorphic, hence analytic (corollary 3.2.7), on the disc P_k , and it vanishes on Q_k , since the corresponding points lie in $P^{(k-1)}$. By corollary 2.5.4, $h = 0$ on P_k , so $F(w) = G(w)$. By induction $F = G$ on $P^{(n)} = P$, so $f = g$ on the neighbourhood $\varphi^{-1}(P)$ of a , and $a \in U$, as we sought to show.

Proposition 6.2.4: Maximum Modulus For Holomorphic Mappings

Let $f : X \rightarrow \mathbb{C}$ be a holomorphic function on a connected complex manifold X . Then if $|f|$ has a relative maximum at a point $a \in X$, then f is constant.

Proof:

Let n be the dimension of X . In a chart, f becomes a holomorphic function F of n complex variables, and in particular F is holomorphic in each variable separately when the others are held fixed. For a holomorphic function g of one variable on a disc and a point z of that disc, theorem 3.2.6 expands g in a power series at z converging on every disc around z contained in

the given disc, so the mean value property (definition 3.3.10, the $p = 0$ case of (3.2)) holds on every circle around z inside the disc.

Step 1: a local maximum forces local constancy. Let F be holomorphic in each variable on a polydisc $P = \{w \in \mathbb{C}^n \mid |w_k - p_k| < r \text{ for all } k\}$ with $|F(w)| \leq |F(p)|$ for all $w \in P$. For $0 \leq k \leq n$ let P_k be the set of $w \in P$ with $w_j = p_j$ for every $j > k$. We show $F = F(p)$ on P_k by induction on k , the case $P_0 = \{p\}$ being trivial. Let $w \in P_k$ and put $g(\zeta) = F(w_1, \dots, w_{k-1}, \zeta, p_{k+1}, \dots, p_n)$ for $|\zeta - p_k| < r$. Then $g(p_k) = F(p)$, as the corresponding point lies in P_{k-1} , and $|g| \leq |F(p)| = |g(p_k)|$ on the disc. So $|g|$ has a relative maximum at p_k , and theorem 3.3.11 makes g constant near p_k . Since g is analytic (corollary 3.2.7) on the connected disc, corollary 2.5.4 gives $g = F(p)$ on the whole disc, and in particular $F(w) = g(w_k) = F(p)$. Hence $F = F(p)$ on $P_n = P$.

Step 2: propagation. Let $c = f(a)$ and let S be the set of $x \in X$ such that $f = c$ on a neighbourhood of x . Then $a \in S$ by Step 1 applied in a chart around a , on a polydisc where $|F|$ is at most its value at the image of a . The constant map c is holomorphic, so S is open and closed by proposition 6.2.3. As X is connected and S is nonempty, $S = X$, that is, f is constant, as we sought to show.

Corollary 6.2.5: Holomorphic Maps On Compact Manifolds

Let X be a compact, connected, complex manifold. Then any holomorphic function $f: X \rightarrow \mathbb{C}$ is constant.

Proof:

Since $|f|$ is continuous on the compact space X , it attains its maximum at some point, which is a relative maximum, so f is constant by proposition 6.2.4.

Note how this is another way of showing that holomorphic doubly-periodic functions must be constant: The map $f \mapsto f \circ \pi$ where $\pi: \mathbb{C} \rightarrow \mathbb{C}/\Omega$ is a bijection. As a final important definition, we give a meromorphic function on a manifold:

Definition 6.2.6: Meromorphic Maps

Let M be a complex manifold. Then a *meromorphic function* on M is a holomorphic map $f: M \rightarrow S^2$, in other words a meromorphic function is a continuous function which can take the value ∞ and which in a neighborhood of each $a \in X$ can be expressed as a meromorphic function of a local coordinate in the neighborhood of a .

For example, composing with the canonical projection $\pi: \mathbb{C} \rightarrow \mathbb{C}/\Omega$ gives a bijection between the meromorphic functions on $\mathbb{C}/\Omega$ and the doubly periodic meromorphic functions on $\mathbb{C}$. Finally, the open mapping theorem generalizes to injective maps between complex manifolds of dimension one.

Theorem 6.2.7: Injective Maps Between Complex Manifolds

Let M and N be complex manifolds of dimension one and let $f : M \rightarrow N$ be an injective holomorphic map. Then $f(M)$ is open in N and f is a biholomorphism of M onto $f(M)$. In particular, a bijective holomorphic map $M \rightarrow N$ is an isomorphism.

The inclusion of the unit disc into $\mathbb{C}$ shows that $f(M)$ need not be all of N .

Proof:

Let $p \in M$. Choose a chart ψ of N around $f(p)$ and a chart φ of M around p whose domain U_p is mapped by f into the domain of ψ , which is possible by continuity. The representative $F = \psi \circ f \circ \varphi^{-1}$ is an injective holomorphic function on the open set $\varphi(U_p) \subseteq \mathbb{C}$, so $F'(\varphi(p)) \neq 0$ by corollary 3.3.7. By theorem 2.2.27, F maps an open neighbourhood $W \subseteq \varphi(U_p)$ of $\varphi(p)$ onto an open set V , with holomorphic inverse $G : V \rightarrow W$. So f maps the open neighbourhood $\varphi^{-1}(W)$ of p onto the open set $\psi^{-1}(V)$, and on $\psi^{-1}(V)$ the inverse of f is $\varphi^{-1} \circ G \circ \psi$. Every open subset of M is a union of such neighbourhoods, contained in it, of its points, so f maps open sets to open sets, and $f(M)$ is open in N . The bijection $f : M \rightarrow f(M)$ is continuous and open, so its inverse is continuous, and it is holomorphic, being $\varphi^{-1} \circ G \circ \psi$ near each point of $f(M)$. Hence f is a biholomorphism of M onto the open set $f(M)$, completing the proof.

Example 6.2.8: Practical Example

Take $z \mapsto e^{2\pi iz}$ that maps $(\mathbb{C}, +)$ to $(\mathbb{C}^\times, \cdot)$. Taking the quotient, we get a holomorphic map $\mathbb{C}/\mathbb{Z} \rightarrow \mathbb{C}^\times$, which in the charts of example 6.2.2 is $z \mapsto e^{2\pi iz}$. It is injective, since $e^{2\pi iz} = e^{2\pi iw}$ exactly when $z - w \in \mathbb{Z}$, and onto $\mathbb{C}^\times$, as every nonzero complex number has a logarithm. So theorem 6.2.7 gives

$$\mathbb{C}/\mathbb{Z} \cong \mathbb{C}^\times$$

it can even be shown to be an isomorphism of topological groups by verifying the homomorphism condition.

A result that is worth mentioning, but not proved in this book, is the classification of all simply connected Riemann surfaces:

Theorem 6.2.9: Uniformization Theorem

Let M be a simply connected Riemann surface. Then M is isomorphic to one of the following:

1. Riemann sphere $\mathbb{CP}^1$
2. The plane $\mathbb{C}$
3. The unit disk $D = \{z \in \mathbb{C} \mid |z| < 1\}$

The theorem concerns complex dimension 1: the complex manifold $\mathbb{C}^2$ is simply connected, and it is not isomorphic to any of the three surfaces above, which have complex dimension 1. Its proof is not given in this book. The proof needs the Dirichlet problem on relatively compact domains of a Riemann surface, solved by Perron's method of subharmonic functions, together with holomorphic functions with prescribed zeros on a noncompact Riemann surface, and this book solves the Dirichlet problem only for discs in the plane (chapter 7). A complete proof is in Forster's *Lectures on Riemann*

Surfaces [11, Theorem 27.9], whose hypothesis that every holomorphic 1-form has a primitive follows from the hypothesis of theorem 6.2.9 by [11, p. 27.1].

Proof Key ideas:

If M is compact, simple connectivity gives every holomorphic 1-form on M a primitive, which is a holomorphic function on a compact surface and hence constant, so M carries no nonzero holomorphic 1-form. Then M has genus 0 and is isomorphic to $\mathbb{CP}^1$. If M is not compact, exhaust it by an increasing sequence of relatively compact Runge domains, that is, domains whose complements in M have no relatively compact connected components, with boundaries regular for the Dirichlet problem ([11, Theorem 23.9]). Each of them again has the property that every holomorphic 1-form has a primitive ([11, Lemma 27.8]). On each of them the solution of a Dirichlet problem, combined with a holomorphic function on M having a single simple zero at a fixed point a , gives a biholomorphism onto a disc centred at 0, normalized to send a to 0 with derivative 1 at a in a fixed chart. The radii of these discs increase, and a normal-family argument in the spirit of theorem 4.1.23 produces a limit map from M biholomorphically onto the disc of radius equal to the limit of the radii. That limit is finite, giving a disc isomorphic to D , or infinite, giving $\mathbb{C}$, which completes the sketch.

Now going to integration of complex manifolds. Let ω be a holomorphic 1-form on a Riemann surface M . As every holomorphic 1-form is locally exact, every representative of a holomorphic function is of the form:

$$\omega_i = f_i(z_i)dz_i$$

where f_i is holomorphic, with change of coordinate rule satisfying $f_j(z_i) = f_i(f_{ij}(z_j))f'_{ij}(z_j)$ for appropriate transition function f_{ij} . Just like before, a primitive of ω , if it exists, is a function g such that $dg = \omega$. In general, a global primitive does not exist. If X is simply connected, then any holomorphic differential form on X has a primitive, and if X is not simply connected then the integral of ω along a closed path of X is not always zero, and the integral has the same value for two homotopic paths. Often, the value of the integral along such a path is called the *period* of the integral $\int \omega$. If M is an orientable manifold, then we may take an orientable boundary Γ . Then we see that if Γ is the oriented boundary of a compact subset of M , then $\int_{\Gamma} \omega$ is zero for any holomorphic differential form ω on M .

Lemma 6.2.10: Holomorphic Forms Are Closed

Let ω be a holomorphic 1-form on an open subset of a complex manifold of dimension one. Then ω and $\bar{\omega}$ are smooth, and $d\omega = 0$ and $d\bar{\omega} = 0$.

Proof:

In a local coordinate z , $\omega = h(z)dz$ with h holomorphic, hence infinitely differentiable, being analytic (corollary 3.2.7). So ω and $\bar{\omega} = \bar{h}d\bar{z}$ are smooth. Writing $dh = \frac{\partial h}{\partial z}dz + \frac{\partial h}{\partial \bar{z}}d\bar{z}$, as computed after definition 2.2.11, and using $\frac{\partial h}{\partial \bar{z}} = 0$ for holomorphic h ,

$$d\omega = dh \wedge dz = \frac{\partial h}{\partial z} dz \wedge dz = 0$$

and $d\bar{\omega}$ is the conjugate of $d\omega$, hence 0, completing the proof.

Moving on to residue theory for complex manifolds, let's say $E \subseteq M$ is a discrete subset of M and ω is holomorphic on $M \setminus E$. Let $a \in E$ and let z be local coordinate of a which is zero at a . Then in some neighborhood of a , the form ω can be written as $f(z)dz$, where f is holomorphic in the neighborhood of 0 except perhaps at $z = 0$. Then the Laurent expansion of $f(z)$ shows that in a neighbourhood of a , the ω for can be written as

$$\omega = \omega_1 + \left(\frac{c_1}{z} + \frac{c_2}{z^2} + \cdots \right) dz$$

where ω_1 is a holomorphic differential form in a neighborhood of a (including a). If γ is a closed path is situated in a small neighbourhood of a which does not pass through a and whose inside with respect to a is equal to 1 (i.e. it winds around only once, then by the classical residue theorem:

$$\int_{\gamma} \omega = 2\pi i c_1$$

Importantly, c_1 does not depends on the choice of local coordinates z , which is zero at a . This is the residue of the differential form ω at a . Then we can naturally generalize the residue theorem to the following:

Theorem 6.2.11: Residue Theorem For Complex Manifolds

Let M be a complex manifold of dimension one, let E be a discrete closed subset of M , and let ω be a holomorphic 1-form on the complement $M \setminus E$. Let $K \subseteq M$ be a compact region with corners ([23, Region with Corners]) in the oriented smooth surface underlying M , let Γ be its oriented boundary, and suppose that Γ does not contain any point of E . Then the integral $\int_{\Gamma} \omega$ is equal to $2\pi i$ times the sum of the residues of ω at the points of E situated in K . In particular, if M is compact, then $K = M$ is allowed, with Γ empty, and the sum of all the residues of ω is 0.

Proof:

The set $E \cap K$ is finite. Indeed, every point of K has an open neighbourhood meeting E in at most one point: for a point not in E take the complement of the closed set E , and for a point of E take a neighbourhood witnessing that the point is isolated in E . Finitely many such neighbourhoods cover the compact set K . Write $E \cap K = \{a_1, \dots, a_r\}$. Since Γ contains no point of E , each a_k lies in the interior of K .

For each k , choose a local coordinate z_k in a neighbourhood of a_k which is zero at a_k , and let S_k be the preimage of a closed disc $|z_k| \leq \rho_k$. Shrinking the radii, we may take the S_k pairwise disjoint, contained in the interior of K , and small enough that S_k meets E only in a_k . Let γ_k be the oriented boundary of S_k , that is, the circle $|z_k| = \rho_k$ described in the positive sense. Set

$$K' = K \setminus \bigcup_k \text{int}(S_k)$$

Then K' is a compact region with corners, as the S_k are closed coordinate discs in the interior of K , its oriented boundary is Γ together with the circles γ_k reversed, and $K' \cap E = \emptyset$. The open set $M \setminus E$ is itself a complex manifold containing K' , and ω is holomorphic on it, so ω is smooth and $d\omega = 0$ there (lemma 6.2.10). The complex charts make M an oriented smooth surface, their transition maps having Jacobian determinant $|\varphi'|^2 > 0$, so [23, Stokes' Theorem

on a Region with Corners], applied to the real and imaginary parts of ω , gives

$$\int_{\Gamma} \omega - \sum_k \int_{\gamma_k} \omega = 0 \quad \text{i.e.} \quad \int_{\Gamma} \omega = \sum_k \int_{\gamma_k} \omega$$

In the coordinate z_k , the form ω is written $f_k(z_k)dz_k$ with f_k holomorphic in a neighbourhood of the closed disc $|z_k| \leq \rho_k$ except at $z_k = 0$, and γ_k is the positively oriented boundary of that disc. Theorem 3.5.4 applies in this coordinate and gives

$$\int_{\gamma_k} \omega = \int_{|z_k|=\rho_k} f_k(z_k)dz_k = 2\pi i \text{Res}(f_k, 0)$$

and $\text{Res}(f_k, 0) = c_1$ is precisely the residue of ω at a_k as defined in the paragraph preceding this theorem, a quantity independent of the chosen coordinate. Summing over k ,

$$\int_{\Gamma} \omega = 2\pi i \sum_{k=1}^r \text{Res}(\omega, a_k)$$

the sum running over the points of E situated in K , completing the proof.

6.2.1 Riemann Surfaces

Recall when we said that we shall introduce the notion of Riemann surface to make the many-to-one functions $z^{1/2}$ and $\log(z)$ well-defined. We now cover these notions:

Definition 6.2.12: Riemann Surface

Let Y be a complex manifold. Then a *Riemann surface* spread over Y (or simply a *Riemann surface* over Y) is defined to be a connected complex manifold X and a non-constant holomorphic mapping $\varphi : X \rightarrow Y$. Usually, we consider the case where $Y \in \{\mathbb{C}, \mathbb{CP}^1\}$.

If the derivative of a local representative $\tilde{\varphi}$ of φ vanishes at a point, then by corollary 3.3.7 the map φ is *not* injective around the point. We hence keep track of such values:

Definition 6.2.13: Ramification Index

Let $\varphi : M \rightarrow N$ be a non-constant holomorphic mapping between connected Riemann surfaces and $a \in M$. In local coordinates z centred at a and w centred at $\varphi(a)$, the representative $\tilde{\varphi} = w \circ \varphi$ is a holomorphic function with $\tilde{\varphi}(0) = 0$, not identically zero near 0. If its zero at 0 has order p (proposition 2.5.7), so that $\tilde{\varphi}'$ vanishes at 0 to order $p - 1$, we say that a has *ramification index* p . If $p > 1$, we say that a is *ramified*, and *unramified* otherwise. The corresponding point $\varphi(a)$ is called a *branching point*.

The ramification index is independent of the coordinates, since the transition maps have nonzero derivative, and composing a holomorphic function with a function having a simple zero, on either side, does not change the order of its zero.

Lemma 6.2.14: Local Normal Form

Let $\varphi: M \rightarrow N$ be a non-constant holomorphic mapping between connected Riemann surfaces, let $a \in M$ have ramification index p , and let w be a local coordinate centred at $\varphi(a)$. Then there is a local coordinate z centred at a in which $w \circ \varphi$ reads $z \mapsto z^p$. Consequently there are open neighbourhoods V of a and U of $\varphi(a)$ with $\varphi(V) \subseteq U$ such that every point of $U \setminus \{\varphi(a)\}$ has exactly p preimages in V , each of ramification index 1. In particular, φ is injective near a if and only if $p = 1$.

Proof:

Choose a local coordinate z_1 centred at a . By definition 6.2.13 and proposition 2.5.7, $w \circ \varphi = z_1^p g(z_1)$ with g holomorphic near 0 and $g(0) \neq 0$. On a disc around 0 on which g has no zeros, g'/g has a holomorphic primitive ℓ (corollary 3.2.3). Then $(ge^{-\ell})' = 0$, so $ge^{-\ell}$ is a nonzero constant (proposition 2.2.22), and after adding a constant to ℓ we have $g = e^\ell$. The function $z = z_1 e^{\ell(z_1)/p}$ satisfies $z^p = w \circ \varphi$, and its derivative at 0 is $e^{\ell(0)/p} \neq 0$, so it is a local coordinate centred at a by theorem 2.2.27. Choose $\epsilon > 0$ so small that $V = \{|z| < \epsilon\}$ lies in the domain of z and $U = \{|w| < \epsilon^p\}$ in that of w . Then $\varphi(V) \subseteq U$, and for $0 < |w_0| < \epsilon^p$ the equation $z^p = w_0$ has exactly p distinct solutions, all in V (corollary 3.3.5), at each of which the derivative pz^{p-1} is nonzero. If $p \geq 2$, these p points show that φ is not injective on any neighbourhood of a , and if $p = 1$, then φ is injective on V , completing the proof.

Note that if the ramification index of each $a \in X$ is 1, then φ is a local biholomorphism, by the holomorphic inverse function theorem (theorem 2.2.27) in coordinates, and if φ is moreover injective, it is a biholomorphism onto its image, which is open. If the ramification index is greater than 1, it would happen at isolated points. Note too that φ may not be injective and have no ramifications (think $z \mapsto e^{iz}$ on S^1 , or $\mathbb{C} \rightarrow \mathbb{C}^\times$ mapping $t \mapsto e^{it}$). Furthermore, if $S \subseteq X$ is the set of ramified points, then $f(S)$ need not be discrete. On the other hand, it may be that $f(S)$ maps to a single point, even if there are infinitely many S (that is, f need not be proper).

For reference purposes, we define the following:

Definition 6.2.15: Unramified Riemann Surface

An *Unramified Riemann Surface* over Y is a Riemann surface (X, φ) where the mapping φ is unramified, that is all points $x \in X$ have ramification index 1.

Example of unramified Riemann surfaces are local homeomorphisms or covering spaces. In fact, we have the following result, which reduces to the existence of universal covers in algebraic topology:

Theorem 6.2.16: Universal Covering Of Open Sets

Let $U \subseteq M$ be any connected open set of a complex manifold M . Then U has a simply connected covering space.

Proof:

By [8, Existence of a Universal Cover], it suffices to check that U is path-connected, locally path-connected and semilocally simply connected. Let n be the dimension of M .

Local path-connectedness. Let $x \in U$ and let $W \subseteq U$ be open with $x \in W$. Choose a chart φ of M around x , a homeomorphism from an open neighbourhood of x onto an open subset of $\mathbb{C}^n$, and an open ball B around $\varphi(x)$ inside the image of the part of that neighbourhood lying in W . The ball B is convex, hence path-connected, so $\varphi^{-1}(B)$ is a path-connected open set with $x \in \varphi^{-1}(B) \subseteq W$. Thus U has a basis of path-connected open sets.

Path-connectedness. By [8, Locally Path-Connected If And Only If Open Components], each path component of U is open. The path components partition U , so if there were two or more, one of them and the union of the others would be disjoint nonempty open sets covering U , contradicting the connectedness of U . Hence U is path-connected.

Semilocal simple connectivity. With x , φ and B as above, taken with $W = U$, let $V = \varphi^{-1}(B)$. A loop γ in V based at x is path homotopic in V to the constant loop by the straight-line homotopy

$$H(t, s) = \varphi^{-1}((1-s)\varphi(\gamma(t)) + s\varphi(x))$$

which stays in V because B is convex and fixes the base point because $\gamma(0) = \gamma(1) = x$. Hence the homomorphism $\pi_1(V, x) \rightarrow \pi_1(U, x)$ induced by the inclusion is trivial, which is semilocal simple connectivity at x . Thus U satisfies the three hypotheses of that theorem, as claimed.

We now introduce our main object of interest

Definition 6.2.17: Holomorphic Function On Riemann Surface

Let (X, φ) be a Riemann surface over Y . Then a Holomorphic (resp. meromorphic) function on (X, φ) is a holomorphic (resp. meromorphic) function on X .

Example 6.2.18: Riemann Surfaces

1. Let $f(z) = \sqrt{z}$. Let us make this a well-defined function on some Riemann surface X so that $f : X \rightarrow \mathbb{C}$ is well-defined. First, notice the relation $z^{1/2} = w$ so $z = w^2$. Take:

$$X \subseteq \mathbb{C} \times \mathbb{C} \quad X = \{(z, w) \mid z = w^2\}$$

Notice that $X = F^{-1}(\{0\})$ where $F(z, w) = z - w^2$. This is thus a perfect setup for the implicit function theorem (theorem 2.2.28). Since $\frac{\partial F}{\partial w} = -2w$, if $w \neq 0$, by the implicit function theorem there exists a function g with $w = g(z)$ on a sufficiently small neighbourhood (namely a branch $g(z) = \sqrt{z}$ of the square root). Since $\frac{\partial F}{\partial z} = 1$ never vanishes, the same theorem gives functions h with $z = h(w)$ (namely $h(w) = w^2$)^a. It is in fact enough to just take the graphs with respect to w as these cover X and the transitions are certainly biholomorphic, but for practice we shall show that all these transition maps are compatible. When $z_0, w_0 \neq 0$, we can represent the two coordinate charts as:

$$\varphi(z, w) = \varphi(z, \sqrt{z}) = z \quad \text{and} \quad \psi(z, w) = \psi(w^2, w) = w$$

Then

$$\psi \circ \varphi^{-1}(z) = \sqrt{z}$$

for the branch of the square root taking the value w_0 at z_0 , which is biholomorphic as $z \neq 0$ in the region we are considering, and so we have a well-defined Riemann surface.

The following commutative diagram visualizes the lifting:

$$\begin{array}{ccc} X & & \\ \pi \downarrow & \searrow \varphi & \\ \mathbb{C} & \xrightarrow{\sqrt{z}} & \mathbb{C} \end{array}$$

And X can be visualized as:

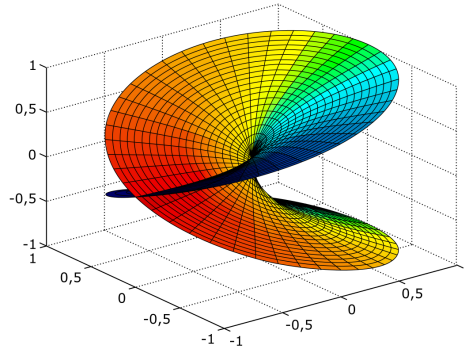


Figure 6.1: Branch Covering for $\sqrt{-}$

2. We shall show how to consider $\log(z)$ as a single-valued function. Consider $(\mathbb{C}, \varphi)$ covering $\mathbb{C}^\times$ where $\varphi(t) = e^{it}$. Since φ is unramified, we see that φ can be expressed as local coordinate in a neighborhood of each $x \in X$, say coordinates z so that $z = e^{it}$. Then any holomorphic function $f : \mathbb{C} \rightarrow Y$ can be *locally expressed* as a holomorphic function of z . But since different points of X can be mapped by φ onto the same points of $\mathbb{C}^\times$, f is not in general a global holomorphic function in the variable $z \neq 0$. The function $t \mapsto it$ is an example: it is holomorphic on $X = \mathbb{C}$, and near each point it is a branch of $\log z$, since $e^{it} = z$. So $\log$ becomes a single-valued holomorphic function on the covering $(\mathbb{C}, \varphi)$, although it is not a function of z on $\mathbb{C}^\times$, the points t and $t + 2\pi$ having the same image.
3. Take $f(z) = \sqrt[3]{1 - z^3}$. First, given $w = (1 - z^3)^{1/3}$, we get the equation:

$$z^3 + w^3 = 1$$

giving us the surface:

$$X \subseteq \mathbb{C} \times \mathbb{C} \quad X = \{(z, w) \mid z^3 + w^3 = 1\}$$

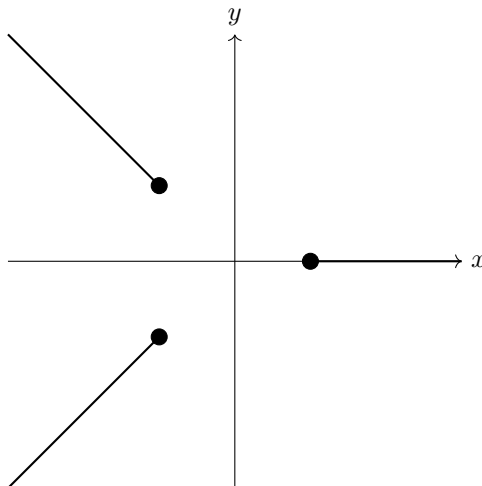
$F(z, w) = z^3 + w^3 - 1$, we again see that $\frac{\partial F}{\partial z} = 3z^2$ and $\frac{\partial F}{\partial w} = 3w^2$, and for $z_0 \neq 0$ or $w_0 \neq 0$, we can use the Implicit Function Theorem to get charts.

From this, we get the following map $\varphi : X \rightarrow \mathbb{C}$ such that the following diagram commutes:

$$\begin{array}{ccc} X & & \\ (z,w) \mapsto z \downarrow & \searrow \varphi & \\ \mathbb{C} & \xrightarrow{\sqrt[3]{1-z^3}} & \mathbb{C} \end{array}$$

For each $z \in \mathbb{C}$, the fibre of the projection $(z, w) \mapsto z$ over z contains 3 points, except when $1 - z^3 = 0$. The exceptional points of X are $(1, 0)$, $(e^{2\pi i/3}, 0)$ and $(e^{4\pi i/3}, 0)$. We often identify these points with their image in the domain of $\sqrt[3]{1 - z^3}$ and write them as 1 , $e^{2\pi i/3}$ and $e^{4\pi i/3}$.

To visualize X , we can imagine three sheets of $\mathbb{C}$ each with cuts that look like:



These three sheets can be seen as “stacked” on top of each other and glued at :

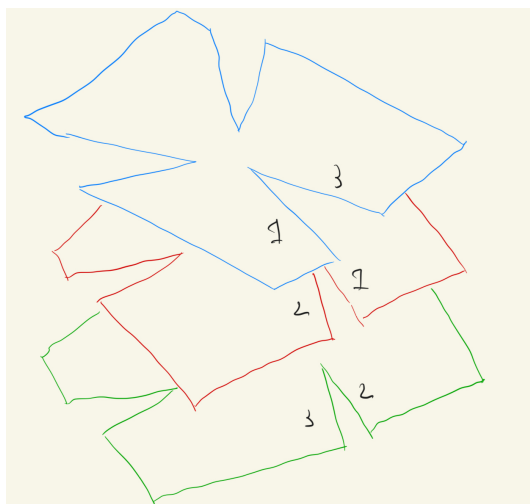


Figure 6.2: 3 sheets with numbers represent which sheet to go to after crossing

Naturally, the surface has no intersections, but when visualizing it in three real dimensions shall see some overlap.

4. Let us extend the above result with codomain $\mathbb{CP}^1$. We can take the natural embedding of $\mathbb{C}^2$ into $\mathbb{CP}^2$ by mapping $(z, w) \mapsto [z : w : 1]$. For an arbitrary point $[z : w : t] = [z/t : w/t : 1]$,

we see that on this chart the equation is:

$$\left(\frac{z}{t}\right)^3 + \left(\frac{w}{t}\right)^3 = 1$$

i.e. $z^3 + w^3 = t^3$ (the homogenization of the original equation). We now let $\bar{X} \subseteq \mathbb{CP}^2$ be the (homogeneous) points^b that satisfy $z^3 + w^3 = t^3$, which for similar reasons still defined a Riemann surface. Let us see if extending to $\mathbb{CP}^2$ added more points at infinity, i.e. points where $t = 0$, so $[z : w : 0] = [z/w : 1 : 0]$. This would have to satisfy:

$$z^3 + w^3 = 0$$

This equation does have 3 solutions:

$$[-1 : 1 : 0] \quad [-e^{2\pi i/3} : 1 : 0] \quad [-e^{4\pi i/3} : 1 : 0]$$

Hence, we get that:

$$\bar{X} = X \cup \{[-1 : 1 : 0], [-e^{2\pi i/3} : 1 : 0], [-e^{4\pi i/3} : 1 : 0]\}$$

Now, $\varphi : X \rightarrow \mathbb{C}$ extends to $\bar{\varphi} : \bar{X} \rightarrow \mathbb{CP}^1$ by mapping the new points to ∞ (i.e. $[1 : 0]$). Let us check whether $\bar{\varphi}$ is holomorphic at the points at infinity. To do this, we shall check in a chart containing the points at infinity. Let us choose the chart where $w \neq 0$. Then in this chart, the points of the curve are given by:

$$(z')^3 - (t')^3 - 1 = 0$$

where $z' = z/w$ and $t' = t/w$. Now, notice that the partial derivative with respect to z' is nonzero at the points at infinity, so t' acts as local coordinates, namely we have:

$$z' = \sqrt[3]{1 - (t')^3}$$

Finally, we must compute this map in coordinates to check for holomorphic transitions:

$$\mathbb{C} \setminus \{0\} \supset U \longrightarrow X' \longrightarrow S^2 \setminus \{S\} \longrightarrow \mathbb{C}$$

$$t' \longrightarrow \left[\sqrt[3]{1 - (t')^3}, 1, t' \right] \longrightarrow \varphi \left(\left[\frac{\sqrt[3]{1 - (t')^3}}{t'}, \frac{1}{t'}, 1 \right] \right) \longrightarrow \frac{t'}{\sqrt[3]{1 - (t')^3}}$$

which we see maps t' to $\frac{t'}{\sqrt[3]{1 - (t')^3}}$ and hence is holomorphic in a neighbourhood around $t' = 0$. More specifically, this is three different functions (accounting for the cube root).

5. From this, we can better understand integrals on $\sqrt[3]{1 - z^3}$. This could not be done before because this was a “multi-valued” function, but it is properly single valued in the Riemann sphere! Suppose γ is a curve enclosing the (image of) the branch points, and let δ be the lift of this map to X . Notice that δ need not be closed, it may not wind enough time to “reconnect”. This is fine, as the path shall measure the amount “rotation” we have accumulated. We use the argument principle by integrating $f'(z)/f(z)$, or in the language of differential forms:

$$\frac{1}{2\pi i} \int_{\delta} \frac{dw}{w}$$

To integrate it, we will need to pullback via φ^* where φ is the lift defined earlier. As $\varphi(z, w) = w$, let us first re-write it in x -charts. By going to the defining equation, we have:

$$3z^2 dz + 3w^2 dw = 0$$

so:

$$\frac{dw}{w} = \frac{-z^2 dz}{1 - z^3}$$

Then, letting the coordinate of in the pullback be x , we get:

$$\begin{aligned} \frac{1}{2\pi i} \int_{\delta} \frac{dw}{w} &= \frac{1}{2\pi i} \int_{\delta} \frac{-z^2 dz}{1 - z^3} \\ &= \frac{1}{2\pi i} \int_{\delta} \varphi^* \left(\frac{-x^2}{1 - x^3} dx \right) \\ &= \frac{1}{2\pi i} \int_{\varphi(\delta)} \frac{-x^2}{1 - x^3} dx \\ &= \frac{1}{2\pi i} \int_{\gamma} \frac{-x^2}{1 - x^3} dx \end{aligned}$$

Finally, we use the residue theorem to compute this integral. Note the poles are the cube-roots of 1, each with residue $1/3$, so:

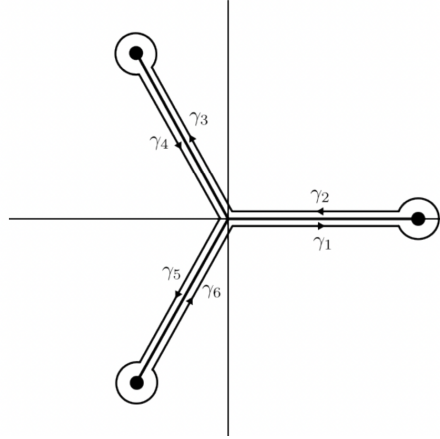
$$\frac{1}{2\pi i} \int_{\delta} \frac{dw}{w} = \frac{1}{2\pi i} \int_{\gamma} \frac{-x^2}{1 - x^3} dx = 3 \cdot \frac{1}{3} = 1$$

Notice the answer is an integer, so δ in this case is a closed curve. This though happens since γ encircles all three branch points; if γ only contained one of the branch points or two branch points we would have ended up with $1/3$ or $2/3$ respectively!

6. Finally, let us compute a real integral with our new found techniques. Let's take:

$$\int_0^1 \frac{1}{(1 - x^3)^{1/3}} dx$$

We see that the integral ends at the image of a branch point. We shall thus integrate around all the branch points, which we know should produce a value of 1, and then isolate the value of one of our paths:



Importantly, the value of the circles shall become negligible, but we must go around them to pick up an argument of $2\pi/3$. Let I be the value when integrating γ_1 , and let $j = e^{2\pi i/3}$. Then:

$$\int_{\gamma_1} \frac{1}{(1-z^3)^{1/3}} dz + \int_{\gamma_2} \frac{1}{(1-z^3)^{1/3}} dz = I - \frac{1}{j}I = I - j^2I$$

To compute γ_3 , first notice that $\gamma_3(t) = j\gamma_1(t)$. Furthermore, the denominator picks up a j term when going around 1. Thus:

$$\int_{\gamma_3} \frac{1}{(1-z^3)^{1/3}} dz = \int_{\gamma_1} \frac{1}{j(1-(jz^3)^{1/3})} d(jz) = \int_{\gamma_1} \frac{1}{(1-z^3)^{1/3}} dz = I$$

Similarly:

$$\int_{\gamma_4} \frac{1}{(1-z^3)^{1/3}} dz = -j^2I$$

and same for γ_5, γ_6 . Combining all the results, we get:

$$\int_{\gamma} \frac{1}{(1-z^3)^{1/3}} dz = 3(I - j^2I)$$

Finally, to compute I , we shall also reverse orientation and compute the residues at infinity, which shall then let us isolate I . To that end, we do the usual $u = 1/z$ coordinate change to get:

$$\frac{dz}{(1-z^3)^{1/3}} = \frac{d(1/u)}{(1-(1/u)^3)^{1/3}} = -\frac{du}{u(u^3-1)^{1/3}}$$

From this, we see the residue at $u = 0$ is $(-1)^{1/3}$, which naturally is 3 different answers given the 3 different poles (because the path in the Riemann surface can circle on a different sheet, depending on the exact path being taken). This, we get by Residue Theory that:

$$3(I - j^2I) = \int_{\gamma} \frac{1}{(1-z^3)^{1/3}} dz = 2\pi i \cdot \{1, j, j^2\}$$

Usually here there could be more work to do, but we are solving a real integral, and hence there is only one choice here that gives a real value, namely j . Manipulating, we finally can conclude:

$$I = \int_0^1 \frac{1}{(1-x^3)^{1/3}} dx = \frac{2\pi}{3\sqrt{3}}$$

^aThis makes sense, as the square-root function has no problem being defined locally, it breaks because of the more “global” behaviour of defining it on the entire complex plane

^bthis is well-defined; the points in the zero-set are independent of scaling

6.3 Divisors And Riemann-Roch Theorem

This section requires a lot of clean-up work; I’m keeping it in this incomplete state for ease of reference

Let X be a Riemann surface. We may ask how many meromorphic functions are there on X . Similarly, a closed tied question is how many meromorphic differential forms are on X . The Riemann-Roch theorem answers these questions for a general compact Riemann manifold. We can get an idea by answering the question in some simple cases. Starting with $X = \mathbb{C}$ (not compact, but an easy starting point), consider the free-abelian group generated by the points of X , so that each element of this group is of the form:

$$\sum_i n_i p_i \quad n_i \in \mathbb{Z}, p_i \in X$$

We may ask if there is a meromorphic function associated to this function. As we are interested in algebraic constructions, we limit ourselves to functions with a finite number of poles and zeros, that is all *rational functions*. Then we get that for each $\sum_i n_i p_i$, we can associate to it the meromorphic function;

$$(z - p_i)^{n_i}$$

Let us call the set of meromorphic functions the *principal divisors*, and the finite formal $\mathbb{Z}$ -linear combinations the *divisors*. Then the quotient of these two groups is zero:

$$\frac{\text{Divisors}}{\text{Principal Divisors}} = 0$$

Let us now consider X to be the Riemann surface. We shall once again take the divisor group to be the free group. This time, the principal divisor group (the group of meromorphic functions) has a fundamental limitation placed on it: recall that the sum of the zeros and poles of a meromorphic function is zero². In this case, the sum of each divisor is some integer $n \in \mathbb{Z}$. Using this, you can show that:

$$\frac{\text{Divisors}}{\text{Principal Divisors}} \cong \mathbb{Z}$$

Finally, consider an elliptic curve given by $\mathbb{C}/L$ where L is a lattice. **Richard Borcherds, finish off!**

6.3.1 Build-up: Normalization and Ramification

- Start with a general plane curve $F = 0$. Its singular points are those where the three partial derivatives of F vanish, so that the equation has no linear term when the point is moved to the origin $[1 : 0 : 0]$ of the chart $(x, y) \mapsto [1 : x : y]$. We can then consider any curve $f(x, y) = F(1, x, y)$ when considering the canonical embedding $(x, y) \mapsto [1 : x : y]$. A singular point is double, triple, or of multiplicity k if the lowest-degree terms of f at it have degree 2, 3, or k . It is ordinary if all the tangent lines are distinct.

²Recall that by adding the point at infinity, we shall add the “complement” of each zero/pole; convince yourself that z^2 has two zeros and two poles

Lemma 6.3.1: Smooth Locus of a Plane Curve

Let $C \subseteq \mathbb{CP}^2$ be the zero set of a nonzero homogeneous polynomial F of degree $d \geq 1$, and let S be the set of its singular points. Let $z \in C \setminus S$, and let (x, y) be the coordinates of an affine chart $\xi^i \neq 0$ of $\mathbb{CP}^2$ containing z , in which C is the zero set of the dehomogenization f of F . Then $f_x(z)$ and $f_y(z)$ are not both 0. If $f_y(z) \neq 0$, there are discs $U \ni x(z)$ and $V \ni y(z)$ and a holomorphic $\varphi: U \rightarrow V$ with $C \cap (U \times V) = \{(x, \varphi(x)) \mid x \in U\}$, and symmetrically if $f_x(z) \neq 0$. The coordinates x or y , restricted to such graphs, form an atlas making $C \setminus S$ a complex manifold of dimension one, for which every holomorphic map into $\mathbb{CP}^2$ with values in $C \setminus S$ is holomorphic into $C \setminus S$. Moreover C is compact.

Proof:

Permuting the homogeneous coordinates, we may take the chart $\xi^0 \neq 0$, with $x = \xi^1/\xi^0$, $y = \xi^2/\xi^0$ and $f(x, y) = F(1, x, y)$, so that f_x and f_y are the values of $\partial F/\partial \xi^1$ and $\partial F/\partial \xi^2$ at $(1, x, y)$. Euler's relation $\sum_i \xi^i \frac{\partial F}{\partial \xi^i} = dF$ at the point $(1, x(z), y(z))$, where $F = 0$, shows that if $f_x(z) = f_y(z) = 0$, then $\partial F/\partial \xi^0$ vanishes there too, and $z \in S$. If $f_y(z) \neq 0$, the implicit function theorem (theorem 2.2.28), applied to the polynomial f , gives the discs U, V and the function φ . On the graph, x is a homeomorphism onto U with inverse $x \mapsto (x, \varphi(x))$. Two such charts differ on overlaps by compositions of the graph functions with the coordinate changes of $\mathbb{CP}^2$, such as $(x, y) \mapsto (1/x, y/x)$, which are holomorphic where defined. So these charts form a complex atlas. The space $C \setminus S$ is Hausdorff and paracompact, being a subspace of the compact metrizable space $\mathbb{CP}^2$, and C is closed in $\mathbb{CP}^2$, hence compact. Finally, a holomorphic map into $\mathbb{CP}^2$ has holomorphic affine coordinates, and these are its representatives in the charts above, completing the proof.

- We can now consider irreducible components. If it is over a UFD, then the GCD is well-defined and so we get that f, g have no common factors if $(f) + (g) = (1)$ so that if $(f) + (g) = (h)$ then $f = Fh$ and $g = Gh$ so that $fG = gF$. We can detect this using the resultant, and hence we can detect singular points over non-characteristic 0 fields using the discriminant. We can then proceed to show that if C is a curve, S the singular points, and $C^* = C \setminus S$ the complement of the singular points, then C^* is connected, not necessarily compact, S is finite.
- Next comes normalization, defined as follows:

Definition 6.3.2: Normalization

Suppose C is a irreducible plane algebraic curve, S the set of its singular points. Then there exists a compact Riemann surface $\tilde{C}$ and a holomorphic mapping

$$\sigma : \tilde{C} \rightarrow \mathbb{CP}^2$$

such that

1. $\sigma(\tilde{C}) = C$
2. $\sigma^{-1}(S)$ is a finite set
3. $\sigma : \tilde{C} \setminus \sigma^{-1}(S) \rightarrow C \setminus S$ is injective

Then $(\tilde{C}, \sigma)$ is the normalization of C , and we write just $\tilde{C}$ if it is unambiguous in context.

- We get the following result equivalent to birational maps:

Lemma 6.3.3: Dense Injective, then Biholomorphic

Suppose $\tilde{C}, \tilde{C}'$ are Riemann surfaces and

$$h : \tilde{C} \rightarrow \tilde{C}'$$

is a surjective holomorphic mapping which is injective on a dense open subset of $\tilde{C}$. Then h is a biholomorphic mapping.

Proof:

Let $U \subseteq \tilde{C}$ be the dense open set on which h is injective. In local coordinates around a point of $\tilde{C}$ and around its image, h is represented by a holomorphic function on a disc.

Step 1: h is open. A nonempty open $W \subseteq \tilde{C}$ meets U in a nonempty open set, which is infinite, being locally homeomorphic to a disc, and h is injective on it. So h is constant on no nonempty open set, and each local representative of h on a coordinate disc is a nonconstant holomorphic function on a connected open set, hence an open map by theorem 3.5.15. Therefore h is open.

Step 2: h is injective. Suppose $h(p) = h(q)$ with $p \neq q$, and choose disjoint open neighbourhoods V_p, V_q . The sets $h(V_p)$ and $h(V_q)$ are open by Step 1, so $W = h(V_p) \cap h(V_q)$ is an open neighbourhood of $h(p)$. The open set $h(V_p \cap U)$ is dense in $h(V_p)$: if $O \subseteq h(V_p)$ is open and nonempty, then $h^{-1}(O) \cap V_p$ is open and nonempty, so it meets U . Hence $W' = W \cap h(V_p \cap U)$ is open and nonempty. The density argument applied to V_q shows that $h(V_q \cap U)$ is dense in $h(V_q) \supseteq W'$, so there is a point $w \in W' \cap h(V_q \cap U)$. Then $w = h(p') = h(q')$ with $p' \in V_p \cap U$ and $q' \in V_q \cap U$, and $p' \neq q'$ because $V_p \cap V_q = \emptyset$. This contradicts the injectivity of h on U .

Step 3: h^{-1} is holomorphic. By Step 2 and the hypothesis, h is a bijection. Let H be a local representative of h near a point z_0 in coordinates. It is injective, and it is nonconstant by Step

1. If $H'(z_0) = 0$, then z_0 is a root of order $k \geq 2$ of $H(z) = H(z_0)$, the first derivative vanishing, and proposition 3.5.11 gives $k \geq 2$ distinct solutions of $H(z) = b$ near z_0 for b close to $H(z_0)$ with $b \neq H(z_0)$, contradicting injectivity. So $H'(z_0) \neq 0$, and theorem 2.2.27 gives a holomorphic inverse of H near $H(z_0)$. As h is bijective, h^{-1} agrees near $h(z_0)$ with this local inverse expressed in the coordinates, so h^{-1} is holomorphic and h is biholomorphic, as claimed.

• Then

Theorem 6.3.4: Uniqueness of Normalization

The normalization of an algebraic curve C is unique up to isomorphism, compatibly with the maps to $\mathbb{CP}^2$: if $(\tilde{C}, \sigma)$ and $(\tilde{C}', \sigma')$ are normalizations of C , then there is exactly one biholomorphic mapping $h: \tilde{C} \rightarrow \tilde{C}'$ with $\sigma' \circ h = \sigma$.

Proof:

Let S be the set of singular points of C , and put $E = \sigma^{-1}(S)$ and $E' = \sigma'^{-1}(S)$, finite sets by definition 6.3.2. A finite set is nowhere dense in a Riemann surface, every nonempty open set being infinite, so $\tilde{C} \setminus E$ and $\tilde{C}' \setminus E'$ are dense. By definition 6.3.2, σ' maps $\tilde{C}' \setminus E'$ bijectively onto $C \setminus S$, and σ maps $\tilde{C} \setminus E$ bijectively onto $C \setminus S$.

Step 1: the map off E . Define $h_0: \tilde{C} \setminus E \rightarrow \tilde{C}' \setminus E'$ by $h_0 = (\sigma'|_{\tilde{C}' \setminus E'})^{-1} \circ \sigma$, so that $\sigma' \circ h_0 = \sigma$ and h_0 is a bijection. It is holomorphic. Indeed, the smooth locus $C \setminus S$ is a complex manifold of dimension one, and σ and σ' are holomorphic into it (lemma 6.3.1). They are bijective from $\tilde{C} \setminus E$ and $\tilde{C}' \setminus E'$ onto it, hence biholomorphic by theorem 6.2.7, so h_0 is a composite of biholomorphisms.

Step 2: extension across E . Let $p \in E$ and $s = \sigma(p)$, and let $q_1, \dots, q_m$ be the points of $\sigma'^{-1}(s) \subseteq E'$. Choose pairwise disjoint coordinate discs $N_1, \dots, N_m$ around $q_1, \dots, q_m$, each identified with the unit disc. The set $\tilde{C}' \setminus (N_1 \cup \dots \cup N_m)$ is compact, so its image under σ' is compact, hence closed, and it does not contain s . Choose a coordinate disc Δ around p with $\Delta \cap E = \{p\}$ and $\sigma(\Delta)$ disjoint from that image. For $z \in \Delta \setminus \{p\}$, $\sigma'(h_0(z)) = \sigma(z)$ lies outside it, so $h_0(z) \in N_1 \cup \dots \cup N_m$. The punctured disc $\Delta \setminus \{p\}$ is connected and the N_j are disjoint and open, so $h_0(\Delta \setminus \{p\})$ lies in a single N_j . In the coordinate of N_j , h_0 is a holomorphic function on the punctured disc of modulus less than 1, so it extends holomorphically across p by proposition 3.4.6, with value of modulus at most 1 at p . The value has modulus less than 1, since otherwise the modulus of the extension would have a maximum at p and theorem 3.3.11 would make it constant near p , against the modulus being less than 1 off p . So h_0 extends to a holomorphic map on Δ with values in N_j . Doing this at each point of E gives a holomorphic $h: \tilde{C} \rightarrow \tilde{C}'$ extending h_0 , and $\sigma' \circ h = \sigma$ holds on $\tilde{C}$ by continuity, both sides being continuous and equal on the dense set $\tilde{C} \setminus E$.

Step 3: h is biholomorphic. The image $h(\tilde{C})$ is compact, hence closed, and it contains the dense set $h_0(\tilde{C} \setminus E) = \tilde{C}' \setminus E'$, so h is surjective. It is injective on the dense open set $\tilde{C} \setminus E$, where it equals h_0 . By lemma 6.3.3, h is biholomorphic.

Step 4: uniqueness. If h_1, h_2 both satisfy $\sigma' \circ h_k = \sigma$, then for $z \in \tilde{C} \setminus E$ both $h_1(z)$ and

$h_2(z)$ lie off E' , as σ' sends them to $\sigma(z) \notin S$, and σ' is injective there, so $h_1(z) = h_2(z)$. Two continuous maps into the Hausdorff space $\tilde{C}'$ that agree on a dense set agree everywhere, so $h_1 = h_2$, completing the proof.

The goal now is to prove a normalization always exists.

- Let $\mathbb{C}\{x\}$ be the ring of *convergent* power series $\sum_k a_k x^k$, a subring of $\mathbb{C}[[x]]$, and let $\mathcal{O} = \mathbb{C}\{x, y\}$ be the ring of convergent power series $\sum_{i,j} a_{ij} x^i y^j$. A *Weierstrass polynomial* is a polynomial of the form

$$w = y^d + a_1(x)y^{d-1} + \cdots + a_d(x)$$

where

$$a_j(x) \in \mathbb{C}\{x\} \quad a_j(0) = 0 \quad j = 1, \dots, d$$

The *Weierstrass preparation theorem* says that, after a generic linear change of coordinates, any nonzero $f \in \mathcal{O}$ can be written as $f = u \cdot w$ where u is a unit (so $1/u \in \mathcal{O}$) and w is a Weierstrass polynomial, possibly of degree 0.

- This gives the following corollary. Its proof needs the Weierstrass preparation theorem, which this book does not prove, and a complete proof is in Huybrechts [16, Proposition 1.1.15].

Corollary 6.3.5: Ring Structure of Analytic Func

$\mathcal{O}$ is a UFD, hence any $f \in \mathcal{O}$ can be expressed up to unit u as:

$$f = u f_1 f_2 \cdots f_n$$

where each f_i is irreducible in $\mathcal{O}$.

Proof Key ideas:

The ring $\mathbb{C}\{x\}$ is a UFD, since every nonzero $f(x)$ equals $x^m h(x)$ with $h(0) \neq 0$, and such an h is a unit, $1/h$ being analytic near 0. For $f \in \mathbb{C}\{x, y\}$, after a linear change of coordinates the Weierstrass preparation theorem writes $f = u \cdot w$ with u a unit and $w \in \mathbb{C}\{x\}[y]$ a Weierstrass polynomial. By Gauss's lemma, $\mathbb{C}\{x\}[y]$ is a UFD because $\mathbb{C}\{x\}$ is, and a factorization of w into irreducible Weierstrass polynomials is a factorization into irreducible elements of $\mathbb{C}\{x, y\}$, unique up to units by the uniqueness in the preparation theorem, which completes the sketch.

- An important realization is that being irreducible in $\mathbb{C}\{x\}[y]$ and $\mathbb{C}\{x, y\}$ is the same thing.
- Next, let's take an irreducible curve, center the point of interest at $(0, 0)$, represent it as

$$y^n + a_1(x)y^{n-1} + \cdots + a_d(x)$$

Then locally at 0, we may have multiple local irreducible components. Let's take the curve over $\mathbb{C}^2$. Then around 0, if there are multiple components, it will look like multiple disks that are glued together at a point.

- Using this, we can locally normalize, and then use that to globally normalize
- Intersection numbers and Bézout's theorem come next.

Ramification Divisors and the Riemann-Hurwitz Formula

Let $\nu_f(p)$ be the order of a point of a holomorphic mapping $f : C \rightarrow C'$ between Riemann surfaces. As such a map is always surjective if it is non-constant, we have $\nu_f(p) \geq 1$. It ramifies if $\nu_f(p) > 1$, and as ramified points are isolated we know that $\{p \in C \mid \nu_f(p) > 1\}$ is finite. Thus we have the well-defined:

Definition 6.3.6: Ramification Divisor

The *ramification divisor* of f is :

$$R = \sum_{p \in C} (\nu_f(p) - 1)p \in \text{Div}(C)$$

Lemma 6.3.7: Degree of a Holomorphic Map

Let $f : C \rightarrow C'$ be a nonconstant holomorphic map between compact connected Riemann surfaces. Then for every $q \in C'$ the fibre $f^{-1}(q)$ is finite, and the number

$$d_f(q) = \sum_{p \in f^{-1}(q)} \nu_f(p)$$

does not depend on q . It is called the *degree* of f .

Proof:

By proposition 6.2.3, applied to f and a constant map, f is constant on no nonempty open set. So in local coordinates the points of a fibre are isolated zeros of a representative (proposition 2.5.7), and the fibre $f^{-1}(q)$, a closed subset of the compact C , is finite, say $\{p_1, \dots, p_k\}$. Let $\nu_i = \nu_f(p_i)$. By lemma 6.2.14, there are open neighbourhoods V_i of p_i , which we may take pairwise disjoint, and neighbourhoods U_i of q , such that every $q' \in U_i \setminus \{q\}$ has exactly ν_i preimages in V_i , each of ramification index 1. The set $K = C \setminus (V_1 \cup \dots \cup V_k)$ is compact, so $f(K)$ is compact, hence closed, and it does not contain q . On the open neighbourhood $U = (U_1 \cap \dots \cap U_k) \setminus f(K)$ of q , every $q' \neq q$ has all its preimages in $V_1 \cup \dots \cup V_k$, so $d_f(q') = \nu_1 + \dots + \nu_k = d_f(q)$. Thus d_f is locally constant on the connected space C' , hence constant, as claimed.

Recall that if $D = \sum_{p \in C} n_p \cdot p$ is a divisor, $\deg D = \sum_{p \in C} n_p$. A meromorphic 1-form ω on C is written $\omega = h(z)dz$ in a local coordinate z centred at p , with h meromorphic, and if $\omega \neq 0$ its order $\nu_\omega(p)$ at p is the order of the zero of h at 0, or minus the order of its pole. It does not depend on the coordinate, and its divisor is $(\omega) = \sum_p \nu_\omega(p)p$, of degree $\deg(\omega)$. The genus g of a compact Riemann surface C is its topological genus, the number of handles, so that its Euler characteristic is $2 - 2g$.

Theorem 6.3.8: Degree of a Meromorphic Differential

Let C be a compact Riemann surface of genus g . Then C carries a nonzero meromorphic 1-form, and every nonzero meromorphic 1-form ω on C satisfies

$$\deg(\omega) = 2g - 2$$

The proof needs two tools this book does not develop: the existence of a nonconstant meromorphic function on a compact Riemann surface, proved in Forster [11, Corollary 14.13], and the triangulation of surfaces behind the Euler characteristic. A complete proof is in Griffiths and Harris [13, Chapter 2, Section 1, p. 219].

Proof Key ideas:

A nonconstant meromorphic function f on C is a nonconstant holomorphic map $f: C \rightarrow \mathbb{CP}^1$, of some degree n , and df is a nonzero meromorphic 1-form. Two nonzero meromorphic 1-forms differ by a factor which is a nonzero meromorphic function, whose divisor has degree 0, so all of them have the same degree. Triangulating $\mathbb{CP}^1$ with the branching points of f among the vertices, and lifting the triangulation to C , gives the topological Riemann-Hurwitz formula $\chi(C) = n\chi(\mathbb{CP}^1) - \deg R$, where R is the ramification divisor of f . Near a point of ramification index ν , where f reads $w = z^\nu u(z)$ with $u(0) \neq 0$, the pullback of a 1-form of order m has order $\nu m + \nu - 1$. So the pullback of the 1-form dw on $\mathbb{CP}^1$, whose divisor is -2∞ , has divisor $f^{-1}((dw)) + R$, of degree $-2n + \deg R = -\chi(C) = 2g - 2$, which completes the sketch.

Theorem 6.3.9: Riemann-Hurwitz Formula

Let C, C' be compact Riemann surfaces where:

$$\text{genus}(C) = g \quad \text{and} \quad \text{genus}(C') = g'$$

Let $f: C \rightarrow C'$ be a nonconstant holomorphic mapping of degree n . Let R be the ramification divisor of f . Then:

$$\deg R = 2(g + n - ng' - 1)$$

Proof:

By theorem 6.3.8, C' carries a nonzero meromorphic 1-form ω , with $\deg(\omega) = 2g' - 2$, and its pullback $f^*\omega$ is a nonzero meromorphic 1-form on C . Let $p \in C$, $q = f(p)$ and $\nu = \nu_f(p)$. In local coordinates z centred at p and w centred at q , the map f reads $w = z^\nu u(z)$ with u holomorphic and $u(0) \neq 0$ (definition 6.2.13, proposition 2.5.7), and $\omega = h(w)dw$ with $h(w) = w^m k(w)$, where $m = \nu_\omega(q)$ and $k(0) \neq 0$. Then

$$f^*\omega = h(z^\nu u(z)) d(z^\nu u(z)) = z^{\nu m} u(z)^m k(z^\nu u(z)) z^{\nu-1} (\nu u(z) + z u'(z)) dz$$

and the factors other than the powers of z are nonzero at 0, so

$$\nu_{f^*\omega}(p) = \nu_f(p) \nu_\omega(f(p)) + \nu_f(p) - 1$$

Summing over $p \in C$ and grouping the points by their image,

$$\deg(f^*\omega) = \sum_{q \in C'} \nu_\omega(q) \sum_{f(p)=q} \nu_f(p) + \deg R = n \deg(\omega) + \deg R$$

since $\sum_{f(p)=q} \nu_f(p) = n$ for every $q \in C'$ (lemma 6.3.7). By theorem 6.3.8 on C and on C' ,

$$2g - 2 = n(2g' - 2) + \deg R$$

and rearranging gives $\deg R = 2(g + n - ng' - 1)$, as claimed.

From this we can immediately conclude:

Corollary 6.3.10: Holomorphic Map and Genus

Let $f : C \rightarrow C'$ be a nonconstant holomorphic map of degree $n \geq 1$ between compact Riemann surfaces of genus g, g' . Then:

$$g \geq g'$$

Proof:

The formula $\deg R = 2(g + n - ng' - 1)$ of theorem 6.3.9 rearranges to $2g - 2 = n(2g' - 2) + \deg R$, where $R = \sum_p (\nu_f(p) - 1)p$ is the ramification divisor of f . Each coefficient $\nu_f(p) - 1$ is at least 0, so $\deg R \geq 0$ and $2g - 2 \geq n(2g' - 2)$. If $g' \geq 1$, then $2g' - 2 \geq 0$ and $n \geq 1$ give $n(2g' - 2) \geq 2g' - 2$, hence $g \geq g'$. If $g' = 0$, then $g \geq 0 = g'$ holds because a genus is never negative, as claimed.

Corollary 6.3.11: Compact Riemann Surface As Algebraic Equation

Let C be a compact Riemann surface and let

$$x, y : C \rightarrow \mathbb{CP}^1$$

be holomorphic mappings, with x nonconstant of degree $\deg x = d$ and y not identically ∞ . Then there is a nonzero polynomial $f \in \mathbb{C}[X, Y]$ of degree d in Y such that

$$f(x(t), y(t)) = 0$$

at every $t \in C$ where x and y are both finite.

Proof:

The degree of x is the common value

$$\sum_{t \in x^{-1}(\lambda)} \nu_x(t) = d \quad \lambda \in \mathbb{CP}^1$$

which does not depend on λ by lemma 6.3.7, and which is the count that theorem 6.3.9 uses. Let $B \subseteq \mathbb{CP}^1$ consist of the branching points of x , the images under x of the poles of y , and the point ∞ . Ramified points of x and poles of y are isolated in the compact surface C , so both sets are finite and B is finite.

Symmetric functions of the fibre. Let $\lambda \in \mathbb{CP}^1 \setminus B$. Every $t \in x^{-1}(\lambda)$ has $\nu_x(t) = 1$, so the fibre is a set of exactly d points $t_1, \dots, t_d$, at each of which y is finite. Put

$$a_j(\lambda) = (-1)^j e_j(y(t_1), \dots, y(t_d)) \quad j = 1, \dots, d$$

where e_j is the j -th elementary symmetric polynomial. A symmetric polynomial does not see the enumeration of the fibre, so each a_j is a well-defined function on $\mathbb{CP}^1 \setminus B$.

Fix $\lambda_0 \notin B$ with fibre $t_1, \dots, t_d$. Each t_i has ramification index 1, so by definition 6.2.13 x carries some neighbourhood V_i of t_i biholomorphically onto a neighbourhood of λ_0 , and the V_i may be taken pairwise disjoint. Some neighbourhood U of λ_0 has $x^{-1}(U) \subseteq V_1 \cup \dots \cup V_d$.

Otherwise there are $\lambda_n \rightarrow \lambda_0$ and points $u_n \in x^{-1}(\lambda_n)$ in the closed set $C \setminus (V_1 \cup \dots \cup V_d)$, which is compact, so a subsequence converges to some u in that closed set with $x(u) = \lim \lambda_n = \lambda_0$, and this contradicts $x^{-1}(\lambda_0) = \{t_1, \dots, t_d\}$. Shrinking U into every $x(V_i)$ produces holomorphic sections $s_i : U \rightarrow V_i$ inverse to x , and then

$$a_j(\lambda) = (-1)^j e_j(y(s_1(\lambda)), \dots, y(s_d(\lambda)))$$

exhibits a_j as a polynomial in holomorphic functions on U . So each a_j is holomorphic on $\mathbb{CP}^1 \setminus B$.

At worst a pole at each point of B . Let $\lambda_0 \in B$ with $x^{-1}(\lambda_0) = \{t_1, \dots, t_k\}$. The same compactness argument produces a neighbourhood U of λ_0 and pairwise disjoint coordinate discs $W_i \ni t_i$ with $x^{-1}(U) \subseteq W_1 \cup \dots \cup W_k$. Let z_i be a local coordinate on W_i vanishing at t_i and let w be a local coordinate on U vanishing at λ_0 , and shrink U so that $|w| < 1$ on U and $U \cap B = \{\lambda_0\}$. Writing $\nu_i = \nu_x(t_i)$, the order of vanishing of $w \circ x$ at t_i is ν_i , so

$$w \circ x = z_i^{\nu_i} h_i$$

on W_i with h_i holomorphic and nowhere zero after shrinking W_i . On the same disc y is meromorphic and not identically ∞ , so $|y| \leq c_i |z_i|^{-m_i}$ for some integer $m_i \geq 0$ and some $c_i > 0$. Shrinking once more bounds $|h_i|$ above by some M_i , whence $|z_i| \geq c'_i |w \circ x|^{1/\nu_i}$ and

$$|y| \leq c''_i |w \circ x|^{-m_i/\nu_i}$$

Set $M = \max_i m_i/\nu_i$ and $c = \max_i c''_i$. For $\lambda \in U$ different from λ_0 , every point of $x^{-1}(\lambda)$ lies in one of the W_i , so every value of y on that fibre is bounded by $c|w(\lambda)|^{-M}$. Each a_j is a sum of $\binom{d}{j}$ products of j such values, so

$$|a_j(\lambda)| \leq \binom{d}{j} c^j |w(\lambda)|^{-jM}$$

Pick an integer $N > dM$. Then $w^N a_j$ is holomorphic on U minus the point λ_0 by the previous step and is bounded near λ_0 , hence extends holomorphically across λ_0 by proposition 3.4.6. So a_j is meromorphic at λ_0 .

The polynomial. Every a_j is now meromorphic on all of $\mathbb{CP}^1$, hence rational by proposition 4.3.1. Let $q \in \mathbb{C}[X]$ be a nonzero common denominator, so that $qa_j \in \mathbb{C}[X]$ for each j , and put

$$f(X, Y) = q(X) (Y^d + a_1(X)Y^{d-1} + \dots + a_d(X)) \in \mathbb{C}[X, Y]$$

Its coefficient of Y^d is $q \neq 0$, so f is nonzero of degree d in Y . If $t \in C$ has $\lambda = x(t) \notin B$, then t is one of the d points enumerating $x^{-1}(\lambda)$, and the definition of the a_j reads

$$Y^d + a_1(\lambda)Y^{d-1} + \dots + a_d(\lambda) = \prod_{i=1}^d (Y - y(t_i))$$

so substituting $Y = y(t)$ gives $f(x(t), y(t)) = 0$. Finally B is finite and each fibre of x is finite, so $x^{-1}(B)$ is finite and its complement is dense in the surface C . At any t where x and y are both finite, $f(x, y)$ is continuous near t and vanishes on a dense set, so it vanishes at t as well.

Lemma 6.3.12: Projection From a General Point

Let $C \subseteq \mathbb{CP}^2$ be an irreducible algebraic curve of degree $d \geq 2$ whose singular points $s_1, \dots, s_\delta$ are ordinary double points, let F be an irreducible homogeneous polynomial of degree d with zero set C , let $S = \{s_1, \dots, s_\delta\}$, and let $\sigma: \tilde{C} \rightarrow C$ be a normalization of C (definition 6.3.2).

1. There are points $p \in \mathbb{CP}^2$ lying off C and off each of the 2δ tangent lines to the two branches of C at its nodes.
2. Let $p = [0 : 0 : 1]$ be such a point, let $\rho: \mathbb{CP}^2 \setminus \{p\} \rightarrow \mathbb{CP}^1$, $[\xi^0 : \xi^1 : \xi^2] \mapsto [\xi^0 : \xi^1]$, be the projection away from p , and put $\pi = \rho \circ \sigma: \tilde{C} \rightarrow \mathbb{CP}^1$. For $i \in \{0, 1\}$ call the chart $\xi^i \neq 0$, with coordinates $u = \xi^{1-i}/\xi^i$ and $v = \xi^2/\xi^i$, a p -adapted chart, and let g be the dehomogenization of F in it. Then σ is a biholomorphism from $\tilde{C} \setminus \sigma^{-1}(S)$ onto $C \setminus S$, and for $z \in C \setminus S$ and $r = \sigma^{-1}(z)$, the ramification index $\nu_\pi(r)$ is 1 if and only if $g_v(z) \neq 0$ in a p -adapted chart containing z .
3. The map π is nonconstant of degree d .
4. Every point $r \in \tilde{C}$ above a node s is unramified for π , and σ maps a neighbourhood of r into a branch of C at s which, in a p -adapted chart, is a graph $v = \varphi(u)$ whose tangent line at s is one of the two tangent lines of the node.

Proof:

The two p -adapted charts cover $\mathbb{CP}^2 \setminus \{p\}$, in each of them ρ reads $(u, v) \mapsto u$ and a line through p reads $u = \text{constant}$. Translating a p -adapted chart changes neither of these shapes.

1. The curve C and the 2δ lines are the zero set of the product of F and 2δ linear forms, a nonzero homogeneous polynomial, which does not vanish on all of $\mathbb{CP}^2$.
2. By definition 6.3.2 the map σ is a holomorphic bijection from $\tilde{C} \setminus \sigma^{-1}(S)$ onto $C \setminus S$, which is a complex manifold of dimension one by lemma 6.3.1, so by theorem 6.2.7 the map σ is a biholomorphism there. Let $z \in C \setminus S$, $r = \sigma^{-1}(z)$, and work in a p -adapted chart containing z . As π reads u , the index $\nu_\pi(r)$ is the order of vanishing at z , along C , of $u - u(z)$. If $g_v(z) \neq 0$, the implicit function theorem (theorem 2.2.28) presents C near z as a graph $v = \varphi(u)$, so $u - u(z)$ is a local coordinate on C at z and $\nu_\pi(r) = 1$. If instead $g_v(z) = 0$, then $g_u(z) \neq 0$ by lemma 6.3.1, the implicit function theorem presents C near z as a graph $u = \psi(v)$, and $v - v(z)$ is a local coordinate on C at z . Then $\nu_\pi(r) = \text{ord}_{v(z)}(\psi - \psi(v(z)))$, and differentiating $g(\psi(v), v) = 0$ gives $g_u(\psi(v), v)\psi'(v) + g_v(\psi(v), v) = 0$, so $\psi'(v(z)) = 0$ and $\nu_\pi(r) \geq 2$.
3. The map π is nonconstant because an irreducible curve of degree $d \geq 2$ is not contained in a line. All but finitely many lines through p miss the δ nodes. Fix such a line ℓ , with linear form L . For $z \in C \cap \ell$, in a p -adapted chart containing z the line reads $u = u(z)$, so $\nu_\pi(\sigma^{-1}(z))$ is the order of vanishing of $u - u(z)$ along C at z , which is $I(z, F \cap L)$ by property (8) of the local intersection number in [26, section 3.3], z being a smooth point. The fibre of π over ℓ is $\sigma^{-1}(C \cap \ell)$, and F and L share no component, so by lemma 6.3.7 and Bézout's theorem ([26, Bézout's Theorem])

$$\deg \pi = \sum_{z \in C \cap \ell} I(z, F \cap L) = d$$

4. Fix a node s and a p -adapted chart translated so that s is the origin. That s is an ordinary double point says

$$g = \ell_1 \ell_2 + (\text{terms of degree at least } 3)$$

with $\ell_1 = \alpha_1 u + \beta_1 v$ and $\ell_2 = \alpha_2 u + \beta_2 v$ linearly independent. The line $\ell_i = 0$ passes through p , the point at infinity of the v -direction, exactly when $\beta_i = 0$, so the choice of p gives $\beta_1 \neq 0$ and $\beta_2 \neq 0$. The restriction $g(0, v) = \beta_1 \beta_2 v^2 + O(v^3)$ is v -regular of order 2, so the Weierstrass preparation theorem (Huybrechts [16, Proposition 1.1.6]) writes g as a unit times $v^2 + a_1(u)v + a_0(u)$ with $a_i(0) = 0$. Writing $\ell_1 \ell_2 = \beta_1 \beta_2 (v - r_1 u)(v - r_2 u)$ with $r_1 \neq r_2$, the discriminant $a_1^2 - 4a_0 = (r_1 - r_2)^2 u^2 + O(u^3)$ is u^2 times a unit, hence a square in $\mathbb{C}\{u\}$, a unit having a holomorphic square root near 0. The quadratic therefore splits into two factors of order 1, giving $g^{(1)}, g^{(2)}$ with lowest forms ℓ_1 and ℓ_2 up to scalars. As $\beta_i \neq 0$, the implicit function theorem presents the branch $B_i = \{g^{(i)} = 0\}$ near s as a graph $v = \varphi_i(u)$, with tangent line $\ell_i = 0$ at s , so u is injective on B_i , and after shrinking $B_1 \cap B_2 = \{s\}$. Now take $r \in \sigma^{-1}(s)$ and a disc $\Delta \ni r$ small enough that $\sigma(\Delta)$ lies in the chart and $\sigma(\Delta \setminus \{r\})$ misses S . Then $\sigma(\Delta \setminus \{r\})$ is a connected subset of $(B_1 \cup B_2) \setminus \{s\}$, hence lies in one branch, on which u is injective. Combined with the injectivity of σ off $\sigma^{-1}(S)$ this makes the holomorphic function $u \circ \sigma - u(s)$ injective on $\Delta \setminus \{r\}$. By lemma 6.2.14, a holomorphic function vanishing to order k at r has k preimages near r of each nearby nonzero value, so $k = 1$ and $\nu_\pi(r) = 1$, completing the proof.

Theorem 6.3.13: Genus Formula

Let $C \subseteq \mathbb{CP}^2$ be an irreducible algebraic curve of degree d with δ singular points all being ordinary double points, and let $\sigma : \tilde{C} \rightarrow C$ be its normalization where $\text{genus}(\tilde{C}) = g$. Then:

$$g = \frac{(d-1)(d-2)}{2} - \delta$$

Proof:

Let F be an irreducible homogeneous polynomial of degree d with zero set C , let $S = \{s_1, \dots, s_\delta\}$ be the set of nodes, and let R be the ramification divisor of definition 6.3.6. A line in $\mathbb{CP}^2$ is smooth and a linear change of coordinates carries it onto $\mathbb{CP}^1$, so for $d = 1$ we get $\delta = 0$ and $g = 0$, which is the asserted formula. Take $d \geq 2$ from here on.

Set-up. Choose p as in lemma 6.3.12(1), put $p = [0 : 0 : 1]$ in homogeneous coordinates, and use the projection ρ , the map $\pi = \rho \circ \sigma : \tilde{C} \rightarrow \mathbb{CP}^1$ and the p -adapted charts of lemma 6.3.12(2). Write

$$P = \frac{\partial F}{\partial \xi^2}$$

for the polar of F at p , a form of degree $d - 1$. If g denotes the dehomogenization of F in a p -adapted chart then g_v is the dehomogenization of P .

The polar count. The form P is not identically zero: it vanishes identically only when F is a form in ξ^0 and ξ^1 , and such a form factors into linear forms over $\mathbb{C}$, which an irreducible F of degree $d \geq 2$ does not. Since $\deg P = d - 1 < d$ and F is irreducible, F and P share no component, so Bézout's theorem ([26, Bézout's Theorem]) gives

$$\sum_{z \in \mathbb{CP}^2} I(z, F \cap P) = d(d-1)$$

where I is the local intersection number of [26, section 3.3] and only finitely many terms are nonzero.

Smooth points. Let $z \in C \setminus S$ and let r be the point of $\tilde{C}$ above it (lemma 6.3.12(2)). Then $\nu_\pi(r)$ is the order of vanishing at z , along C , of $u - u(z)$ in a p -adapted chart containing z , and the claim is

$$I(z, F \cap P) = \nu_\pi(r) - 1$$

Work in that chart. If $g_v(z) \neq 0$, then $\nu_\pi(r) = 1$ by lemma 6.3.12(2), while z lies off P and $I(z, F \cap P) = 0$. If instead $g_v(z) = 0$, then $g_u(z) \neq 0$, the implicit function theorem presents C near z as a graph $u = \psi(v)$, and $v - v(z)$ is a local coordinate on C at z . Put $m = \text{ord}_{v(z)}(\psi - \psi(v(z)))$, which is $\nu_\pi(r)$. Differentiating the identity $g(\psi(v), v) = 0$ gives

$$g_u(\psi(v), v)\psi'(v) + g_v(\psi(v), v) = 0$$

and $g_u(\psi(v), v)$ is nonzero at $v(z)$, so $g_v(\psi(v), v)$ has the same order there as ψ' , namely $m - 1$. Since z is a smooth point of g , property (8) of the intersection number in [26, section 3.3] identifies $I(z, F \cap P)$ with the order of vanishing of g_v along C at z , which is $m - 1$. The claim follows.

The degree of π . By lemma 6.3.12(3), $\deg \pi = d$.

Nodes. Fix a node s and a p -adapted chart translated so that s is the origin. That s is an ordinary double point says

$$g = \ell_1 \ell_2 + (\text{terms of degree at least 3})$$

with $\ell_1 = \alpha_1 u + \beta_1 v$ and $\ell_2 = \alpha_2 u + \beta_2 v$ linearly independent. The line $\ell_i = 0$ passes through p , the point at infinity of the v -direction, exactly when $\beta_i = 0$, so the choice of p gives $\beta_1 \neq 0$ and $\beta_2 \neq 0$. The lowest form of g_v is

$$\frac{\partial(\ell_1 \ell_2)}{\partial v} = \beta_1 \ell_2 + \beta_2 \ell_1$$

which is nonzero because the ℓ_i are independent and the β_i are not zero. So $m_s(g_v) = 1$ and the tangent cone of g_v at s is the line $\beta_1 \ell_2 + \beta_2 \ell_1 = 0$. That line is neither $\ell_1 = 0$ nor $\ell_2 = 0$, since $\beta_1 \ell_2 + \beta_2 \ell_1 = c \ell_1$ would force $\beta_1 \ell_2 = (c - \beta_2) \ell_1$ and make ℓ_2 proportional to ℓ_1 , and symmetrically. With $m_s(g) = 2$ and no tangent line in common, property (5) of the intersection number in [26, section 3.3] gives

$$I(s, F \cap P) = m_s(g) m_s(g_v) = 2$$

and every point of $\tilde{C}$ above s is unramified for π by lemma 6.3.12(4).

Conclusion. Every point of $\tilde{C}$ lies over a smooth point of C or over a node. Over a node the ramification index is 1, and over a smooth point z it is $I(z, F \cap P) + 1$, so

$$\deg R = \sum_{r \in \tilde{C}} (\nu_\pi(r) - 1) = \sum_{z \in C \setminus S} I(z, F \cap P)$$

Points off C contribute nothing to the Bézout count, so subtracting the node terms from it leaves

$$\deg R = d(d-1) - \sum_{j=1}^{\delta} I(s_j, F \cap P) = d(d-1) - 2\delta$$

Applying theorem 6.3.9 to $\pi : \tilde{C} \rightarrow \mathbb{CP}^1$, of degree d onto a target of genus 0, gives $\deg R = 2(g + d - 1)$. Equating the two expressions,

$$2g + 2d - 2 = d(d - 1) - 2\delta$$

so $2g = d^2 - 3d + 2 - 2\delta = (d - 1)(d - 2) - 2\delta$, which is the stated formula.

6.3.2 Proof of Riemann Roch Theorem

Let C be a compact Riemann surface. Define the divisor to be the formal sum:

$$D = \sum_i^k n_i p_i \in \text{Div}(C)$$

A divisor is called *effective* if each coefficient is ≥ 0 . and we write $D \geq 0$. Any divisor can be expressed as the difference of two effective divisors. We now define the Riemann-Roch space:

$$\mathcal{L}(D) = \{f \in K(C) \mid (f) \geq -D\}$$

That is, $\mathcal{L}(D)$ contains all $f \in K(C)$ whether f has a pole of multiplicity at most n_i at p_i if $n_i > 0$ and a zero of multiplicity at least $|n_i|$ at p_i if $n_i < 0$. We can then define:

$$l(D) = \dim \mathcal{L}(D)$$

We then define a similar notion for divisors:

$$K^1(D) = \{\omega \in K^1(C) \mid (\omega) \geq D\}$$

where now, if $n_i < 0$ then ω can have a p_i as a pole of multiplicity at most $|n_i|$ and if $n_i > 0$ then ω must have a p_i as a zero of multiplicity at least n_i . Again, $K^1(D)$ is a vector space and :

$$\dim K^1(D) = i(D)$$

We shall usually be focusing on effect divisors; $D \geq 0$; in that case we have that $\mathbb{C} \subseteq \mathcal{L}(D)$. Furthermore, to be more align with the usual differential geometry notation, we shall write $\Omega^1(D)$ instead of $K^1(D)$. Then for $D \geq 0$, $\omega \in \Omega^1(D)$ means ω is holomorphic and has a zero of appropriate multiplicity at p_i for appropriate points given by D .

Definition 6.3.14: Linearly Equivalent

Let $D, E \in \text{Div}(C)$. Then D, E are *linearly equivalent* if there exists a $f \in K(C)$ such that

$$(f) = D - E$$

and we write $D \sim E$.

We can also write $D = (f) + E$ for some (f) , and hence we are really saying that $D \sim E$ if they are equal up to principal divisor. Hence, two divisor are linearly equivalent if they belong to the same coset in $\text{Div}(C)/\text{PDiv}(C)$.

Proposition 6.3.15: Dimension and Linear Equivalence

Let $D, E \in \text{Div}(C)$, $D \sim E$. Then:

$$\mathcal{L}(D) \cong \mathcal{L}(E) \quad \Omega^1(D) \cong \Omega^1(E) \quad \deg D = \deg E$$

Proof:

Say $D \sim E$. Then there exists a nonzero $f \in K(C)$ such that

$$(f) = D - E$$

Now pick $g \in \mathcal{L}(D)$ so that $(g) + D \geq 0$. Then:

$$(fg) + E = (f) + (g) + E = D - E + (g) + E = (g) + D \geq 0$$

Thus, there is a mapping:

$$\mathcal{L}(D) \rightarrow \mathcal{L}(E) \quad g \mapsto fg$$

This mapping is linear, and it is bijective, as the same computation with $-(f) = E - D$ shows that its inverse is:

$$\mathcal{L}(E) \rightarrow \mathcal{L}(D) \quad g \mapsto \frac{g}{f}$$

Thus, $\mathcal{L}(D) \cong \mathcal{L}(E)$. For $\omega \in \Omega^1(D)$, that is $(\omega) \geq D$, we have $(\omega/f) = (\omega) - (f) \geq D - (D - E) = E$, so $\omega \mapsto \omega/f$ maps $\Omega^1(D)$ linearly to $\Omega^1(E)$, and $\omega \mapsto f\omega$ is its inverse, by the same computation with $-(f)$. Thus $\Omega^1(D) \cong \Omega^1(E)$. To show the degrees are the same, recall that for any nonzero $f \in K(C)$:

$$\sum_{p \in C} \nu_p(f) = 0$$

If f is constant, both sides are 0. Otherwise f is a nonconstant holomorphic map $C \rightarrow \mathbb{CP}^1$, and the sum of the orders of its zeros is $d_f(0)$, while the sum of the orders of its poles is $d_f(\infty)$, the order of a pole being the ramification index at it in the coordinate $1/w$ around ∞ . These agree by lemma 6.3.7. Then:

$$\sum \nu_p(f) = \deg(f) = \deg D - \deg E$$

hence $\deg E = \deg D$, as we sought to show.

Proposition 6.3.16: Brill-Noether Reciprocity

Let $\omega \in K^1(C)$ be a nonzero meromorphic 1-form and $(\omega) = D + E$. Then:

$$\mathcal{L}(D) \cong \Omega^1(E) \quad \mathcal{L}(E) \cong \Omega^1(D)$$

Proof:

Let $f \in \mathcal{L}(D)$ so that $(f) \geq -D$. Then:

$$(f\omega) = (f) + (\omega) \geq -D + D + E = E$$

Thus define:

$$\mathcal{L}(D) \rightarrow \Omega^1(E) \quad f \mapsto f\omega$$

which is a vector-space homomorphism. For $\varphi \in \Omega^1(E)$, the quotient φ/ω of two meromorphic 1-forms, ω being nonzero, is a meromorphic function, with $(\varphi/\omega) = (\varphi) - (\omega) \geq E - D - E = -D$. So the inverse is:

$$\Omega^1(E) \rightarrow \mathcal{L}(D) \quad \varphi \mapsto \frac{\varphi}{\omega}$$

showing they are isomorphic. As $(\omega) = E + D$, the same argument with E and D swapped gives the second isomorphism, completing the proof.

Definition 6.3.17: Laurent Principal Part

Let $\sum_{i=-n}^{\infty} a_i z^i$ be a Laurent series. Then its singular part

$$\sum_{i=-n}^{-1} a_i z^i$$

is the *Laurent principal part*, or just *principal part*. If $a_{-n} \neq 0$, then N is called the order of the Laurent principal part.

Now suppose C is a Riemann surface, D an effective divisor. Let $z_i, i = 1, 2, \dots, k$ be local coordinates near p_i and $z_i(p_i) = 0$. Then if for every p_i , there is given a Laurent principal part of order n_i :

$$\eta_i = \frac{a_{in_i}}{z_i^{n_i}} + \dots + \frac{a_{i1}}{z_i}$$

we can ask the following question: under what condition does there exist an $f \in K(C)$ such that in a neighbourhood of each p_i , f has η_i as a Laurent principal part? This is called the *Mittag-Leffler problem*, and answering it will allow us to deduce the Riemann-Roch Theorem.

First off, a meromorphic function on a compact Riemann surface C is up to a constant uniquely determined by its Laurent principal part, for if $f_1, f_2 \in K(C)$ that have the same Laurent principal part then $f_1 - f_2$ is holomorphic on C , and hence is constant. Thus, we may identify $\mathcal{L}(D)/\mathbb{C}$ as a subspace of $\mathbb{C}^d$, $d = \deg D$, as follows. Fix $D = \sum_i^k n_i p_i \geq 0$, and let z_i be a coordinate function around p_i where $z_i(p_i) = 0$. Then for $f \in \mathcal{L}(D)$, let η_i be the principal part at p_i . Then we know that:

$$\eta_i = \frac{a_{in_i}}{z_i^{n_i}} + \dots + \frac{a_{i2}}{z_i^2} + \frac{a_{i1}}{z_i}$$

where $a_{in_i}, \dots, a_{i1} \in \mathbb{C}$. Now let P_i be the n_i -dimensional vector space over $\mathbb{C}$ spanned by $\{1/z_i^{n_i}, \dots, 1/z_i^2, 1/z_i\}$. Then we have the injective vector-space homomorphism:

$$\frac{\mathcal{L}(D)}{\mathbb{C}} \rightarrow P_1 \oplus \dots \oplus P_k \quad [f] \mapsto (\eta_1, \dots, \eta_k)$$

Then by identifying each P_i with $\mathbb{C}^{n_i}$, we get that this space is a subspace of $\mathbb{C}^d$. Note that surjectivity is not so obvious or even necessary because we do not know if these maps are compatible (ex. the gluing conditions can impose restrictions).

Dimension of $\Omega^1(C)$

The goal is to show that $\dim \Omega^1(C) = g$ for every compact Riemann surface C of genus g , which is theorem 6.3.23 below. For the Riemann sphere, of genus 0, this says that the only holomorphic 1-form is 0. Indeed, if $f(z)dz$ is holomorphic on $\mathbb{CP}^1$, then in the coordinate $w = 1/z$ it reads $-f(1/w)w^{-2}dw$, so the entire function f satisfies $|f(z)| \leq c|z|^{-2}$ for $|z|$ large. It is bounded, hence constant by Liouville's theorem (theorem 3.3.3), and the constant is 0. The proof will proceed in two parts:

Proposition 6.3.18: Dimension of Differential Form: I

Let C be a compact Riemann surface of genus g . Then:

$$\dim \Omega^1(C) \geq g$$

The key idea is that we can “embed” the space of homogeneous functions of the right degree (minus the singularities) into $\Omega^1(C)$ by essentially taking its differential with a bit of normalization. The proof starts from a plane model of C : a holomorphic map $\pi: C \rightarrow \mathbb{P}^2$ whose image C' is an irreducible plane algebraic curve of some degree d whose only singular points are ordinary double points, and which is injective off the finitely many points above them, so that (C, π) is a normalization of C' (definition 6.3.2). Such a map exists. It is obtained by embedding C in a projective space by the sections of a line bundle of large degree and projecting generically to $\mathbb{P}^2$, which needs line bundles on compact Riemann surfaces, a tool this book does not develop. A complete account is in Griffiths and Harris [13, Chapter 2, Section 1, pp. 214, 215].

Proof:

Let $\pi: C \rightarrow \mathbb{P}^2$ be a plane model as above, with image C' of degree d and nodes $p_1, \dots, p_\delta$, and let F be an irreducible homogeneous polynomial of degree d with zero set C' . By the genus formula (theorem 6.3.13), $g = \frac{(d-1)(d-2)}{2} - \delta$. If $d \leq 2$, this gives $g = 0$, and there is nothing to prove, so assume $d \geq 3$.

Step 1: coordinates. Choose $p \in \mathbb{P}^2$ off C' and off the 2δ tangent lines to the branches of C' at its nodes (lemma 6.3.12(1)), and homogeneous coordinates in which $p = [0 : 0 : 1]$. Let $\varpi: C \rightarrow \mathbb{P}^1$ be π followed by the projection from p . Apply lemma 6.3.12 to the normalization (C, π) of C' . By its items (2) to (4), π is a biholomorphism from C minus the points above the nodes onto the smooth locus of C' , the map ϖ has degree d , and the points of C above a node are unramified for ϖ , each having a neighbourhood mapped by π into a branch of C' which is a graph over the first coordinate of a p -adapted chart. The map ϖ has finitely many ramification points, these being isolated in the compact surface C , so all but finitely many lines through p pass through no node and through no image of a ramification point. Fix such a line ℓ . The fibre of ϖ over ℓ consists of d distinct unramified points (lemma 6.3.7), so ℓ meets C' in d distinct smooth points. Changing coordinates by a linear map fixing p , we may assume $\ell = \{\xi^0 = 0\}$. Put $x = \xi^1/\xi^0$ and $y = \xi^2/\xi^0$ on $U_0 = \{\xi^0 \neq 0\}$, and $u = \xi^0/\xi^1$ and $v = \xi^2/\xi^1$ on $U_1 = \{\xi^1 \neq 0\}$. These are the two p -adapted

charts, and they cover C' as $p \notin C'$. The points of $C' \cap \ell$ lie in U_1 , where $\ell = \{u = 0\}$, and as they are unramified for ϖ , lemma 6.3.12(2) shows that the derivative in v of the equation of C' in U_1 is nonzero at each of them. Let $f(x, y) = F(1, x, y)$ and $k(u, v) = F(u, 1, v)$. As $p \notin C'$, the coefficient of $(\xi^2)^d$ in F is nonzero, so f has degree d in y with constant leading coefficient. All but finitely many vertical lines $x = x_0$ meet C' in d distinct points, by the argument just given, so for such x_0 the polynomial $f(x_0, \cdot)$ of degree d has d distinct roots, all simple.

Step 2: adjoint polynomials. Let S^{d-3} be the vector space of homogeneous polynomials of degree $d-3$ in three variables, of dimension $\binom{d-1}{2} = \frac{(d-1)(d-2)}{2}$, the number of monomials of that degree, and let

$$S^{d-3}(-\Gamma) = \{G \in S^{d-3} \mid G(p_i) = 0, \forall i = 1, 2, \dots, \delta\}$$

Vanishing at p_i is one linear condition on the coefficients, so

$$\dim S^{d-3}(-\Gamma) \geq \frac{(d-1)(d-2)}{2} - \delta = g$$

Step 3: the forms. Regard x, y, u and v as meromorphic functions on C through π . The function $f_y(x, y)$ is not identically zero on C , since $f_y(x_0, y) \neq 0$ at the simple roots of $f(x_0, \cdot)$ from Step 1. For $G \in S^{d-3}(-\Gamma)$ put $G_0(x, y) = G(1, x, y)$ and

$$\omega_G = \frac{G_0(x, y)}{f_y(x, y)} dx$$

a meromorphic 1-form on C . The map $G \mapsto \omega_G$ is linear, and it is injective: if $\omega_G = 0$, then $G_0(x_0, \cdot)$, a polynomial of degree at most $d-3$, vanishes at the d distinct roots of $f(x_0, \cdot)$, so it is zero, for all but finitely many x_0 . Each coefficient of G_0 as a polynomial in y is then a polynomial in x with infinitely many roots, so $G_0 = 0$, and $G = 0$, as $G = (\xi^0)^{d-3} G_0(\xi^1/\xi^0, \xi^2/\xi^0)$ on the dense set $\xi^0 \neq 0$.

Step 4: ω_G is holomorphic. Let $r \in C$ and $z = \pi(r)$.

1. If $z \in U_0$ is smooth and $f_y(z) \neq 0$, then C' is a graph $y = \varphi(x)$ near z (lemma 6.3.1), and x is a local coordinate at r , π being a biholomorphism near r . In it, $\omega_G = \frac{G_0(x, \varphi(x))}{f_y(x, \varphi(x))} dx$ is holomorphic.
2. If $z \in U_0$ is smooth and $f_y(z) = 0$, then $f_x(z) \neq 0$ by lemma 6.3.1, so C' is a graph $x = \psi(y)$ near z , y is a local coordinate at r , and differentiating $f(\psi(y), y) = 0$ gives $f_x dx + f_y dy = 0$ on C near r . Hence $\omega_G = -\frac{G_0(x, y)}{f_x(x, y)} dy$ near r , off the isolated zeros of f_y , and the right-hand side is holomorphic at r .
3. If $z = p_i$ is a node, then $G_0(z) = 0$. By Step 1, r is unramified for ϖ , which reads x in U_0 , so $t = x - x(z)$ is a local coordinate at r , and π maps a neighbourhood of r into a branch $y = \varphi(x)$ of C' through z , whose tangent line at z is one of the two tangent lines of the node (lemma 6.3.12(4)). Write $f(z+w) = \ell_1(w)\ell_2(w) + O(|w|^3)$ with $\ell_j(w) = \alpha_j w_1 + \beta_j w_2$ linearly independent, as z is an ordinary double point. Then $\beta_j \neq 0$, since p is not on the tangent lines $\ell_j = 0$ at z , and the branch has tangent $\ell_1 = 0$, say, so $\varphi(x(z)+t) - y(z) = mt + O(t^2)$ with $\alpha_1 + \beta_1 m = 0$. Now $f_y(z+w) = \beta_1 \ell_2(w) + \beta_2 \ell_1(w) + O(|w|^2)$, so along the branch

$f_y = \beta_1 \ell_2(1, m) t + O(t^2)$. Here $\ell_2(1, m) \neq 0$, as $(1, m)$ lies in the kernel of ℓ_1 and the two forms are independent. So f_y has a simple zero at r in the coordinate t , while G_0 vanishes there and $dx = dt$, and $\omega_G = \frac{G_0}{f_y} dt$ is holomorphic at r .

4. If $z \in \ell$, then $z = (0, v_0)$ in U_1 is a smooth point with $k_v(z) \neq 0$, by Step 1, so C' is a graph $v = \chi(u)$ near z (lemma 6.3.1), and u is a local coordinate at r . Near r , $x = 1/u$ and $y = v/u$, so $dx = -u^{-2} du$, and by homogeneity $G_0(x, y) = u^{-(d-3)} G(u, 1, v)$ and $f_y(x, y) = \frac{\partial F}{\partial \xi^2}(1, x, y) = u^{-(d-1)} k_v(u, v)$. Hence

$$\omega_G = -\frac{u^{-(d-3)} G(u, 1, v)}{u^{-(d-1)} k_v(u, v)} \cdot \frac{du}{u^2} = -\frac{G(u, 1, v)}{k_v(u, v)} du$$

which is holomorphic at r .

Every point of C' lies in U_0 or on ℓ , so $\omega_G \in \Omega^1(C)$ for every G . The map $G \mapsto \omega_G$ is an injective linear map $S^{d-3}(-\Gamma) \rightarrow \Omega^1(C)$, and $\dim \Omega^1(C) \geq \dim S^{d-3}(-\Gamma) \geq g$, as we sought to show.

Let us now show the other direction of $\dim \Omega^1(C) \leq g$. This direction shall require some differential geometry reasoning.

Let C be a compact Riemann surface of genus g . Its first singular homology group $H_1(C, \mathbb{Z})$ is free abelian of rank $2g$. Let $\gamma_1, \gamma_2, \dots, \gamma_{2g-1}, \gamma_{2g}$ be closed piecewise smooth curves whose classes form a $\mathbb{Z}$ -basis of it. A closed piecewise smooth curve γ , cut at its break points, is a smooth singular 1-cycle, with homology class $[\gamma]$. Every class is a combination $\sum_i n_i [\gamma_i]$ with $n_i \in \mathbb{Z}$. A smooth 1-form λ on C is called closed if $d\lambda = 0$.

Proposition 6.3.19: Natural Dual of Homology Group

Let λ be a closed differential one-form on C . Then the mapping:

$$\eta_\lambda : H_1(C, \mathbb{Z}) \rightarrow \mathbb{C} \quad [\gamma] \mapsto \int_\gamma \lambda$$

is well-defined and additive. More generally, $\int_c \lambda$ depends only on the class of the smooth singular 1-cycle c in $H_1(C; \mathbb{R})$, and it is $\mathbb{R}$ -linear in that class.

Proof:

The underlying smooth surface of C is compact, so [23, Perfection of the Period Pairing], applied to the real and imaginary parts of λ , shows that $\int_c \lambda$ depends only on the class of c in $H_1(C; \mathbb{R})$ and is linear in it. The class in $H_1(C; \mathbb{R})$ of a cycle with integer coefficients depends only on its class in $H_1(C, \mathbb{Z})$, which gives the claims.

Proposition 6.3.20: Exactness

Let λ be a closed differential one-form on C . Then if for all i ,

$$\int_{\gamma_i} \lambda = 0$$

then λ is exact.

Proof:

By the universal coefficient theorem ([20, Universal Coefficient Theorem For Homology]), $H_1(C; \mathbb{R}) \cong H_1(C; \mathbb{Z}) \otimes \mathbb{R}$, the term $\text{Tor}(H_0(C; \mathbb{Z}), \mathbb{R})$ vanishing because $H_0(C; \mathbb{Z}) \cong \mathbb{Z}$ is free. So the classes $[\gamma_i]$ span $H_1(C; \mathbb{R})$ over $\mathbb{R}$, and every smooth singular 1-cycle c has $[c] = \sum_i r_i [\gamma_i]$ with $r_i \in \mathbb{R}$. By proposition 6.3.19, $\int_c \lambda = \sum_i r_i \int_{\gamma_i} \lambda = 0$. Thus every period of the real and imaginary parts of λ vanishes, and they are exact by [23, Perfection of the Period Pairing], so $\lambda = df$ for a smooth function f , as we sought to show.

Proposition 6.3.21: Cauchy Riemann For Differential Forms

Let C be a compact Riemann surface, and $\omega, \varphi \in \Omega^1(C)$. Then if:

$$\omega + \bar{\varphi} = df$$

then

$$\omega = \varphi = 0$$

Proof:

Let $\omega = h(z)dz$ and $\varphi = g(z)dz$. Then as C is one-dimensional:

$$\omega \wedge \varphi = 0$$

and

$$\frac{i}{2} \varphi \wedge \bar{\varphi} = |g(z)|^2 \frac{i}{2} dz \wedge d\bar{z} = |g(z)|^2 du \wedge dv$$

We shall show $\varphi = 0$. For the sake of contradiction suppose $\varphi \neq 0$. Then:

$$\frac{i}{2} \iint_C \varphi \wedge \bar{\varphi} = \iint_C |g(z)|^2 du \wedge dv > 0$$

But then:

$$\varphi \wedge \bar{\varphi} = \varphi \wedge \omega + \varphi \wedge \bar{\varphi} = \varphi \wedge (\omega + \bar{\varphi}) = \varphi \wedge df$$

If we show $\iint_C \varphi \wedge df = 0$, we are done. Indeed, notice that:

$$d(f\varphi) = df \wedge \varphi + f d\varphi$$

and $d\varphi = 0$ by lemma 6.2.10. Hence:

$$\varphi \wedge df = -df \wedge \varphi = -d(f\varphi)$$

Note that $f\varphi$ is a smooth differential one-form on the compact surface C , which has empty boundary, so Stokes' theorem ([23, Stokes' Theorem]) gives:

$$\iint_C d(f\varphi) = 0$$

But then:

$$\iint_C \varphi \wedge df = - \iint_C d(f\varphi) = 0$$

Conjugating the hypothesis gives $\varphi + \bar{\omega} = d\bar{f}$, and the argument just given, with the roles of ω and φ exchanged, shows $\omega = 0$, completing the proof.

Proposition 6.3.22: Dimension of Differential Form: II

Let C be a compact Riemann surface of genus g . Then:

$$\dim \Omega^1(C) \leq g$$

Proof:

For the sake of contradiction suppose $\dim \Omega^1(C) \geq g + 1$ so that there are $g + 1$ elements $\omega_1, \dots, \omega_{g+1} \in \Omega^1(C)$ that are $\mathbb{C}$ -linearly independent. Take the equations:

$$\int_{\gamma_i} \sum_{j=1}^{g+1} (\lambda^j \omega_j + \eta^j \bar{\omega}_j) = 0 \quad i = 1, 2, \dots, 2g$$

in the unknowns $\lambda^1, \dots, \lambda^{g+1}, \eta^1, \dots, \eta^{g+1} \in \mathbb{C}$. They are $2g$ homogeneous linear equations in $2g + 2$ unknowns, so they have a nontrivial solution $\lambda_0^1, \dots, \lambda_0^{g+1}, \eta_0^1, \dots, \eta_0^{g+1}$. Put:

$$\omega = \sum_{j=1}^{g+1} \lambda_0^j \omega_j \in \Omega^1(C) \quad \varphi = \sum_{j=1}^{g+1} \overline{\eta_0^j} \omega_j \in \Omega^1(C)$$

so that $\bar{\varphi} = \sum_j \eta_0^j \bar{\omega}_j$, and for all $i = 1, 2, \dots, 2g$:

$$\int_{\gamma_i} (\omega + \bar{\varphi}) = \int_{\gamma_i} \sum_{j=1}^{g+1} (\lambda_0^j \omega_j + \eta_0^j \bar{\omega}_j) = 0$$

The form $\omega + \bar{\varphi}$ is smooth and closed, holomorphic 1-forms and their conjugates being smooth and closed (lemma 6.2.10). By proposition 6.3.20, there exists a C^∞ function f such that

$$\omega + \bar{\varphi} = df$$

But then by proposition 6.3.21 $\omega = \varphi = 0$, and the linear independence of $\omega_1, \dots, \omega_{g+1}$ gives $\lambda_0^j = 0$ and $\overline{\eta_0^j} = 0$ for all j . This contradicts the choice of a nontrivial solution, as we sought to show.

Theorem 6.3.23: Dimension of Differential Forms

Let C be a compact Riemann surface of genus g . Then:

$$\dim \Omega^1(C) = g$$

Proof:

The inequality $\dim \Omega^1(C) \geq g$ is proposition 6.3.18, and $\dim \Omega^1(C) \leq g$ is proposition 6.3.22, which together give the claim.

Two Important Theorems

Let $H_1(C, \mathbb{Z})$ be the first singular homology group of a genus g Riemann surface, a free abelian group on $2g$ generators. We have a natural dual definition:

Definition 6.3.24: 1st Cohomology Group

Let C be a compact Riemann surface. Then the first cohomology group of C is:

$$H^1(C, \mathbb{C}) = \text{Hom}_{\mathbb{Z}}(H_1(C, \mathbb{Z}), \mathbb{C})$$

Then we saw in proposition 6.3.19 that we have a map:

$$\begin{aligned} \eta : \{\text{closed differential one-forms on } C\} &\rightarrow H^1(C, \mathbb{C}) \\ \lambda &\mapsto \eta_\lambda \end{aligned}$$

and furthermore this map is injective by proposition 6.3.20. Let $H_{\text{DR}}(C)$ be the equivalence classes of closed one forms mod exact one forms. Then by proposition 6.3.20, we have in fact that the above η is a map from $H_{\text{DR}}(C) \hookrightarrow H^1(C, \mathbb{C})$. Then, by the “Hodge” condition, we have that:

$$\begin{aligned} \Omega^1(C) \oplus \overline{\Omega^1(C)} &\hookrightarrow H_{\text{DR}}(C) \\ \omega \oplus \bar{\varphi} &\mapsto \omega + \bar{\varphi} \end{aligned}$$

is also injective. We thus get a triple of injective maps:

$$\Omega^1(C) \oplus \overline{\Omega^1(C)} \hookrightarrow H_{\text{DR}}(C) \hookrightarrow H^1(C, \mathbb{C})$$

Note that the vector spaces on both ends are $2g$, and hence all these spaces are isomorphic! The first isomorphism is a special case of the *Hodge Theorem*, and the second isomorphism is a special case of *DeRham’s Theorem*.

Definition 6.3.25: Period Vector

Let C be a compact Riemann surface of genus g , $\omega_1, \dots, \omega_g$ a basis for $\Omega^1(C)$, and $\gamma_1, \dots, \gamma_{2g}$ a $\mathbb{Z}$ -basis for $H_1(C, \mathbb{Z})$. Then the $2g$ g-dimensional vector:

$$\pi_i = \begin{pmatrix} \int_{\gamma_i} \omega_1 \\ \vdots \\ \int_{\gamma_i} \omega_g \end{pmatrix}$$

are called a *system of period vectors of C* . The matrix:

$$\Omega = (\pi_1, \dots, \pi_{2g})_{g \times 2g}$$

is called the *period matrix* of C

Proposition 6.3.26: Period Matrix Nonsingular

The $2g$ period vectors are $\mathbb{R}$ -linearly independent.

Proof:

For the sake of contradiction, suppose $\pi_1, \dots, \pi_{2g}$ are $\mathbb{R}$ -linearly dependent so that there exists $a_1, \dots, a_{2g} \in \mathbb{R}$ not all zero st

$$a_1 \pi_1 + \dots + a_{2g} \pi_{2g} = 0$$

We make take the complex conjugate of this (note conjugating an integral brings the conjugation to the inside) and get the matrix:

$$\Omega^* \begin{pmatrix} \pi_1 & \cdots & \pi_{2g} \\ \bar{\pi}_1 & \cdots & \bar{\pi}_{2g} \end{pmatrix}_{2g \times 2g}$$

By our assumption, the columns of the square matrix Ω^* satisfy a nontrivial linear relation, so $\text{Rank}(\Omega^*) < 2g$, and the transpose of Ω^* is singular as well. Hence there is a nonzero vector $(\lambda^1, \dots, \lambda^g, \eta^1, \dots, \eta^g) \in \mathbb{C}^{2g}$ with

$$\int_{\gamma_i} \left(\sum_{j=1}^g \lambda^j \omega_j + \sum_{j=1}^g \eta^j \bar{\omega}_j \right) = 0 \quad (i = 1, \dots, 2g)$$

Let $\omega = \sum_{j=1}^g \lambda^j \omega_j$ and $\varphi = \sum_{j=1}^g \bar{\eta}^j \omega_j$, both in $\Omega^1(C)$, so that $\bar{\varphi} = \sum_{j=1}^g \eta^j \bar{\omega}_j$. Then:

$$\int_{\gamma_i} (\omega + \bar{\varphi}) = 0 \quad (i = 1, \dots, 2g)$$

The form $\omega + \bar{\varphi}$ is smooth and closed, since $d\omega = 0$ and $d\bar{\varphi} = 0$ for holomorphic 1-forms (lemma 6.2.10). By proposition 6.3.20 it is exact, $\omega + \bar{\varphi} = df$, and proposition 6.3.21 gives $\omega = \varphi = 0$. As $\omega_1, \dots, \omega_g$ are linearly independent, all λ^j and η^j vanish, a contradiction, as we sought to show.

Riemann Roch Theorem

Theorem 6.3.27: Riemann Inequality

Let C be a compact Riemann surface of genus g , $D \in \text{Div}(C)$, and $\deg D = d$. Then;

$$l(D) \geq d - g + 1$$

The proof needs the cohomology of sheaves on a compact Riemann surface, which this book does not develop. A complete proof is in Forster [11, Theorem 16.9]. There $H^0(C, \mathcal{O}_D)$ is the space $\mathcal{L}(D)$, and the genus is $\dim H^1(C, \mathcal{O})$, which equals $\dim \Omega^1(C)$ by [11, p. 17.10], hence equals g by theorem 6.3.23.

Proof Key ideas:

Let $\mathcal{O}_D$ be the sheaf of meromorphic functions f with $(f) \geq -D$ on the open set where f is defined. Forster's form of the Riemann-Roch theorem states that $H^0(C, \mathcal{O}_D) = \mathcal{L}(D)$ and $H^1(C, \mathcal{O}_D)$ are finite dimensional, with

$$l(D) - \dim H^1(C, \mathcal{O}_D) = 1 - g + d$$

For $D = 0$ this holds because the holomorphic functions on C are the constants and $\dim H^1(C, \mathcal{O}) = g$. Adding one point P to D , the exact sequence $0 \rightarrow \mathcal{O}_D \rightarrow \mathcal{O}_{D+P} \rightarrow \mathbb{C}_P \rightarrow 0$, whose last map reads off one Laurent coefficient at P , shows that the alternating sum of the two dimensions changes by 1, as does the degree, so the formula passes from D to $D + P$ and back. As $\dim H^1(C, \mathcal{O}_D) \geq 0$, it gives $l(D) \geq d - g + 1$, which completes the sketch.

Corollary 6.3.28: Comparing Degree and Genus

Let C be a compact Riemann surface of genus g and $D \in \text{Div}(C)$ with $\deg D = d$. If $d \geq g$, then

$$l(D) \neq 0$$

Hence if $l(D) = 0$, then:

$$d \leq g - 1$$

Proof:

By theorem 6.3.27, $l(D) \geq d - g + 1$. If $d \geq g$, the right-hand side is at least 1, so $l(D) \neq 0$. Consequently $l(D) = 0$ forces $d < g$, that is $d \leq g - 1$ as d and g are integers, as claimed.

Proposition 6.3.29: Meromorphic Differentials and Riemann Inequality

Let C be a compact Riemann surface of genus g and $D \in \text{Div}(C)$ with $\deg D = d$. Then

$$i(D) \geq g - d - 1$$

Proof:

By theorem 6.3.8 there is a nonzero $\omega \in K^1(C)$, and its divisor $K = (\omega)$ has degree $2g - 2$. Since $(\omega) = D + (K - D)$, proposition 6.3.16 gives $\Omega^1(D) \cong \mathcal{L}(K - D)$, so $i(D) = l(K - D)$. The divisor $K - D$ has degree $2g - 2 - d$, and theorem 6.3.27 gives

$$i(D) = l(K - D) \geq (2g - 2 - d) - g + 1 = g - d - 1$$

as claimed.

Corollary 6.3.30: Comparing Degree and Genus – Meromorphic Differential

Let C be a compact Riemann surface of genus g and $D \in \text{Div}(C)$ with $\deg D = d$. If $d \leq g - 2$, then $i(D) \neq 0$, hence if $i(D) = 0$ then

$$d \geq g - 1$$

Proof:

By proposition 6.3.29, $i(D) \geq g - d - 1$. If $d \leq g - 2$, the right-hand side is at least 1, so $i(D) \neq 0$. Consequently $i(D) = 0$ forces $d > g - 2$, that is $d \geq g - 1$ as d and g are integers, as claimed.

Finally this leads us to:

Theorem 6.3.31: Riemann-Roch Theorem

Let C be a compact Riemann surface of genus g and let

$$D = \sum_{i=1}^k n_i p_i \in \text{Div}(C)$$

have degree d . Then:

$$l(D) = d - g + i(D) + 1$$

Proof:

Let us first assume $D \geq 0$ and prove

$$l(D) \leq d - g + i(D) + 1$$

And then take cases and judiciously use the Riemann inequality and its corollary to get the results. Suppose $\eta = (\eta_1, \dots, \eta_k)$ where:

$$\eta_i = \frac{a_{in_i}}{z_i^{n_i}} + \dots + \frac{a_{i1}}{z_i}$$

is a Laurent principal part near p_i . The linear space consisting of all these η is isomorphic to $\mathbb{C}^d$. Then thinking of η as a vector in $\mathbb{C}^d$, consider:

$$\begin{aligned} \text{Res} : \mathbb{C}^d \times \Omega^1(C) &\rightarrow \mathbb{C} \\ (\eta, \omega) &\mapsto \sum_{i=1}^k \text{Res}_{p_i}(\eta_i \omega) \end{aligned}$$

This is certainly a bilinear mapping. Let us show that:

$$\text{Res}(\eta, \omega) = 0, \forall \eta \in \mathbb{C}^d \quad \text{if and only if} \quad \omega \in \Omega^1(D)$$

If $\omega \in \Omega^1(D)$, then $\eta_i \omega$ does not have p_i as a pole for all $i = 1, \dots, k$, hence:

$$\text{Res}_{p_i}(\eta_i \omega) = 0$$

and so the sum of residues is zero, giving the $\Leftarrow$ direction. For the $\Rightarrow$ direction, assume otherwise so that $\omega \in \Omega^1(C)$ such that $\text{Res}(\eta, \omega) = 0$ for all $\eta \in \mathbb{C}^d$ but $\omega \notin \Omega^1(D)$. Then near p_i , let

$$\omega = (b_{i0}z_i^{m_i} + b_{i1}z_i^{m_i+1} + \dots) dz_i$$

where $m_i \geq 0$, $b_{i0} \neq 0$, and by assumption $m_i < n_i$ for some i . Then there is a Laurent Principal part near p_i

$$\xi_i = (\xi_{i1}, \dots, \xi_{ik})$$

such that

$$\xi_{ii} = \frac{1}{z_i^{m_i+1}} \quad \xi_{ij} = 0 \quad (i \neq j)$$

As $m_i + 1 \leq n_i$, $\mathbb{C}^d$, by treating these as $\mathbb{C}^d$ we have $\xi_i \in \mathbb{C}^d$. However:

$$0 = \text{Res}(\xi_i, \omega) = \text{Res}_{p_i} \frac{b_{i0}}{z_i} = b_{i0} \neq 0$$

a contradiction. Thus, $m_i \geq n_i$, and so $\omega \in \Omega^1(D)$.

Let $B : V \times W \rightarrow \mathbb{C}$ be a bilinear map between two finite dimensional vector spaces, in particular $\dim V = n$ and $\dim W = m$. Then B is called non-degenerate if given $w \in W$, $B(v, w) = 0$ for all $v \in V$ implies $w = 0$. If B is non-degenerate and $L \subseteq V$ is a subspace with $B(L \times W) = 0$, then:

$$\dim L \leq n - m$$

For the proof, choose a basis $w_1, \dots, w_m$ of W and let $\Phi : V \rightarrow \mathbb{C}^m$ be the linear map $v \mapsto (B(v, w_1), \dots, B(v, w_m))$. If Φ were not onto, its image would be a proper subspace of $\mathbb{C}^m$, hence contained in the kernel of some nonzero linear functional $(y_1, \dots, y_m) \mapsto \sum_j c_j y_j$. Then $B(v, \sum_j c_j w_j) = \sum_j c_j B(v, w_j) = 0$ for every $v \in V$, and non-degeneracy would give $\sum_j c_j w_j = 0$, contradicting the linear independence of the w_j . So Φ is onto, its kernel has dimension $n - m$ by the rank-nullity theorem, and $L \subseteq \ker \Phi$, which gives $\dim L \leq n - m$. Then by the property we just found earlier about the residue map, we have a non-degenerate bilinear map:

$$\text{Res}_1 : \mathbb{C}^d \times \frac{\Omega^1(C)}{\Omega^1(D)} \rightarrow \mathbb{C}$$

The map sending $f \in \mathcal{L}(D)$ to its principal parts at the p_i is linear, and its kernel consists of the functions in $\mathcal{L}(D)$ without poles, which are the constants (corollary 6.2.5). So we can interpret $\mathcal{L}(D)/\mathbb{C}$ as a subspace of $\mathbb{C}^d$ through principal parts, and this subspace pairs to zero with everything:

$$\text{Res}_1 \left(\frac{\mathcal{L}(D)}{\mathbb{C}} \times \frac{\Omega^1(C)}{\Omega^1(D)} \right) = 0$$

Let $f \in \mathcal{L}(D)$ have principal parts $\eta = (\eta_1, \dots, \eta_k)$ and let $\omega \in \Omega^1(C)$. Near p_i the function $f - \eta_i$ is holomorphic, so $(f - \eta_i)\omega$ is holomorphic at p_i and $\text{Res}_{p_i}(\eta_i \omega) = \text{Res}_{p_i}(f\omega)$. Since

$D \geq 0$ and $(f) \geq -D$, the poles of f lie among the p_i , so the meromorphic differential $f\omega$ is holomorphic off $\{p_1, \dots, p_k\}$. By theorem 6.2.11, applied with $K = C$, whose oriented boundary is empty, the sum of the residues of $f\omega$ is 0. Hence $\text{Res}(\eta, \omega) = \sum_i \text{Res}_{p_i}(f\omega) = 0$. Then:

$$\dim \left(\frac{\mathcal{L}(D)}{\mathbb{C}} \right) \leq \dim \mathbb{C}^d - \dim \left(\frac{\Omega^1(C)}{\Omega^1(D)} \right)$$

That is:

$$l(D) - 1 \leq d - \dim \Omega^1(C) + i(D)$$

By theorem 6.3.23, $\dim \Omega^1(C) = g$, and so we get:

$$l(D) \leq d - g + i(D) + 1$$

It remains to consider cases on $l(D)$ and $i(D)$:

1. Consider $l(D) \neq 0, i(D) = 0$. As $l(D) \neq 0$, there exists an $f \in K(C)$ such that $(f) + D \geq 0$. Let $E = (f) + D$. Then $(f) = E - D$, so $E \sim D$. Then

$$l(D) = l(E) \quad i(D) = i(E) \quad \deg D = \deg E$$

Thus, we may assume $D \geq 0$. Thus we get our inequality, and by the Riemann inequality:

$$d - g + i(D) + 1 = d - g + 1 \stackrel{\text{R.}}{\leq} l(D) \leq d - g + 1$$

giving equality

2. Consider $l(D) \neq 0, i(D) \neq 0$. As $l(D) \neq 0$, we may do the same trick to assume $D \geq 0$. Since $i(D) \neq 0$, there exists a nonzero $\omega \in K^1(D)$, that is

$$(\omega) \geq D$$

and ω is holomorphic, as $D \geq 0$. Let $E = (\omega) - D$ so that $E \geq 0$. Then by proposition 6.3.16 and the inequality proved above, applied to the effective divisor E , we get:

$$i(D) = l(E) \leq \deg E - g + 1 + i(E)$$

By theorem 6.3.8, $\deg(\omega) = 2g - 2$ so

$$\deg E = \deg(\omega) - \deg(D) = 2g - 2 - d$$

As $i(E) = l(D)$, we get:

$$i(D) \leq 2g - 2 - d - g + 1 + l(D)$$

re-arranging we get:

$$l(D) \geq d - g + i(D) + 1$$

which, combined with the inequality for $D \geq 0$, gives equality

3. Consider $l(D) = 0$, $i(D) \neq 0$. Since $i(D) \neq 0$, there exists a nonzero $\omega \in K^1(D) \subseteq K^1(C)$, a meromorphic 1-form with $(\omega) = D + E$ where $E \geq 0$. Here ω need not be holomorphic, as D need not be effective. Then by proposition 6.3.16

$$l(E) = i(D) \neq 0 \quad i(E) = l(D) = 0$$

Thus, by case (1) applied to E we get:

$$l(E) = \deg E - g + i(E) + 1$$

Note that, by theorem 6.3.8,

$$\deg E = \deg(\omega) - \deg D = 2g - 2 - d$$

Then substituting and rewriting $l(E)$, $i(E)$ as $i(D)$ and $l(D)$, we get the equality.

4. Consider $l(D) = 0$, $i(D) = 0$. Then by corollary 6.3.28 and corollary 6.3.30

$$d = g - 1$$

which is the desired equation.

6.4 Application of Riemann-Roch Theorem

It shall cover genus 0, 1, 2, 3, 4.

Genus 0

Proposition 6.4.1: Genus 0 then Rational

Let C be a compact Riemann surface of genus 0. Then

$$C \cong \mathbb{P}^1$$

Proof:

Let $p \in C$, $D = (p) \in \text{Div}(C)$. Then:

$$0 \leq i(D) \leq i(0) = \dim \Omega^1(C) = 0$$

Thus $i(D) = 0$. By the Riemann-Roch theorem:

$$l(D) = d - g + i(D) + 1 = 1 - 0 + 0 + 1 = 2$$

Thus, $\mathcal{L}(p)$ contains two linearly independent function, one of which can be chosen to be $f_1 = 1$. The other function must be a meromorphic function (as C is a compact Riemann surface) and must have p as a pole of order 1; call this function f_2 . Then by definition $f_2^{-1}(\infty) = p$. Hence it follows that this is a holomorphic mapping:

$$f_2 : C \rightarrow \mathbb{P}^1$$

where $\deg f_2 = \#\{f_2^{-1}(\infty)\} = 1$ implying that f_2 is in fact biholomorphic (recall lemma 6.3.3). Thus:

$$C \cong \mathbb{P}^1$$

as we sought to show.

Genus 1

Lemma 6.4.2: Involution of a Degree Two Map

Let $x: C \rightarrow \mathbb{P}^1$ be a holomorphic map of degree 2 from a compact connected Riemann surface C . For $t \in C$ let $j(t)$ be the other point of the fibre $x^{-1}(x(t))$ if t is unramified for x , and $j(t) = t$ if t is ramified. Then j is a holomorphic automorphism of C with $j \circ j = \text{id}$ and $x \circ j = x$, whose fixed points are the ramified points. Near a ramified point t_0 , in a local coordinate z centred at t_0 in which $w \circ x$ reads $z \mapsto z^2$ for a local coordinate w centred at $x(t_0)$ (lemma 6.2.14), j reads $z \mapsto -z$.

Proof:

By lemma 6.3.7, every fibre of x consists of two points counted with ramification index. So an unramified t shares its fibre with exactly one other point, also unramified, and a ramified t is alone in its fibre. Hence j is well defined, $x \circ j = x$, $j \circ j = \text{id}$, and the fixed points of j are the ramified points. Let t_0 be unramified and $t_1 = j(t_0)$. By lemma 6.2.14 with $p = 1$, there are disjoint neighbourhoods $V_0 \ni t_0$ and $V_1 \ni t_1$ mapped by x biholomorphically onto the same neighbourhood U of $x(t_0)$. For $t \in V_0$, the fibre of $x(t)$ contains t and the point $(x|_{V_1})^{-1}(x(t)) \in V_1$, and these are its two points, so $j = (x|_{V_1})^{-1} \circ x$ on V_0 , which is holomorphic. Near a ramified t_0 , in the coordinate z , a value $w \neq 0$ near 0 has the two distinct preimages $\pm\sqrt{w}$ in the coordinate disc, and these are all its preimages, so $j(z) = -z$, also at $z = 0$. So j is holomorphic, and it is its own inverse, completing the proof.

Proposition 6.4.3: Genus 1 Weierstrass Equation Representation

Let C be a compact Riemann surface of genus 1. Then C can be represented by a smooth algebraic curve of degree 3 in $\mathbb{P}^2$

Proof:

We shall construct a holomorphic bijection $f: C \rightarrow C'$ onto a smooth algebraic curve $C' \subseteq \mathbb{P}^2$ of degree 3.

Step 1: a function with a double pole. By theorem 6.3.23, $\dim \Omega^1(C) = g = 1$, so there is a nonzero $\omega \in \Omega^1(C)$, and by theorem 6.3.8 we have:

$$\deg(\omega) = 2g - 2 = 2 - 2 = 0$$

As ω is holomorphic, $(\omega) \geq 0$, so $(\omega) = 0$ and ω has no zeros. For an effective divisor $D \neq 0$, a nonzero $\alpha \in K^1(D)$ would have $\deg(\alpha) \geq \deg D > 0$, against theorem 6.3.8, so $i(D) = 0$. Fix

$p \in C$. By the Riemann-Roch theorem we have that:

$$l(2p) = d - g + i(2p) + 1 = 2 - 1 + 0 + 1 = 2 \quad l(p) = 1 - 1 + 0 + 1 = 1$$

So $\mathcal{L}(p)$ contains only the constants, and $\mathcal{L}(2p)$ contains a meromorphic function x whose only pole is p , of order exactly 2. Write $\omega = g(t)dt$ in a local coordinate t centred at p , with $g(0) \neq 0$. A holomorphic primitive G of g near 0 with $G(0) = 0$ (corollary 3.2.3) has $G'(0) \neq 0$, so $s = G(t)$ is a local coordinate centred at p (theorem 2.2.27) with $\omega = ds$. In it,

$$x = \frac{c_2}{s^2} + \frac{c_1}{s} + h_1(s)$$

with $c_2 \neq 0$ and h_1 holomorphic. The meromorphic form $x\omega = x ds$ has its only pole at p , with residue c_1 , so theorem 6.2.11, applied with $K = C$, whose oriented boundary is empty, gives $c_1 = 0$. With $z = s/\sqrt{c_2}$ for a square root of c_2 , we get:

$$x = \frac{1}{z^2} + h(z)$$

Note that this x has many similar properties to the Weierstrass function, as we shall see later.

Step 2: the ramification of x . Regard x as a holomorphic map $C \rightarrow \mathbb{P}^1$. Its only pole is p , of order 2, so $\deg x = 2$ (lemma 6.3.7). Every ramification index is therefore at most 2, and the ramification divisor R of x has all its coefficients equal to 0 or 1. By the Riemann-Hurwitz formula (theorem 6.3.9) with $g = 1$, $n = 2$ and $g' = 0$,

$$\deg R = 2(1 + 2 - 0 - 1) = 4$$

As p is ramified, there are 3 other ramification points, and

$$R = p_1 + p_2 + p_3 + p$$

with p_1, p_2, p_3, p mutually distinct. Let $x(p_i) = a_i$. The values a_1, a_2, a_3 are distinct, since each fibre of x consists of 2 points counted with multiplicity. Let q be one of p_1, p_2, p_3, p , and let w be the coordinate $w = x - a_i$ around a_i if $q = p_i$, and $w = 1/x$ around ∞ if $q = p$. The function $w \circ x$ has a zero of order 2 at q , so by lemma 6.2.14 there is a local coordinate t centred at q with $w \circ x = t^2$.

Step 3: the divisor of y . Let

$$y = \frac{dx}{\omega}$$

a meromorphic function, well defined as ω has no zeros. As $(\omega) = 0$, we have $(y) = (dx)$. To be succinct with tracking roots and poles, for a meromorphic function or differential form α , let $(\alpha)_0$ and $(\alpha)_\infty$ represent the zeros and poles (including multiplicity) respectively. Away from p , in a local coordinate z , $dx = x'(z)dz$ vanishes exactly where $x' = 0$, that is at the ramified points p_1, p_2, p_3 . Near p_i , in the coordinate t of Step 2, $x - a_i = t^2$, so $dx = 2t dt$ and p_i is a simple zero of dx . Near p , $dx = (-2z^{-3} + h'(z))dz$ has a pole of order 3. So we found:

$$(y)_0 = p_1 + p_2 + p_3 \quad (y)_\infty = 3p \quad (y) = p_1 + p_2 + p_3 - 3p$$

Step 4: the equation. Let $G(x) = (x - a_1)(x - a_2)(x - a_3)$, a meromorphic function on C . The

fibre of x over a_i is p_i with multiplicity 2, so $x - a_i$ has a double zero at p_i , no other zero, and a double pole at p . Hence

$$(G)_0 = 2p_1 + 2p_2 + 2p_3 \quad (G)_\infty = 6p$$

so the divisor of G is the divisor of y^2 , and $(y^2/G) = 0$. The meromorphic function y^2/G has no poles, so it is a holomorphic function on the compact C , hence a constant a (corollary 6.2.5), and $a \neq 0$ as y^2/G has no zeros. Replacing ω by $\sqrt{a}\omega$, which replaces y by $y/\sqrt{a}$, we get:

$$y^2 = G(x)$$

Define

$$\begin{aligned} f : C &\rightarrow \mathbb{P}^2 \\ q &\mapsto [1 : x(q) : y(q)] \quad (p \mapsto [0 : 0 : 1]) \end{aligned}$$

Away from p the map f is holomorphic. Near p , $f = [1/y : x/y : 1]$, where $1/y$ and x/y are holomorphic, vanishing at p to orders 3 and 1, so f is holomorphic there with $f(p) = [0 : 0 : 1]$. The image lies in the curve of degree 3

$$C' = \{[1 : x : y] \in \mathbb{P}^2 \mid y^2 - G(x) = 0\} \cup \{[0 : 0 : 1]\}$$

the zero set of $F = \xi^0(\xi^2)^2 - (\xi^1 - a_1\xi^0)(\xi^1 - a_2\xi^0)(\xi^1 - a_3\xi^0)$, whose only point with $\xi^0 = 0$ is $[0 : 0 : 1]$.

Step 5: the involution j . Let j be the involution of the degree 2 map x (lemma 6.4.2). It is a holomorphic automorphism of C with $x \circ j = x$ and $j^2 = \text{id}$, fixing exactly p_1, p_2, p_3, p , and $j(t) = -t$ in the coordinate t of Step 2 near each of these points. By definition of j :

$$j^*(x) = x \quad j^*(dx) = d(x \circ j) = dx$$

As $\dim \Omega^1(C) = g = 1$, we get $\Omega^1(C) = \mathbb{C}\omega$, and j^* is a linear map of this line with $(j^*)^2 = 1$, so $j^*\omega = \omega$ or $j^*\omega = -\omega$. Suppose $j^*\omega = \omega$, and look near the ramification point p_1 , in the coordinate t of Step 2, where $j(t) = -t$. Writing $\omega = u(t)dt$ there, with u holomorphic and $u(0) \neq 0$ because ω has no zeros, we get $j^*\omega = u(-t)d(-t) = -u(-t)dt$, and $j^*\omega = \omega$ gives $u(t) = -u(-t)$. Letting $t \rightarrow 0$ gives $u(0) = -u(0)$, so $u(0) = 0$, which is a contradiction. Hence $j^*\omega = -\omega$, and

$$j^*y = \frac{j^*(dx)}{j^*(\omega)} = \frac{dx}{-\omega} = -y$$

Step 6: f is a bijection onto C' . Let $q_1, q_2 \in C$ with $f(q_1) = f(q_2)$. If either is p , then both images are $[0 : 0 : 1]$, which is the image of p only, so $q_1 = q_2 = p$. Otherwise

$$\begin{aligned} f(q_1) &= [1 : x(q_1) : y(q_1)] \\ f(q_2) &= [1 : x(q_2) : y(q_2)] \end{aligned}$$

and as these are equal $x(q_1) = x(q_2)$ and $y(q_1) = y(q_2)$. If $q_1 \neq q_2$, then $q_2 = j(q_1)$, so $y(q_2) = y(j(q_1)) = -y(q_1) = -y(q_2)$ and $y(q_2) = 0$. But $(y)_0 = p_1 + p_2 + p_3$, so q_2 is a ramified point and $q_1 = j(q_2) = q_2$, a contradiction. So f is injective. For surjectivity, let $[1 : \lambda : \mu] \in C'$, so $\mu^2 = G(\lambda)$. The fibre of x over λ is nonempty by lemma 6.3.7. Let q lie in

it. Then $y(q)^2 = G(\lambda) = \mu^2$, so $y(q) = \pm\mu$. If $y(q) = \mu$, then $f(q) = [1 : \lambda : \mu]$, and otherwise $f(j(q)) = [1 : \lambda : -y(q)] = [1 : \lambda : \mu]$. Together with $f(p) = [0 : 0 : 1]$, this gives $f(C) = C'$.

Step 7: C' is smooth. We have:

$$\begin{aligned}\frac{\partial F}{\partial \xi^0} &= (\xi^2)^2 + a_1(\xi^1 - a_2\xi^0)(\xi^1 - a_3\xi^0) \\ &\quad + a_2(\xi^1 - a_1\xi^0)(\xi^1 - a_3\xi^0) \\ &\quad + a_3(\xi^1 - a_1\xi^0)(\xi^1 - a_2\xi^0)\end{aligned}$$

$$\begin{aligned}\frac{\partial F}{\partial \xi^1} &= -(\xi^1 - a_2\xi^0)(\xi^1 - a_3\xi^0) \\ &\quad - (\xi^1 - a_1\xi^0)(\xi^1 - a_3\xi^0) \\ &\quad - (\xi^1 - a_1\xi^0)(\xi^1 - a_2\xi^0)\end{aligned}$$

$$\frac{\partial F}{\partial \xi^2} = 2\xi^0\xi^2$$

Suppose that $\partial F/\partial \xi^i$ ($i = 0, 1, 2$) all vanish at a point $[\xi^0 : \xi^1 : \xi^2]$. By the Euler formula, $F = 0$ there. From $2\xi^0\xi^2 = 0$ we get $\xi^0(\xi^2)^2 = 0$, and substituting this into $F = 0$ gives

$$(\xi^1 - a_1\xi^0)(\xi^1 - a_2\xi^0)(\xi^1 - a_3\xi^0) = 0$$

If exactly one of the three factors vanished, $\partial F/\partial \xi^1$ would be minus the product of the other two, which is nonzero. So at least two factors vanish, and then $\xi^0 = \xi^1 = 0$ because a_1, a_2, a_3 are mutually distinct. Now $\partial F/\partial \xi^0 = (\xi^2)^2 = 0$ gives $\xi^2 = 0$, which is impossible for a point of $\mathbb{P}^2$. Thus C' is smooth, so by lemma 6.3.1 it is a compact complex manifold of dimension one, and f is holomorphic into it. It is connected, being the continuous image $f(C)$ of the connected surface C . So $f: C \rightarrow C'$ is a holomorphic bijection between complex manifolds of dimension one, hence biholomorphic by theorem 6.2.7, and C is biholomorphic to a smooth degree 3 curve, as we sought to show.

Genus ≥ 2 : Canonical Map

In higher dimensions, there are generally many kinds of mapping from a compact Riemann surface into an n -dimensional projective space. However all compact Riemann surface of genus $g \geq 2$, they exists as special holomorphic mapping from this compact Riemann surface into $\mathbb{P}^{g-1}$ that is called he *canonical map*.

Definition 6.4.4: Canonical Map

Let C be a compact Riemann surface of genus $g \geq 2$ an $\{\omega_1, \dots, \omega_g\}$ a basis for $\Omega^1(C)$. Then:

$$\begin{aligned}\varphi_K : C &\rightarrow \mathbb{P}^{g-1} \\ p &\mapsto [\omega_1(p) : \dots : \omega_g(p)]\end{aligned}$$

Note that this is well-defined. It is certainly not coordinate dependent, and if, for the sake of contradiction, there is a p where $\omega_i(p) = 0$ for all p (which implies p maps to $[0 : \cdots : 0]$ which does not exist), then notice that $\Omega^1(p) = \Omega^1(C)$ so

$$i(p) = \dim \Omega^1(C) = g$$

and by the Riemann Roch-theorem

$$l(p) = 1 - g + g + 1 = 2$$

and so there exists nonconstant meromorphic function $f \in \mathcal{L}(p)$ where $f^{-1}(\infty) = p$. Thus

$$\deg f = 1$$

giving us that $f : \mathbb{P}^1 \rightarrow \mathbb{P}^1$ is an isomorphism, but that is not possible as $\mathbb{P}^1$ has genus 0.

Proposition 6.4.5: Non-degeneracy of Canonical Map

Let φ_K be the canonical map. Then φ_K is non-degenerate.

Proof:

Assume φ_K is degenerate so that there exists $\lambda_\alpha \in \mathbb{C}$ not all zero st for all $p \in C$

$$\sum_{\alpha}^g \lambda_{\alpha} \omega_{\alpha}(p) = 0$$

Then:

$$\sum_{\alpha}^g \lambda_{\alpha} \omega_{\alpha}$$

contradicting that $\{\omega_1, \dots, \omega_g\}$ is a basis of $\Omega^1(C)$, as we sought to show.

Proposition 6.4.6: Degree 2 Mapping Condition

Let C be a Riemann surface of genus $g \geq 2$ and let φ_K be the canonical mapping. Then if

$$\varphi_K(p) = \varphi_K(q)$$

for some distinct $p, q \in C$, then there exists a holomorphic mapping of degree 2 from C into $\mathbb{P}^1$

Proof:

If $\varphi_K(p) = \varphi_K(q)$, then $\omega_{\alpha}(p) = \lambda \omega_{\alpha}(q)$ for all $\alpha = 1, \dots, g$ for some $0 \neq \lambda \in \mathbb{C}$. Let $D = p + q$. Let us find $i(D)$. We shall start by showing that:

$$\Omega^1(D) = \Omega^1(p)$$

Indeed, for any $\omega \in \Omega^1(C)$, if

$$\omega = \sum_{\alpha=1}^g \mu_{\alpha} \omega_{\alpha}$$

for $\mu_{\alpha} \in \mathbb{C}$, $\alpha = 1, \dots, g$, then:

$$\begin{aligned} \omega(p) = 0 &\Leftrightarrow \sum_{\alpha=1}^g \mu_{\alpha} \omega_{\alpha}(p) = 0 \\ &\Leftrightarrow \sum_{\alpha=1}^g \lambda_{\alpha} \mu_{\alpha} \omega_{\alpha}(q) = 0 \\ &\Leftrightarrow \omega(q) = 0 \end{aligned}$$

Thus, $\Omega^1(p) = \Omega^1(p+q) = \Omega^1(D)$. Hence, to calculate $i(D)$ it suffices to calculate $i(p)$.

To that end, consider :

$$\lambda_1 \omega_1(p) + \lambda_2 \omega_2(p) + \dots + \lambda_g \omega_g(p) = 0$$

for $\lambda_1, \dots, \lambda_g \in \mathbb{C}$. Note not all these could be zero or else we would have a point map to $[0 : \dots : 0]$. Hence, this linear equation *must* have $g-1$ linearly independent solution vectors $(\lambda_1, \dots, \lambda_g) \in \mathbb{C}^g$, giving rise to $g-1$ linearly independent elements in $\Omega^1(p)$. Thus:

$$i(D) = i(p) = g-1$$

Applying the Riemann-Roch there to $D = p+q$, we get:

$$l(D) = 2 - g + (g-1) + 1 = 2$$

Thus, there exist a nonconstant meromorphic function $f \in \mathcal{L}(p+q)$. Thus there is a holomorphic $f : C \rightarrow \mathbb{P}^1$ where $f^{-1}(\infty) = p+q$. Note that $f^{-1}(\infty)$ cannot be a single point p or q otherwise we get $C \cong \mathbb{P}^1$, contradicting the fact that the genus g of C is ≥ 2 . Thus, it must be that:

$$\deg f = 2$$

as we sought to show.

This gives a rough division of Riemann surface of genus $g > 1$:

Definition 6.4.7: Hyperelliptic and Nonhyperelliptic

Let C be a compact Riemann surface of genus $g \geq 2$. Then it is *hyperelliptic* if there exists a holomorphic mapping of degree 2 $f : C \rightarrow \mathbb{P}^1$. Otherwise, C is said to be *nonhyperelliptic*.

We can deal with the nonhyperelliptic example holistically and then see the hyperelliptic examples by examining different genera:

Definition 6.4.8: Canonical Curve

Let C be a nonhyperelliptic compact Riemann surface of genus $g \geq 2$ with canonical map $\varphi_K : C \rightarrow \mathbb{P}^{g-1}$. Then

$$\varphi_K(C) \subseteq \mathbb{P}^{g-1}$$

is called the *canonical curve*.

Note that we put $g \geq 2$, but we shall show shortly that all $g = 2$ compact Riemann surface are hyperelliptic, but this shall not necessarily be the case for $g \geq 3$.

Proposition 6.4.9: Canonical Curve Smooth

Let C be a nonhyperelliptic compact Riemann surface. Then the canonical map φ_K is injective and $\varphi_K(C)$ is smooth.

Proof:

Let $g \geq 2$ be the genus of C (definition 6.4.7). If $\varphi_K(p) = \varphi_K(q)$ for distinct p, q , then proposition 6.4.6 gives a holomorphic mapping of degree 2 from C into $\mathbb{P}^1$, which is impossible as C is nonhyperelliptic. So φ_K is injective, and it remains to show that $\varphi_K(C)$ is smooth, that is, near each of its points it is the graph of holomorphic functions of one affine coordinate of $\mathbb{P}^{g-1}$.

Fix $p \in C$ and a local coordinate t centred at p , and write $\omega_\alpha = h_\alpha(t)dt$ near p , so that $\varphi_K = [h_1 : \dots : h_g]$ there. As shown after definition 6.4.4, the vector $h(0) = (h_1(0), \dots, h_g(0))$ is nonzero. So $\Omega^1(p)$, the kernel of the nonzero linear form $\sum_\alpha \mu_\alpha \omega_\alpha \mapsto \sum_\alpha \mu_\alpha h_\alpha(0)$ on $\Omega^1(C)$, has dimension $i(p) = g - 1$. By theorem 6.3.31, $l(p) = 1 - g + (g - 1) + 1 = 1$, so $\mathcal{L}(p)$ consists of the constants.

Step 1: $h(0)$ and $h'(0) = (h'_1(0), \dots, h'_g(0))$ are linearly independent. Suppose instead that $h'(0) = ch(0)$ for some $c \in \mathbb{C}$. If $\omega = \sum_\alpha \mu_\alpha \omega_\alpha$ vanishes at p , then $\sum_\alpha \mu_\alpha h'_\alpha(0) = c \sum_\alpha \mu_\alpha h_\alpha(0) = 0$, so ω vanishes to order at least 2 at p . Hence $\Omega^1(2p) = \Omega^1(p)$, $i(2p) = g - 1$, and theorem 6.3.31 gives $l(2p) = 2 - g + (g - 1) + 1 = 2$. A nonconstant $f \in \mathcal{L}(2p)$ has p as its only pole, of order at most 2, and not at most 1, since $\mathcal{L}(p)$ consists of constants. So $f : C \rightarrow \mathbb{P}^1$ is a holomorphic mapping with $f^{-1}(\infty) = \{p\}$ and $\nu_f(p) = 2$, hence of degree 2, and C would be hyperelliptic. This contradiction proves Step 1.

Step 2: the local graph. Choose α with $h_\alpha(0) \neq 0$. In the affine chart of $\mathbb{P}^{g-1}$ where the α -th homogeneous coordinate equals 1, the map φ_K is given near p by the functions $F_\beta = h_\beta/h_\alpha$ for $\beta \neq \alpha$, with

$$F'_\beta(0) = \frac{h'_\beta(0)h_\alpha(0) - h_\beta(0)h'_\alpha(0)}{h_\alpha(0)^2}$$

If all of these vanished, then $h'(0) = (h'_\alpha(0)/h_\alpha(0))h(0)$, against Step 1. So $F'_\beta(0) \neq 0$ for some β , and by theorem 2.2.27 the function F_β maps a neighbourhood N of p biholomorphically onto a disc around $F_\beta(0)$. On $\varphi_K(N)$ every other affine coordinate is the holomorphic function $F_\gamma \circ (F_\beta|_N)^{-1}$ of the coordinate w_β . The map φ_K is a continuous bijection from the compact space C onto the Hausdorff space $\varphi_K(C)$, hence a homeomorphism by [8, making continuous function homeomorphisms], so $\varphi_K(N)$ is a neighbourhood of $\varphi_K(p)$ in $\varphi_K(C)$. Near $\varphi_K(p)$ the

curve $\varphi_K(C)$ is therefore the graph of holomorphic functions of w_β . As p was arbitrary, $\varphi_K(C)$ is smooth, completing the proof.

Hyperelliptic Compact Riemann Surfaces

Proposition 6.4.10: Algebraic Characterization of Hyperelliptic Riemann Surface

Let C be a hyperelliptic compact Riemann surface. Then it can be represented as the normalization of a plane algebraic curve of degree $2g + 2$ where g is the genus of C . By a suitable linear projection, this plane curve will be a 2-sheeted covering of a straight line (i.e. $\mathbb{C}$).

Proof:

Let C be a hyperelliptic compact Riemann surface of genus g . We shall construct a holomorphic mapping $f : C \rightarrow \mathbb{P}^2$ such that $f(C)$ can be given by a plane algebraic curve of degree $2g + 2$.

Step 1: the branch points. As C is hyperelliptic, there exists a degree 2 holomorphic mapping $x : C \rightarrow \mathbb{P}^1$. Composing x with a Möbius transformation (lemma 4.3.10), we may assume that ∞ is not a branching point, so $x^{-1}(\infty) = p + q$ with $p \neq q$. Let R be the ramification divisor of x . By the Riemann-Hurwitz formula (theorem 6.3.9) with $n = 2$ and $g' = 0$:

$$\deg R = 2(g + 2 - 1) = 2g + 2$$

As x has degree 2 (lemma 6.3.7), every ramification index is at most 2, so the coefficients of the points in R are equal to 1, hence:

$$R = p_1 + p_2 + \cdots + p_{2g+2}$$

where each p_i is distinct on C . Let $x(p_i) = a_i$. The fibre over a_i is the single point p_i , counted twice, so the a_i are mutually distinct. By lemma 6.2.14, there is a local coordinate z_i centred at p_i with $x - a_i = z_i^2$.

Step 2: the involution. Let j be the involution of the degree 2 map x (lemma 6.4.2). Then $x \circ j = x$, $j^2 = \text{id}$, $j(z_i) = -z_i$ near p_i in the coordinate z_i , and $j(p) = q$, the fibre over ∞ being $\{p, q\}$. So $j \in \text{Aut}(C)$, and $j^*\omega = \omega \circ j$ defines a linear involution of $\Omega^1(C)$, so $\Omega^1(C)$ is the direct sum of the eigenspaces for 1 and -1 , an element ω splitting as $\frac{1}{2}(\omega + j^*\omega) + \frac{1}{2}(\omega - j^*\omega)$. Suppose $\omega \neq 0$ satisfies $j^*\omega = \omega$. Near p_i write $\omega = u(z_i)dz_i$. Then $j^*\omega = u(-z_i)d(-z_i)$, so $u(-z_i) = -u(z_i)$, and u is odd, hence it vanishes at 0. So ω , which is holomorphic, has a zero at each of the $2g + 2$ points p_i , and $\deg(\omega) \geq 2g + 2 > 2g - 2$, against theorem 6.3.8. Thus it must be that :

$$j^* : \Omega^1(C) \rightarrow \Omega^1(C) \quad \omega \mapsto -\omega$$

Step 3: the function y . Let

$$D = (g + 1)p + (g + 1)q \in \text{Div}(C)$$

Recall that for any nonzero $\omega \in K^1(C)$, theorem 6.3.8 gives $\deg(\omega) = 2g - 2 < \deg D$, and so it cannot be that $(\omega) - D \geq 0$, and so $i(D) = 0$. Then by Riemann-Roch (theorem 6.3.31):

$$l(D) = (2g + 2) - g + 1 = g + 3$$

Since $j(p) = q$ and $j(q) = p$, j preserves D , so $h \mapsto h \circ j$ is a linear involution of $\mathcal{L}(D)$, and we can split $\mathcal{L}(D)$ into eigenspaces:

$$\mathcal{L}(D) = \mathcal{L}(D)^+ \oplus \mathcal{L}(D)^-$$

where $\mathcal{L}(D)^+$ and $\mathcal{L}(D)^-$ are the eigenspace for eigenvalue $+1$ and -1 respectively. The functions $1, x, \dots, x^{g+1}$ lie in $\mathcal{L}(D)^+$, and they are linearly independent, x taking infinitely many values. They span it. Indeed, let $h \in \mathcal{L}(D)^+$, and define H on $\mathbb{P}^1$ by $H(\lambda) = h(t)$ for $t \in x^{-1}(\lambda)$, which is well defined as $h \circ j = h$. Near a value λ_0 over which x is unramified, $H = h \circ s$ for a holomorphic local inverse s of x (theorem 2.2.27), and near a_i , in the coordinate z_i , h is an even function, being invariant under $z_i \mapsto -z_i$, so $h = \sum_k c_k z_i^{2k} = \sum_k c_k (x - a_i)^k$ and $H(\lambda) = \sum_k c_k (\lambda - a_i)^k$. So H is holomorphic on $\mathbb{C}$. Near p , where x is unramified and $1/x$ is a local coordinate, h has a pole of order at most $g + 1$, so H has a pole of order at most $g + 1$ at ∞ . By proposition 4.3.1, H is a rational function without finite poles, hence a polynomial, of degree at most $g + 1$, and $h = H(x)$ is a combination of $1, x, \dots, x^{g+1}$. Thus:

$$\dim \mathcal{L}(D)^+ = g + 2$$

As $l(D) = g + 3$, it follows that

$$\dim \mathcal{L}(D)^- = 1$$

In particular, there exists a nonzero meromorphic function $y \in \mathcal{L}(D)$ such that

$$j^*(y) = -y$$

and y is not constant, a constant equal to its negative being 0.

Step 4: the equation. We compare divisors to show that, with $G(x) = (x - a_1)(x - a_2) \cdots (x - a_{2g+2})$,

$$y^2 = cG(x)$$

where $0 \neq c \in \mathbb{C}$. Since

$$j^*(y) = -y \quad j(p_i) = p_i \quad (i = 1, 2, \dots, 2g + 2)$$

and y is finite at p_i , we get $y(p_i) = -y(p_i)$, and so $y(p_i) = 0$. Thus

$$\deg(y)_0 \geq \deg \left(\sum_{i=1}^{2g+2} p_i \right) = 2g + 2$$

The degree of y as a map $C \rightarrow \mathbb{P}^1$ is both $\deg(y)_0$ and $\deg(y)_\infty$ (lemma 6.3.7), and $(y)_\infty \leq D$ has degree at most $2g + 2$, so all these degrees equal $2g + 2$. Thus

$$(y)_0 = \sum_{i=1}^{2g+2} p_i \quad (y)_\infty = D$$

For $G(x)$: the fibre of x over a_i is p_i counted twice, and x has simple poles at p and q , so $(x - a_i) = 2p_i - p - q$, and

$$(G(x)) = \sum_{i=1}^{2g+2} 2p_i - (2g+2)(p+q) = 2(y)$$

So $(y^2/G(x)) = 0$, and $y^2/G(x)$ is a meromorphic function without poles on the compact surface C , hence a constant c (corollary 6.2.5), with $c \neq 0$ as it has no zeros. By replacing y with $y/\sqrt{c}$ we get

$$y^2 = G(x)$$

Step 5: the plane curve. Define the map:

$$\begin{aligned} f : C &\rightarrow \mathbb{P}^2 \\ t &\mapsto [1 : x(t) : y(t)] \\ p, q &\mapsto [0 : 0 : 1] \end{aligned}$$

Near p , $f = [1/y : x/y : 1]$ with $1/y$ and x/y holomorphic, vanishing at p to orders $g+1$ and g , and likewise near q , so f is holomorphic. Its image lies in the curve

$$C' = \left\{ [\xi^0 : \xi^1 : \xi^2] \in \mathbb{P}^2 \mid (\xi^2)^2 (\xi^0)^{2g} = \prod_{i=1}^{2g+2} (\xi^1 - a_i \xi^0) \right\}$$

of degree $2g+2$, whose only point with $\xi^0 = 0$ is $[0 : 0 : 1]$. The polynomial $y^2 - G(x)$ is irreducible, G having simple roots and hence not being a square. The map f is onto C' : for $[1 : \lambda : \mu] \in C'$, take $t \in x^{-1}(\lambda)$. Then $y(t)^2 = G(\lambda) = \mu^2$, so $y(t) = \pm\mu$, and $f(t)$ or $f(j(t)) = [1 : \lambda : -y(t)]$ is $[1 : \lambda : \mu]$. It is injective off $\{p, q\}$: if $f(t_1) = f(t_2)$ with $t_1 \neq t_2$ there, then $x(t_1) = x(t_2)$, so $t_2 = j(t_1)$, and $y(t_1) = y(j(t_1)) = -y(t_1)$ gives $y(t_1) = 0$, so t_1 is some p_i and $t_2 = j(p_i) = p_i = t_1$, a contradiction. In the affine part $\mathbb{C}^2 = \{[1 : x : y] \in \mathbb{P}^2\}$, the partial derivatives $2y$ and $-G'(x)$ of $y^2 - G(x)$ vanish together on C' only where $y = 0$ and x is a multiple root of G , which does not happen, so C' is smooth there. Hence the singular points of C' lie in $\{[0 : 0 : 1]\}$, whose preimage $\{p, q\}$ is finite, and (C, f) is a normalization of C' (definition 6.3.2). The curve is singular at $[0 : 0 : 1]$, its equation in the chart $\xi^2 = 1$ having lowest-degree part $(\xi^0)^{2g}$, of degree $2g \geq 2$, unlike the smooth cubic of proposition 6.4.3. A linear projection $(x, y) \rightarrow x$ gives the 2-sheeted covering, giving everything asked for by the proposition, completing the proof.

Proposition 6.4.11: Canonical Map – Hyperelliptic Case

Let C be a hyperelliptic curve and let x, y be as in proposition 6.4.10. Then

$$\omega_1 = \frac{dx}{y}, \omega_2 = \frac{(xdx)}{y}, \dots, \omega_g = \frac{(x^{g-1}dx)}{y}$$

forms a basis for $\Omega^1(C)$.

Proof:

Keep the notation of proposition 6.4.10: the ramification points of x are $p_1, \dots, p_{2g+2}$ with $x(p_i) = a_i$, the fibre over ∞ is $x^{-1}(\infty) = p + q$, and

$$(y) = \sum_{i=1}^{2g+2} p_i - (g+1)(p+q)$$

The divisor of dx . The map x is ramified at p_i with index 2, so by lemma 6.2.14 there is a local coordinate z_i with $z_i(p_i) = 0$ and

$$x - a_i = z_i^2$$

Differentiating gives $dx = 2z_i dz_i$, so $\nu_{dx}(p_i) = 1$. At a point t with $x(t) = b \neq \infty$ and $\nu_x(t) = 1$, the function $x - b$ is itself a local coordinate at t , so dx generates there and $\nu_{dx}(t) = 0$. As $\deg x = 2$ and $x^{-1}(\infty) = p + q$, the map x is unramified at p and at q , so $s = 1/x$ is a local coordinate at either point and $dx = -s^{-2}ds$ gives $\nu_{dx}(p) = \nu_{dx}(q) = -2$. Collecting,

$$(dx) = \sum_{i=1}^{2g+2} p_i - 2(p+q)$$

of degree $(2g+2) - 4 = 2g - 2$, as theorem 6.3.8 requires.

The divisors of the ω_k . Subtracting the two divisors,

$$\left(\frac{dx}{y}\right) = (dx) - (y) = (g-1)(p+q)$$

Let $(x)_0$ be the zero divisor of x , effective of degree 2, so that $(x) = (x)_0 - (p+q)$. Then for $k = 1, \dots, g$,

$$(\omega_k) = \left(\frac{x^{k-1}dx}{y}\right) = (k-1)(x)_0 + (g-k)(p+q)$$

Both coefficients are nonnegative in that range, so each ω_k has effective divisor and is therefore a holomorphic differential, that is $\omega_k \in \Omega^1(C)$. The coefficient of p turns negative at $k = g+1$, which is where the list stops.

Independence. Suppose $c_1\omega_1 + \dots + c_g\omega_g = 0$ with $c_k \in \mathbb{C}$. Factoring out gives

$$(c_1 + c_2x + \dots + c_gx^{g-1}) \frac{dx}{y} = 0$$

and dx/y is a nonzero meromorphic differential, so the meromorphic function $c_1 + c_2x + \dots + c_gx^{g-1}$ vanishes identically. Now x is nonconstant, hence takes infinitely many values, so the one-variable polynomial $c_1 + c_2X + \dots + c_gX^{g-1}$ has infinitely many roots and every c_k is zero.

So $\omega_1, \dots, \omega_g$ are g linearly independent elements of $\Omega^1(C)$, and $\dim \Omega^1(C) = g$ by theorem 6.3.23. A linearly independent family whose size equals the dimension is a basis.

The canonical map therefore has a very simple expression:

$$\begin{aligned} \varphi_K : C &\rightarrow \mathbb{P}^{g-1} \\ t &\mapsto [1 : x(t) : \dots : x(t)^{g-1}] \end{aligned}$$

Hence, φ_K factors in the following way:

$$\begin{array}{ccc} & & \mathbb{P}^{g-1} \\ & \nearrow \varphi_k & \uparrow r \\ C & \xrightarrow{x} & \mathbb{P}^1 \end{array}$$

where

$$\begin{aligned} r : \mathbb{P}^1 &\rightarrow \mathbb{P}^{g-1} \\ x &\mapsto [1 : x : \dots : x^{g-1}] \end{aligned}$$

Note that the converse of proposition 6.4.10 is true: given any $2g + 2$ mutually distinct complex numbers $a_1, a_2, \dots, a_{2g+2}$, and letting:

$$C' = \{y^2 = \prod_{i=1}^{2g+2} (x - a_i) = 0\} \cup [0 : 0 : 1] \subseteq \mathbb{P}^2$$

then the normalization of C of C' is a hyperelliptic compact Riemann surface of genus g , giving an equivalence. Combining this with the commutative diagram given above, we see that C is uniquely determined by the equivalence of vectors in $\mathbb{C}^{2g+2}$ with (pairwise) distinct coordinates:

$$\{(a_1, \dots, a_{2g+2})\}$$

where $(a_1, \dots, a_{2g+2}) \sim (b_1, \dots, b_{2g+2})$ if each coordinate is related by the same linear automorphisms of $\mathbb{P}^1$, that is there exists $a, b, c, d \in \mathbb{C}$ ($ad - bc \neq 0$) such that

$$\frac{aa_i + b}{ca_i + d} = b_i \quad (i = 1, 2, \dots, 2g + 2)$$

(recall the classification of $\text{Aut}(\mathbb{P}^1)$). Naturally, if replaced by $\lambda a, \lambda b, \lambda c, \lambda d$ the result is unchanged, hence without loss of generality we may assume one of them is fixed.

Genus 2 always hyperelliptic

Proposition 6.4.12: Genus 2 is Hyperelliptic

Let C be a compact Riemann surface of genus 2. Then C is hyperelliptic.

Proof:

Let C be a compact Riemann surface of genus 2. Then by proposition 6.4.6 it suffices to show that φ_K is not injective. But if $\varphi_K : C \rightarrow \mathbb{P}^{g-1} = \mathbb{P}^1$ were injective, C would have genus 0, completing the proof.

Thus C is always given over $\mathbb{P}^1$ as:

$$y^2 = \prod_{i=1}^6 (x - a_i)$$

for pairwise distinct $a_1, \dots, a_6$ and forms

$$\omega_1 = \frac{dx}{y} \quad \omega_2 = \frac{x dx}{y}$$

where:

$$\varphi: C \rightarrow \mathbb{P}^1 \quad t \mapsto [\omega_1 : \omega_2] = [1 : x(t)]$$

Genus 3

A compact Riemann surface of genus 3 which is not hyperelliptic is identified, through its canonical map, with a smooth plane curve of degree $2g - 2 = 4$ (proposition 6.4.16 below). Conversely, a smooth plane curve of degree 4 has genus 3 by the genus formula (theorem 6.3.13).

let $H \subseteq \mathbb{P}^n$ be a hyperplane given by:

$$L(\xi) = \sum_{i=0}^n l_i \xi^i = 0$$

Let C be a smooth algebraic curve in $\mathbb{P}^n$. Let H not contain any irreducible branches of C . Then pick the elements of $\text{Div}(C)$ in the following way. Choose a generic linear function:

$$L'(\xi) = \sum_{i=0}^n l'_i \xi^i$$

where by generic we mean:

$$\{L' = 0\} \cap C \cap \{L = 0\} = \emptyset$$

Then we get:

$$\frac{L}{L'} \in K(\mathbb{P}^n) \quad \frac{L}{L'}|_C \in K(C)$$

Then the set of all the zeros of $(L/L')|_C$ is given a name

We give a name to this set:

Definition 6.4.13: Hyperplane Section of Riemann Surface

Let C, H be as above. Then:

$$H \cdot C = \left(\frac{L}{L'} \Big|_C \right)_0$$

is the *hyperplane section* of C .

Intuitively:

$$H \cdot C = C \cap \{L(\xi) = 0\}$$

Certainly this definition is independent of choice of L' .

Definition 6.4.14: Degree of Smooth Curve

Let C be a smooth curve in $\mathbb{P}^n$. Then $\deg(H \cdot C)$ is called the *degree* of C and is denoted $\deg C$.

This too is well-defined as it is independent of choice of hyperplane H since if there exists another H_1 :

$$\deg(H_1 \cdot C) = \deg \left(\frac{L_1}{L'} \Big|_C \right)_0 = \deg \left(\frac{L_1}{L'} \Big|_C \right)_\infty = \deg \left(\frac{L}{L'} \Big|_C \right)_\infty = \deg \left(\frac{L}{L'} \Big|_C \right)_0 = \deg(H \cdot C)$$

Note that when $n = 2$, then H is a straight line in $\mathbb{P}^2$ not containing any irreducible curve components of C , and so by Bézout's theorem we know the curve degree is just the same as the plane curve degree given earlier.

In this section, a nonhyperelliptic compact Riemann surface of genus $g > 1$ is identified with its image under the canonical map, the canonical curve in $\mathbb{P}^{g-1}$, which is smooth. This is possible because the canonical map of a nonhyperelliptic surface is injective with smooth image (proposition 6.4.9).

Proposition 6.4.15: Degree and Genus

Let C be a canonical curve of genus g , given by $\varphi_K = [\omega_1 : \cdots : \omega_g]$ for a basis $\omega_1, \dots, \omega_g$ of $\Omega^1(C)$. For a hyperplane $H = \{L = 0\}$ with $L = \sum_i l_i \xi^i$, the hyperplane section is the divisor of the holomorphic form $L(\omega_1, \dots, \omega_g) = \sum_i l_i \omega_i$:

$$H \cdot C = \left(\sum_i l_i \omega_i \right)$$

Consequently

$$\deg C = 2g - 2$$

Proof:

Let $\omega = \sum_i l_i \omega_i$, and let $L' = \sum_i l'_i \xi^i$ define a hyperplane H' with $H' \cap C \cap H = \emptyset$, as in definition 6.4.13, and $\omega' = \sum_i l'_i \omega_i$. In a local coordinate z near a point $p \in C$, write $\omega_i = h_i dz$. The functions h_i have no common zero, as φ_K is well defined, and $\varphi_K = [h_1 : \cdots : h_g]$ near p . So $(L/L') \circ \varphi_K = \omega/\omega' = \sum_i l_i h_i / \sum_i l'_i h_i$ near p . If $\omega(p) = 0$, then $\varphi_K(p) \in H$, hence $\varphi_K(p) \notin H'$ and $\omega'(p) \neq 0$, so the order of the zero of L/L' at p is the order of the zero of ω at p . Hence $H \cdot C = (\omega)_0$, and as $\omega \in \Omega^1(C)$ is holomorphic, $(\omega)_0 = (\omega)$:

$$H \cdot C = (\omega)$$

Thus, by theorem 6.3.8,

$$\deg C = \deg(H \cdot C) = \deg(\omega) = 2g - 2$$

as we sought to show.

Proposition 6.4.16: Genus 3 By Degree 4 Curve

Let C be a canonical curve of genus 3. Then it is a smooth plane algebraic curve of degree 4: the zero set in $\mathbb{P}^2$ of an irreducible homogeneous polynomial of degree 4, smooth at each of its points.

Proof:

Write $C' = \varphi_K(C) \subseteq \mathbb{P}^{g-1} = \mathbb{P}^2$, with $\varphi_K = [\omega_1 : \omega_2 : \omega_3]$. By proposition 6.4.9, φ_K is injective and C' is smooth. By proposition 6.4.15,

$$\deg(C') = 2g - 2 = 2 \cdot 3 - 2 = 4$$

For a line $H = \{L = 0\}$ and another linear form L' with $L(z) = L'(z) = 0$ only at one point $z \notin C'$, the meromorphic function L/L' on C has zero divisor $H \cdot C$, of degree 4, so it has degree 4 (lemma 6.3.7). Its fibres are the intersections of C' with the lines through z , and all but finitely many of them consist of 4 distinct points, the ramification points being finitely many. So all but finitely many lines through a point $z \notin C'$ meet C' in 4 distinct points.

A quartic through C' . Fix a line $H = \{L = 0\}$ and put $D = H \cdot C$, the divisor of the holomorphic form $\omega = L(\omega_1, \omega_2, \omega_3)$ by proposition 6.4.15. For a homogeneous polynomial Q of degree 4, $Q(\omega_1, \omega_2, \omega_3)/L(\omega_1, \omega_2, \omega_3)^4$ is a meromorphic function on C in $\mathcal{L}(4D)$, and this defines a linear map from the space of such Q , of dimension 15, to $\mathcal{L}(4D)$. Here $\deg 4D = 16 > 2g - 2$, so $i(4D) = 0$ by theorem 6.3.8, and the Riemann-Roch theorem (theorem 6.3.31) gives $l(4D) = 16 - 3 + 1 = 14$. So some nonzero Q maps to 0, that is, Q vanishes on C' .

Q is irreducible and $C' = \{Q = 0\}$. Factor $Q = Q_1 \cdots Q_k$ into irreducible homogeneous polynomials. The meromorphic functions $Q_i(\omega_1, \omega_2, \omega_3)/L(\omega_1, \omega_2, \omega_3)^{\deg Q_i}$ have product 0 on the connected surface C . On the open dense set where none of them has a pole, the union of their finitely many closed zero sets is everything, so one of these zero sets has nonempty interior, and that function vanishes identically by proposition 6.2.3, say the one for Q_1 . Take w with $Q_1(w) \neq 0$, so $w \notin C'$, and a line H through w meeting C' in 4 distinct points. The restriction of Q_1 to $H \cong \mathbb{P}^1$ is a binary form of degree $\deg Q_1 \leq 4$, nonzero at w , with 4 distinct zeros, so $\deg Q_1 = 4$ and Q is a constant multiple of the irreducible Q_1 . Finally, suppose $z \in \{Q = 0\} \setminus C'$. All but finitely many lines through z meet C' in 4 distinct points, and on each such line Q has at least 5 zeros, so it vanishes on the line. These lines cover $\mathbb{P}^2$ apart from finitely many lines through z , a set with empty interior, so $Q = 0$ by continuity, a contradiction. Hence $C' = \{Q = 0\}$.

Smoothness. Let $a \in C'$. By proposition 6.4.9, C' is near a the graph of a holomorphic function of one affine coordinate: after exchanging the coordinates x, y of an affine chart containing a , there are discs $U \ni x_0 = x(a)$ and $V \ni y(a)$ and a holomorphic $\varphi: U \rightarrow V$ with $C' \cap (U \times V) = \{(x, \varphi(x)) \mid x \in U\}$. Let q be the dehomogenization of Q in this chart, so that $C' = \{q = 0\}$ there. It is a polynomial of degree at most 4 in y , so

$$q(x, \varphi(x) + t) = \sum_{k=0}^4 c_k(x) t^k, \quad c_k(x) = \frac{1}{k!} \frac{\partial^k q}{\partial y^k}(x, \varphi(x))$$

with each c_k holomorphic on U , and $c_0 = 0$ as the graph lies in C' . If all the c_k vanished identically, q would vanish on $U \times \mathbb{C}$, and $C' \cap (U \times V)$ would be all of $U \times V$. So there is a least $m \geq 1$ with $c_m \not\equiv 0$, and $q(x, \varphi(x) + t) = t^m u(x, t)$ with $u(x, t) = \sum_{k=m}^4 c_k(x) t^{k-m}$.

First, $u(x_0, 0) \neq 0$. If the polynomial $u(x_0, \cdot)$ were identically zero, then $q(x_0, y) = 0$ for every y , and the points (x_0, y) with $y \in V$ would all lie in the graph $C' \cap (U \times V)$. So suppose that $u(x_0, \cdot)$

is a nonzero polynomial with $u(x_0, 0) = 0$, and choose $\epsilon > 0$ with $\{y \mid |y - \varphi(x_0)| \leq 2\epsilon\} \subseteq V$. Then $c_m(x_0) = u(x_0, 0) = 0$, and c_m does not vanish identically near x_0 (theorem 2.5.3), so by proposition 2.5.7 there are $x_n \rightarrow x_0$ in U with $c_m(x_n) \neq 0$. The coefficients $c_k(x_n)$ converge to $c_k(x_0)$, so the polynomials $u(x_n, \cdot)$ converge to $u(x_0, \cdot)$ uniformly on the disc $|t| < \epsilon$. If infinitely many of them had no zero there, Hurwitz's theorem (proposition 4.1.12), applied to these, would make $u(x_0, \cdot)$ zero-free on the disc or identically zero on it, and it is neither. So for n large there is t_n with $|t_n| < \epsilon$ and $u(x_n, t_n) = 0$, hence $q(x_n, \varphi(x_n) + t_n) = 0$. For n large also $|\varphi(x_n) + t_n - \varphi(x_0)| < 2\epsilon$, so $(x_n, \varphi(x_n) + t_n)$ lies in $C' \cap (U \times V)$, on the graph, and $t_n = 0$. This gives $c_m(x_n) = u(x_n, 0) = 0$, a contradiction.

Next, $m = 1$. If $m \geq 2$, then $q(x, y) = (y - \varphi(x))^m u(x, y - \varphi(x))$ for $x \in U$, and both partial derivatives of q vanish at every point $(x, \varphi(x))$ with $x \in U$. By lemma 6.3.1, applied to $F = Q$, these infinitely many points of $\{Q = 0\}$ are singular: all three partial derivatives of Q vanish at them. One of these partial derivatives, $\partial Q / \partial \xi^i$ say, is nonzero, by Euler's relation $\sum_j \xi^j \partial Q / \partial \xi^j = 4Q$, and it is homogeneous of degree 3. The curves $\{Q = 0\}$ and $\{\partial Q / \partial \xi^i = 0\}$ have infinitely many common points, so by Bézout's theorem ([26, Bézout's Theorem]) they have a common component, and the irreducible Q divides $\partial Q / \partial \xi^i$, which is impossible for degree reasons. So $m = 1$, and $\frac{\partial q}{\partial y}(a) = u(x_0, 0) \neq 0$. This is the value of a partial derivative of Q at a representative of a , so a is a smooth point of $\{Q = 0\}$, as we sought to show.

Note that if F is a degree 4 algebraic curve, then by the genus formula:

$$g = \frac{(d-1)(d-2)}{2} = \frac{3 \cdot 2}{2} = 3$$

From this we can do the same classification trick as in section 6.4 to show that the converse is mostly true (having to take into account that some canonical embeddings are not injective). Thus one may verify that a smooth quartic such as

$$F(x, y, z) = x^4 + y^4 + z^4 + \lambda x^2 y^2 + \mu y^2 z^2 + \nu z^2 x^2$$

is a canonical curve.

Genus 4

Canonical genus 4 curves have the amazing property that they can be characterized by the intersection of quadric³ and a cubic hypersurface in $\mathbb{P}^3$.

Just like before, suppose C is a smooth algebraic curve in $\mathbb{P}^n$ and $H = \{\xi^0 = 0\}$ is a hypersurface. Let $D = H \cdot C \in \text{Div}(C)$. Notice that $\bigoplus_{k=0}^{\infty} \mathcal{L}(kD)$ and $\mathbb{C}[\xi^0, \dots, \xi^n]$ are graded rings. Let $F(\xi) \in \mathbb{C}[\xi^0, \dots, \xi^n]$ be a homogeneous polynomial and define:

$$r(F(\xi)) = \frac{F(\xi)}{(\xi^0)^k} \Big|_C \in \mathcal{L}(k(D))$$

Then extended r linearly to all of $\mathbb{C}[\xi^0, \dots, \xi^n]$, we get ring homomorphism:

$$r : \mathbb{C}[\xi^0, \dots, \xi^n] \rightarrow \bigoplus_{k=0}^{\infty} \mathcal{L}(kD)$$

³I have not seen the word "quadric" before. It means a degree 2 hypersurface embedded in D dimensions. Quadratic seems perfectly fine, but it seems that quadric is just a little more specific

A homogeneous polynomial F of degree k lies in the kernel of r exactly when it vanishes on C , so comparing the dimension of the space of homogeneous polynomials of degree k with $l(kD)$ produces hypersurfaces containing C . For a canonical curve of genus 4 this gives the characterization announced above.

Proposition 6.4.17: Canonical Curves of Genus 4

Let $C \subseteq \mathbb{P}^3$ be a canonical curve of genus 4. Then C lies on an irreducible quadric surface Q and on a cubic surface Q' not containing Q , and $C = Q \cap Q'$.

The last step needs Bézout's theorem for surfaces in $\mathbb{P}^3$, which this book does not prove. A complete proof is in Griffiths and Harris [13, Chapter 2, Section 3, p. 258].

Proof Key ideas:

The hyperplane section $D = H \cdot C$ is the divisor of a holomorphic 1-form, by proposition 6.4.15, so $\deg D = 6$, and for $k \geq 2$ the divisor kD has $i(kD) = l(D - kD) = 0$ by proposition 6.3.16, as $D - kD$ has negative degree. The Riemann-Roch theorem gives $l(2D) = 12 - 4 + 1 = 9$ and $l(3D) = 18 - 4 + 1 = 15$. Quadratic forms in four variables form a space of dimension $10 > 9$, so some nonzero quadratic form vanishes on C , defining a quadric Q . It is irreducible, since a reducible quadric is a union of two planes and C lies in no plane, the ω_i being linearly independent. Cubic forms form a space of dimension 20, so those vanishing on C form a space of dimension at least 5, while the cubic forms divisible by the equation of Q form a space of dimension 4. So some cubic Q' contains C and not Q . By Bézout's theorem $Q \cap Q'$ is a curve of degree 6 containing the curve C of degree 6, so $C = Q \cap Q'$, which completes the sketch.

6.4.1 Abel's Theorem and its Applications

Let C be a compact Riemann surface of genus $g \geq 1$. Then $H_1(C, \mathbb{Z})$ is a free abelian group of rank $2g$. Choose a canonical basis $\gamma_1, \gamma_2, \dots, \gamma_{2g-1}, \gamma_{2g}$ for $H_1(C, \mathbb{Z})$ satisfying the intersection requirements:

$$\gamma_i \cdot \gamma_{g+i} = 1 \quad \gamma_{g+i} \cdot \gamma_i = -1$$

for all i , and 0 otherwise; that is it is the usual vertical and horizontal circles around the handles when visualized in $\mathbb{R}^3$. Recall the period vector:

$$\pi_j = \begin{pmatrix} \int_{\gamma_j} \omega_1 \\ \vdots \\ \int_{\gamma_j} \omega_g \end{pmatrix} \in \mathbb{C}^g$$

for $j = 1, \dots, 2g$. Recall that we proved that this vector is well-defined as the paths are closed and that the period matrix is nonsingular (proposition 6.3.26). Thus, with $\mathbb{Z}$ -coordinates, we get a lattice in $\mathbb{C}^g$:

$$\Lambda = \left\{ \sum_{j=1}^{2g} m_j \pi_j \mid m_j \in \mathbb{Z} \right\} \subseteq \mathbb{C}^g$$

We thus can define the following:

Definition 6.4.18: Jacobian Variety

Let C be a compact Riemann surface. Then the g -dimensional complex torus $\mathbb{C}^g/\Lambda$ is called the *Jacobian variety* of C and will be denoted $J(C)$.

This variety will be useful for finding when a divisor is principal, give a measure of the lack thereof. Namely, given a $D \in \text{Div}(C)$, does there exist a $f \in K^*(C)$ such that

$$(f) = D$$

First off, as we are working over compact Riemann surfaces, we can look at divisor of degree 0. Let:

$$\begin{aligned} \deg : \text{Div}(C) &\rightarrow \mathbb{Z} \\ D &\mapsto \deg D \end{aligned}$$

Let $\text{Div}^0(C) = \ker(\deg)$. Then if $(f) = D$, it must be that $D \in \text{Div}^0(C)$. The converse is not always true, and Abel's theorem will outline exactly how to associate these two.

To that end, let us start by defining a map from $\text{Div}(C)$ to $J(C)$. Let $q \in C$ be any fixed point and let $\omega \in \Omega^1(C)$ be a holomorphic differential. For a point $p \in C$, consider:

$$\int_q^p \omega \tag{6.1}$$

Since there is nontrivial winding, the difference of two paths from q to p is a homology class that can be written as

$$\sum_{j=1}^{2g} m_j \gamma_j$$

hence two values of equation (6.1) will differ by:

$$\sum_{j=1}^{2g} m_j \int_{\gamma_j} \omega$$

Letting ω run through the basis of $\Omega^1(C)$ gives the following:

Definition 6.4.19: Abel-Jacobi Map

Let C be a compact Riemann surface, $q \in C$. Then the *Abel-Jacobi map*

$$u : \text{Div}(C) \rightarrow J(C)$$

is given by:

$$u(D) = \begin{pmatrix} \sum_{i=1}^k n_i \int_q^{p_i} \omega_1 \\ \vdots \\ \sum_{i=1}^k n_i \int_q^{p_i} \omega_g \end{pmatrix}$$

where $D = \sum_{i=1}^k n_i p_i$.

Certainly this is a homomorphism for if $D = \sum_{i=1}^k p_i - \sum_{i=1}^k q_i \in \text{Div}^0(C)$ then by the rules of integration bounds we get the homomorphism rule. We shall see that $(f) = D$ for $D \in \text{Div}^0(C)$ if and only if

$$u(D) = 0$$

Or, slightly more generally:

Theorem 6.4.20: Abel-Jacobi Theorem

Let C be a compact Riemann surface. Then the sequence:

$$K^*(C) \xrightarrow{(-)} \text{Div}^0(C) \xrightarrow{u} J(C) \rightarrow 0$$

is exact

Its proof needs harmonic differential forms on a compact Riemann surface and the Jacobi inversion problem, which this book does not develop. A complete proof is in Forster: exactness at $\text{Div}^0(C)$, which says that a divisor D of degree 0 is principal exactly when $u(D) = 0$, is Abel's theorem [11, Theorem 20.7], and the surjectivity of u on $\text{Div}^0(C)$ is [11, Theorem 21.7].

Proof Key ideas:

If $D = (f)$ with f nonconstant, then for $\lambda \in \mathbb{CP}^1$ the divisors $D_\lambda = f^{-1}(\lambda) - f^{-1}(\infty)$, with multiplicities, have degree 0 by lemma 6.3.7, and $\lambda \mapsto u(D_\lambda)$ is a holomorphic map from $\mathbb{CP}^1$ to $J(C)$. It lifts to a holomorphic map from the simply connected compact $\mathbb{CP}^1$ to $\mathbb{C}^g$, which is constant, so $u(D) = u(D_0) = u(D_\infty) = 0$. Conversely, if $u(D) = 0$, one integrates a meromorphic 1-form with simple poles of residue ± 1 at the points of D , corrected by harmonic forms so that all its periods lie in $2\pi i\mathbb{Z}$, and the exponential of the integral is a function f with $(f) = D$. For surjectivity, the map $(x_1, \dots, x_g) \mapsto u(x_1 + \dots + x_g - gq)$ from C^g to $J(C)$ has invertible differential at a suitable point, so, after subtracting the value at that point, the image of $\text{Div}^0(C)$ under the homomorphism u contains a neighbourhood of 0. Every point of $J(C)$ is a positive integer multiple of a point of that neighbourhood, which completes the sketch.

From this we get:

Definition 6.4.21: Picard Variety

Let C be a compact Riemann surface and $(-) : K^*(C) \rightarrow \text{Div}^0(C)$. Then:

$$\text{Pic}(C) = \frac{\text{Div}^0(C)}{\text{im}((-)}$$

The Abel-Jacobi map then becomes an isomorphism.

(the Riemann bilinear relations puts important constrictions on the period matrix that shows how it reflects some of its topological properties!) (this gives us:)

Theorem 6.4.22: Torelli Theorem

Let C, C' be compact Riemann surfaces of the same genus g . Then a necessary and sufficient condition for C and C' to be biholomorphic is that they have the same normalized period matrix under a suitable choice of canonical homology bases.

Proof:

The forward implication. Let $\varphi : C \rightarrow C'$ be a biholomorphism. Choose a $\mathbb{Z}$ -basis $\gamma'_1, \dots, \gamma'_{2g}$ of $H_1(C', \mathbb{Z})$ and a basis $\omega'_1, \dots, \omega'_g$ of $\Omega^1(C')$, and transport them to C by

$$\gamma_i = (\varphi^{-1})_* \gamma'_i \quad \omega_j = \varphi^* \omega'_j$$

Both transports are isomorphisms, so these are again bases, and the change of variables formula gives

$$\int_{\gamma_i} \omega_j = \int_{(\varphi^{-1})_* \gamma'_i} \varphi^* \omega'_j = \int_{\gamma'_i} \omega'_j$$

for all i and j . The two period matrices of definition 6.3.25 therefore agree entry by entry. A biholomorphism is an orientation-preserving homeomorphism, so φ_* carries the intersection pairing on first homology to itself and the transported basis is canonical exactly when the original one is. The two matrices then normalize in the same way.

The converse is deferred. Three inputs are missing, and each is absent from this book, not omitted as routine.

1. The two terms in the statement have no definition above. Definition 6.3.25 fixes the period matrix relative to an arbitrary $\mathbb{Z}$ -basis of $H_1(C, \mathbb{Z})$, but no intersection pairing on $H_1(C, \mathbb{Z})$ is constructed, so “canonical homology basis” is undefined here, and so is the normalization that turns the first $g \times g$ block of the period matrix into the identity. The Riemann bilinear relations, which are what make that block invertible, are named in the surrounding text but nowhere stated or proved. This also leaves the last step of the forward implication above resting on the same undefined vocabulary.
2. Theorem 6.4.20 is stated without proof, and the Jacobi inversion theorem asserting that $u : \text{Div}^0(C) \rightarrow J(C)$ is surjective is not proved either. Every route to the converse runs through both.
3. Riemann’s theorem, which identifies the theta divisor of $J(C)$ with a translate of the image W_{g-1} of the $(g-1)$ -fold symmetric product of C under u , needs the theta function and its divisor, and neither is constructed above. Andreotti’s argument then recovers the canonical curve of definition 6.4.8 from the branch locus, in the dual of $\mathbb{P}^{g-1}$, of the Gauss map on the smooth locus of that divisor. That calls for a ramification theory on a g -dimensional complex torus which this book does not develop.

The argument is carried out in full in [13, Chapter 2, Section 7, Torelli’s Theorem], in the equivalent form that C and C' are biholomorphic once their Jacobians are isomorphic as principally polarized Abelian varieties.

The Jacobi inversion theorem, that $u : \text{Div}^0(C) \rightarrow J(C)$ is surjective, is part of theorem 6.4.20.

Applications of Abel's Theorem

Let C be a nonhyperelliptic compact Riemann surface of genus 3; without loss of generality we may identify C with its canonical curve in $\mathbb{P}^2$. The nC is a fourth-degree smooth plane algebraic curve. Let L be a straight line in $\mathbb{P}^2$. Then:

$$D = L \cdot C = p_1 + p_2 + p_3 + p_4 \in C^{(4)}$$

is a hyperplane section. Let $E \in \mathbb{C}^{(4)}$. Then $u(E) = u(D)$ if and only if there exists a line L' such that

$$E = L' \cdot C$$

hence, E must be a hyperplane. Indeed, if $E = L' \cdot C$, let L and L' be given by linear equations l, l' so that $L = \{l = 0\}$ and $L' = \{l' = 0\}$. Letting:

$$f = \frac{l'}{l} \Big|_C \in K(C)$$

then $(f) = E - D$, and so $0 = u(E) - u(D)$, hence $u(E) = u(D)$. Conversely, if $u(D) = u(E)$, then we need to show that:

$$E \equiv q_1 + q_2 + q_3 + q_4$$

is a hyperplane section. Suppose that L' is the unique line passing through q_1, q_2 . Then:

$$L' \cdot C = q_1 + q_2 + q' + q''$$

If $L' \cdot C \neq E$, then it must be that :

$$q_3 + q_4 - q' - q'' \neq 0$$

If $u(D) = u(E)$, then $u(E) = u(L' \cdot C)$ (as D and $L' \cdot C$ are both hyperplanes sections), and so:

$$u(q_3 + q_4 - q' - q'') = 0$$

Thus by the Abel-Jacobi Theorem there exists an f such that

$$(f) = q_3 + q_4 - q' - q''$$

Now, consider $f : C \rightarrow \mathbb{P}^1$ as a holomorphic mapping. Then f has either degree 1 or 2 (depending on whether one of $\{q_3, q_4\}$ is equal to one of $\{q', q''\}$). Implying that C is either Riemann sphere or a hyperelliptic Riemann surface, but that contradicts the fact that C is nonhyperelliptic of genus 3. Thus:

$$E = L' \cdot C$$

Thus, given a hyperplane section D of a canonical curve of genus 3, $|D|$ can be identified with the set of hyperplanes of $\mathbb{P}^2$, that is:

$$|D| \cong \mathbb{P}^2$$

Giving us a full geometric picture!

Next, if $g = 1$, we can rather easily classify all such surfaces up to biholomorphism. We first have:

Theorem 6.4.23: Jacobi Variety in Genus 1

Let C be a compact Riemann surface of genus 1. Then it is biholomorphic to its Jacobian variety

$$C \cong J(C)$$

Given a normalized period matrix

$$\Pi = (1 \ z)_{1 \times 2}$$

Then $C \cong \mathbb{C}/\Lambda$, where Λ is the lattice spanned by 1 and z .

Proof:

Since $g = 1$, theorem 6.3.23 gives $\Omega^1(C) = \mathbb{C}\omega$ for a nonzero holomorphic differential ω , and by proposition 6.3.26 its two periods π_1, π_2 are $\mathbb{R}$ -linearly independent. So $\Lambda = \mathbb{Z}\pi_1 + \mathbb{Z}\pi_2$ is a lattice and $J(C) = \mathbb{C}/\Lambda$. Rescaling ω so that $\pi_1 = 1$ makes the period matrix $(1 \ z)$ with $z = \pi_2$, and then Λ is spanned by 1 and z . Fix $q \in C$. It remains to show that the Abel-Jacobi map on points, $u(p) = \int_q^p \omega$ modulo Λ (definition 6.4.19), is a biholomorphism $C \rightarrow J(C)$.

Holomorphy. Near a point p_0 , in a coordinate disc N , we have $u(p) = \int_q^{p_0} \omega + \int_{p_0}^p \omega$ modulo Λ , with the second integral taken along a path in N . On the disc N it is a primitive of ω , hence holomorphic, so u is holomorphic.

Injectivity. If $u(p_1) = u(p_2)$, then $u(p_1 - p_2) = 0$, as u is additive on divisors, and by theorem 6.4.20, whose proof this book defers to Abel's theorem in Forster [11, Theorem 20.7], the degree 0 divisor $p_1 - p_2$ is principal, $p_1 - p_2 = (f)$. Then $f \in \mathcal{L}(p_2)$. The holomorphic form $\omega \neq 0$ has $(\omega) \geq 0$ of degree $2g - 2 = 0$ (theorem 6.3.8), so ω has no zeros, and $i(p_2) = 0$, as every nonzero element of $\Omega^1(C) = \mathbb{C}\omega$ has no zeros. So theorem 6.3.31 gives $l(p_2) = 1 - 1 + 0 + 1 = 1$. So f is constant, $(f) = 0$, and $p_1 = p_2$.

Surjectivity. The torus $J(C) = \mathbb{C}/\Lambda$ is a complex manifold of dimension one (example 6.2.2), and u is an injective holomorphic map into it, so $u(C)$ is open and u is a biholomorphism of C onto $u(C)$ (theorem 6.2.7). The image is also compact, hence closed in the Hausdorff space $J(C)$. The torus $J(C)$ is connected, being the image of $\mathbb{C}$ under the quotient map, so $u(C) = J(C)$. Hence u is biholomorphic and $C \cong J(C) = \mathbb{C}/\Lambda$, as claimed.

Theorem 6.4.24: Up To Projective Special Automorphism

Let $C = \mathbb{C}/\Lambda$ and $C' = \mathbb{C}/\Lambda'$ where:

$$\begin{aligned}\Lambda &= \{m + nz \mid m, n \in \mathbb{Z}\} & \operatorname{Im} z > 0 \\ \Lambda' &= \{m + nz' \mid m, n \in \mathbb{Z}\} & \operatorname{Im} z' > 0\end{aligned}$$

Then $C \cong C'$ if and only if

$$z = \frac{\alpha z' + \beta}{\gamma z' + \delta}$$

where $\alpha, \beta, \gamma, \delta \in \mathbb{Z}$ and

$$\alpha\delta - \beta\gamma = 1$$

Proof:

Let $\pi: \mathbb{C} \rightarrow C$ and $\pi': \mathbb{C} \rightarrow C'$ be the quotient maps. The tori carry the complex structure of example 6.2.2, whose charts are local inverses of π and of π' , any two of which differ on overlaps by translations by elements of the lattice. Two pairs of complex numbers that are $\mathbb{Z}$ -bases of the same lattice are related by an integer matrix whose inverse is also an integer matrix, so its determinant is ± 1 .

Necessity, Step 1: a constant derivative. Let $\varphi: C \rightarrow C'$ be biholomorphic and put $F = \varphi \circ \pi$, a holomorphic map $\mathbb{C} \rightarrow C'$. In a chart s of C' , given by a local inverse of π' , the map F reads as the holomorphic function $s \circ F$, and two charts give readings differing by a constant of Λ' , so $a(w) = (s \circ F)'(w)$ is a well-defined entire function. It is doubly periodic with respect to Λ , because $F(w + \lambda) = F(w)$ for $\lambda \in \Lambda$. By proposition 5.1.2, a is a constant, again written a . If $a = 0$, then F would be locally constant, hence constant on the connected space $\mathbb{C}$, whereas F is surjective. So $a \neq 0$.

Step 2: F is affine. Choose $b \in \mathbb{C}$ with $\pi'(b) = F(0)$, and put $G(w) = \pi'(aw + b)$. In a chart s of C' around $F(0) = G(0)$, both $s \circ F$ and $s \circ G$ are holomorphic near 0 with derivative a and with the same value at 0, so they agree near 0. By proposition 6.2.3, the set of points near which F and G coincide is open, closed and nonempty in $\mathbb{C}$, hence equal to $\mathbb{C}$. Thus

$$\varphi(\pi(w)) = \pi'(aw + b) \quad w \in \mathbb{C}$$

Step 3: $a\Lambda = \Lambda'$. For $\lambda \in \Lambda$, Step 2 gives $\pi'(aw + a\lambda + b) = \varphi(\pi(w + \lambda)) = \varphi(\pi(w)) = \pi'(aw + b)$, so $a\lambda \in \Lambda'$. For $\lambda' \in \Lambda'$, Step 2 gives $\varphi(\pi(w + \lambda'/a)) = \pi'(aw + b + \lambda') = \pi'(aw + b) = \varphi(\pi(w))$, so $\pi(w + \lambda'/a) = \pi(w)$ by injectivity of φ , and $\lambda' \in a\Lambda$. Hence $a\Lambda = \Lambda'$.

Step 4: the integer matrix. Multiplication by a carries the $\mathbb{Z}$ -basis $z, 1$ of Λ to the $\mathbb{Z}$ -basis az, a of Λ' , and $z', 1$ is another $\mathbb{Z}$ -basis of Λ' . So $az = \alpha z' + \beta$ and $a = \gamma z' + \delta$ with $\alpha, \beta, \gamma, \delta \in \mathbb{Z}$ and $\alpha\delta - \beta\gamma = \pm 1$, and dividing gives $z = (\alpha z' + \beta)/(\gamma z' + \delta)$. Since $\alpha, \beta, \gamma, \delta$ are real,

$$\operatorname{Im} z = \frac{\operatorname{Im}((\alpha z' + \beta)(\gamma \overline{z'} + \delta))}{|\gamma z' + \delta|^2} = \frac{(\alpha\delta - \beta\gamma)\operatorname{Im} z'}{|\gamma z' + \delta|^2}$$

and $\text{Im} z > 0$, $\text{Im} z' > 0$ force $\alpha\delta - \beta\gamma = 1$.

Sufficiency. Suppose $z = (\alpha z' + \beta)/(\gamma z' + \delta)$ with integer entries and $\alpha\delta - \beta\gamma = 1$, and put $a = \gamma z' + \delta$, which is nonzero as $\text{Im} z' > 0$ and γ, δ are not both 0. The integer matrix with rows (α, β) and (γ, δ) has determinant 1, so its inverse is an integer matrix, and $z', 1$ are integer combinations of $az = \alpha z' + \beta$ and a . Hence $a\Lambda = \mathbb{Z}az + \mathbb{Z}a = \Lambda'$. Then $\pi(w) \mapsto \pi'(aw)$ is a well-defined bijection $C \rightarrow C'$, since $aw_1 - aw_2 \in \Lambda'$ exactly when $w_1 - w_2 \in \Lambda$, with inverse $\pi'(w) \mapsto \pi(w/a)$. In the charts given by local inverses of π and π' , both maps read as $w \mapsto aw$ or $w \mapsto w/a$ up to translations by lattice elements, so they are holomorphic, and $C \cong C'$, as we sought to show.

Further topics in this direction are elliptic integrals and the “usual” representation $p(x) = 4x^3 + ax + b$, the number $3n(n-2)$ of flex points of a smooth plane curve of degree n , counted with multiplicity, which comes from Bézout’s theorem, the original connection to integrating ellipses, and the algebraic properties of $x(u)$, as in the Weierstrass equation section of a complex analysis course.

6.4.2 Riemann Surfaces and Elliptic Curves

Recall that any $\wp$ -function induces the following algebraic relation

$$y^2 = 4x^3 - 20a_2x - 28a_4 \quad (6.2)$$

where $(x, y) = (\wp', \wp)$. Let $p(x)$ represent the polynomial on the right hand side. We may choose a_2, a_4 so that the right hand side has 3 distinct roots, and hence we have a smooth curve (namely since $p'(x) \neq 0$ for any value of x such that $p(x) = 0$). From this, we shall define the following manifold: Let $X \subseteq \mathbb{C} \times \mathbb{C}$ to be the pair of points $(x, y) \in \mathbb{C} \times \mathbb{C}$ that satisfy equation (6.2). To show X is a complex manifold, for any point $(x_0, y_0) \in X$ where $y_0 \neq 0$, as shall take a neighborhood sufficiently small so that we may project onto the first coordinate; the x are the “local coordinates”. If at a point $(x_0, y_0) \in X$ we have $y = 0$, then notice that $P'(x_0) \neq 0$ by assumption of smoothness. Hence by the implicit function theorem (theorem 2.2.28), equation (6.2) is equivalent to a relation of the form $x = f(y)$ in a sufficiently small neighborhood of $(x_0, 0)$ where $f(0) = x_0$. In such a neighborhood, we take y as the local coordinates.

We thus have a mapping $\varphi : X \rightarrow \mathbb{C}$ which maps (x, y) onto $x \in \mathbb{C}$. Certainly, this is a holomorphic map, and hence (X, φ) is a Riemann surface over $\mathbb{C}$. This surface has two “sheets” corresponding to the two values of y that correspond to x ; if $p(x) \neq 0$, they are distinct. Furthermore, the map $X \rightarrow \mathbb{C}$ mapping (x, y) onto y is also holomorphic on X ; we shall write these coordinates as y . We may then define the differentiable form $\omega = dx/y$ in a neighborhood of $(x_0, y_0) \in X$ such that $y_0 \neq 0$ and

$$\omega = \frac{dy}{6x^2 - 10a_2}$$

in a neighbourhood of $(x_0, 0) \in X$. Then this is a holomorphic differential form on X , hence a closed form. In particular, it has a primitive in a neighbourhood of each point of X , globally this primitive is a *many-valued* function z which is holomorphic in a neighbourhood of each point X .

We now define another Riemann surface over $\mathbb{CP}^1$. First, we must make our polynomial a homogeneous polynomial so that it’s well-defined in projective space. If we consider $[x, y, t] \in \mathbb{CP}^1$, then $[x/t, y/t, 1] = [x, y, t]$. Substituting these values in the equation and re-arranging, we get

$$y^2t = 4x^3 - 20a_2xt^2 - 28a_4t^3 \quad (6.3)$$

giving us a Homogeneous polynomial, hence on that is well-defined on $\mathbb{CP}^1$. Hence, we shall consider the subset of points $X' \subseteq \mathbb{CP}^1$ that satisfy equation (6.3). The space X' can be seen as an extension of X as follows: we can identify X as a subspace of X' by associating each $(x, y) \in X$ with a point X' whose homogeneous coordinates are $[x, y, 1]$. Then the compliment $X' \setminus X$ consists only of $[0, 1, 0]$. We shall denote this point as ∞ , and take the local coordinates $x/y = x'$ since x' defines a homeomorphism of a neighborhood of the point ∞ onto a neighborhood 0 in $\mathbb{C}$ (put $t/y = t'$ in equation (6.3)). Then we get

$$t' = 4(x')^3 - 20a_2x'(t')^2 - 28a_4(t')^3$$

in some neighbourhood $x' = 0, t' = 0$ and the implicit function theorem gives t' as a holomorphic function of x' :

$$t' = 4(x')^3 - 320a^2(x')^7 + \dots \quad (6.4)$$

The complex manifold structure on X' is now defined since we have chosen x' as the local coordinates at ∞ , and the mapping φ' is defined to be equal to φ on X and take the point ooo of X' onto the point at infinity of $\mathbb{CP}^1$. The holomorphic differential form ω defined on X extends to a holomorphic differential form on X' : in a neighborhood of ∞ , we use the local coordinates x' and the holomorphic function t' of x' defined in equation (6.4) to get

$$\omega = t'd(x'/t') = dx' - x'\frac{dt'}{t'} = dx' - \frac{12(x')^2 + \dots}{4(x')^2 + \dots}dx' = -2dx'(1 + g(x'))$$

where g is a holomorphic function in a neighborhood of $x' = 0$ and is zero for $x' = 0$. The form ω is thus defined on the compact space X' , hence has a local primitive which is many-valued function on X' and serves so a local coordinate at each point of X' .

Now, if we get a_2, a_4 via a Weiestrass $\wp$ -function defined given a discrete group Ω , then we have that ht meromorphic transformation

$$x = \wp(z) \quad y = \wp'(z)$$

defines an isomorphism of the complex manifold $\mathbb{C}/\Omega$ onto the complex manifold X' . The inverse isomorphism defines z as a holomorphic many-valued function on X' whose (local) branches differ in value by a constant belonging to Ω .

(there is still more in Cartan p. 202, I will get back to it)

(here is from lecture; I believe this is linking to the original motivation of the term “elliptic”)

Recall we have the following implicit equation defining a circle: $y^2 = 1 - x^2$. Then if

$$x = \cos(\theta) = \sin'(\theta) \quad y = \sin(\theta) \quad dy \sin'(\theta) d\theta = x d\theta$$

Then differentiating we get

$$x dx + y dy = 0 \implies d\theta = \frac{dy}{x} = -\frac{dx}{y}$$

Then

$$\int d\theta = - \int \frac{dx}{y} = \int \frac{dy}{\sqrt{1-y^2}}$$

We can invert $\int \frac{dy}{\sqrt{1-y^2}}$ in a neighborhood of $(1, 0)$ by defining trigonometric functions by

$$\theta = \int_{(1,0)}^{(\cos(\theta), \sin(\theta))} \frac{dy}{x} = \int_0^{\sin(\theta)} \frac{dy}{\sqrt{1-y^2}}$$

Now, if $dx = \wp'(z)dz = ydz$, then $dz = \frac{dx}{y}$ where x has local coordinates (given $y \neq 0$). From the curve, we then have

$$2ydy = (12x^2 - 20a_2)dx \quad \frac{dy}{6x^2 - 10a_2} = \frac{dx}{y} = dz$$

Then dz is an extension to all of X' of the holomorphic differential form

$$\frac{dx}{\sqrt{4x^3 - 20a_2x - 28a_4}}$$

and

$$z = \wp^{-1}(x) = \int_{[0,1,0]}^{\wp(z), \wp'(z), 1} \frac{dx}{\sqrt{4x^3 - 20a_2x - 28a_4}}$$

Harmonic Functions

Recall that all holomorphic functions are harmonic, but not all harmonic functions are holomorphic. In this section, we shall study harmonic functions, see their connection to holomorphic functions, and understand better their behavior.

One motivation for the study of harmonic functions is that many of the properties we have seen about holomorphic functions apply to harmonic functions, for example that an entire holomorphic function is bounded implies it is constant can be proven by only working with harmonic functions. We shall explore more of such results in this chapter.

7.1 Harmonic Functions and Dirichlet Problem

Recall that a (two dimensional) function is harmonic if

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$$

which as a Wirtinger derivative we may write as:

$$\frac{\partial^2 f}{\partial z \partial \bar{z}} = 0$$

which shows us that holomorphic functions are harmonic. However, not all harmonic functions are holomorphic (ex. $\operatorname{Re} z$). In this section, we shall look over general (real and complex) harmonic functions and see how they relate to holomorphic functions. First, if we have any complex harmonic function, then by linearity of the derivative its real and imaginary parts are harmonic. Hence, looking at real harmonic functions, we shall show that they are locally equal to a holomorphic function. First,

suppose g is a real harmonic function. Then by definition we have

$$\frac{\partial^2 g}{\partial z \partial \bar{z}} = 0$$

hence, $\frac{\partial g}{\partial \bar{z}}$ is holomorphic, and so the differential form $\frac{\partial g}{\partial \bar{z}} dz$ locally has a holomorphic primitive $f(z)$. In particular, this means that locally:

$$df = \frac{\partial g}{\partial z} dz$$

taking the conjugate of both sides, then similar to the calculation we have done on page 47, we get

$$d\bar{f} = \frac{\partial g}{\partial \bar{z}} d\bar{z}$$

Thus:

$$df + d\bar{f} = d(f + \bar{f}) = \frac{\partial g}{\partial z} dz + \frac{\partial g}{\partial \bar{z}} d\bar{z} = dg$$

Hence:

$$g = 2\operatorname{Re}(f) + c$$

for some constant c . We can in fact explicitly compute f . Since it's holomorphic, we know f has a power series representation:

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

Without loss of generality we can assume a_0 is real (add a constant to make it real since f was only defined up to a constant). Let's say this power series has radius R , so consider $r < R$. Then doing a u -substitution $z = e^{i\theta}$, we can work out the real part of $f(re^{i\theta})$ to find $g(r \cos(\theta), r \sin(\theta))$. In particular:

$$g(r \cos \theta, r \sin \theta) = \operatorname{Re}(f(z)) = a_0 + \frac{1}{2} \sum_{n=1}^{\infty} r^n a_n (e^{in\theta} + e^{-in\theta})$$

Then integrating both sides from 0 to 2π we get

$$\frac{1}{2\pi} \int_0^{2\pi} g(r \cos \theta, r \sin \theta) d\theta = a_0$$

and we can use the usual trick to get the Fourier coefficients by multiplying by $e^{in\theta}$:

$$\frac{1}{2\pi} \int_0^{2\pi} g(r \cos \theta, r \sin \theta) d\theta e^{-in\theta} d\theta = \frac{1}{2} r^n a_n$$

where $a_n = \frac{1}{\pi} \int_0^{2\pi} g(r \cos \theta, r \sin \theta) r^{-n} e^{in\theta} d\theta$. The power series representation of $f(z)$ thus becomes:

$$\begin{aligned} f(z) &= a_0 + \sum_{n=1}^{\infty} a_n z^n \\ &= \frac{1}{2\pi} \int_0^{2\pi} g(r \cos \theta, r \sin \theta) d\theta + \sum_{n=1}^{\infty} \left(\frac{1}{\pi} \int_0^{2\pi} g(r \cos \theta, r \sin \theta) r^{-n} e^{in\theta} d\theta \right) z^n \\ &= \frac{1}{2\pi} \int_0^{2\pi} g(r \cos \theta, r \sin \theta) \left[1 + 2 \sum_{n=1}^{\infty} \left(\frac{z}{r e^{i\theta}} \right)^n \right] d\theta \end{aligned}$$

where the inner sum in the last equality is the usual geometric series. Simplifying the last line out, we get:

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} g(r \cos \theta, r \sin \theta) \frac{re^{i\theta} + z}{re^{i\theta} - z} d\theta$$

We can also re-write g to be of the above form. To see this, first note that

$$\frac{re^{i\theta} + z}{re^{i\theta} - z} \cdot \frac{re^{i\theta} - \bar{z}}{re^{i\theta} - \bar{z}} = \frac{r^2 - |z|^2}{|re^{i\theta} - z|^2} + \frac{-re^{i\theta}\bar{z} + zre^{i\theta}}{|re^{i\theta} - z|^2}$$

The first term is real and is called the *Poisson kernel*. The second term is purely imaginary since the difference between a complex number and its conjugate. Thus, for $|z| < r$, we get

$$g(z) = \frac{1}{2\pi} \int_0^{2\pi} g(r \cos \theta, r \sin \theta) \frac{r^2 - |z|^2}{|re^{i\theta} - z|^2} d\theta$$

The reason this is called the Poisson kernel is if take $g \equiv 1$ then

$$\frac{1}{2\pi} \int_0^{2\pi} \frac{r^2 - |z|^2}{|re^{i\theta} - z|^2} d\theta = 1 \quad (7.1)$$

which if you have done some PDEs or Fourier Analysis should ring some bells

7.2 Dirichlet Problems for a Disk

In mathematics, extension questions are very prevalent, for example Tietze extension theorem or Hahn-Banach Theorem. Another famous extension theorem comes from algebraic geometry and asks when a continuous function $f : S^1 \rightarrow \mathbb{R}$ (or $\mathbb{C}$) can be extended to a continuous function on B^1 . Dirichlet problem is a question about the extensionality of a continuous function on S^1 into a harmonic function. If f is periodic and defined on a circle, then it can be extended to a harmonic function on the disk. We first need a lemma:

Lemma 7.2.1: Dirichlet Problem On Disk Lemma

Let $\eta > 0$ and $\theta_0 \in [0, 2\pi)$. Let γ denote the arc of the circle with radius r where $|\arg(z) - \theta_0| > \eta$. Then

$$\frac{1}{2\pi} \int_{\gamma} \frac{r^2 - |z|^2}{|re^{i\theta} - z|^2} d\theta \xrightarrow{(z \rightarrow re^{i\theta_0})} 0$$

that is, the above integral tends to zero as $z \rightarrow re^{i\theta_0}$.

Proof:

Choose $\rho < r$ and let $z = \rho e^{i\alpha}$. We want to bound the denominator of the integral $|re^{i\theta} - z|$ for z close to $re^{i\theta_0}$ so that we can factor it out of the integral, leaving us with $r^2 - |z|^2 = r^2 - \rho^2$. Then clearly as $z \rightarrow re^{i\theta_0}$, the integral will go to zero.

First, by the triangle inequality, if $|\theta - \theta_0| > \eta$ and $|\alpha - \theta_0| < \frac{\eta}{2}$, then $|\alpha - \theta| \geq \frac{\eta}{2}$. Then by a simple geometric argument, it is easy to see that z and $re^{i\theta}$ are separated by a distance of d where d is the distance between two radii spanning the sector of angle $\eta/2$. In particular, z is on one

side of the sector and $re^{i\theta}$ is on the other. Thus, this distance is necessarily $r \sin(\eta/2)$. Thus:

$$|z - re^{i\theta}| \geq r \sin\left(\frac{\eta}{2}\right)$$

for all $re^{i\theta}$ on γ . Thus:

$$\frac{1}{2\pi} \int_{\gamma} \frac{r^2 - |z|^2}{|re^{i\theta} - z|^2} d\theta \leq \frac{1}{2\pi} \int_{\gamma} \frac{r^2 - |z|^2}{r^2 \sin^2(\eta/2)} d\theta < \frac{r^2 - \rho^2}{r^2 \sin^2(\eta/2)}$$

Thus, as $z \rightarrow re^{i\theta_0}$, $\rho \rightarrow r$, and so the integral tends to 0, as we sought to show.

Theorem 7.2.2: Dirichlet Problem On A Disk

Let $f(\theta)$ be a continuous, periodic function defined the circle of radius r centered at 0 with period 2π . Then there exists a function $F(z)$ that is continuous on the closed disk $|z| \leq r$ and harmonic in the interior $|z| < r$ such that

$$F(re^{i\theta}) = f(\theta)$$

Moreover, F is unique.

Proof:

It suffices to show this for real-valued f since harmonicity is closed under linearity. Uniqueness is also easy: if F_1, F_2 are two harmonic extensions of f , then $F_1 - F_2 = 0$ on $|z| = r$. By the Maximum Modulus principle, $F_1 - F_2 = 0$ on $|z| \leq r$, which implies $F_1 = F_2$.

For existence, we shall take advantage of what we found in the beginning of last section, namely define

$$F(z) = \frac{1}{2i} \int_0^{2\pi} f(\theta) \frac{r^2 - |z|^2}{|re^{i\theta} - z|^2} d\theta$$

We know that F is the real part of the holomorphic function:

$$\frac{1}{2i} \int_0^{2\pi} f(\theta) \frac{re^{i\theta} + z}{re^{i\theta} - z} d\theta$$

and hence is Harmonic. Certainly $F(z) = f(z)$ on the boundary, so we need to check that it is continuous on the boundary. First, recall equation (7.1)

$$\frac{1}{2\pi} \int_0^{2\pi} \frac{r^2 - |z|^2}{|re^{i\theta} - z|^2} d\theta = 1$$

Now, consider:

$$\begin{aligned} &= F(z) - f(\theta_0) \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(\theta) \frac{r^2 - |z|^2}{|re^{i\theta} - z|^2} d\theta - \frac{1}{2\pi} \int_0^{2\pi} f(\theta_0) \frac{r^2 - |z|^2}{|re^{i\theta} - z|^2} d\theta \\ &= \frac{1}{2\pi} \int_{|\theta - \theta_0| \leq \eta} (f(\theta) - f(\theta_0)) \frac{r^2 - |z|^2}{|re^{i\theta} - z|^2} d\theta + \frac{1}{2\pi} \int_{|\theta - \theta_0| > \eta} (f(\theta) - f(\theta_0)) \frac{r^2 - |z|^2}{|re^{i\theta} - z|^2} d\theta \end{aligned}$$

This can be done for any η , in particular we want to choose one so that the value of the equation goes to zero. Since f is continuous, we know $\sup_{|\theta-\theta_0|\leq\eta} |f(\theta) - f(\theta_0)|$ can be arbitrarily small by choosing η appropriately. In particular, choose η so that the first integral is less than $\epsilon/2$ for some $\epsilon > 0$ (that is, we will integrate over a sufficiently small arc). Then:

$$\left| \frac{1}{2\pi} \int_{|\theta-\theta_0|>\epsilon} (f(\theta) - f(\theta_0)) \frac{r^2 - |z|^2}{|re^{i\theta} - z|^2} d\theta \right| \leq M \cdot \frac{1}{2\pi} \int_{|\theta-\theta_0|>\epsilon} \frac{r^2 - |z|^2}{|re^{i\theta} - z|^2} d\theta$$

Then by lemma 7.2.1, the integral on the right hand side can be made arbitrarily small, in particular we can make it smaller than $\epsilon/2$, hence giving us continuity of F on the boundary, hence completing the proof.

This gives us a new way to characterize harmonic functions without needing to rely on differentiability, showing they can be defined in a much weaker context¹

Theorem 7.2.3: Mean Value Property, Then Harmonic

Let f be a continuous function on an open set Σ that satisfies the mean value property. Then f is harmonic.

Proof:

It suffices to show that f is locally harmonic. Let $z \in \Omega$ be arbitrary and D a disk centered at z such that $D \subseteq \Omega$. Then $f|_{\partial D}$ is continuous and thus by the above theorem there exists a continuous F that agrees on f on the boundary and is harmonic on the interior of D . Since F and f satisfy the maximum modulus principle, so does $F - f$. But $F - f$ is zero on the boundary, and hence must be identically zero on D . Thus, $f = F$ is harmonic on D .

(perhaps use Harnack's inequality to prove Harnack's Theorem? When the convergence of harmonic functions is harmonic, commented)

Exercise 7.2.1

1. Let $\Omega \subseteq \mathbb{C}$ be a bounded simply connected domain whose boundary is a Jordan curve. Show that the Dirichlet problem can be solved on Ω (use the Riemann mapping theorem together with the Carathéodory boundary theorem, theorem 6.1.13)

¹However, the mean value property implies smoothness, and hence it turns out that this is not a weakening of the conditions

Higher Dimensional Complex Analysis

In this chapter, we shall go over the theory for higher dimensions. In many ways, this theory is not very central to complex analysis, as it ventures more into the realm of differential geometry and away from many of the interesting results afforded to complex differentiable functions. Nonetheless, many results from one dimensions naturally generalize to higher dimensions (for example, Cauchy's Theorem), while on the other hand many key results do not (are the zeros of two dimensions holomorphic functions still isolated? What consequences does that bring?)

Note that this chapter is more of an overview than a deep-dive into the matter touching on some very important results but not exploring the matters in great depth as compared to the exposition in earlier chapters.

8.1 Basic Definitions

The formal power series on two variables is defined to be $k[[x, y]]$ where for any element $f(x, y) \in k[[x, y]]$ we have

$$f(x, y) = \sum_{i,j} a_{i,j} x^i y^j$$

Note that $k[[x]][y] \neq k[y][[x]]$. Addition, scalar multiplication, and multiplication make $k[[x, y]]$ into a k -algebra. The *order* of a formal power series is the smallest n such that

$$\sum_{i+j=n} a_{i,j} x^i y^j \neq 0$$

The order of the product of two non-zero series is the sum of the orders of the series, hence $k[[x, y]]$ is an integral domain. If $k \in \{\mathbb{R}, \mathbb{C}\}$, we may consider for any $f(x, y) \in k[[x, y]]$ the series

$$|f|(x, y) = \sum_{i, j \geq 0} |a_{i, j}| (r_1)^i (r_2)^j$$

where $r_1, r_2 \geq 0$. We want to find all positive r_p that satisfy the equation. Let Γ be the set of points where $r_1, r_2 \geq 0$ where

$$|f|(x, y) = \sum_{i, j \geq 0} |a_{i, j}| (r_1)^i (r_2)^j < \infty$$

Then using Γ , we can find the set of all points in $\mathbb{C}$ where f is an *absolutely convergent series*. The set Γ is certainly non-empty since $(0, 0) \in \Gamma$. This set is given a name:

Definition 8.1.1: Domain Of Convergence

Given a series $f(x, y) \in k[[x, y]]$, is defined to be

$$\Delta = \text{int}(\Gamma)$$

that is, it is the interior of Γ .

Note that Δ may be empty, just recall the example for the one-dimensional case, and is always open. In the one dimensional case, Δ would simply be $(0, \rho)$ where ρ is the radius of convergence.

Proposition 8.1.2: Openness Of Domain Of Convergence

If Δ is the domain of convergence, then $(r_1, r_2) \in \Delta$ if and only if there exists $r'_1 > r_1, r'_2 > r_2$ such that $(r'_1, r'_2) \in \Gamma$

Proof:

if:

$$\sum_{i, j} |a_{i, j}| (r'_1)^i (r'_2)^j < \infty$$

then certainly the same works for (r_1, r_2) , and the converse follows from continuity.

With this, we may ask when a series converges. Just like for the one dimensional case, we need Abels lemma

Lemma 8.1.3: Abel's Lemma Higher Dimension

If $|a_{i, j}| (r'_1)^i (r'_2)^j \leq M$, where M is independent of i, j , and if $r_1 < r'_1, r_2 < r'_2$, then the series $\sum_{i, j} a_{i, j} z_1^i z_2^j$ converges absolutely and uniformly for $|z_1| \leq r_1$ and $|z_2| \leq r_2$

Proof:

The argument is that of Abel's Theorem (theorem 2.3.2): the terms are bounded by those of a double geometric progression. Put $q_1 = r_1/r'_1 < 1$ and $q_2 = r_2/r'_2 < 1$. For $|z_1| \leq r_1$ and

$$|z_2| \leq r_2,$$

$$|a_{i,j} z_1^i z_2^j| \leq |a_{i,j}| (r'_1)^i (r'_2)^j q_1^i q_2^j \leq M q_1^i q_2^j$$

so every finite partial sum of $\sum_{i,j} |a_{i,j} z_1^i z_2^j|$ is at most

$$M \sum_{i \geq 0} q_1^i \sum_{j \geq 0} q_2^j = \frac{M}{(1 - q_1)(1 - q_2)}$$

The series of absolute values has nonnegative terms and bounded partial sums, so it converges, and the series $\sum_{i,j} a_{i,j} z_1^i z_2^j$ converges absolutely, hence to a sum independent of the order of summation. Since the bound $M q_1^i q_2^j$ does not depend on z_1, z_2 , the tail beyond any finite set of indices containing all (i, j) with $i, j \leq N$ is at most $M \sum_{\max(i,j) > N} q_1^i q_2^j$, which tends to 0 as $N \rightarrow \infty$, so the convergence is uniform on $|z_1| \leq r_1, |z_2| \leq r_2$, as claimed.

using this, we get that $(r_1, r_2) \in \Delta$ then $f(z_1, z_2)$ converges for $|z_1| \leq r_1$ and $|z_2| \leq r_2$ and that if $(|z_1|, |z_2|) \notin \bar{\Gamma}$, then $f(z_1, z_2)$ is divergent.

Note that we will sometimes abuse language and say a point (z_1, z_2) is in the domain of convergence if $(|z_1|, |z_2|)$ is in the domain of convergence. We do this all the time in 1-dimensions, where we say the domain of convergence is the open disk $|z| < \rho$ (where ρ is the radius of convergence).

Consider now a function $f : \mathbb{C}^n \rightarrow \mathbb{C}$. Then:

Definition 8.1.4: Holomorphic Over Several Variables

Let $D \subseteq \mathbb{C}^n$ be open and let $f : D \rightarrow \mathbb{C}$ be a complex valued function. Then f is *holomorphic* if f is continuous and holomorphic in each variable separately: for each k and each fixed value of the other coordinates, $z_k \mapsto f(z_1, \dots, z_n)$ is holomorphic on the open set of z_k for which the point lies in D . This is the notion used for the transition maps in definition 6.2.1.

We shall see that a holomorphic function in this sense is analytic, that is, locally the sum of a convergent power series in $z_1, \dots, z_n$ (contrary to the real case), see theorem 8.1.7. To that end, we shall go down the same path as the one-dimensional path and show the higher-dimensional analogue of Cauchy's integral formula. It extends from disks to polydisks around an arbitrary centre by iterating the one-variable formula over each coordinate in turn.

Proposition 8.1.5: Polydisk Cauchy Integral Formula

Let $a = (a_1, \dots, a_n) \in \mathbb{C}^n$ and $r = (r_1, \dots, r_n)$ with each $r_k > 0$, and let f be holomorphic on a neighbourhood of the closed polydisk $\prod_{k=1}^n \{|z_k - a_k| \leq r_k\}$. Then for every z with $|z_k - a_k| < r_k$ for all k ,

$$f(z_1, \dots, z_n) = \frac{1}{(2\pi i)^n} \int_{|\zeta_1 - a_1| = r_1} \cdots \int_{|\zeta_n - a_n| = r_n} \frac{f(\zeta_1, \dots, \zeta_n)}{(\zeta_1 - z_1) \cdots (\zeta_n - z_n)} d\zeta_1 \cdots d\zeta_n$$

each circle $|\zeta_k - a_k| = r_k$ being traversed once in the positive direction.

Proof:

Argue by induction on n , the case $n = 1$ being Cauchy's integral formula (theorem 3.2.5). Fix z in the open polydisk. Holding $z_2, \dots, z_n$ fixed, the function $\zeta_1 \mapsto f(\zeta_1, z_2, \dots, z_n)$ is holomorphic on a neighbourhood of the closed disc $|\zeta_1 - a_1| \leq r_1$, so the one-variable formula gives

$$f(z_1, z_2, \dots, z_n) = \frac{1}{2\pi i} \int_{|\zeta_1 - a_1| = r_1} \frac{f(\zeta_1, z_2, \dots, z_n)}{\zeta_1 - z_1} d\zeta_1$$

Applying the formula for $n - 1$ variables to $f(\zeta_1, z_2, \dots, z_n)$ in the variables $z_2, \dots, z_n$, with ζ_1 held on its circle, and substituting into the line above, the joint continuity of f lets the iterated integral be combined into the stated integral over the product of circles, completing the induction.

Expanding each Cauchy kernel as a geometric series turns this representation into a power series, again around an arbitrary centre and now in any number of variables.

Proposition 8.1.6: Higher-dimensional Power Series Expansion

Let f be holomorphic on the open polydisk $\Delta = \prod_{k=1}^n \{|z_k - a_k| < r_k\}$ around $a = (a_1, \dots, a_n) \in \mathbb{C}^n$. Then f has a power series expansion

$$f(z) = \sum_{p \in \mathbb{N}^n} c_p (z - a)^p, \quad (z - a)^p := \prod_{k=1}^n (z_k - a_k)^{p_k}$$

converging normally, that is, uniformly on every compact subset of Δ , with coefficients

$$c_p = \frac{1}{(2\pi i)^n} \int_{|\zeta_1 - a_1| = \rho_1} \cdots \int_{|\zeta_n - a_n| = \rho_n} \frac{f(\zeta)}{(\zeta_1 - a_1)^{p_1+1} \cdots (\zeta_n - a_n)^{p_n+1}} d\zeta_1 \cdots d\zeta_n$$

independent of the radii ρ_k with $0 < \rho_k < r_k$.

Proof:

Fix a compact subset of Δ . It lies in a closed subpolydisk $\prod_k \{|z_k - a_k| \leq r'_k\}$ with $r'_k < \rho_k < r_k$. For z there and ζ on the product of circles $|\zeta_k - a_k| = \rho_k$, expand each Cauchy kernel as a geometric series

$$\frac{1}{\zeta_k - z_k} = \frac{1}{(\zeta_k - a_k) - (z_k - a_k)} = \sum_{p_k \geq 0} \frac{(z_k - a_k)^{p_k}}{(\zeta_k - a_k)^{p_k+1}}$$

which converges because $|z_k - a_k| \leq r'_k < \rho_k = |\zeta_k - a_k|$, uniformly in ζ_k and in z on the compact set. Multiplying the n kernels and substituting into the polydisk Cauchy formula (proposition 8.1.5), the product of series converges uniformly on the product of circles, so it may be integrated term by term, yielding $f(z) = \sum_{p \in \mathbb{N}^n} c_p (z - a)^p$ with c_p as stated. The convergence is uniform on the chosen compact set, hence normal on Δ . Independence of the ρ_k follows from Cauchy's theorem, the integrand being holomorphic in each ζ_k on the annulus between two admissible radii.

The above is a special case of something called the Bochner-Martinelli formula which is the higher-dimensional analogue of Cauchy's integral formula within harmonic analysis. For those interested,

they may checkout [17].

Theorem 8.1.7: Component-wise Holomorphic, then Holomorphic

Let f be a continuous function on an open set D which is holomorphic in each component. Then f is analytic in D .

Furthermore, if f is once complex-differentiable, it is infinity complex differentiable.

Proof:

Follows immediately from proposition 8.1.6.

8.1.8 Note In fact the continuity hypothesis can be dropped: a function on an open subset of $\mathbb{C}^n$ which is holomorphic in each variable separately is automatically continuous, hence analytic. This is Hartogs' theorem on separately holomorphic functions, which is harder to prove and is not done here.

A basic source of compact complex manifolds in dimension g comes from quotienting $\mathbb{C}^g$ by a lattice, generalizing the one-dimensional torus $\mathbb{C}/\Gamma$ of example 6.2.2.

Proposition 8.1.9: The Complex Torus $\mathbb{C}^g/\Lambda$

Let $\Lambda \subseteq \mathbb{C}^g$ be a lattice, that is, a discrete subgroup of rank $2g$, acting on $\mathbb{C}^g$ by translation. Then the quotient $X = \mathbb{C}^g/\Lambda$ carries a unique structure of complex manifold of dimension g for which the projection $\pi: \mathbb{C}^g \rightarrow X$ is holomorphic and locally biholomorphic, and X is compact.

Proof:

Because Λ is discrete, its smooth translation action on $\mathbb{C}^g$ is free and properly discontinuous: for each $v \in \mathbb{C}^g$ there is an open neighbourhood U with $(U + \lambda) \cap U = \emptyset$ for every nonzero $\lambda \in \Lambda$, and the orbit space is Hausdorff. Consequently X is a smooth manifold and π is a smooth covering map ([23, Covering Quotient Theorem]), with $\pi|_U: U \rightarrow \pi(U)$ a homeomorphism for each such U . Take as charts the pairs $(\pi(U), (\pi|_U)^{-1})$. On an overlap $\pi(U) \cap \pi(U')$ the transition map $(\pi|_{U'})^{-1} \circ \pi|_U$ agrees, on each connected component, with a translation $w \mapsto w + \lambda$ of $\mathbb{C}^g$ by some $\lambda \in \Lambda$, and translations are biholomorphic, so the atlas is holomorphic and X is a complex manifold of dimension g (definition 6.2.1), with π holomorphic and locally biholomorphic. This is the only such structure, since any two making π locally biholomorphic agree on every chart $\pi(U)$. Finally Λ has rank $2g$, so a closed fundamental parallelopete for the action is compact and surjects onto X under π , whence X is compact.

Exercise 8.1.1

1. Find the higher dimensional equivalent of the following:
 - a) Cauchy inequalities
 - b) the composition of holomorphic functions is holomorphic

- c) the implicit function theorem and inverse function theorem
- d) uniform convergence on compact sets is holomorphic

8.2 From Roots to Zero Sets: Hartog's Theorems

In the case of one dimensions, it was shown that roots are isolated, in particular if $f(z_0) = 0$ then some open neighbourhood $w \in B_\epsilon(z_0) \setminus \{z_0\}$ we have $f(w) \neq 0$. This is certainly no longer the case in higher dimensions! This is immediate if we just consider a polynomial example such as

$$f(z, w) = z^2 + w^2 - 1$$

which naturally has as zero set the complex circle $z^2 + w^2 = 1$. The reader may recall that we have encountered zero sets of $C^n(M)$ or $C^\infty(M)$ functions for some real (differentiable or smooth) manifold M when we used the inverse function theorem or implicit function theorem to construct the manifold using zero sets of functions F where the derivative F is nonzero. The higher-dimensional analogue of the inverse function theorem (exercise 8.1.1 (1)) shows that the same is true for the complex case, and hence as long as the determinant of the derivative is nonzero we shall get a $n - 1$ dimensional manifold (given an n -dimensional complex function). However, the zero-set of a function may contain points where the derivative is zero, the critical points, and hence the zero-sets need not be manifolds (for example $F(z, w) = z^2 - w^3$). In the real case, this is a complicated area of study as the behaviour could be a good deal more complicated.

Theorem 8.2.1: Hartogs Extension Theorem

Let $\Omega \subseteq \mathbb{C}^n$ be open with $n \geq 2$, and let $K \subseteq \Omega$ be a compact subset such that $\Omega \setminus K$ is connected. Then every holomorphic function f on $\Omega \setminus K$ (see definition 8.1.4) extends uniquely to a holomorphic function on all of Ω .

The proof of the general statement needs the solution of the equation $\bar{\partial}u = v$ for $(0, 1)$ -forms with compact support in $\mathbb{C}^n$, $n \geq 2$, through the Bochner-Martinelli kernel, which this book does not develop. A complete proof is [9, Chapter I, (3.28)]. The special case of a Hartogs figure in a polydisc is proved in lemma 8.2.2 below.

Proof Key ideas:

Uniqueness holds because $\Omega \setminus K$ is a nonempty open subset of each component of Ω that it meets, and two holomorphic functions agreeing on it agree on that component (proposition 6.2.3). For existence, choose a smooth function ψ with compact support in Ω , equal to 1 near K , and put $f_0 = (1 - \psi)f$, extended by 0 on K , a smooth function on Ω . The form $v = \bar{\partial}f_0 = -f \bar{\partial}\psi$ extends by 0 to a smooth $(0, 1)$ -form with compact support in $\mathbb{C}^n$ satisfying $\bar{\partial}v = 0$, and for $n \geq 2$ the equation $\bar{\partial}u = v$ has a smooth solution u with compact support. Then $\tilde{f} = f_0 - u$ is holomorphic on Ω . The function u is holomorphic off the support of ψ and vanishes on the unbounded component of its complement, so $\tilde{f} = f$ on a nonempty open subset of the connected set $\Omega \setminus K$, hence on all of it, which completes the sketch.

Write $\mathbb{D}(c, \rho)$ for the open disc of radius ρ centred at $c \in \mathbb{C}$.

Lemma 8.2.2: Hartogs Figure Extension

Let $n \geq 2$ and write points of $\mathbb{C}^n$ as (z', w) with $z' \in \mathbb{C}^{n-1}$ and $w \in \mathbb{C}$.

1. Let $V \subseteq \mathbb{C}$ be an open set containing the circle $|\zeta - c| = t$, let $U \subseteq \mathbb{C}^m$ be open, and let φ be holomorphic on $V \times U$ (definition 8.1.4). Then $g(u) = \int_{|\zeta-c|=t} \varphi(\zeta, u) d\zeta$ is holomorphic on U .
2. Let $P' \subseteq \mathbb{C}^{n-1}$ be an open polydisc, $Q' \subseteq P'$ a nonempty connected open set, $c \in \mathbb{C}$ and $0 < s < r$, and put

$$H = (Q' \times \mathbb{D}(c, r)) \cup \{(z', w) \mid z' \in P', s < |w - c| < r\}$$

Then every holomorphic function on H extends uniquely to a holomorphic function on $P' \times \mathbb{D}(c, r)$.

Proof:

1. The function g is continuous, since φ is uniformly continuous on the product of the circle with a compact neighbourhood of a given point of U . Fix $u_0 \in U$. Each point ζ_0 of the circle has a closed polydisc neighbourhood $B_{\zeta_0} \times B'_{\zeta_0} \subseteq V \times U$ of (ζ_0, u_0) on which φ is the uniform limit of the partial sums of a power series in $(\zeta - \zeta_0, u - u_0)$, by theorem 8.1.7 and proposition 8.1.6. Finitely many of the interiors of the discs B_{ζ_0} cover the circle, so it splits into finitely many arcs $\sigma_1, \dots, \sigma_N$, each lying in one of these discs, and we let B' be the intersection of the corresponding finitely many polydiscs B'_{ζ_0} . On σ_j , integrating the partial sums in ζ gives polynomials in u , which converge uniformly on B' to $\int_{\sigma_j} \varphi(\zeta, u) d\zeta$ by the length estimate of proposition 3.1.1. A uniform limit of polynomials is continuous and, by lemma 4.1.3 applied in each variable separately, holomorphic in each variable on the interior of B' . Adding over j (proposition 3.1.1), g is holomorphic near u_0 .
2. Choose $s < \tau < t < r$, put $U = P' \times \mathbb{D}(c, \tau)$, and for $(z', w) \in U$ let

$$F(z', w) = \frac{1}{2\pi i} \int_{|\zeta-c|=t} \frac{f(z', \zeta)}{\zeta - w} d\zeta$$

The integrand is holomorphic in (ζ, z', w) on $\{\zeta \mid \tau < |\zeta - c| < r\} \times U$, since the points (z', ζ) lie in H and $\zeta \neq w$ there, so F is holomorphic on U by (1). For $z' \in Q'$, the function $\zeta \mapsto f(z', \zeta)$ is holomorphic on the whole disc $\mathbb{D}(c, r)$, and Cauchy's integral formula (theorem 3.2.5) gives $F(z', w) = f(z', w)$ for $|w - c| < \tau$. So $F = f$ on the nonempty open set $Q' \times \mathbb{D}(c, \tau)$. The set

$$H \cap U = (Q' \times \mathbb{D}(c, \tau)) \cup \{(z', w) \mid z' \in P', s < |w - c| < \tau\}$$

is connected: the two pieces are products of connected sets ([8, Finite Product Of Connected Spaces]), the annulus being path-connected ([8, Path-Connected $\Rightarrow$ Connected]), and they share a point ([8, points in common between connected sets]). By proposition 6.2.3, $F = f$ on $H \cap U$. Every point of $P' \times \mathbb{D}(c, r)$ outside U has $|w - c| \geq \tau > s$, so it lies in H , and the function equal to f on H and to F on U is well defined and holomorphic on $P' \times \mathbb{D}(c, r)$. It is unique by proposition 6.2.3, as H is a nonempty open subset of the connected set $P' \times \mathbb{D}(c, r)$, completing the proof.

This result is false in 1 dimension: simply take $f(z) = 1/z$ on $\mathbb{C} \setminus \overline{B_1(0)}$. Let us start with a weaker statement that is a special case of the above

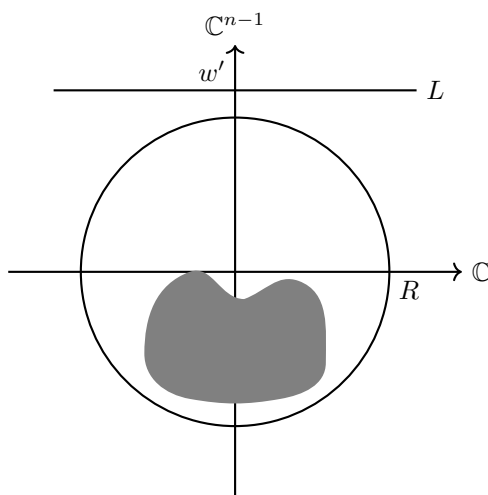
Lemma 8.2.3: Stronger Liouville Theorem

Let $K \subseteq \mathbb{C}^n$ be a compact subset where $n \geq 2$. Then every bounded holomorphic function $f \in \mathcal{O}(\mathbb{C}^n \setminus K)$ is constant on unbounded components, and hence admits a holomorphic extension from that component to all of $\mathbb{C}^n$

Note this is not true in 1 dimension, $f(z) = 1/z$ being such an example!

Proof:

Let f be the given bounded function and K the compact in $\mathbb{C}^n$. As K is bounded, let R be large enough so that $K \subseteq B_R(0)$. Take $w_0 \in \mathbb{C}^{n-1}$ where $|w_0| > R$ and consider the line $L = \{(z_1, w_0) \mid z_1 \in \mathbb{C}\}$:



Notice how we took advantage of the that we were in more than one dimension to pick w_0 . We can now define $f(z_1, w_0)$ to be a 1-dimensional holomorphic function. Then by the usual Liouville's Theorem it is constant. Repeating the same for each variable (potentially adjusting the radius if needed) we see that f is constant in each variable $z_1, \dots, z_n$, and hence is constant out side some ball $B_R(0)$ for appropriate R . But then by the higher dimensional analogue of theorem 2.5.3 f is constant, completing the proof.

The proof of lemma 8.2.2 uses the Cauchy integral formula in the same way, one variable at a time. The maximal extension is given a name:

Definition 8.2.4: Domain of Holomorphy

Let f be a holomorphic function on a domain $D \subseteq \mathbb{C}^n$. Then the *domain of holomorphy* is the maximal extension of D for which f is holomorphic.

Another way of looking at the domain of holomorphic is that D is such a domain if no $f \in \mathcal{O}(D)$ can extend by an open subset on any boundary point of ∂D ; this is called *weak holomorphy* and it is a non-trivial result to show these two are the same. A good deal of work goes into finding properties of domain of Holomorphy. For example, all euclidean convex domains are domains of Holomorphy.

8.2.5 Cannot Extend Riemann Mapping Theorem Let D be a convex euclidean domain and $p \in \partial D$. After a translation and a rotation, $p = 0$ and, due to convexity, D lies completely on one side of the hyperplane $\operatorname{Re} z_1 = 0$. It follows that $f_0(z) = 1/z_1$ defines a function in $\mathcal{O}(D)$, and $|f_0(z)| \rightarrow \infty$ as $z \rightarrow 0$ inside D .

Let $H = (\{z \mid |z| < \frac{1}{2}\} \times \mathbb{D}) \cup (\mathbb{D} \times \{w \mid \frac{1}{2} < |w| < 1\}) \subseteq \mathbb{C}^2$, the Hartogs figure of lemma 8.2.2 with $P' = \mathbb{D}$, $Q' = \{z \mid |z| < \frac{1}{2}\}$, $c = 0$, $s = \frac{1}{2}$ and $r = 1$, inside the bidisc $P = \mathbb{D} \times \mathbb{D}$. Suppose that $\Phi: H \rightarrow D$ were a biholomorphism onto a convex domain D . By lemma 8.2.2, the components of Φ and $f \circ \Phi$ for every $f \in \mathcal{O}(D)$ extend holomorphically to P . The point $q = (\frac{1}{2}, 0) \in P$ is not in H , but the points $h_k = ((1 - \frac{1}{k})\frac{1}{2}, 0)$ are, and $\Phi(h_k)$ tends to the value y at q of the extension of Φ . If y were in D , then $h_k = \Phi^{-1}(\Phi(h_k)) \rightarrow \Phi^{-1}(y) \in H$, so $q \in H$, which is false. So $y \in \partial D$, and with f_0 built at y as above, $|f_0(\Phi(h_k))| \rightarrow \infty$, while $f_0 \circ \Phi$ extends holomorphically to P and so stays bounded near q . Hence there is no biholomorphic map from H onto any convex domain. In particular, even though H is simply connected, being contractible through the maps $(z, w) \mapsto ((1-t)z, w)$, $t \in [0, 1]$, which keep H inside itself and end in the disc $\{0\} \times \mathbb{D}$, H is *not biholomorphically equivalent* to a bi-disc or to a ball! Hence there is no analogue of the Riemann mapping theorem in higher dimensions!

By the observation at the beginning of this section and the Hartogs extension theorem, all isolated singularities must in fact be removable:

Corollary 8.2.6: Higher Dimensions: Singularity, then Removable Singularity

Let $n \geq 2$, let $D \subseteq \mathbb{C}^n$ be a domain and $p \in D$, and let f be holomorphic on $D \setminus \{p\}$, so that p is an isolated singularity of f . Then p is a removable singularity: f extends to a holomorphic function on D .

Proof:

Write $p = (p', c)$ with $p' \in \mathbb{C}^{n-1}$ and $c \in \mathbb{C}$, and choose an open polydisc $B = P' \times \mathbb{D}(c, r) \subseteq D$ centred at p . Choose a point $q' \in P'$ with $q' \neq p'$ and an open polydisc $Q' \subseteq P'$ around q' not containing p' , and let $0 < s < r$. The Hartogs figure $H = (Q' \times \mathbb{D}(c, r)) \cup \{(z', w) \mid z' \in P', s < |w - c| < r\}$ of lemma 8.2.2 does not contain p , so f is holomorphic on H , and lemma 8.2.2 gives a holomorphic F on B equal to f on H . The set $B \setminus \{p\}$ is connected. Indeed, let $x, y \in B \setminus \{p\}$. If the segment $[x, y]$ misses p , it joins x to y inside the convex set B . Otherwise p lies on the real line L through x and y , and since $\mathbb{C}^n = \mathbb{R}^{2n}$ with $2n \geq 4$ there is a point $u \in B$ off L , for instance $u = p + \epsilon v$ with v not in the direction of L and $\epsilon > 0$ small. The segments $[x, u]$ and $[u, y]$ lie in B and meet L only at x and at y , so they miss p and join x to y in $B \setminus \{p\}$. So F and f , which are holomorphic on the connected set $B \setminus \{p\}$ and agree on its nonempty open subset H , agree on $B \setminus \{p\}$ (proposition 6.2.3). Setting $f(p) = F(p)$ then defines a function on D which agrees with F on B and with f on $D \setminus \{p\}$, both open sets on which it is holomorphic, so it is holomorphic on D , as we sought to show.

Theorem 8.2.7: Higher Dimensions - Riemann Removable Singularity Theorem

Let $D \subseteq \mathbb{C}^n$ be a domain. Let $E \subseteq D$ be a closed subset of D with the following property: for each $p \in E$, there exists an open subset $U_p \subseteq \mathbb{C}^n$ containing p such that there exists a $g_p \in \mathcal{O}(U_p)$ which is zero on $U_p \cap E$ but not identically zero on any neighbourhood of p .

Then if f is bounded in $\mathcal{O}(D \setminus E)$ (similarly to the removable singularity case), then f extends to a holomorphic function in $\mathcal{O}(D)$.

Proof:

Let M bound $|f|$ on $D \setminus E$, which is open since E is closed in D .

Let h be holomorphic on an open $V \subseteq \mathbb{C}^n$ and let $\ell(t) = a + tb$ be an affine map $\mathbb{C} \rightarrow \mathbb{C}^n$. Then $h \circ \ell$ is holomorphic on the open set $\{t \in \mathbb{C} \mid \ell(t) \in V\}$. Indeed, h is analytic by theorem 8.1.7, hence continuously differentiable, and it is holomorphic in each variable, so each barred Wirtinger derivative $\partial h / \partial \bar{z}_j$ vanishes, being the Cauchy-Riemann equations in the j -th variable as computed after definition 2.2.11. The chain rule then gives

$$\frac{\partial(h \circ \ell)}{\partial \bar{t}} = \sum_{j=1}^n \frac{\partial h}{\partial z_j} \frac{\partial(a_j + tb_j)}{\partial \bar{t}} + \sum_{j=1}^n \frac{\partial h}{\partial \bar{z}_j} \frac{\partial(\overline{a_j + tb_j})}{\partial \bar{t}} = 0 + \sum_{j=1}^n \frac{\partial h}{\partial \bar{z}_j} \bar{b}_j = 0$$

so $h \circ \ell$ is a C^1 function satisfying the Cauchy-Riemann equations, hence holomorphic by theorem 2.2.9. Consequently, for any invertible affine map $A : \mathbb{C}^n \rightarrow \mathbb{C}^n$ the composite $h \circ A$ is holomorphic on $A^{-1}(V)$: it is continuously differentiable, and each of its partial maps $t \mapsto h(A(z) + (t - z_j)(A(e_j) - A(0)))$ is an affine slice of h , hence holomorphic.

Fix $p \in E$, with U_p and $g := g_p$ as in the hypothesis. Choose an open ball $B \subseteq U_p \cap D$ centered at p . Since g is not identically zero on B , there is $q \in B$ with $g(q) \neq 0$, and $q \neq p$ because $g(p) = 0$, the point p lying in $U_p \cap E$. Set $b := q - p$. The function $h(t) := g(p + tb)$ is holomorphic on the disc $\{t \mid |t| < R_0\}$, where $R_0 > 1$ is the radius of B divided by $|b|$, by the slicing device. We have $h(0) = g(p) = 0$ and $h(1) = g(q) \neq 0$. If h vanished identically on a neighbourhood of 0, then theorem 2.5.3 on the connected disc would force $h(1) = 0$. So 0 is an isolated zero of h by proposition 2.5.7: there is $R_1 \in (0, 1)$ with $h(t) \neq 0$ for $0 < |t| \leq R_1$.

Choose an invertible linear map $L : \mathbb{C}^n \rightarrow \mathbb{C}^n$ with $Le_n = b$ and set $A(w) := p + Lw$, an invertible affine map with $A(0) = p$. By the slicing device, $\tilde{g} := g \circ A$ is holomorphic on $A^{-1}(U_p \cap D)$ and $\tilde{f} := f \circ A$ is holomorphic on $A^{-1}(D \setminus E)$, with $|\tilde{f}| \leq M$. Set $\tilde{E} := A^{-1}(E)$, so that $\tilde{E} \cap A^{-1}(U_p) \subseteq \{w \mid \tilde{g}(w) = 0\}$ and $0 \in \tilde{E}$. Write $w = (w', w_n) \in \mathbb{C}^{n-1} \times \mathbb{C}$. Then $\tilde{g}(0', t) = g(p + tb) = h(t) \neq 0$ for $0 < |t| \leq R_1$.

The compact set $\{0'\} \times \{\zeta \mid |\zeta| \leq R_1\}$ lies in the open set $A^{-1}(U_p \cap D)$, and the compact annulus $\{0'\} \times \{\zeta \mid R_1/2 \leq |\zeta| \leq R_1\}$ lies in the open set where $\tilde{g} \neq 0$. By compactness there is $\delta > 0$ such that, writing $r := R_1/2$,

$$Q := \{(w', \zeta) \mid |w'| \leq \delta, |\zeta| \leq R_1\} \subseteq A^{-1}(U_p \cap D)$$

and

$$\tilde{g} \neq 0 \text{ on } \{(w', \zeta) \mid |w'| \leq \delta, r \leq |\zeta| \leq R_1\}$$

Here $|w'| \leq \delta$ abbreviates $|w_j| \leq \delta$ for $j = 1, \dots, n-1$. Two consequences. First, no point (w', ζ) of Q with $r \leq |\zeta| \leq R_1$ lies in $\tilde{E}$, since $\tilde{E} \cap Q$ is contained in the zero set of $\tilde{g}$. Second, for each

fixed w' with $|w'| \leq \delta$, the partial map $\zeta \mapsto \tilde{g}(w', \zeta)$ is holomorphic on an open set containing the disc $\{\zeta \mid |\zeta| \leq R_1\}$ and is nonzero for $r \leq |\zeta| \leq R_1$, hence is not identically zero, so by proposition 2.5.7 its zeros in that disc form a finite set contained in $\{\zeta \mid |\zeta| < r\}$. Therefore the slice of $\tilde{E}$,

$$N_{w'} := \{\zeta \mid |\zeta| < R_1, (w', \zeta) \in \tilde{E}\}$$

is a finite subset of $\{\zeta \mid |\zeta| < r\}$.

Fix w' with $|w'| < \delta$. The partial map $\varphi(\zeta) := \tilde{f}(w', \zeta)$ is defined and holomorphic at every ζ with $|\zeta| < R_1$ and $\zeta \notin N_{w'}$, and $|\varphi| \leq M$. Around each of the finitely many points of $N_{w'}$, the function φ is bounded and holomorphic on a punctured disc, so proposition 3.4.6 extends it holomorphically across that point. Patching these extensions with φ yields $\hat{\varphi}$ holomorphic on the disc $\{\zeta \mid |\zeta| < R_1\}$ with $\hat{\varphi} = \varphi$ off $N_{w'}$. The circle $|\zeta| = r$ is nullhomotopic in that disc and winds once around each point of $\{\zeta \mid |\zeta| < r\}$, so Cauchy's integral formula (theorem 3.2.5) gives, for $|w_n| < r$,

$$\hat{\varphi}(w_n) = \frac{1}{2\pi i} \int_{|\zeta|=r} \frac{\hat{\varphi}(\zeta)}{\zeta - w_n} d\zeta = \frac{1}{2\pi i} \int_{|\zeta|=r} \frac{\tilde{f}(w', \zeta)}{\zeta - w_n} d\zeta$$

the second equality because the circle avoids $N_{w'} \subseteq \{\zeta \mid |\zeta| < r\}$. Define, on the polydisc $\Delta := \{(w', w_n) \mid |w'| < \delta, |w_n| < r\}$,

$$F(w', w_n) := \frac{1}{2\pi i} \int_{|\zeta|=r} \frac{\tilde{f}(w', \zeta)}{\zeta - w_n} d\zeta$$

By the two displays, $F(w', w_n) = \hat{\varphi}(w_n) = \tilde{f}(w', w_n)$ whenever $(w', w_n) \in \Delta \setminus \tilde{E}$.

The function F is holomorphic on Δ . The integration set $T := \{(w', \zeta) \mid |w'| \leq \delta, |\zeta| = r\}$ is a compact subset of the open set $A^{-1}(D \setminus E)$, on which $\tilde{f}$ is holomorphic, hence continuous, so $\tilde{f}$ is uniformly continuous on T . Continuity of F : for (w', w_n) ranging over a compact block $|w'| \leq \delta' < \delta$, $|w_n| \leq r' < r$, the integrand is a uniformly continuous function of (w', w_n, ζ) , the denominator being bounded below by $r - r' > 0$, and integration over the fixed circle preserves uniform limits. Holomorphy of F in each variable separately follows from theorem 4.1.6, applied with X the circle $|\zeta| = r$ carrying its arc-length measure and with all variables but one held fixed, the free variable s ranging over $|s| < \delta$ (if s is one of $w_1, \dots, w_{n-1}$) or $|s| < r$ (if $s = w_n$). The integrand $\zeta \mapsto \tilde{f}(w', \zeta)/(\zeta - w_n)$ is continuous on the circle, hence measurable. For each fixed ζ on the circle the integrand is holomorphic in s : for $s = w_j$ the numerator is a partial map of $\tilde{f}$ at points of $A^{-1}(D \setminus E)$ and the denominator does not involve s , while for $s = w_n$ the numerator does not involve s and $s \mapsto 1/(\zeta - s)$ is holomorphic for $|s| < r = |\zeta|$. Near any fixed s_0 the integrand is dominated by a constant, namely M divided by the distance from the circle to a compact neighbourhood of s_0 when $s = w_n$, and $M/(r - |w_n|)$ when $s = w_j$, and constants are integrable on the circle. Being continuous and holomorphic in each variable, F is analytic, hence holomorphic, on Δ by theorem 8.1.7.

The set $\tilde{E} \cap \Delta$ has empty interior in Δ : an open subset of $\tilde{E} \cap \Delta$ would contain, for some fixed w' , a nonempty open subset of the slice $\{\zeta \mid (w', \zeta) \in \Delta\}$, hence infinitely many points of the finite set $N_{w'}$. Consequently $\Delta \setminus \tilde{E}$ is dense in Δ .

Now return through A . Set $\Omega_p := A(\Delta)$, an open neighbourhood of $p = A(0)$ contained in $U_p \cap D$, and $F_p := F \circ A^{-1}$, holomorphic on Ω_p by the slicing device applied to the invertible affine map A^{-1} . Since A maps $\Delta \setminus \tilde{E}$ onto $\Omega_p \setminus E$, we have $F_p = f$ on $\Omega_p \setminus E$, and since A is a homeomorphism, $E \cap \Omega_p$ has empty interior, so $\Omega_p \setminus E$ is dense in Ω_p .

Finally, define $F : D \rightarrow \mathbb{C}$ by $F := f$ on $D \setminus E$ and $F := F_p$ on Ω_p for each $p \in E$. The clauses are consistent. On $\Omega_p \setminus E$ the two prescriptions agree, by the previous paragraph. On an overlap $\Omega_p \cap \Omega_q$, the functions F_p and F_q are continuous and agree with f on $(\Omega_p \cap \Omega_q) \setminus E$. That set is dense in $\Omega_p \cap \Omega_q$, since any nonempty open subset of $\Omega_p \cap \Omega_q$ meets the complement of E , the set $E \cap \Omega_p$ having empty interior. Continuous functions agreeing on a dense subset of an open set agree on all of it, so $F_p = F_q$ there. Thus F is well defined, agrees with f on $D \setminus E$, and is holomorphic on a neighbourhood of each point of D : near points of $D \setminus E$ it is f , and near points of E it is some F_p . Since holomorphy is a local condition, $F \in \mathcal{O}(D)$ is the required extension, completing the proof.

Corollary 8.2.8: Nonzero Set is Connected

let f, D, E be as described in theorem 8.2.7 with the extra condition that $E \subsetneq D$. Then $D \setminus E$ is connected.

Proof:

Suppose for contradiction $D \setminus E = U_1 \coprod U_2$ with U_1, U_2 open and disjoint. Without loss of generality suppose $U_1 \neq \emptyset$. Then, define a function where $f \equiv 0$ on U_1 and $f \equiv 1$ on U_2 . By theorem 8.2.7, such an f must holomorphically extend to all of D . It must then be that $U_2 = \emptyset$ (or else there would be a contradiction), completing the proof.

Lemma 8.2.9: Bi-disc Extension Lemma

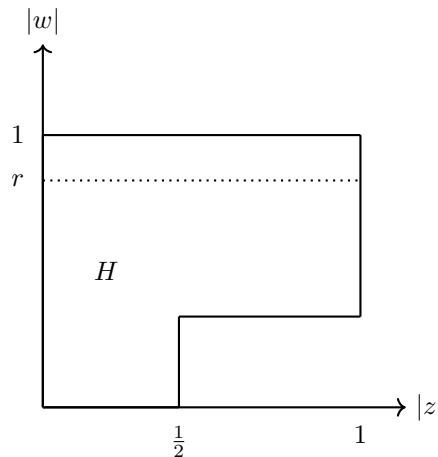
Let

$$H = \left\{ (z, w) \in \mathbb{C}^2 \mid |z| < 1, \frac{1}{2} < |w| < 1 \right\} \cup \left\{ |z| < \frac{1}{2}, |w| < 1 \right\}$$

Then every $f \in \mathcal{O}(H)$ has a holomorphic extension to the bi-disc

$$P = \{ (z, w) \in \mathbb{C}^2 \mid |z| < 1, |w| < 1 \}$$

H can be visualized as:



Proof:

This is lemma 8.2.2(2) with $n = 2$, $z' = z$, $P' = \{z \mid |z| < 1\}$, $Q' = \{z \mid |z| < \frac{1}{2}\}$, $c = 0$, $s = \frac{1}{2}$ and $r = 1$, as the set H above is the Hartogs figure of that lemma. There, for $\frac{1}{2} < t < 1$, the extension is given on $|w| < t$ by the Cauchy integral

$$F(z, w) = \frac{1}{2\pi i} \int_{|\zeta|=t} \frac{f(z, \zeta)}{\zeta - w} d\zeta$$

which equals f for $|z| < \frac{1}{2}$, as we sought to show.

Definition 8.2.10: Analytic Subset

an analytic subset $A \subseteq \mathbb{C}^n$ is the zero set of a system of (complex) analytic functions.

Theorem 8.2.11: Codimension Two Singularities

Let $D \subseteq \mathbb{C}^n$ where $n \geq 2$ be a domain and $E \subseteq D$ a closed subset which is analytic, that is, locally the zero set of finitely many holomorphic functions, and of complex dimension at most $n - 2$ at each of its points (or codimension at least 2). Then every $f \in \mathcal{O}(D \setminus E)$ extends holomorphically to D .

The dimension of an analytic set at a point, and the local structure theory behind it, are not developed in this book. Their definition is in Demailly [9, Chapter II, (4.25)], and the proof below rests on his Proposition (6.1) [9, Chapter II, (6.1)], which states that every $f \in \mathcal{O}(D \setminus E)$ is locally bounded near E when E has codimension at least 2.

Proof Key ideas:

By [9, Chapter II, (6.1)], every point of E has a neighbourhood B , which we may take to be a ball in D , with f bounded on $B \setminus E$. Near each $p \in E$, the set E is the common zero set of finitely many holomorphic functions, and one of them is not identically zero near p , since E has dimension less than n and so has empty interior. So $E \cap B$ satisfies the hypothesis of theorem 8.2.7 on B , which extends f holomorphically across $E \cap B$. The extensions on overlapping balls agree with f on the dense set off E , hence agree, and they glue to the required extension, which completes the sketch.

Another result that seems to be a consequence but is in fact different is the following:

Theorem 8.2.12: Totally Real Set Extension

Let $n \geq 2$. Then every $f \in \mathcal{O}(\mathbb{C}^n \setminus \mathbb{R}^n)$ extends holomorphically to $\mathbb{C}^n$.

This result is not the same as theorem 8.2.11 as $\mathbb{R}^n$ cannot be the zero-set of a complex function (notice it does not contain any non-trivial complex subspaces). It is a “totally real subset”. Its proof needs Bochner’s tube theorem, which this book does not prove. Demailly states it as an exercise with hints [9, Chapter I, Exercise 8.15, p. 63].

Proof Key ideas:

Multiplying by i carries $\mathbb{C}^n \setminus \mathbb{R}^n$ onto $\mathbb{C}^n \setminus i\mathbb{R}^n = \omega + i\mathbb{R}^n$, the tube over the base $\omega = \mathbb{R}^n \setminus \{0\}$, which is connected as $n \geq 2$. Bochner's tube theorem extends every function holomorphic on a connected tube to the tube over the convex hull of its base. The convex hull of ω is $\mathbb{R}^n$, and the tube over it is $\mathbb{C}^n$, which completes the sketch.

We looked at zero sets of higher-dimensional holomorphic functions, what about higher-dimensional poles-sets? These are in fact more “algebraic” in nature rather than analytic! They are not so easily analyzed with the tools we have developed as can be demonstrated with the following example: take $f(z_1, z_2) = \frac{z_1}{z_2}$. Then the point $(c, 0)$ a singularity. However, the limit-set of $f(z_1, z_2)$ as $(z_1, z_2) \rightarrow 0$ is the entire Riemann sphere! In the one-dimensional case, this is the behaviour of *essential singularities*, which we certainly would not think the given example here should be classified as. Instead, this example (and most example) are what's called *points of indeterminacy*. For a deeper exploration of these ideas when restricting to only complex polynomials, see[1].

Recall that if the degree of the numerator and denominator match for a meromorphic function in one dimension then we either have a pole or a removable singularity, showing again the difference between a single dimension and multiple dimensions.

8.3 an Overview of the Cousin Problem

Recall the Mittag-Leffler Theorem (theorem 4.2.7) and the Weierstrass Factorization Theorem (theorem 4.2.16) for decomposition a holomorphic function into sums or products. These decompositions are readily doable for complex polynomials, and require some massaging of the terms in the sequence to insure it converges in the general holomorphic case.

The higher-dimensional case does not readily generalize this strategy, namely since it is no longer the case, partly because the singularities are no longer isolated and need not even be compact anymore, but more important this “local” information of zero-sets is no longer capture by *global* holomorphic functions (the terms of the summand or product in the decomposition) but rather capture by *local* holomorphic functions.

To demonstrate this, let $U \subseteq \mathbb{C}^n$ be a domain and consider $f, g \in \mathcal{O}(U)$. Then f, g would describe the same set of zeros if and only if $f/g \in \mathcal{O}^*(U)$ (i.e. they belong to the units of the ring of holomorphic functions on U , i.e. nowhere vanishing holomorphic function).

Let us say we are trying to understand the zero sets in D of certain functions. Let us do this by decomposing $D = U_1 \cup U_2$ and considering the *local* zero set provided by two new functions $f_i \in \mathcal{O}(U_i)$ with $i = 1, 2$. We would need to “glue” these zero-sets together, which means that on $U_{12} = U_1 \cap U_2$, we must have $f_1/f_2 \in \mathcal{O}^*(U_{12})$. Let us assume this is the case; i.e. we have given the appropriate *local data*. Then we can now ask if this is sufficient (or even necessary) to find a function $f \in \mathcal{O}(D)$ whose zero-sets with each f_i (not necessarily agreeing outside the zero-sets), namely $\frac{f}{f_i} \in \mathcal{O}^*(U_i)$ for $i = 1, 2$. If $f_1 = f_2$ on U_{12} then the answer is immediate by the usual strategy of $f_1 - f_2 \equiv 0$ on U_{12} we use analytic continuation to not only glue the zero sets of f_1, f_2 but on the functions themselves (simply define $f(z) = f_i(z)$). Naturally, this is not necessarily the case. The natural next best thing is to ask whether we can find units $u_i \in \mathcal{O}^*(U_i)$ such that

$$\frac{f_1}{u_1} = \frac{f_2}{u_2} \text{ on } U_{12}$$

If this is possible, then we can define $f = \frac{f_i}{u_i}$ on each U_i to give a holomorphic function on D which shall give the desired zero-set. As u_i is invertible, we can write $u_i = \exp(g_i)$ for appropriate g_i (see exercise 3.2.1[3.2.1 (1)]) and so by manipulating to get $u_2/u_1 = f_2/f_1$, we can formulate the problem to be equivalent to finding:

$$g_2 - g_1 = \log \left(\frac{f_2}{f_1} \right) \text{ on } U_{12} \quad g_i \in \mathcal{O}(U_i)$$

The advantage to this additive notation is that we may see this now as a generalization the Mittag-Leffler Theorem. In particular, we can think of the local data as a collection of meromorphic functions f_i on U_i that must agree on intersections up to some functions g_{12} :

$$f_2 - f_1 = g_{12} \text{ in } \mathcal{O}(U_{12})$$

The functions f_i are called the *principal parts* on U_i . We thus see that we can often build *global* analytic objects that appropriately piecing together compatible *local* analytic objects.

The following is the correct formulation for the infinite case:

1: Cousin's Additive Problem

Let $D \subseteq \mathbb{C}^n$ be an open subset and $D = \bigcup_i^\infty U_i$ an open covering such that the functions $g_{ij} \in \mathcal{O}(U_i \cap U_j)$ are given such that on $U_i \cap U_j \cap U_k$ whenever this intersection is nonzero:

$$g_{ij} + g_{jk} + g_{ki} = 0$$

Then can find functions $g_i \in \mathcal{O}(U_i)$ such that $g_i - g_j = g_{ij}$ on $U_i \cap U_j$?

The extra condition is sometimes called the *cocycle condition* and is a necessary compatibility condition once there is more than 2 sets involved in the decomposition. Note that in the process of choosing such functions, when we take the logarithm we shall often have some obstructions to the cocycle condition due to the existence of these non-isolated zeros. This is a topological property that introduces an obstruction to finding such problems, and would be measure using cohomology, namely by looking at $H^2(D, \mathbb{Z})$. Note that in one dimension, $H^2(D, \mathbb{Z}) = 0$, showing the lack of concern in one-dimension. This shows that the limit of the generalization of Weierstrass Factorization Theorem. For examples, the zero set problem may fail to have a solution on the product of two annuli (see [15]).

The Cousin problem was first solved by *Pierre Cousin* in 1895 where $D = D_1 \times D_2 \times \cdots \times D_n$ where each D_i is a planar domain, essentially by systematically iterating over the domains and judiciously applying Cauchy's integral formula. It is a much harder problem to extend it more generally, as can already be surmised by remembering that we do not even have a simple classification of simply connected open subsets as noted in box 8.2.5. The problem has been extended greatly by a mathematician called *Kiyoshi Oka* from 1936. He put in the labour to be able to understand well these results, and then *Henri Cartan* was able to vastly expand upon the range of Oka's ideas, as well as make streamline his ideas using the at-the-time (1940s) new concept of *coherent analytic sheaf* and *sheaf cohomology theory* (first described by *Jean Leray*). By the 1960s and 70s, the problem has been considered solved problem over any complex manifold M ! Many of these ideas went onto inspire mathematicians such as *Alexander Grothendieck* and *Jean-Pierre Serre* to find the "right" algebraic object to unify a vast number of algebraic objects that were being studied, namely he define the notion of a *scheme* whose key property heavy use of *structure sheaves* to keep track of the function information that can be defined on open subsets on Spectra (see [1]).

**Further Explorations*

This textbook introduced the beginning of complex analysis. The journey of complex analysis has had huge leads and bounds throughout the 20th century. In fact, many of the problems of complex analysis have been “solved” and a good deal of the situation in which complex analysis occurs is now within the domain of complex/algebraic/Kähler geometry and harmonic analysis. Nevertheless, the reader who may wish to explore further the topics covered in this book may consider consulting the following list of references:

1. Value Distribution Theory:
 - (a) Nevanlinna Theory: this theory studies the asymptotic distribution of solutions of the equation $f(z) = a$ for holomorphic f as a varies https://en.wikipedia.org/wiki/Nevanlinna_theory
 - (b) Ahlfors Theory: the geometric counter-part to Nevanlinna theory https://en.wikipedia.org/wiki/Ahlfors_theory.
2. Uniformization Theorem: for the geometrically minded who want to see a generalization of the Riemann mapping theorem, they may be interested in the Uniformization Theorem (often covered in a standard course in Riemann Geometry). Note that the Riemann mapping theorem does not immediately extend to higher dimension such as the results from exercise 8.1.1 (1).
3. Complex Analysis in Operator Theory or Harmonic Analysis: For those in the functional analysis aspect of complex analysis, they may continue to study Harmonic analysis (a very rich field of study) or the complex analysis of operator theory which has some interesting applications in K -theory and real analysis.
4. Hodge Theory: Complex manifolds are even dimensional Real manifolds, but they certainly have more structure. Hodge Theory answers such questions.

5. Teichmüller Theory: this theory provides a framework for understanding the space of all possible complex structures on a surface of a given genus, up to conformal equivalence.

Bibliography

- [1] Nathanael Chwojko-Srawley. *Everything You Need to Know about Algebraic Geometry*. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algGeo.pdf.
- [2] Nathanael Chwojko-Srawley. *Everything You Need To Know About Analytic Number Theory*. URL: <https://nathanaelsrawley.com/pdf-viewer?src=.%2F.%2Fassets%2Fpdfs%2Fbooks%2FEYNTKA-analytic-number-theory.pdf&name=Analytic%20Number%20Theory&source=books>.
- [3] Nathanael Chwojko-Srawley. *Everything You Need To Know About Elementary Analysis*. 2024. URL: <https://nathanaelsrawley.com/assets/pdfs/books/EYNTKA-elementary-analysis.pdf>.
- [4] Nathanael Chwojko-Srawley. *Everything You Need To Know About Elliptic Curves*. 1st ed. 2025. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_elliptic_curves.pdf.
- [5] Nathanael Chwojko-Srawley. *Everything You Need To Know About Metric and Function Spaces*. Bootstrapped 2026-07-29 to metric-function-spaces/ from NateElemAnalysis Part II. Series home for metric completeness, the Banach contraction principle, Stone-Weierstrass, Arzela-Ascoli and Baire category; see DECISIONS 2026-07-29. 2026.
- [6] Nathanael Chwojko-Srawley. *Everything You Need To Know About Modular Forms*. 1st ed. Planned volume; stubbed so forward citations resolve (series rule: cite books, never bare prose). 2026.
- [7] Nathanael Chwojko-Srawley. *Everything You Need To Know About Riemannian Geometry*.
- [8] Nathanael Chwojko-Srawley. *Everything You Need To Know About Topology*.
- [9] Jean-Pierre Demailly. *Complex Analytic and Differential Geometry*. version of June 21, 2012. Grenoble: OpenContent book, Université de Grenoble I, 2012. URL: <https://www-fourier.ujf-grenoble.fr/~demailly/manuscripts/agbook.pdf>.
- [10] D. B. A. Epstein. “Prime Ends”. In: *Proceedings of the London Mathematical Society* s3-42.3 (May 1981), pp. 385–414. ISSN: 00246115. DOI: [10.1112/plms/s3-42.3.385](https://doi.org/10.1112/plms/s3-42.3.385). URL: <http://doi.wiley.com/10.1112/plms/s3-42.3.385> (visited on 04/02/2025).

- [11] Otto Forster. *Lectures on Riemann Surfaces*. New York, NY: Springer New York, 1981. ISBN: 978-1-4612-5963-3. DOI: [10.1007/978-1-4612-5961-9](https://doi.org/10.1007/978-1-4612-5961-9). URL: <https://doi.org/10.1007/978-1-4612-5961-9>.
- [12] John B. Garnett and Donald Eddy Marshall. *Harmonic Measure*. New Mathematical Monographs 2. Cambridge: Cambridge university press, 2005. ISBN: 978-0-521-47018-6.
- [13] Phillip Griffiths and Joseph Harris. *Principles of Algebraic Geometry*. 1st ed. John Wiley & Sons, Ltd, 1978. 824 pp. ISBN: 978-0-471-05059-9. DOI: [10.1002/9781118032527](https://doi.org/10.1002/9781118032527). URL: <https://onlinelibrary.wiley.com/doi/book/10.1002/9781118032527>.
- [14] Jack Hall. *GAGA Theorems*. May 17, 2022. DOI: [10.48550/arXiv.1804.01976](https://doi.org/10.48550/arXiv.1804.01976). arXiv: [1804.01976 \[math\]](https://arxiv.org/abs/1804.01976). URL: <http://arxiv.org/abs/1804.01976> (visited on 05/27/2025). Pre-published.
- [15] Sin Hitotumatu. “Cousin Problems for Ideals and the Domain of Regularity”. In: *Kodai Mathematical Seminar Reports* 3.1–2 (Jan. 1951), pp. 26–32. ISSN: 0023-2599. DOI: [10.2996/kmj/1138843066](https://doi.org/10.2996/kmj/1138843066). URL: <https://projecteuclid.org/journals/kodai-mathematical-seminar-reports/volume-3/issue-1-2/Cousin-problems-for-ideals-and-the-domain-of-regularity/10.2996/kmj/1138843066.full> (visited on 05/04/2025).
- [16] Daniel Huybrechts. *Complex Geometry: An Introduction*. Universitext. Berlin, Heidelberg: Springer, 2005. ISBN: 978-3-540-21290-4. DOI: [10.1007/b137952](https://doi.org/10.1007/b137952).
- [17] Ingo Lieb and Joachim Michel. “The Bochner-Martinelli-Koppelman Formula”. In: *The Cauchy-Riemann Complex*. Vieweg+Teubner Verlag, 2002, pp. 9–57. ISBN: 978-3-322-91608-2. DOI: [10.1007/978-3-322-91608-2_2](https://doi.org/10.1007/978-3-322-91608-2_2). URL: https://link.springer.com/chapter/10.1007/978-3-322-91608-2_2 (visited on 05/04/2025).
- [18] Michael Spivak. *Calculus*.
- [19] Michael Spivak. *Calculus on Manifolds*. Vol. 1. Boca Raton, 1965. URL: <https://doi.org/10.1201/9780429501906>.
- [20] Nathanael Chwojko-Srawley. *Everything You Need To Know About Algebraic Topology*. 1st ed. 126 pp. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algebraic_topology.pdf.
- [21] Nathanael Chwojko-Srawley. *Everything You Need To Know About Class Field Theory*. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_class_field_theory.pdf.
- [22] Nathanael Chwojko-Srawley. *Everything You Need To Know About Core Algebra*. 1st ed. 1210 pp. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algebra.pdf.
- [23] Nathanael Chwojko-Srawley. *Everything You Need To Know About Differential Topology*. 1st ed. 233 pp. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_differential_geometry.pdf.
- [24] Nathanael Chwojko-Srawley. *Everything You Need To Know About Functional Analysis*. 1st ed. 62 pp. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_functional_analysis.pdf.
- [25] Nathanael Chwojko-Srawley. *Everything You Need To Know About Real Analysis*. 2026. URL: <https://nathanaelsrawley.com/assets/pdfs/books/EYNTKA-real-analysis.pdf>.
- [26] William Fulton. *Algebraic Curves. An Introduction to Algebraic Geometry*. URL: <https://dept.math.lsa.umich.edu/~wfulton/CurveBook.pdf>.